School of Molecular and Life Sciences
Faculty of Science and Engineering

The development and application of eDNA metabarcoding for aquatic biomonitoring in Australia's Indian Ocean region

Katrina Marie West

This thesis is presented for the Degree of
Doctor of Philosophy
of
Curtin University

March 2020

Declaration

To the best of my knowledge and belief this thesis contains no material previously published by any other person except where due acknowledgment has been made.

This thesis contains no material which has been accepted for the award of any other degree or diploma in any university.

SIGNED | Katrina West

DATED | 5 March 2020

Abstract

The speed and low cost of next-generation sequencing (NGS) platforms have enabled the mass parallel sequencing of complex environmental samples. This has facilitated the development of environmental DNA (eDNA) metabarcoding, a technique which characterises DNA shed by both aquatic and terrestrial organisms into their immediate surroundings to rapidly identify complex species assemblages. The use of eDNA to discover biodiversity or investigate temporal/spatial patterns in marine environments has considerable potential, but there are few studies that have applied these technologies in Australia.

The research described within this thesis focuses on the development and application of eDNA metabarcoding techniques across a variety of aquatic ecosystems and spatial scales within Australia's Indian Ocean region. I investigate the localisation of marine eDNA across a coral reef atoll and compare metabarcoding efficiency to conventional survey techniques (Chapter 2). I then test the ability of eDNA metabarcoding to discern fine-scale patterns across large-scale bioregions along the coastline of north-western Australia, develop a broad-taxonomic assay to detect aquatic reptiles and update distribution information for a number of endangered and elusive taxa, including potential biosecurity risks (Chapter 3). In my last data chapter, I investigate the utility of eDNA metabarcoding in subterranean environments and combine eDNA-derived community composition and environmental (water quality) data to explore potential underground interconnectivity between an island's cave systems (Chapter 4).

A multi-trophic survey of coral reef composition at the Cocos (Keeling) Islands (CKI), Australia revealed that site composition varied significantly over a small spatial scale, reflecting habitat partitioning across the island ecosystem and demonstrating the localisation of eDNA signals, despite extensive tidal and oceanic movements through the atoll. The eDNA approach proved to be equally efficient in detecting fish and elasmobranch diversity when compared to recent baited remote underwater video (BRUV) surveys at the CKI. Interestingly, there was no overlap in new occurrence records between the BRUV and eDNA surveys, demonstrating their potential utility to provide complementary baseline data (Chapter 2).

In order to assess the applicability over large-scale regions, a multi-marker eDNA metabarcoding survey was conducted over 700 km of north-western Australian (NWA) coastline. Bioregional and depth effects on inshore bony fish and shark/ray compositional data support a purported marine biogeographic break across Cape Leveque, demarcating the border between the Kimberley and Canning bioregions. In addition, eDNA detected elusive species and updated distribution information for critically endangered and data-deficient taxa such as sawfish (*Pristis sp.*), northern river shark (*Glyphis garricki*), sea snakes (*Hydrophis stokesii* and *Fordonia leucobalia*) and wedgefish (*Rhynchobatus sp.*) (Chapter 3).

Finally, to explore the biodiversity in a poorly characterised ecosystem, an eDNA metabarcoding survey was executed across the subterranean karst landscape of Christmas Island (CI), Australia. This revealed that high-quality metabarcoding compositional data can be gleaned from understudied ecosystems, despite suboptimal databases. Aquatic subterranean community composition was found to vary between cave and spring sites and across salinity gradients. Cave sites were additionally influenced by dissolved oxygen and longitudinal effects. Hierarchical clustering of community composition and environmental data identified three cave/spring groups with a high probability of underground interconnection (Chapter 4).

Collectively the research undertaken as part of this thesis genetically profiled in excess of 700 water samples across three study regions and applied up to six eDNA metabarcoding assays generating ~155 million metabarcoding reads. This resulted in the assignment of ~1000 (primarily marine) taxa and the establishment of multiple baselines upon which regular eDNA biomonitoring can be planned and executed. The studies presented here address the viability of eDNA metabarcoding across a variety of applications within Australia. In a broader context, this thesis data adds to a growing body of literature that demonstrates the viability of eDNA as a tool to study aquatic biodiversity at a time when climate and anthropogenic influences necessitate careful management and monitoring of biota.

— Acknowledgements —

The undertaking of this PhD has been both challenging and rewarding. It has allowed me to explore remote and remarkable locations, meet a wide variety of people, expand my scientific and adventuring skills and immerse myself in the extraordinary ecology of Australia's Indian Ocean region. This PhD has taught me the power of persistence and the importance of collaboration which underpin all scientific endeavours. I could not have completed this PhD without the support of my collaborators, supervisors, colleagues, friends and loved ones.

I'd firstly like to thank everyone at the Trace and Environmental DNA (TrEnD) Laboratory, Curtin University for their unwavering support over the last three and a half years. In particular, to my extensive supervisory panel of Prof. Mike Bunce, Prof. Euan Harvey, Dr. Michael Stat, Dr. Zoe Richards and Dr. Joey DiBattista. Your guidance and approach to good science has been invaluable and has imparted me with the knowledge and skills to build my own career.

To my colleagues and friends at TrEnD—thank you for your good humour and deliveries of chocolate and beer during the challenging times! I would like to thank Matt Power, Megan Coghlan, Alicia Grealy, Tina Berry and Nicole White for teaching and kickstarting my foray into the world of eDNA. I would also like to give special mention to my office and lab friends: Adam Koziol, Matt Heydenrych, Frederik Seersholm, Miwa Takahashi, Mieke Van Der Heyde, Kristen Fernandes, Arne Adam, Georgia Nester, Sophie Preston, Jackie Jones, Alishum Ali, David Juszkiewicz, Jason Alexander and Sarah Crump. Our ability to troubleshoot QIAGEN robots is second to none.

The input of my collaborators and co-authors has been invaluable over the duration of my PhD. I would firstly like to acknowledge Mike Travers, Craig Skepper and Steve Newman at the Fisheries Division of the WA Department of Primary Industries and Regional Development (DPIRD). The Cocos (Keeling) Islands and Northwestern Australian projects would not have been possible without your guidance and skill sets. The support of the Western Australian Speleological Group (WASG) during the

Christmas Island project was instrumental to its success (and safety). Thank you so much for the cave and abseiling training sessions in the lead up, and the support and cave navigation skills from the WASG team during the two-week cave expedition on Christmas Island. To both DPIRD and WASG, it was an honour to visit these remarkable places alongside you.

Above all, my sincerest appreciation goes to my parents (Alicja and Graham West), siblings (Angela, Tim, Russell and Sophie) and friends (Emily, Kate, Llewelyn, Kat and more) who have supported me during my time at university. It is bittersweet knowing that my university journey is coming to an end, but I am looking forward to new adventures and scientific conundrums.

Finally, I would like to recognise financial support for my PhD projects from the Australian Research Council, DPIRD, BMT Oceanica and the Holsworth Wildlife Research Endowment. I was personally supported by an Australian Government Research Training Program (RTP) Scholarship over the duration of my PhD research.

— Table of Contents —

Declaration ..... **ii**

Abstract ..... **iii**

Acknowledgements ..... **v**

Table of Contents ..... **vii**

Statement of Contributions ..... **xi**

List of Figures ..... **xii**

List of Tables ..... **xv**

Abbreviations ..... **xix**

Chapter 1 ..... **1**

1.1 Australia's Indian Ocean region *2*
1.1.1 The biodiversity and economic value of Australia's Indian Ocean region 2
1.1.2 Threats to management, sustainability and conservation 5

1.2 eDNA metabarcoding: an emerging tool for biomonitoring *8*
1.2.1 Background of eDNA research 8
1.2.2 eDNA metabarcoding workflow 10
1.2.3 The preliminary use of eDNA metabarcoding in Australian waters 14

1.3 eDNA metabarcoding in practice *17*
1.3.1 Benefits of applied eDNA metabarcoding 17
1.3.2 Further applications and methodological testing 18

1.4 Research question and thesis structure *22*

1.5 References *26*

Chapter 2 ..... **41**

2.1 Prologue *42*
2.1.1 Chapter acknowledgements 44
2.1.2 Data accessibility 45

2.2 Abstract *46*

2.3 Introduction *47*

2.4 Methods *50*
2.4.1 Field sampling 50
2.4.2 Laboratory processing 50
2.4.3 Bioinformatics and taxonomic assignments 52
2.4.4 Statistical analyses 54

2.5 Results *55*
2.5.1 Sampling and sequencing statistics 55

2.5.2 Overall diversity 56
2.5.3 Community composition 63
2.5.4 Comparison of eDNA to previous CKI monitoring techniques 66

2.6 Discussion *69*
2.6.1 eDNA metabarcoding baseline of a tropical island reef ecosystem 69
2.6.2 Bony fish and elasmobranch diversity and community composition 69
2.6.3 Crustacean diversity and community composition 72
2.6.4 Mollusc diversity and community composition 73
2.6.5 Echinoderm diversity and community composition 74
2.6.6 Future integration in reef monitoring 75

2.7 Conclusion *78*

2.8 References *79*

2.9 Supplementary information *88*
S2.9.1 Habitat information 90
S2.9.2 Laboratory processing 91
S2.9.3 Bioinformatics and taxonomic assignments 92
S2.9.4 Statistical analyses 102
S2.9.5 Assay attributes 105
S2.9.6 Similarity percentage and indicator species analyses 137
S2.9.7 Bony fish community composition 153
S2.9.8 Supplementary references 155

Chapter 3 158

3.1 Prologue *159*
3.1.1 Chapter acknowledgements 161
3.1.2 Data accessibility 162

3.2 Abstract *163*

3.3 Introduction *164*

3.4. Methods *167*
3.4.1 Field sampling 167
3.4.2 DNA extraction 167
3.4.3 Metabarcoding assay design, amplification & library sequencing 167
3.4.4 Bioinformatics 170
3.4.5 Statistics 171

3.5 Results *173*
3.5.1 Sampling and sequencing statistics 173
3.5.2 Overall diversity 174
3.5.3 Community composition 178
3.5.4 Haplotype (ASV) diversity in stripey snapper (*L. carponotatus*) 184

3.6 Discussion *187*
3.6.1 Bony fish compositional transitions across NWA 187
3.6.2 Shark and ray compositional transitions across NWA 189
3.6.3 Haplotype community variation in stripey snapper across NWA 190
3.6.4 A new approach for surveying endangered, elusive and data-deficient taxa in NWA 192

3.7 Conclusion *195*

3.8 References *196*

3.9 Supplementary information *206*
S3.9.1 Laboratory processing 209
S3.9.2 Bioinformatics processing 210
S3.9.3 Primer development note for 16S Reptile 250
S3.9.4 Supplementary references 262

Chapter 4 ..... **267**

4.1 Prologue *268*
4.1.1 Chapter acknowledgements 270
4.1.2 Data accessibility 271

4.2 Abstract *272*

4.3 Introduction *273*

4.4 Methods *276*
4.4.1 Field sampling 276
4.4.2 Laboratory processing 280
4.4.3 Bioinformatics 280
4.4.4 Statistical analyses 282

4.5 Results *284*
4.5.1 Sampling and sequencing statistics 284
4.5.2 Overall diversity 285
4.5.3 Community composition and clustering 289

4.6 Discussion *295*
4.6.1 A new approach to biospeleological surveying 295
4.6.2 Subterranean composition 296
4.6.3 Subterranean connectivity 297

4.7 Conclusion *300*

4.8 References *302*

4.9 Supplementary information *308*
S4.9.1 qPCR reagents and conditions 309
S4.9.2 Bioinformatic parameters 309
S4.9.3 Supplementary references 328

Chapter 5 ..... **329**

5.1 Summary of findings *330*

5.2 Limitations *335*

5.3 Significance of thesis *338*

5.4 Future directions *340*
5.4.1 New aims and questions arising from this thesis 340
5.4.2 New technologies for eDNA metabarcoding 342

5.5 Thesis conclusion *347*

5.6 References *348*

Appendices ..... **xxi**

Appendix I. Copyright statements ..... *xxii*

Appendix II. Research output ..... *xxvii*
II.I Publications xxvii
II.II Awards and conference presentations xxviii
II.III Public outreach xxix

── Statement of Contributions ──

Chapter 2: West K.M, Stat M, Harvey E.S, Skepper C.L, DiBattista J.D, Richards Z.T, Travers M.J, Newman S.J, Bunce M. (2020). eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem. *Molecular Ecology*. https://doi.org/10.1111/mec.15382.

Author contributions: Funding acquisition, M.B, M.S, E.S.H, J.D.D, S.J.N; conceptualisation, M.B, M.S, E.S.H, J.D.D, S.J.N, K.M.W; fieldwork, K.M.W, E.S.H, C.L.S; laboratory processing and analysis, K.M.W, M.S, J.D.D, Z.T.R, M.J.T, M.B; writing and editing of manuscript, K.M.W, M.S, E.S.H, J.D.D, S.J.N, Z.T.R, M.J.T, M.B.

Chapter 3: West K.M, Travers M.J, Stat M, Harvey E.S, Richards Z.T, DiBattista J.D, Skepper C.L, Heydenrych M, Newman S.J, Bunce M. Large-scale eDNA metabarcoding survey reveals marine biogeographic break and transitions over tropical north-western Australia. *[in preparation]*

Author contributions: Funding acquisition, M.B, M.S, E.S.H, J.D.D, S.J.N; conceptualisation, M.B, M.S, E.S.H, J.D.D, S.J.N, M.J.T K.M.W; fieldwork, K.M.W, M.J.T, C.L.S; laboratory processing and analysis, K.M.W, M.H, M.J.T; writing and editing of manuscript, K.M.W, M.J.T, E.S.H, Z.T.R, J.D.D, M.S, S.J.N, M.H, M.B.

Chapter 4: West K.M, Richards Z.T, Harvey E.S, Susac R, Grealy A, Bunce M. Under the karst: detecting hidden subterranean assemblages using eDNA metabarcoding in the caves of Christmas Island, Australia. *[in preparation]*

Author contributions: Funding acquisition, K.M.W; conceptualisation, K.M.W, A.G; fieldwork, K.M.W, R.S; laboratory processing and analysis, K.M.W, A.G, Z.T.R, E.S.H, M.B; writing and editing of manuscript, K.M.W, A.G, Z.T.R, E.S.H, R.S, M.B.

List of Figures

Figure 1.1 The Australian Government's official interpretation of the Indian Ocean and other ocean boundaries surrounding Australia (Australian Hydrographic Office, 2019). 3

Figure 1.2 Conceptual flow diagram of the general thesis structure..... 25

Figure 2.1 A specimen of *Scarus chlorodon* (now *Scarus prasiognathos*) collected by Charles Darwin at the Cocos (Keeling) Islands in 1836. 43

Figure 2.2 (A) Location of the Cocos (Keeling) Islands (CKI) in the eastern Indian Ocean. (B) Satellite imagery of the main CKI atoll with an inset of North Keeling atoll (not to scale). (C) Marine habitat zone map of the CKI, based on previous habitat strata divisions reported by Parks Australia (2005), Hender et al. (2001) and Williams (1994). 51

Figure 2.3 Order level dendrogram of bony fish, elasmobranch, crustacean, mollusc and echinoderm diversity detected at the CKI using eDNA metabarcoding. 57

Figure 2.4 Taxa of interest detected by eDNA at the CKI. 61

Figure 2.5 Non-metric multi-dimensional scaling (nMDS) ordination plots of a) bony fish and elasmobranch, b) crustaceans, c) molluscs and d) echinoderm community compositions. 64

Figure S2.1 Rarefaction analyses of sequencing depth for the Fish 16S, Crustacean 16S, Elasmobranch COI and Universal 18S assays. 104

Figure S2.2 Proportion of sequencing reads from the 16S Fish assay, 16S Crustacean assay, 18S Universal assay, and COI Elasmobranch assay, attributed to various taxonomic classes. 107

Figure S2.3 Taxonomic accumulation curves for the targeted taxonomic groups (bony fish & elasmobranchs, crustaceans, molluscs and echinoderms). 108

Figure S2.4 Overall taxonomic richness across marine habitat zones. 125

Figure S2.5 Taxonomic richness across the marine habitat zones by taxonomic group. 128

Figure S2.6 Overall taxonomic richness across lagoon zones. 132

Figure S2.7 Taxonomic richness across the lagoon zones by taxonomic group..... 133

Figure 3.1 The first scientific drawings of Australian marine life as illustrated by William Dampier during the *Roebuck* voyage in 1699. 160

Figure 3.2 Location of sampling sites across the Canning and Kimberley bioregions in north-western Australia (NWA). 168

Figure 3.3 Order level dendrogram of bony fish, sharks and reptile diversity detected across the Canning and Kimberley bioregions using eDNA metabarcoding. 175

Figure 3.4 Principal coordinates analysis (PCO) of bony fish composition in a) inshore, b) coastal and c) nearshore estuarine sites. 181

Figure 3.5 Principal coordinates analysis (PCO) of shark and ray composition in a) inshore, b) coastal and c) nearshore estuarine sites..... 183

Figure 3.6 Median-joining (MJ) network of 16S rDNA haplotypes and corresponding haplotype distribution map for the stripey snapper (*Lutjanus carponotatus*) in NWA. 186

Figure S3.1 ASV accumulation curves by sampling replicate for the 16S Fish assay. 213

Figure S3.2 ASV accumulation curves by sampling replicate for the COI Elasmobranch assay. 214

Figure S3.3 ASV accumulation curves by sampling replicate for the 16S Reptile assay. 215

Figure S3.4 ASV rarefaction curve by sequencing depth for the Fish 16S assay.... 216

Figure S3.5 Ratio between rarefied and observed number of ASVs for the Fish 16S assay. 217

Figure S3.6 ASV accumulation curve by site for the 16S Fish assay. 217

Figure S3.7 Accumulation curve of ASVs by sequencing depth for the Elasmobranch COI assay. 218

Figure S3.8 Ratio between rarefied and observed number of ASVs for the Elasmobranch COI assay. 219

Figure S3.9 ASV accumulation curve by site for the COI Elasmobranch assay. 219

Figure S3.10 Accumulation curve of ASVs by sequencing depth for the Reptile 16S assay. 220

Figure S3.11 Ratio between rarefied and observed number of ASVs for the Reptile 16S assay. 221

Figure S3.12 ASV accumulation curve by site for the 16S Reptile assay. 221

Figure S3.13 Taxonomic richness in inshore bony fish across the bioregions. 241

Figure S3.14 Taxonomic richness in inshore shark and ray taxa across the subregions. 246

Figure S3.15 Average proportion of sequencing reads per sample for the 16S Reptile assay. 260

Figure 4.1 Depiction of Christmas Island by Captain Daniel Beeckman in 1714. .. 269

Figure 4.2 Location of eDNA sampling sites on Christmas Island..... 277

Figure 4.3 Imagery from eDNA sampling localities across Christmas Island..... 279

Figure 4.4 A total ordinal-level dendrogram of chordate, cnidarian, porifera, arthropod, mollusc, annelid and bryozoan taxa detected by multi-assay metabarcoding on 159 eDNA samples collected across Christmas Island. 286

Figure 4.5 Principal coordinates analysis (PCO) of bony fish composition in a) all sites, b) cave sites only and c) spring sites only..... 292

Figure 4.6 Cluster analysis of CI community composition similarity. 293

Figure 4.7 Cluster analysis of CI abiotic environmental similarity. 294

Figure S4.1 Taxa accumulation curve by the addition of replicates (six water and one sediment). 312

Figure S4.2 Principal coordinates analysis (PCO) of cave faunal composition by substrate (water and sediment). 314

Figure S4.3 Boxplots of total taxa per sediment and water replicate..... 315

Figure S4.4 Taxa accumulation curve by the addition of sites (cave/spring). 317

Figure S4.5 Taxa richness in subterranean fauna across the two site types..... 324

List of Tables

Table 1.1 Environmental DNA metabarcoding studies that have been conducted in Australian aquatic ecosystems. 15
Table 2.1 PCR assay information for marine eDNA metabarcoding at CKI. 53
Table 2.2 Taxa of interest 58
Table 2.3 Summary table of the distance based linear model (DistLM) analyses. 65
Table 2.4 Overview of bony fish, elasmobranch and crustacean taxonomic surveys conducted at the CKI, ordered by decreasing year. 67
Table S2.1 Site location information at the Cocos (Keeling) Islands. 88
Table S2.2 An *in-silico* analysis of average percent identities for 16S Fish amplicons. 95
Table S2.3 An *in-silico* analysis of average percent identities for COI Elasmobranch amplicons. 96
Table S2.4 An *in-silico* analysis of average percent identities for 16S Crustacean amplicons. 97
Table S2.5 An *in-silico* analysis of average percent identities for 18S Universal amplicons. 98
Table S2.6 In-house 16S rDNA fish database. 99
Table S2.7 Mean reads per sample in bioinformatic processing. 103
Table S2.8 Consolidated table of eDNA detections at the CKI. 109
Table S2.9 Sequencing statistics of new occurrence records. 122
Table S2.10 ANOVA of Taxon richness between marine habitat zones. 126
Table S2.11 Tukey multiple comparisons of taxonomic richness between marine habitat zones across all taxonomic groups. 127
Table S2.12 Tukey multiple comparisons of bony fish and elasmobranch taxonomic richness. 129
Table S2.13 Tukey multiple comparisons of crustacean taxonomic richness. 130
Table S2.14 Tukey multiple comparisons of echinoderm taxonomic richness. 131
Table S2.15 ANOVA of Taxon richness between lagoon zones. 134
Table S2.16 Tukey multiple comparisons of echinoderms. 134
Table S2.17 Distance based linear model (DistLM) analysis of bony fish and elasmobranch community variation. 135

Table S2.18 Distance based linear model (DistLM) analysis of crustacean community variation..... 135
Table S2.19 Distance based linear model (DistLM) analysis of mollusc community variation..... 136
Table S2.20 Distance based linear model (DistLM) analysis of echinoderm community variation..... 136
Table S2.21 Similarity percentage analysis (SIMPER) of bony fish and elasmobranch taxa across the lagoon zones. 139
Table S2.22 Similarity percentage analysis (SIMPER) of bony fish and elasmobranch taxa across the marine habitat zones. 141
Table S2.23 Similarity percentage analysis (SIMPER) of crustacean taxa across the lagoon zones. 143
Table S2.24 Similarity percentage analysis (SIMPER) of crustacean taxa across the marine habitat zones. 143
Table S2.25 Similarity percentage analysis (SIMPER) of mollusc taxa across the lagoon zones. 144
Table S2.26 Similarity percentage analysis (SIMPER) of mollusc taxa across the marine habitat zones. 144
Table S2.27 Similarity percentage analysis (SIMPER) of echinoderm taxa across the lagoon zones. 145
Table S2.28 Indicator species analyses of lagoon zones. 146
Table S2.29 Indicator species analyses of marine habitat zones. 148
Table S2.30 Harvey et al. (unpublished) recorded species list for BRUV surveys conducted in 2016 at CKI. 149
Table S2.31 Harvey et al. (unpublished) recorded species list for BRUV surveys conducted in 2017 at CKI. 151
Table 3.1 PCR assay information for marine eDNA metabarcoding across the Canning/Kimberley bioregions. 169
Table 3.2 Selected taxa of conservation, biosecurity, cultural, recreation and/or commercial importance detected via eDNA. 176
Table 3.3 Summary table of the distance based linear model (DistLM) analyses for bony fish. 180
Table 3.4 Summary table of the distance based linear model (DistLM) analyses for sharks and rays. 182

Table S3.1 Site location information across the Canning/Kimberley bioregions. ... 206
Table S3.2 Mean reads per sample in bioinformatic processing. ... 212
Table S3.3 Consolidated taxa of eDNA detections across NWA. ... 222
Table S3.4 Distance based linear model (DistLM) analysis of bony fish community variation across the inshore sites. ... 239
Table S3.5 ANOVA of taxonomic richness of inshore bony fish between bioregions. ... 240
Table S3.6 Similarity percentage analysis (SIMPER) of bony fish in inshore sites. ... 242
Table S3.7 Distance based linear model (DistLM) analysis of bony fish community variation across the coastal sites. ... 243
Table S3.8 Distance based linear model (DistLM) analysis of bony fish community variation across the nearshore estuarine sites. ... 243
Table S3.9 Distance based linear model (DistLM) analysis of shark and ray community variation across the inshore sites. ... 244
Table S3.10 ANOVA of taxonomic richness of inshore shark and ray between subregions. ... 245
Table S3.11 Similarity percentage analysis (SIMPER) of sharks and rays in inshore sites. ... 247
Table S3.12 Distance based linear model (DistLM) analysis of shark and ray community variation across the coastal sites. ... 248
Table S3.13 Distance based linear model (DistLM) analysis of shark and ray community variation across the nearshore estuarine sites. ... 248
Table S3.14 Distance based linear model (DistLM) analysis of stripey snapper variation. ... 249
Table S3.15 List of targeted Australian aquatic and semi-aquatic reptile species for in-silico primer analysis. ... 259
Table S3.16 Reptile taxa detected from environmental samples in northwestern Australia. ... 261
Table 4.1 eDNA site information on Christmas Island. ... 278
Table 4.2 Metabarcoding assay information for subterranean eDNA surveys on Christmas Island. ... 281
Table 4.3 Taxa of conservation, biodiversity and biosecurity importance. ... 287

Table 4.4 Summary table of the distance based linear model (DistLM) analyses for subterranean fauna. 291
Table S4.1 Environmental parameter data collected at the time of eDNA sampling. 308
Table S4.2 Mean reads per replicate sample in bioinformatic processing. 311
Table S4.3 Two-way crossed PERMANOVA testing for the effect of site and substrate (water and sediment). 313
Table S4.4 Similarity percentage analysis (SIMPER) of CI subterranean taxa. 316
Table S4.5 Consolidated taxa of eDNA detections across Christmas Island. 318
Table S4.6 Distance based linear model (DistLM) analysis of subterranean faunal community variation. 322
Table S4.7 One-way ANOVA testing for the effect of site type (cave or spring) on site richness. 323
Table S4.8 Similarity percentage analysis (SIMPER) of CI subterranean taxa. 325
Table S4.9 Similarity percentage analysis (SIMPER) of CI subterranean taxa. 326

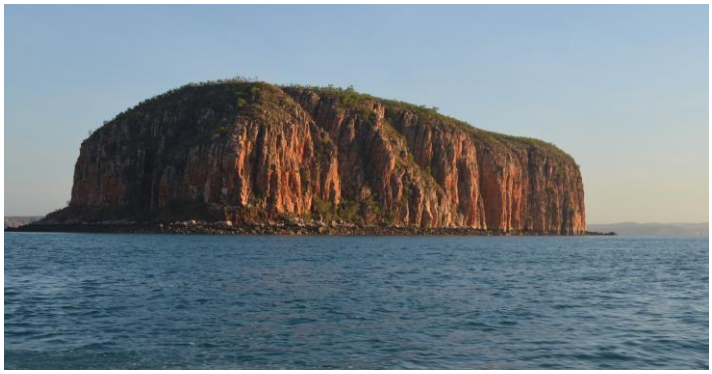
Abbreviations

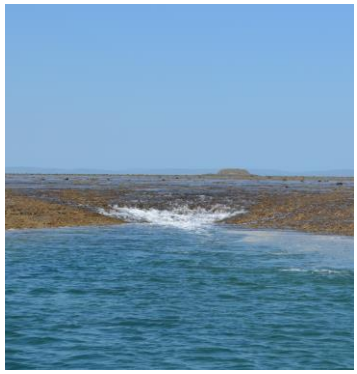
αHL	α-haemolysin
ΔRn	Normalised reporter value
AET	Australian External Territory
ANOVA	Analysis of variance
ASV	Amplicon sequence variant
AUD	Australian Dollar
BLAST	Basic Local Alignment Search Tool
BOLD	Barcode of Life Database
bp	Base pairs
BRUV	Baited remote underwater video
cDNA	Complementary DNA
CI	Christmas Island
CKI	Cocos (Keeling) Islands
COI	Mitochondrial cytochrome c oxidase I
Ct	Fluorescence threshold
ddPCR	Droplet digital PCR
DistLM	Distance-based linear modelling
DNA	Deoxyribonucleic acid
dNTPs	Deoxynucleotide triphosphates
DPIRD	Department of Primary Industries and Regional Development
eDNA	Environmental DNA
EEZ	Exclusive economic zone
eRNA	Environmental RNA
GBRMP	Great Barrier Reef Marine Park
GBS	Genotype by sequencing
GDP	Gross domestic product
HVR1	Hypervariable region 1
iBOL	International Barcode of Life
IHO	International Hydrographic Organization
ITS	Internal transcribed spacer
IUCN	International Union for Conservation of Nature

IUU	Illegal, unreported and unregulated (fishing)
MEGAN	Metagenome Analyzer
MJ	Median-joining (network)
mtDNA	Mitochondrial DNA
NCBI	National Center for Biotechnology Information
NGS	Next-generation sequencing
nMDS	Non-metric multidimensional scaling
NMP	Ningaloo Reef Marine Park
NRSMPA	National Representative System of Marine Protected Areas
NWA	North-western Australia
OTU	Operational taxonomic unit
PCO	Principal coordinates analysis
PCR	Polymerase chain reaction
PERMANOVA	Permutational multivariate analysis of variance
PERMDISP	Permutational analysis of multivariate dispersions
qPCR	Quantitative PCR
RDP	Ribosomal Database Project
RNA	Ribonucleic acid
rRNA	Ribosomal RNA
RTP	Australian Government Research Training Program Scholarship
RT-PCR	Real-time PCR
SIMPER	Similarity percentage analyses
SIMPROF	Similarity profile test
SMRT	Single molecule real-time sequencing
TGS	Third-generation sequencing
ToL	Tree-of-Life (metabarcoding)
TrEnD	Trace and Environmental DNA Laboratory
UNESCO	United Nations Educational, Scientific and Cultural Organization
WASG	Western Australian Speleological Group
WoRMS	World Register of Marine Species
ZOTU	Zero-radius operational taxonomic unit

Chapter 1

Introduction











1.1 Australia's Indian Ocean region

1.1.1 The biodiversity and economic value of Australia's Indian Ocean region

Australia's Indian Ocean region comprises approximately 15,000 km of tropical to temperate coastline (10.58° S, 142.07° E to 34.36° S, 115.13° E; Figure 1.1), over 4000 islands, three Australian External Territories (AET) and extensive offshore and coastal reef systems, including the World Heritage-listed Ningaloo Reef (Abbott & Wills, 2016; Australian Hydrographic Office, 2019). This region exhibits a variety of marine, coastal, estuarine and subterranean aquatic landscapes that are in part, connected through the Leeuwin Current, a poleward flowing surface stream set up by the South Indian Ocean Counter Current and the Indonesian Throughflow (Lambert, Le Bars, & de Ruijter, 2016; Yit Sen Bull & Van Sebille, 2016). This current directs warm, low-salinity tropical water southwards from northern Australia to Cape Leeuwin in the southwest, and beyond into the Great Australian Bight. The current and its associated eddies, act as a southwards mechanism to distribute larvae (Caputi, Fletcher, Pearce, & Chubb, 1996) and facilitate migration of juvenile fish such as southern bluefin tuna (*Thunnus maccoyii*) and Australian salmon (*Arripis trutta*) (Cresswell, 1986; Hobday, 2010). The Leeuwin Current enables the presence of coral reefs as far south as the Houtman Abrolhos (29°S) and has dispersed tropical reef fishes into temperate localities, such as Rottnest Island (32°S) (Hutchins & Pearce, 1994). As such, Australia's Indian Ocean region is a global hotspot of coastal and oceanic biodiversity and endemism, and has some of the least impacted marine ecosystems in the world (Halpern et al., 2008; Tittensor et al., 2010). The western coastline of Australia boasts a higher number of marine fish (over 3000 species) than any other Indian Ocean rim country, including coral-rich Indonesia (Hutchins, 2001; Keesing & Irvine, 2005). Importantly, this region presents a refuge for vulnerable and critically endangered taxa, such as the dugong (*Dugong dugon*), flatback turtle (*Natator depressus*), bottlenose wedgefish (*Rhynchobatus australiae*), the largetooth sawfish (*Pristis pristis*), hard corals (order: Scleractinia) and giant clams (genus: *Tridacna*) (Kyne, Rigby, Dharmadi, & Jabado, 2019; Kyne, Carlson, & Smith, 2013; Last & Stevens, 2009; Rees, Colquhoun, Smith, & Heyward, 2003; Richards et al., 2019; Wells, 1997; Whiting, Thomson, Chaloupka, & Limpus, 2009).

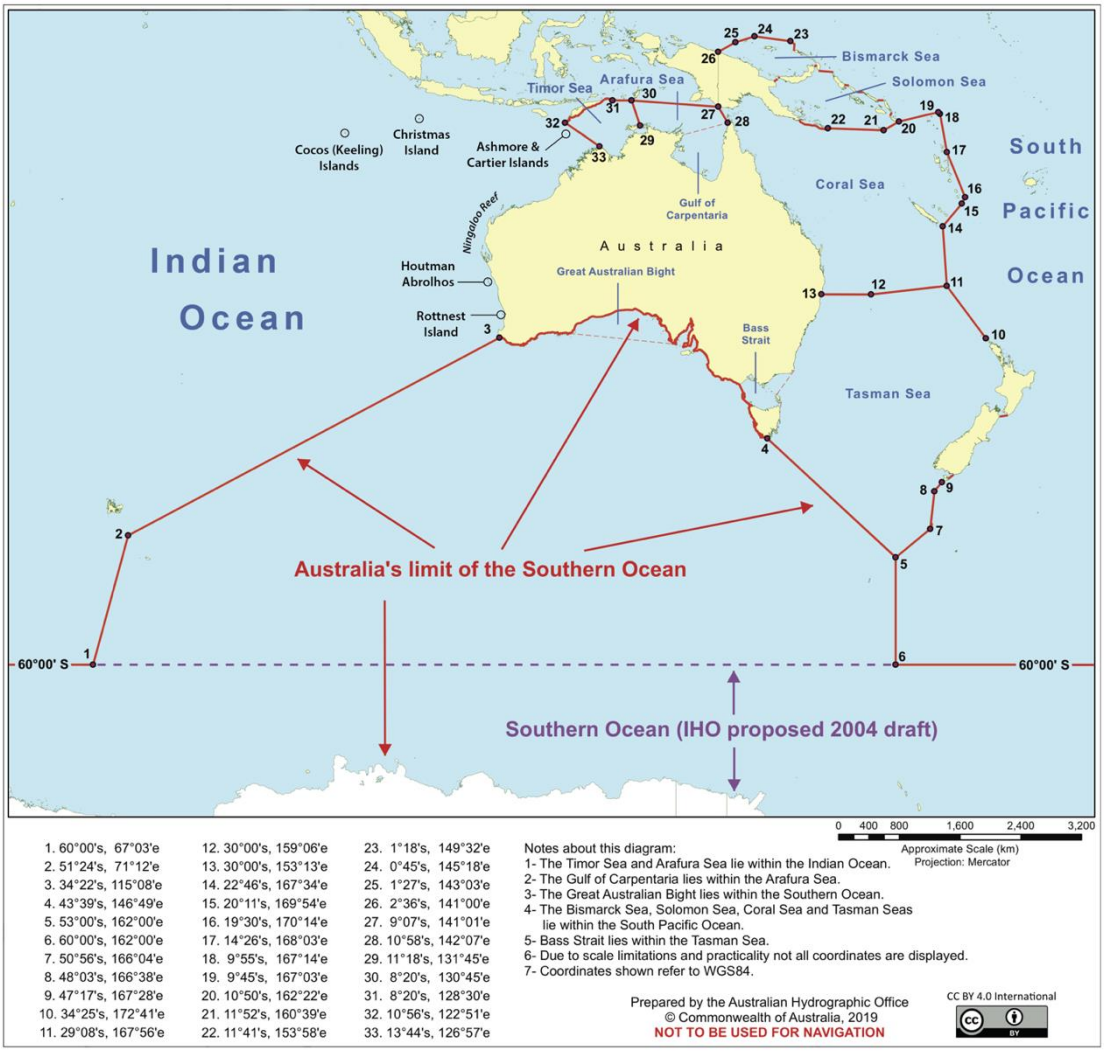


Figure 1.1 The Australian Government's official interpretation of the Indian Ocean and other ocean boundaries surrounding Australia (Australian Hydrographic Office, 2019).

Australia's delimitation of the Indian Ocean extends from Cape York (28) in northern Queensland, to Cape Leeuwin (3) in southwest Western Australia. In Australia, this supersedes the International Hydrographic Organization's (IHO) charting guidelines (1953) which extends the Indian Ocean into the Great Australian Bight. In this thesis, I acknowledge the Australian Government's official interpretation of the Indian Ocean boundary. Australia's External Territories (AET) in the Indian Ocean—Cocos (Keeling) Islands, Christmas Island, and the Ashmore and Cartier Islands—in addition to the Houtman Abrolhos and Rottnest Island, are marked with open circles. The World Heritage-listed Ningaloo Reef located off mainland Western Australia is marked in italics.

Australia has a number of security, strategic and economic interests in the Indian Ocean (Brewster, 2019). It holds a larger area of marine jurisdiction in this region than any other Indian Ocean rim country, with an exclusive economic zone (EEZ; extending 200 nautical miles off the coastline and the AETs) of 3.88 million km^2 and an extended continental shelf area (beyond the EEZ limit) of 2.02 million km^2 (Bateman & Bergin, 2010). Australia's blue economy (i.e. marine and maritime industries such as tourism, ports, shipbuilding, fisheries, undersea mining and offshore oil and gas) was estimated to have contributed $68.1 billion AUD to Australia's gross domestic product (GDP) in 2015/16 (Australian Institute of Marine Science, 2018). A significant component of this contribution was attributed to offshore oil and gas developments, which are largely located in the North West Shelf province of Western Australia. This industry is expected to expand with the North West Shelf Project Extension (Woodside, 2018) and increasing exploration of deep water acreage in this region (Bussell, Jablonski, Enman, Wilson, & Bint, 2001).

Commercial (wild capture) fisheries in Western Australia and the Northern Territory generated in excess of $594 million AUD in 2017/18 (Fisheries Research and Development Corporation (FRDC), 2019). Western Australia is a premium-quality seafood producer, largely selling and exporting low-volume, high-quality products such as western rock lobster, abalone and Exmouth Gulf prawn. The Northern Territory commercial fishers largely target Northern prawn, demersal finfish, mud crab and barramundi. Achieving ecologically sustainable fisheries whilst maintaining pristine waters is therefore a high priority to sustain industry standards. In addition, recreational fishing provides social and economic benefits to local Australian communities (Henry & Lyle, 2003); the economic value of recreational fishing in Western Australia alone is estimated to exceed $2.4 billion AUD a year (McLeod & Lindner, 2018). Lastly, indigenous fishing is of subsistence, economic and cultural importance for many coastal Indigenous Australians (Henry & Lyle, 2003). Subsequently, there is a pressing need to maintain sustainable fisheries for commercial, recreational and indigenous fishers. This is underpinned by the need to protect important coastlines (e.g. mangrove areas for fish nursery habitats), restrict pollution and conserve multi-trophic marine communities.

1.1.2 Threats to management, sustainability and conservation

The major threats to the management, sustainability and conservation of Australia's Indian Ocean region, and more broadly across the Indian Ocean, are linked to climate change, overexploitation of living and non-living resources, population pressures on coastal regions and knowledge gaps of marine diversity in the Indian Ocean (Keesing & Irvine, 2005; Roy, 2019; Zinke et al., 2018). Many Indian Ocean rim countries are developing nations and among the world's most populous. With a lower GDP per capita than their Pacific and Atlantic counterparts and increasing coastal populations that extensively rely on marine resources, Indian Ocean rim countries are expected to have the most impact on marine environments (Keesing & Irvine, 2005; Van der Elst et al., 2005). Illegal, unreported and unregulated (IUU) fishing targeting globally important pelagic stocks by foreign flagged vessels is of major concern for many Indian Ocean rim countries (Agnew et al., 2009; Cabral et al., 2018). Australia's fishing interests in the Indian Ocean region are also likely to be affected by broad-ranging shifts in pelagic fishes, as a result of changes in ocean temperature (Hobday, 2010).

It has been argued that Australia has a greater responsibility to contribute to biodiversity knowledge and conservation in the Indian Ocean region, given that it may be one of the few Indian Ocean rim countries that can afford to implement biomonitoring and management measures in the near future (Preen, 1998). A National Representative System of Marine Protected Areas (NRSMPA) was established by Australian, state and territory governments in 1998 (TFMPA, 1998). In 2012, the Australian Government expanded the coverage of the NRSMPA to 3.3 million km^2 (Director of National Parks, 2018). The objective was to develop a "comprehensive, adequate and representative system" of marine protected areas to conserve a variety of natural, cultural, socio-economic and heritage values. This led to the development of new Australian Commonwealth marine parks. There are now 26 national marine parks covering a total of 591 thousand km^2, that are located within Australia's Indian Ocean region—these are regionally managed under the North Marine Parks Network, the North-west Marine Parks Network and the South-west Marine Parks Network. These marine parks are located within Australian Commonwealth waters and extend from the outer edge of state and territory waters (3 nautical miles or approximately 5.5km

from the shoreline) to the outer boundary of Australia's EEZ (Director of National Parks, 2018). In addition, there are 15 state-managed marine parks along the Indian Ocean coastline of Western Australia.

Whilst the development of the NRSMPA was a significant step towards marine conservation in Australia, there is criticism of the Australian Government's selection and designation of marine protected areas and reserves within the marine parks under this initiative (Barr & Possingham, 2013; Hobbs, 2014). A key principle in the development of the NRSMPA was "representativeness", whereby the zoning of each marine protected area would encapsulate a full range of marine ecosystems and their diversity (TFMPA, 1998). However, outside of the Coral Sea Marine Park (which includes the Great Barrier Reef), the designation of the marine protected areas and reserves is not representative of bioregions and regional ecosystems (Barr & Possingham, 2013). In addition, some areas of high diversity importance, such as the Cocos (Keeling) Islands and Christmas Island, were omitted entirely from the NRSMPA and receive no marine protection (Hobbs, 2014). This has been attributed to a lack of quantitative targets in the representative zoning process, i.e. protection of *x*% of bioregional diversity (Barr & Possingham, 2013). Additionally, some of the marine parks and reserves, such as within the remote North-west Marine Park, were poorly documented prior to their creation (Moore et al., 2016), and present an absence of baseline data for future monitoring. There is also a concern that these marine protected areas have not been zoned to adequately protect threatened marine species, such as sawfish (family: Pristidae) (Devitt, Adams, & Kyne, 2015). With a significant increase in heat stress and coral bleaching across north-west Australia (Gilmour et al., 2019) regardless of marine park networks, there is an increasing need for monitoring to inform management strategies for aquatic biota.

However, with growing global pressures on marine, freshwater and terrestrial populations, the majority of countries, including Australia, have limited conservation and management resources and will increasingly require spatial cost-benefit analyses to ensure optimal delivery and return on biodiversity protection measures (Weeks, Mason, Ausseil, & Herzig, 2014). A progression towards whole-ecosystem surveying may provide a more time and cost-efficient approach in collecting multi-trophic distribution data, which will facilitate spatial cost-benefit analyses to ultimately better

manage key conservation areas and populations. Recent advances in environmental DNA (eDNA) technologies may provide a genetic avenue to whole-ecosystem surveying across a variety of environments (Deiner, Fronhofer, Mächler, Walser, & Altermatt, 2016; Port, O'Donnell, Romero-Maraccini, et al., 2016; Stat et al., 2017).

1.2 eDNA metabarcoding: an emerging tool for biomonitoring

1.2.1 Background of eDNA research

Environmental DNA (eDNA) refers to any traces of animal or plant DNA that have been shed into the environment. This could be through the loss or excretion of skin cells, mucous, blood, sperm, roots and leaves (Taberlet, Coissac, Hajibabaei, & Rieseberg, 2012). These traces of DNA are usually short, degraded fragments less than 500bp, and can then be recovered from an environmental sample, such as from water, soil, marine sediments, scat, ice cores, snow, and even air (Andersen et al., 2012; Bohmann et al., 2014; Folloni et al., 2012; Thomsen, Kielgast, Iversen, Møller, et al., 2012). This can also extend to the recovery of eDNA from gut-contents of a selected animal, which may be indistinguishable in morphological analyses (Blankenship & Yayanos, 2005; Oyafuso, Toonen, & Franklin, 2016; Paula et al., 2016).

Following the extraction of eDNA from an environmental substrate, researchers can either target and amplify an individual species or a broader taxonomic group by applying species-specific or universal taxonomic-wide assays, respectively (Ficetola, Miaud, Pompanon, & Taberlet, 2008; Hajibabaei, Shokralla, Zhou, Singer, & Baird, 2011; Jerde, Mahon, Chadderton, & Lodge, 2011; Medinger et al., 2010; Thomsen, Kielgast, Iversen, Wiuf, et al., 2012). The former approach is solely referred to as eDNA and has been used to detect individual taxa in a variety of applications, such as locating the presence of invasive species (Dougherty et al., 2016; Ficetola et al., 2008) and elusive and/or endangered taxa (Brozio et al., 2017; Nevers et al., 2018). The latter approach is referred to as eDNA metabarcoding and through the simultaneous amplification of multiple taxa can be used to provide community composition data. eDNA metabarcoding has been used, for example, to undertake aquatic biodiversity surveys monitoring a variety of eukaryotic taxa, such as bony fish, elasmobranchs, arthropods, amphibians and molluscs, from within marine or freshwater ecosystems (Lacoursière-Roussel et al., 2018; Stat et al., 2017; Valentini et al., 2016).

The emergence of eDNA metabarcoding as a biomonitoring tool has been facilitated by recent advances in sequencing technologies (Shokralla, Spall, Gibson, & Hajibabaei, 2012). Early eukaryotic eDNA research relied upon the use of Sanger

DNA sequencing (Ficetola et al., 2008; Willerslev et al., 2003), which provided a robust method for single-specimen sequencing and thus primarily the development of reference sequence libraries (Hajibabaei et al., 2005; Ivanova, Zemlak, Hanner, & Hebert, 2007). However, it did not provide a feasible approach for the application of eDNA metabarcoding which requires the sequencing of complex environmental samples containing DNA from potentially thousands of organisms, representing hundreds of species (Hajibabaei et al., 2011). Next-generation sequencing (NGS) technologies were introduced from 2005 and provided a new suite of PCR-based high-throughput sequencing platforms, such as Roche 454 genome sequencers, Illumina sequencers, the Applied Biosystems SOLiD sequencer and Life Technologies Ion Torrent (Shokralla et al., 2012). These platforms provide mass parallel sequencing of samples, generating millions of DNA sequencing reads from multiple species simultaneously (Shokralla et al. 2012). Large amounts of multi-species eDNA metabarcoding data can then be quality filtered and matched against reference databases, providing multiple taxonomic assignments. These can range from species to phylum-level assignments depending on how well reference databases have been curated.

The development of eDNA and NGS sequencing technologies have coincided with an expansion of reference databases, through movements such as the International Barcode of Life (iBOL; https://ibol.org/) which aims to generate a comprehensive "barcode" library of the animal kingdom. Barcoding initiatives have largely targeted a 650 base-pair fragment of the 5' end of the mitochondrial cytochrome oxidase I (COI) gene region in order to standardise the development of reference databases (Hajibabaei, Singer, Hebert, & Hickey, 2007). The first phase of the iBOL initiative (2010-2015) successfully barcoded the COI gene for 500 thousand species, which was submitted to globally accessible reference databases such as the Barcode of Life Data System (BOLD) and the National Center for Biotechnology Information's (NCBI) GenBank. It's second phase is currently underway and aims to extend its barcoding initiative to 2.5 million species by 2026 (Hobern & Hebert, 2019).

Mitochondrial DNA (mtDNA) is typically targeted for the development of barcode markers, as there can be up to several thousand copies within a cell, which is substantially more than most nuclear genes. Subsequently mitochondrial DNA persists

at a greater proportion than nuclear DNA in the environment (Rees, Maddison, Middleditch, Patmore, & Gough, 2014). Therefore, there is a higher likelihood of detecting a species in its environment through targeting mitochondrial eDNA. The mitochondrial COI marker has generally been effective in discerning congeneric species across the animal kingdom (Hajibabae 2006). However, primer binding sites around the COI marker are not highly conserved, requiring the design of a greater number of primer assays to amplify various animal taxonomic groups (Deagle, Jarman, Coissac, Pompanon, & Taberlet, 2014). As such, a number of other marker genes have been utilised to amplify broader taxonomic groups and to gain higher levels of resolution, afforded by the degree of variation within respective gene regions. Another commonly used marker is the 16S ribosomal RNA (rRNA) gene which is routinely used for bacteria and archaea (Sogin et al., 2006; Taberlet, Bonin, Zinger, & Coissac, 2018); its mitochondrial homologue increasingly used for fish, amphibian, mollusc, echinoderm and arthropod identification (Berry et al., 2017; Deagle et al., 2007; Taberlet et al., 2018; Vences et al., 2016). Additionally, the mitochondrial 12S rRNA gene is used for bird, mammal, fish and elasmobranch identification (Coghlan et al., 2013; Miya et al., 2015; Riaz et al., 2011). Nuclear markers are also utilised in eDNA research, and include the ITS for fungi (Schoch et al., 2012) and the 18S and 28S rRNA genes as universal eukaryotic markers (Hardy, Krull, Hartley, & Oliver, 2010; Machida & Knowlton, 2012; Pochon, Bott, Smith, & Wood, 2013). The plastid coding genes *matK* and *rbcL* have been recommended as the standard plant barcodes (Hollingsworth et al., 2009).

1.2.2 eDNA metabarcoding workflow

An eDNA metabarcoding analysis involves the 1) collection of samples, 2) eDNA extraction and 3) amplification of eDNA using polymerase chain reaction (PCR) steps, prior to 4) library preparation and sequencing, 5) bioinformatics and lastly 6) statistical analyses. There is an increasing number of metabarcoding protocols as outlined in Rees et al., (2014). The commonly applied methods in an aquatic eDNA metabarcoding workflow are summarised below.

For aquatic eDNA biomonitoring, water is the routinely sampled substrate, detecting a broader spectrum of fish species than marine sediment (Shaw et al., 2016).

Nevertheless, the use of marine sediment, settlement plates and planktonic tows can provide a higher detection rate of benthic taxa such as dinoflagellates, tunicates, molluscs and crustaceans (Koziol et al., 2018). For water, sample sizes usually vary from 15 mL to 20 L (Ficetola et al., 2008; Kelly, Port, Yamahara, & Crowder, 2014). However, on average, water samples are within 1-2 L and can be collected from a variety of waterbodies, e.g. ocean, lakes, rivers (Rees et al., 2014). Intracellular and extracellular eDNA is then captured through the use of precipitation (Dejean et al., 2012), centrifugation (Klymus, Richter, Chapman, & Paukert, 2015) or filtration of the water samples through a peristaltic or in-line pump (Kelly et al., 2014). The yield of eDNA can vary widely based on environmental variables (e.g. pH, inorganic and organic particles in water), capture volume and technique, and the pore size and composition of filter membranes (Eichmiller, Miller, & Sorensen, 2016). Recent research advocates filtering through mixed cellulose-ester or cellulose-nitrate membranes, followed by preserving the filters by freezing either dry or in lysis buffer to maximise eDNA yields (Jeunen et al., 2019; Majaneva et al., 2018).

It is recommended that the extraction of eDNA takes place in state-of-the-art laboratory facilities, ideally a dedicated trace DNA laboratory that is physically separated from areas of PCR amplification (Goldberg et al., 2016). There are numerous eDNA extraction methods for water samples, however they largely use and modify the following commercial kits: the DNeasy Blood and Tissue Kit, the DNeasy PowerWater DNA Isolation kit and the QIAamp DNA Micro Kit (Qiagen, Inc., Valencia, California, USA); and the Quick-DNA kits (Zymo Research Corporation, Irvine, California, USA) (Rees et al., 2014).

The next step of the workflow involves the *in vitro* amplification of eDNA by PCR. This requires the use of primer pairs (synthetic oligonucleotides) which anneal to and amplify the DNA of targeted taxa during PCR. Therefore, *in-silico* analyses of pre-validated metabarcoding primers is recommended to certify that targeted taxa will be amplified, and if not, the design of new primers may be necessary. Target specificity is influenced by the number and placement of base pair mismatches between the template DNA and primer (Wilcox et al., 2013). It is therefore recommended to minimise base pair mismatches between the primer and target taxa DNA, whilst incorporating as many differences as possible with non-target taxa (Rees et al., 2014).

A typical PCR reaction contains a small amount (usually 2 μL) of the sample extract containing template DNA, a primer pair, deoxynucleoside triphosphates (dNTPs) which consist of the four basic nucleotides (adenine, thymine, cytosine and guanine), magnesium ion (Mg^{2+}) and a thermostable DNA polymerase, such as *Taq* DNA polymerase, in an appropriate buffer (Taberlet et al., 2018). PCR amplification is achieved by the repeating of three major steps—denaturation, annealing and extension (or elongation)—where at the end of each round, the quantity of the target DNA fragments doubles. This results in the exponential amplification of the target DNA (herein referred to as amplicons) and usually reaches a plateau between 35-40 rounds. However, this is dependent on initial concentration of target DNA and PCR efficiency (Taberlet et al., 2018). It is recommended that amplification of eDNA samples is conducted with real-time PCR (RT-PCR), also referred to as quantitative PCR (qPCR), in order to observe amplification above a fluorescence threshold (Ct).

Next-generation sequencing platforms were initially designed, and are therefore optimised for sequencing genomic DNA. However, it is possible to conduct amplicon sequencing for eDNA metabarcoding purposes, which comprises "multiplexed" samples in the same sequencing lane (Taberlet et al., 2018). For Illumina sequencing platforms, which are commonly utilised for metabarcoding studies, amplicons must be flanked by two different adapters which are complementary to the sequencing surface of Illumina flow cells. Additionally, an index, acting as a unique tag for each sample, must be incorporated into the adapter. This allows samples to be bioinformatically identified post-sequencing, amidst the multiplexing of up to several hundred samples within the same sequencing lane. The addition of adapters to amplicons, with a unique index for each sample, can be achieved in one of three ways: 1) a two-step PCR protocol whereby PCR is conducted with gene-specific primers with modified Illumina adapter tails, and Illumina adapters and sample index are ligated in a second-round PCR using Nextera library construction kits; 2) a single-step PCR protocol whereby PCR is conducted with normal gene-specific primers, and Illumina adapters and sample index are additionally ligated using TruSeq PCR-free kits; and 3) a single-step PCR with fusion tagged primers, containing a forward or reverse primer that is already ligated to a sample index and Illumina adapter (Pochon et al. 2019; Taberlet et al., 2018). Additional preparation of an amplicon library (containing multiplexed indexed samples) prior to sequencing, can include fragment analysis, purification and

/or size selection (and thus removal of primer dimer) and final quantification to ensure optimal load concentration into a sequencing kit (Lindahl et al., 2013; Stat et al., 2017).

Post-sequencing bioinformatics is necessary to filter out molecular artifacts (such as from PCR and sequencing error) from sequencing datasets. A range of software and R packages are available to implement bioinformatic metabarcoding tools, such as USEARCH (Edgar, 2010), OBITools (Boyer, Mercier, Bonin, Taberlet, & Coissac, 2014), QIIME (Caporaso et al., 2010), DADA2 (Callahan et al., 2016) and MOTHUR (Schloss et al., 2009). Bioinformatic metabarcoding processing can include quality-based filtering of the sequencing data, quality trimming, paired-end read merging (if used paired-end sequencing), sequence demultiplexing (using the unique sample-specific tags), dereplication, denoising and chimera removal. Sequences can then be clustered into operational taxonomic units (OTUs), or remain as unique amplicon sequence variants (ASVs or zero-radius OTUs, ZOTUs), prior to the querying of sequences against publicly-accessible reference databases, such as BOLD, GenBank SILVA, PR^2, the Ribosomal Database Project (RDP) and UniEUK. This produces taxonomic assignments that depending on the percent identity match could range from a phylum to species level assignment. The culmination of multiple taxonomic assignments provides community composition data which can be used to statistically examine alpha and beta diversity between samples, sites and more broadly, ecosystems.

There are, however, some limitations associated with the distinct steps of the metabarcoding workflow that undermine the accuracy and reproducibility of the resulting taxonomic assignments. The right choice of marker is critical to reliably amplify and gain enough resolution to distinguish closely-related species within the targeted taxonomic group. Whilst the mitochondrial COI region is often a universal marker of choice, the primer-binding sites are not highly conserved, resulting in unreliable amplification (Deagle et al. 2014). Quality filtering parameters are often subjective—a common variation between published papers is the minimum count or proportion of sequencing reads for a taxonomic assignment to be retained. To avoid a false positive from sequencing error (often resulting in a singleton), a minimum count of two reads in a singular sample or one read across multiple samples is recommended (Callahan et al. 2016). OTU clustering is typically applied at a 97% identity threshold,

however the optimal identity threshold reflecting a species cluster varies between gene regions and taxonomic groups (Edgar 2018). Therefore, an ASV approach with no clustering may be more appropriate. Lastly, the extent to which a reference database is developed for a chosen gene region and targeted taxonomic group will provide greater accuracy and confidence in the resulting taxonomic assignments. Successive eDNA studies conducted at the same location should therefore consider variability in the availability of reference sequences at the time of analysis and potentially retrospectively analyse previous eDNA data using updated reference databases and bioinformatic processes (see 5.2 for an in-depth discussion on the limitations of eDNA metabarcoding).

1.2.3 The preliminary use of eDNA metabarcoding in Australian waters

Fourteen eDNA metabarcoding studies targeting a wide range of eukaryotic and prokaryotic taxa from marine and freshwater ecosystems have been conducted in Australian waters to date (Table 1.1). It should be noted that only one study (Shaw et al. 2016) had been published at the outset of this PhD research. Environmental DNA from these Australian studies have been sourced from a variety of environmental substrates including water, marine sediment, biological material from settlement plates, and debris from planktonic tows and kick-nets, in addition to gut-contents of marine fish, sea lions and fur seals. Six of these studies have been recently conducted in Australia's Indian Ocean region and demonstrate the applicability of eDNA metabarcoding, particularly with a multi-marker approach, to characterise marine multi-trophic community composition (Alexander et al., 2020; Berry et al., 2017, 2019; Jones et al., 2018; Koziol et al., 2018; Stat et al., 2017). A few of the Australian studies have addressed methodological questions in regards to environmental substrate selection and replication (Bylemans, Gleeson, Lintermans, et al., 2018; Koziol et al., 2018), and present preliminary comparisons to underwater visual surveillance and netting techniques (Alexander et al., 2020; Shaw et al., 2016; Stat et al., 2019). However, further applications and methodological testing of eDNA metabarcoding is required in aquatic environments before widespread implementation alongside conventional biomonitoring tools in Australia's Indian Ocean region.

Table 1.1 Environmental DNA metabarcoding studies that have been conducted in Australian aquatic ecosystems.
A publication arising from chapter 2 of this thesis has been omitted from this table.

Study	Location	Ecosystem	Targeted taxa	Molecular marker/s	Main findings
Shaw et al., 2016	North Para River and Lower River Murray Wetlands, South Australia	Freshwater	Vertebrates, fish	12S rRNA, 16S rRNA	eDNA metabarcoding detected 100% of fyke net-caught species. Water samples provided a greater representation of fyke net-caught species than sediment.
Stat et al., 2017	Coral Bay, Western Australia	Marine	Eukaryotes, metazoans, fish, mammals, crustaceans, cephalopods, dinoflagellates, plants, prokaryotes	COI, 18S rRNA, 16S rRNA, 23S rRNA, trnL	Shotgun sequencing is inefficient for assessing eukaryotic diversity. Multi-marker eDNA metabarcoding provided greater detection across major divisions of eukaryotes.
Berry et al., 2017	Southwest coast, Western Australia	Gut-contents (marine)	Mammals, fish, crustaceans, cephalopods, birds	COI, 12S rRNA, 16S rRNA	eDNA metabarcoding identified over 80 taxa in the gut-contents of Australian sea lion (*Neophoca cinereal*). Confirmed that the Australian sea lion is a wide-ranging opportunistic predator.
Reef et al., 2017	Moreton Bay, Queensland	Marine, estuarine	Plants	rbcL	Identified 150 plant OTUs that contribute to seagrass sediment organic carbon.
Hardy et al., 2017	Jervis Bay and Montague Island, New South Wales	Gut-contents (marine)	Birds, mammals, fish, cephalopods, crustaceans	12S rRNA, 16S rRNA	Identified a higher level of prey diversity in Australian (*Arctocephalus pusillus doriferus*) and long-nosed fur seals *(A. forsteri*) than previously reported. Observed strong trophic linkages with coastal ecosystems.
Bylemans et al., 2018	Murrumbidgee River, New South Wales	Freshwater	Fish	12S rRNA	Optimal sampling strategies (i.e. sampling intensity and spatial replication) are influenced by river morphology, species richness and abundance.
Bylemans, Gleeson, Hardy, & Furlan, 2018	Murray–Darling Basin, New South Wales	Freshwater	Fish	12S rRNA	Advocates for *in-silico* analyses of primers on target fish taxa in order to select suitable metabarcoding assays. Designed a new primer pair for freshwater fish in the Murray-Darling Basin that allows for more robust taxonomic assignments.

Koziol et al., 2018	Garden Island and Albany, Western Australia	Marine	Fish, crustaceans, eukaryotes	16S rRNA, 18S rRNA	Taxonomic composition varied between four substrates (surface water, marine sediment, settlement plates and planktonic tows) taken at the same location for eDNA metabarcoding. Advocates for substrate selection best suited to the detection of target taxa.
Jones et al., 2018	Coastline from Marmion Marine Park to Coral Bay, Western Australia	Gut-contents (marine)	Bacteria	16S rRNA	Identified 20 bacterial OTUs that form the core microbiome for black rabbitfish (*Siganus fuscescens*). Also observed distinct hindgut microbial communities in different rabbitfish populations.
Stat et al., 2019	Jurien Bay Marine Park, Western Australia	Marine	Fish	16S rRNA	eDNA metabarcoding and baited remote underwater videos (BRUVs) yielded 30% more fish genera than either method used alone.
Berry et al., 2019	Rottnest Island, Western Australia	Marine	Cnidarians, copepods, crustaceans, molluscs, fish, eukaryotes	COI, 16S rRNA, 18S rRNA	Detected consistent seasonal assemblages of zooplankton species, and also a clear departure from seasonal compositions during an extreme marine heatwave.
Bylemans, Gleeson, Duncan, Hardy, & Furlan, 2019	Blakney Creek and Urumwalla Creek, New South Wales	Freshwater	Fish	12S rRNA	The seasonal sampling of river systems does not influence the performance of eDNA metabarcoding. When investigating a target taxon, eDNA metabarcoding was found to be less sensitive than a species-specific eDNA approach.
Nichols et al., 2019	Goodradigbee River, New South Wales	Freshwater	Invertebrates	COI	The presence of debris in freshwater samples obtained from kick-nets did not inhibit the detection of six target macroinvertebrate taxa. Various sample storage methods post-collection did not affect the detection of these taxa either.
Alexander et al., 2020	Cocos (Keeling) Islands, Australia	Marine	Corals, sponges	ITS2, 16S rRNA	OTU richness obtained from eDNA metabarcoding was comparable to species richness from visual surveys.

1.3 eDNA metabarcoding in practice

1.3.1 Benefits of applied eDNA metabarcoding

The emergence of eDNA metabarcoding is revolutionising our approach to biodiversity surveying, conservation, biosecurity and invasive species management. Its use in taxa detection is complementary, and in some cases, provides a viable alternative to conventional aquatic surveying techniques (Port, O'Donnell, Romero-Maraccini, et al., 2016; Shaw et al., 2016; Stat et al., 2019; Thomsen, Kielgast, Iversen, Møller, et al., 2012; Thomsen et al., 2016). Currently, marine, freshwater and estuarine monitoring utilise numerous surveying methods, each optimised to target specific groups of taxa (Birindelli, Meza-Vargas, Sousa, & Hidalgo, 2016; Caldwell, Zgliczynski, Williams, & Sandin, 2016). The development of broad eDNA metabarcoding assays however, can target and amplify various groups of taxa, from bacteria, fungi, plants and animals, from the same environmental sample. This may provide a more cost-effective and timely approach to surveying a wide variety of taxa from the same location. In addition, eDNA metabarcoding offers a number of advantages over conventional survey approaches. Firstly, eDNA metabarcoding requires no specialised taxonomic skill in identifying and distinguishing taxa (Yoccoz, 2012). This is particularly useful in detecting cryptic species and taxa in juvenile life stages, which may be difficult to morphologically assess (Thomsen & Willerslev, 2015). Secondly, the collection of samples for eDNA metabarcoding can be conducted in a non-invasive manner from sites that may be logistically difficult to survey (Thomsen & Willerslev, 2015). Thirdly, it can be used to detect unknown or previously unobserved taxa (Boussarie et al., 2018), which can then be used to guide observational and capture-based surveying to sites with hidden biodiversity.

Nevertheless, conventional methods still have some advantages over eDNA metabarcoding, such as the ability to distinguish different life stages (Rees et al., 2014) and provide quantitative estimates in relation to population sizes or biomass, which at present, is limited for eDNA metabarcoding. Alternatively, however, eDNA metabarcoding may be able to provide estimates of genetic diversity, effective population size, and population and landscape connectivity (Sigsgaard et al., 2016). Given its practical applications, it is anticipated that eDNA metabarcoding will soon

become a key tool in biodiversity detection, monitoring and biosecurity applications (Bohmann et al., 2014; Bott et al., 2010; Shokralla et al., 2012; Thomsen & Willerslev, 2015; Valentini et al., 2016).

1.3.2 Further applications and methodological testing

eDNA metabarcoding has great potential to be implemented alongside conventional aquatic surveying techniques for monitoring and management purposes in Australia's Indian Ocean region. However, at present, there remains a number of applied and methodological questions that need to be thoroughly addressed.

One area that requires further investigation, is the localisation of eDNA in aquatic environments, particularly in areas of high-water movement. The diffusion and decay of eDNA affects its localisation, i.e. the distance that we can detect an eDNA signal from its point of origin. This can subsequently obscure the exact locality where the organism shed DNA and confound estimates of diversity and community composition if the eDNA signal is actually representative of an upstream locality. Preliminary studies investigating eDNA degradation in seawater have shown that eDNA could be detected for as long as a few days in Danish coastal waters (Thomsen, Kielgast, Iversen, Møller, et al., 2012) to as low as 10 hours in the Adriatic Sea (Dell'Anno & Corinaldesi, 2004) before the DNA degrades beyond the detection threshold. Thomsen et al. (2012) also conducted decay experiments within mesocosms and reported that it takes 0.9 days for European flounder (*Platichthys flesus*) and 6.7 days for three-spined stickleback (*Gasterosteus aculeatus*) eDNA to fall below the detection threshold, from initial DNA concentrations of 214 and 48 DNA molecules per 400 ml seawater, respectively. Successive mesocosm studies have reported hourly eDNA decay rates of 0.104 ± 0.047 for bluegill sunfish (*Lepomis macrochirus*) (Maruyama, Nakamura, Yamanaka, Kondoh, & Minamoto, 2014), 0.101 ± 0.011 for northern anchovy (*Engraulis mordax*), 0.068 ± 0.004 for Pacific sardine (*Sardinops sagax*) and 0.070 ± 0.004 for Pacific chub mackerel (*Scomber japonicus*) (Sassoubre, Yamahara, Gardner, Block, & Boehm, 2016). Decay rates would be highly dependent upon abiotic (e.g. daylight cycle, temperature, acidity and salinity) and biotic factors (e.g. microorganism and extracellular enzymes), and consequently would be expected to vary between aquatic localities and also experimental setups.

Another consideration in the localisation of eDNA is the diffusion of DNA molecules by water volume and movement in open water environments, such as in the ocean or large lakes. It is possible that eDNA localisation is largely influenced by diffusion and dissipated beyond the detection threshold before eDNA decay naturally occurs (Yamamoto et al., 2016). Localisation research in coastal environments have indicated that eDNA derived from seawater can discriminate habitats within a small spatial area (<5 km) (Jeunen et al., 2018; Koziol et al., 2018; Murakami et al., 2019; Stat et al., 2019). Further research is required to examine eDNA localisation amidst extensive tidal and oceanic movements.

The efficiency of eDNA metabarcoding compared to conventional aquatic survey techniques is dependent upon the sensitivity of metabarcoding to detect various taxonomic groups in a given ecosystem, in addition to time and cost-efficiency to implement. As discussed previously, the ability to apply universal (e.g. eukaryotic) or multiple metabarcoding assays to the same environmental sample, circumvents the need for specific taxonomic expertise and various field methods for sampling different taxonomic groups. This can reduce fieldwork and taxonomic expertise-associated costs. However, eDNA research can entail high start-up costs in relation to the development of trace DNA laboratories and the development of project-specific primers (Smart et al., 2016). The majority of studies that have examined efficiency in eDNA research have done so using a species-specific approach. It was reported to be cheaper to detect a highly endangered fish (*Misgurnus fossilis*), a trematode (*Ribeiroia ondatrae*) and a newt species (*Triturus cristatus*) using species-specific assays than a combination of traditional survey approaches (Biggs et al., 2015; Huver, Koprivnikar, Johnson, & Whyard, 2015; Sigsgaard, Carl, Møller, & Thomsen, 2015). In some circumstances however, traditional sampling methods are more cost-efficient than species-specific eDNA assays (Smart et al., 2016).

At present, there are limited eDNA metabarcoding studies that have examined efficiency compared to conventional approaches. Bálint et al., (2018) found that visual and audio encounter surveys were less costly in regions of low diversity, however eDNA quickly became more cost-efficient when scaled up in regions of high diversity. This indicates that eDNA metabarcoding may be particularly well-suited for broad-

scale surveying in high biodiversity areas. In regards to sensitivity, it has been reported that eDNA metabarcoding can match and even outcompete conventional aquatic survey approaches (Thomsen, Kielgast, Iversen, Møller, et al., 2012). It has also been shown that eDNA metabarcoding used in conjunction with select survey tools can produce a higher biodiversity yield than applied solely on its own (Kelly et al., 2017; Port, O'Donnell, Romero-Maraccini, et al., 2016; Stat et al., 2019). Further research into the efficiency of eDNA metabarcoding compared to conventional aquatic survey techniques is required for the cost-effective implementation of this tool.

eDNA metabarcoding has demonstrated its applicability across small, yet highly sensitive, spatial (Jeunen et al., 2018; O'Donnell et al., 2017; Port, O'Donnell, Romero-Maraccini, et al., 2016) and temporal scales (Berry et al., 2019) in aquatic ecosystems. However, it is only in a preliminary stage of being scaled up and tested across broader regional scales (Fukumoto, Ushimaru, & Minamoto, 2015). There remains a question of whether eDNA metabarcoding is applicable across large-scale regions as a rapid-assessment broad-scale biomonitoring tool.

The ability of eDNA to detect endangered and elusive taxa in aquatic environments is well documented, albeit largely with species-specific primers (Day et al., 2019; Piaggio et al., 2014; Thomsen, Kielgast, Iversen, Wiuf, et al., 2012; Weltz et al., 2017). There is particular interest by Australian state government and research departments in using eDNA (both species-specific and metabarcoding) to detect and update distribution records for critically endangered and elusive taxa, such as sawfish (genus: *Pristis*) and northern river shark (*Glyphis garricki*). Further research is required to evaluate the efficiency of detecting endangered taxa with eDNA metabarcoding, as opposed to designing and applying species-specific primers. The use of eDNA for aquatic reptile detection has also largely been through the application of species-specific primers (Baker et al., 2018; Davy, Kidd, & Wilson, 2015; De Souza, Godwin, Renshaw, & Larson, 2016; Feist et al., 2018; Halstead et al., 2017). Interestingly, there have been mixed results for the detection of aquatic reptiles, even with species-specific primers, which may reflect an emerging challenge in regards to the shedding of reptilian skin cells and subsequent detection with eDNA (Adams, Hoekstra, Muell, & Janzen, 2019). Further research, particularly with the application of eDNA

metabarcoding, is required to assess the sensitivity of eDNA in detecting aquatic reptile distributions.

eDNA metabarcoding has been extensively applied in marine, freshwater and terrestrial environments and has demonstrated its power in providing high-resolution taxonomic assignments (i.e. to a species or genus-level) (Evans et al., 2016; Olds et al., 2016; Stat et al., 2017; Thomsen, Kielgast, Iversen, Møller, et al., 2012). This has coincided with other scientific research and barcoding initiatives within these ecosystems. Subterranean environments however, are notoriously underexplored and subsequently present suboptimal reference databases. The application of eDNA metabarcoding in poorly-surveyed environments may therefore be limited in its ability to provide high-resolution taxonomic assignments. Preliminary research has demonstrated the viability of eDNA metabarcoding in detecting aquatic "stygofauna" taxa from underground water samples (Humphreys et al., 2012; Korbel et al., 2017), however has not yet been thoroughly tested in its ability to provide a variety of high-resolution eukaryotic assignments.

Lastly, there are a number of additional applications for eDNA metabarcoding that are in various levels of development. There has been a considerable global focus within the eDNA community to provide estimates of quantification (e.g. population size and biomass) from eDNA sequencing data (Atkinson et al., 2018; Turner, Miller, Coyne, & Corush, 2014; Ushio et al., 2018). At present, eDNA (both species-specific and metabarcoding) can only reliably provide presence/absence data (Lamb et al., 2019). Quantification research has largely focused on individual species and thus utilised specific-specific primers, given that eDNA shedding rates vary between different species (Sassoubre et al., 2016). An alternative to quantification estimates is the extraction of population genetic information from eDNA data, which may be able to provide estimates on genetic diversity, effective population size, and population and landscape connectivity. This application however, is only in a very preliminary stage (Sigsgaard et al., 2016; Stat et al., 2017). Additionally, the integration of eDNA metabarcoding compositional data with environmental abiotic data may be used to assess overall ecosystem health (Chariton et al., 2016) and biotic interconnectivity between adjacent sites and regions.

1.4 Research question and thesis structure

The primary question of this thesis is: "Is eDNA metabarcoding a viable aquatic biomonitoring tool for use in Australia's Indian Ocean ecosystems?" (Figure 1.2). This will be achieved by addressing a series of specific questions and aims across three data chapters (i.e. Chapters 2, 3 and 4; Figure 1.2). Each chapter is self-contained and is comprised of a prologue (providing background information on the study site), abstract, introduction, methods, results, discussion, conclusion, chapter references and supplementary information. See below for descriptions pertaining to Chapters 2, 3 and 4 (1.4.1, 1.4.2 and 1.4.3, respectively). The data chapters in this thesis are written and formatted as three journal articles. The final discussion (Chapter 5), synthesises the thesis' research, limitations and significance, and explores future directions of eDNA metabarcoding in aquatic environments. Ultimately, this thesis provides an insight into the use of eDNA metabarcoding as an aquatic biomonitoring tool and will aid integration of this technique for species monitoring and management in Australia's Indian Ocean region.

1.4.1 eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem (Chapter 2)

In chapter two, I investigate the localisation of eDNA signals from seawater at the Cocos (Keeling) Islands, a remote, island atoll in the Indian Ocean. Previous coastal research has indicated that eDNA derived from seawater can discriminate habitats within a small spatial area (<5 km) (Jeunen et al., 2018; Koziol et al., 2018; Murakami et al., 2019; Stat et al., 2019). However, this has not yet been tested amidst extensive tidal and oceanic movements. Using a multi-marker metabarcoding approach, I characterised coral reef assemblages across 42 sites within a 140 km^2 area at the Cocos (Keeling) Islands. This allowed me to analyse whether compositional variation reflects fine-scale habitat partitioning and subsequently the extent of eDNA localisation. Additionally, I compared the sensitivity of the eDNA approach to previously-applied conventional marine methods at this well-studied island atoll. This study is the first to conduct a multi-trophic eDNA survey of an entire coral reef atoll and demonstrates the utility of multi-assay Tree-of-life (ToL) eDNA metabarcoding approaches in assessing

whole-ecosystem biodiversity. Chapter two has been published in the peer-reviewed journal *Molecular Ecology* (West et al. 2020).

1.4.2 Large-scale eDNA metabarcoding survey reveals marine biogeographic break and transitions over tropical north-western Australia (Chapter 3)

In chapter three, I investigate the applicability of eDNA metabarcoding as a broad-scale biomonitoring tool which can discern fine-scale patterns across large-scale bioregions. Previous broad-scale biomonitoring in north-western Australia has identified a marine biogeographic break at Cape Leveque—the tip of the Dampier Peninsula—demarcating the border between the Canning and Kimberley bioregions (Travers, Potter, Clarke, Newman, & Hutchins, 2010; Wilson, 2014). I conducted a broad-scale multi-marker eDNA metabarcoding survey across 700km of the Canning and Kimberley coastlines to assess how well this technique can detect the biogeographic break. In addition, I examined the ability of eDNA metabarcoding to detect a number of endangered and elusive marine taxa, which are under-detected using conventional methods. Lastly, I explore whether eDNA metabarcoding can be used for the detection of aquatic reptile taxa, such as crocodiles, turtles and sea snakes, the latter of which, is data-deficient across this vast and remote region.

1.4.3 Under the karst: detecting hidden subterranean assemblages using eDNA metabarcoding in the caves of Christmas Island, Australia (Chapter 4)

In chapter four, I investigate the potential of using eDNA metabarcoding to detect and characterise eukaryotic community compositions in subterranean aquatic ecosystems. Subterranean ecosystems are understudied and challenging to conventionally survey because of the inaccessibility of underground voids and networks. Flowing stream and conduit water could potentially act as a conveyor belt of aquatic "stygofauna" DNA and facilitate the use of eDNA metabarcoding as an alternative to capture-based sampling. However, these poorly-surveyed ecosystems present sub-optimal reference databases which may hinder the ability to resolve taxonomic assignments and produce high-quality compositional data. I investigate this by applying a multi-marker metabarcoding survey in 23 cave and spring systems on Christmas Island, located in the Indian Ocean. Additionally, I examine whether resulting community composition,

in conjunction with environmental data, can be used to identify potential underground interconnectivity between these systems.

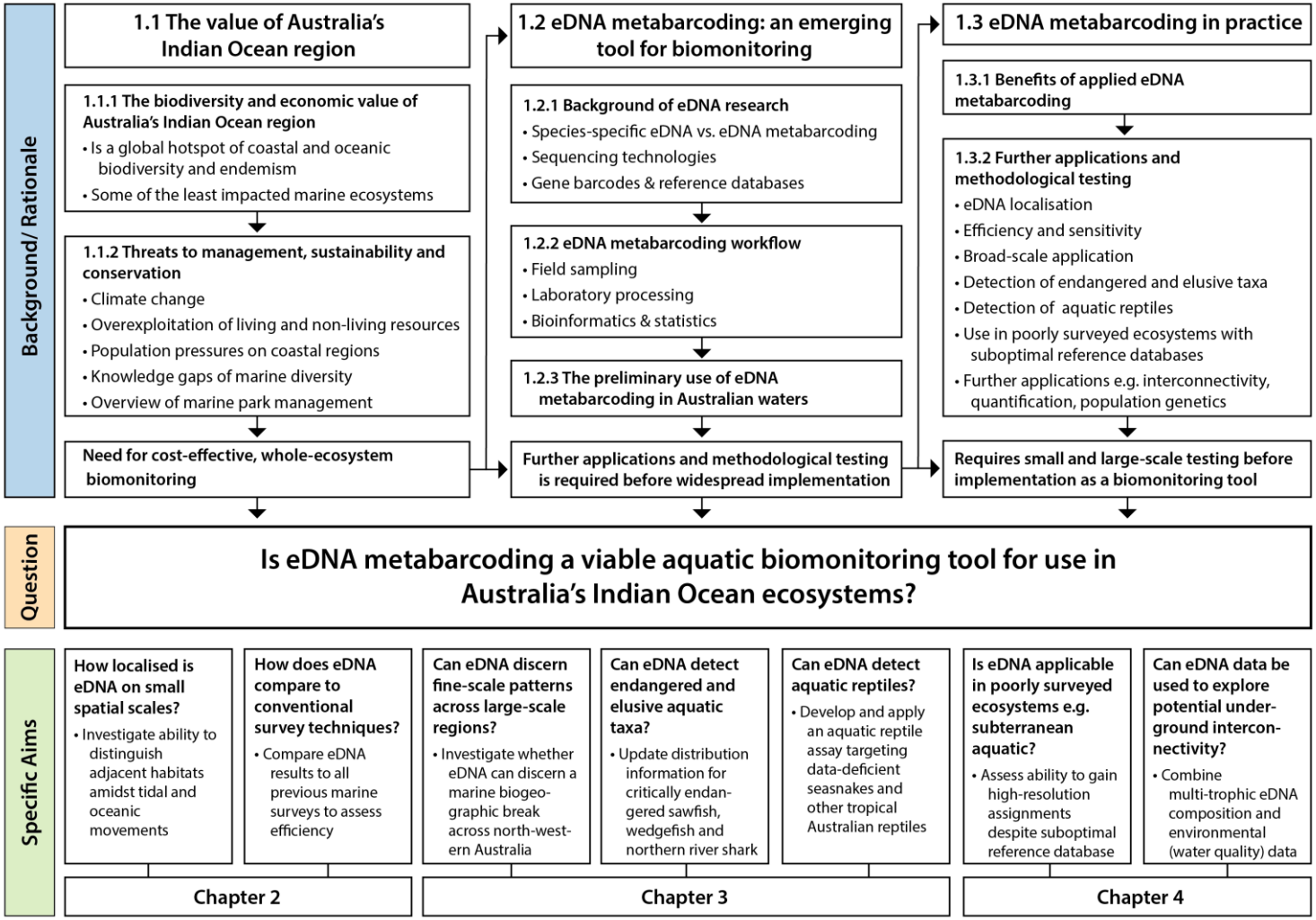


Figure 1.2 Conceptual flow diagram of the general thesis structure

1.5 References

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Abbott, I., & Wills, A. (2016). Review and synthesis of knowledge of insular ecology, with emphasis on the islands of Western Australia. *Conservation Science Western Australia*, *11*, 1–209.

Adams, C. I. M., Hoekstra, L. A., Muell, M. R., & Janzen, F. J. (2019). A Brief Review of Non-Avian Reptile Environmental DNA (eDNA), with a Case Study of Painted Turtle (*Chrysemys picta*) eDNA Under Field Conditions. *Diversity*, *11*(4), 50.

Agnew, D. J., Pearce, J., Pramod, G., Peatman, T., Watson, R., Beddington, J. R., & Pitcher, T. J. (2009). Estimating the worldwide extent of illegal fishing. *PLoS One*, *4*(2), e4570.

Alexander, J. B., Bunce, M., White, N., Wilkinson, S. P., Adam, A. A. S., Berry, T., … Dugal, L. (2020). Development of a multi-assay approach for monitoring coral diversity using eDNA metabarcoding. *Coral Reefs*, *39*(1), 159–171.

Andersen, K., Bird, K. L., Rasmussen, M., Haile, J., Breuning-Madsen, H., Kjaer, K. H., … Willerslev, E. (2012). Meta-barcoding of 'dirt'DNA from soil reflects vertebrate biodiversity. *Molecular Ecology*, *21*(8), 1966–1979.

Atkinson, S., Carlsson, J. E. L., Ball, B., Egan, D., Kelly-Quinn, M., Whelan, K., & Carlsson, J. (2018). A quantitative PCR-based environmental DNA assay for detecting Atlantic salmon (*Salmo salar L.*). *Aquatic Conservation: Marine and Freshwater Ecosystems*, *28*(5), 1238–1243.

Australian Hydrographic Office. (2019). *Names and Limits of Oceans and Seas around Australia*, Department of Defence, Australian Government, Canberra.

Australian Institute of Marine Science (AIMS). (2018). *The AIMS Index of Marine Industry December 2018*.

Baker, S. J., Niemiller, M. L., Stites, A. J., Ash, K. T., Davis, M. A., Dreslik, M. J., & Phillips, C. A. (2018). Evaluation of environmental DNA to detect *Sistrurus catenatus* and *Ophidiomyces ophiodiicola* in crayfish burrows. *Conservation Genetics Resources*, 1–3.

Bálint, M., Nowak, C., Márton, O., Pauls, S. U., Wittwer, C., Aramayo, J. L., … Jansen, M. (2018). Accuracy, limitations and cost efficiency of eDNA-based community survey in tropical frogs. *Molecular Ecology Resources*, *18*(6), 1415–1426.

Barr, L. M., & Possingham, H. P. (2013). Are outcomes matching policy commitments in Australian marine conservation planning? *Marine Policy*, *42*, 39–48.

Bateman, S., & Bergin, A. (2010). *Our western front: Australia and the Indian Ocean*. Australian Strategic Policy Institute (ASPI).

Berry, T. E., Osterrieder, S. K., Murray, D. C., Coghlan, M. L., Richardson, A. J., Grealy, A. K., … Bunce, M. (2017). DNA metabarcoding for diet analysis and biodiversity: A case study using the endangered Australian sea lion (*Neophoca cinerea*). *Ecology and Evolution*, *7*(14), 5435–5453.

Berry, T. E., Saunders, B. J., Coghlan, M. L., Stat, M., Jarman, S., Richardson, A. J., … Bunce, M. (2019). Marine environmental DNA biomonitoring reveals seasonal patterns in biodiversity and identifies ecosystem responses to anomalous climatic events. *PLOS Genetics*, *15*(2), e1007943. Retrieved from https://doi.org/10.1371/journal.pgen.1007943

Biggs, J., Ewald, N., Valentini, A., Gaboriaud, C., Dejean, T., Griffiths, R. A., … Brotherton, P. (2015). Using eDNA to develop a national citizen science-based monitoring programme for the great crested newt (*Triturus cristatus*). *Biological Conservation*, *183*, 19–28.

Birindelli, J., Meza-Vargas, V., Sousa, L., & Hidalgo, M. (2016). Standardized rapid biodiversity protocols: freshwater fishes. In Larsen T. H. (Eds.), *Core Standardized Methods for Rapid Biological Field Assessment* (pp. 127–138).

Blankenship, L. E., & Yayanos, A. A. (2005). Universal primers and PCR of gut contents to study marine invertebrate diets. *Molecular Ecology*, *14*(3), 891–899.

Bohmann, K., Evans, A., Gilbert, M. T. P., Carvalho, G. R., Creer, S., Knapp, M., … De Bruyn, M. (2014). Environmental DNA for wildlife biology and biodiversity monitoring. *Trends in Ecology & Evolution*, *29*(6), 358–367.

Bott, N. J., Ophel-Keller, K. M., Sierp, M. T., Rowling, K. P., McKay, A. C., Loo, M. G. K., … Deveney, M. R. (2010). Toward routine, DNA-based detection methods for marine pests. *Biotechnology Advances*, *28*(6), 706–714.

Boussarie, G., Bakker, J., Wangensteen, O. S., Mariani, S., Bonnin, L., Juhel, J.-B., … Robbins, W. D. (2018). Environmental DNA illuminates the dark diversity of

sharks. *Science Advances*, *4*(5), eaap9661.

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a Unix-inspired software package for DNA metabarcoding. *Molecular Ecology Resources*, 16(1), 176–182.

Brewster, D. (2019). *Australia's second sea: Facing our multipolar future in the Indian Ocean*, Australian Strategic Policy Institute (ASPI).

Brozio, S., Manson, C., Gourevitch, E., Burns, T. J., Greener, M. S., Downie, J. R., & Hoskisson, P. A. (2017). Development and application of an eDNA method to detect the critically endangered Trinidad golden tree frog (*Phytotriades auratus*) in bromeliad phytotelmata. *PLoS One*, *12*(2).

Bussell, M. R., Jablonski, D., Enman, T., Wilson, M. J., & Bint, A. N. (2001). Deepwater exploration: North Western Australia compared with Gulf of Mexico and Mauritania. *The APPEA Journal*, *41*(1), 289–320.

Bylemans, J., Gleeson, D. M., Duncan, R. P., Hardy, C. M., & Furlan, E. M. (2019). A performance evaluation of targeted eDNA and eDNA metabarcoding analyses for freshwater fishes. *Environmental DNA*, *1*(4), 402–414.

Bylemans, J., Gleeson, D. M., Hardy, C. M., & Furlan, E. (2018). Toward an ecoregion scale evaluation of eDNA metabarcoding primers: A case study for the freshwater fish biodiversity of the Murray–Darling Basin (Australia). *Ecology and Evolution*, *8*(17), 8697–8712.

Bylemans, J., Gleeson, D. M., Lintermans, M., Hardy, C. M., Beitzel, M., Gilligan, D. M., & Furlan, E. M. (2018). Monitoring riverine fish communities through eDNA metabarcoding: determining optimal sampling strategies along an altitudinal and biodiversity gradient. *Metabarcoding and Metagenomics*, *2*, e30457.

Cabral, R. B., Mayorga, J., Clemence, M., Lynham, J., Koeshendrajana, S., Muawanah, U., … Zulbainarni, N. (2018). Rapid and lasting gains from solving illegal fishing. *Nature Ecology & Evolution*, *2*(4), 650–658.

Caldwell, Z. R., Zgliczynski, B. J., Williams, G. J., & Sandin, S. A. (2016). Reef fish survey techniques: assessing the potential for standardizing methodologies. *PLoS One*, *11*(4).

Callahan, B. J., McMurdie, P. J., Rosen, M. J., Han, A. W., Johnson, A. J. A., & Holmes, S. P. (2016). DADA2: high-resolution sample inference from Illumina amplicon data. *Nature Methods*, *13*(7), 581.

Caporaso, J. G., Kuczynski, J., Stombaugh, J., Bittinger, K., Bushman, F. D., Costello,

E. K., … Gordon, J. I. (2010). QIIME allows analysis of high-throughput community sequencing data. *Nature Methods*, *7*(5), 335.

Caputi, N., Fletcher, W. J., Pearce, A., & Chubb, C. F. (1996). Effect of the Leeuwin Current on the recruitment of fish and invertebrates along the Western Australian coast. *Marine and Freshwater Research*, *47*(2), 147–155.

Chariton, A. A., Sun, M., Gibson, J., Webb, J. A., Leung, K. M. Y., Hickey, C. W., & Hose, G. C. (2016). Emergent technologies and analytical approaches for understanding the effects of multiple stressors in aquatic environments. *Marine and Freshwater Research*, *67*(4), 414–428.

Coghlan, M. L., White, N. E., Murray, D. C., Houston, J., Rutherford, W., Bellgard, M. I., … Bunce, M. (2013). Metabarcoding avian diets at airports: implications for birdstrike hazard management planning. *Investigative Genetics*, *4*(1), 27.

Cresswell, G. (1986). The role of the Leeuwin current in the life cycles of several marine creatures. *UNESCO Technical Papers in Marine Science*, *49*, 50–54.

Davy, C. M., Kidd, A. G., & Wilson, C. C. (2015). Development and validation of environmental DNA (eDNA) markers for detection of freshwater turtles. *PLoS One*, *10*(7), e0130965.

Day, K., Campbell, H., Fisher, A., Gibb, K., Hill, B., Rose, A., & Jarman, S. N. (2019). Development and validation of an environmental DNA test for the endangered Gouldian finch. *Endangered Species Research*, *40*, 171–182.

De Souza, L. S., Godwin, J. C., Renshaw, M. A., & Larson, E. (2016). Environmental DNA (eDNA) detection probability is influenced by seasonal activity of organisms. *PLoS One*, *11*(10), e0165273.

Deagle, B. E., Gales, N. J., Evans, K., Jarman, S. N., Robinson, S., Trebilco, R., & Hindell, M. A. (2007). Studying seabird diet through genetic analysis of faeces: a case study on macaroni penguins (*Eudyptes chrysolophus*). *PLoS One*, *2*(9), e831.

Deagle, B. E., Jarman, S. N., Coissac, E., Pompanon, F., & Taberlet, P. (2014). DNA metabarcoding and the cytochrome c oxidase subunit I marker: not a perfect match. *Biology Letters*, *10*(9), 20140562.

Deiner, K., Fronhofer, E. A., Mächler, E., Walser, J.-C., & Altermatt, F. (2016). Environmental DNA reveals that rivers are conveyer belts of biodiversity information. *Nature Communications*, *7*(1), 1–9.

Dejean, T., Valentini, A., Miquel, C., Taberlet, P., Bellemain, E., & Miaud, C. (2012).

Improved detection of an alien invasive species through environmental DNA barcoding: the example of the American bullfrog *Lithobates catesbeianus.* *Journal of Applied Ecology*, *49*(4), 953–959.

Dell'Anno, A., & Corinaldesi, C. (2004). Degradation and turnover of extracellular DNA in marine sediments: ecological and methodological considerations. *Applied and Environmental Microbiology*, *70*(7), 4384–4386.

Devitt, K. R., Adams, V. M., & Kyne, P. M. (2015). Australia's protected area network fails to adequately protect the world's most threatened marine fishes. *Global Ecology and Conservation*, *3*, 401–411.

Director of National Parks. (2018). *North Marine Parks Network Management Plan 2018*. Canberra.

Dougherty, M. M., Larson, E. R., Renshaw, M. A., Gantz, C. A., Egan, S. P., Erickson, D. M., & Lodge, D. M. (2016). Environmental DNA (eDNA) detects the invasive rusty crayfish *Orconectes rusticus* at low abundances. *Journal of Applied Ecology*, *53*(3), 722–732.

Edgar, R. C. (2010). Search and clustering orders of magnitude faster than BLAST. *Bioinformatics*, *26*(19), 2460–2461. Retrieved from http://dx.doi.org/10.1093/bioinformatics/btq461

Edgar, R. C. (2018). Updating the 97% identity threshold for 16S ribosomal RNA OTUs. *Bioinformatics*, *34*(14), 2371-2375.

Eichmiller, J. J., Miller, L. M., & Sorensen, P. W. (2016). Optimizing techniques to capture and extract environmental DNA for detection and quantification of fish. *Molecular Ecology Resources*, *16*(1), 56–68.

Evans, N. T., Olds, B. P., Renshaw, M. A., Turner, C. R., Li, Y., Jerde, C. L., … Lodge, D. M. (2016). Quantification of mesocosm fish and amphibian species diversity via environmental DNA metabarcoding. *Molecular Ecology Resources*, *16*(1), 29–41.

Feist, S. M., Jones, R. L., Copley, J. L., Pearson, L. S., Berry, G. A., & Qualls, C. P. (2018). Development and validation of an environmental DNA method for detection of the Alligator Snapping Turtle (*Macrochelys temminckii*). *Chelonian Conservation and Biology*, *17*(2), 271–279.

Ficetola, G. F., Miaud, C., Pompanon, F., & Taberlet, P. (2008). Species detection using environmental DNA from water samples. *Biology Letters*, *4*(4), 423–425. https://doi.org/10.1098/rsbl.2008.0118

Fisheries Research and Development Corporation (FRDC). (2019). *Australian Fisheries and Aquaculture Industry 2017/18: Economic Contributions Estimates Report*, Adelaide.

Folloni, S., KAGKLI, D., Rajcevic, B., Guimaraes, N. C. C., Van Droogenbroeck, B., Valicente, F. H., … Van den Bulcke, M. (2012). Detection of airborne genetically modified maize pollen by real-time PCR. *Molecular Ecology Resources*, *12*(5), 810–821.

Fukumoto, S., Ushimaru, A., & Minamoto, T. (2015). A basin-scale application of environmental DNA assessment for rare endemic species and closely related exotic species in rivers: A case study of giant salamanders in Japan. *Journal of Applied Ecology*, *52*(2), 358–365.

Gilmour, J. P., Cook, K. L., Ryan, N. M., Puotinen, M. L., Green, R. H., Shedrawi, G., … Buckee, J. (2019). The state of Western Australia's coral reefs. *Coral Reefs*, *38*(4), 651–667.

Goldberg, C. S., Turner, C. R., Deiner, K., Klymus, K. E., Thomsen, P. F., Murphy, M. A., … Cornman, R. S. (2016). Critical considerations for the application of environmental DNA methods to detect aquatic species. *Methods in Ecology and Evolution*, *7*(11), 1299–1307.

Hajibabaei, M., deWaard, J. R., Ivanova, N. V, Ratnasingham, S., Dooh, R. T., Kirk, S. L., … Hebert, P. D. N. (2005). Critical factors for assembling a high volume of DNA barcodes. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *360*(1462), 1959–1967.

Hajibabaei, M., Shokralla, S., Zhou, X., Singer, G. A. C., & Baird, D. J. (2011). Environmental barcoding: a next-generation sequencing approach for biomonitoring applications using river benthos. *PLoS One*, *6*(4).

Hajibabaei, M., Singer, G. A. C., Hebert, P. D. N., & Hickey, D. A. (2007). DNA barcoding: how it complements taxonomy, molecular phylogenetics and population genetics. *TRENDS in Genetics*, *23*(4), 167–172.

Halpern, B. S., Walbridge, S., Selkoe, K. A., Kappel, C. V, Micheli, F., D'Agrosa, C., … Fox, H. E. (2008). A global map of human impact on marine ecosystems. *Science*, *319*(5865), 948–952.

Halstead, B. J., Wood, D. A., Bowen, L., Waters, S. C., Vandergast, A. G., Ersan, J. S., … Casazza, M. L. (2017). *An evaluation of the efficacy of using environmental DNA (eDNA) to detect giant gartersnakes (Thamnophis gigas)*. U.S. Geological

Survey Open-File Report 2017-1123, 41 p., https://doi.org/10.3133/ofr20171123.

Hardy, C. M., Krull, E. S., Hartley, D. M., & Oliver, R. L. (2010). Carbon source accounting for fish using combined DNA and stable isotope analyses in a regulated lowland river weir pool. *Molecular Ecology*, *19*(1), 197–212.

Hardy, N., Berry, T., Kelaher, B. P., Goldsworthy, S. D., Bunce, M., Coleman, M. A., … Figueira, W. (2017). Assessing the trophic ecology of top predators across a recolonisation frontier using DNA metabarcoding of diets. *Marine Ecology Progress Series*, *573*, 237–254.

Henry, G. W., & Lyle, J. M. (2003). *National recreational and indigenous fishing survey*. Department of Agriculture, Fisheries and Forestry, Australian Government, Canberra.

Hobbs, J.-P. A. (2014). A glaring omission in Australia's marine conservation planning. *Marine Policy*, *44*, 149–151.

Hobday, A. J. (2010). Ensemble analysis of the future distribution of large pelagic fishes off Australia. *Progress in Oceanography*, *86*(1–2), 291–301.

Hobern, D., & Hebert, P. (2019). BIOSCAN-revealing eukaryote diversity, dynamics, and interactions. *Biodiversity Information Science and Standards*, *3*, e37333.

Hollingsworth, P. M., Forrest, L. L., Spouge, J. L., Hajibabaei, M., Ratnasingham, S., van der Bank, M., … Erickson, D. L. (2009). A DNA barcode for land plants. *Proceedings of the National Academy of Sciences*, *106*(31), 12794–12797.

Humphreys, W., Tetu, S., Elbourne, L., Gillings, M., Seymour, J., Mitchell, J., & Paulsen, I. (2012). Geochemical and microbial diversity of Bundera sinkhole, an anchialine system in the eastern Indian Ocean. *Natura Croatica: Periodicum Musei Historiae Naturalis Croatici*, *21*(1), 59–63.

Hutchins, J. B. (2001). Checklist of fishes of Western Australia. *Records of the Western Australian Museum Supplement*, *63*, 9–50.

Hutchins, J. B., & Pearce, A. F. (1994). Influence of the Leeuwin Current on recruitment of tropical reef fishes at Rottnest Island, Western Australia. *Bulletin of Marine Science*, *54*(1), 245–255.

Huver, J. R., Koprivnikar, J., Johnson, P. T. J., & Whyard, S. (2015). Development and application of an eDNA method to detect and quantify a pathogenic parasite in aquatic ecosystems. *Ecological Applications*, *25*(4), 991–1002.

Ivanova, N. V, Zemlak, T. S., Hanner, R. H., & Hebert, P. D. N. (2007). Universal primer cocktails for fish DNA barcoding. *Molecular Ecology Notes*, *7*(4), 544–

548.

Jerde, C. L., Mahon, A. R., Chadderton, W. L., & Lodge, D. M. (2011). “Sight-unseen” detection of rare aquatic species using environmental DNA. *Conservation Letters*, *4*(2), 150–157.

Jeunen, G., Knapp, M., Spencer, H. G., Lamare, M. D., Taylor, H. R., Stat, M., … Gemmell, N. J. (2018). Environmental DNA (eDNA) metabarcoding reveals strong discrimination among diverse marine habitats connected by water movement. *Molecular Ecology Resources, 19*(2), 426-438.

Jeunen, G., Knapp, M., Spencer, H. G., Taylor, H. R., Lamare, M. D., Stat, M., … Gemmell, N. J. (2019). Species-level biodiversity assessment using marine environmental DNA metabarcoding requires protocol optimization and standardization. *Ecology and Evolution*, *9*(3), 1323–1335.

Jones, J., DiBattista, J. D., Stat, M., Bunce, M., Boyce, M. C., Fairclough, D. V, … Huggett, M. J. (2018). The microbiome of the gastrointestinal tract of a range-shifting marine herbivorous fish. *Frontiers in Microbiology*, *9*, 2000.

Keesing, J., & Irvine, T. (2005). Coastal biodiversity in the Indian Ocean: The known, the unknown. *Indian Journal of Marine Sciences*, *34*(1), 11-26.

Kelly, R. P., Closek, C. J., O’Donnell, J. L., Kralj, J. E., Shelton, A. O., & Samhouri, J. F. (2017). Genetic and manual survey methods yield different and complementary views of an ecosystem. *Frontiers in Marine Science*, *3*, 283.

Kelly, R. P., Port, J. A., Yamahara, K. M., & Crowder, L. B. (2014). Using environmental DNA to census marine fishes in a large mesocosm. *PLoS One*, *9*(1), e86175.

Klymus, K. E., Richter, C. A., Chapman, D. C., & Paukert, C. (2015). Quantification of eDNA shedding rates from invasive bighead carp *Hypophthalmichthys nobilis* and silver carp *Hypophthalmichthys molitrix*. *Biological Conservation*, *183*, 77–84.

Korbel, K., Chariton, A., Stephenson, S., Greenfield, P., & Hose, G. C. (2017). Wells provide a distorted view of life in the aquifer: implications for sampling, monitoring and assessment of groundwater ecosystems. *Scientific Reports*, *7*, 40702.

Koziol, A., Stat, M., Simpson, T., Jarman, S., DiBattista, J. D., Harvey, E. S., … Bunce, M. (2018). Environmental DNA metabarcoding studies are critically affected by substrate selection. *Molecular Ecology Resources, 19*(2).

Kyne, P.M, Rigby, C. ., Dharmadi, & Jabado, R. . (2019). *Rhynchobatus australiae*. The IUCN Red List of Threatened Species 2019: e.T41853A68643043.

Kyne, Peter M, Carlson, J., & Smith, K. (2013). *Pristis pristis* (errata version published in 2019). The IUCN Red List of Threatened Species 2013: e.T18584848A141788242.

Lacoursière-Roussel, A., Howland, K., Normandeau, E., Grey, E. K., Archambault, P., Deiner, K., … Bernatchez, L. (2018). eDNA metabarcoding as a new surveillance approach for coastal Arctic biodiversity. *Ecology and Evolution*, *8*(16), 7763–7777.

Lamb, P. D., Hunter, E., Pinnegar, J. K., Creer, S., Davies, R. G., & Taylor, M. I. (2019). How quantitative is metabarcoding: A meta-analytical approach. *Molecular Ecology*, *28*(2), 420–430.

Lambert, E., Le Bars, D. M., & de Ruijter, W. P. M. (2016). The connection of the Indonesian Throughflow, South Indian Ocean Countercurrent and the Leeuwin Current. *Ocean Science*, *12*(3), 771–780.

Last, P. R., & Stevens, J. D. (2009). *Sharks and Rays of Australia. Second Edition.* Collingwood, Australia: CSIRO Publishing.

Lindahl, B. D., Nilsson, R. H., Tedersoo, L., Abarenkov, K., Carlsen, T., Kjøller, R., … Stenlid, J. (2013). Fungal community analysis by high-throughput sequencing of amplified markers–a user's guide. *New Phytologist*, *199*(1), 288–299.

Machida, R. J., & Knowlton, N. (2012). PCR primers for metazoan nuclear 18S and 28S ribosomal DNA sequences. *PLoS One*, *7*(9).

Majaneva, M., Diserud, O. H., Eagle, S. H. C., Boström, E., Hajibabaei, M., & Ekrem, T. (2018). Environmental DNA filtration techniques affect recovered biodiversity. *Scientific Reports*, *8*(1), 1–11.

Maruyama, A., Nakamura, K., Yamanaka, H., Kondoh, M., & Minamoto, T. (2014). The release rate of environmental DNA from juvenile and adult fish. *PLoS One*, *9*(12).

McLeod, P., & Lindner, R. (2018). *Economic Dimension of Recreational Fishing in Western Australia*, Perth.

Medinger, R., Nolte, V., Pandey, R. V., Jost, S., Ottenwaelder, B., Schloetterer, C., & Boenigk, J. (2010). Diversity in a hidden world: potential and limitation of next-generation sequencing for surveys of molecular diversity of eukaryotic microorganisms. *Molecular Ecology*, *19*, 32–40.

Miya, M., Sato, Y., Fukunaga, T., Sado, T., Poulsen, J. Y., Sato, K., … Araki, H. (2015). MiFish, a set of universal PCR primers for metabarcoding environmental DNA from fishes: detection of more than 230 subtropical marine species. *Royal Society Open Science*, *2*(7), 150088.

Moore, C. H., Radford, B. T., Possingham, H. P., Heyward, A. J., Stewart, R. R., Watts, M. E., … Fisher, R. (2016). Improving spatial prioritisation for remote marine regions: optimising biodiversity conservation and sustainable development trade-offs. *Scientific Reports*, *6*, 32029.

Murakami, H., Yoon, S., Kasai, A., Minamoto, T., Yamamoto, S., Sakata, M. K., … Yamashita, Y. (2019). Dispersion and degradation of environmental DNA from caged fish in a marine environment. *Fisheries Science*, 1–11.

Nevers, M. B., Byappanahalli, M. N., Morris, C. C., Shively, D., Przybyla-Kelly, K., Spoljaric, A. M., … Roseman, E. F. (2018). Environmental DNA (eDNA): A tool for quantifying the abundant but elusive round goby (*Neogobius melanostomus*). *PLoS One*, *13*(1).

Nichols, S. J., Kefford, B. J., Campbell, C. D., Bylemans, J., Chandler, E., Bray, J. P., … Furlan, E. M. (2019). Towards routine DNA metabarcoding of macroinvertebrates using bulk samples for freshwater bioassessment: Effects of debris and storage conditions on the recovery of target taxa. *Freshwater Biology*.

O'Donnell, J. L., Kelly, R. P., Shelton, A. O., Samhouri, J. F., Lowell, N. C., & Williams, G. D. (2017). Spatial distribution of environmental DNA in a nearshore marine habitat. *PeerJ*, 5. https://doi.org/10.7717/peerj.3044

Olds, B. P., Jerde, C. L., Renshaw, M. A., Li, Y., Evans, N. T., Turner, C. R., … Shirey, P. D. (2016). Estimating species richness using environmental DNA. *Ecology and Evolution*, *6*(12), 4214–4226.

Oyafuso, Z. S., Toonen, R. J., & Franklin, E. C. (2016). Temporal and spatial trends in prey composition of wahoo *Acanthocybium solandri*: a diet analysis from the central North Pacific Ocean using visual and DNA barcoding techniques. *Journal of Fish Biology*, *88*(4), 1501–1523.

Paula, D. P., Linard, B., Crampton-Platt, A., Srivathsan, A., Timmermans, M. J. T. N., Sujii, E. R., … Vogler, A. P. (2016). Uncovering trophic interactions in arthropod predators through DNA shotgun-sequencing of gut contents. *PLoS One*, *11*(9).

Piaggio, A. J., Engeman, R. M., Hopken, M. W., Humphrey, J. S., Keacher, K. L., Bruce, W. E., & Avery, M. L. (2014). Detecting an elusive invasive species: a

diagnostic PCR to detect Burmese python in Florida waters and an assessment of persistence of environmental DNA. *Molecular Ecology Resources*, *14*(2), 374–380.

Pochon, X., Bott, N. J., Smith, K. F., & Wood, S. A. (2013). Evaluating detection limits of next-generation sequencing for the surveillance and monitoring of international marine pests. *PLoS One*, *8*(9), e73935.

Pochon, X., Wecker, P., Stat, M., Berteaux-Lecellier, V., & Lecellier, G. (2019). Towards an in-depth characterization of Symbiodiniaceae in tropical giant clams via metabarcoding of pooled multi-gene amplicons. *PeerJ*, *7*, e6898.

Port, J. A., O'Donnell, J. L., Romero-Maraccini, O. C., Leary, P. R., Litvin, S. Y., Nickols, K. J., … Kelly, R. P. (2016). Assessing vertebrate biodiversity in a kelp forest ecosystem using environmental DNA. *Molecular Ecology*, *25*(2). https://doi.org/10.1111/mec.13481

Port, J. A., O'Donnell, J. L., Romero-Maraccini, O. C., Leary, P. R., Litvin, S. Y., Nickols, K. J., … Kelly, R. P. (2016). Assessing vertebrate biodiversity in a kelp forest ecosystem using environmental DNA. *Molecular Ecology*, *25*(2), 527–541.

Preen, A. (1998). PROFILE: Marine Protected Areas and Dugong Conservation Along Australia's Indian Ocean Coast. *Environmental Management*, *22*(2), 173–181.

Reef, R., Atwood, T. B., Samper-Villarreal, J., Adame, M. F., Sampayo, E. M., & Lovelock, C. E. (2017). Using eDNA to determine the source of organic carbon in seagrass meadows. *Limnology and Oceanography*, *62*(3), 1254–1265.

Rees, H. C., Maddison, B. C., Middleditch, D. J., Patmore, J. R. M., & Gough, K. C. (2014). The detection of aquatic animal species using environmental DNA–a review of eDNA as a survey tool in ecology. *Journal of Applied Ecology*, *51*(5), 1450–1459.

Rees, M., Colquhoun, J., Smith, L. ., & Heyward, A. . (2003). *Survey of trochus, holothuria, giant clams, and the coral communities of Ashmore, Cartier Reef and Mermaid Reef, northwestern Australia.* Townsville, Queensland.

Riaz, T., Shehzad, W., Viari, A., Pompanon, F., Taberlet, P., & Coissac, E. (2011). ecoPrimers: inference of new DNA barcode markers from whole genome sequence analysis. *Nucleic Acids Research*, *39*(21), e145–e145.

Richards, Z. T., Garcia, R., Moore, G., Fromont, J., Kirkendale, L., Bryce, M., … Gomez, O. (2019). A tropical Australian refuge for photosymbiotic benthic fauna. *Coral Reefs*, *38*(4), 669–676.

Roy, A. (2019). Blue Economy in the Indian Ocean: Governance Perspectives for Sustainable Development in the Region. *Occasional Paper, January*.

Sassoubre, L. M., Yamahara, K. M., Gardner, L. D., Block, B. A., & Boehm, A. B. (2016). Quantification of environmental DNA (eDNA) shedding and decay rates for three marine fish. *Environmental Science & Technology*, *50*(19), 10456–10464.

Schloss, P. D., Westcott, S. L., Ryabin, T., Hall, J. R., Hartmann, M., Hollister, E. B., … Robinson, C. J. (2009). Introducing mothur: open-source, platform-independent, community-supported software for describing and comparing microbial communities. *Applied and Environmental Microbiology.*, *75*(23), 7537–7541.

Schoch, C. L., Seifert, K. A., Huhndorf, S., Robert, V., Spouge, J. L., Levesque, C. A., … Consortium, F. B. (2012). Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for Fungi. *Proceedings of the National Academy of Sciences*, *109*(16), 6241–6246.

Shaw, J. L. A., Clarke, L. J., Wedderburn, S. D., Barnes, T. C., Weyrich, L. S., & Cooper, A. (2016). Comparison of environmental DNA metabarcoding and conventional fish survey methods in a river system. *Biological Conservation*, *197*, 131–138.

Shokralla, S., Spall, J. L., Gibson, J. F., & Hajibabaei, M. (2012). Next-generation sequencing technologies for environmental DNA research. *Molecular Ecology*, *21*(8). https://doi.org/10.1111/j.1365-294X.2012.05538.x

Sigsgaard, E. E., Carl, H., Møller, P. R., & Thomsen, P. F. (2015). Monitoring the near-extinct European weather loach in Denmark based on environmental DNA from water samples. *Biological Conservation*, *183*, 46–52.

Sigsgaard, E. E., Nielsen, I. B., Bach, S. S., Lorenzen, E. D., Robinson, D. P., Knudsen, S. W., … Thomsen, P. F. (2016). Population characteristics of a large whale shark aggregation inferred from seawater environmental DNA. *Nature Ecology & Evolution*, *1*(1). https://doi.org/10.1038/s41559-016-0004

Smart, A. S., Weeks, A. R., van Rooyen, A. R., Moore, A., McCarthy, M. A., & Tingley, R. (2016). Assessing the cost-efficiency of environmental DNA sampling. *Methods in Ecology and Evolution*, *7*(11), 1291–1298.

Sogin, M. L., Morrison, H. G., Huber, J. A., Welch, D. M., Huse, S. M., Neal, P. R., … Herndl, G. J. (2006). Microbial diversity in the deep sea and the underexplored

"rare biosphere." *Proceedings of the National Academy of Sciences*, *103*(32), 12115–12120.

Stat, M., Huggett, M. J., Bernasconi, R., DiBattista, J. D., Berry, T. E., Newman, S. J., … Bunce, M. (2017). Ecosystem biomonitoring with eDNA: metabarcoding across the tree of life in a tropical marine environment. *Scientific Reports*, *7*(1), 12240.

Stat, M., John, J., DiBattista, J. D., Newman, S. J., Bunce, M., & Harvey, E. S. (2019). Combined use of eDNA metabarcoding and video surveillance for the assessment of fish biodiversity. *Conservation Biology*, *33*(1), 196-205.

Taberlet, P, Coissac, E., Hajibabaei, M., & Rieseberg, L. H. (2012). Environmental DNA. *Molecular Ecology*, *21*, 1789–1793. https://doi.org/10.1111/j.1365-294X.2012.05542.x

Taberlet, Pierre, Bonin, A., Zinger, L., & Coissac, E. (2018). *Environmental DNA: For Biodiversity Research and Monitoring*. Oxford University Press.

TFMPA, A. (1998). *Guidelines for establishing the national representative system of marine protected areas*. Canberra: Australia and New Zealand Environment and Conservation Council Task Force on Marine Protected Areas.

Thomsen, P. F., Kielgast, J., Iversen, L. L., Møller, P. R., Rasmussen, M., & Willerslev, E. (2012). Detection of a diverse marine fish fauna using environmental DNA from seawater samples. *PLoS One*, *7*(8). https://doi.org/10.1371/journal.pone.0041732

Thomsen, P. F., Kielgast, J. O. S., Iversen, L. L., Wiuf, C., Rasmussen, M., Gilbert, M. T. P., … Willerslev, E. (2012). Monitoring endangered freshwater biodiversity using environmental DNA. *Molecular Ecology*, *21*(11), 2565–2573.

Thomsen, P. F., Møller, P. R., Sigsgaard, E. E., Knudsen, S. W., Jørgensen, O. A., & Willerslev, E. (2016). Environmental DNA from Seawater Samples Correlate with Trawl Catches of Subarctic, Deepwater Fishes. *PLoS One*, 1–22. https://doi.org/10.1371/journal.pone.0165252

Thomsen, P. F., & Willerslev, E. (2015). Environmental DNA - An emerging tool in conservation for monitoring past and present biodiversity. *Biological Conservation*, *183*, 4-18. https://doi.org/10.1016/j.biocon.2014.11.019

Tittensor, D. P., Mora, C., Jetz, W., Lotze, H. K., Ricard, D., Berghe, E. Vanden, & Worm, B. (2010). Global patterns and predictors of marine biodiversity across taxa. *Nature*, *466*(7310), 1098–1101.

Travers, M. J., Potter, I. C., Clarke, K. R., Newman, S. J., & Hutchins, J. . (2010). The inshore fish faunas over soft substrates and reefs on the tropical west coast of Australia differ and change with latitude and bioregion. *Journal of Biogeography*, *37*(1), 148–169.

Turner, C. R., Miller, D. J., Coyne, K. J., & Corush, J. (2014). Improved methods for capture, extraction, and quantitative assay of environmental DNA from Asian bigheaded carp (*Hypophthalmichthys* spp.). *PLoS One*, *9*(12).

Ushio, M., Murakami, H., Masuda, R., Sado, T., Miya, M., Sakurai, S., … Kondoh, M. (2018). Quantitative monitoring of multispecies fish environmental DNA using high-throughput sequencing. *Metabarcoding and Metagenomics*, *2*, e23297.

Valentini, A., Taberlet, P., Miaud, C., Civade, R., Herder, J., Thomsen, P. F., … Dejean, T. (2016). Next-generation monitoring of aquatic biodiversity using environmental DNA metabarcoding. *Molecular Ecology, 25*(4), 929-942. https://doi.org/10.1111/mec.13428

Van der Elst, R., Everett, B., Jiddawi, N., Mwatha, G., Afonso, P. S., & Boulle, D. (2005). Fish, fishers and fisheries of the Western Indian Ocean: their diversity and status. A preliminary assessment. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *363*(1826), 263–284.

Vences, M., Lyra, M. L., Perl, R. G. B., Bletz, M. C., Stanković, D., Lopes, C. M., … Haddad, C. F. B. (2016). Freshwater vertebrate metabarcoding on Illumina platforms using double-indexed primers of the mitochondrial 16S rRNA gene. *Conservation Genetics Resources*, *8*(3), 323–327.

Weeks, E. S., Mason, N., Ausseil, A.-G. E., & Herzig, A. (2014). Prioritising land-use decisions for the optimal delivery of ecosystem services and biodiversity protection in productive landscapes. *Biodiversity–the Dynamic Balance of the Planet. Rijeka, Croatia: InTech*, 1–32.

Wells, S. M. (1997). *Giant clams: Status, trade and mariculture, and the role of CITES in management*. IUCN-The World Conservation Union.

Weltz, K., Lyle, J. M., Ovenden, J., Morgan, J. A. T., Moreno, D. A., & Semmens, J. M. (2017). Application of environmental DNA to detect an endangered marine skate species in the wild. *PLoS One*, *12*(6).

Whiting, A. U., Thomson, A., Chaloupka, M., & Limpus, C. J. (2009). Seasonality,

abundance and breeding biology of one of the largest populations of nesting flatback turtles, *Natator depressus*: Cape Domett, Western Australia. *Australian Journal of Zoology*, *56*(5), 297–303.

Wilcox, T. M., McKelvey, K. S., Young, M. K., Jane, S. F., Lowe, W. H., Whiteley, A. R., & Schwartz, M. K. (2013). Robust detection of rare species using environmental DNA: the importance of primer specificity. *PLoS One*, *8*(3).

Willerslev, E., Hansen, A. J., Binladen, J., Brand, T. B., Gilbert, M. T. P., Shapiro, B., … Cooper, A. (2003). Diverse plant and animal genetic records from Holocene and Pleistocene sediments. *Science*, *300*(5620), 791–795.

Wilson, B. (2014). Kimberley marine biota. History and environment. *Records of the Western Australian Museum Supplement*, *84*, 1–18.

Woodside. (2018). *Woodside's North West Shelf Project Extension*.

Yamamoto, S., Minami, K., Fukaya, K., Takahashi, K., Sawada, H., Murakami, H., … Horiuchi, T. (2016). Environmental DNA as a 'snapshot'of fish distribution: A case study of Japanese jack mackerel in Maizuru Bay, Sea of Japan. *PLoS One*, *11*(3).

Yit Sen Bull, C., & Van Sebille, E. (2016). Sources, fate, and pathways of Leeuwin Current water in the Indian Ocean and Great Australian Bight: A Lagrangian study in an eddy-resolving ocean model. *Journal of Geophysical Research: Oceans*, *121*(3), 1626–1639.

Yoccoz, N. G. (2012). The future of environmental DNA in ecology. *Molecular Ecology*, *21*(8), 2031–2038.

Zinke, J., Gilmour, J. P., Fisher, R., Puotinen, M., Maina, J., Darling, E., … Beger, M. (2018). Gradients of disturbance and environmental conditions shape coral community structure for south-eastern Indian Ocean reefs. *Diversity and Distributions*, *24*(5), 605–620.

— Chapter 2 —

eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem









2.1 Prologue

Chapter one of this thesis introduced the development of eDNA metabarcoding as a biomonitoring tool in aquatic ecosystems. However, at present, there remains a number of applied and methodological questions that need to be thoroughly addressed. One area that requires further investigation, is the localisation of eDNA in aquatic environments, particularly in areas of high-water movement. A second area, is the efficiency of eDNA metabarcoding compared to conventional survey techniques in a well-studied aquatic system. The following chapter addresses these topics by conducting a marine eDNA metabarcoding survey across the Cocos (Keeling) Islands, a remote, albeit highly surveyed, island group situated in Australia's Indian Ocean region.

The Cocos (Keeling) Islands (12.1642° S, 96.8710° E) are located approximately 1000 km southwest of Java, Indonesia, in the tropical eastern Indian Ocean. The uninhabited islands were first reported by Captain William Keeling of the East India Company in 1609. However, they were not settled until 1826, with the arrival of Englishman Alexander Hare and 50 enslaved Malay men and women (Ackrill, 1984). The arrival of Captain John Clunies-Ross a year later, was the beginning of the Clunies-Ross family dynasty which exerted control over the Cocos (Keeling) Islands for 121 years. John Clunies-Ross established a long-standing copra (coconut) plantation in 1829 with imported indentured labour from the Malay peninsula and Indonesia (Ackrill, 1984)—their island descendants now referred to as the Cocos Malays. Notably, Charles Darwin visited the islands in 1836 as part of the *HMS Beagle* voyage. This is where he developed his theory on coral atoll formation and proclaimed that the coral reefs of Cocos "*surely rank high amongst the wonderful objects of this world*" (Darwin, Fitzroy, & King, 1839). The islands were annexed by the British Crown in 1857, however, Queen Victoria granted them in perpetuity to the Clunies-Ross family (Stanhope, Wettenhall, & Bhusal, 2016). Administration of the Cocos (Keeling) Islands was transferred from the British Colony of Singapore to Australia in 1954-55. The Australian Government officially purchased the islands off the Clunies-Ross family, under the threat of compulsory acquisition, in 1978 for the sum of AU$6.5 million (Stanhope et al., 2016). The island group is now managed as one of Australia's external Indian Ocean territories.

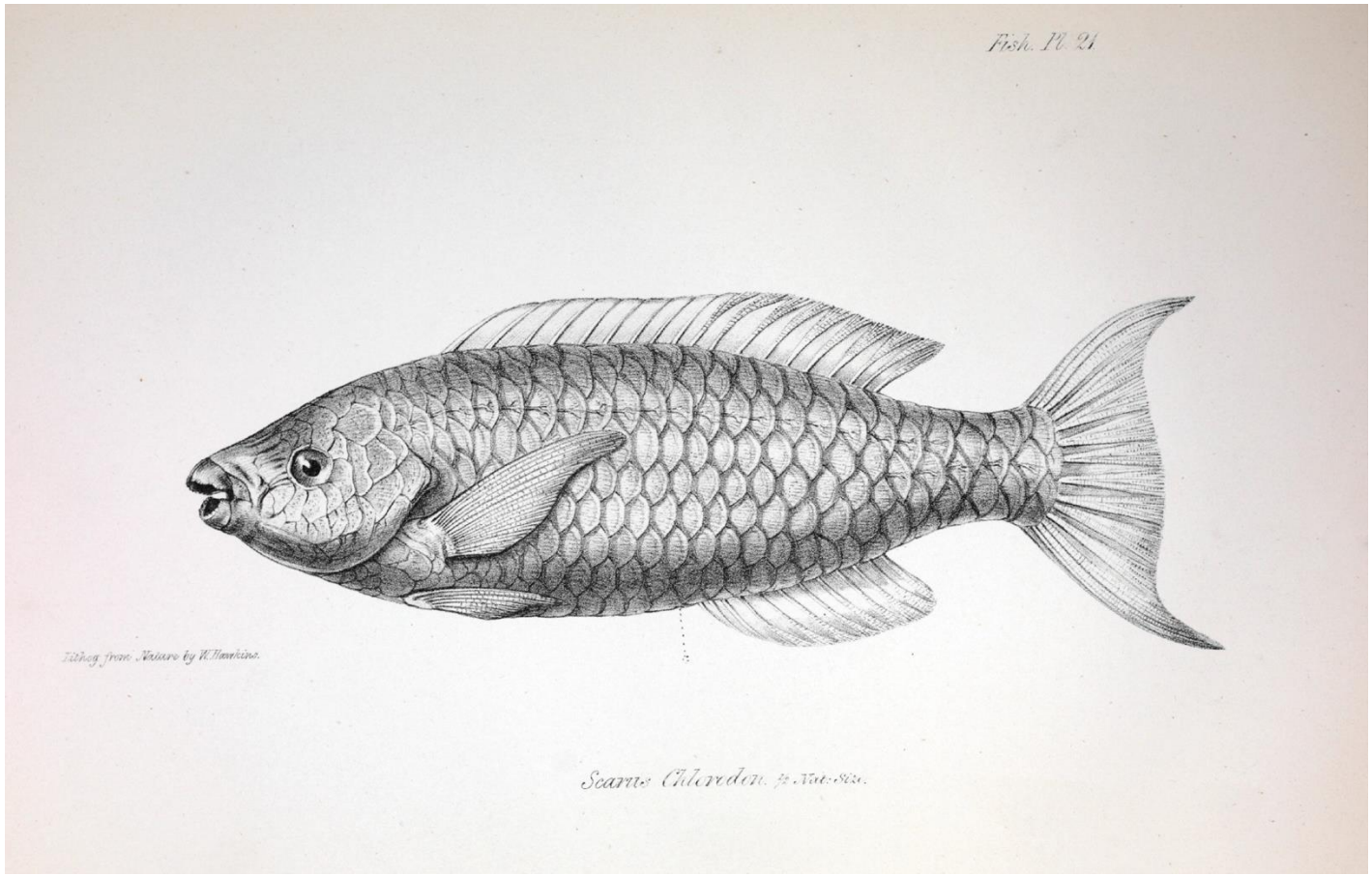


Figure 2.1 A specimen of *Scarus chlorodon* (now *Scarus prasiognathos*) collected by Charles Darwin at the Cocos (Keeling) Islands in 1836.
Illustrated in The Zoology of the Voyage of the Beagle Part IV, Fish (1842), Plate XXI by Leonard Jenyns. Image courtesy of the Biodiversity Heritage Library. Contributed by the Natural History Museum Library, London. | http://www.biodiversitylibrary.org

The Cocos (Keeling) Islands are comprised of two low lying atolls; a singular island atoll in the north, whilst the (main) southern atoll is comprised of 24 islets surrounding a lagoon. There is extensive oceanic and tidal movement flushing through and around the main atoll in a south-east to north-west direction (Hobbs & Macrae, 2012). Cocos receives the tail end of the Indonesian Throughflow, which brings an upwelling of nutrient-rich water from the Java coast (Brewer et al., 2009). Over 150 terrestrial species, largely birds, and 1700 marine species (largely of Indo-West Pacific origin and including hybrid reef fish) have been recorded (Hobbs, Frisch, Allen, & Van Herwerden, 2008; Hobbs et al., 2014; Johnstone & Darnell, 2017; Marsh, 1994; Morgan, 1994; Richards & Hobbs, 2014; Tan & Low, 2014). Due to its recent emergence approximately 3000 years ago (Woodroffe, McLean, Polach, & Wallensky, 1990), there is a lack of endemicity on the islands, despite their oceanic isolation. There is only one endemic terrestrial species (the Cocos Buff-banded Rail, *Gallirallus philippensis andrewsi*), and one endemic marine species (the Cocos pygmy angelfish, *Centropyge joculator*). Nevertheless, the Cocos (Keeling) Islands, particularly Pulu Keeling National Park on the northern atoll, are of high marine and terrestrial biodiversity importance.

2.1.1 Chapter acknowledgements

I acknowledge the Department of Primary Industries and Regional Development (WA), BMT Oceanica and the Australian Research Council (LP160100839) for project funding. For fieldwork assistance I thank Chynna Cahill and Parks Australia. In the development of the 16S fish database I would like to thank R. Czarnik, E. Fenny, C. Wakefield, D. Fairclough, S. Payet, N. Jarvis, S. Moyle, D. Boddington, and K. Smith. For valuable assistance in the TrEnD laboratory I thank Megan Coghlan, Matt Power, Adam Koziol, Frederik Seersholm, Matthew Heydenrych, Kristen Fernandes, Alicia Grealy & Tina Berry. I acknowledge bioinformatic assistance from the Pawsey Supercomputing Centre (WA). For fish distribution data at the Cocos (Keeling) Islands, I would like to thank J.P. Hobbs. Finally, I give thanks to Tane Sinclair-Taylor for providing photos taken in the Cocos (Keeling) Islands. Cover photos include those from Sophie Preston and Zoe Richards.

2.1.2 Data accessibility

Demultiplexed (unfiltered) metabarcoding sequencing data and taxonomic presence/absence matrices from this study are available for download on Dryad Digital Repository (https://doi.org/10.5061/dryad.4qrfj6q6h).

This chapter resulted in a manuscript that was published in *Molecular Ecology* (West et al. 2020, DOI: doi:10.1111/mec.15382); a facsimile of which can be located at the end of the Appendices. As first author, permission is automatically granted to reproduce this copyright material if it constitutes less than half of the total material in such publication. This chapter is a reproduction of the aforementioned manuscript.

2.2 Abstract

Environmental DNA (eDNA) metabarcoding, a technique for retrieving multi-species DNA from environmental samples, can detect a diverse array of marine species from filtered seawater samples. There is a growing potential to integrate eDNA alongside existing monitoring methods in order to establish or improve the assessment of species diversity. Remote island reefs are increasingly vulnerable to climate-related threats and as such there is a pressing need for cost-effective whole-ecosystem surveying to baseline biodiversity, study assemblage changes and ultimately develop sustainable management plans. We investigated the utility of eDNA metabarcoding as a high-resolution, multi-trophic biomonitoring tool at the Cocos (Keeling) Islands, Australia (CKI)—a remote tropical coral reef atoll situated within the eastern Indian Ocean. Metabarcoding assays targeting the mitochondrial 16S rRNA and CO1 genes, as well as the 18S rRNA nuclear gene, were applied to 252 surface seawater samples collected from 42 sites within a 140 km^2 area. Our assays successfully detected a wide range of bony fish and elasmobranchs (244 taxa), crustaceans (88), molluscs (37) and echinoderms (7). Assemblage composition varied significantly between sites, reflecting habitat partitioning across the island ecosystem and demonstrating the localisation of eDNA signals, despite extensive tidal and oceanic movements. In addition, we document putative new occurrence records for 46 taxa and compare the efficiency of our eDNA approach to visual survey techniques at CKI. Our study demonstrates the utility of a multi-marker metabarcoding approach in capturing multi-trophic biodiversity across an entire coral reef atoll and sets an important baseline for ongoing monitoring and management.

2.3 Introduction

Tropical islands harbour at least 40% of the world's coral reefs (Spalding, Ravilious, & Green, 2001) and support a diverse range of marine ecosystems and communities. Such high island reef diversity results from a complex interplay of biogeographic and oceanographic processes, along with historical and contemporary factors such as disturbance history (Briggs, 1966; Cortés et al., 2017; Graham et al., 2018; Houk, Benavente, Iguel, Johnson, & Okano, 2014). Isolated archipelagos and peripheral islands at the borders of biogeographic provinces, such as those of the Hawaiian archipelago, Mascarene Islands, Galapagos Islands, Pitcairn Island group and Easter Island, are a source of evolutionary novelty and high endemism (Allen, 2008; Budd & Pandolfi, 2010; Friedlander et al., 2013; Hourigan & Reese, 1987). Other peripheral island reef ecosystems such as Christmas Island, the Cocos (Keeling) Islands and the Socotra Archipelago lie in suture zones where marine biogeographical provinces collide, leading to regions of secondary contact that drive hybridisation and diversification (DiBattista et al., 2015; Hobbs, Frisch, Allen, & Van Herwerden, 2008). Documenting and monitoring biodiversity is particularly important at remote reefs because in many cases, these locations can provide baseline information about how 'natural' systems are structured and composed with fewer (or less intense) anthropogenic influences (Knowlton & Jackson, 2008).

The establishment of island reef baselines and subsequent monitoring, that can inform both local and global management decisions, is intrinsically reliant upon cost-effective and robust surveying techniques. The emergence of environmental DNA (eDNA) metabarcoding, an innovative detection technique profiling multi-species genetic material in an environmental sample, is revolutionising our approach to non-invasive, efficient, whole-ecosystem surveying (Bista et al., 2017; Port et al., 2016; Stat et al., 2017). After widespread global testing, this technique has demonstrated its ability to detect a wide variety of biota from aquatic ecosystems, including dark diversity—taxa that are not easily identified using observation-based approaches (Boussarie et al., 2018; Jerde, Mahon, Chadderton, & Lodge, 2011; Vörös, Márton, Schmidt, Gál, & Jelić, 2017). As such, eDNA metabarcoding is anticipated to become a key tool in marine biodiversity discovery and monitoring, as recently outlined in a decadal plan for taxonomy and biosystematics in Australia and New Zealand (Taxonomy Decadal

Plan Working Group, 2018). Likewise, the method is capable of responding to UNESCO's 'Decade of Ocean Science for Sustainable Development (2021-2030) which calls for improved ways to monitor the world's oceans (IOC-UNESCO, 2018).

With high biodiversity values, varied marine environments and a remote geographical position, the Cocos (Keeling) Islands (CKI; 12° 12″ S, 96° 54″ E) are an ideal model system for testing the utility of eDNA metabarcoding in surveying the multi-trophic biodiversity of an entire island reef system. The CKI are one of Australia's external territories, located approximately 1000km southwest of Java, Indonesia, in the tropical eastern Indian Ocean. This remote island group is comprised of a southern and northern coral atoll; the southern atoll (considered the main CKI atoll) is comprised of 24 islets (including the inhabited isles of West Island and Home Island), whilst the northern atoll (referred to as North Keeling) is a singular island atoll protected as a national park. The CKI lie in a suture zone along the marine biogeographic boundaries of the Indian and Pacific Oceans and as such are comprised of species of Indo-West Pacific origin, with little endemism (Hobbs *et al.*, 2014; Woodroffe & Berry, 1994). The islands are renowned for their marine biodiversity; with a total of 602 recorded fish species (Hobbs, Newman, et al., 2014), over 100 species of hard coral (Richards & Hobbs, 2014), 700 mollusc species (Tan & Low, 2014) and 200 crustacean species (Mendoza, Lasley Jr., & Ng, 2014; Morgan, 1994).

Over the last two decades, disturbance events such as mass fish and coral die-offs from deoxygenation, coral bleaching, crowns-of-thorns starfish (*Acanthaster planci*) outbreaks, and cyclonic storm surges (Bunce, 1988; Hobbs & McDonald, 2010; Woodroffe & Berry, 1994), have collectively impacted marine communities at the CKI. The susceptibility of the marine life at the CKI to increasing climate-related threats is exacerbated by the isolation of the island group, as recruitment is restricted from external locations such as from neighbouring Christmas Island (approximately 980 km distance) and Indonesia (approximately 1,600 km), to species that have a long-lived pelagic larval phase (Hourston, 2010). The application of eDNA methods, which has the potential to simultaneously characterise a variety of reef communities, provides an efficient approach for documenting the increasingly vulnerable marine biodiversity of the CKI.

In this study, we conduct a multi-marker eDNA metabarcoding survey of the various marine environments surrounding the main CKI atoll in order to: (i) evaluate the spatial sensitivity of an eDNA signal retrieved from seawater, by assessing its ability to delineate adjacent habitats amidst tidal and oceanic movements within and around an entire island ecosystem, (ii) compare the efficiency of our eDNA approach against extensive visual/video, line fishing and ichthyocide survey techniques previously employed at the CKI, and (iii) provide the first molecular baselines for CKI bony fish, elasmobranch, crustacean, mollusc and echinoderm diversity and community composition, which we anticipate will provide the basis for inclusion into longer term monitoring programs. Lastly, in order to provide a roadmap for surveying tropical island ecosystems, we also explore assay performance, the required levels of replication and the taxonomic resolution that can be achieved at present with our chosen assays and publicly accessible reference databases. The overarching aim was to better evaluate the overall utility of a multi-marker metabarcoding approach towards whole-ecosystem surveying of island reef ecosystems.

2.4 Methods

2.4.1 Field sampling

Six one-litre seawater replicates were sampled from 42 sites in and around the main CKI atoll (Figure 2.2, Table S2.1) in April 2017, totalling 252 samples across a 140 km^2 area. Samples were obtained from six different habitats zones based on habitat classifications generated from prior environmental surveys (Hender, McDonald, & Gilligan, 2001; Parks Australia, 2005; Williams, 1994). Thirteen sites were located within the designated low complexity lagoon area, nine within the medium complexity lagoon, four within the high complexity lagoon, four on the outer reef terrace, 11 within the intertidal reef and one within a lagoonlet site on the West Island (see S2.9.1 for further habitat zone information). In addition, sample sites were further classified based on whether they were in the outer lagoon, inner lagoon or within one of the major surge channels, where oceanic water from the outside of the atoll is flushed through the lagoon at high tide, in a south-east to north-west direction (Hobbs and Macrae, 2012; S2.9.1).

Seawater samples were collected at the surface of each site using bleach sterilised Nalgene bottles and then immediately stored on ice. Each sample was individually filtered across Pall 0.2μm Supor® polyethersulfone membranes using a Pall Sentino® Microbiology pump (Pall Corporation, Port Washington, USA), within three hours of collection. Samples of the 10% bleach solution—used to clean filtration equipment over the course of each sampling day—were included as filtration controls. These filtration controls serve to detect any potential cross-contamination in water filtering between successive sites and during handling post-filtering. The filter membranes were immediately frozen and stored at -20°C prior to their transportation to a quarantine facility within the Trace & Environmental DNA (TrEnD) Laboratory in Perth, Western Australia; filters were stored long-term at -80°C.

2.4.2 Laboratory processing

DNA was extracted from the filter membranes, within two weeks of collection, using a DNeasy Blood and Tissue Kit (Qiagen; Venlo, Netherlands) with the following

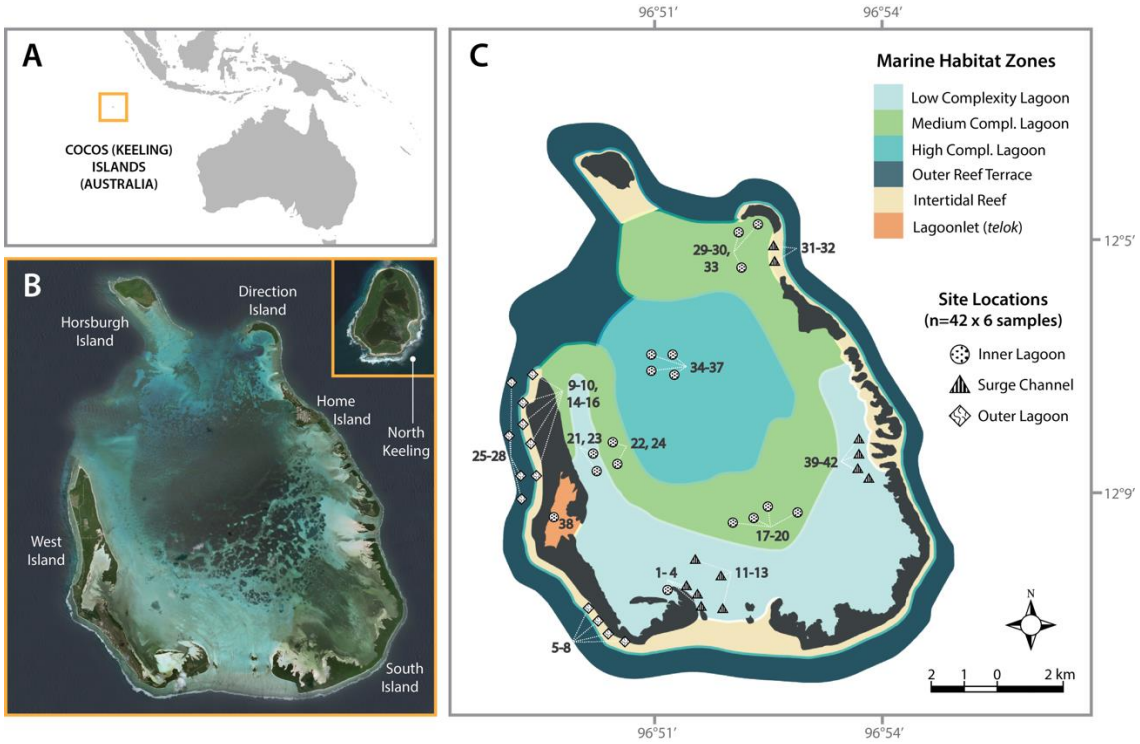


Figure 2.2 (A) Location of the Cocos (Keeling) Islands (CKI) in the eastern Indian Ocean. (B) Satellite imagery of the main CKI atoll with an inset of North Keeling atoll (not to scale). (C) Marine habitat zone map of the CKI, based on previous habitat strata divisions reported by Parks Australia (2005), Hender et al. (2001) and Williams (1994).

Site locations indicate where 6x1L water replicates were collected per site for eDNA surveying (for a total of 252 water samples across 42 sites) and were additionally classified as being situated either within the Inner Lagoon, Surge Channel or Outer Lagoon. Number of sites per marine habitat zone are as follows: Low Complexity n=13, Medium Complexity n=9, High Complexity n=4, Outer Reef Terrace n=4, Intertidal Reef n=11, and Lagoonlet (*telok*) n=1. Number of sites per lagoon zone as are follows: Inner Lagoon n=17, Surge Channel n=12, and Outer Lagoon n=13.

modifications: 540µl of ATL lysis buffer and 60µl of Proteinase K during the cell digestion phase. Negative controls, containing no sample (or filters), were extracted and processed alongside all samples in order to detect any cross-contamination. Previously published primers were sourced to amplify bony fish, elasmobranchs, crustaceans and other eukaryotes (including molluscs and echinoderms), respectively, from mixed environmental samples. The four applied PCR assays, herein referred to as 16S Fish, COI Elasmobranch, 16S Crustacean and 18S Universal (Table 2.1), were incorporated into our eDNA metabarcoding workflow. The multiplex COI Elasmobranch assay (forward primers: Fish F2 and V52 (Ward, Zemlak, Innes, Last, & Hebert, 2005) and reverse primer: Shark COI-MINIR (Fields, Abercrombie, Eng, Feldheim, & Chapman, 2015) had two degenerate positions (Ns) added into each primer to increase the variability of elasmobranch amplicons. Quantitative PCR (qPCR) protocols are described in depth in S2.9.2.

Pooled libraries were then size-selected using a Pippin Prep (Sage Science, Beverly, USA), purified with a Qiaquick PCR Purification Kit (Qiagen, Venlo, Netherlands), quantified using a Qubit 4.0 Fluorometer (Invitrogen, Carlsbad, USA), and loaded onto either a 300 cycle (for unidirectional sequencing) or 500 cycle (for paired-end sequencing) MiSeq® V2 Standard Flow Cell on an Illumina MiSeq platform (Illumina, San Diego, USA), housed in the TrEnD Laboratory at Curtin University (S2.9.2).

2.4.3 Bioinformatics and taxonomic assignments

Sequences were merged, demultiplexed and filtered using a combination of AdapterRemoval (v2; Schubert, Lindgreen, & Orlando, 2016), Geneious® (10.0.6; Kearse *et al.*, 2012), USEARCH (v9.0.2132; Edgar, 2010) and OBITools (v1.2.9; Boyer, Mercier, Bonin, Taberlet, & Coissac, 2014). Quality filtering parameters included a minimum length of 100bp, a maximum expected error rate of 0.5, no ambiguous bases (N's), a minimum read count of 5, a read threshold ratio of 0.05% and no chimeras (see S2.9.3 for pipeline details). Quality filtered dereplicated sequences were then queried against the National Centre for Biotechnology Information's (NCBI) GenBank nucleotide database (accessed in 2017/18; Benson *et al.*, 2005) and an in-house 16S rDNA fish database (Table S2.6) via Zeus, an SGI

Table 2.1 PCR assay information for marine eDNA metabarcoding at CKI.
Four primer sets: 16S Fish, 16S Crustacean, COI Elasmobranch and 18S Universal, corresponding to the following gene regions, respectively: mitochondrial 16S rDNA, mitochondrial COI, and nuclear 18S rDNA, were applied to all collected seawater samples. In the primer name, "F" refers to the forward primer and "R" to the reverse primer.

PCR Assay	Target taxa	Primer Name	Oligonucleotide Sequence (5'-3')	Target Length (bp)	Annealing temp (°C)	Primer Reference
16S Fish	Fish	16SF/D	GACCCTATGGAGCTTTAGAC	178-228	54	Berry et al., 2017
		16S2R-degenerate	CGCTGTTATCCCTADRGTAACT			Deagle et al., 2007
COI Elasmobranch	Sharks, Skates & Rays	FishF1-degenerate	ACCAACCACAAAGANATNGGCAC	110-241	52	**Original:** Ward et al., 2005 **Degenerate:** This study
		FishF2-degenerate	TCNACNAATCATAAAGATATCGGCAC			
		Shark COI-MINIR-degenerate	GATTATTACNAAAGCNTGGGC			**Original:** Fields et al., 2015 **Degenerate:** This study
16S Crustacean	Crustacean	Crust16S_F (short)	GGGACGATAAGACCCTATA	90-213	51	Berry et al., 2017
		Crust16S_R (short)	ATTACGCTGTTATCCCTAAAG			
18S Universal	Eukaryotes	18S_1F	GCCAGTAGTCATATGCTTGTCT	336-423	52	Pochon, Bott, Smith, & Wood, 2013
		18S_400R	GCCTGCTGCCTTCCTT			

cluster, based at the Pawsey Supercomputing Centre in Kensington, Western Australia. Taxonomic assignments were made in MEGAN (MEtaGenome ANalyzer v5.11.3; (Huson *et al.*, 2007; see S2.9.3) and classified based on geographic distribution and habitat association data obtained from FishBase (Froese & Pauly, 2018) and the World Register of Marine Species (WoRMS; Horton *et al.*, 2018).

To present putative new occurrence records at the CKI, we required that all global congeneric taxa (of the putative species in question) had been barcoded for the gene region of the respective assay; this removes the possibility that a new occurrence record is actually that of a closely-related (not yet barcoded) species. Rarefaction analyses were conducted using QIIME (v1; Caporaso *et al.*, 2010), and assay datasets subsampled in USEARCH (S2.9.3). Taxa lists were converted into presence/absence matrices for taxonomic groups of interest (Fish & Elasmobranchs, Crustaceans, Molluscs, and Echinoderms) using fuzzySim (Barbosa, 2015) in RStudio (v1.1.423; RStudio Team, 2015).

2.4.4 Statistical analyses

Subsampled datasets were employed for univariate and multivariate statistics comparing diversity and community composition between sites/habitats at the CKI (see S2.9.4). Observed taxonomic diversity (equivalent to species richness) at each site was calculated using the package 'BAT' (Biodiversity Assessment Tools; Cardoso, Rigal and Carvalho, 2015) and graphed using ggplot2 (Wickham, 2016) in RStudio. Community composition was visualised by non-metric multi-dimensional scaling (nMDS) and tested for the amount of variation explained by spatial predictor variables (distance - latitude/longitude, depth, lagoon and marine habitat) by distance-based linear modelling (DistLM) in the PERMANOVA+ add-on (Anderson, Gorley, & Clarke, 2008) of PRIMER v7 (Clarke & Gorley, 2015). Permutational analysis of multivariate dispersions (PERMDISP; Anderson et al., 2008) was used to test homogeneity of variance within marine habitat zones – there were no significant differences for any of the taxonomic groups. Similarity percentage analyses (SIMPER) were also conducted in PRIMER, and an indicator species analysis performed using the R package 'indicspecies' (De Caceres & Jansen, 2016).

2.5 Results

2.5.1 Sampling and sequencing statistics

The four metabarcoding assays run across 252 seawater eDNA extracts yielded a total of 61,219,212 sequencing reads. The mean number of quality filtered sequences per sample was 35,931 for the 16S Fish assay; 30,390 for the 16S Crustacean assay; 42,476 for the COI Elasmobranch assay; and 28,123 for the 18S Universal assay (Table S2.7). The rarefaction analyses determined suitable subsampling cut-offs of 8000 reads for the 16S Fish assay, 4000 for the 16S Crustacean assay, 4000 for the COI Elasmobranch assay, and 8000 for the 18S Universal assay (Figure S2.1). Assay performance is reported in S2.9.5 and Figure S2.2.

After examining taxa accumulation based on the addition of sampling replicates at each site, it was observed that six water replicates (chosen a priori) was not quite sufficient to fully sample the fish and elasmobranch diversity, as well as the crustacean and mollusc diversity from each of the respective sites at the CKI (Figure S2.3). On fitting polynomial curves to the taxa accumulation data, it was extrapolated that an average of 8.38 ± 1.33, 11.75 ± 8.73 and 8.23 ± 3.32 water replicates would be required to sample the fish and elasmobranch, crustacean and mollusc diversity at each site, respectively. However, in using six replicates, we were still able to detect on average 90.1% of fish and elasmobranch, 79.2% of crustacean and 89.0% of mollusc diversity at each site. Six water replicates were adequate to sample echinoderms with the applied assays, with an average of 5.91 ± 2.39 water replicates required.

Two filtration controls amplified (with qPCR C_T values lower than 40 cycles) for the 18S Universal assay; all other assays did not amplify controls. The controls were subsequently sequenced and produced detection hits for algae (order: Pleurostomatida and infrakingdom: Heterokonta), fungi (genus: Cladosporium), insects (superfamily: Membracoidea), demosponges (order: Haplosclerida) and plants (clade: Rosids). Given that these groups were not targeted or of interest for this project, they were omitted from further analyses for all samples. One sample contained a detection hit for the subfamily Salmoninae; this taxon is sporadically detected as a reagent

contamination in our workflows and has been reported elsewhere as fish contamination (Thomsen et al., 2016)—accordingly, it was removed from the fish dataset.

2.5.2 Overall diversity

The four metabarcoding assays collectively yielded 372 identifiable taxa, representing 110 families within 40 orders (Figure 2.3, Table S2.8) across the targeted taxonomic groups at the CKI. The following diversity and community composition data provides new baselines for bony fish, elasmobranch, crustacean, mollusc and echinoderm groups across the sampled CKI habitat zones.

Two hundred and forty fish and elasmobranch taxa were detected (219 at a genus or species level of assignment) from 48 families within 17 orders (Table S2.8). Of these, 84.2% of the bony fish and elasmobranch taxa are widely distributed across the Indo-Pacific region, 2.1% distributed in the Pacific Ocean only, 2.1% distributed in the Indian Ocean only, 11.2% are circumglobal and 0.4% have uncertain distributions. There were no species detected at the CKI that are endemic. The majority of detected bony fish and elasmobranch species (90%) are associated with coral reefs (1-60m depth), 0.8% with deep reef (>60m depth), 3.8% with coral and deep reef, 4.6% are pelagic, 0.4% with coral and deep reef and are also pelagic, and 0.4% with coral and intertidal reef. Predominant bony fish families included Labridae (32 taxa), Pomacentridae (23), Blenniidae (15), Gobiidae (14) and Muraenidae (13). The only elasmobranch families detected were Mobulidae (1), Myliobatidae (1) and Carcharhinidae (4). Particular taxa of interest that have subsistence, recreation or conservation importance at the CKI are presented in Table 2.2 and Figure 2.4. In addition, we report 18 putative new fish occurrence records at the CKI (see Table S2.9 for sequencing statistics and evaluation of the new occurrence records), including goldband snapper (*Pristipomoides multidens*) and a number of deep-sea fishes including lanternfish (*Lampanyctus*, *Lampadena*, order Myctophiformes) and dragonfish (order Stomiiformes).

Within the Crustacea, we focussed on decapods, krill (order Euphausiacea) and mantis shrimp (order Stomatopoda) for this study. This resulted in a total of 88 crustacean taxa (77 assigned at a genus or species level) from 28 families within the orders of

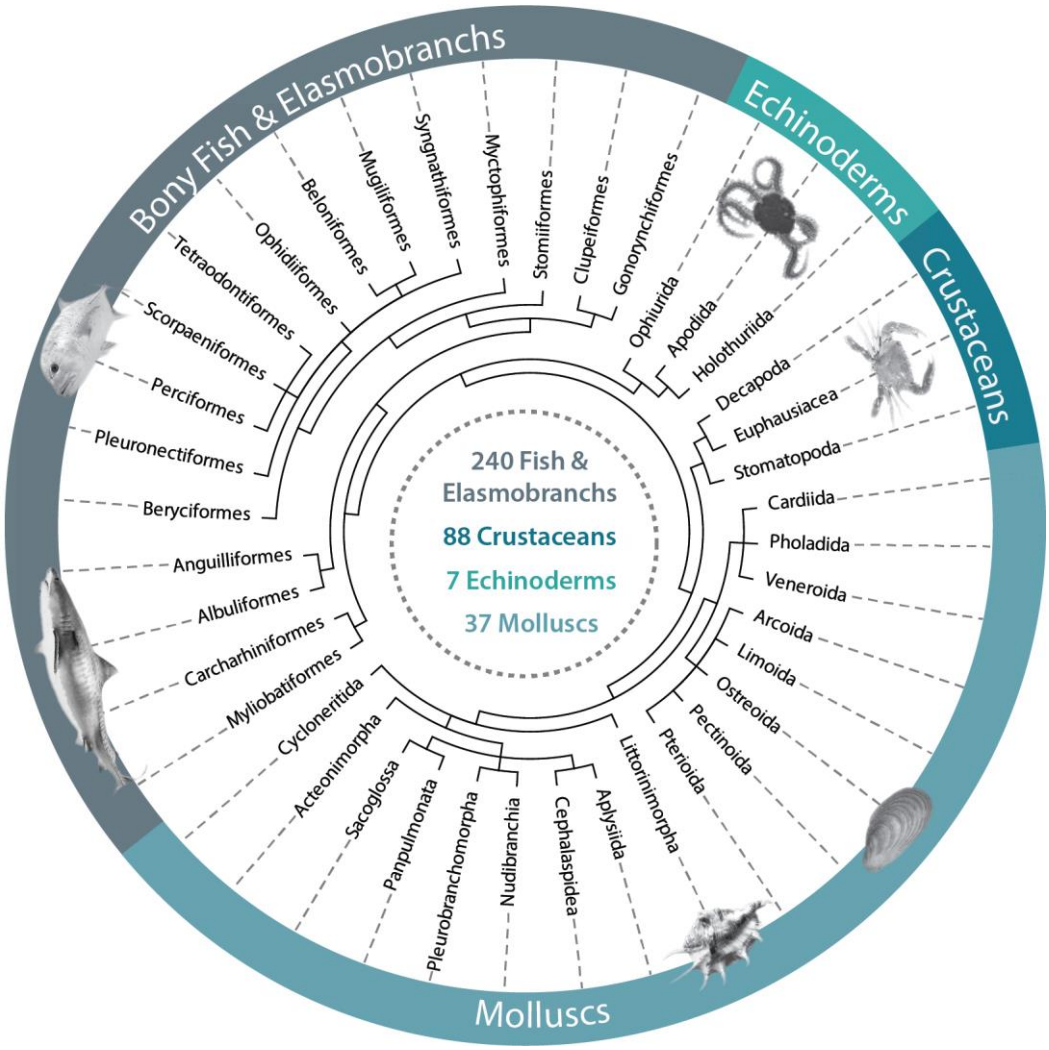


Figure 2.3 Order level dendrogram of bony fish, elasmobranch, crustacean, mollusc and echinoderm diversity detected at the CKI using eDNA metabarcoding.
The number of taxa depicted in the central circle refers to the overall number of unique assignments in each taxonomic group (i.e. at a species, genus, family level etc).

Table 2.2 Taxa of interest.
Marine habitat abbreviations are as follows: LCL, Low Complexity Lagoon; MCL, Medium Complexity Lagoon; HCL, High Complexity Lagoon; ORT, Outer Reef Terrace; IR, Intertidal Reef; L, Lagoonlet.

Common Name	Scientific Name	Importance	Total sites	Total reads	Marine Habitat Zone	Distribution
Bony Fish & Elasmobranchs						
Bonefish	*Albula*	Recreational fishing species	15	27,017	LCL, MCL, ORT, IR, L	Uncertain
White banded cod	*Anyperodon leucogrammicus*	Subsistence and recreational fishing species	1	157	HC	Indo-Pacific
Green humphead parrotfish	*Bolbometopon muricatum*	Subsistence fishing species	20	41,444	LCL, MCL, HCL, ORT, IR	Indo-Pacific
Lanternfish	*Bolinichthys*	New record at CKI. Genus comprised of epi- and mesopelagic species.	1	49	IR	Circumglobal
Goatsbeard brotula	*Brotula multibarbata*	Recreational fishing species	4	11,098	LCL, ORT, IR	Indo-Pacific
Giant trevally	*Caranx ignobilis*	Recreational fishing species	6	1,915	LCL, MCL, ORT, IR	Indo-Pacific
Bluefin trevally	*Caranx melampygus*	Recreational fishing species	1	177	MCL	Indo-Pacific
Grey reef shark	*Carcharhinus amblyrhynchos*	Requiem shark, not fished or targeted	12	120,664	LCL, MCL, HCL, IR	Indo-Pacific
Blacktip reef shark	*Carcharhinus melanopterus*	Requiem shark, not fished or targeted	19	49,427	LCL, MCL, HCL, ORT, IR, L	Indo-Pacific
Milkfish	*Chanos chanos*	Subsistence and recreational fishing species	18	108,492	LCL, MCL, ORT, IR, L	Indo-Pacific
Humphead Maori wrasse	*Cheilinus undulatus*	Subsistence fishing species	10	34,252	LCL, MCL, HCL, ORT, IR	Indo-Pacific
Rainbow runner	*Elagatis bipinnulata*	Recreational fishing species	1	132	LCL	Circumglobal
Squaretail mullet	*Ellochelon vaigiensis*	Subsistence fishing species	6	75,038	LCL, IR, L	Indo-Pacific
Grouper	*Epinephelus*	Subsistence and recreational fishing species	38	131,075	LCL, MCL, HCL, IR	Indo-Pacific

Tiger shark	*Galeocerdo cuvier*	Requiem shark, not fished or targeted	1	111	HCL	Circumglobal
Strongspine silver-biddy	*Gerres longirostris*	Subsistence fishing species	9	1527	LCL, IR, L	Indo-Pacific
Luminous lanternfish	*Lampadena luminosa*	New record at CKI. Bathypelagic species.	1	739	MCL	Circumglobal
Lanternfish	*Lampanyctus*	New record at CKI. Genus comprised of epi-, meso- and bathypelagic species.	1	40	MCL	Circumglobal
Emperors	*Lethrinus*	Subsistence fishing species. Historical concerns regarding overexploitation (Hender et al. 2001).	31	68,884	LCL, MCL, HCL, IR, L	Indo-Pacific
Two-spot red snapper	*Lutjanus bohar*	Subsistence and recreational fishing species	12	63,742	LCL, MCL, HCL, ORT, IR	Indo-Pacific
Blacktail snapper	*Lutjanus fulvus*	Subsistence and recreational fishing species	24	84,016	LCL, MCL, ORT, IR, L	Indo-Pacific
One-spot snapper	*Lutjanus monostigma*	Subsistence fishing species	1	940	HCL	Indo-Pacific
Giant oceanic manta ray	*Mobula birostris* (formerly *Manta birostris*)	Not fished or targeted. *Manta* has recently been revised as a subgroup under *Mobula* (see section 2.6.2; White et al. 2018)	11	695	LCL, MCL, HCL, ORT, IR	Circumglobal
Lanternfish & blackchins	Myctophiformes	New record at CKI. Order comprised of bathypelagic species. Requires reference sequences to resolve assignments.	4	4,208	LCL, MCL, IR	Circumglobal
Hornlip mullet	*Oedalechilus labiosus*	New mullet record at CKI	1	218	LCL	Indo-Pacific
Passionfruit coral trout	*Plectropomus areolatus*	Subsistence and recreational fishing species. Historical concerns regarding overexploitation (Fletcher & Santoro, 2009). Recently permitted to take small catch, given no current concerns (S. Newman, personal communication, February 2020).	5	4,563	MCL, HCL	Indo-Pacific
Goldband snapper	*Pristipomoides multidens*	New snapper record at CKI	1	313	IR	Indo-Pacific
Dragonfishes & relatives	Stomiiformes	New record at CKI. Order comprised of epi-, meso- and bathypelagic species. Requires reference sequences to resolve assignments.	1	252	IR	Circumglobal
Snubnose dart	*Trachinotus blochii*	Recreational fishing species	5	3,764	LCL, MCL, L	Indo-Pacific

Whitetip reef shark	*Triaenodon obesus*	Requiem shark, not fished or targeted	9	29,617	LCL, MCL, HCL, IR	Indo-Pacific
Crustaceans						
Fourspine rock lobster	*Panulirus penicillatus*	Subsistence fishing species	1	5,144	LCL	Indo-Pacific
Prawns	Penaeidae	New record at CKI	2	51,845	LCL, MCL	Circumglobal
Mangrove swimming crab	*Thalamita crenata*	Subsistence fishing species	2	9,544	LCL, L	Indo-Pacific
Molluscs						
Spider conch	*Lambis lambis*	Subsistence fishing species	1	19	MCL	Indo-Pacific
Oysters	Ostreidae	Subsistence fishing species	2	54	MCL, IR	Circumglobal
Pearl oysters	*Pinctada*	Potential commercial interest	2	53	LCL	Circumglobal
Echinoderms						
Sea cucumbers	Holothuriidae	Ecologically important species	13	1,454	LCL, MCL, IR	Circumglobal

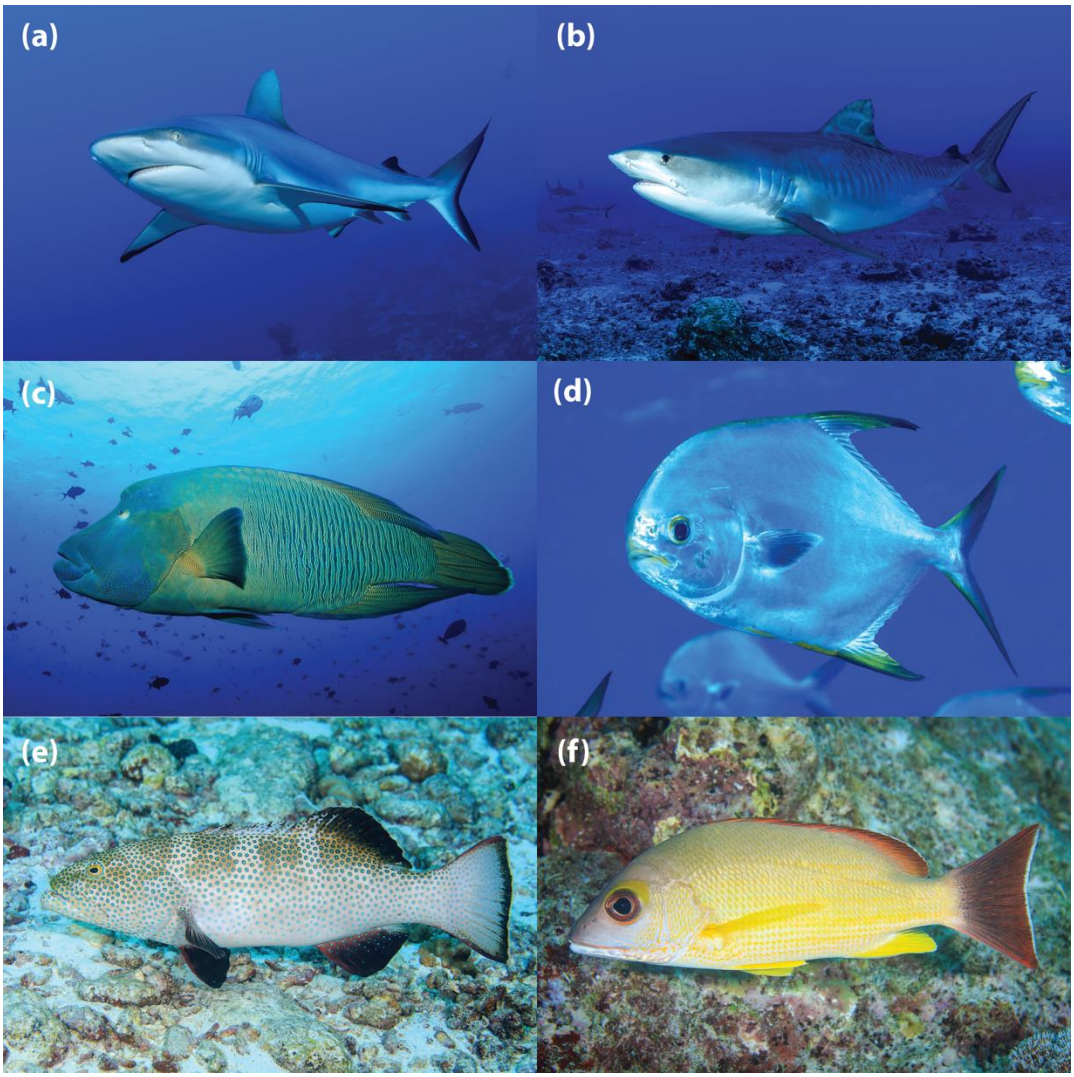


Figure 2.4 Taxa of interest detected by eDNA at the CKI.
Shark species**: (a)** Grey reef shark (*Carcharhinus amblyrhynchos*) and **(b)** Tiger shark (*Galeocerdo cuvier*). Subsistence and recreational fishing species: **(c)** Humphead Maori wrasse (*Cheilinus undulatus*); **(d)** Snubnose dart (*Trachinotus blochii*); **(e)** Passionfruit coral trout (*Plectropomus areolatus*); **(f)** Blacktail snapper (*Lutjanus fulvus*). Photos: Tane Sinclair-Taylor.

Decapoda, Euphausiacea and Stomatopoda (Table S2.8). Of these crustacean taxa, 60% are widely distributed across the Indo-Pacific region, 6% distributed in the Pacific Ocean only and 34% are circumglobal. There were no taxa solely of Indian Ocean origin or endemic to CKI. Predominate families include Xanthidae (34 taxa), Alpheidae (8), Diogenidae (6) and Grapsidae (4). Taxa of interest (Table 2.2) include subsistence species such as the mangrove swimming crab (*Thalamita crenata*) and the fourspine rock lobster (*Panulirus penicillatus*), in addition to prawns (Penaeidae), which were recorded for the first time at CKI. Overall, we detected 25 crustaceans not previously recorded at the CKI, largely xanthid crabs and snapping shrimp (Table S2.9).

Thirty-seven mollusc taxa were detected (23 at a genus or species level of assignment) from 29 families within 17 orders (Table S2.8). Of these mollusc taxa, 13.5% are widely distributed across the Indo-Pacific region, 5.4% distributed in the Pacific Ocean only, 8.1% distributed in the Indian Ocean only and 73% are circumglobal. No taxa were endemic to CKI. No singular family dominated the diversity of molluscs detected at CKI; all families were represented with only one or two taxa. This included the gastropod families Bullidae (2 taxa), Littorinidae (2), Aplysiidae (2) and Strombidae (2), and the bivalve mollusc families such as Limidae (2) and Mytilidae (2). Taxa of interest (Table 2.2) included subsistence species such as the spider conch (*Lambis lambis*) and oysters (family Ostreidae), as well as pearl oysters (*Pinctada sp.*). There were two putative new mollusc occurrence records at the CKI (Table S2.9), including the sea slug genus *Stiliger* and the bivalve genus *Basterotia.*

Only 7 echinoderm taxa were detected (5 at a genus or species level of assignment) from 5 families within 3 orders (Table S2.8). Of these echinoderm taxa, 14.3% are associated with an Indian Ocean distribution; the remaining 85.7% are circumglobal. Similar to the molluscs, no single echinoderm family dominated the biodiversity. The five families detected were the brittle star families Ophiocomidae (3), Ophiodermatidae (1) and Amphiuridae (1), and the sea cucumber families Synaptidae (1) and Holothuriidae (1). There was one putative new echinoderm occurrence record at the CKI, with the detection of *Ophioderma,* a genus of brittle star (Table S2.9).

2.5.3 Community composition

Taxon richness varied across habitat zones (Figure S2.4) with significant differences in diversity detected between intertidal reef and high complexity lagoon habitats; the latter with the lowest taxon richness ($P < \alpha$ H-B; Tables S2.10 and S2.11). The highest diversity of bony fish/elasmobranchs and crustaceans was detected in the intertidal reef (average of 47.2 ± 5.4 bony fish/elasmobranchs and 9.7 ± 1.7 crustaceans per site), however this was not significantly different for bony fish/elasmobranchs ($P > \alpha$ H-B; Figure S2.5; Tables S2.10, S2.12 and S2.13). Echinoderm richness was highest in the low complexity lagoon (average of 3.6 ± 0.4 taxa per site; Figure S2.5; Tables S2.10 and S2.14), whilst mollusc richness revealed no significant differences between habitat types (Table S2.10). The proportion of the major taxonomic groups amplified and represented at each site was consistent between marine habitat zones. The only deviations were a lack of echinoderms in the high complexity lagoon area and an increased proportion of crustaceans in the lagoonlet compared to all other marine habitat zones (Figure S2.5).

Overall taxonomic richness was also investigated for variation between sites within the outer lagoon, surge channels and inner lagoon (Figures S2.6 & S2.7). There were no significant differences in richness between lagoon zones, except when examining echinoderms independently (Table S2.15). A significant difference was detected in echinoderm richness between the surge channel (4.1 ± 0.3 taxa) and the inner lagoon zone (1.9 ± 0.5 taxa per site; $P < \alpha$ H-B; Table S2.16).

The taxa composition of bony fish, elasmobranch and crustacean communities is clearly seen to transition across the habitat and lagoon zones (nMDS; Figure 2.5). Within the mollusc and echinoderm ordinations however, very little clustering based on lagoon and marine habitat zones was observed. Distance based linear model (DistLM) analyses confirmed that community variation based on marine habitat was highly significant between sites ($P < 0.05$) and explained the highest proportion of fitted variance for all taxonomic groups (Table 2.3; Tables S2.17-S2.20), except echinoderms where habitat was a non-significant predictor variable. Conversely, the

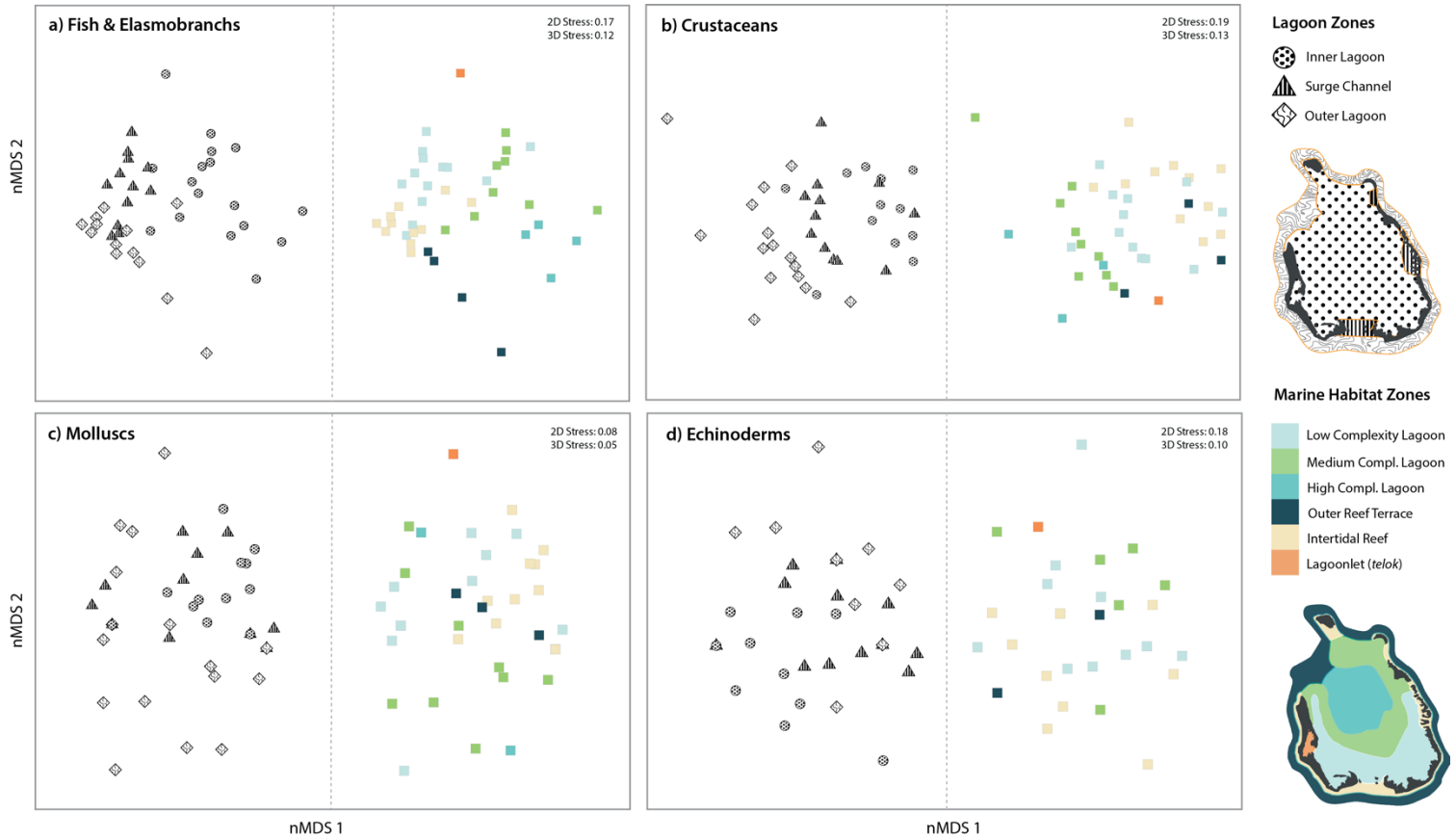


Figure 2.5 Non-metric multi-dimensional scaling (nMDS) ordination plots of a) bony fish and elasmobranch, b) crustaceans, c) molluscs and d) echinoderm community compositions.
Each plot is divided into two parts with the left and right sides representing individual sites colour coded according to lagoon and marine habitat zone, respectively.

Table 2.3 Summary table of the distance based linear model (DistLM) analyses. These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Under the spatial predictor variables, 'Lagoon' refers to the lagoon zones, whilst 'Habitat' refers to the marine habitat zones. Significant codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. The variables highlighted in bold are significant (P<0.05). Full DistLM results, including marginal tests, are provided in Tables S2.17-S2.20.

Response	Predictor	Adj R^2	Pseudo-*F*	Proportion	Cumulative Proportion	*P*	
Bony Fish & Elasmobranchs	**Habitat**	0.187	2.885	0.286	0.286	**0.000**	***
	Lagoon	0.210	1.531	0.059	0.345	**0.007**	**
	Longitude	0.233	1.995	0.037	0.382	**0.000**	***
	Latitude	0.247	1.640	0.030	0.412	**0.009**	**
	Depth	0.258	1.476	0.027	0.439	**0.047**	*
Crustaceans	**Habitat**	0.097	1.817	0.216	0.216	**0.000**	***
	Lagoon	0.119	1.407	0.065	0.281	**0.042**	*
	Longitude	0.121	1.073	0.024	0.306	0.358	
Molluscs	**Habitat**	0.076	1.661	0.192	0.192	**0.001**	***
	Latitude	0.111	2.408	0.053	0.245	**0.006**	**
	Longitude	0.139	2.050	0.044	0.289	**0.020**	*
	Lagoon	0.185	1.945	0.079	0.369	**0.008**	**
Echinoderms	**Lagoon**	0.129	3.453	0.182	0.182	**0.000**	***
	Latitude	0.176	2.754	0.069	0.251	**0.020**	*
	Depth	0.189	1.478	0.036	0.287	0.210	
	Longitude	0.200	1.418	0.034	0.322	0.232	
	Habitat	0.235	1.313	0.122	0.443	0.186	

lagoon zones (outer lagoon, surge channel, inner lagoon) explained the highest proportion of fitted variance for echinoderms; additionally, it was also a highly significant predictor variable for all taxonomic groups. Distance between sites (latitude/longitude) formed significant predictor variables for bony fish and elasmobranchs as well as molluscs and echinoderms; however, it only explained a small proportion of the variability for these taxonomic groups. For crustaceans, distance between sites did not influence their community composition. Finally, depth explained a small, yet significant, proportion of fish and elasmobranch community composition. Cumulatively, the spatial predictor variables tested in the DistLM analyses formed models that explained between 30-44% of total fitted variance between sites. Where the predictor variables of marine habitat and lagoon zones were significant in the DistLM analyses, similarity percentage analyses (SIMPER) were used to identify prominent taxa contributing most to the similarity between sites within each zone. These assignments are reported in S2.9.6 and in Tables S2.21-S2.27, in addition to a variety of indicator species in Tables S2.28 and S2.29.

2.5.4 Comparison of eDNA to previous CKI monitoring techniques

An extensive set of fish collections and site surveys conducted at the CKI between 1973-2017, have recorded in total, 626 bony fish and elasmobranch species from 85 families (Hobbs, Newman, et al., 2014; Harvey et al. (unpublished)). Our eDNA metabarcoding survey from a single sampling point identified 240 bony fish and elasmobranch taxa from 48 families, including 18 putative new fish occurrence records. This raises the total fish record at the CKI to 644 taxa from 88 families, with our single eDNA survey capturing 37.3% of the total CKI fish fauna. This level of detection is comparable to recent baited remote underwater video (BRUV) surveys conducted in 2016 and 2017 (Table 2.4) which captured 41.5% and 38.0% of the total CKI fish fauna, respectively (Harvey et al. (unpublished); see SI Tables 30 and 31 for BRUV species lists). Our eDNA survey however, was not as effective in capturing CKI bony fish and elasmobranch diversity as previously used invasive techniques, such as rotenone, explosives, quinaldine, angling, dipnets and spearing, which were collectively applied in two successive surveys in 1973 and 1989 (Table 2.4; Allen & Smith-Vaniz, 1994). Although, it is worth noting only a single (16S) bony fish assay was employed and additional assays would have the potential to detect further fishes.

Table 2.4 Overview of bony fish, elasmobranch and crustacean taxonomic surveys conducted at the CKI, ordered by decreasing year.
The results from this survey are in bold. The percentage of taxa detected over total diversity is calculated based on the updated diversity count of 644 taxa for bony fish and elasmobranchs, and 236 taxa for crustaceans (221 Decapoda, 14 Stomatopoda and 1 Euphausiacea). New records refer to either new occurrence records at the CKI, or new species descriptions. Harvey et al. (unpublished) recorded species list for BRUV surveys in 2016 and 2017 are provided in Tables S2.30 and S2.31.

Year	Trips	Method	Taxa detected	Taxa detected/ total diversity	New records	Reference
Bony Fish & Elasmobranchs						
2017	**1**	**eDNA metabarcoding**	**240**	**37.3%**	**18 new occurrence**	**This study**
2017	1	Baited remote underwater video (BRUV)	245	38.0%	21 new occurrence	Harvey et al. (unpublished)
2016	1	Baited remote underwater video (BRUV)	267	41.5%	3 new occurrence	Harvey et al. (unpublished)
2014 – 2001	11	Underwater visual observations and line-fishing	N/A	N/A	67 new occurrence	Hobbs, Newman, et al., 2014
2004	1	Underwater visual observations (large >30 TL cm fish only)	58	9.0%	0	Bennett et al., 2018
2001	1	Underwater visual observations (commercial and recreational target species only)	74	11.5%	0	Hender et al., 2001
1989	1	Rotenone, spear and dipnets	448	69.6%	N/A	Allen & Smith-Vaniz, 1994
1973	1	Rotenone, explosives, quinaldine, angling and spearing	425	65.9%	12 new species	Allen & Smith-Vaniz, 1994
Crustaceans						
2017	**1**	**eDNA metabarcoding**	**88**	**37.3%**	**25 new occurrence**	**This study**
1994	1	Above and underwater observations (Decapoda only)	198	89.6%	78 new occurrence	Morgan, 1994

A total of 198 species of Decapoda (e.g. crabs, shrimps, lobsters) and 13 species of Stomatopoda (mantis shrimp) have previously been recorded at the CKI (Ahyong, 2014; Morgan, 1994). Our eDNA survey detected a total of 88 crustacean taxa; 83 Decapoda, 4 Stomatopoda and the first official record of Euphausiacea (krill) at the CKI. The detection and verification of 25 putative new crustacean occurrence records from the eDNA survey raises the total Decapoda recorded at the CKI to 221 taxa, the Stomatopoda recorded to 14 taxa and the Euphausiacea recorded to one taxon. However, an above and underwater observation survey for crustaceans (Morgan, 1994) still detected a greater fraction of crustacean groups (89.6% of recorded Decapoda) than eDNA metabarcoding (37.3%) at the CKI (see Table 2.4).

2.6 Discussion

2.6.1 eDNA metabarcoding baseline of a tropical island reef ecosystem

This study is the first to conduct a multi-trophic eDNA survey of an entire coral reef atoll and demonstrates the utility of multi-assay Tree-of-life (ToL) eDNA metabarcoding approaches in assessing whole-ecosystem biodiversity (Stat et al., 2017). This non-destructive technique exhibited a high detection capability that is particularly suitable for characterising island reef habitats. There is a pressing need to efficiently survey remote, island ecosystems given that they support an exceptionally high level of marine biodiversity and/or endemism (Roberts et al., 2002), yet through their isolation are increasingly vulnerable to climate-related threats (Whittaker & Fernández-Palacios, 2007). ToL metabarcoding approaches (Stat et al., 2017) are of particular use for environmental management in Small Island Developing States (SIDS), which typically share resource challenges, yet are at the forefront of the downstream effects of global climate change (United Nations, 2012). While our survey is restricted to a single time-point, the data provides novel information on the biodiversity present at the CKI, demonstrates the value of non-destructive approaches to assess community composition, simultaneously assesses multitrophic diversity, establishes a roadmap for ongoing monitoring, and raises new biogeographic hypotheses regarding range extensions and habitat health at the CKI. The value of long-term eDNA chronosequences was recently demonstrated (Berry et al., 2019) and this study will establish a baseline upon which regular eDNA biomonitoring can take place.

2.6.2 Bony fish and elasmobranch diversity and community composition

The community composition of the fish and elasmobranch taxa detected by eDNA was largely congruent with the previously established fish diversity profile at the CKI. The predominant fish families of the eDNA dataset were all in the top ten recorded families with greatest species richness at the CKI and also neighbouring Indonesia (Hobbs, Newman, et al., 2014). The highest taxonomic diversity of fish was observed in the intertidal reef and the lowest in the high complexity lagoon. This disparity was surprising given previous surveys have documented a high diversity and abundance of

butterflyfish and herbivorous fish species in the outer reef terrace, as well as high and medium complexity lagoon zones, which incidentally were noted in previous surveys to feature the most developed and healthy reefs of the CKI (Hender et al., 2001; Parks Australia, 2005). Recent surveys of coral reef health and distribution at the CKI however, noted a 50-60% reduction of hard coral cover and genera abundance in the lagoon between 2010-2014 (Evans, Konzewitsch, & Bellchambers, 2016) and substantial numbers of dead branching and plating coral across the high complexity lagoon zone in 2018 (Z Richards, personal communication, December 2018).

Habitat heterogeneity at CKI, as defined by Hender, McDonald and Gilligan (2001) in their classification of habitat zones (S2.9.1), was a significant spatial predictor variable in our survey, explaining the highest proportion of fish and elasmobranch variability. The position of each site either in the outer lagoon, surge channel or inner lagoon, alongside the distance and depth between sites were also found to be significant predictors, yet explained a lower proportion of variability in community composition. This indicates that fish and elasmobranch community composition across the CKI is largely shaped by marine habitat preferences, however the small, yet significant effect of the lagoonal zones, does indicate that species move across habitat zones. This is likely influenced by tidal changes, whereby at low tide, the low complexity area at the southern end of the inner lagoon partially dries (Woodroffe & Biribo, 2011). Likewise, in the outer lagoon, species would be expected to migrate off the intertidal reef amidst exposure with daily tidal changes.

There is a high level of daily water movement via surge channels at CKI, hence it is also possible that an eDNA signal from the inner lagoon may contain DNA shed in the outer lagoon and subsequently transported; thus, the habitat profile of inner lagoon sites would be confounded. However, there was no significant difference between the taxonomic diversity of fish in the three lagoon zones. In addition, we observed a clear transition of fish community composition between sites situated in the outer lagoon, surge channel and inner lagoon (see S2.9.7 for further fish community composition information). This indicates that eDNA signals from seawater are localised enough to discriminate between sites over small spatial scales and can provide a reproducible profile of their community composition. This data adds to a growing body of evidence that there is strong spatial fidelity in eDNA signals, attributed to dispersion and

degradation in seawater (Jeunen et al., 2018; Koziol et al., 2018; Murakami et al., 2019; Stat et al., 2019).

The detection of unresolved deep-sea lanternfish and dragonfish assignments (order: Myctophiformes and Stomiiformes) illustrate the potential of using eDNA metabarcoding to identify cryptic biodiversity, particularly of taxonomic groups that inhabit pelagic zones and are largely inaccessible with other monitoring techniques. The expansion of reference sequence databases remains essential to resolve assignments to a high taxonomic resolution. Whilst it was not possible to elucidate the unresolved Myctophiformes and Stomiiformes sequences beyond order level (exempting the lanternfish *Lampadena luminosa*, *Lampanyctus* and *Bolinichthys*), it may be relevant that these taxonomic groups consist of small to medium-sized predatory fishes known to undertake diel vertical migration; whereby they inhabit meso- and bathypelagic zones during the day before migrating to surface waters at sunset to feed (Watanabe, Moku, Kawaguchi, Ishimaru, & Ohno, 1999). Additionally, there are reports of Stomiiforme larvae developing in near-surface layers, prior to descending to the meso- and bathypelagic zones (Fahay, 2007). It is therefore not unexpected that we detected these groups in the intertidal and adjacent medium complexity reefs. The detection of deep-sea fish, which we attribute to diel-vertical migration and the surfacing of larvae, does however, raise a potential issue with the crossover detection of eggs/larvae and mature fish, as at present there is no way to discriminate them using an eDNA metabarcoding approach. During spawning periods, there is likely a sizeable increase in (gamete derived) mitochondrial DNA, and to a greater extent nuclear DNA (Bylemans et al., 2017), which in conjunction with water movement could confound community profiles. Therefore, some caution must be taken when analysing eDNA metabarcoding data on small spatial scales that are subject to extensive tidal and current movement, or even cyclical spawning events.

Six of the nine elasmobranch taxa previously recorded at the CKI were detected using our eDNA method. The genus of *Mobula* (devil rays) was the only new elasmobranch occurrence record, however recent phylogenetic research indicates that *Mobula* is paraphyletic to *Manta* (manta rays) and subsequently it was recommended that *Manta* be revised as a subgroup under *Mobula* (White et al., 2018). *Mobula birostris* (formerly *Manta birostris*) is a known elasmobranch species at the CKI, commonly

found in cleaning stations between Home Island and Direction Island (Z. Richards, personal communication, December 2018) and on the southern edge of West Island (Brewer et al., 2009). This species was widely detected by eDNA at 11 sites across the atoll, including the intertidal reef and outer reef terrace.

2.6.3 Crustacean diversity and community composition

The Xanthidae are the predominant decapod group at the CKI, comprising approximately 38% (34 species) of crustaceans detected in our survey. This is congruent with previous surveys whereby xanthid taxa comprised 36% of Decapoda and Stomatopoda records at the CKI (Ahyong, 2014; Morgan, 1994) and validates that the 16S Crustacean primers (Berry et al., 2017) have the capability to accurately detect these crustacean groups. However, we only detected six of a possible 48 species in the Diogenidae family (marine hermit crabs), which after xanthid crabs, are thought to be one of the most diverse decapod groups at the CKI (Morgan, 1994). An *in-silico* analysis established that the 16S Crustacean primer assay has the ability to amplify the majority of taxa within Diogenidae. Thus, there seems to be no obvious explanation for this reduction in taxonomic diversity, as we surveyed extensively across their preferred habitats of sheltered, shallow intertidal reef and sandy areas. That the Diogenidae species were largely detected on the outer side of the lagoon is consistent with Morgan's (1994) survey indicating an absence of marine hermit crabs in the northern and southern part of the lagoon off West Island.

Another significant finding in the crustacean analysis, was the detection and first official occurrence record of Penaeidae (prawns) at the CKI. There is only one recorded sighting of Penaeidae at the CKI, with reference to an unidentified prawn being fed on by a *Thalamita crenata* (McKillup & McKillup, 1996). The lack of Penaeidae in any formal survey at the CKI is surprising, given the widespread abundance of the family, particularly around Australia and Indonesia and its presence at neighbouring Christmas Island (Morgan, 2000). Interestingly, our detection could only be assigned at a family level (Penaeidae) because of a low identity percentage, indicating that we do not have a reference sequence for the detected taxa. This largely excludes the possibility that our detection is of imported prawns from mainland Australia for the purposes of human consumption, as Australian commercial prawn

species (e.g. tiger prawns, king prawns and banana prawns) have numerous publicly-accessible reference sequences. The detection of Penaeidae at two sites (4, 30) at the CKI can now be used to guide further surveys in an effort to recover specimens for taxonomic and reference sequencing purposes.

The community composition of crustaceans is similar to that of the bony fish and elasmobranchs, with the habitat and lagoon zones explaining the most variation. The genus of *Thalamita* (Portunidae, Decapoda) was the most widespread crustacean taxa, representative in all habitat and lagoonal zones, except for the high complexity lagoon zone. This genus contains a plethora of species found at the CKI both in the shallows of the lagoon and on the outer edge of the atoll (Tweedie, 1950). The genus of *Cyclodius* (Xanthidae, Decapoda), in particular *Cyclodius ungulatus*, typified the high complexity lagoon zone and is largely associated with dead coral and coral rock, although has also been recorded among living *Pocillopora* coral (Tweedie, 1950). The omnivorous *Daira perlata* (Daridae, Decapoda) was widespread both in the low complexity lagoon and intertidal reef, dominating the latter. *Daira perlata* has been previously reported in crevices and under rocks on the outer edge of the CKI atoll (Tweedie, 1950). Lastly, *Chlorodiella barbata* (Xanthidae, Decapoda) was widespread in the low and medium complexity lagoon zones, found in both the surge channel and inner lagoon. Whilst its habitat distribution has not been formally reported at the CKI, *Chlorodiella barbata* has been reported in a number of lagoon, fringing reef and barrier reef sites in French Polynesia (Monteforte, 1987; Peyrot-Clausade, 1989).

2.6.4 Mollusc diversity and community composition

A total of 757 mollusc species from 126 families have been previously recorded at the CKI; 80.6% of the class Gastropoda (snails and slugs) and 18.4% Bivalvia (bivalve molluscs) (Tan & Low, 2014). Our mollusc analysis, however, only detected a small fraction of these reported species with a total of 37 mollusc taxa across 29 families. This low level of detection likely reflects the fact that we did not apply a specifically-designed mollusc assay, but rather extracted mollusc assignments from a ‘universal’ 18S assay. As discussed previously, the 18S universal assay amplified 72 classes with the majority of BLAST hits at a family level. Therefore, it is recommended that for an

in-depth eDNA mollusc analysis or to monitor threatened species such as giant clams, a more specific primer assay must be developed and applied. From the molluscs that were amplified however, we were able to discriminate community composition variation between sites. This was again largely attributed to habitat variation, in conjunction with distance (latitude/longitude) between sites and the site's position within the inner lagoon, surge channel or outer lagoon.

Lambis lambis, known as the common spider conch or *gong gong* locally, is a gastropod mollusc that is extensively fished using artisanal techniques in the CKI where it is considered a delicacy. Preferring algal-dominated lagoon flats, the southeast area of the lagoon once supported a high density of *L. lambis*. However, recent surveys indicate a change in their distribution, which may be due to macroalgal habitat declines (Bellchambers, Meeuwig, Evans, & Legendre, 2011) and an overall reduction in stock abundance between 2008-2014 (Evans et al., 2016). Not many of the sites surveyed overlapped with those of Evans, Konzewitsch and Bellchambers (2016); nonetheless we only detected *L. lambis* at one site (20), in the medium complexity lagoon area. This may reflect low population size in this area, however further eDNA and visual surveying across their preferred habitats is necessary.

2.6.5 Echinoderm diversity and community composition

The last updated checklist of echinoderms at the CKI reported a total of 89 species across 5 classes (Crinoidea, Asteroidea, Ophiuroidea, Echinoidea and Holothuroidea; Marsh, 1994). We however, only detected 7 echinoderm taxa from 5 families across 2 classes. Limited detection is again attributed to the use of the 18S universal assay, rather than applying a more specific echinoderm assay. Unlike the bony fish and elasmobranch, crustacean and mollusc communities, habitat, in addition to depth, was not a significant predictor variable for echinoderm community composition. This may reflect the ability of the detected echinoderm taxa to occupy mixed habitats, however, there was a lack of echinoderm DNA detected in the four sites located in the high complexity lagoon area of the inner lagoon. Whilst it could be argued that by sampling surface water we may be excluding benthic species, we did detect echinoderms on the outer reef terrace, which had the greatest depth profile by far (average depth ~ 13 m).

Therefore, our lack of detection likely reflects a low abundance of echinoderms in the high complexity lagoon area.

2.6.6 Future integration in reef monitoring

As demonstrated in this survey, eDNA metabarcoding is a powerful detection tool, capable of producing localised profiles of marine ecosystem composition amidst tidal and oceanic movements, which are reflective of habitat partitioning on an island scale. The eDNA approach proved to be equally efficient in detecting fish and elasmobranch diversity to recent BRUV surveys at the CKI. Interestingly there was no overlap in new occurrence records between the BRUV and eDNA surveys, demonstrating their potential utility to provide complementary baseline data, as previously reported by Stat et al., (2019). Whilst both the eDNA and BRUV surveys fall short of the efficiency of line and spearfishing as well as ichthyocides (Allen & Smith-Vaniz, 1994) as a fish survey tool at the CKI, they are preferable in that they are non-lethal; the added benefit of eDNA metabarcoding is that it is completely non-invasive. However, like visual surveillance approaches, putative new occurrence records derived from eDNA will require further verification by specimen collection. Despite this, eDNA will be useful for directing targeted surveying, not only to verify new occurrence records, but also in an effort to physically recover unresolved taxa that may represent new species.

In capturing the crustacean diversity at the CKI, our eDNA crustacean primer assay fell short of traditional above and underwater observation surveys (Morgan, 1994). However, the application of this assay did contribute 25 new occurrence records for crustaceans at the CKI, demonstrating that there remain cryptic and/or elusive taxa that have yet to be surveyed in this region. The low percentage of known crustacean diversity captured by the 16S Crustacean primers (Berry et al., 2017), in comparison to previous survey techniques, may reflect a need for further primer optimisation in addition to the continual development of reference databases, given that the majority proportion of 16S Crustacean amplicons were attributed as no hits (S2.9.5). Alternatively, this may reflect preliminary research indicating lower eDNA detection rates from crustaceans, in contrast to fish and amphibian species (Forsström & Vasemägi, 2016; Tréguier et al., 2014).

Whilst eDNA metabarcoding is now widely employed in a number of settings, there remains some uncertainties in regard to experimental design, assay choice and laboratory/bioinformatic workflows that need to be optimised before this technique can be fully integrated into monitoring toolkits (see Cristescu and Hebert, 2018). In testing the utility of eDNA metabarcoding to survey the marine biodiversity of the CKI, we encountered some of these uncertainties, which we wish to address here, in anticipation that our findings will guide further eDNA surveying in tropical islands and more broadly, coral reef ecosystems. Our sampling design (six water replicates per site) was determined *post hoc* to be suboptimal with eight samples being the optimal replication required to completely sample the bony fish, elasmobranch, crustacean and mollusc diversity at sites within the CKI. Given that coral reefs support an exceptionally high level of marine biodiversity (Roberts et al., 2002), it is recommended that where possible, a pilot study is undertaken prior to a large survey in order to evaluate the level of replication required (for the given question). Alternatively, preference should be given to collecting a high yet feasible number of water replicates in the field, not all of which might progress through metabarcoding workflows.

Whilst there was some overlap in the detection of taxonomic groups, the four assays applied in this eDNA survey each produced unique taxon hits, which further supports their use as stand-alone metabarcoding assays. It is therefore recommended that for metabarcoding, the 16S Fish assay is used to amplify a wide range of bony fish at a fine-scale level of taxonomic resolution (family to species level), the 16S Crustacean assay is used to amplify crustaceans, whilst the COI Elasmobranch assay is primarily used for the amplification of elasmobranchs. The majority of BLAST hits extracted from the 18S Universal assay were at a family-level resolution; the power of this assay lies in its ability to anneal to a wide range of taxonomic groups. We therefore recommend that the 18S Universal assay is employed to explore the range of taxonomic groups present in an environmental sample, prior to using a secondary assay(s) to increase resolution of target taxa.

Global research into the use of eDNA metabarcoding widely supports its implementation as a marine detection tool alongside traditional, observation-based monitoring techniques (Kelly et al., 2017; Stat et al., 2019; Taberlet, Bonin, Zinger, &

Coissac, 2018). All traditional techniques in current use have biases and limitations ranging from inabilities to detect small and cryptic species, requirements for the use of bait, invasiveness (e.g. lethal methods), as well as logistical and cost-constraints. Whilst eDNA metabarcoding bypasses many of these concerns, it is not without its own limitations. This is largely in regard to its reliance on incomplete reference databases, where many species have yet to be barcoded, and, at present, its limited use for quantification above and beyond presence-absence surveys (McInerney & Rees, 2018). It is therefore highly recommended that eDNA metabarcoding is used in conjunction with widely employed morphological assessments, such as through line fishing, trawling and visual/video surveillance, which additionally provide information on size profiles and biomass. Ultimately, the method employed will depend on the primary biomonitoring question at hand, but a combined approach with eDNA metabarcoding has proven to yield substantially greater taxonomic richness than with either a sole genetic or sole morphological method (Kelly et al., 2017; Stat et al., 2019). Building on rapid developments in eDNA metabarcoding, it is expected that this genetic approach will soon be integrated into best-practice marine resource management practices.

2.7 Conclusion

The application of a fine-tuned suite of eDNA metabarcoding assays can characterise whole marine ecosystem diversity, reveal fine-scale reef communities and provide putative new occurrence records that can efficiently establish or update marine baselines. This study demonstrated the localisation of eDNA signals from seawater, providing community composition data that reflected habitat partitioning across an entire coral reef atoll. It provided high resolution data (i.e. to genus and species level) that is complementary to recent stereo-video surveillance research on bony fish and elasmobranchs at the CKI. An updated multi-trophic baseline, with the discovery of 46 new occurrence records - notably species of mullet, snapper and deep-sea fish - will provide the basis for inclusion into longer term monitoring programs. Repeated observations of eDNA measurements across the georeferenced sites herein will facilitate assessment of the status and ultimately trends in biodiversity, particularly in response to disturbance events at CKI (e.g. deoxygenation, coral bleaching). This can be integrated alongside traditional surveying methods to yield greater taxonomic richness and provide additional information in relation to size profiles and biomass. There is also the added benefit that eDNA samples can be archived, and as new primer assays for improved detection of echinoderms or molluscs, or any other taxa of interest are developed, samples can be retrospectively analysed. We strongly advocate for the ongoing development and application of multi-marker eDNA metabarcoding approaches to efficiently characterise and establish multi-trophic island reef baselines. In addition, continual barcoding of endemic and cryptic taxa, particularly deep-sea marine species, will aid in resolving ambiguous taxonomic assignments, which will be beneficial in future island reef metabarcoding projects.

2.8 References

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Ackrill, M. (1984). The origins and nature of the first permanent settlement on the Cocos-Keeling Islands. *Australian Historical Studies*, *21*(83), 229–244.

Ahyong, S. T. (2014). Stomatopod Crustacea of Christmas Island and the Cocos (Keeling) Islands. *Raffles Bulletin of Zoology*, *30*, 246-254.

Allen, G. R. (2008). Conservation hotspots of biodiversity and endemism for Indo-Pacific coral reef fishes. *Aquatic Conservation: Marine and Freshwater Ecosystems*, *18*(5), 541–556.

Allen, G. R., & Smith-Vaniz, W. F. (1994). Fisheries of the Cocos (Keeling) Islands, *Atoll Research Bulletin, 412*.

Anderson, M., Gorley, R. N., & Clarke, R. K. (2008). *Permanova+ for Primer: Guide to Software and Statisticl Methods*. Primer-E Limited.

Barbosa, A. M. (2015). fuzzySim: applying fuzzy logic to binary similarity indices in ecology. *Methods in Ecology and Evolution*, *6*(7), 853–858.

Bellchambers, L. M., Meeuwig, J. J., Evans, S. N., & Legendre, P. (2011). Modelling habitat associations of the common spider conch in the Cocos (Keeling) Islands. *Marine Ecology Progress Series*, *432*, 83–90.

Bennett, S., Halford, A. R., Choat, J. H., Hobbs, J. A., Santana-Garcon, J., Ayling, A. M., … Newman, S. J. (2018). Geography and island geomorphology shape fish assemblage structure on isolated coral reef systems. *Ecology and Evolution*, *8*(12), 6242–6252.

Benson, D. A., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Wheeler, D. L. (2005). GenBank. *Nucleic Acids Research*, *33*, D34–D38. https://doi.org/10.1093/nar/gki063

Berry, T. E., Osterrieder, S. K., Murray, D. C., Coghlan, M. L., Richardson, A. J., Grealy, A. K., … Bunce, M. (2017). DNA metabarcoding for diet analysis and biodiversity: A case study using the endangered Australian sea lion (*Neophoca cinerea*). *Ecology and Evolution*, *7*(14), 5435–5453.

Berry, T. E., Saunders, B. J., Coghlan, M. L., Stat, M., Jarman, S., Richardson, A. J.,

… Bunce, M. (2019). Marine environmental DNA biomonitoring reveals seasonal patterns in biodiversity and identifies ecosystem responses to anomalous climatic events. *PLOS Genetics*, *15*(2), e1007943. https://doi.org/10.1371/journal.pgen.1007943

Bista, I., Carvalho, G. R., Walsh, K., Seymour, M., Hajibabaei, M., Lallias, D., … Creer, S. (2017). Annual time-series analysis of aqueous eDNA reveals ecologically relevant dynamics of lake ecosystem biodiversity. *Nature Communications*, *8*(1), 1-11.

Boussarie, G., Bakker, J., Wangensteen, O. S., Mariani, S., Bonnin, L., Juhel, J.-B., … Robbins, W. D. (2018). Environmental DNA illuminates the dark diversity of sharks. *Science Advances*, *4*(5).

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a Unix-inspired software package for DNA metabarcoding. *Molecular Ecology Resources* 16(1), 176–182.

Brewer, D. T., Skewes, T. D., Lyne, V. D., Davies, C., Taranto, T. J., Fuller, M., … Andersen, J. (2009). Conservation values in Commonwealth waters of the Christmas and Cocos (Keeling) Island remote Australian territories. Report to Department of Environment and Water Resources, CSIRO, Cleveland.

Briggs, J. C. (1966). Oceanic islands, endemism, and marine paleotemperatures. *Systematic Zoology*, *15*(2), 153–163.

Budd, A. F., & Pandolfi, J. M. (2010). Evolutionary novelty is concentrated at the edge of coral species distributions. *Science*, *328*(5985), 1558–1561.

Bunce, P. (1988). *The Cocos (Keeling) Islands: Australian Atolls in the Indian Ocean*. Jacaranda Press for the Cocos (Keeling) Islands Council.

Bylemans, J., Furlan, E. M., Hardy, C. M., McGuffie, P., Lintermans, M., & Gleeson, D. M. (2017). An environmental DNA-based method for monitoring spawning activity: A case study, using the endangered Macquarie perch (*Macquaria australasica*). *Methods in Ecology and Evolution*, *8*(5), 646–655.

Caporaso, J. G., Kuczynski, J., Stombaugh, J., Bittinger, K., Bushman, F. D., Costello, E. K., … Gordon, J. I. (2010). QIIME allows analysis of high-throughput community sequencing data. *Nature Methods*, *7*(5), 335.

Cardoso, P., Rigal, F., & Carvalho, J. C. (2015). BAT–Biodiversity Assessment Tools, an R package for the measurement and estimation of alpha and beta taxon, phylogenetic and functional diversity. *Methods in Ecology and Evolution*, *6*(2),

232–236.

Clarke, K. R., & Gorley, R. N. (2015). *PRIMER v7: User Manual/Tutorial*. Plymouth, UK.

Cortés, J., Enochs, I. C., Sibaja-Cordero, J., Hernández, L., Alvarado, J. J., Breedy, O., … Hermosillo, A. (2017). Marine biodiversity of Eastern Tropical Pacific coral reefs. In *Coral Reefs of the Eastern Tropical Pacific* (pp. 203–250). Springer.

Cristescu, M. E., & Hebert, P. D. N. (2018). Uses and Misuses of Environmental DNA in Biodiversity Science and Conservation. *Annual Review of Ecology, Evolution, and Systematics*, *49*, 209–230.

Darwin, C., Fitzroy, R., & King, P. P. (1839). *Narrative of the Surveying Voyages of His Majesty's Ships Adventure and Beagle, Between the Years 1826 and 1836: Describing Their Examination of the Southern Shores of South America, and the Beagle's Circumnavigation of the Globe*. Henry Colburn.

De Caceres, M., & Jansen, F. (2016). Package 'indicspecies.' *Relationship between species and groups of sites.* R package version, *1*(6).

Deagle, B. E., Gales, N. J., Evans, K., Jarman, S. N., Robinson, S., Trebilco, R., & Hindell, M. A. (2007). Studying seabird diet through genetic analysis of faeces: a case study on macaroni penguins (*Eudyptes chrysolophus*). *PLoS One*, *2*(9), e831.

DiBattista, J. D., Rocha, L. A., Hobbs, J. A., He, S., Priest, M. A., Sinclair-Taylor, T. H., … Berumen, M. L. (2015). When biogeographical provinces collide: hybridization of reef fishes at the crossroads of marine biogeographical provinces in the Arabian Sea. *Journal of Biogeography*, *42*(9), 1601–1614.

Edgar, R. C. (2010). Search and clustering orders of magnitude faster than BLAST. *Bioinformatics*, *26*(19), 2460–2461. http://dx.doi.org/10.1093/bioinformatics/btq461

Evans, S. N., Konzewitsch, N., & Bellchambers, L. (2016). *An update of the Department of Fisheries, Western Australia, Invertebrate and Reef Health Research and Monitoring at Cocos (Keeling) Islands*. Fisheries Research Division, Western Australian Fisheries and Marine Research Laboratories, Perth.

Fahay, M. P. (2007). Stomiiformes and Ateleopodiformes. In *Early Stages of Fishes in the Western North Atlantic Ocean (Davis Strait, Southern Greenland and Flemish Cap to Cape Hatteras)* (pp. 252–305). Northwest Atlantic fisheries

organization.

Fields, A. T., Abercrombie, D. L., Eng, R., Feldheim, K., & Chapman, D. D. (2015). A novel mini-DNA barcoding assay to identify processed fins from internationally protected shark species. *PLoS One*, *10*(2), e0114844.

Fletcher, W. J., & Santoro, K. (2009). *State of the Fisheries Report 2008/09*. W.A. Department of Fisheries, Perth, Western Australia.

Forsström, T., & Vasemägi, A. (2016). Can environmental DNA (eDNA) be used for detection and monitoring of introduced crab species in the Baltic Sea? *Marine Pollution Bulletin*, *109*(1), 350–355.

Friedlander, A. M., Ballesteros, E., Beets, J., Berkenpas, E., Gaymer, C. F., Gorny, M., & Sala, E. (2013). Effects of isolation and fishing on the marine ecosystems of Easter Island and Salas y Gómez, Chile. *Aquatic Conservation: Marine and Freshwater Ecosystems*, *23*(4), 515–531.

Froese, R., & Pauly, D. (Eds.) (2018). FishBase. In Species 2000 & ITIS Catalogue of Life. Retrieved from http://www.fishbase.org.

Graham, N. A. J., Wilson, S. K., Carr, P., Hoey, A. S., Jennings, S., & MacNeil, M. A. (2018). Seabirds enhance coral reef productivity and functioning in the absence of invasive rats. *Nature*, *559*(7713), 250.

Hender, J., McDonald, C. ., & Gilligan, J. . (2001). *Baseline surveys of the marine environments and stock size estimates of marine resources of the south Cocos (Keeling) Atoll (0-15m), eastern Indian Ocean*. Draft Report to Fisheries Resources Research Fund. Canberra.

Hobbs, J.P.A., & McDonald, C. A. (2010). Increased seawater temperature and decreased dissolved oxygen triggers fish kill at the Cocos (Keeling) Islands, Indian Ocean. *Journal of Fish Biology*, *77*(6), 1219–1229. https://doi.org/10.1111/j.1095-8649.2010.02726.x

Hobbs, J. P. A., & Macrae, H. (2012). Unusual weather and trapped coral spawn lead to fish kill at a remote coral atoll. *Coral Reefs*, *31*(4), 961. https://doi.org/10.1007/s00338-012-0918-8

Hobbs, J.P.A, Frisch, A. J., Allen, G. R., & Van Herwerden, L. (2008). Marine hybrid hotspot at Indo-Pacific biogeographic border. *Biology Letters*, *5*(2), 258–261.

Hobbs, J.P.A, Newman, S. J., Mitsopoulos, G. E. A., Travers, M. J., Skepper, C. L., Gilligan, J. J., … Ayling, A. M. (2014). Fishes of the Cocos (Keeling) Islands: new records, community composition and biogeographic significance. *Raffles*

Bulletin of Zoology, (30), 203–219.

Horton, T., Kroh, A., Ahyong, S., Bailly, N., Boyko, C. B., Brandão, S. N., … Zhao, Z. (2018). World Register of Marine Species (WoRMS). WoRMS Editorial Board. Retrieved from http://www.marinespecies.org

Houk, P., Benavente, D., Iguel, J., Johnson, S., & Okano, R. (2014). Coral reef disturbance and recovery dynamics differ across gradients of localized stressors in the Mariana Islands. *PLoS One*, *9*(8), e105731.

Hourigan, T. F., & Reese, E. S. (1987). Mid-ocean isolation and the evolution of Hawaiian reef fishes. *Trends in Ecology & Evolution*, *2*(7), 187–191.

Hourston, M. (2010). Review of the exploitation of marine resources of the Australian Indian Ocean Territories: the implications of biogeographic isolation for tropical island fisheries. *Fisheries Research Report*, *208*. Western Australian Fisheries and Marine Research Laboratories.

Huson, D. H., Auch, A. F., Qi, J., & Schuster, S. C. (2007). MEGAN analysis of metagenomic data. *Genome Research*, *17*(3), 377–386. https://doi.org/10.1101/gr.5969107

IOC-UNESCO. (2018). *The United Nations Decade of Ocean Science for Sustainable Development (2021-2030)*. Retrieved from http://unesdoc.unesco.org/images/0026/002619/261962e.pdf

Jerde, C. L., Mahon, A. R., Chadderton, W. L., & Lodge, D. M. (2011). "Sight-unseen" detection of rare aquatic species using environmental DNA. *Conservation Letters*, *4*(2), 150–157.

Jeunen, G., Knapp, M., Spencer, H. G., Lamare, M. D., Taylor, H. R., Stat, M., … Gemmell, N. J. (2018). Environmental DNA (eDNA) metabarcoding reveals strong discrimination among diverse marine habitats connected by water movement. *Molecular Ecology Resources, 19*(2), 426-438.

Johnstone, R. ., & Darnell, J. . (2017). *Checklist of the Birds of Christmas Island and Cocos (Keeling) Islands*. Perth.

Kearse, M., Moir, R., Wilson, A., Stones-Havas, S., Cheung, M., Sturrock, S., … Drummond, A. (2012). Geneious Basic: An integrated and extendable desktop software platform for the organization and analysis of sequence data. *Bioinformatics*, *28*(12), 1647–1649. http://dx.doi.org/10.1093/bioinformatics/bts199

Kelly, R. P., Closek, C. J., O'Donnell, J. L., Kralj, J. E., Shelton, A. O., & Samhouri,

J. F. (2017). Genetic and manual survey methods yield different and complementary views of an ecosystem. *Frontiers in Marine Science, 3*, 283.

Knowlton, N., & Jackson, J. B. C. (2008). Shifting baselines, local impacts, and global change on coral reefs. *PLoS Biology, 6*(2), e54.

Koziol, A., Stat, M., Simpson, T., Jarman, S., DiBattista, J. D., Harvey, E. S., … Bunce, M. (2018). Environmental DNA metabarcoding studies are critically affected by substrate selection. *Molecular Ecology Resources, 19*(2), 366-376.

Marsh, L. M. (1994). Echinoderms of the Cocos (Keeling) Islands. *Atoll Research Bulletin, 411*, 1–12.

McInerney, P. J., & Rees, G. N. (2018). More (or less?) bounce for the ounce: a comparison of environmental DNA and classical approaches for bioassessment. *Marine and Freshwater Research, 69*(6), 992–996.

McKillup, S. C., & McKillup, R. V. (1996). The feeding behaviour of *Thalamita crenata* (Portunidae, Decapoda), a cannibalistic marine scavenger. *Marine & Freshwater Behaviour & Physiology*, *28*(4), 255–267.

Mendoza, J. C. E., Lasley Jr., R. M., & Ng, P. K. L. (2014). New rock crab records (Crustacea: Brachyura: Xanthidae) from Christmas and Cocos (Keeling) Islands, Eastern Indian Ocean. *Raffles Bulletin of Zoology, 30*, 274-300.

Monteforte, M. (1987). The decapod reptantia and stomatopod crustaceans of a typical high island coral reef complex in French Polynesia (Tiahura, Moorea Island): zonation, community composition and trophic structure. *Atoll Research Bulletin, 309*, 1-37.

Morgan. (2000). Decapod Crustacea of Christmas Island. *History, 10*(16), 629–637.

Morgan, G. J. (1994). Decapod crustaceans of the Cocos (Keeling) Islands. *Atoll Research Bulletin, 414*, 1-11.

Murakami, H., Yoon, S., Kasai, A., Minamoto, T., Yamamoto, S., Sakata, M. K., … Yamashita, Y. (2019). Dispersion and degradation of environmental DNA from caged fish in a marine environment. *Fisheries Science, 85*(2), 327-337.

Nations, U. (2012). The future we want. In *Outcome document of the United Nations Conference on Sustainable Development*. UN, New York.

Parks Australia. (2005). Status of the coral reefs at the Cocos (Keeling) Islands: a report on the status of the marine community at Cocos (Keeling) Islands, east Indian Ocean, 1997-2005. *Department of the Environment and Heritage, Australian Government,* Canberra.

Peyrot-Clausade, M. (1989). Crab cryptofauna (Brachyura and Anomura) of Tikehau, Tuamotu Archipelago, French Polynesia. *Coral Reefs*, *8*(3), 109–117.

Pochon, X., Bott, N. J., Smith, K. F., & Wood, S. A. (2013). Evaluating detection limits of next-generation sequencing for the surveillance and monitoring of international marine pests. *PLoS One*, *8*(9), e73935.

Port, J. A., O'Donnell, J. L., Romero-Maraccini, O. C., Leary, P. R., Litvin, S. Y., Nickols, K. J., … Kelly, R. P. (2016). Assessing vertebrate biodiversity in a kelp forest ecosystem using environmental DNA. *Molecular Ecology*, *25*(2), 527–541.

Richards, Z. T., & Hobbs, J.-P. A. (2014). The status of hard coral diversity at Christmas Island and Cocos (Keeling) Islands. *Raffles Bulletin of Zoology*, *30*(30), 376–398.

Roberts, C. M., McClean, C. J., Veron, J. E. N., Hawkins, J. P., Allen, G. R., McAllister, D. E., … Wells, F. (2002). Marine biodiversity hotspots and conservation priorities for tropical reefs. *Science*, *295*(5558), 1280–1284.

R Studio Team (2015). RStudio: integrated development for R. *RStudio Inc*, Boston, MA. Retrieved from *http://www.rstudio.com*

Schubert, M., Lindgreen, S., & Orlando, L. (2016). AdapterRemoval v2: rapid adapter trimming, identification, and read merging. *BMC Research Notes*, *9*, 88. https://doi.org/10.1186/s13104-016-1900-2

Spalding, M. D., Ravilious, C., & Green, E. P. (2001). *World Atlas of Coral Reefs*. Univ of California Press.

Stanhope, J., Wettenhall, R., & Bhusal, T. (2016). Governance challenge: Australia's Indian Ocean island territories. *Asia Pacific Journal of Public Administration*, *38*(2), 87–102.

Stat, M., Huggett, M. J., Bernasconi, R., DiBattista, J. D., Berry, T. E., Newman, S. J., … Bunce, M. (2017). Ecosystem biomonitoring with eDNA: metabarcoding across the tree of life in a tropical marine environment. *Scientific Reports*, *7*(1), 1-11.

Stat, M., John, J., DiBattista, J. D., Newman, S. J., Bunce, M., & Harvey, E. S. (2019). Combined use of eDNA metabarcoding and video surveillance for the assessment of fish biodiversity. *Conservation Biology*, *33*(1), 196-205.

Taberlet, P., Bonin, A., Zinger, L., & Coissac, E. (2018). *Environmental DNA: For Biodiversity Research and Monitoring*. Oxford University Press.

Tan, S. K., & Low, M. E. Y. (2014). Checklist of the Mollusca of Cocos (Keeling) /

Christmas Island ecoregion. *Raffles Bulletin of Zoology*, (*30*), *313-375*.

Taxonomy Decadel Plan Working Group. (2018). *Discovering Diversity: A decadel plan for taxonomy and biosystematics in Australia and New Zealand 2018-2028*. Australian Academy of Science and Royal Society Te Apārangi, Canberra and Wellington. https://doi.org/10.1111/j.1469-8137.2010.03389.x

Thomsen, P. F., Møller, P. R., Sigsgaard, E. E., Knudsen, S. W., Jørgensen, O. A., & Willerslev, E. (2016). Environmental DNA from Seawater Samples Correlate with Trawl Catches of Subarctic , *Deepwater Fishes*, 1–22. https://doi.org/10.1371/journal.pone.0165252

Tréguier, A., Paillisson, J., Dejean, T., Valentini, A., Schlaepfer, M. A., & Roussel, J. (2014). Environmental DNA surveillance for invertebrate species: advantages and technical limitations to detect invasive crayfish *P rocambarus* clarkii in freshwater ponds. *Journal of Applied Ecology*, *51*(4), 871–879.

Tweedie, M. W. F. (1950). The fauna of the Cocos-Keeling Islands. Brachyura and Stomatopoda. In *Bulletin of the Raffles Museum*. Citeseer.

Vörös, J., Márton, O., Schmidt, B. R., Gál, J. T., & Jelić, D. (2017). Surveying Europe's only cave-dwelling chordate species (*Proteus anguinus*) using environmental DNA. *PLoS One*, *12*(1), e0170945.

Ward, R. D., Zemlak, T. S., Innes, B. H., Last, P. R., & Hebert, P. D. N. (2005). DNA barcoding Australia's fish species. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *360*(1462), 1847–1857.

Watanabe, H., Moku, M., Kawaguchi, K., Ishimaru, K., & Ohno, A. (1999). Diel vertical migration of myctophid fishes (Family Myctophidae) in the transitional waters of the western North Pacific. *Fisheries Oceanography*, *8*(2), 115–127.

White, W. T., Corrigan, S., Yang, L. E. I., Henderson, A. C., Bazinet, A. L., Swofford, D. L., & Naylor, G. J. P. (2018). Phylogeny of the manta and devilrays (Chondrichthyes : Mobulidae), with an updated taxonomic arrangement for the family, *Zoological Journal of the Linnean Society*, 182, 50–75. https://doi.org/10.1093/zoolinnean/zlx018.

Whittaker, R. J., & Fernández-Palacios, J. M. (2007). *Island Biogeography: Ecology, Evolution, and Conservation*. Oxford University Press.

Wickham, H. (2016). *ggplot2: elegant graphics for data analysis*. Springer.

Williams, D. G. (1994). Marine habitats of the Cocos (Keeling) Islands. *Atoll Research Bulletin*, (406), 1-10.

Woodroffe, C. D., & Berry, P. F. (1994). Scientific studies in the Cocos (Keeling) islands: an introduction. *Atoll Research Bulletin,* (399), 1-16.

Woodroffe, C. D., & Biribo, N. (2011). Atolls. In D. Hopley (Ed.), *Encyclopedia of Modern Coral Reefs: Structure, Form and Process* (pp. 51–71). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-90-481-2639-2_4

Woodroffe, C., McLean, R., Polach, H., & Wallensky, E. (1990). Sea level and coral atolls: Late Holocene emergence in the Indian Ocean. *Geology*, *18*(1), 62–66.

2.9 Supplementary information

Table S2.1 Site location information at the Cocos (Keeling) Islands.
Six water replicates (1L each) were taken from each site for eDNA metabarcoding. Marine habitat abbreviations are as follows: LCL, Low Complexity Lagoon; MCL, Medium Complexity Lagoon; HCL, High Complexity Lagoon; ORT, Outer Reef Terrace; IR, Intertidal Reef; L, Lagoonlet.

Site number	Coordinates Latitude	Longitude	Depth (m)	Marine Habitat Zone	Lagoon Zone	Date Collected
1	12.196842	96.864803	0.5	LCL	Surge Channel	23/04/2017
2	12.192553	96.862819	0.5	LCL	Surge Channel	23/04/2017
3	12.190375	96.858636	0.5	LCL	Surge Channel	23/04/2017
4	12.192678	96.853008	0.5	LCL	Inner Lagoon	23/04/2017
5	12.207150	96.843333	1	IR	Outer Lagoon	24/04/2017
6	12.205025	96.839125	1	IR	Outer Lagoon	24/04/2017
7	12.201278	96.836408	1	IR	Outer Lagoon	24/04/2017
8	12.197483	96.834094	1	IR	Outer Lagoon	24/04/2017
9	12.143050	96.816950	1	IR	Outer Lagoon	25/04/2017
10	12.150603	96.817739	1	IR	Outer Lagoon	25/04/2017
11	12.187597	96.870856	1	LCL	Surge Channel	25/04/2017
12	12.181667	96.861183	1	LCL	Surge Channel	25/04/2017
13	12.195750	96.869617	1	LCL	Surge Channel	25/04/2017
14	12.134767	96.819717	1	IR	Outer Lagoon	26/04/2017
15	12.147600	96.817500	1	IR	Outer Lagoon	26/04/2017
16	12.158717	96.820317	1	IR	Outer Lagoon	26/04/2017
17	12.173547	96.873139	1	MCL	Inner Lagoon	26/04/2017
18	12.172097	96.879981	1	MCL	Inner Lagoon	26/04/2017
19	12.166472	96.884531	1.5	MCL	Inner Lagoon	26/04/2017
20	12.168200	96.893400	1.5	MCL	Inner Lagoon	26/04/2017
21	12.156122	96.838331	3.3	LCL	Inner Lagoon	27/04/2017
22	12.153319	96.842900	5.7	MCL	Inner Lagoon	27/04/2017
23	12.150522	96.836833	4.1	LCL	Inner Lagoon	27/04/2017
24	12.148383	96.841611	7.4	MCL	Inner Lagoon	27/04/2017
25	12.139531	96.813733	12.2	ORT	Outer Lagoon	27/04/2017
26	12.148644	96.813447	13.6	ORT	Outer Lagoon	27/04/2017
27	12.158061	96.815628	16	ORT	Outer Lagoon	27/04/2017
28	12.166897	96.817122	9.5	ORT	Outer Lagoon	27/04/2017
29	12.091794	96.882667	5.5	MCL	Inner Lagoon	28/04/2017
30	12.097836	96.876711	14	MCL	Inner Lagoon	28/04/2017
31	12.101986	96.883764	1	IR	Surge Channel	28/04/2017
32	12.106692	96.883528	1.5	IR	Surge Channel	28/04/2017
33	12.108275	96.878086	3.5	MCL	Inner Lagoon	28/04/2017
34	12.133892	96.851025	5.8	HCL	Inner Lagoon	28/04/2017
35	12.128472	96.851856	6.6	HCL	Inner Lagoon	28/04/2017
36	12.128783	96.856525	11.2	HCL	Inner Lagoon	28/04/2017
37	12.135022	96.857061	4.1	HCL	Inner Lagoon	28/04/2017
38	12.171106	96.823053	0.5	L	Inner Lagoon	1/05/2017
39	12.166381	96.907214	0.5	LCL	Surge Channel	1/05/2017
40	12.162139	96.904689	0.5	LCL	Surge Channel	1/05/2017

41	12.157300	96.904450	0.5	LCL	Surge Channel	1/05/2017
42	12.152753	96.903414	0.5	LCL	Surge Channel	1/05/2017

S2.9.1 Habitat information

The CKI atoll stretching over 190 km^2 comprises a mid-sized low-lying lagoon, with only 33% of the lagoon area exceeding 2 m and <8% exceeding 3 m above mean sea level, respectively (C. D. Woodroffe & Biribo, 2011). There are a number of distinct habitat areas around and within the CKI lagoon; from the coral reef drop-offs and intertidal reef on the outer lagoon side, to the shallow seagrass and sand flats, reticulate reef amidst blue holes, massive coral outcrops, and shallow muddy lagoonlets (Williams, 1994). Habitat strata classifications were based on previous habitat strata divisions from extensive marine environmental surveys conducted at CKI (Parks Australia, 2005; Hender, McDonald, & Gilligan, 2001; Williams, 1994). Thirteen sites were located within a low complexity lagoon area, nine within the medium complexity lagoon, four within the high complexity lagoon, four on the outer reef terrace, 11 within the intertidal reef and one within a lagoonlet on the West Island. The low complexity lagoon area is characterised by shallow sand and seagrass beds (1-3m deep), devoid of any reef structure; the medium complexity lagoon by a comparably high level of sand/seagrass with increasing live hard coral cover (up to 11.2%) in small patch reefs and blue holes (that can reach depths of more than 20m); and the high complexity lagoon by a high percentage of hard coral cover (up to 50.7%) within extensive reef structures (but no blue holes), moderate level of sand/rubble and the highest plain depth in the lagoon (up to 20m deep) (Hender et al., 2001). The outer reef terrace has the highest cover of soft coral (up to 22.4%) and dead hard coral (up to 45.5%) and is characterized by a major slope change at 18m depth, beyond which the slope falls sharply away at >45-degree angles at depths of over 50m (Hender et al., 2001). The intertidal reef on the outer side of the islands drains twice a day with low tides, and typically contains a mix of sparse corals, sand, rock and seagrass. The lagoonlet situated on West Island is a shallow inlet, that with incomplete flushing from the lagoon, has predominately developed into a mud flat. This sampling strategy was devised to assess community composition and provide new baselines between sites that are in close proximity, yet are situated in varying marine habitat zones.

In addition, sample sites were also classified whether they occur in the outer lagoon, inner lagoon or within one of the major surge channels, where water from the outer side of the atoll is flushed through the lagoon in a south-east to north-west direction

(Hobbs & Macrae, 2012). The surge channels are typically less than 1.5m deep and by facilitating the influx of ocean water, which is then counteracted by lagoonal tides, forms large sand deposits on the lagoon end of the channels (see Figure 2.2 (B); Harper, Gourlay, & Jones, 2001). If the persistence of eDNA in seawater affects the localisation of an eDNA signal at a given site, then we would expect to see a carryover of taxa detections from outside the lagoon into the major surge channels and finally into the inner lagoon.

S2.9.2 Laboratory processing

DNA was extracted from the filter membranes, within two weeks of collection, using a DNeasy Blood and Tissue Kit (Qiagen; Venlo, Netherlands) with the following modifications: 540µl of ATL lysis buffer and 60µl of Proteinase K. Negative controls, containing no sample, were extracted and processed alongside all samples in order to detect any cross-contamination. Four PCR assays were optimized for use in eDNA metabarcoding workflows: 16S Fish, COI Elasmobranch, 16S Crustacean and 18S Universal (Table 2.1) that successfully amplify ray-finned fish, elasmobranchs, crustaceans and other eukaryotes (including molluscs and echinoderms) from mixed environmental samples.

Quantitative PCR (qPCR) amplification was performed in a single step to minimize excessive rounds of PCR and the possibility of mistagging events (Esling, Lejzerowicz, & Pawlowski, 2015), by using fusion tagged primers that consist of an Illumina sequencing adaptor, a unique index (8bp in length) and a primer sequence from each respective assay. All qPCR reactions were prepared in a dedicated ultra-clean laboratory facility, designed for trace DNA work. Input DNA was optimised via qPCR to maximise input template free of inhibition. Each qPCR reaction was carried out in 25µl containing: 1X AmpliTaq Gold® PCR buffer (Life Technologies, Massachusetts, USA), 2mM $MgCl_2$, 0.1mM dNTPs, 0.2µM each of forward and reverse primers (Integrated DNA Technologies, Australia), 10ug BSA (Fisher Biotec, Australia), 0.6µl of 5X SYBR® Green (Life Technologies), 1U AmpliTaq Gold® DNA Polymerase (Life Technologies), 4µl of eDNA template, and made to volume with Ultrapure™ Distilled Water (Life Technologies). As the COI Elasmobranch assay is a multiplex reaction containing two forward primers and one reverse primer (Fields

et al., 2015), the volume of forward primers was adjusted to 0.1μM each, whilst the volume of reverse primer remained unchanged at 0.2μM. Each qPCR was performed on a StepOnePlus Real-Time PCR System (Applied Biosystems, Massachusetts, USA) under the following conditions: initial denaturation at 95 °C for 5 min, followed by 50 cycles of 30 s at 95 °C, 51-54 °C for 30 s (see respective annealing temperatures in Table 2.1) and 45 s at 72 °C, with a final extension for 10 min at 72 °C.

Duplicate PCRs amplified from the same eDNA template were combined and then pooled into a library with all amplicons at equimolar ratios based on qPCR ΔRn values. Each library was size-selected (retaining amplicons between 160-450bp for the 16S and COI assays, and 160-600bp for the 18S assay) using a Pippin Prep (Sage Science, Beverly, USA) and purified using the Qiaquick PCR Purification Kit (Qiagen, Venlo, Netherlands). Final libraries were quantified using a Qubit 4.0 Fluorometer (Invitrogen, Carlsbad, USA) and if necessary, diluted to 2nM, prior to loading on either a 300 cycle (for unidirectional sequencing) or 500 cycle (for paired-end sequencing) MiSeq® V2 Standard Flow Cell on an Illumina MiSeq platform (Illumina, San Diego, USA), housed in the TrEnD Laboratory at Curtin University, Western Australia. We aimed for a sequencing depth of 50,000 reads per sample, per assay (see Table S2.7 for more information).

S2.9.3 Bioinformatics and taxonomic assignments

Reads obtained from paired-end sequencing were initially merged using AdapterRemoval (v2; Schubert, Lindgreen, & Orlando, 2016), requiring a minimum overlap of 20bp with a maximum of three mismatches. All sequencing reads obtained from both unidirectional and paired-end sequencing were then subject to the same demultiplexing and quality filtering steps. Based on their unique indexes, reads were assigned to their sample of origin using Geneious® (10.0.6; Kearse et al., 2012), however only reads with a complete match to the index, assay-specific primer (allowing up to three mismatches) and Illumina sequencing adapters were retained for further processing. After demultiplexing and trimming of fusion tagged primers, remaining reads were quality filtered using a combination of USEARCH (v9.0.2132; Edgar, 2010) and OBITools (v1.2.9; Boyer et al., 2014) programs.

USEARCH's fastq filter was implemented with a maximum error of 0.5, minimum length of 100bp and maximum ambiguous bases (N's) as 0. OBITools' obiuniq package was implemented to dereplicate reads, obigrep to retain unique sequences with a minimum read count of 5, and obiclean to denoise the dataset applying a threshold ratio (-r) of 0.05%. Finally, chimeras were identified and discarded using USEARCH's uchime2_denovo command.

Quality filtered dereplicated sequences were queried against the National Centre for Biotechnology Information's (NCBI) GenBank nucleotide database (accessed in 2017/18; Benson et al., 2005) using BLASTn with default settings and a reward value of 1. This was conducted on Zeus, an SGI cluster, based at the Pawsey Supercomputing Centre in Kensington, Western Australia. Taxonomic assignments were made by applying the following LCA parameters: a min score of 250 and the 16S Percent Identity Filter in MEGAN (METaGenome Analyzer v5.11.3; (Huson et al., 2007). The 16S Percent Identity Filter implements a conserved assignment criterion where a species-level assignment must meet or exceed a 99% match; genus, a 97% match; family, a 95% match; order, a 90% match; class, an 85% match; and phylum, an 80% match. If, for example, a unique sequence does not meet/exceed a 99% match to a species, and/or is too closely matched to another species, the taxonomic assignment will be collapsed back to a genus-level assignment. These percent cut-offs function to restrict a high level of taxonomic assignment at a low % match (e.g. species assignment at 95%). A species/genus may legitimately contain a lot of SNPs in the targeted gene region and therefore a 95% sequence match to a species may be legitimate. However, we feel we cannot give a high taxonomic assignment unless it is to a high % match. This requires more targeted barcoding of the species in various geographic localities in order for it to be widely represented in public databases. We provide average percent identity cutoffs for select taxonomic groups for each of the four assays in Tables S2.2, S2.3, S2.4 and S2.5. To present putative new occurrence records at the CKI, we required that all global congeneric taxa (of the species in question) had been barcoded for the gene region of the respective assay; this removes the possibility that a new occurrence record is actually that of a related (not yet barcoded) species. Barcode searches for congeneric taxa were conducted on the NCBI Taxonomy database. See Table S2.9 for sequencing statistics and evaluation of the new occurrence records. Taxonomic nomenclature was validated

using the World Register of Marine Species (WoRMS; http://www.marinespecies.org/).

To identify the sequencing depth at which to subsample reads in each respective assay, rarefaction analyses were conducted using QIIME (v1; Caporaso et al., 2010). The multiple_rarefactions script was executed with the following parameters: a minimum number of sequences (-m) of 1000, a maximum number of sequences (-x) of 60,000, a step-size (-s) of 1000, and a default number of iterations (10) per sequencing depth. The scripts alpha_diversity and collate_alpha were used to calculate and collate alpha diversity for each sample at each level of sequencing depth, in each respective assay. Rarefaction output was imported into RStudio (v1.1.423; RStudio Team, 2015) and graphed using ggplot2 (Wickham, 2016). We determined sequencing depth cutoffs for each assay by finding a suitable depth that maximised the number of samples reaching plateau, whilst minimising the exclusion of samples through inadequate sequencing depth. We then incorporated USEARCH's fastx_subsample command to subsample quality filtered reads (post-chimera filtering) to the following depths: Fish 16S Assay, 8000 reads; Crustacean 16S Assay, 4000 reads; Elasmobranch COI Assay, 4000 reads; and Universal 18S Assay, 8000 reads. At the respective cutoffs, there was a loss of 1 sample for the Fish 16S assay, 2 samples for the Crustacean 16S assay, 10 samples for the Elasmobranch COI assay and 11 samples for the 18S assay. Querying against GenBank was then repeated for subsampled datasets. Taxa lists were converted into presence/absence matrices for taxonomic group of interest (bony fish & elasmobranchs, crustaceans, molluscs, and echinoderms) using the splist2presabs function in the fuzzySim package (Barbosa, 2015) in RStudio.

Table S2.2 An *in-silico* analysis of average percent identities for 16S Fish amplicons.
Three bony fish families found in high abundance at CKI were selected to examine inter-species, genera and family variation within the assay's amplified region. Mitochondrial sequences were sourced from NCBI Genbank; accession numbers: KY815372.1, NC_037707.1, KY815373.1, JF457357.1, KY815374.1, MH688052.1, JF457481.1, JF457478.1, AY279701.1, MG665354.1, JF457498.1, MG704838.1, KC879146.1, FJ617076.1, KF242363.1, KF242366.1, EF463076.1, FJ616422.1, FJ616425.1, JQ707073.1, JQ707068.1, FJ616492.1, JQ707094.1, JF457654.1, FJ616499.1

16S Fish		**Labridae**		**Gobiidae**		**Pomacentridae**			
		Cheilinus	*Halichoeres*	*Bathygobius*	*Gobiodon*	*Chromis*	*Stegastes*	**Variation**	**Avg % similarity**
Labridae	*Cheilinus*	90.0	71.7	62.2	67.3	68.5	68.9	Inter-species	79.4 ± 12.2
	Halichoeres		82.8	61.8	61.4	67.7	68.3	Inter-genera	68.7 ± 6.9
Gobiidae	*Bathygobius*			57.4	60.8	59.5	58.3	Inter-family	65.4 ± 5.8
	Gobiodon				89.7	63.8	63.0		
Pomacentridae	*Chromis*					81.7	73.7		
	Stegastes						74.8		

Table S2.3 An *in-silico* analysis of average percent identities for COI Elasmobranch amplicons.
Three elasmobranch families were selected to examine inter-species, genera, and family variation within the assay's amplified region. Mitochondrial sequences were sourced from NCBI Genbank; accession numbers: NC_023948.1, NC_026696.1, NC_027081.1, NC_025520.1, KJ865755.1, NC_024284.1, NC_023361.1, NC_021768.2, MF537044.1, NC_024101.1, NC_024269.1, NC_033911.1, NC_037969.1, JQ034406.1, NC_037968.1, NC_037972.1, NC_037973.1, MH899131.1, NC_021964.1

COI Elasmobranch		**Carcharhinidae**		**Lamnidae**		**Rajidae**			
		Carcharhinus	*Glyphis*	*Isurus*	*Lamna*	*Leucoraja*	*Raja*	**Variation**	**Avg % similarity**
Carcharhinidae	*Carcharhinus*	94.9	92.3	81.6	81.6	77.7	78.3	Inter-species	93.8 ± 3.3
	Glyphis		98.4	81.4	81.4	77.9	79.1	Inter-genera	92.5 ± 2.8
Lamnidae	*Isurus*			91.3	91.3	79.7	81.2	Inter-family	80.2 ± 1.7
	Lamna				95.2	79.5	82.4		
Rajidae	*Leucoraja*					88.9	85.8		
	Raja						94.0		

Table S2.4 An *in-silico* analysis of average percent identities for 16S Crustacean amplicons.
Three crustacean families were selected to examine inter-species, genera, and family variation within the assay's amplified region. Mitochondrial sequences were sourced from NCBI Genbank; accession numbers: DQ666843.1, MH300636.1, KP256206.1, HM798481.1, HM798482.1, AF192086.1, KT959508.1, AF255057.1, U96085.1, EU882886.1, EU882881.1, U96086.1, LC191209.1, KU950453.1, MF563556.1, LC150888.1, EU863382.2, HM798560.1, HM798561.1, HM798562.1, HM798563.1

16S Crustacean		**Penaeidae**		**Xanthidae**		**Nephropidae**			
		Penaeus	*Litopenaeus*	*Leptodius*	*Xanthias*	*Homarus*	*Nephropsis*	**Variation**	**Avg % similarity**
Penaeidae	*Penaeus*	81.3	80.8	57.3	55.2	56.2	58.5	Inter-species	88.7 ± 7.9
	Litopenaeus		93.7	56.3	55.3	56.9	59.4	Inter-genera	78.5 ± 2.3
Xanthidae	*Leptodius*			90.3	78.5	55.9	55.7	Inter-family	56.2 ± 1.5
	Xanthias				77.0	53.5	53.8		
Nephropidae	*Homarus*					97.5	76.2		
	Nephropsis						92.1		

Table S2.5 An *in-silico* analysis of average percent identities for 18S Universal amplicons.
Three mollusc families were selected to examine inter-species, genera, and family variation within the assay's amplified region. Mitochondrial sequences were sourced from NCBI Genbank; accession numbers: JF899202.1, JF899203.1, JF899204.1, JF899205.1, AM774541.1, AM774542.1, JF899206.1, JF899212.1, KU201120.1, AF117739.1, JF899214.1, AY192698.1, EF613240.1, KX713346.1, AM774544.1

18S Universal		**Teredinidae**		**Pholadidae**		**Myidae**			
		Bankia	*Lyrodus*	*Barnea*	*Pholas*	*Mya*	*Sphenia*	**Variation**	**Avg % similarity**
Teredinidae	*Bankia*	95.5	97.2	92.6	92.7	92.3	92.3	Inter-species	97.7 ± 2.2
	Lyrodus		99.7	93.2	93.3	92.7	92.8	Inter-genera	95.5 ± 2.0
Pholadidae	*Barnea*			99.0	93.3	90.6	91.4	Inter-family	92.4 ± 0.7
	Pholas				97.5	91.7	92.3		
Myidae	*Mya*					99.7	95.8		
	Sphenia						94.5		

Table S2.6 In-house 16S rDNA fish database.
Assembled by Joseph D. DiBattista (2017; unpublished), this database contains 361 sequences (296 bp average fragment length) from 306 recognised taxa collected off the Western Australian coastline, including Christmas Island and the Cocos (Keeling) Islands. This database is available on request from the TrEnD Laboratory, for non-commercial use only.

Taxa represented in TrEnD 16S rDNA fish database		
Abalistes stellatus	*Caranx bucculentus*	*Dascyllus aruanus*
Abudefduf bengalensis	*Caranx ignobilis*	*Decapterus maruadsi*
Abudefduf septemfasciatus	*Centroberyx gerrardi*	*Decapterus muroadsi*
Abudefduf sexfasciatus	*Centroberyx australis*	*Diagramma pictum*
Abudefduf vaigiensis	*Cephalopholis argus*	*Diploprion bifasciatum*
Acanthistius serratus	*Cephalopholis aurantia*	*Elates ransonnettii*
Acanthocybium solandri	*Cephalopholis boenak*	*Engraulis australis*
Acanthurus grammoptilus	*Cephalopholis miniata*	*Enoplosus armatus*
Acanthurus olivaceus	*Cephalopholis sexmaculata*	*Epinephelus amblycephalus*
Acanthurus tennentii	*Cephalopholis sonnerati*	*Epinephelus bilobatus*
Acanthurus tristis	*Chaetoderma penicilligerus*	*Epinephelus chlorostigma*
Alepes apercna	*Chaetodon assarius*	*Epinephelus cyanopodus*
Amphiprion clarkii	*Chaetodon aureofaciatus*	*Epinephelus fasciatus*
Anampses lennardi	*Chaetodon mitratus*	*Epinephelus fuscoguttatus*
Anodontostoma chacunda	*Chaetodontoplus duboulayi*	*Epinephelus miliaris*
Anyperodon leucogrammicus	*Cheilinus chlorurus*	*Epinephelus morrhua*
Aphareus furca	*Cheilinus undulatus*	*Epinephelus multinotatus*
Aphareus rutilans	*Cheilodactylus gibbosus*	*Epinephelus poecilonotus*
Apistus carinatus	*Chelidonichthys kumu*	*Epinephelus polyphekadion*
Apogon apogonides	*Chironemus maculosus*	*Epinephelus quoyanus*
Apogonichthyoides nigripinnis	*Chlorurus microrhinos*	*Epinephelus radiatus*
Aprion virescens	*Chlorurus sordidus*	*Epinephelus retouti*
Argyrosomus hololepidotus	*Chlorurus sp.*	*Epinephelus tauvina*
Arothron hispidus	*Choerodon cauteroma*	*Equulites cf. leuciscus*
Arripis truttaceus	*Choerodon cephalotes*	*Equulites cf. moretoniensis*
Bodianus frenchii	*Choerodon cyanodus*	*Equulites sp.*
Bodianus vulpinus	*Choerodon jordani*	*Erythrocles schlegelii*
Caesio caerulaurea	*Choerodon schoenleinii*	*Etelis carbunculus*
Caesio cuning	*Choerodon vitta*	*Etelis radiosus*
Carangoides cf. hedlandensis	*Chromis klunzingeri*	*Etelis sp*
Carangoides chrysophrys	*Chromis lineata*	*Eubalichthys mosaicus*
Carangoides equula	*Chromis nigrura*	*Feroxodon multistriatus*
Carangoides fulvoguttatus	*Chrysophrys auratus*	*Fistularia commersonii*
Carangoides gymnostethus	*Cookeolus japonicus*	*Gazza minuta*
Carangoides humerosus	*Coris cf. pictoides*	*Gephyroberyx darwini*
Carangoides malabaricus	*Coris pictoides*	*Gerres longirostris*
Carangoides talamparoides	*Dactyloptena cf. orientalis*	*Gerres oyena*

Glaucosoma buergeri
Glaucosoma magnificum
Gnathanodon speciosus
Gracila albomarginata
Gymnapistes marmoratus
Gymnocranius grandoculis
Gymnosarda unicolor
Gymnothorax cribroris
Gymnothorax thyrsoideus
Halichoeres nigrescens
Hemigymnus melapterus
Herklotsichthys lippa
Heteroclinus adelaidae
Hipposcarus cf. longiceps
Hoplostethus gigas
Hyporhamphus melanochir
Hyporhamphus quoyi
Inegocia japonica
Iniistius jacksonensis
Iniistius sp.
Inimicus cf. sinensis
Inimicus sinensis
Istiompax indica
Jaydia cf. ellioti
Jaydia cf. poecilopterus
Kathetostoma nigrofasciatum
Katsuwonus pelamis
Labroides dimidiatus
Lagocephalus lunaris
Lagocephalus sceleratus
Lates calcarifer
Lactoria cornuta
Latropiscis purpurissatus
Lepidotrigla papilio
Leptoscarus vaigiensis
Lethrinus erythracanthus
Lethrinus genivittatus
Lethrinus laticaudis
Lethrinus miniatus
Lethrinus nebulosus
Lethrinus obsoletus
Lethrinus punctulatus
Lethrinus rubrioperculatus
Lethrinus variegatus
Lethrinus xanthochilus
Leviprora inops
Lutjanus argentimaculatus
Lutjanus bitaeniatus
Lutjanus bohar
Lutjanus carponotatus
Lutjanus erythropterus
Lutjanus johnii
Lutjanus malabaricus
Lutjanus rufolineatus
Lutjanus russellii
Lutjanus sebae
Melichthys niger
Meuschenia flavolineata
Meuschenia freycineti
Microcanthus strigatus
Mobula japanica
Mola mola
Monacanthus chinensis
Mugil cephalus
Mulloidichthys cf. vanicolensis
Naso cf. tonganus
Neoglyphidodon melas
Neopomacentrus filamentosus
Neosebastes nigropunctatus
Neosebastes pandus
Neosebastes thetidis
Netuma thalassina
Nuchequula gerreoides
Onigocia oligolepis
Oplegnathus woodwardi
Opthalmolepis lineolata
Ostorhinchus aureus
Ostorhinchus cf. apogonoides
Ostracion cubicus
Paracaesio xanthura
Parachaetodon ocellatus
Paracirrhites arcatus
Paracirrhites forsteri
Paracirrhites hemistictus
Paramonacanthus choirocephalus
Paramonacanthus filicauda
Parapercis nebulosa
Parma mccullochi
Parma occidentalis
Parma victoriae
Parupeneus heptacanthus
Parupeneus indicus
Pelates octolineatus
Pempheris sp.
Petroscirtes breviceps
Pictilabrus laticlavius
Plagiogeneion macrolepis
Plagiogeneion rubiginosum
Platax batavianus
Platycephalus australis
Platycephalus chauliodous
Platycephalus conatus
Platycephalus endrachtensis
Platycephalus longispinis
Platycephalus speculator
Platycephalus westraliae
Plectorhinchus flavomaculatus
Plectorhinchus gibbosus
Plectorhinchus polytaenia
Plectorhynchus polytaenia
Plectropomus areolatus
Plectropomus leopardus
Plectropomus maculatus
Plectropomus oligacanthus
Polydactylus cf. multiradiatus
Polyprion cf. americanus
Polyprion oxygeneios
Pomadasys argyreus
Pomadasys cf. maculatus
Pomadasys kaakan
Pomadasys maculatus
Pomatomus saltatrix
Priacanthus macracanthus
Priacanthus tayenus
Prionobutis microps
Pristipomoides auricilla
Pristipomoides filamentosus
Pristipomoides multidens
Pristipomoides sieboldii

Pristipomoides typus	*Saurida undosquamis*	*Terapon cf. puta*
Pristipomoides zonatus	*Scarus ghobban*	*Terapon cf. theraps*
Pristotis obtusirostris	*Scarus prasiognathus*	*Thalassoma cf. hardwicke*
Protonibea diacanthus	*Scarus rivulatus*	*Thalassoma lunare*
Psammoperca waigiensis	*Scatophagus argus*	*Thunnus tonggol*
Psettodes erumei	*Scomberomorus commerson*	*Thyrsitoides marleyi*
Pseudorhombus argus	*Scomberomorus queenslandicus*	*Tilodon sexfasciatus*
Pseudorhombus arsius	*Selaroides cf. leptolepis*	*Torquigener pallimaculatus*
Pseudorhombus cf. jenynsii	*Selaroides leptolepis*	*Torquigener whitleyi*
Pseudorhombus jenynsii	*Seriola dumerili*	*Trachinocephalus myops*
Pseudorhombus spinosus	*Seriolina nigrofasciata*	*Trachinotus baillonii*
Ptereleotris microlepis	*Siganus canaliculatus*	*Tragulichthys jaculiferus*
Pterocaesio digramma	*Siganus cf. margaritiferus*	*Upeneus asymmetricus*
Pterois mombasae	*Siganus virgatus*	*Upeneus sulphureus*
Rachycentron canadum	*Sillago cf. sihama*	*Upeneus sundaicus*
Randallichthys filamentosus	*Sillago ingenua*	*Upeneus tragula*
Rhynchostracion nasus	*Sillago robusta*	*Variola albimarginata*
Rogadius asper	*Sillago schomburgkii*	*Variola louti*
Saloptia powelli	*Sillago sp.*	*Yongeichthys cf. nebulosus*
Sardinella brachysoma	*Sphyraena pinguis*	*Zabidius novemaculeatus*
Sardinella gibbosa	*Stegastes cf. obreptus*	*Zeus faber*
Saurida tumbil	*Symphorus nemaptophorus*	

S2.9.4 Statistical analyses

Taxonomic detections from unsubsampled data were collated into a taxonomic table providing site/assay information and were categorised by associated habitats (coral reef, deep reef, pelagic, intertidal, fresh/brackish water) and distributions (widespread Indo-Pacific, Pacific Ocean only, Indian Ocean only, circumglobal, endemic or "uncertain"). Subsampled datasets were employed for univariate and multivariate statistics comparing diversity and community composition between sites/habitats. Taxa accumulation, based on the addition of sampling replicates at each site, was calculated and graphed using the package 'vegan' (v 2.4-3; Oksanen et al., 2017). Polynomial curves were fitted to the taxa accumulation data in order to calculate the minimum number of replicates required to sample the total extrapolated diversity at a site. Observed taxonomic diversity (equivalent to species richness) at each site was calculated using the package 'BAT' (Biodiversity Assessment Tools; Cardoso, Rigal, & Carvalho, 2015) and graphed using ggplot2 in RStudio. Normal distribution and homogeneity of variances were tested using the Shapiro-Wilk test and Levene's test respectively, in RStudio. If ANOVA assumptions were met, a one-way ANOVA and Tukey multiple pairwise-comparisons was used to test for significance between habitats. A Holm-Bonferroni correction to alpha values (α H-B) were made for the Tukey multiple pairwise-comparisons.

To assess beta diversity between sites/habitats, presence/absence matrices were imported into the software package PRIMER v7 (Clarke & Gorley, 2015). Sample replicates were summed by site and Jaccard similarity was performed on the matrices after a presence/absence transformation. Similarity patterns were visualised by non-metric multi-dimensional scaling (nMDS) and tested for variation explained by spatial predictor variables (distance - latitude/longitude, depth, lagoon and marine habitat) by distance-based linear modelling (DistLM) in the PERMANOVA+ add-on (Anderson et al., 2008). DistLM analyses were constructed by both marginal tests and by a sequential step-wise selection procedure (9999 permutations) with adjusted R^2 as the selection criterion. Similarity percentage analysis (SIMPER) was also conducted in PRIMER to identify taxa contributing to the similarity of sites categorised within the marine habitat and lagoon zones, if either were found to be a significant predictor variable in the DistLM. Finally, an indicator species analysis performed using the

Table S2.7 Mean reads per sample in bioinformatic processing.

Assay	Demultiplexed Input	Quality filtered	Dereplicated unique sequences	BLAST Hits
Fish 16S	48,941	35,931	40	31
Crustacean 16S	40,611	30,390	46	9
Universal 18S	61,051	28,123	220	144
Elasmobranch COI	109,466	42,476	77	6

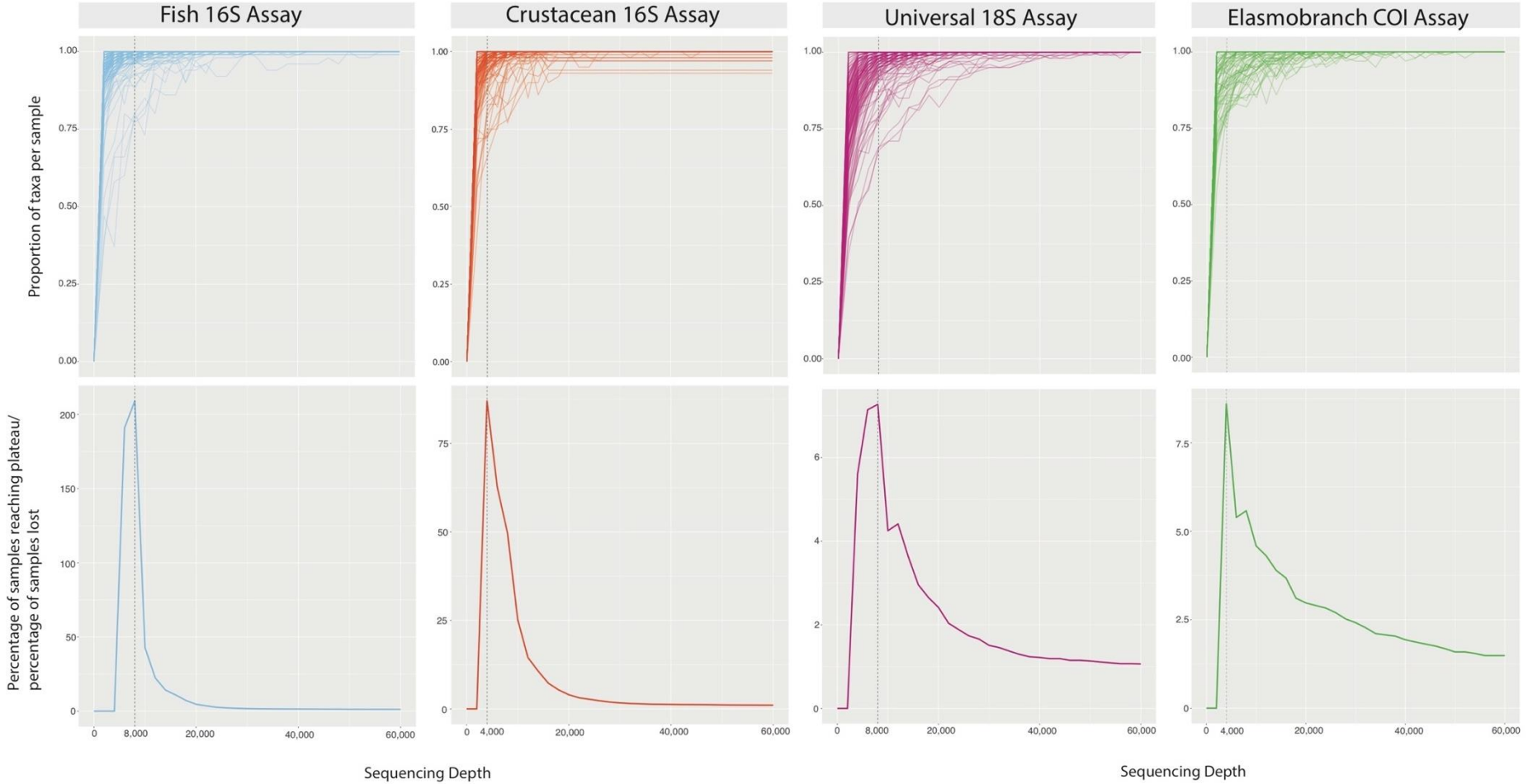


Figure S2.1 Rarefaction analyses of sequencing depth for the Fish 16S, Crustacean 16S, Elasmobranch COI and Universal 18S assays. Suitable subsampling cut-offs indicated by the dotted line were determined by maximising the number of samples that reached a detection plateau, whilst minimising the number of samples that would be excluded through inadequate sequencing depth.

package ‘indicspecies’ (De Caceres & Jansen, 2016) in RStudio, identified taxa uniquely associated with each marine habitat and lagoon group/s.

S2.9.5 Assay attributes

The 16S Fish assay was primarily used to target and amplify taxa within Actinopterygii (bony fish), which comprised 89.3% of the assay’s sequencing reads (Figure S2.2). In addition, this assay also amplified some Chondrichthyes (cartilaginous fish including sharks and rays) at 0.009% and Enteropneusta (acorn worms) at 0.0007% of the sequencing reads. Whilst the 16S Fish assay did amplify some shark taxa, mismatches between the primer binding sites and specific shark taxa such as *Galeocerdo cuvier, Carcharhinus amblyrhynchos* and *Carcharhinus melanopterus* prevented their amplification with this assay. The COI Elasmobranch degenerate assay employed in this study was designed to amplify all known elasmobranch species of the CKI and successfully amplified six elasmobranch species at 4.45% of the assay’s sequencing reads (Figure S2.2). Other amplified taxonomic groups included Actinopterygii (7.8%) and Mammalia (5.3%). A large proportion of the reads (77.5%) provided no BLAST hits, which was attributed to the degeneracy of the primer assay. There were many shared fish families amplified by both 16S Fish and COI Elasmobranch, which included Acanthuridae, Balistidae, Blenniidae, Carangidae, Chaetodontidae, Chanidae, Fistularidae, Holocentridae, Labridae, Mugilidae, Muraenidae, Pomacentridae and Tetraodontidae. There were some fish taxa, including a new occurrence record for the deepwater snapper, *Pristipomoides multidens*, which were solely amplified by COI Elasmobranch, however the majority of assigned fish taxa was amplified solely by the 16S Fish assay.

The 16S Crustacean assay amplified the classes of Malacostraca (the largest crustacean class), which comprised 20.2% of the sequencing reads, and Ophiuroidea (brittle stars) which comprised 4.4% of the sequencing reads (Figure S2.2). Similar to the COI Elasmobranch assay, the 16S Crustacean assay had a large proportion of sequencing reads attributed as no BLAST hits (68.9%). There was some overlap between the 16S Crustacean and the 18S Universal assay with the amplification of two crustacean families – Alpheidae and Callianassidae, however a greater detection of crustacean families at higher taxonomic resolution was achieved with the 16S Crustacean assay.

The 18S Universal assay amplified 72 classes in total, including Polychaeta (bristle worms, 22.7%), Gastropoda (snails and slugs, 13.7%), Mamiellophyceae (green algae, 4.5%), Gregarinasina (alveolates, 3.9%), Maxillopoda (crustacean order including barnacles and copepods, 3.5%), Demospongiae (sponges, 3.4%) and Bivalvia (bivalve molluscs, 3.0%). Whilst however a large number of classes were amplified, the majority of taxonomic assignments extracted from the 18S assay were restricted to a family level, as the gene region did not afford enough variation to distinguish genera and species within families.

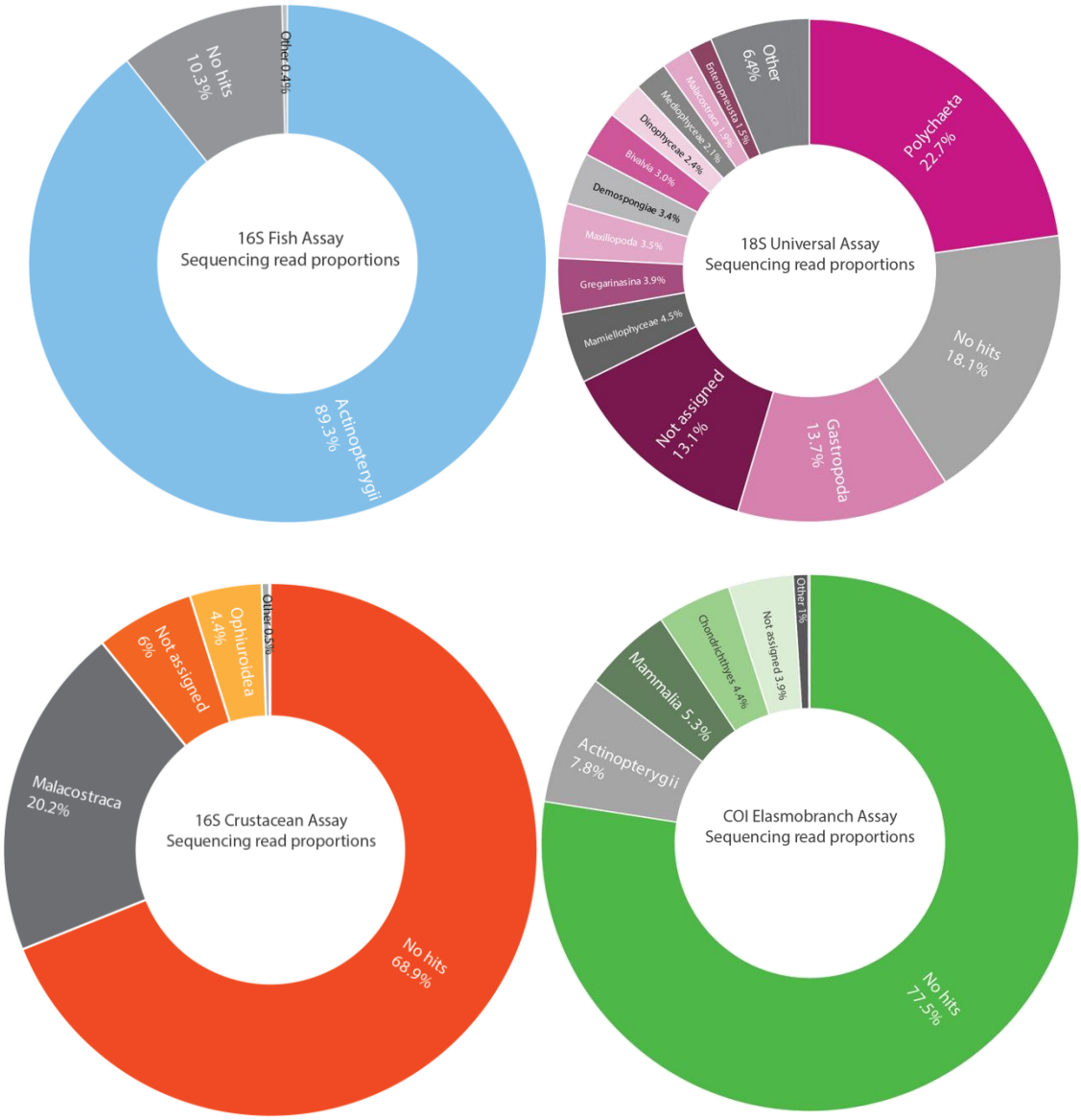


Figure S2.2 Proportion of sequencing reads from the 16S Fish assay, 16S Crustacean assay, 18S Universal assay, and COI Elasmobranch assay, attributed to various taxonomic classes.

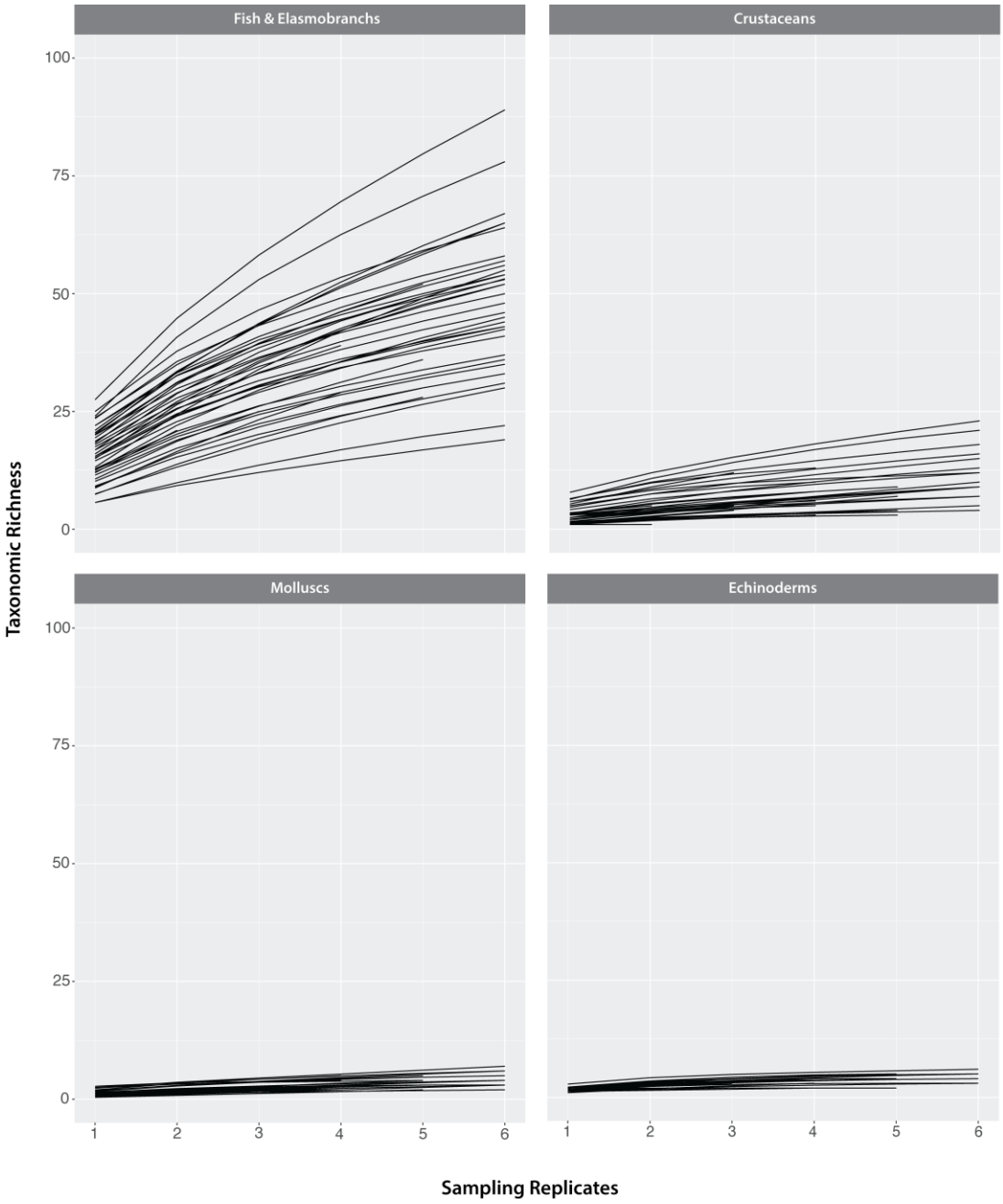


Figure S2.3 Taxonomic accumulation curves for the targeted taxonomic groups (bony fish & elasmobranchs, crustaceans, molluscs and echinoderms).
The taxa accumulation is based on the addition of sampling replicates at each site. It was extrapolated that an average of 8.38 ± 1.33, 11.75 ± 8.73, 8.23 ± 3.32 and 5.91 ± 2.39 water replicates are required to sample the total diversity at each site, for the respective taxonomic groups.

Table S2.8 Consolidated table of eDNA detections at the CKI.
NR refers to new occurrence record.

Order	Family	Genus/Species	NR	Low Complexity	Medium Complexity	High Complexity	Outer Reef Terrace	Intertidal Reef	Lagoon -let	eDNA Assay	Total reads
Perciformes	Pomacentridae	*Abudefduf*		11,40				15		16S Fish	481
Perciformes	Pomacentridae	*Abudefduf septemfasciatus*		12,42				5,15,16,32		16S Fish	3614
Perciformes	Pomacentridae	*Abudefduf sordidus*		39,42				7,9,32		16S Fish	2202
Perciformes	Acanthuridae	-		3,11,12,13,40	17,19,22,30,33	35,36,37	28	6,7,8,9,10,16,32		16S Fish	37862
Perciformes	Acanthuridae	*Acanthurus*		12,40,42			26	5,6,7,8,9,10,15,16,31,32		16S Fish	65824
Perciformes	Acanthuridae	*Acanthurus lineatus*		12				5,6,7,8,9,10,15		16S Fish	22463
Perciformes	Acanthuridae	*Acanthurus nigrofuscus*		12,42	30,33	36	25,28	14,15,16,31		16S Fish	11400
Perciformes	Acanthuridae	*Acanthurus triostegus*		1,2,3,4,11,12,13,23, 39,40,41,42	22,24,29,30	35	25,26,27,28	5,6,7,8,9,10,14,15,16,31,32		16S Fish, COI Elasmobranch	399159
Albuliformes	Albulidae	*Albula*		4,39,40,41,42	19,20,30		26	6,7,8,9,32	38	16S Fish	27017
Perciformes	Pomacentridae	*Amblyglyphidodon*			17,33	36				16S Fish	2362
Perciformes	Gobiidae	*Amblygobius*		23,39,40,41,42	17,18,19,20,22			32		16S Fish	79913
Perciformes	Labridae	*Anampses*						8,14,15,16,31		16S Fish	4882
Perciformes	Serranidae	-		40						16S Fish	635
Perciformes	Serranidae	*Anyperodon leucogrammicus*				37				16S Fish	157
Perciformes	Apogonidae	*Apogon*			18,22			14		16S Fish	642
Perciformes	Apogonidae	-		3,4,11,12,13,23,39	17,20,22,33		28	6,15,16		16S Fish	7268
Perciformes	Serranidae	*Aporops*						6		16S Fish	54
Perciformes	Serranidae	*Aporops bilinearis*					25	6,15		16S Fish	460
Tetraodontiformes	Tetraodontidae	*Arothron*		13						16S Fish	344
Tetraodontiformes	Tetraodontidae	*Arothron hispidus*		1,2,3,4,11,13,39	20,29,30			5,6,7,8,31,32	38	16S Fish, COI Elasmobranch	20250
Syngnathiformes	Aulostomidae	*Aulostomus chinensis*					28			16S Fish	115
Tetraodontiformes	Balistidae	*Balistapus undulatus*						31		16S Fish	16
Tetraodontiformes	Balistidae	-		4,11,12,13,21,23	17,20,22,29,30,33	34,35		5,6,7,15,16,31,32		16S Fish, COI Elasmobranch	5414
Perciformes	Gobiidae	*Bathygobius*		3,4,40,42	29		28	6,7,8,9,32	38	16S Fish	1557
Perciformes	Blenniidae	*Blenniella*		39						16S Fish	36
Perciformes	Blenniidae	*Blenniella chrysospilos*		2,3,11,12,13	30		28	5,6,7,8,10,15,16,31,32		16S Fish	13669
Perciformes	Blenniidae	*Blenniella periophthalmus*		1,2,3,4,11,13,39,40, 41	30			5,7,8,9,16,31		16S Fish	7140

Perciformes	Labridae	*Bodianus axillaris*				28		16S Fish	1664
Perciformes	Labridae	*Bolbometopon muricatum*	11,23	17,18,19,20,22,33	36,37	25,28	5,8,9,10,14,15,16,31	16S Fish	41444
Perciformes	Caesionidae	-	11,40,41,42	29,30	35,37	25,26,27,28	5,10,15,31	16S Fish	29151
Perciformes	Labridae	*Calotomus carolinus*	1	30		28	7,10,15	16S Fish	11186
Perciformes	Labridae	*Calotomus spinidens*		19			32	16S Fish	28
Tetraodontiformes	Monacanthidae	*Cantherhines*	42				8	16S Fish	127
Tetraodontiformes	Monacanthidae	*Cantherhines dumerilii*		30				16S Fish	28
Tetraodontiformes	Tetraodontidae	*Canthigaster*					16	16S Fish	89
Tetraodontiformes	Tetraodontidae	*Canthigaster amboinensis*					5	16S Fish	227
Scorpaeniformes	Scorpaenidae	*Caracanthus maculatus*	11					16S Fish	17
Perciformes	Carangidae	*Caranx*	4,23,40,42	17,18,22,30	35	28	6,7,8,32	16S Fish	15662
Perciformes	Carangidae	*Caranx ignobilis*	23,40	17,30		28	7	16S Fish	1915
Perciformes	Serranidae	*Cephalopholis*	3,11,13,42	30		28	6	16S Fish	1210
Perciformes	Serranidae	*Cephalopholis argus*	13					16S Fish	2793
Perciformes	Chaetodontidae	*Chaetodon*	3,12,13,21,23	17,22,29,33	35,37	25,28	8,31,32	16S Fish	19512
Perciformes	Chaetodontidae	*Chaetodon auriga*	2,3,11,12,40	17,18,29			6,8,9,14,16,32	16S Fish, COI Elasmobranch	8928
Perciformes	Chaetodontidae	*Chaetodon citrinellus*	12			28		16S Fish	545
Perciformes	Chaetodontidae	*Chaetodon ephippium*		18,33			5,14,31,32	16S Fish	2950
Perciformes	Chaetodontidae	*Chaetodon meyeri*					31	16S Fish	74
Perciformes	Chaetodontidae	*Chaetodon vagabundus*	12					16S Fish	688
Gonorynchiformes	Chanidae	*Chanos*	3,4,23,39,40					16S Fish	89
Gonorynchiformes	Chanidae	*Chanos chanos*	3,4,21,23,39,40,41,42	19,20,22,30,33		26,28	31,32 38	16S Fish, COI Elasmobranch	108492
Perciformes	Labridae	*Cheilinus*	3,21,23	17,18,20,24,30,33	35,37	25,26,28	14,32	16S Fish	5011
Perciformes	Labridae	*Cheilinus fasciatus*	21	24,33	36,37			16S Fish, COI Elasmobranch	22665
Perciformes	Labridae	*Cheilinus trilobatus*	1,3,4,12,13,21,23,39	17,18,19,20,30	37	25,28	6,14,15,31,32	16S Fish, COI Elasmobranch	26297
Perciformes	Labridae	*Cheilinus undulatus*	4	18,19,22,24,33	37	25,28	31	16S Fish, COI Elasmobranch	34252
Perciformes	Labridae	*Cheilio inermis*	2,23				7	16S Fish	3386
Perciformes	Labridae	*Chlorurus*	1,2,3,4,11,12,13,23,40,42	17,18,19,20,22,29,30,33	34,35,36,37	25,26,28	5,6,7,8,9,10,14,15,16,31,32	16S Fish	260197
Perciformes	Pomacentridae	*Chromis*	39	29				16S Fish	221

Perciformes	Pomacentridae	*Chromis nigrura*		42			25,26,28	5,8,15,16		16S Fish, COI Elasmobranch	1252
Perciformes	Pomacentridae	*Chromis viridis*		21,23	17,18,20,22,29,30,33			14		16S Fish	33187
Perciformes	Pomacentridae	*Chrysiptera*		1,11,13,40,41,42	30			8,32		16S Fish	955
Perciformes	Pomacentridae	*Chrysiptera unimaculata*	NR	1,2,3,4,13,21,23,40	17,19,22,30			14,31	38	16S Fish	7584
Perciformes	Blenniidae	*Cirripectes*		11	30	37	25,26,28	5,6,7,8,9,14,15,16,31		16S Fish, COI Elasmobranch	47
Perciformes	Blenniidae	*Cirripectes castaneus*		1,3,4,11,12,13,39,40,41	22,30,33	34,35	25,26,28	5,6,7,8,9,14,15,16,31		16S Fish	16362
Perciformes	Acanthuridae	*Ctenochaetus*		11,12,21,23	17,22,30,33	34,35,36,37	25,26,28	6,7,8,10,15,16,31,32		16S Fish	152025
Perciformes	Gobiidae	*Ctenogobiops*	NR	13	17,20,29,33			31	38	16S Fish	10635
Perciformes	Labridae	*Cymolutes praetextatus*		3,23,39,40,41	17,19,30			31,32		16S Fish, COI Elasmobranch	161278
Perciformes	Labridae	*Cymolutes torquatus*	NR		29	37				16S Fish	2112
Perciformes	Pomacentridae	*Dascyllus*			30			15		16S Fish	135
Perciformes	Pomacentridae	*Dascyllus aruanus*		3,4,13,21,23	17,20,29,33	35		14,32		16S Fish	12299
Perciformes	Carangidae	*Decapterus macarellus*		11	18,22,24,29	35,37	26,27,28	10		16S Fish, COI Elasmobranch	3859
Tetraodontiformes	Diodontidae	*Diodon hystrix*		1,3,13			25	8,9,16		16S Fish	5540
Syngnathiformes	Syngnathidae	*Doryrhamphus*						9		16S Fish	28
Anguilliformes	Muraenidae	*Echidna nebulosa*		1,2,3,4,11,13,21,40	17,30		25	5,6,8,9,14,16,32		16S Fish, COI Elasmobranch	17483
Anguilliformes	Muraenidae	*Echidna polyzona*		3,40	30					16S Fish, COI Elasmobranch	7438
Perciformes	Blenniidae	*Ecsenius*				35				16S Fish	5
Perciformes	Blenniidae	*Ecsenius bicolor*			22,24	35				16S Fish	255
Perciformes	Carangidae	*Elagatis bipinnulata*		1						16S Fish	132
Mugiliformes	Mugilidae	*Ellochelon vaigiensis*		4,39				6,14,32	38	16S Fish, COI Elasmobranch	75038
Anguilliformes	Muraenidae	*Enchelynassa*		2,13,39,40,41				6,8		16S Fish	1002
Perciformes	Blenniidae	*Entomacrodus*		11,12,13,39,42	29		25,28	5,6,7,8,9,10,15,16,31,32		16S Fish	41146
Perciformes	Blenniidae	*Entomacrodus striatus*						31,32		16S Fish	173
Perciformes	Labridae	*Epibulus insidiator*		42	22,30	35,36	26	16		16S Fish	7744
Perciformes	Serranidae	*Epinephelus*		1,2,3,4,13,21,23,39,40,41,42	17,18,19,20,22,24,29,30,33	34,35,36,37		6,8,9,14,15,16,31,32		16S Fish	102649
Perciformes	Serranidae	*Epinephelus hexagonatus*		1,11,12,13,40,42	29			5,6,7,8,9,10,15,16,31,32		16S Fish	26342

Perciformes	Serranidae	*Epinephelus spilotoceps*						14		16S Fish	1228
Perciformes	Serranidae	*Epinephelus tauvina*		12,39			26	14		16S Fish	856
Perciformes	Blenniidae	*Exallias*						16		16S Fish	35
Perciformes	Blenniidae	*Exallias brevis*		4,12			25	15,16		16S Fish, COI Elasmobranch	320
Beloniformes	Exocoetidae	-		11,12,13,39,41,42	29,30		25,26,27,28	7,9,10,15,16,31, 32		16S Fish	36440
Syngnathiformes	Fistulariidae	*Fistularia commersonii*		3,21,23				5,6,32		16S Fish, COI Elasmobranch	4412
Perciformes	Blenniidae	*Glyptoparus delicatulus*		11,12				31		16S Fish	2207
Perciformes	Gobiidae	*Gnatholepis*		3,23,40,41	17,29					16S Fish	217
Perciformes	Gobiidae	*Gnatholepis anjerensis*		1,3,4,21,23,39,40,41 ,42	17,18,22,29			32	38	16S Fish	131032
Perciformes	Gobiidae	-		13	18,20,29				38	16S Fish	2416
Perciformes	Gobiidae	*Gobiodon*						31		16S Fish	30
Perciformes	Gobiidae	*Gobiodon ater*	NR					31		16S Fish	258
Perciformes	Gobiidae	*Gobiodon citrinus*	NR					14		16S Fish	99
Perciformes	Labridae	*Gomphosus*		1	19					16S Fish	5404
Perciformes	Microdesmidae	*Gunnellichthys*			30					16S Fish	63
Anguilliformes	Muraenidae	*Gymnomuraena zebra*	NR					32		16S Fish	164
Anguilliformes	Muraenidae	*Gymnothorax*		2,3,23,39,42	20,30			5,7,14,32		16S Fish, COI Elasmobranch	3024
Anguilliformes	Muraenidae	*Gymnothorax buroensis*			24,29			6,14,15,31		16S Fish, COI Elasmobranch	1779
Anguilliformes	Muraenidae	*Gymnothorax fimbriatus*		4	30					16S Fish	79
Anguilliformes	Muraenidae	*Gymnothorax flavimarginatus*		3,4,11						16S Fish	844
Anguilliformes	Muraenidae	*Gymnothorax richardsonii*	NR	21	33			32	38	16S Fish	2652
Anguilliformes	Muraenidae	*Gymnothorax rueppellii*		13				8		16S Fish	110
Anguilliformes	Muraenidae	*Gymnothorax thyrsoideus*		2,4,12,13,40			25	7,32		16S Fish	4576
Anguilliformes	Muraenidae	*Gymnothorax undulatus*		4,21,23	18,20,29	37		14		16S Fish	4404
Perciformes	Haemulidae	-					28			16S Fish	111
Perciformes	Labridae	*Halichoeres*		1,2,3,4,11,12,13,39, 40,41	29,30		25,26,28	5,6,7,8,9,10,14,15,16,31,3 2		16S Fish, COI Elasmobranch	25684

Perciformes	Labridae	*Halichoeres hortulanus*						8,16,31		16S Fish	2019
Perciformes	Labridae	*Halichoeres marginatus*		3,11,12	29,30		28	6,8,10,14,31		16S Fish, COI Elasmobranch	15644
Perciformes	Labridae	*Halichoeres trimaculatus*		1,2,3,4,13,21,23,39,40,41,42	17,18,20,22,30,33		25	6,8,14,31,32	38	16S Fish	405876
Perciformes	Labridae	*Hemigymnus melapterus*		13	18,20,30	35	25	5,14		16S Fish, COI Elasmobranch	21446
Perciformes	Labridae	*Hipposcarus*		12				10,14		16S Fish	7853
Perciformes	Labridae	*Hipposcarus harid*		1,2,3,4,11,12,13,23,40,42	17,18,19,20,22,29,30,33	34,35,37	25,27,28	5,6,7,8,9,10,14,15,16,31,32		16S Fish	298472
Beryciformes	Holocentridae	-		23,39,42	22,29,30		26,28	9,10,14,15,32		16S Fish	7436
Perciformes	Blenniidae	*Istiblennius edentulus*		4,12,41,42				31,32		16S Fish, COI Elasmobranch	793
Perciformes	Kuhliidae	*Kuhlia*		12,39						16S Fish	652
Perciformes	Kyphosidae	*Kyphosus*		4,11,12,23,39,40,42	18,20,22,29,30	34	25,26,28	6,7,9,10,14,15,16,31,32		16S Fish	217893
Perciformes	Labridae	*Labrichthys unilineatus*			30					16S Fish	1888
Perciformes	Labridae	-		1,2,3,11,12,13,23,42	30,33		25	5,6,7,8,9,10,14,15,16,31,32		16S Fish, COI Elasmobranch	24663
Perciformes	Labridae	*Labroides dimidiatus*		13	22,30			8		16S Fish, COI Elasmobranch	6444
Myctophiformes	Myctophidae	*Lampadena luminosa*	NR		30					16S Fish	739
Perciformes	Lethrinidae	-		23	24,29					16S Fish	5479
Perciformes	Lethrinidae	*Lethrinus*		2,3,4,11,12,13,21,23,39,40,42	17,18,19,20,22,29,30,33	35		10,15,16,32	38	16S Fish	36759
Perciformes	Lethrinidae	*Lethrinus xanthochilus*		1,11,13,23,39,41	17,18,22,30	37	28	5,6,8,9,15,16,31,32		16S Fish	32125
Perciformes	Lutjanidae	-		11	30		28	7,8,31		16S Fish	3042
Perciformes	Lutjanidae	*Lutjanus*		4,11,12,21,23,40,42	19,22			9,31	38	16S Fish	23754
Perciformes	Lutjanidae	*Lutjanus bohar*		13,23	20,22,24,30,33	35,37	27	16,32		16S Fish	63742
Perciformes	Lutjanidae	*Lutjanus fulvus*		2,3,4,11,12,23,39,40,41,42	17,18,20,22,30		28	5,6,9,10,16,31,32	38	16S Fish	84016
Perciformes	Lutjanidae	*Lutjanus monostigma*				36				16S Fish	940
Perciformes	Labridae	*Macropharyngodon meleagris*						5,31		16S Fish, COI Elasmobranch	379
Myliobatiformes	Myliobatidae	*Manta birostris* (revised to *Mobula birostris*)		42	18,19,20,22,29,33	37	28	5,9		16S Fish, COI Elasmobranch	695

Tetraodontiformes	Balistidae	*Melichthys*	3,12,13,39,40,42	30		25,28	5,6,7,8,9,10,15,16,31,32	16S Fish, COI Elasmobranch	9969
Tetraodontiformes	Balistidae	*Melichthys niger*	3,12,13,39	30		28	6,7,8,10,16,31,32	16S Fish	4208
Tetraodontiformes	Balistidae	*Melichthys vidua*					10,16,31	16S Fish	601
Perciformes	Lethrinidae	*Monotaxis grandoculis*		30	34			16S Fish	1237
Anguilliformes	Moringuidae	*Moringua javanica*	39	29				16S Fish	272
Perciformes	Mugilidae	-	3,4,11,12,39,40,42	20,29,30,33			6,7,8,9,14,15,16,32 38	16S Fish	176156
Perciformes	Mullidae	-	2,4,11,12,13,23,39,42	17,18,20,22,24,29,30,33	37	28	8,10,14,16,32 38	16S Fish	126712
Perciformes	Mullidae	*Mulloidichthys*	4,21,23,39,42	17,18,22,29		27	32	16S Fish	51421
Perciformes	Mullidae	*Mulloidichthys vanicolensis*	39	29,30			6	16S Fish	13107
Myctophiformes	-	- NR	2,42	30			9	16S Fish	4208
Beryciformes	Holocentridae	*Myripristis kuntee*	11	29,30		25	10	16S Fish	694
Perciformes	Acanthuridae	*Naso*	1,4,11,12,13,21,23,40,41,42	17,18,19,22,24,29,30,33	34,35,36,37	25,26,27,28	5,6,7,8,9,10,15,16,31,32	16S Fish	735097
Perciformes	Labridae	*Novaculichthys taeniourus*	1,2,3,4,11,12,13				6,7,8	16S Fish	54689
Mugiliformes	Mugilidae	*Oedalechilus labiosus* NR	12					16S Fish	218
Perciformes	Labridae	*Oxycheilinus unifasciatus*					6	16S Fish	250
Perciformes	Cirrhitidae	*Paracirrhites*	12	30			5,8,9,15	16S Fish	4778
Perciformes	Cirrhitidae	*Paracirrhites arcatus*	12	29			5,6,7,8,31	16S Fish	7238
Perciformes	Cirrhitidae	*Paracirrhites forsteri*					8	16S Fish	1035
Perciformes	Mullidae	*Parupeneus multifasciatus*	23				15	16S Fish	985
Perciformes	Pempheridae	-				28	15	16S Fish	513
Perciformes	Ephippidae	*Platax*		20	35			16S Fish	2417
Beloniformes	Belonidae	*Platybelone argala*	2,11,13,41,42	29,30			8,14,15,31	16S Fish	7582
Perciformes	Haemulidae	*Plectorhinchus*	23,39,40	29		26,28	15	16S Fish	11103
Perciformes	Pomacentridae	*Plectroglyphidodon dickii*	11,12	30		28	9,10,15,31	16S Fish	415
Perciformes	Pomacentridae	*Plectroglyphidodon imparipennis*	3,11,12	30		28	5,6,7,8,9,15,16,31	16S Fish	3886
Perciformes	Pomacentridae	*Plectroglyphidodon lacrymatus*		30	35,36,37	26		16S Fish	24371
Perciformes	Pomacentridae	*Plectroglyphidodon leucozonus*	11,13				7,8,9,16	16S Fish	7539
Perciformes	Pomacentridae	*Plectroglyphidodon phoenixensis*	1,12,13,40	30			5,6,7,8,9,15,16	16S Fish	4289

Perciformes	Serranidae	*Plectropomus areolatus*			19,22,24,29	37			16S Fish	4563
Perciformes	Pomacentridae	*Pomacanthus imperator*					26		16S Fish	591
Perciformes	Pomacentridae	-		11,12,40	17,30	36	25,28	5,7,8,9,10,15,16,31,32	16S Fish, COI Elasmobranch	10084
Perciformes	Pomacentridae	*Pomacentrus pavo*		2,23	22,24,29,30	34,35,36	26	9,16	16S Fish	15991
Perciformes	Labridae	*Pseudocheilinus hexataenia*			29			8	16S Fish	2454
Perciformes	Serranidae	*Pseudogramma polyacantha*			30			8,9,14	16S Fish	1044
Perciformes	Microdesmidae	*Ptereleotris*		23					16S Fish	809
Perciformes	Microdesmidae	*Ptereleotris microlepis*		21,23	17,22				16S Fish	26238
Perciformes	Microdesmidae	*Ptereleotris zebra*						31	16S Fish	2042
Perciformes	Caesionidae	*Pterocaesio*						31	16S Fish	1323
Tetraodontiformes	Balistidae	*Rhinecanthus*		1,2,3,4,11,12,13,23	22,29,30		26,28	5,6,7,8,9,10,14,15,16,31,32	16S Fish	29890
Perciformes	Blenniidae	-		1,2,3,4,11,12,13,39			25,28	5,6,7,8,9,10,15, 16,31	16S Fish	19158
Beryciformes	Holocentridae	*Sargocentron*				36	27	8,10,15	16S Fish, COI Elasmobranch	589
Beryciformes	Holocentridae	*Sargocentron punctatissimum*		11,12,13,39,40			27,28	5,9,10,15,16,32	16S Fish	9967
Perciformes	Labridae	*Scarus*		1,2,11,12,21,23,39,41	17,18,19,20,22,29,30,33	34,35,36,37	25,26,27,28	7,10,14,15,16,31,32	16S Fish	190850
Perciformes	Labridae	*Scarus psittacus*		1,2,3,4,11,12,13,21, 23,39,41,42	22,24,29,30,33	35,37	25,26,28	5,6,8,10,15,16,31,32	16S Fish	102304
Perciformes	Schindleriidae	*Schindleria praematura*	NR	21,23	22,24				16S Fish	2325
Perciformes	Scombridae	-			29				16S Fish	1369
Perciformes	Siganidae	*Siganus*			17,18,19,20				16S Fish	11216
Perciformes	Siganidae	*Siganus argenteus*		2,11,12,13,23	17,18,19,29	36	25,28	14,15,16,31	16S Fish	45912
Clupeiformes	Clupeidae	*Spratelloides*		1,4,13,39,40,41,42	17,18,19,20,22,29,30	35		5,6,7,14,32	16S Fish	4303
Clupeiformes	Clupeidae	*Spratelloides delicatulus*		1,2,3,4,13,23,39,40, 41,42	17,18,19,20,22,29	35,36		5,6,7,8,14,32	38 16S Fish	5869
Perciformes	Pomacentridae	*Stegastes*		1,2,3,4,13,21,23	17,18,19,20,22,29,30,33	37		14,32	16S Fish	224357
Perciformes	Pomacentridae	*Stegastes albifasciatus*		1,3,4,11,12,13,21,23	29,30			5,14,15,16,31	16S Fish	31138
Perciformes	Pomacentridae	*Stegastes nigricans*		23	17,22,24,29,30,33	34,35,36,37		14	16S Fish	11541
Perciformes	Labridae	*Stethojulis*		1,2,3,4,11,12,13,21, 23,40,42	17,18,19,29,30		25,28	5,6,7,8,9,10,14, 15,16,31	38 16S Fish	710716

Perciformes	Labridae	*Stethojulis strigiventer*		2,3,4,13,21,23,39,40	17,18,19,20,22			32	38	16S Fish	92929
Stomiiformes	-	-	NR					9		16S Fish	252
Syngnathiformes	Syngnathidae	-						9,15		16S Fish	316
Perciformes	Labridae	*Thalassoma*		1,2,3,4,11,12,13,39, 40,41,42	22,29,30	37	25,26,28	5,6,7,8,9,10,14,15,16,31,3 2		16S Fish, COI Elasmobranch	211190
Perciformes	Labridae	*Thalassoma amblycephalum*		11,13	29		25,28	5,6,8,10,31		16S Fish, COI Elasmobranch	122540
Perciformes	Carangidae	*Trachinotus blochii*		39,40	17,20				38	16S Fish, COI Elasmobranch	3764
Carcharhiniformes	Carcharhinidae	*Triaenodon obesus*		21,23	18,22,30,33	35		8,31		16S Fish, COI Elasmobranch	29617
Beloniformes	Belonidae	*Tylosurus acus*	NR		30		27	6		16S Fish	17868
Beloniformes	Belonidae	*Tylosurus crocodilus*		21,39,40,42	17,19			5,6,15,31		16S Fish	4320
Perciformes	Gobiidae	*Vanderhorstia*		2,4,21,23,39,40,41	24					16S Fish	602
Perciformes	Gobiidae	*Vanderhorstia ornatissima*		2,3,4,21,23,39,40,41	18,22,24,29	36,37		32	38	16S Fish	90104
Perciformes	Acanthuridae	*Zebrasoma*			22,24,30,33	35,36,37				16S Fish	15678
Perciformes	Acanthuridae	*Zebrasoma desjardinii*		12,21,23	17,18,19,20,22,24,2 9,33	34,35,36,37				16S Fish	98792
Decapoda	Alpheidae			23	29	35		14		16S Crust, COI Elasmobranch, 18S Universal	4046
Decapoda	Alpheidae	*Alpheus*		1,4		35		14,15		16S Crust	64537
Ophiurida	Amphiuridae	*Amphipholis squamata*		1,2,4,11,12,13,39,40 ,41,42	17,18,19,20		28	5,6,7,8,9,10,15, 32	38	16S Crust	58021
Decapoda	Calappidae	*Calappa*		4						16S Crust	15
Decapoda	Calappidae	*Calappa hepatica*		4,39,40	29					16S Crust	31072
Decapoda	Diogenidae	*Calcinus*					26	15,32		16S Crust	505
Decapoda	Diogenidae	*Calcinus elegans*		12				6		16S Crust	769
Decapoda	Diogenidae	*Calcinus latens*			18,30			14	38	16S Crust	1882
Decapoda	Diogenidae	*Calcinus morgani*						8		16S Crust	271
Decapoda	Callianassidae				33					16S Crust, 18S Universal	27996
Decapoda	Gecarcinidae	*Cardisoma carnifex*		39,42				7,16	38	16S Crust	8961
Decapoda	Carpiliidae	*Carpilius convexus*					28	9		16S Crust	316
Decapoda	Carpiliidae	*Carpilius maculatus*		11						16S Crust	130
Decapoda	Xanthidae	*Chlorodiella*		1,2,4,13,23,39,40	17,20,30,33			14		16S Crust	42185
Decapoda	Xanthidae	*Chlorodiella barbata*		1,2,3,4,13,21,23,39, 40,41	17,18,19,20,22,29,3 0,33	37	26	14,32	38	16S Crust, COI Elasmobranch	291560

Decapoda	Xanthidae	*Chlorodiella cytherea*		13,39			28	5,14		16S Crust	1043
Decapoda	Xanthidae	*Chlorodiella nigra*	NR	1				14		16S Crust	444
Decapoda	Callianassidae	*Corallianassa*			33				38	16S Crust	139604
Decapoda	Xanthidae	*Cyclodius*		3,11,39,40		35,37		5,14,31,32		16S Crust	16149
Decapoda	Xanthidae	*Cyclodius ungulatus*		1,3,4,13,40	19,22,29	35,37		14,32		16S Crust	121487
Decapoda	Xanthidae	*Cymo*		4,11,13	33			5,9,14		16S Crust	18482
Decapoda	Dairidae							6,8		16S Crust	722
Decapoda	Dairidae	*Daira*		11,12	29		25	6,9,16		16S Crust	16036
Decapoda	Dairidae	*Daira perlata*		1,2,3,11,12,13,39,40,41,42			25,28	5,6,7,8,9,10,15, 16,32		16S Crust	251781
Decapoda	Xanthidae	*Danielea noelensis*	NR					10		16S Crust	43
Decapoda	Diogenidae	*Dardanus*		4			25			16S Crust	964
Decapoda	Diogenidae	*Dardanus lagopodes*					28	15		16S Crust	1143
Decapoda	Xanthidae	*Etisus albus*	NR	4				5		16S Crust	26
Decapoda	Xanthidae	*Etisus anaglyptus*	NR					32		16S Crust	365
Decapoda	Xanthidae	*Etisus bifrontalis*		40						16S Crust	9
Decapoda	Xanthidae	*Etisus frontalis*		3,39,40,41	18,30			7,8	38	16S Crust	1910
Euphausiacea	Euphausiidae		NR				28			16S Crust	83
Decapoda	Galatheidae	*Galathea*	NR	13			26,28	6		16S Crust	1612
Decapoda	Xanthidae	*Garthiella aberrans*	NR				28			16S Crust	1543
Stomatopoda	Gonodactylidae			1,4				5,8,10,32		16S Crust	2994
Decapoda	Grapsidae	*Grapsus albolineatus*	NR	42						16S Crust	249
Decapoda	Grapsidae	*Grapsus intermedius*						32		16S Crust	15
Decapoda	Hippidae	*Hippa*	NR	4						16S Crust	27
Decapoda	Xanthidae	*Lachnopodus subacutus*		11,12				6,10,15,32		16S Crust	1765
Decapoda	Xanthidae	*Leptodius*		4						16S Crust	30
Decapoda	Xanthidae	*Liocarpilodes*	NR	11				15		16S Crust	664
Decapoda	Xanthidae	*Liocarpilodes integerrimus*	NR	11,12				32		16S Crust	128
Decapoda	Xanthidae	*Liomera bella*		13,23	24			14,31		16S Crust	2956
Decapoda	Xanthidae	*Lybia leptochelis*	NR					31		16S Crust	165
Decapoda	Macrophthalmidae			4						16S Crust	72
Decapoda	Majoidea	*Majidae*			18					16S Crust	16

Decapoda	Alpheidae	*Metalpheus*	NR	11,39			28	9		16S Crust	19783
Decapoda	Callianassidae	*Neocallichirus*		4	33			32	38	16S Crust	991
Decapoda	Ocypodidae	*Ocypode*							38	16S Crust	34
Decapoda	Ocypodidae								38	16S Crust	55
Ophiurida	Ophiocomidae	*Ophiocoma*		1,2,3,4,11,13,39,40,42	17,29,30		25,28	5,8,9,14,15,31,32		16S Crust	32261
Ophiurida	Ophiocomidae	*Ophiocoma scolopendrina*		2,3,4,40				32		16S Crust	5771
Ophiurida	Ophiocomidae	*Ophiocomella*		1,2,3,4,11,12,39,40,42	18,19,20,29,30		28	6,7,8,10,14,15,16,31,32		16S Crust	100242
Decapoda	Grapsidae	*Pachygrapsus minutus*		4,13				8,14,31,32		16S Crust	538
Decapoda	Grapsidae	*Pachygrapsus plicatus*		4				32		16S Crust	871
Decapoda	Xanthidae	*Palapedia*		1						16S Crust	74
Decapoda	Palinuridae	*Panulirus penicillatus*		12						16S Crust	5144
Decapoda	Xanthidae	*Paractaea rufopunctata*				35				16S Crust	44
Decapoda	Xanthidae	*Paraxanthias*	NR					7,15		16S Crust	1082
Decapoda	Xanthidae	*Paraxanthias notatus*	NR					32		16S Crust	485
Decapoda	Penaeidae		NR	4	30					16S Crust	51845
Decapoda	Plagusiidae	*Percnon*						10		16S Crust	8
Decapoda	Xanthidae	*Pilodius*			30			14		16S Crust	20
Decapoda	Xanthidae	*Pilodius areolatus*		4,11,39	30		28	7,8,9,14,31,32	38	16S Crust	40485
Decapoda	Xanthidae	*Pilodius scabriculus*		11,13,39,40	20,30		25	5,6,8,9,14,16,31,32		16S Crust	13519
Decapoda	Plagusiidae	*Plagusia*						6		16S Crust	113
Decapoda	Plagusiidae	*Plagusia speciosa*	NR					5		16S Crust	51
Decapoda	Xanthidae	*Platypodia*			18,20					16S Crust	224
Decapoda	Portunidae			1,4,13,39,40				32		16S Crust	1426
Decapoda	Xanthidae	*Psaumis cavipes*						14		16S Crust	2884
Decapoda	Xanthidae	*Pseudoliomera*					28			16S Crust	62
Stomatopoda	Pseudosquillidae	*Pseudosquilla ciliata*		4,13,40						16S Crust	462
Stomatopoda	Nannosquillidae	*Pullosquilla*	NR		29				38	16S Crust	5829
Stomatopoda	Pseudosquillidae	*Raoulserenea*					28	14		16S Crust	1279
Decapoda	Alpheidae	*Rugathanas*	NR	39	17,24			32		16S Crust	509
Decapoda	Alpheidae	*Salmoneus*		13						16S Crust	2070
Decapoda	Hippolytidae	*Saron*	NR	12			28	5,6	38	16S Crust	599

Decapoda	Xanthidae	*Sulcodius deflexus*	NR	23	17,18	35		32		16S Crust	39130
Decapoda	Alpheidae	*Synalpheus*						5,8,9		16S Crust	3551
Decapoda	Alpheidae	*Synalpheus charon*	NR					7		16S Crust	14
Decapoda	Alpheidae	*Synalpheus tumidomanus*	NR	4,11,12				7,9,15,16,31		16S Crust	21146
Decapoda	Tetraliidae	*Tetralia*			29					16S Crust	199
Decapoda	Portunidae	*Thalamita*		1,2,3,4,11,12,13,23, 39,40,41,42	17,18,19,22,29,33	37	25,26,28	5,6,7,8,9,10,14, 15,32	38	16S Crust	104486
Decapoda	Portunidae	*Thalamita crenata*		39					38	16S Crust	9544
Decapoda	Varunidae	*Thalassograpsus harpax*		4						16S Crust	8
Decapoda	Trapeziidae	*Trapezia*						15		16S Crust	12
Decapoda	Xanthidae	*Tweedieia*						32		16S Crust	657
Decapoda	Xanthidae	*Vellodius*	NR		17					16S Crust	1455
Decapoda	Xanthidae			1,4,13,23,39,40	17,18,20,22,30			5,10,14,32		16S Crust	19291
Decapoda	Xanthidae	*Zosimus aeneus*						7		16S Crust	450
Tetraodontiformes	Monacanthidae	*Aluterus scriptus*				35				COI Elasmo	41
Perciformes	Gobiidae	*Amblygobius phalaena*		39,40,41	20					COI Elasmo	212
Myctophiformes	Myctophidae	*Bolinichthys*	NR					6		COI Elasmo	49
Pleuronectiformes	Bothidae	*Bothus pantherinus*		39,42	30					COI Elasmo	27
Ophidiiformes	Ophidiidae	*Brotula multibarbata*		12,39			28	8		COI Elasmo	11098
Cephalaspidea	Bullidae	*Bulla orientalis*						32		COI Elasmo	9
Cephalaspidea	Bullidae	*Bulla vernicosa*		23						COI Elasmo	166
Perciformes	Gobiidae	*Cabillus tongarevae*		1,2,3,4,21,40	18,29,30			5,6,8		COI Elasmo	3257
Tetraodontiformes	Monacanthidae	*Cantherhines*		12						COI Elasmo	26
Perciformes	Carangidae	*Carangoides ferdau*		4,21,39,40,41	30			6,7,8		COI Elasmo	7033
Perciformes	Carangidae	*Carangoides orthogrammus*			30			6,8		COI Elasmo	26913
Perciformes	Carangidae	*Caranx melampygus*			30					COI Elasmo	177
Carcharhiniformes	Carcharhinidae	*Carcharhinus amblyrhynchos*		12,23,40	17,24,29,30,33	35,36		8,32		COI Elasmo	120664
Carcharhiniformes	Carcharhinidae	*Carcharhinus melanopterus*		3,4,23,39,40,42	18,19,22,29,30,33	34,35	28	9,15,32	38	COI Elasmo	49427
Perciformes	Chaetodontidae	*Chaetodon unimaculatus*						5,31		COI Elasmo	293
Perciformes	Labridae	*Cheilinus chlorourus*			18,19	35		32		COI Elasmo	60
Perciformes	Labridae	*Chlorurus sordidus*		12	17			8,16,31		COI Elasmo	153
Perciformes	Blenniidae	*Cirripectes quagga*					28	5,6,7,8,15		COI Elasmo	265
Perciformes	Labridae	*Coris*			30					COI Elasmo	40
Tetraodontiformes	Diodontidae	*Diodon*						9		COI Elasmo	20

Perciformes	Apogonidae	*Fowleria variegata*		23					32		COI Elasmo	15150
Carcharhiniformes	Carcharhinidae	*Galeocerdo cuvier*				37					COI Elasmo	111
Perciformes	Gerridae	*Gerres longirostris*		3,4,12,42					6,7,9,15	38	COI Elasmo	1527
Perciformes	Microdesmidae	*Gunnellichthys monostigma*			29						COI Elasmo	8
Perciformes	Apogonidae	*Gymnapogon urospilotus*		23					6		COI Elasmo	6120
Anguilliformes	Muraenidae	*Gymnothorax pictus*							14,32		COI Elasmo	133
Myctophiformes	Myctophidae	*Lampanyctus*	NR		30						COI Elasmo	40
Littorinimorpha	Littorinidae	*Littoraria undulata*		4,42							COI Elasmo	105
Tetraodontiformes	Balistidae	*Melichthys indicus*							32		COI Elasmo	27
Myliobatiformes	Mobulidae	*Mobula*	NR						6,16		COI Elasmo	1628
Littorinimorpha	Cypraeidae	*Monetaria annulus*			29						COI Elasmo	65
Perciformes	Acanthuridae	*Naso hexacanthus*			30		26				COI Elasmo	338
Perciformes	Acanthuridae	*Naso unicornis*		11,12	30				5,7,8,9,31		COI Elasmo	222
Decapoda	Palaemonidae	*Palaemonella*	NR	23		34					COI Elasmo	7260
Perciformes	Haemulidae	*Plectorhinchus chaetodonoides*			20	37	28				COI Elasmo	1088
Perciformes	Lutjanidae	*Pristipomoides multidens*	NR						7		COI Elasmo	313
Tetraodontiformes	Balistidae	*Rhinecanthus rectangulus*			30				14		COI Elasmo	1438
Perciformes	Nemipteridae	*Scolopsis*		1,2,3,4,11,21,39,42	18,29,30				6,8,9,14,15,16,31,32		COI Elasmo	132507
Perciformes	Carangidae	*Scomberoides lysan*			18						COI Elasmo	28
Perciformes	Siganidae	*Siganus puellus*			18,19	35					COI Elasmo	151
Perciformes	Siganidae	*Siganus stellatus*			22						COI Elasmo	1357
Perciformes	Pomacentridae	*Stegastes fasciolatus*		11							COI Elasmo	5
Perciformes	Labridae	*Thalassoma hardwicke*							14		COI Elasmo	71
Perciformes	Gobiidae	*Valenciennea sexguttata*							31		COI Elasmo	10
Perciformes	Labridae	*Wetmorella nigropinnata*	NR			35					COI Elasmo	87
Aplysiida	Aplysiidae			1,3,11,13,23	22,24,30,33		25,26		9,10,15,16,31,32		18S Universal	17568
Aplysiida	Aplysiidae	*Aplysia*		21							18S Universal	238
Arcoida	Arcidae	*Arca*							6		18S Universal	5
Decapoda	Axiidae				33						18S Universal	6
Pholadida	Sportellidae	*Basterotia*	NR				28		10		18S Universal	162
Pleurobranchomorpha	Pleurobranchidae	*Berthella martensi*			17						18S Universal	282

Limoida	Limidae	*Ctenoides*			24					18S Universal	63
Nudibranchia	Dendrodoridida e	*Dendrodoris*					28	32		18S Universal	10
Nudibranchia	Discodoridídae				29			15		18S Universal	87
Cardiida	Cardiidae	*Fragum*						6		18S Universal	13
Holothuriida	Holothuriidae			1,2,3,4,12,13,42	17,18,20,30			31,32		18S Universal	1454
Ostreoida	Gryphaeidae	*Hyotissa*		40,41						18S Universal	12
Sacoglossa	Juliidae	*Julia*					25	9,10,15		18S Universal	113
Littorinimorpha	Strombidae	*Lambis lambis*			20					18S Universal	19
Limoida	Limidae	*Limaria*		3,4,12,40,42	17,18,19,29,30	35,37		32		18S Universal	357
Littorinimorpha	Littorinidae								38	18S Universal	676
Veneroida	Lucinidae				33					18S Universal	46
Ostreoida	Mytilidae			2,3						18S Universal	30
Ostreoida	Mytilidae	*Modiolus*		2,3,12,42			25,28	5,6,7,8,9,10,15, 16		18S Universal	1166
Cycloneritida	Neritidae			4						18S Universal	7
Ophiurida	Ophiodermatid ae	*Ophioderma*	NR	3,11,12,39,41,42	30,33		25	5,6,7,8,10,15,16 ,32		18S Universal	655
Ostreoida	Ostreidae				22			9		18S Universal	54
Sacoglossa	Oxynoidae			13	18,19					18S Universal	474
Sacoglossa	Oxynoidae	*Oxynoe*			17,18,19,20,33	35,36		16		18S Universal	7309
Cephalaspidea	Haminoeidae	*Phanerophthalmus*		1,11,12,13,40,42	18,19,33		26	7,8,10,15,32		18S Universal	11256
Pterioida	Pteriidae	*Pinctada*		1,3						18S Universal	53
Pterioida	Pinnidae	*Pinna*			24,30,33		28	32		18S Universal	1251
Pleurobranchomorp ha	Pleurobranchid ae			1						18S Universal	11
Acteonimorpha	Acteonidae	*Pupa nitidula*			18,19,20					18S Universal	62
Panpulmonata	Pyramidellidae			3,11	19,20			9,32		18S Universal	1971
Pectinoida	Spondylidae	*Spondylus*		40						18S Universal	5
Sacoglossa	Limapontiidae	*Stiliger*	NR		20					18S Universal	56
Littorinimorpha	Strombidae			13						18S Universal	3632
Apodida	Synaptidae			13,21,23,39	17,18,20,29,30,33		28	6,7,32		18S Universal	745
Veneroida	Tellinidae			4,39,40,41,42	29,30	34	25,27,28	15,32		18S Universal	9291
Veneroida	Veneridae			40,41	20					18S Universal	1966
Sacoglossa	Volvatellidae	*Volvatella*			19,22	36				18S Universal	141

Table S2.9 Sequencing statistics of new occurrence records.
For a putative new occurrence record to be evaluated below, we required that all global congeneric taxa (of the putative species in question) had been barcoded for the gene region of the respective assay; this removes the possibility that a new occurrence record is actually that of a related (not yet barcoded) species. Sequencing statistics are provided to assess whether new occurrence records may be the product of sequencing error and therefore a false positive detection. A low possibility of a false positive corresponds to the detection of the taxa in >1 site and >2 nucleotide differences between the taxa and its closest match in this dataset. A medium possibility corresponds to >1 site or >2 nucleotide differences; a high possibility corresponds to the detection of the taxa in only 1 site and 2 or less nucleotide differences to the closest match in the dataset.

Scientific name	No. unique sequences	No. sites detected	Total reads	Closest match in dataset	No. nucleotide differences	Possibility of false positive
Bony Fish & Elasmobranchs						
Bolinichthys	1	1	49	*Lampanyctus*	10	MEDIUM
Chrysiptera unimaculata	1	15	7584	*Chrysiptera glauca*	13	LOW
Ctenogobiops	3	7	10635	*Cheilinus fasciatus*	53	LOW
Cymolutes torquatus	2	2	2112	*Cymolutes praetextatus*	13	LOW
Gobiodon ater	1	1	258	*Gobiodon citrinus*	22	MEDIUM
Gobiodon citrinus	1	1	99	*Gobiodon ater*	22	MEDIUM
Gymnomuraena zebra	1	1	164	*Gymnothorax flavimarginatus*	43	MEDIUM
Gymnothorax richardsonii	4	4	2652	*Echidna polyzona*	35	LOW
Lampadena luminosa	1	1	739	Myctophidae	57	MEDIUM
Lampanyctus	1	1	40	*Bolinichthys*	10	MEDIUM
Oedalechilus labiosus	2	1	218	*Ellochelon vaigiensis*	10	MEDIUM
Pristipomoides multidens	1	1	313	*Halichoeres margaritaceus*	14	MEDIUM
Mobula	1	2	1628	Eukaryota	17	LOW
Schindleria praematura	3	4	2325	*Amblygobius*	51	LOW
Stomiiformes	3	1	252	*Salmo*	91	MEDIUM

Tylosurus acus	6	3	17868	*Platybelone argala*	37	LOW
Wetmorella nigropinnata	1	1	87	*Cheilinus undulatus*	11	MEDIUM
Crustaceans						
Chlorodiella nigra	2	2	444	*Chlorodiella cytherea*	27	LOW
Danielea noelensis	1	1	43	*Paraxanthias notatus*	39	MEDIUM
Etisus albus	1	2	26	*Pseudoliomera*	33	LOW
Etisus anaglyptus	1	1	365	*Etisus*	25	MEDIUM
Euphausiidae	1	1	83	Euphausiacea	70	MEDIUM
Galathea	5	4	1612	*Calcinus morgani*	68	LOW
Garthiella aberrans	1	1	1543	*Carpilius maculatus*	57	MEDIUM
Grapsus albolineatus	1	1	249	*Grapsus intermedius*	29	MEDIUM
Hippa	1	1	27	*Galathea*	63	MEDIUM
Liocarpilodes integerrimus	4	3	128	Xanthidae	26	LOW
Lybia leptochelis	1	1	165	*Cyclodius*	41	MEDIUM
Metalpheus	1	4	19783	*Alpheus*	55	LOW
Palaemonella	1	2	7260	*Decapoda*	25	LOW
Paraxanthias notatus	2	1	485	*Etisus frontalis*	35	MEDIUM
Penaeidae	10	2	51845	*Thalassograpsus harpax*	67	LOW
Plagusia speciosa	1	1	51	*Plagusia*	28	MEDIUM
Pullosquilla	1	2	5829	*Pseudosquilla ciliata*	37	LOW
Rugathanas	2	4	509	*Calappa hepatica*	72	LOW
Saron	1	5	599	Euphausiidae	64	LOW
Sulcodius deflexus	5	5	39130	*Pilodius areolatus*	38	LOW
Synalpheus charon	6	1	14	*Synalpheus tumidomanus*	52	MEDIUM

Synalpheus tumidomanus	2	8	21146	*Synalpheus charon*	52	LOW
Vellodius	1	1	1445	Xanthidae	26	MEDIUM
Molluscs						
Basterotia	1	2	162	Bivalvia	5	LOW
Stiliger	1	1	56	Gastropoda	11	MEDIUM
Echinoderms						
Ophioderma	3	17	655	Eukaryota	21	LOW

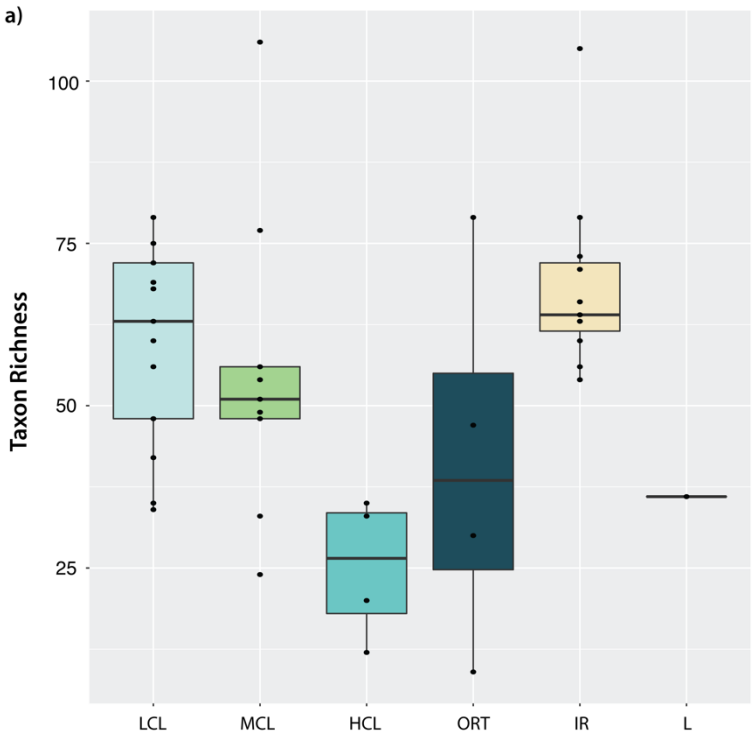


Figure S2.4 Overall taxonomic richness across marine habitat zones.
Marine habitat abbreviations are as follows: LCL, Low Complexity Lagoon; MCL, Medium Complexity Lagoon; HCL, High Complexity Lagoon; ORT, Outer Reef Terrace; IR, Intertidal Reef; L, Lagoonlet.

Table S2.10 ANOVA of Taxon richness between marine habitat zones.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Taxonomic Group		d.f	Sum of squares	Mean Sq	F value	*P*	
All Taxa	Habitat Zone	5	6989	1398	4.052	**0.005**	**
	Residuals	36	12419	345			
Fish & Elasmobranchs	Habitat Zone	5	3435	687.1	3.345	**0.013**	*
	Residuals	36	7394	205.4			
Crustaceans	Habitat Zone	5	1.629	0.3257	4.161	**0.004**	**
	Residuals	36	2.818	0.0783			
Molluscs	Habitat Zone	5	26.65	5.330	2.183	0.077	
	Residuals	36	87.92	2.442			
Echinoderms	Habitat Zone	5	1.492	0.29831	6.607	**0.000**	***
	Residual	36	1.625	0.04515			

Table S2.11 Tukey multiple comparisons of taxonomic richness between marine habitat zones across all taxonomic groups.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. ^ Indicates significant *P* value after Holm-Bonferroni correction to alpha values.

(I) Group	(J) Group	Mean Difference (I-J)	95% Confidence Interval		*P*		Holm-Bonferroni adjustment (α H-B)	
			Lower Bound	Upper Bound				
Intertidal Reef	High Complexity	43.545	10.919	76.171	**0.003**	**	0.00333333	^
Lagoonlet	High Complexity	11.000	-51.474	73.474	0.994		0.01666667	
	Intertidal Reef	-32.545	-90.909	25.818	0.554		0.005	
Low Complexity	High Complexity	34.461	2.511	66.411	**0.028**	*	0.00357143	
	Intertidal Reef	-9.083	-31.976	13.808	0.836		0.01	
	Lagoonlet	23.461	-34.526	81.449	0.825		0.00833333	
Medium Complexity	High Complexity	30.333	-3.245	63.912	0.096		0.00384615	
	Intertidal Reef	-13.212	-38.327	11.903	0.614		0.00555556	
	Lagoonlet	19.333	-39.568	78.234	0.918		0.0125	
	Low Complexity	-4.128	-28.358	20.102	0.995		0.025	
Outer Reef Terrace	High Complexity	16.250	-23.262	55.762	0.815		0.00714286	
	Intertidal Reef	-27.295	-59.921	5.330	0.146		0.00416667	
	Lagoonlet	5.250	-57.224	67.724	0.999		0.05	
	Low Complexity	-18.211	-50.161	13.738	0.531		0.00454545	
	Medium Complexity	-14.083	-47.662	19.495	0.803		0.00625	

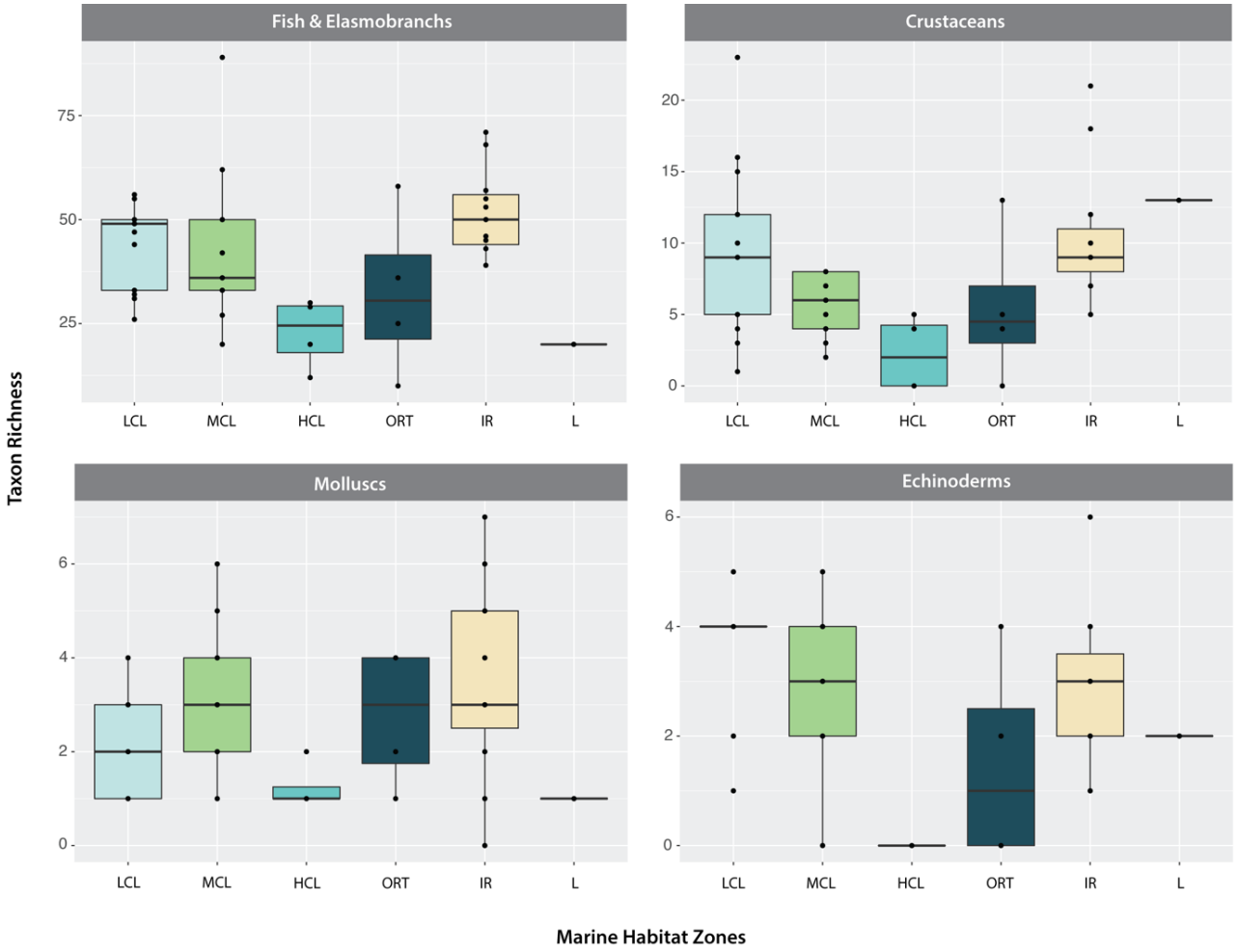


Figure S2.5 Taxonomic richness across the marine habitat zones by taxonomic group.

Table S2.12 Tukey multiple comparisons of bony fish and elasmobranch taxonomic richness.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. ^ Indicates significant *P* value after Holm-Bonferroni correction to alpha values.

(I) Group	(J) Group	Mean Difference (I-J)	95% Confidence Interval		*P*		Holm-Bonferroni adjustment (α H-B)
			Lower Bound	Upper Bound			
Intertidal Reef	High Complexity	28.704	3.529	53.880	**0.017**	*	0.00333333
Lagoonlet	High Complexity	-2.750	-50.957	45.457	0.999		0.05
	Intertidal Reef	-31.454	-76.489	13.580	0.309		0.00454545
Low Complexity	High Complexity	21.711	-2.942	46.365	0.111		0.00357143
	Intertidal Reef	-6.993	-24.657	10.671	0.838		0.00833333
	Lagoonlet	24.461	-20.284	69.207	0.575		0.005
Medium Complexity	High Complexity	21.138	-4.771	47.049	0.165		0.00384615
	Intertidal Reef	-7.565	-26.945	11.814	0.845		0.01
	Lagoonlet	23.888	-21.561	69.339	0.615		0.00555556
	Low Complexity	-0.572	-19.269	18.124	0.999		0.025
Outer Reef Terrace	High Complexity	9.500	-20.989	39.989	0.933		0.0125
	Intertidal Reef	-19.204	-44.380	5.970	0.222		0.00416667
	Lagoonlet	12.250	-35.957	60.457	0.971		0.01666667
	Low Complexity	-12.211	-36.865	12.442	0.672		0.00625
	Medium Complexity	-11.638	-37.549	14.271	0.754		0.00714286

Table S2.13 Tukey multiple comparisons of crustacean taxonomic richness.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. ^ Indicates significant *P* value after Holm-Bonferroni correction to alpha values.

(I) Group	(J) Group	Mean Difference (I-J)	95% Confidence Interval		*P*		Holm-Bonferroni adjustment (α H-B)	
			Lower Bound	Upper Bound				
Intertidal Reef	High Complexity	0.663	0.172	1.155	**0.003**	**	0.00333333	^
Lagoonlet	High Complexity	0.776	-0.164	1.717	0.156		0.00416667	
	Intertidal Reef	0.113	-0.765	0.992	0.998		0.05	
Low Complexity	High Complexity	0.555	0.074	1.036	**0.015**	*	0.00357143	
	Intertidal Reef	-0.108	-0.452	0.236	0.932		0.0125	
	Lagoonlet	-0.221	-1.095	0.652	0.972		0.025	
Medium Complexity	High Complexity	0.427	-0.078	0.932	0.139		0.00384615	
	Intertidal Reef	-0.236	-0.614	0.141	0.430		0.005	
	Lagoonlet	-0.349	-1.237	0.537	0.840		0.00833333	
	Low Complexity	-0.128	-0.493	0.236	0.894		0.01	
Outer Reef Terrace	High Complexity	0.286	-0.308	0.881	0.698		0.00714286	
	Intertidal Reef	-0.376	-0.868	0.114	0.217		0.00454545	
	Lagoonlet	-0.490	-1.431	0.450	0.624		0.00625	
	Low Complexity	-0.268	-0.750	0.212	0.552		0.00555556	
	Medium Complexity	-0.140	-0.646	0.365	0.958		0.01666667	

Table S2.14 Tukey multiple comparisons of echinoderm taxonomic richness.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. ^ Indicates significant *P* value after Holm-Bonferroni correction to alpha values.

(I) Group	(J) Group	Mean Difference (I-J)	95% Confidence Interval		*P*		Holm-Bonferroni adjustment (α H-B)	
			Lower Bound	Upper Bound				
Intertidal Reef	High Complexity	0.580	0.207	0.953	**0.000**	***	0.00333333	^
Lagoonlet	High Complexity	0.477	-0.237	1.191	0.357		0.005	
	Intertidal Reef	-0.103	-0.770	0.564	0.997		0.025	
Low Complexity	High Complexity	0.638	0.273	1.004	**0.000**	***	0.00357143	^
	Intertidal Reef	0.058	-0.203	0.320	0.983		0.01666667	
	Lagoonlet	0.161	-0.501	0.825	0.976		0.0125	
Medium Complexity	High Complexity	0.492	0.108	0.876	**0.005**	**	0.00384615	
	Intertidal Reef	-0.087	-0.375	0.199	0.938		0.00833333	
	Lagoonlet	0.015	-0.658	0.689	0.999		0.05	
	Low Complexity	-0.146	-0.423	0.130	0.609		0.00625	
Outer Reef Terrace	High Complexity	0.294	-0.158	0.746	0.386		0.00555556	
	Intertidal Reef	-0.286	-0.659	0.086	0.217		0.00454545	
	Lagoonlet	-0.183	-0.897	0.531	0.970		0.01	
	Low Complexity	-0.344	-0.710	0.020	0.073		0.00416667	
	Medium Complexity	-0.198	-0.582	0.185	0.633		0.00714286	

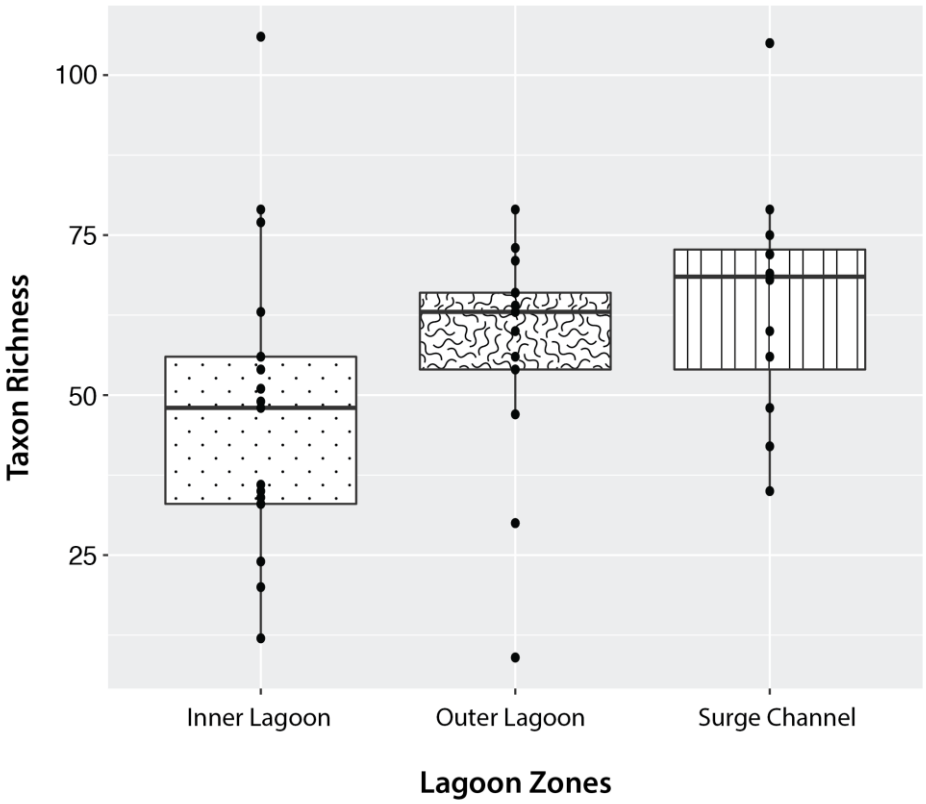


Figure S2.6 Overall taxonomic richness across lagoon zones.

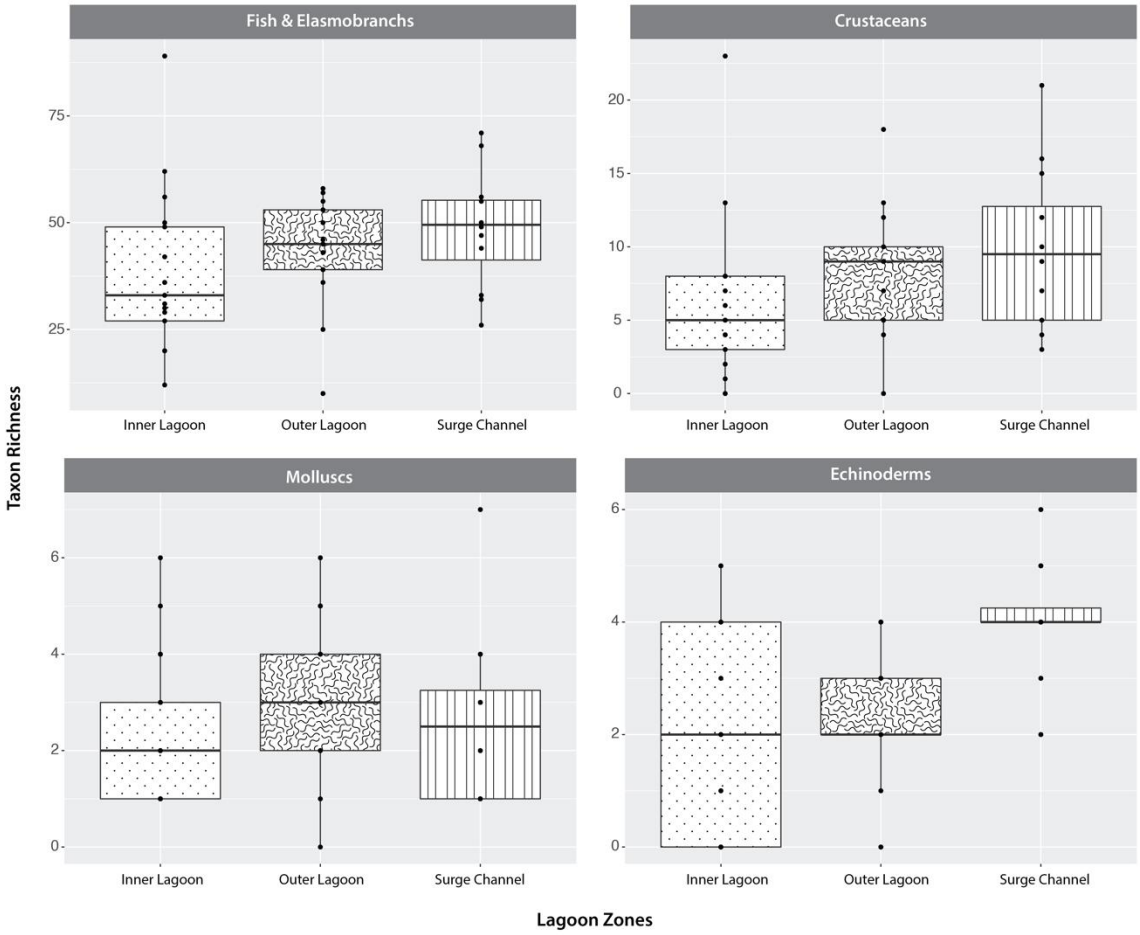


Figure S2.7 Taxonomic richness across the lagoon zones by taxonomic group.

Table S2.15 ANOVA of Taxon richness between lagoon zones.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Taxonomic Group		d.f	Sum of squares	Mean Sq	F value	*P*	
All Taxa	Lagoon Zone	2	2164	1081.9	2.447	0.099	
	Residuals	39	17244	442.2			
Bony Fish & Elasmobranchs	Lagoon Zone	2	801	400.7	1.558	0.223	
	Residuals	39	10028	257.1			
Crustaceans	Lagoon Zone	2	114.3	57.13	2.094	0.137	
	Residuals	39	1064.1	27.29			
Molluscs	Lagoon Zone	2	5.71	2.857	1.024	0.369	
	Residuals	39	108.86	2.791			
Echinoderms	Lagoon Zone	2	34.71	17.353	7.468	**0.001**	**
	Residual	39	90.63	2.324			

Table S2.16 Tukey multiple comparisons of echinoderms.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. ^ Indicates significant *P* value after Holm-Bonferroni correction to alpha values.

(I) Group	(J) Group	Mean Difference (I-J)	95% Confidence Interval		*P*		Holm-Bonferroni adjustment (α H-B)	
			Lower Bound	Upper Bound				
Outer Lagoon	Inner Lagoon	0.3665158	-1.0018219	1.734854	0.792		0.00384615	
Surge Channel	Inner Lagoon	2.1421569	0.7418838	3.542430	**0.001**	**	0.00333333	^
	Outer Lagoon	1.7756410	0.2888959	3.262386	**0.016**	*	0.00357143	

Table S2.17 Distance based linear model (DistLM) analysis of bony fish and elasmobranch community variation.
These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Adj R^2	Pseudo-*F*	Proportion	Cumulative Proportion	*P*	
Marginal	Latitude		2.377	0.056		**0.002**	**
	Longitude		2.036	0.048		**0.006**	**
	Depth		2.791	0.065		**0.000**	***
	Lagoon		4.025	0.171		**0.000**	***
	Habitat		2.885	0.286		**0.000**	***
Sequential (Step-wise)	Habitat	0.187	2.885	0.286	0.286	**0.000**	***
	Lagoon	0.210	1.531	0.059	0.345	**0.007**	**
	Longitude	0.233	1.995	0.037	0.382	**0.000**	***
	Latitude	0.247	1.640	0.030	0.412	**0.009**	**
	Depth	0.258	1.476	0.027	0.439	**0.047**	*

Table S2.18 Distance based linear model (DistLM) analysis of crustacean community variation.
These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Adj R^2	Pseudo-*F*	Proportion	Cumulative Proportion	*P*	
Marginal	Latitude		1.107	0.029		0.306	
	Longitude		1.443	0.038		0.077	
	Depth		0.872	0.023		0.641	
	Lagoon		2.782	0.134		**0.000**	***
	Habitat		1.817	0.216		**0.000**	***
Sequential (Step-wise)	Habitat	0.097	1.817	0.216	0.216	**0.000**	***
	Lagoon	0.119	1.407	0.065	0.281	**0.042**	*
	Longitude	0.121	1.073	0.024	0.306	0.358	

Table S2.19 Distance based linear model (DistLM) analysis of mollusc community variation.
These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Adj R^2	Pseudo-*F*	Proportion	Cumulative Proportion	*P*	
Marginal	Latitude		2.373	0.057		**0.006**	**
	Longitude		2.061	0.050		**0.019**	*
	Depth		1.140	0.028		0.296	
	Lagoon		2.412	0.113		**0.000**	***
	Habitat		1.661	0.192		**0.002**	**
Sequential (Step-wise)	Habitat	0.076	1.661	0.192	0.192	**0.001**	***
	Latitude	0.111	2.408	0.053	0.245	**0.006**	**
	Longitude	0.139	2.050	0.044	0.289	**0.020**	*
	Lagoon	0.185	1.945	0.079	0.369	**0.008**	**

Table S2.20 Distance based linear model (DistLM) analysis of echinoderm community variation.
These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Adj R^2	Pseudo-*F*	Proportion	Cumulative Proportion	*P*	
Marginal	Latitude		2.427	0.070		0.034	*
	Longitude		1.688	0.050		0.143	
	Depth		1.882	0.056		0.088	
	Lagoon		3.453	0.182		**0.000**	***
	Habitat		1.226	0.145		0.228	
Sequential (Step-wise)	Lagoon	0.129	3.453	0.182	0.182	**0.000**	***
	Latitude	0.176	2.754	0.069	0.251	**0.020**	*
	Depth	0.189	1.478	0.036	0.287	0.210	
	Longitude	0.200	1.418	0.034	0.322	0.232	
	Habitat	0.235	1.313	0.122	0.443	0.186	

S2.9.6 Similarity percentage and indicator species analyses

Where the predictor variables of marine habitat and lagoon zones were significant in the DistLM analyses, similarity percentage analyses (SIMPER) were used to identify prominent taxa contributing most to the similarity between sites within each zone. This identified the fish taxa *Acanthurus triostegus* and *Hipposcarus harid* as contributing the highest to similarity within the outer lagoon, *Acanthurus triostegus* and *Thalassoma* in the surge channel, and *Epinephelus* and *Zebrasoma desjardinii* in the inner lagoon (Table S2.21). In examining similarity within each of the marine habitat zones (Table S2.22), the following fish were dominant: *Scarus psittacus* and *Acanthurus triostegus* in the low complexity lagoon; *Epinephelus* and *Zebrasoma desjardinii* in the medium complexity lagoon; *Chlorurus* and *Epinephelus* in the high complexity lagoon; *Acanthurus triostegus* and Caesionidae in the outer reef terrace; and *Acanthurus triostegus* and *Chlorurus* in the intertidal reef. For crustacean taxa, *Thalamita* and *Daira perlata* contributed the highest to similarity within the outer lagoon and surge channel, whilst *Chlorodiella barbata* contributed highest to similarity in the inner lagoon (Table S2.23). In examining the marine habitat zones, *Thalamita* and *Chlorodiella barbata* were predominate in the low and medium complexity lagoon; *Cyclodius* and in particular, *Cyclodius ungulates*, in the high complexity lagoon; *Thalamita* and *Galathea* in the outer reef terrace; and *Daira perlata* and *Thalamita* in the intertidal reef (Table S2.24). Finally, the mollusc and echinoderm taxa of *Modiolus* and *Ophioderma* was predominate in the outer lagoon, *Aplysiidae* and *Amphipholis squamata* in the surge channel, and *Limaria* and Synaptidae in the inner lagoon (Tables S2.25-S2.28).

Indicator species analyses revealed taxa uniquely associated with each marine habitat and lagoon group/s. Likewise to the SIMPER analyses, only predictor variables deemed significant in the DistLM analyses were included in the indicator species analyses. In examining the marine habitat zones, *Chromis viridis* was identified as a highly significant ($P < 0.05$) fish indicator species in the medium complexity lagoon zone; *Acanthurus* (including *A. lineatus*) on the intertidal reef; and *Gymnothorax richardsonii, Trachinotus blochii, Ellochelon vaigiensis,* and *Lutjanus* (unidentified species) in the lagoonlet (Table S2.29). In addition, the following crustacean taxa were identified as indicator species in the lagoonlet: *Ocypode*, Ocypodidae, *Thalamita*

crenata, Corallianassa, Pullosquilla, Neocallichirus, Cardisoma carnifex and *Etisus frontalis* ($P < 0.05$). In examining the lagoon zones, the fish taxa of *Acanthurus lineatus* and *Chromis nigrura* were identified as highly significant ($P < 0.01$) indicator species in the outer lagoon, *Chrysiptera* in the surge channel, and *Zebrasoma desjardinii*, *Stegastes nigricans* and *Chromis viridis* in the inner lagoon (Table S2.28). In addition, the crustacean and mollusc genera of *Synalpheus* and *Modiolus* respectively, were identified as indicator species ($P < 0.05$) in the outer lagoon, the crustacean family of Portunidae in the surge channel, and the mollusc genus of *Oxynoe* in the inner lagoon.

Table S2.21 Similarity percentage analysis (SIMPER) of bony fish and elasmobranch taxa across the lagoon zones.

Lagoon Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim/SD	Contrib. %	Cum. %
Outer Lagoon	43.99	*Acanthurus triostegus*	1.00	2.47	3.58	5.62	5.62
		Hipposcarus harid	0.92	2.01	1.96	4.56	10.18
		Chlorurus	0.92	1.90	2.17	4.32	14.50
		Halichoeres	0.92	1.90	2.17	4.32	18.82
		Thalassoma	0.92	1.90	2.17	4.32	23.13
		Cirripectes castaneus	0.85	1.56	1.48	3.55	26.69
		Rhinecanthus	0.85	1.55	1.48	3.53	30.21
		Stethojulis	0.85	1.51	1.51	3.42	33.64
		Kyphosus	0.77	1.31	1.13	2.98	36.61
		Labridae	0.77	1.26	1.15	2.86	39.47
		Entomacrodus	0.77	1.23	1.14	2.79	42.26
		Melichthys	0.77	1.23	1.14	2.79	45.05
		Salariinae	0.77	1.23	1.14	2.79	47.84
		Scarus	0.69	1.23	0.85	2.79	50.63
		Exocoetidae	0.69	1.22	0.85	2.78	53.41
		Scarus psittacus	0.69	1.04	0.90	2.35	55.76
		Ctenochaetus	0.69	1.02	0.90	2.33	58.09
		Bolbometopon muricatum	0.69	0.98	0.91	2.23	60.32
		Caesionidae	0.54	0.78	0.56	1.78	62.10
		Acanthurus	0.62	0.76	0.74	1.72	63.83
		Epinephelus hexagonatus	0.62	0.76	0.74	1.72	65.55
		Pomacentridae	0.62	0.75	0.74	1.71	67.26
		Blenniella chrysospilos	0.62	0.74	0.74	1.67	68.93
		Plectroglyphidodon imparipennis	0.62	0.72	0.74	1.64	70.57
Surge Channel	46.04	*Acanthurus triostegus*	1.00	2.14	5.06	4.65	4.65
		Thalassoma	1.00	2.14	5.06	4.65	9.31
		Scarus psittacus	0.92	1.79	1.98	3.90	13.20
		Halichoeres	0.92	1.78	1.99	3.86	17.06
		Epinephelus	0.83	1.50	1.36	3.25	20.32
		Halichoeres trimaculatus	0.83	1.50	1.36	3.25	23.57
		Lutjanus fulvus	0.83	1.43	1.38	3.11	26.68
		Chlorurus	0.83	1.39	1.39	3.01	29.69
		Hipposcarus harid	0.83	1.39	1.39	3.01	32.70
		Blenniella periophthalmus	0.75	1.24	1.05	2.70	35.40
		Cirripectes castaneus	0.75	1.17	1.05	2.54	37.94
		Stethojulis	0.75	1.16	1.06	2.51	40.45
		Lethrinus	0.75	1.10	1.07	2.39	42.84
		Spratelloides delicatulus	0.67	0.98	0.82	2.14	44.98
		Scarus	0.67	0.93	0.81	2.01	46.99
		Salariinae	0.67	0.89	0.83	1.94	48.92
		Arothron hispidus	0.67	0.86	0.83	1.88	50.80
		Rhinecanthus	0.67	0.85	0.83	1.85	52.64
		Exocoetidae	0.67	0.83	0.83	1.81	54.46
		Epinephelus hexagonatus	0.67	0.82	0.84	1.77	56.23
		Melichthys	0.67	0.78	0.84	1.70	57.93
		Gnatholepis anjerensis	0.58	0.73	0.66	1.58	59.51
		Spratelloides	0.58	0.71	0.66	1.54	61.05
		Lethrinus xanthochilus	0.58	0.66	0.65	1.44	62.49
		Chanos chanos	0.58	0.65	0.66	1.41	63.90
		Labridae	0.58	0.64	0.67	1.40	65.30
		Mullidae	0.58	0.63	0.67	1.38	66.68
		Mugilidae	0.58	0.61	0.67	1.34	68.02
		Blenniella chrysospilos	0.58	0.60	0.67	1.30	69.32
		Kyphosus	0.58	0.58	0.67	1.27	70.58

Inner Lagoon	38.16	*Epinephelus*	0.94	2.52	1.95	6.61	6.61
		Zebrasoma desjardinii	0.82	2.09	1.26	5.47	12.09
		Scarus	0.82	1.88	1.23	4.92	17.01
		Chlorurus	0.82	1.82	1.22	4.77	21.78
		Hipposcarus harid	0.76	1.48	1.04	3.89	25.67
		Lethrinus	0.76	1.43	1.07	3.75	29.42
		Mullidae	0.71	1.22	0.89	3.19	32.60
		Stegastes	0.71	1.15	0.92	3.01	35.61
		Stegastes nigricans	0.65	1.14	0.73	3.00	38.61
		Spratelloides delicatulus	0.65	1.11	0.78	2.92	41.53
		Balistidae	0.65	1.00	0.76	2.61	44.14
		Vanderhorstia ornatissima	0.59	0.96	0.66	2.52	46.66
		Ctenochaetus	0.59	0.95	0.64	2.48	49.14
		Cheilinus	0.59	0.89	0.67	2.34	51.48
		Scarus psittacus	0.59	0.80	0.66	2.09	53.57
		Bolbometopon muricatum	0.53	0.75	0.58	1.97	55.54
		Stethojulis strigiventer	0.53	0.72	0.58	1.89	57.44
		Chanos chanos	0.53	0.67	0.57	1.74	59.18
		Stethojulis	0.53	0.63	0.56	2.65	60.84
		Spratelloides	0.53	0.61	0.57	1.60	62.43
		Halichoeres trimaculatus	0.53	0.60	0.58	1.56	63.99
		Pomacentrus pavo	0.47	0.58	0.45	1.52	65.51
		Chromis viridis	0.53	0.58	0.58	1.52	67.03
		Acanthurinae	0.47	0.56	0.48	1.47	68.50
		Lutjanus bohar	0.47	0.53	0.48	1.38	69.88
		Chaetodon	0.47	0.51	0.50	1.33	71.21

Table S2.22 Similarity percentage analysis (SIMPER) of bony fish and elasmobranch taxa across the marine habitat zones.
The lagoonlet was exempt from this analysis as it had less than 2 sites in the zone.

Marine Habitat Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim /SD	Contrib. %	Cum. %
Low Complexity Lagoon	44.88	*Scarus psittacus*	0.92	1.97	2.13	4.40	4.40
		Acanthurus triostegus	0.92	1.90	2.16	4.23	8.63
		Epinephelus	0.85	1.69	1.47	3.76	12.39
		Halichoeres trimaculatus	0.85	1.69	1.47	3.76	16.15
		Thalassoma	0.85	1.62	1.47	3.60	19.75
		Stethojulis	0.85	1.58	1.48	3.51	23.26
		Lethrinus	0.85	1.51	1.50	3.37	26.63
		Halichoeres	0.77	1.32	1.12	2.95	29.58
		Lutjanus fulvus	0.77	1.28	1.13	2.85	32.43
		Chlorurus	0.77	1.24	1.14	2.77	35.20
		Hipposcarus harid	0.77	1.24	1.14	2.77	37.97
		Gnatholepis anjerensis	0.69	1.11	0.90	2.47	40.44
		Blenniella periophthalmus	0.69	1.09	0.90	2.42	42.86
		Spratelloides delicatulus	0.69	1.08	0.90	2.41	45.27
		Cirripectes castaneus	0.69	1.03	0.90	2.28	47.56
		Scarus	0.62	0.89	0.72	1.98	49.54
		Vanderhorstia ornatissima	0.62	0.87	0.73	1.93	51.47
		Chanos chanos	0.62	0.83	0.73	1.86	53.33
		Chrysiptera unimaculata	0.62	0.83	0.73	1.86	55.19
		Stethojulis strigiventer	0.62	0.79	0.73	1.76	56.95
		Salariinae	0.62	0.78	0.73	1.75	58.70
		Stegastes albifasciatus	0.62	0.78	0.73	1.73	60.43
		Rhinecanthus	0.62	0.77	0.73	1.72	62.15
		Mullidae	0.62	0.74	0.74	1.64	63.80
		Vanderhorstia	0.54	0.66	0.59	1.47	65.27
		Stegastes	0.54	0.64	0.59	1.42	66.69
		Spratelloides	0.54	0.63	0.59	1.41	68.10
		Arothron hispidus	0.54	0.61	0.60	1.35	69.45
		Novaculichthys taeniourus	0.54	0.60	0.60	1.33	70.77
Medium Complexity Lagoon	41.47	*Epinephelus*	1.00	2.48	3.42	5.98	5.98
		Zebrasoma desjardinii	0.89	2.13	1.67	5.13	11.11
		Mullidae	0.89	1.83	1.58	4.41	15.53
		Chlorurus	0.89	1.77	1.63	4.27	19.80
		Hipposcarus harid	0.89	1.77	1.63	4.27	24.07
		Lethrinus	0.89	1.77	1.63	4.27	28.34
		Scarus	0.89	1.77	1.63	4.27	32.62
		Stegastes	0.89	1.77	1.63	4.27	36.89
		Spratelloides	0.78	1.29	1.08	3.11	40.00
		Chromis viridis	0.78	1.24	1.10	2.99	42.98
		Bolbometopon muricatum	0.67	1.13	0.82	2.73	45.72
		Cheilinus	0.67	1.08	0.78	2.59	48.31
		Spratelloides delicatulus	0.67	1.03	0.81	2.49	50.80
		Halichoeres trimaculatus	0.67	0.93	0.80	2.24	53.03

		Stegastes nigricans	0.67	0.91	0.78	2.19
		Cheilinus undulatus	0.56	0.85	0.60	2.05
		Balistidae	0.67	0.85	0.80	2.04
		Amblygobius	0.56	0.75	0.60	1.81
		Stethojulis strigiventer	0.56	0.75	0.60	1.81
		Lutjanus bohar	0.56	0.68	0.57	1.63
		Chanos chanos	0.56	0.64	0.58	1.55
		Acanthurinae	0.56	0.61	0.58	1.48
		Lutjanus fulvus	0.56	0.59	0.59	1.42
		Scarus psittacus	0.56	0.59	0.57	1.42
High Complexity Lagoon	46.12	*Chlorurus*	1.00	4.56	4.60	9.89
		Ctenochaetus	1.00	4.56	4.60	9.89
		Epinephelus	1.00	4.56	4.60	9.89
		Scarus	1.00	4.56	4.60	9.89
		Stegastes nigricans	1.00	4.56	4.60	9.89
		Zebrasoma desjardinii	1.00	4.56	4.60	9.89
		Pomacentrus pavo	0.75	2.50	0.88	5.42
		Hipposcarus harid	0.75	2.17	0.89	4.71
		Acanthurinae	0.75	1.91	0.91	4.15
Outer Reef Terrace	35.32	*Acanthurus triostegus*	1.00	3.47	2.58	9.82
		Caesionidae	1.00	3.47	2.58	9.82
		Exocoetidae	1.00	3.47	2.58	9.82
		Scarus	1.00	3.47	2.58	9.82
		Hipposcarus harid	0.75	1.57	0.84	4.44
		Cheilinus	0.75	1.30	0.88	3.69
		Chlorurus	0.75	1.30	0.88	3.69
		Chromis nigrura	0.75	1.30	0.88	3.69
		Cirripectes	0.75	1.30	0.88	3.69
		Cirripectes castaneus	0.75	1.30	0.88	3.69
		Ctenochaetus	0.75	1.30	0.88	3.69
		Halichoeres	0.75	1.30	0.88	3.69
		Kyphosus	0.75	1.30	0.88	3.69
Intertidal Reef	51.39	*Acanthurus triostegus*	1.00	1.98	7.51	3.85
		Chlorurus	1.00	1.98	7.51	3.85
		Halichoeres	1.00	1.98	7.51	3.85
		Hipposcarus harid	1.00	1.98	7.51	3.85
		Rhinecanthus	1.00	1.98	7.51	3.85
		Thalassoma	1.00	1.98	7.51	3.85
		Labridae	0.91	1.67	2.03	3.26
		Stethojulis	0.91	1.67	2.03	3.26
		Acanthurus	0.91	1.60	2.00	3.11
		Entomacrodus	0.91	1.60	2.00	3.11
		Epinephelus hexagonatus	0.91	1.60	2.00	3.11
		Melichthys	0.91	1.60	2.00	3.11
		Salariinae	0.82	1.33	1.34	2.58
		Cirripectes castaneus	0.82	1.31	1.34	2.54
		Kyphosus	0.82	1.27	1.33	2.48
		Blenniella chrysospilos	0.82	1.26	1.33	2.46
		Bolbometopon muricatum	0.73	1.03	0.99	2.00
		Plectroglyphidodon imparipennis	0.73	1.00	1.00	1.95
		Mugilidae	0.73	0.98	1.00	1.91
		Scarus psittacus	0.73	0.96	0.99	1.87
		Pomacentridae	0.73	0.96	0.99	1.87
		Ctenochaetus	0.73	0.95	0.99	1.85
		Lethrinus xanthochilus	0.73	0.94	0.99	1.83
		Epinephelus	0.73	0.93	1.00	1.80
		Acanthurus lineatus	0.64	0.82	0.77	1.59
		Plectroglyphidodon phoenixensis	0.64	0.78	0.77	1.53
		Lethrinus xanthochilus	0.73	0.94	0.99	1.83

Table S2.23 Similarity percentage analysis (SIMPER) of crustacean taxa across the lagoon zones.

Lagoon Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim/SD	Contrib. %	Cum. %
Outer Lagoon	30.30	*Thalamita*	0.92	9.36	1.84	30.87	30.87
		Daira perlata	0.83	8.05	1.36	26.57	57.44
		Pilodius scabriculus	0.58	3.54	0.64	11.68	69.12
		Pilodius areolatus	0.42	1.33	0.41	4.37	73.50
Surge Channel	34.63	*Daira perlata*	0.92	9.57	1.49	27.63	27.63
		Thalamita	0.92	9.57	1.49	27.63	55.26
		Chlorodiella barbata	0.67	4.72	0.71	13.62	68.88
		Cyclodius	0.50	2.04	0.51	5.88	74.77
Inner Lagoon	28.95	*Chlorodiella barbata*	0.87	12.71	1.22	43.89	43.89
		Thalamita	0.67	6.38	0.76	22.05	65.94
		Cyclodius ungulatus	0.40	2.42	0.36	8.38	74.32

Table S2.24 Similarity percentage analysis (SIMPER) of crustacean taxa across the marine habitat zones.
The lagoonlet was exempt from this analysis as it had less than 2 sites in the zone.

Marine Habitat Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim/SD	Contrib. %	Cum. %
Low Complexity Lagoon	33.02	*Thalamita*	0.92	10.16	1.52	30.75	30.75
		Chlorodiella barbata	0.77	8.33	0.81	25.24	55.99
		Daira perlata	0.77	7.23	0.98	21.88	77.87
Medium Complexity Lagoon	32.28	*Chlorodiella barbata*	0.89	13.26	1.67	41.09	41.09
		Thalamita	0.67	7.27	0.80	22.53	63.61
		Xanthidae	0.56	4.73	0.60	14.65	78.27
High Complexity Lagoon	14.81	*Cyclodius*	0.67	7.41	0.58	50.00	50.00
		Cyclodius ungulatus	0.67	7.41	0.58	50.00	100.0
Outer Reef Terrace	22.66	*Thalamita*	1.00	15.03	2.41	66.35	66.35
		Galathea	0.67	3.92	0.58	17.31	83.65
Intertidal Reef	30.60	*Daira perlata*	0.82	6.93	1.26	22.65	22.65
		Thalamita	0.82	6.02	1.27	19.68	42.33
		Pilodius scabriculus	0.73	4.89	0.93	15.99	58.31
		Pilodius areolatus	0.55	2.38	0.58	7.78	66.09
		Synalpheus tumidomanus	0.45	2.32	0.46	7.58	73.67

Table S2.25 Similarity percentage analysis (SIMPER) of mollusc taxa across the lagoon zones.

Lagoon Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim/SD	Contrib. %	Cum. %
Outer Lagoon	35.78	*Modiolus*	0.83	18.18	1.36	50.85	50.85
		Aplysiidae	0.50	5.71	0.52	15.97	66.82
		Phanerophthalmus	0.42	3.99	0.41	11.14	77.96
Surge Channel	21.18	Aplysiidae	0.42	7.74	0.37	36.57	36.57
		Tellinidae	0.42	4.75	0.39	22.44	59.01
		Modiolus	0.33	3.49	0.30	16.49	75.45
Inner Lagoon	10.32	*Limaria*	0.29	3.57	0.25	34.54	34.54
		Aplysiidae	0.29	2.47	0.27	23.93	58.48
		Tellinidae	0.24	1.75	0.20	16.97	75.45

Table S2.26 Similarity percentage analysis (SIMPER) of mollusc taxa across the marine habitat zones.

Lagoon Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim/SD	Contrib. %	Cum. %
Low Complexity Lagoon	46.39	*Amphipholis squamata*	0.77	14.37	1.13	30.97	30.97
		Synaptidae	0.46	7.03	0.42	15.15	46.12
		Holothuriidae	0.54	6.31	0.60	13.60	59.72
		Ophiocomella	0.54	6.31	0.60	13.60	73.33
Intertidal Reef	49.66	*Ophiocomella*	0.73	17.09	0.92	34.42	34.42
		Ophioderma	0.73	15.94	0.96	32.09	66.50
		Amphipholis squamata	0.55	9.38	0.59	18.89	85.40
Medium Complexity Lagoon	49.09	Synaptidae	0.86	20.05	1.47	40.84	40.84
		Ophiocomella	0.57	8.99	0.61	18.32	59.15
		Amphipholis squamata	0.57	8.33	0.61	16.98	76.13
Outer Reef Terrace		*Ophiocoma*	1.00	33.33	-	100	100

Table S2.27 Similarity percentage analysis (SIMPER) of echinoderm taxa across the lagoon zones.

Lagoon Zone	Avg. Sim % (Zone)	Taxa	Av. Abund	Av. Sim	Sim/SD	Contrib.%	Cum. %
Outer Lagoon	52.45	*Ophioderma*	0.73	18.30	0.98	34.89	34.89
		Ophiocomella	0.64	13.57	0.73	25.87	60.76
		Amphipholis squamata	0.64	12.54	0.76	23.91	84.67
Surge Channel	56.24	*Amphipholis squamata*	0.75	13.92	1.06	24.75	24.75
		Holothuriidae	0.67	10.14	0.84	18.03	42.79
		Ophiocomella	0.67	10.14	0.84	18.03	60.82
		Ophiocoma	0.67	9.86	0.84	17.53	78.35
Inner Lagoon	46.70	Synaptidae	0.82	25.69	1.11	55.01	55.01
		Amphipholis squamata	0.55	8.16	0.59	17.47	72.48

Table S2.28 Indicator species analyses of lagoon zones.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Lagoon Zone	Taxonomic group	Taxa	A	B	Stat	*P*	
Outer Lagoon	Fish & Elasmobranchs	*Acanthurus lineatus*	0.866	0.538	0.683	**0.002**	**
		Chromis nigrura	0.866	0.538	0.683	**0.001**	***
		Anampses	0.786	0.307	0.492	**0.048**	*
		Aporops bilinearis	1.000	0.230	0.480	**0.044**	*
	Crustaceans	*Synalpheus*	1.000	0.250	0.500	**0.050**	*
	Molluscs	*Modiolus*	0.714	0.833	0.772	**0.001**	***
		Cytheroidea	1.000	0.333	0.577	**0.015**	*
		Julia	1.000	0.333	0.577	**0.008**	**
Surge Channel	Fish & Elasmobranchs	*Chrysiptera*	0.894	0.500	0.669	**0.003**	**
		Enchelynassa	0.730	0.416	0.552	**0.015**	*
		Istiblennius edentulus	0.850	0.333	0.532	**0.024**	*
		Glyptoparus delicatulus	1.000	0.250	0.500	**0.020**	*
	Crustaceans	Portunidae	0.862	0.416	0.599	**0.013**	*
Inner Lagoon	Fish & Elasmobranchs	*Zebrasoma desjardinii*	0.908	0.823	0.865	**0.001**	***
		Stegastes nigricans	0.893	0.647	0.760	**0.001**	***
		Chromis viridis	0.873	0.529	0.680	**0.002**	**
		Zebrasoma	1.000	0.411	0.642	**0.001**	***
		Gymnothorax undulatus	0.842	0.411	0.642	**0.012**	*
		Cheilinus fasciatus	1.000	0.294	0.542	**0.012**	*
		Plectropomus areolatus	1.000	0.294	0.542	**0.004**	**
		Gobiinae	1.000	0.235	0.485	**0.030**	*
		Ptereleotris microlepis	1.000	0.235	0.485	**0.033**	*
		Schindleria praematura	1.000	0.235	0.485	**0.031**	*
		Siganus	1.000	0.235	0.485	**0.024**	*
	Molluscs	*Oxynoe*	1.000	0.235	0.485	**0.029**	*
Outer Lagoon & Surge Channel	Fish & Elasmobranchs	*Halichoeres*	0.912	0.920	0.916	**0.001**	***
		Thalassoma	0.867	0.960	0.912	**0.001**	***
		Acanthurus triostegus	0.829	1.000	0.911	**0.001**	***
		Melichthys	0.960	0.720	0.832	**0.001**	***
		Salariinae	0.960	0.720	0.832	**0.001**	***
		Cirripectes castaneus	0.818	0.800	0.809	**0.007**	**
		Entomacrodus	0.958	0.680	0.807	**0.001**	***
		Rhinecanthus	0.837	0.760	0.798	**0.008**	**
		Exocoetidae	0.920	0.680	0.791	**0.002**	**
		Labridae	0.920	0.680	0.791	**0.001**	***
		Ephinephelus hexagonatus	0.956	0.640	0.782	**0.002**	**
		Blenniella chrysospilos	0.953	0.600	0.756	**0.004**	**
		Acanthurus	1.000	0.520	0.721	**0.001**	***
		Sargocentron punctatissimum	1.000	0.520	0.721	**0.003**	**
		Blenniella periophthalmus	0.906	0.560	0.712	**0.003**	**
		Pomacentridae	0.897	0.520	0.683	**0.019**	*
		Melichthys niger	0.942	0.480	0.673	**0.007**	**
		Plectroglyphidodon imparipennis	0.941	0.480	0.672	**0.008**	**
		Plectroglyphidodon phoenixensis	0.930	0.400	0.610	**0.044**	*
	Crustaceans	*Daira perlata*	1.000	0.875	0.935	**0.001**	***
		Pilodius scabriculus	0.890	0.541	0.694	**0.016**	*
	Echinoderms	*Ophioderma*	0.871	0.608	0.728	**0.025**	*
Outer Lagoon & Inner Lagoon	Fish & Elasmobranchs	*Bolbometopon muricatum*	0.880	0.600	0.727	**0.027**	*

Surge Channel & Inner Lagoon	Fish & Elasmobranchs	*Lethrinus*	0.867	0.758	0.811	**0.001**	***
		Vanderhorstia ornatissima	1.000	0.551	0.743	**0.002**	**
		Stegastes	0.935	0.586	0.741	**0.004**	**
		Gnatholepis anierensis	1.000	0.517	0.719	**0.003**	**
		Stethojulis strigiventer	1.000	0.517	0.719	**0.004**	**
		Chanos chanos	0.878	0.551	0.696	**0.024**	*
		Chrysiptera unimaculata	0.926	0.482	0.669	**0.018**	*
		Amblygobius	1.000	0.379	0.616	**0.016**	*
		Cymolutes praetextatus	1.000	0.344	0.587	**0.026**	*
	Crustaceans	*Chlorodiella barbata*	0.902	0.777	0.838	**0.001**	***
	Echinoderms	*Holothuriidae*	1.000	0.565	0.752	**0.006**	**

Table S2.29 Indicator species analyses of marine habitat zones.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. Echinoderms were not included in these analyses, as the DistLM analyses (Tables S2.11-S2.12) revealed that the marine habitat zones were not a significant predictor variable in explaining echinoderm community variation.

Marine Habitat Zone	Taxonomic group	Taxa	A	B	Stat	*P*	
Medium Complexity	Fish & Elasmobranchs	*Chromis viridis*	0.760	0.777	0.769	**0.006**	**
Intertidal Reef	Fish & Elasmobranchs	*Acanthurus*	0.797	0.909	0.851	**0.003**	**
		Acanthurus lineatus	0.892	0.636	0.753	**0.036**	*
Lagoonlet	Fish & Elasmobranchs	*Gymnothorax richardsonii*	0.781	1.000	0.884	**0.035**	*
		Trachinotus blochii	0.726	1.000	0.852	**0.045**	*
		Ellochelon vaigiensis	0.701	1.000	0.837	**0.012**	*
		Lutjanus	0.514	1.000	0.717	**0.040**	*
	Crustaceans	*Ocypode*	1.000	1.000	1.000	**0.033**	*
		Ocypodidae	1.000	1.000	1.000	**0.033**	*
		Thalamita crenata	0.928	1.000	0.964	**0.019**	*
		Corallianassa	0.900	1.000	0.949	**0.044**	*
		Pullosquilla	0.900	1.000	0.949	**0.049**	*
		Neocallichirus	0.781	1.000	0.884	**0.040**	*
		Cardisoma carnifex	0.748	1.000	0.865	**0.019**	*
		Etisus frontalis	0.647	1.000	0.805	**0.044**	*
	Molluscs	*Littorinidae*	1	1	1	**0.021**	*
Low Complexity & Intertidal Reef	Fish & Elasmobranchs	*Epinephelus hexagonatus*	0.925	0.666	0.785	**0.007**	**
		Blenniella periophthalmus	0.917	0.625	0.757	**0.025**	*
Medium & High Complexity	Fish & Elasmobranchs	*Zebrasoma desjardinii*	0.891	0.923	0.907	**0.001**	***
		Stegastes nigricans	0.908	0.769	0.836	**0.001**	***
Intertidal Reef & Outer Reef Terrace	Fish & Elasmobranchs	*Entomacrodus*	0.739	0.800	0.769	**0.021**	*
	Molluscs	*Modiolus*	0.8086	0.7143	0.76	**0.017**	*

Table S2.30 Harvey et al. (unpublished) recorded species list for BRUV surveys conducted in 2016 at CKI.

2016 survey		
Abudefduf notatus	*Carcharhinus falciformis*	*Coris dorsomacula*
Abudefduf vaigiensis	*Carcharhinus melanopterus*	*Coris gaimard*
Acanthurus blochii	*Centropyge flavissima*	*Crenimugil crenilabis*
Acanthurus guttatus	*Centropyge joculator*	*Ctenochaetus striatus*
Acanthurus leucosternon	*Centropyge vrolikii*	*Dascyllus aruanus*
Acanthurus maculiceps	*Cephalopholis argus*	*Dascyllus reticulatus*
Acanthurus nigricans	*Cephalopholis miniata*	*Dascyllus trimaculatus*
Acanthurus nigricauda	*Cephalopholis nigripinnis*	*Decapterus macarellus*
Acanthurus nigrofuscus	*Cephalopholis spiloparaea*	*Decapterus spp*
Acanthurus olivaceus	*Cephalopholis urodeta*	*Ecsenius bicolor*
Acanthurus pyroferus	*Chaetodon auriga*	*Ecsenius midas*
Acanthurus thompsoni	*Chaetodon bennetti*	*Elagatis bipinnulata*
Acanthurus triostegus	*Chaetodon citrinellus*	*Epibulus insidiator*
Acanthurus xanthopterus	*Chaetodon ephippium*	*Epinephelus fuscoguttatus*
Aethaloperca rogaa	*Chaetodon guttatissimus*	*Epinephelus hexagonatus*
Albula sp.	*Chaetodon kleinii*	*Epinephelus lanceolatus*
Aluterus scriptus	*Chaetodon lineolatus*	*Epinephelus maculatus*
Amblyglyphidodon aureus	*Chaetodon lunula*	*Epinephelus merra*
Amblyglyphidodon curacao	*Chaetodon madagaskariensis*	*Epinephelus ongus*
Amphiprion clarkii	*Chaetodon melannotus*	*Epinephelus polyphekadion*
Anampses meleagrides	*Chaetodon meyeri*	*Epinephelus spilotoceps*
Anampses twistii	*Chaetodon ornatissimus*	*Epinephelus tauvina*
Anyperodon leucogrammicus	*Chaetodon punctatofasciatus*	*Fistularia commersonii*
Aphareus furca	*Chaetodon trifascialis*	*Forcipiger flavissimus*
Apogon crassiceps	*Chaetodon trifasciatus*	*Forcipiger longirostris*
Apolemichthys trimaculatus	*Chaetodon ulietensis*	*Galeocerdo cuvier*
Aprion virescens	*Chaetodon unimaculatus*	*Gerres longirostris*
Arothron hispidus	*Chanos chanos*	*Gnathodentex aureolineatus*
Aulostomus chinensis	*Cheilinus chlorourus*	*Gobiidae spp*
Balistapus undulatus	*Cheilinus fasciatus*	*Gomphosus varius*
Balistoides viridescens	*Cheilinus trilobatus*	*Gracila albomarginata*
Bodianus axillaris	*Cheilinus undulatus*	*Gunnellichthys monostigma*
Bodianus diana	*Chlorurus sordidus*	*Gymnocranius grandoculis*
Bolbometopon muricatum	*Chlorurus strongylocephalus*	*Gymnosarda unicolor*
Caesio teres	*Chromis alpha*	*Gymnothorax javanicus*
Cantherhines dumerilii	*Chromis amboinensis*	*Halichoeres claudia*
Carangoides dinema	*Chromis fieldi*	*Halichoeres hortulanus*
Carangoides ferdau	*Chromis margaritifer*	*Halichoeres marginatus*
Carangoides orthogrammus	*Chromis nigrura*	*Halichoeres melasmapomus*
Carangoides orthogrammus	*Chromis opercularis*	*Halichoeres trimaculatus*
Caranx ignobilis	*Chromis ternatensis*	*Hemigymnus fasciatus*
Caranx lugubris	*Chromis viridis*	*Hemigymnus melapterus*
Caranx melampygus	*Chrysiptera biocellata*	*Hemitaurichthys polylepis*
Caranx sexfasciatus	*Chrysiptera glauca*	*Heniochus chrysostomus*
Carcharhinus amblyrhynchos	*Cirrhitichthys oxycephalus*	*Heniochus monoceros*
Carcharhinus falciformis	*Coris aygula*	*Hipposcarus harid*
Hipposcarus longiceps	*Nemateleotris magnifica*	*Scarus frenatus*

Hologymnosus annulatus	*Novaculichthys taeniourus*	*Scarus ghobban*
Hologymnosus doliatus	*Odonus niger*	*Scarus niger*
Hyporhamphus affinis	*Oxycheilinus unifasciatus*	*Scarus oviceps*
Iniistius aneitensis	*Paracanthurus hepatus*	*Scarus prasiognathos*
Iniistius pavo	*Paracirrhites arcatus*	*Scarus psittacus*
Kyphosus cinerascens	*Paracirrhites forsteri*	*Scarus rubroviolaceus*
Kyphosus vaigiensis	*Paracirrhites hemistictus*	*Scarus schlegeli*
Labrichthys unilineatus	*Parapercis clathrata*	*Scarus tricolor*
Labroides bicolor	*Parupeneus barberinus*	*Scarus xanthopleura*
Labroides dimidiatus	*Parupeneus cyclostomus*	*Scolopsis lineata*
Labroides pectoralis	*Parupeneus macronemus*	*Scomberoides lysan*
Labropsis xanthonota	*Parupeneus multifasciatus*	*Siganus argenteus*
Lethrinus erythropterus	*Parupeneus pleurostigma*	*Siganus puellus*
Lethrinus harak	*Parupeneus trifasciatus*	*Siganus punctatus*
Lethrinus lentjan	*Plagiotremus rhinorhynchos*	*Siganus stellatus*
Lethrinus microdon	*Plagiotremus tapeinosoma*	*Sphyraena barracuda*
Lethrinus obsoletus	*Platax orbicularis*	*Sphyraena sp1*
Lethrinus rubrioperculatus	*Platybelone argalus*	*Stegastes albifasciatus*
Lethrinus xanthochilus	*Plectorhinchus chaetodonoides*	*Stegastes insularis*
Lutjanus bohar	*Plectroglyphidodon dickii*	*Stegastes nigricans*
Lutjanus fulvus	*Plectroglyphidodon johnstonianus*	*Stethojulis bandanensis*
Lutjanus gibbus	*Plectroglyphidodon lacrymatus*	*Stethojulis strigiventer*
Lutjanus monostigma	*Plectroglyphidodon leucozonus*	*Sufflamen bursa*
Macolor niger	*Plectropomus areolatus*	*Sufflamen chrysopterum*
Macropharyngodon meleagris	*Plectropomus laevis*	*Sufflamen fraenatum*
Malacanthus brevirostris	*Plectropomus leopardus*	*Thalassoma amblycephalum*
Manta birostris	*Plectropomus spp*	*Thalassoma hardwicke*
Melichthys indicus	*Pomacanthus imperator*	*Thalassoma jansenii*
Melichthys niger	*Pomacentrus pavo*	*Thalassoma lunare*
Melichthys sp1	*Pseudanthias cooperi*	*Thalassoma lutescens*
Melichthys vidua	*Pseudanthias evansi*	*Thalassoma purpureum*
Monotaxis grandoculis	*Pseudanthias squamipinnis*	*Thalassoma quinquevittatum*
Mulloidichthys flavolineatus	*Pseudocheilinus hexataenia*	*Torquigener sp1*
Mulloidichthys vanicolensis	*Pseudocheilinus octotaenia*	*Trachinotus baillonii*
Myripristis adusta	*Pseudodax moluccanus*	*Trachinotus blochii*
Myripristis berndti	*Ptereleotris evides*	*Triaenodon obesus*
Myripristis chryseres	*Pterocaesio tile*	*Variola albimarginata*
Myripristis kuntee	*Rhinecanthus aculeatus*	*Variola louti*
Myripristis murdjan	*Rhinecanthus rectangulus*	*Xanthichthys auromarginatus*
Naso brevirostris	*Sargocentron punctatissimum*	*Zanclus cornutus*
Naso lituratus	*Sargocentron spiniferum*	*Zebrasoma desjardinii*
Naso unicornis	*Sargocentron tiere*	*Zebrasoma scopas*
Naso vlamingii	*Scarus forsteni*	

Table S2.31 Harvey et al. (unpublished) recorded species list for BRUV surveys conducted in 2017 at CKI.

2017 survey		
Acanthurus blochii	*Cephalopholis leopardus*	*Epinephelus corallicola*
Acanthurus dussumieri	*Cephalopholis miniata*	*Epinephelus fuscoguttatus*
Acanthurus grammoptilus	*Cephalopholis urodeta*	*Epinephelus macrospilos*
Acanthurus guttatus	*Chaetodon aureofasciatus*	*Epinephelus merra*
Acanthurus maculiceps	*Chaetodon auriga*	*Epinephelus polyphekadion*
Acanthurus nigricans	*Chaetodon ephippium*	*Epinephelus spilotoceps*
Acanthurus nigricauda	*Chaetodon guttatissimus*	*Epinephelus tauvina*
Acanthurus nigrofuscus	*Chaetodon lineolatus*	*Forcipiger longirostris*
Acanthurus pyroferus	*Chaetodon lunula*	*Gerres longirostris*
Acanthurus triostegus	*Chaetodon lunulatus*	*Gobiidae spp*
Acanthurus triostegus	*Chaetodon semeion*	*Gomphosus varius*
Acanthurus xanthopterus	*Chaetodon trifascialis*	*Gymnosarda unicolor*
Albula spp	*Chaetodon trifasciatus*	*Gymnothorax javanicus*
Amblyglyphidodon aureus	*Chaetodon ulietensis*	*Gymnothorax undulatus*
Amblyglyphidodon curacao	*Chaetodon unimaculatus*	*Halichoeres claudia*
Anampses caeruleopunctatus	*Chanos chanos*	*Halichoeres hortulanus*
Anampses meleagrides	*Cheilinus chlorourus*	*Halichoeres margaritaceus*
Anampses twistii	*Cheilinus fasciatus*	*Halichoeres marginatus*
Anyperodon leucogrammicus	*Cheilinus trilobatus*	*Halichoeres melasmapomus*
Aphareus furca	*Cheilinus undulatus*	*Halichoeres trimaculatus*
Apogon sp	*Cheilodipterus quinquelineatus*	*Hemigymnus fasciatus*
Apolemichthys trimaculatus	*Chlorurus sordidus*	*Hemigymnus melapterus*
Aprion virescens	*Chlorurus spp*	*Heniochus chrysostomus*
Aulostomus chinensis	*Chlorurus strongylocephalus*	*Heniochus monoceros*
Balistapus undulatus	*Chromis amboinensis*	*Hipposcarus harid*
Balistoides viridescens	*Chromis atripectoralis*	*Hipposcarus longiceps*
Blenniidae spp	*Chromis cinerascens*	*Hoplolatilus cuniculus*
Bodianus axillaris	*Chromis margaritifer*	*Iniistius pavo*
Bolbometopon muricatum	*Chromis nigrura*	*Kyphosus cinerascens*
Caesio cuning	*Chromis opercularis*	*Kyphosus vaigiensis*
Calotomus carolinus	*Chromis ternatensis*	*Labridae spp*
Calotomus spinidens	*Chromis viridis*	*Labroides bicolor*
Cantherhines dumerilii	*Chromis xanthura*	*Labroides dimidiatus*
Cantherhines pardalis	*Chrysiptera biocellata*	*Labroides pectoralis*
Carangoides ferdau	*Chrysiptera glauca*	*Labropsis xanthonota*
Carangoides orthogrammus	*Coris aygula*	*Lethrinus erythropterus*
Carangoides plagiotaenia	*Coris dorsomacula*	*Lethrinus harak*
Caranx ignobilis	*Crenimugil crenilabis*	*Lethrinus lentjan*
Caranx lugubris	*Ctenochaetus striatus*	*Lethrinus microdon*
Caranx melampygus	*Dascyllus aruanus*	*Lethrinus miniatus*
Caranx sexfasciatus	*Dascyllus trimaculatus*	*Lethrinus obsoletus*
Carcharhinus amblyrhynchos	*Decapterus macarellus*	*Lethrinus rubrioperculatus*
Carcharhinus melanopterus	*Echeneis naucrates*	*Lethrinus xanthochilus*
Centropyge flavissima	*Elagatis bipinnulata*	*Lutjanus bohar*
Cephalopholis argus	*Epibulus insidiator*	*Lutjanus erythropterus*
Cephalopholis leopardus	*Epinephelus*	*Lutjanus fulvus*

Lutjanus gibbus	*Plectroglyphidodon dickii*	*Scolopsis lineata*
Lutjanus monostigma	*Plectroglyphidodon johnstonianus*	*Scolopsis trilineata*
Macolor niger	*Plectroglyphidodon lacrymatus*	*Scomberoides commersonnianus*
Malacanthus brevirostris	*Plectroglyphidodon leucozonus*	*Scomberoides lysan*
Manta birostris	*Plectropomus laevis*	*Seriola lalandi*
Melichthys niger	*Plectropomus spp*	*Siganus argenteus*
Melichthys vidua	*Pomacanthus imperator*	*Siganus fuscescens*
Monotaxis grandoculis	*Pomacentridae spp*	*Siganus puellus*
Mulloidichthys flavolineatus	*Pomacentrus chrysurus*	*Siganus punctatus*
Myripristis adusta	*Pomacentrus coelestis*	*Siganus stellatus*
Myripristis kuntee	*Pomacentrus pavo*	*Sphaeramia nematoptera*
Naso brevirostris	*Pseudanthias evansi*	*Sphyraena barracuda*
Naso hexacanthus	*Pseudocheilinus hexataenia*	*Stegastes albifasciatus*
Naso lituratus	*Pseudocheilinus octotaenia*	*Stegastes fasciolatus*
Naso unicornis	*Pseudocoris aurantiofasciata*	*Stegastes insularis*
Naso vlamingii	*Pseudodax moluccanus*	*Stegastes nigricans*
Neocirrhites armatus	*Pterocaesio lativittata*	*Stethojulis bandanensis*
Neopomacentrus azysron	*Pterocaesio spp*	*Stethojulis strigiventer*
Ostorhinchus taeniophorus	*Pterocaesio tile*	*Sufflamen bursa*
Ostracion cubicus	*Rhinecanthus aculeatus*	*Sufflamen fraenatum*
Oxycheilinus digrammus	*Rhinecanthus rectangulus*	*Thalassoma amblycephalum*
Oxycheilinus unifasciatus	*Sardinella melanura*	*Thalassoma hardwicke*
Paracirrhites arcatus	*Sargocentron diadema*	*Thalassoma jansenii*
Paracirrhites forsteri	*Sargocentron spiniferum*	*Thalassoma lunare*
Parapercis schauinslandii	*Sargocentron tiere*	*Thalassoma lutescens*
Parapercis sp1	*Scarus forsteni*	*Thalassoma purpureum*
Parapercis spp	*Scarus frenatus*	*Thalassoma quinquevittatum*
Parupeneus cyclostomus	*Scarus ghobban*	*Trachinotus baillonii*
Parupeneus heptacanthus	*Scarus niger*	*Triaenodon obesus*
Parupeneus macronemus	*Scarus oviceps*	*Tylosurus crocodilus*
Parupeneus multifasciatus	*Scarus prasiognathos*	*Valenciennea sexguttata*
Parupeneus pleurostigma	*Scarus psittacus*	*Variola louti*
Parupeneus trifasciatus	*Scarus rubroviolaceus*	*Zanclus cornutus*
Petroscirtes mitratus	*Scarus schlegeli*	*Zebrasoma desjardinii*
Plagiotremus tapeinosoma	*Scarus viridifucatus*	*Zebrasoma scopas*
Plectorhinchus chaetodonoides	*Scarus xanthopleura*	

S2.9.7 Bony fish community composition

The fish families Acanthuridae, Labridae and Pomacentridae are speciose and form on average 15-25% of fish diversity and biomass on coral reefs (Bakus, 1964) and make up large mixed species aggregations that serve to increase individual protection and feeding efficiency in areas inhabited by territorial fish and predators (Ogden, Lobel, Box, Croix, & Islands, 1978). In our outer lagoon sites, the herbivorous species of convict surgeonfish (*Acanthurus triostegus*; Acanthuridae) and longnose parrotfish (*Hipposcarus harid*; Labridae) dominated across the community assemblages. Both fishes are known to aggregate over coral reefs and reef flats, particularly in algal rich territories, and as grazers, are frequently expelling biological material. The bluelined surgeonfish (*Acanthurus lineatus*) and blacktail chromis (*Chromis nigrura*; Pomacentridae) were identified as indicator species unique to the outer lagoon, the former associated with intertidal zones (Ogden et al., 1978); the latter associated with coral-rich outer reef crests (Lieske & Myers, 1994). In the surge channels, *A. triostegus* once again dominated the site assemblages, alongside a genus of wrasse (*Thalassoma*; Labridae), which are abundant in shallow, lagoon and seaward coral reefs across the tropics (Bernardi, Bucciarelli, Costagliola, Robertson, & Heiser, 2004; Lieske & Myers, 1994). The *Chrysiptera* genus of damselfish (Pomacentridae) was identified as unique to the surge channels, reflecting its habitat preference for shallow (<5m), rubbly lagoon and inner reef flats under mild surge conditions (Lieske & Myers, 1994).

In the protected inner lagoon however, a very different community assemblage was revealed. The grouper genus *Epinephelus* and the Indian sail-fin surgeonfish (*Zebrasoma desjardinii*; Acanthuridae), both of which are associated with clear lagoon and seaward reefs, dominated the inner lagoon sites. It is likely that these sites form protected recruitment nurseries for juvenile *Z. desjardinii*, which are reported to prefer sheltered inner reef areas (Lieske & Myers, 1994). *Zebrasoma desjardinii*, in addition to the dusky gregory (*Stegastes nigricans*; Pomacentridae) and the planktivorous blue-green puller (*Chromis viridis*; Pomacentridae), were identified as indicator species unique to the inner lagoon. Both *S. nigricans* and *C. viridis* dwell in reef flats and shallow lagoon areas, the former a territorial species, guarding and maintaining areas of filamentous algae that propagate on dead coral, whilst the latter aggregates in large schools over branching *Acropora* corals (Lieske & Myers, 1994). Interestingly, *S.*

nigricans was detected and *C. viridis* was absent in the high complexity lagoon, which in conjunction with low taxonomic richness and reports of coral bleaching, may reflect the damage and coral mortality reported in this area.

The addition of a sampling site in the lagoonlet on West Island, produced a unique and distinct taxon profile as visualised in the bony fish and elasmobranch nMDS ordination. With incomplete flushing, this shallow inlet is characterised by highly turbid water and is protected from strong currents and wave action from the lagoon. Backed by mangrove vegetation (Williams, 1994), it has been hypothesised that the lagoonlet may act as an important fish nursery area (Berry, 1989). Unfortunately, it is not possible to determine age profiles of detected species with eDNA metabarcoding. We did however, detect a number of fish species including the snubnose pompano (*Trachinotus blochii*), squaretail mullet (*Ellochelon vaigiensis*), milkfish *(Chanos chanos*) and blacktail snapper (*Lutjanus fulvus*) of which juveniles are reported to inhabit sandy shorelines, estuaries and mangroves before moving onto coral reef and reef flats as adults (Bagarinao, 1999; Fischer, Sousa, Silva, de Freitas, & Poutiers, 1990; Harrison & Senou, 1999; Nakamura et al., 2008). The majority of fish taxa found in this area were also detected in the low and medium complexity lagoon, which is consistent with the expected transition onto the reef flats.

S2.9.8 Supplementary references

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Anderson, M., Gorley, R. N., & Clarke, R. K. (2008). *Permanova+ for Primer: Guide to Software and Statistical Methods*. Primer-E Limited.

Bagarinao, T. (1999). Milkfish. In K. E. Carpenter & V. H. Niem (Eds.), *FAO species identification guide for fishery purposes. The living marine resources of the Western Central Pacific. Volume 3. Batoid fishes, chimaeras and bony fishes part 1 (Elopidae to Linophrynidae).* (pp. 1822–1826). Rome.

Bakus, G. J. (1964). *The effects of fish-grazing on invertebrate evolution in shallow tropical waters*. University of Southern California Press.

Barbosa, A. M. (2015). fuzzySim: applying fuzzy logic to binary similarity indices in ecology. *Methods in Ecology and Evolution*, *6*(7), 853–858.

Benson, D. A., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Wheeler, D. L. (2005). GenBank. *Nucleic Acids Research*, *33*, D34–D38. https://doi.org/10.1093/nar/gki063

Bernardi, G., Bucciarelli, G., Costagliola, D., Robertson, D. R., & Heiser, J. B. (2004). Evolution of coral reef fish Thalassoma spp. (Labridae). 1. Molecular phylogeny and biogeography. *Marine Biology*, *144*(2), 369–375. https://doi.org/10.1007/s00227-003-1199-0

Berry, P. F. (1989). Survey of the marine fauna of Cocos (Keeling) Islands, Indian Ocean. *Western Australian Museum*, Perth.

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a Unix-inspired software package for DNA metabarcoding. *Molecular Ecology Resources* 16(1), 176–182.

Caporaso, J. G., Kuczynski, J., Stombaugh, J., Bittinger, K., Bushman, F. D., Costello, E. K., … Gordon, J. I. (2010). QIIME allows analysis of high-throughput community sequencing data. *Nature Methods*, *7*(5), 335.

Cardoso, P., Rigal, F., & Carvalho, J. C. (2015). BAT–Biodiversity Assessment Tools, an R package for the measurement and estimation of alpha and beta taxon, phylogenetic and functional diversity. *Methods in Ecology and Evolution*, *6*(2),

232–236.

Clarke, K. R., & Gorley, R. N. (2015). *PRIMER v7: User Manual/Tutorial*. Plymouth, UK.

De Caceres, M., & Jansen, F. (2016). Package 'indicspecies.' *Relationship between species and groups of sites*. R package version, 1(6).

Edgar, R. C. (2010). Search and clustering orders of magnitude faster than BLAST. *Bioinformatics*, *26*(19), 2460–2461. http://dx.doi.org/10.1093/bioinformatics/btq461

Esling, P., Lejzerowicz, F., & Pawlowski, J. (2015). Accurate multiplexing and filtering for high-throughput amplicon-sequencing. *Nucleic Acids Research*, *43*(5), 2513–2524. https://doi.org/10.1093/nar/gkv107

Fields, A. T., Abercrombie, D. L., Eng, R., Feldheim, K., & Chapman, D. D. (2015). A novel mini-DNA barcoding assay to identify processed fins from internationally protected shark species. *PLoS One*, *10*(2), e0114844.

Fischer, W., Sousa, I. M., Silva, C., de Freitas, A., & Poutiers, J. M. (1990). Guia de campo das espécies comerciais marinhas e de águas salobras de Moçambique.

Harper, B. A., Gourlay, M. R., & Jones, C. M. (2001). Storm tide risk assessment for the Cocos (Keeling) Islands. *8th Australasian Port and Harbour Conference*, 75–80.

Harrison, I. ., & Senou, H. (1999). Mullets. In K. E. Carpenter & V. H. Niem (Eds.), *FAO species identification guide for fishery purposes. The living marine resources of the Western Central Pacific. Volume 4. Bony fishes part 2 (Mugilidae to Carangidae).* (pp. 2069–2083). Rome: FAO.

Hender, J., McDonald, C. ., & Gilligan, J. . (2001). *Baseline surveys of the marine environments and stock size estimates of marine resources of the south Cocos (Keeling) Atoll (0-15m), eastern Indian Ocean.* Draft Report to Fisheries Resources Research Fund. Canberra.

Hobbs, J. P. A., & Macrae, H. (2012). Unusual weather and trapped coral spawn lead to fish kill at a remote coral atoll. *Coral Reefs*, *31*(4), 961. https://doi.org/10.1007/s00338-012-0918-8

Huson, D. H., Auch, A. F., Qi, J., & Schuster, S. C. (2007). MEGAN analysis of metagenomic data. *Genome Research*, *17*(3), 377–386. https://doi.org/10.1101/gr.5969107

Kearse, M., Moir, R., Wilson, A., Stones-Havas, S., Cheung, M., Sturrock, S., …

Drummond, A. (2012). Geneious Basic: An integrated and extendable desktop software platform for the organization and analysis of sequence data. *Bioinformatics*, *28*(12), 1647–1649. http://dx.doi.org/10.1093/bioinformatics/bts199

Lieske, E., & Myers, R. (1994). *Coral reef fishes: Indo-Pacific & Caribbean including the Red Sea.* Harper Collins Publishers.

Nakamura, Y., Horinouchi, M., Shibuno, T., Tanaka, Y., Miyajima, T., Koike, I., … Sano, M. (2008). Evidence of ontogenetic migration from mangroves to coral reefs by black-tail snapper *Lutjanus fulvus*: stable isotope approach. *Marine Ecology Progress Series*, *355*, 257–266.

Ogden, J. C., Lobel, P. S., Box, P. O. A., Croix, S., & Islands, U. S. V. (1978). The role of herbivorous fishes and urchins in coral reef communities. *Environmental biology of fishes*, *3*(1), 49–63.

Oksanen, J., Guillaume Blanchet, F., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., … Wagner, H. (2017). vegan: Community Ecology Package. R package version 2.4-3.

Parks Australia. (2005). Status of the coral reefs at the Cocos (Keeling) Islands: a report on the status of the marine community at Cocos (Keeling) Islands, east Indian Ocean, 1997-2005. *Department of the Environment and Heritage, Australian Government,* Canberra.

R Studio Team (2015). RStudio: integrated development for R. *RStudio Inc,* Boston, MA. Retrieved from *http://www.rstudio.com*

Schubert, M., Lindgreen, S., & Orlando, L. (2016). AdapterRemoval v2: rapid adapter trimming, identification, and read merging. *BMC Research Notes*, *9*, 88. https://doi.org/10.1186/s13104-016-1900-2

Wickham, H. (2016). *ggplot2: elegant graphics for data analysis*. Springer.

Williams, D. G. (1994). Marine habitats of the Cocos (Keeling) Islands. *Atoll Research Bulletin, 406*.

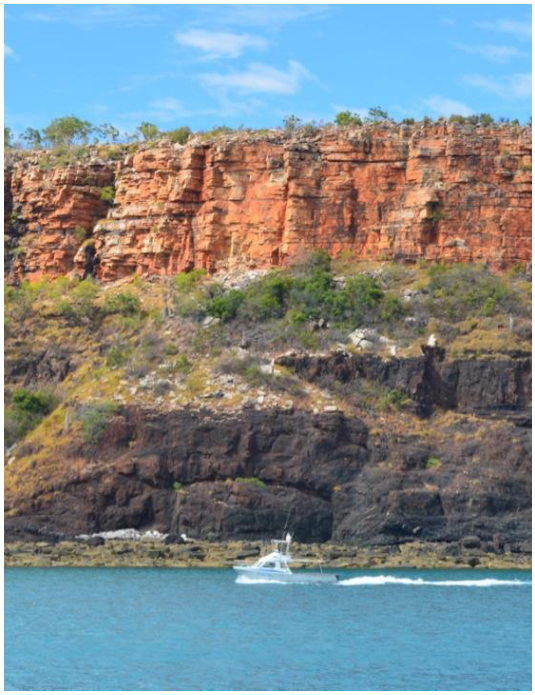
Woodroffe, C. D., & Biribo, N. (2011). Atolls. In D. Hopley (Ed.), *Encyclopedia of Modern Coral Reefs: Structure, Form and Process* (pp. 51–71). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-90-481-2639-2_4

— Chapter 3 —

Large-scale eDNA metabarcoding survey reveals marine biogeographic break and transitions over tropical north-western Australia.









3.1 Prologue

Chapter two applied eDNA metabarcoding across an entire coral reef atoll and set a new baseline for ongoing monitoring and management. Notably, it revealed that eDNA signals from seawater are localised within a small spatial area. It follows that if eDNA can statistically separate habitats over small spatial scales then this pattern should hold true over much broader spatial scales. Despite this, relatively little research has explored eDNA over a wide geographical area (see Fraija-Fernández et al., 2019). The following chapter addresses this topic by conducting one of the largest (on a geographical scale) continuous eDNA metabarcoding surveys across 700 km of north-western Australian coastline. This survey additionally evaluated the ability of eDNA metabarcoding to detect endangered and/or elusive bony fish and elasmobranch taxa, and more broadly, aquatic reptiles, across this vast, remote region.

The Pilbara, Canning and Kimberley regions (22.28° S, 114.51° E to 14.88° S, 128.99° E) comprise north-western Australia. These distinct regions support a diversity of coastal landscapes, ranging from intertidal sand and mudflats, to rocky, intertidal platforms, fringing coral, offshore coral reefs, and substantial mangrove habitats (Wilson, 2014). These vast coastlines have supported 40,000 years of continual subsistence practices and cultural traditions. Aboriginal people in the west Kimberley developed unique double log rafts (referred to as *galwa* or *kalum*) to access offshore islands and other coastal areas (Akerman, 2015; O'Connor, 2017). Collected pearl shell (*Pinctada maxima*) became the most widely traded item across two-thirds of the Australian continent (Department of Sustainability, Environment, Water, Population and Communities, 2011). North-western Australia was also the scene for some of the earliest European exploration of the Australian continent. The Dutch seafarer Abel Tasman mapped the north coast of Australia with a fleet of three ships—the *Limmen*, *Zeemeeuw* and the *Braek* in 1644 (Feeken, 1973). In 1699, the famous English privateer and hydrographer William Dampier sailed the *Roebuck* from Shark Bay (mid-coast of Western Australia) to the later named Dampier Peninsula, in the Kimberley. His extensive descriptions and collection of Australian flora and fauna during this voyage, represents the first scientific research conducted in Australia (George, 1999).

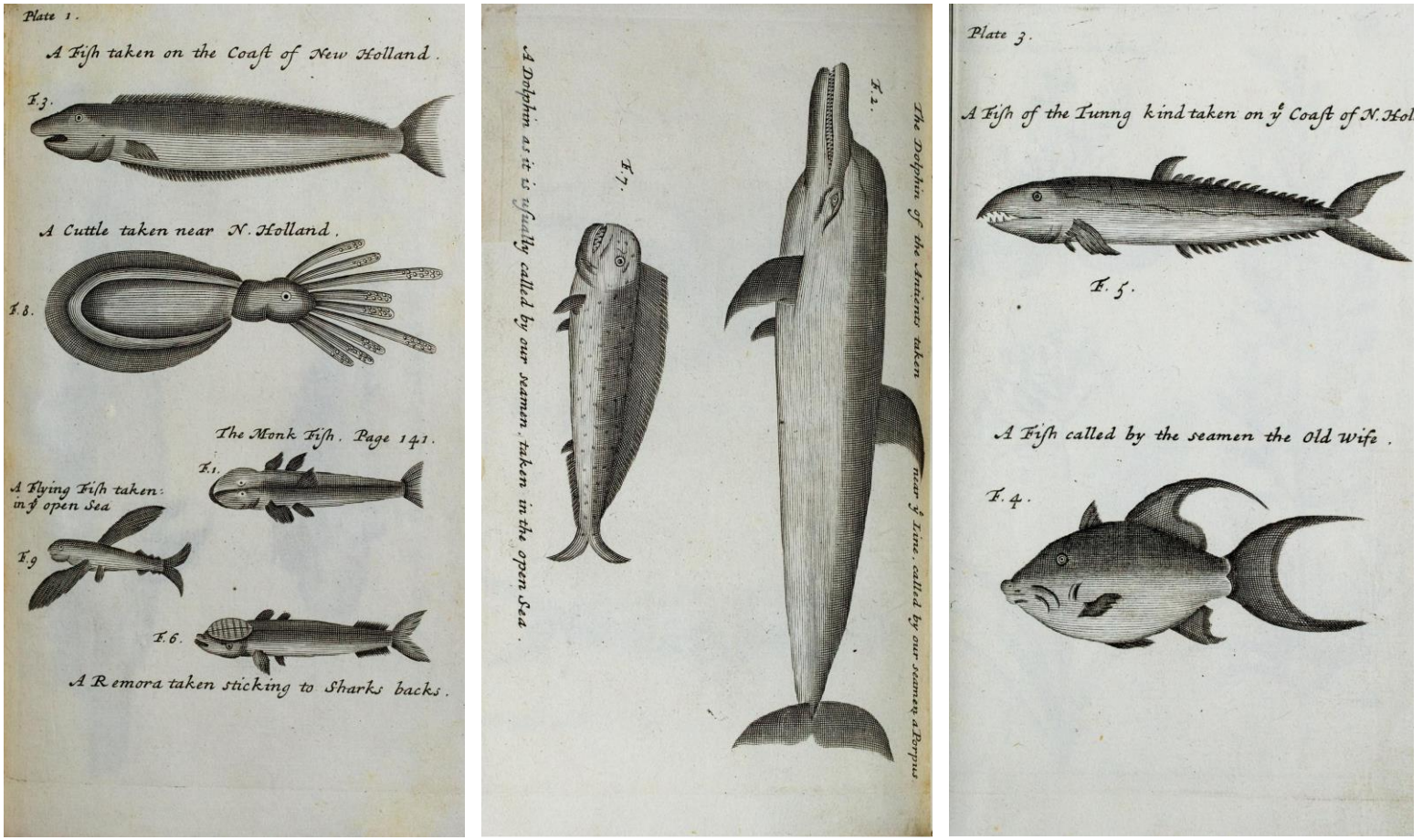


Figure 3.1 The first scientific drawings of Australian marine life as illustrated by William Dampier during the *Roebuck* voyage in 1699. This exploratory and scientific expedition traversed the coastlines of northwestern Australia, from Shark Bay to the Dampier Peninsula. Illustrated in *A voyage to New Holland, etc., in the year 1699* (1703), Plates 1, 2 and 3. Images courtesy of the Biodiversity Heritage Library. Contributed by the State Botanical Collection, Royal Botanic Gardens Victoria. | http://www.biodiversitylibrary.org

Marine environmental conditions and connectivity patterns across north-western Australia are shaped by various oceanic currents, immense tidal systems (macrotides ranging up to 11m in the Kimberley), seasonal discharge and turbidity from major rivers (Semeniuk, 1993; Thackway & Cresswell, 1998). Biological connectivity along the North West Shelf is facilitated by the Indonesian Throughflow-Holloway-Leeuwin Current, dispersing marine larvae from eastern Indonesia to the Ningaloo Reef (Wilson, 2013). Coastal currents are tidal-influenced and multi-directional, creating regionally constrained pockets of benthic shelf and shore communities (Wilson, 2013). North-western Australia is a hotspot of coastal and oceanic biodiversity and endemism, and due to its remoteness, has some of the least impacted marine ecosystems in the world (Halpern et al., 2008; Tittensor et al., 2010). There are over 300 reported species of hard coral, many that have adapted tolerance to the turbidity and immense tidal systems of the Kimberley (Richards, Sampey, & Marsh, 2014). There are over 1,500 reported species of bony fish (Moore, Morrison, & Johnson, 2020; Moore, Morrison, Hutchins, Allen, & Sampey, 2014), 300 species of echinoderms and 1900 species of mollusc (Wilson, 2013). Notably, the coastline of north-western Australia is a vital refuge for endangered elasmobranchs such as sawfish (family: Pristidae) and the northern river shark (*Glyphis garricki*) (Morgan et al., 2011).

3.1.1 Chapter acknowledgements

I acknowledge the Department of Primary Industries and Regional Development (WA), BMT Oceanica and the Australian Research Council (LP160100839) for project funding. I give special mention to the crew of the RV Naturaliste, Sam Moyle, Dion Boddington and Gabby Mitsopoulos for fieldwork assistance. Gratitude is also extended to the Wunambal Gaambera, Dambimangari, Mayala, Bardi Jawi, Nyul Nyul, Jabirr Jabirr and Yawuru people, the traditional owners of the areas sampled and to their indigenous rangers for assistance with sampling. I would like to thank Kate Sanders for providing the sea snake databases for reptile assay development. For access to the Zeus supercomputer, which sped up much of my bioinformatic processing, I would like to thank Pawsey Supercomputing Centre (Kensington, WA). Lastly, I would like to give thanks to everyone at the TrEnD Laboratory for invaluable eDNA assistance across the duration of the project.

3.1.2 Data accessibility

Demultiplexed (unfiltered) metabarcoding sequencing data and taxonomic presence/absence matrices from this study are temporarily available from https://cloudstor.aarnet.edu.au/plus/s/VD3Y9HS7CIMotuX and on request. The manuscript arising from this chapter is currently in preparation for submission to *Diversity and Distributions*. As such, this chapter has been formatted for *Diversity and Distributions*. Data will be uploaded onto Dryad Digital Repository coinciding with the manuscript submission.

3.2 Abstract

The coastlines of north-western Australia (NWA) support a diversity of relatively untouched tropical marine habitats with high species richness, including a number of critically endangered taxa. Baseline surveys and long-term monitoring are crucial for the sustainable management and conservation of this unique marine region. Environmental DNA (eDNA) metabarcoding has demonstrated its applicability as a highly sensitive biomonitoring tool across small spatial and temporal scales in marine ecosystems. However, it has rarely been tested across large spatial scales, or biogeographical barriers. We present data from a broad-scale multi-marker eDNA metabarcoding survey across the Canning and Kimberley coastlines of north-western Australia (NWA) to evaluate the potential of scaling up eDNA metabarcoding for use as a regional biomonitoring tool. We applied metabarcoding assays targeting the mitochondrial 16S rRNA and CO1 genes to 284 surface seawater eDNA samples collected from 71 midshelf, inshore, coastal and nearshore estuarine sites over 700km of the NWA coastline. The assays detected a wide range of bony fish (404 taxa), sharks and rays (44) and aquatic reptiles (5). We found bioregional and depth differentiation within inshore bony fish and shark/ray communities. These findings support the presence of a marine biogeographic break, which is purported to occur in the vicinity of Cape Leveque, demarcating the border between the Kimberley and Canning bioregions. Coastal bony fish and inshore shark/ray assemblages were additionally found to differ between the South and North Kimberley regions indicating previously unrecognised subregional differentiation among these taxa. We also investigated whether eDNA-derived 16S rRNA haplotype data for stripey snapper (*Lutjanus carponotatus*) corroborated previous fine-scale population genetic studies, which identified the influence of the biogeographic break on population level variation for this species. Our findings however, showed no significant differentiation in stripey snapper haplotype composition across our study range. The overall compositional data has been used to update distribution information for a number of endangered, elusive and data-deficient taxa and identified potential biosecurity risks. eDNA metabarcoding demonstrated a high level of sensitivity that was able to discern fine-scale patterns across this large-scale, remote and oceanographically complex bioregion and provides an important foundation for gaining a better understanding of the marine biota in NWA.

3.3 Introduction

Broad-scale biomonitoring approaches in marine environments are integral in assessing human-induced change, stressors, and shifting baselines over large spatial and temporal scales (Dafforn et al., 2016). Typically, these approaches utilise rapid-assessment methods, such as underwater visual census, marine manta tow and baited-remote underwater video (BRUV) surveys (Ellis et al., 2011; Gaertner et al., 2013; Piacenza et al., 2015) that provide information to distinguish broad-scale indicators and subsequently direct further research efforts to areas of interest. The advent of environmental DNA (eDNA) metabarcoding coupled with next-generation sequencing (NGS) has enabled the genetic profiling of a wide range of biota present in environmental samples (e.g. water, scat, soil etc.). eDNA metabarcoding has the potential to be utilised as a sensitive, cost-effective, and rapid broad-scale biomonitoring tool and is particularly well suited to marine environments (Thomsen et al., 2012; Thomsen & Willerslev, 2015; Valentini et al., 2016). eDNA-derived compositional data can provide greater biological coverage to distinguish spatial and habitat variation, identify network associations, trophic structure, biological invasions and the locality of critically-endangered species (Valentini et al., 2016). It is however at present, limited in its ability to provide quantitative data (i.e. population size and biomass). Nevertheless, it can provide finer-scale information such as population connectivity (Parsons, Everett, Dahlheim, & Park, 2018; Sigsgaard et al., 2016), and in conjunction with remote-sensing of physical-chemical components can evaluate whole-ecosystem health (Dafforn et al., 2016). Whilst eDNA metabarcoding has demonstrated its applicability across small, yet highly sensitive, spatial (Jeunen et al., 2018; O'Donnell et al., 2017; Port et al., 2016) and temporal scales (Berry et al., 2019) in marine ecosystems, it is in a preliminary stage of being scaled up and tested across broader regional scales (Fraija-Fernández et al., 2019).

The extensive coastline of north-western Australia (NWA) supports a diverse array of tropical marine habitats and biota, extending from offshore coral reefs on the edge of the continental shelf, to coastal intertidal sand, rock and reef habitats, constituting 12 distinct bioregions (Wilson, 2014). A profound change in the underlying geomorphology of the Canning and Kimberley basins, from Cretaceous-Cainozoic sedimentary (largely sandstone) rocks to Proterozoic metasedimentary, metamorphic

and igneous rocks, have shaped various coastal marine habitats in these bioregions (Wilson, 2014). The Canning bioregion comprises coastlines typified by benthic soft substrates, such as intertidal sand and mudflat habitats with very little coral reefs, whilst the Kimberley bioregion is dominated by rocky, intertidal platforms, fringing coral and offshore coral reefs, and substantial mangrove habitat (Richards, Bryce, & Bryce, 2018; Wilson, 2014). Environmental conditions and connectivity patterns across these bioregions are additionally shaped by various oceanic currents, immense tidal systems (macrotides ranging up to 11m in the Kimberley), seasonal discharge and turbidity from major rivers (Semeniuk, 1993; Thackway & Cresswell, 1998). Temperature varies between the bioregions and also across subregions; semiarid in the Canning, subhumid in the south Kimberley, humid in the north Kimberley and subhumid in the northeast Kimberley (Cresswell & Semeniuk, 2011).

This environmental variation is purported to contribute to a major biogeographic break at Cape Leveque—the tip of the Dampier Peninsula, demarcating the border between the Kimberley and Canning bioregions—see Figure 3.2 (Travers, Potter, Clarke, Newman, & Hutchins, 2010; Wilson, 2014). A significant change in fish composition across Cape Leveque (Hutchins, 2001) likely reflects the latitudinal transition in benthic substrates, overlaid on a strong bioregional effect reflecting various habitat, tidal, and riverine discharge influences (Travers, Newman, & Potter, 2006; Travers et al., 2010). Population connectivity studies in bony fish (stripey snapper; *Lutjanus carponotatus* and blackspotted croaker; *Protonibea diacanthus*) and corals (*Isopora brueggemanni* and *Acropora aspera*) further revealed a genetic transition zone across Cape Leveque, with dispersal and gene flow likely constricted by extreme tidal flushing at the head of King Sound (DiBattista et al., 2017; Taillebois et al., 2017; Underwood, Richards, Berry, & Gilmour, 2017).

The aim of this research is to conduct a broad-scale multi-marker eDNA metabarcoding survey across the extensive coastline of NWA in order to: (i) detect the purported biogeographic break across Cape Leveque using eDNA-derived bony fish, shark and ray, and aquatic reptile taxonomic compositional data, (ii) examine whether eDNA-derived haplotype (amplicon sequence variant, ASV) data for stripey snapper (*L. carponotatus*) reflects previous (genotype by sequencing, GBS) population connectivity research in NWA, (iii) update distributional information for endangered

elasmobranchs, such as sawfish (family Pristidae) and the northern river shark (*Glyphis garricki*), marine turtles (superfamily Chelonioidea), and for data deficient taxa such as sea snakes (subfamily Hydrophiinae), and (iv) evaluate the overall strengths and weaknesses of eDNA metabarcoding as a biomonitoring tool when used across a broad geographic region. Given the remoteness and subsequent low use of the marine environment across NWA, long-term monitoring programs are sparse, particularly of species that are not of commercial value (Evans, Bax, & Smith, 2017). As such, there is a great potential to integrate eDNA metabarcoding as a long-term biomonitoring tool in NWA, capable of providing distribution information on a wide variety of taxa.

3.4. Methods

3.4.1 Field sampling

Four one-litre water replicates were sampled from 71 sites across the Canning/Kimberley bioregions (Figure 3.2; Table S3.1) in September 2017 and July/September 2018, totaling 284 samples over 700km of coastline. Samples were taken on a transect line traversing the purported bioregional break and more widely across the Kimberley region in midshelf, inshore, coastal and nearshore estuarine habitats. Surface water was collected using bleach sterilized 1L Nalgene bottles attached to an extended pole, to avoid close encounters with saltwater crocodiles (*Crocodylus porosus*). Samples were immediately stored on ice and were individually filtered across Pall 0.45µm Supor® polyethersulfone membranes using a Pall Sentino® Microbiology pump (Pall Corporation, Port Washington, USA) within five hours of collection. Filtration equipment was cleaned using 10% bleach – a one litre sample of this was taken at the end of each sampling day as a filtration control to test for any carry over contamination. Membranes with filtrate were immediately frozen at -20°C, prior to their transportation to the Trace & Environmental DNA (TrEnD) Laboratory in Perth, Western Australia, where they were stored at -80°C until extraction—all eDNA was isolated within two months of collection.

3.4.2 DNA extraction

DNA was extracted from half of the membrane using a DNeasy Blood and Tissue Kit (Qiagen; Venlo, Netherlands) with the following modifications: 540µl of ATL lysis buffer, 60µl of Proteinase K and a 3-hour digestion at 56 ºC. This was completed within four weeks of collection. Blank controls were processed in parallel with all samples to detect any cross-contamination. Genomic DNA extracts were then stored at -20°C.

3.4.3 Metabarcoding assay design, amplification & library sequencing

Three PCR metabarcoding assays were employed: 16S Fish, COI Elasmobranch and 16S Reptile (Table 3.1) to amplify bony fish, elasmobranchs and aquatic reptiles,

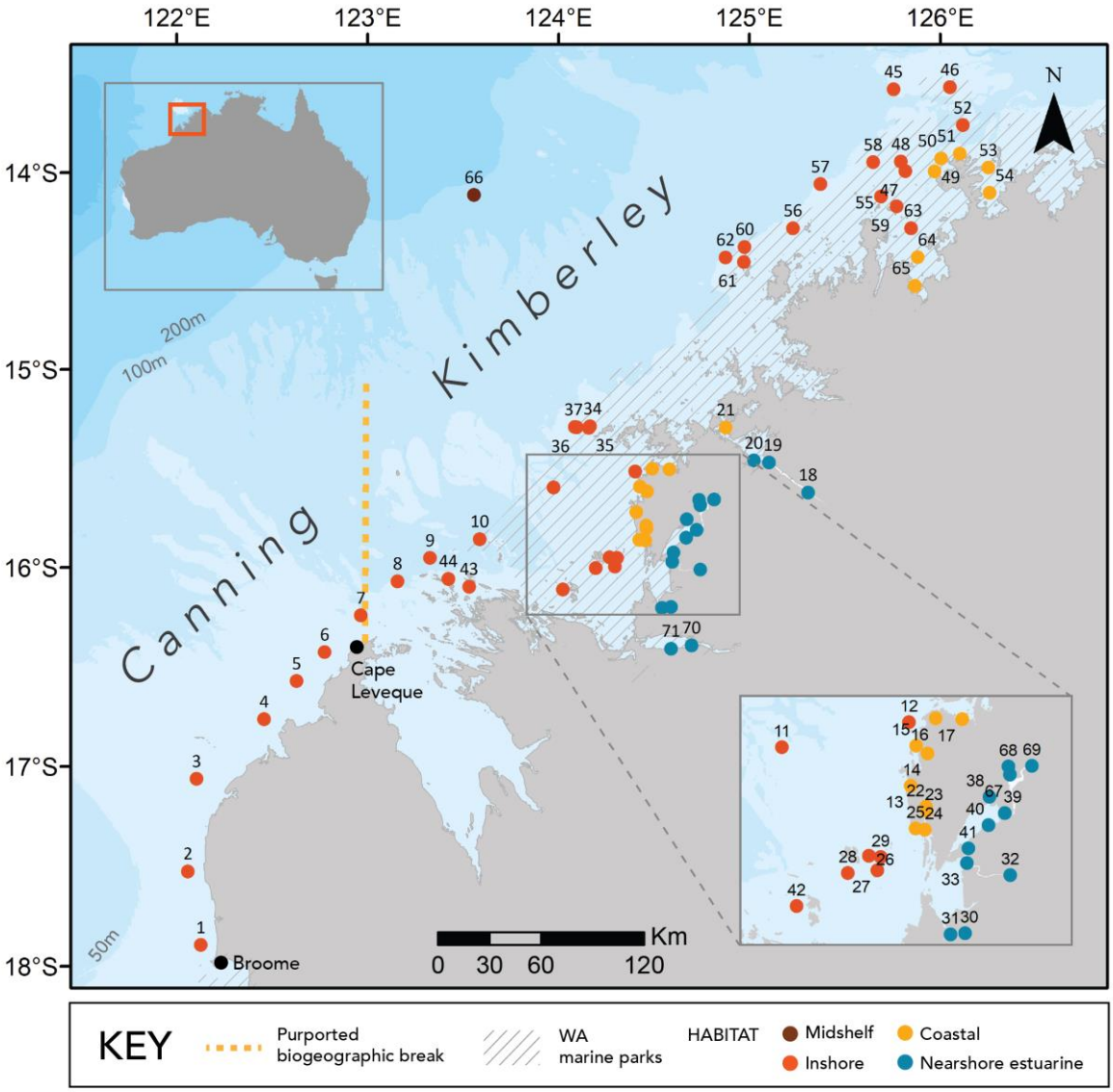


Figure 3.2 Location of sampling sites across the Canning and Kimberley bioregions in north-western Australia (NWA).
The Canning sites (1-7; latitude 16.1° S to 17.5° S) extend up to the purported biogeographic break line off Cape Leveque. The Kimberley sites (8-71) north-west of the purported biogeographic break are additionally categorised into the subregions, South Kimberley (sites 8-44 and 68-71; latitude 15.1° S to 16.4° S) and North Kimberley (sites 45-66; latitude 13.6° S to 14.6° S). Bathymetry data was sourced from Geoscience Australia (Whiteway, 2009).

Table 3.1 PCR assay information for marine eDNA metabarcoding across the Canning/Kimberley bioregions.
Three primer sets: 16S Fish, 16S Reptile and COI Elasmobranch corresponding to the mitochondrial 16S rDNA and COI regions, were applied to all collected seawater samples. In the primer name, "F" refers to the forward primer and "R" to the reverse primer.

PCR Assay	Target taxa	Primer Name	Oligonucleotide Sequence (5'-3')	Target Length (bp)	Annealing temp (°C)	Primer Reference
16S Fish	Bony Fish	16SF/D	GACCCTATGGAGCTTTAGAC	178-228	54	Berry et al., 2017 Deagle et al., 2007
		16S2R-degenerate	CGCTGTTATCCCTADRGTAACT			
COI Elasmobranch	Sharks, Skates & Rays	FishF1-degenerate	ACCAACCACAAAGANATNGGCAC	110-241	52	**Original:** Ward, Zemlak, Innes, Last, & Hebert, 2005 **Degenerate:** West et al., 2020
		FishF2-degenerate	TCNACNAATCATAAAGATATCGGCAC			
		Shark COI-MINIR-degenerate	GATTATTACNAAAGCNTGGGC			**Original:** Fields, Abercrombie, Eng, Feldheim, & Chapman, 2015 **Degenerate:** West et al.; 2020
16S Reptile	Aquatic Reptiles	AqReptileF-degenerate	AGACNAGAAGACCCTGTG	211-277	52	This study
		AqReptileR	CCTGATCCAACATCGAGG			

respectively, from mixed environmental samples. The 16S Fish and COI Elasmobranch PCR assays have been optimised and successfully applied in previous eDNA metabarcoding studies examining fish diversity and trophic interactions (Berry et al., 2017; Stat et al., 2017, 2019; West et al., 2020). The 16S Reptile assay was specifically designed to amplify northern Australian sea snakes, turtles and crocodiles in this study (see S3.9.3: Primer development note for 16S Reptile).

Following quantitative PCR-based (qPCR) quantification to optimise levels of input DNA (Berry et al., 2017), final qPCR was performed in a single step using fusion tagged primer architecture consisting of Illumina compatible sequencing adaptors, a unique index (6-8bp in length) and a respective primer sequence for each assay. All qPCR reactions were prepared in dedicated clean room facilities at the TrEnD Laboratory, Curtin University and are described in detail in S3.9.1. qPCR amplicons were pooled at equimolar ratios based on qPCR ΔRn values and size-selected using a Pippin-Prep (Sage Science, Beverly, USA) to remove any off-target amplicons. Size-selected libraries were then purified using the Qiaquick PCR Purification Kit (Qiagen, Venlo, Netherlands), quantified using a Qubit 4.0 Fluorometer (Invitrogen, Carlsbad, USA) and diluted to 2nM for loading onto a 300 cycle MiSeq® V2 Standard Flow Cell. Sequencing was conducted on an Illumina MiSeq platform (Illumina, San Diego, USA), housed in the TrEnD Laboratory at Curtin University, Western Australia.

3.4.4 Bioinformatics

Sequencing reads were demultiplexed and quality filtered in OBITools (v1.2.9; Boyer et al., 2014) and in RStudio (v1.1.423; RStudio Team, 2015) using the DADA2 bioinformatics package (Callahan et al., 2016; see S3.9.2 for bioinformatic parameter details). Resulting amplicon sequence variants (ASV) were queried using BLASTn against NCBI's GenBank nucleotide database (accessed in 2018/19; Benson et al., 2005) and also against an in-house 16S rRNA Western Australian fish database (see Table S2.6 in chapter two). Taxonomic assignments of ASVs were curated using a lowest common ancestor approach (https://github.com/mahsa-mousavi/LCA_taxonomyAssignment; see S3.9.2); consolidated taxa assignments were then additionally categorised based on habitat association and biogeographic

distribution obtained from Kimberley biodiversity checklists (Moore et al. 2014), FishBase (Froese & Pauly, 2018) and the World Register of Marine Species (WoRMS; Horton *et al.*, 2018).

In order to normalise the dataset, we determined appropriate subsampling depths for each assay (see S3.9.2 for more detail) and conducted consecutive rounds of subsampling using the multiple_rarefactions function in Qiime (Caporaso et al., 2010). ASVs detected in filtration and/or extraction blanks were entirely removed, prior to ASVs being merged by taxonomy using the phyloseq 'tax_glom' function (McMurdie & Holmes, 2013) in RStudio. We then consolidated our assay data into three taxonomic-based datasets: Actinopterygii (bony fish), Chondrichthyes (sharks and rays) and Reptilia (reptiles), which allowed us to examine community composition by discrete taxonomic group, rather than by individual metabarcoding assays which contained some overlap in taxonomic detections. Additionally, we extracted the 16S Fish ASV data for stripey snapper (*L. carponotatus*) to examine 16S rDNA haplotype variation for this species across NWA.

3.4.5 Statistics

Community composition variation was tested between sites within inshore, coastal and nearshore estuarine habitats independently. Presence-absence data was converted to a Jaccard similarity matrix and tested for compositional variation by distance-based linear modelling (DistLM) using a step-wise selection procedure and Akaike Information Criterion (AIC) in the PERMANOVA+ add-on (Anderson, Gorley, & Clarke, 2008) of PRIMER v7 (Clarke & Gorley, 2015). Normalised spatial predictor variables included bioregion (Canning and Kimberley), subregion (Canning, South Kimberley and North Kimberley), latitude and site depth (Table S3.1); longitude was omitted due to collinearity with latitude. Temporal variation in sampling was not included, given earlier research indicated that season only has a small influence on inshore fish composition in this region (Travers et al., 2006). Site variation was visualised by principal coordinates analysis (PCO) using the stats function 'cmdscale' and predictor variables overlaid using the vegan 'ordisurf' function (Oksanen et al., 2019) in R Studio (v1.1.423; RStudio Team, 2015). Observed taxonomic richness between bioregions and subregions were tested for significance using ANOVA and

graphed in ggplot (Wickham, 2016) in RStudio. Additionally, similarity percentage analyses (SIMPER) were conducted in PRIMER to identify the top inshore and coastal taxa that contribute to pairwise dissimilarity between bioregions and subregions, where significant in the DistLM analyses. This elucidated whether variation in community composition between the bioregions and subregions is driven by uneven taxonomic richness and/or variation in compositional diversity.

A median-joining (MJ) network analysis of the 16S rDNA haplotypes for stripey snapper, corresponding to unique ASVs, was generated in PopArt (v 1.7; Leigh & Bryant, 2015) using default settings. Haplotype community variation between sites was tested by a DistLM analysis, employing the spatial predictor variables of bioregion, latitude, and site depth, in addition to habitat.

3.5 Results

3.5.1 Sampling and sequencing statistics

The three eDNA metabarcoding assays yielded a total of 57,311,878 sequencing reads. The mean number of filtered sequences (post-quality, denoising and chimera filtering) was 75,800 ± 41,071 per replicate sample (303,202 ± 143,648 per site) for the 16S Fish assay; 15,838 ± 21,799 per replicate sample (61,347 ± 58,418 per site) for the COI Elasmobranch assay; and 32,569 ± 42,241 per replicate sample (130,278 ± 114,036 per site) for the 16S Reptile assay (Table S3.2). ASV accumulation curves based on the addition of each sampling replicate per site indicated that four 1L water replicates (chosen *a priori* to sampling) was just shy of maximising ASV richness for each of the three assays (Figures S3.1, S3.2, S3.3). On fitting a polynomial curve to the median accumulation curve for each assay, it was extrapolated that an average of 6.9, 6.1 and 6.1 1L water replicates would be required to maximise ASV richness for 16S Fish, COI Elasmobranch and 16S Reptile assays, respectively. The rarefaction analyses determined suitable subsampling cut-offs (after pooling of the four replicates per site) of 30,977 reads for the 16S Fish assay (Figures S3.4, S3.5, S3.6), 4000 reads for the COI Elasmobranch assay (Figures S3.7, S3.8, S3.9), and 4000 reads for the 16S Reptile assay (Figures S3.10, S3.11, S3.12).

Potential cross-contaminant ASVs that were detected in filtration and/or extraction blanks were removed from subsequent analyses. This included ASVs that produced detection hits for giant trevally (*Caranx ignobilis*), snapper (genus: *Lutjanus*), blue threadfin (*Eleutheronema tetradactylum*), barramundi (*Lates calcarifer*), and pilchard (genus: *Sardinops*). These species were targeted for fisheries and/or commercial research on the sampling vessels with exception to pilchard, which was utilised as bait for BRUVs deployments. Only compromised ASVs were removed from subsequent analyses. For example, we retained multiple barramundi ASVs that were not detected in filtration and/or extraction blanks. We also detected salmon (genus: *Salmo*) in the extraction blanks. This has previously been detected as a sporadic reagent contamination in both our workflows and other laboratories (Thomsen et al., 2016), and as such was entirely removed. We also omitted all ASVs that produced detection hits for taxa outside of our targeted taxonomic groups of bony fish, sharks, rays and

reptiles. This included humans (*Homo sapiens*), chicken (*Gallus gallus*) and horse (*Equus caballus*).

3.5.2 Overall diversity

A total of 310 taxa (ranging from family to species-level assignments; 4.9 ± 9.9 ASVs per taxa) were detected with the 16S Fish assay, 139 taxa (1.3 ± 1.0 ASVs per taxa) with the COI Elasmobranch assay, and 181 taxa (2.9 ± 5.2 ASVs per taxa) with the 16S Reptile assay, prior to subsampling (Figure 3.3). Collectively, the three metabarcoding assays yielded 453 identifiable taxa, representing 96 families within 41 orders of bony fish, sharks, rays and aquatic reptiles (Table S3.3). Of these taxa, 63.7% are widely distributed across the Indo-West Pacific, 12.4% circumglobal, and 8.6% endemic to the Australian region (including Indo-Australian, Northern Australian and Western Australian bioregions). The majority of detected taxa are associated with hard substrate habitats (54.0%), followed by soft substrate (38.6%), estuarine (33.6%), pelagic (28%), freshwater (12.4%), mangrove (9.7%) and seagrass (2.6%) habitats (Table S3.3).

Four hundred and four bony fish taxa (class: Actinopterygii) were detected (15 at family level only, 136 at genus level only, and 253 at a species level) from 80 families within 34 orders (Table S3.3). Predominant bony fish families included Gobiidae (gobies; 39 taxa), Labridae (wrasse; 33), Carangidae (jacks and pompanos; 26) and Lutjanidae (snapper; 20), which is in line with the most speciose families in tropical marine environments. Bony fish of conservation, biosecurity, cultural, recreation and/or commercial importance in the Canning and Kimberley bioregions are presented in Table 3.2. This includes the protected Queensland groper (*Epinephelus lanceolatus)*, the highly prized barramundi (*Lates calcarifer*), and potential invasive taxa, such as shokihaze goby (*Tridentiger barbatus*), that present biosecurity risks to Australian aquatic ecosystems. We also report 21 putative new fish occurrence records in the Canning and Kimberley (Table S3.3), however these cannot be fully validated in our study. To verify a new occurrence record based on eDNA metabarcoding, we

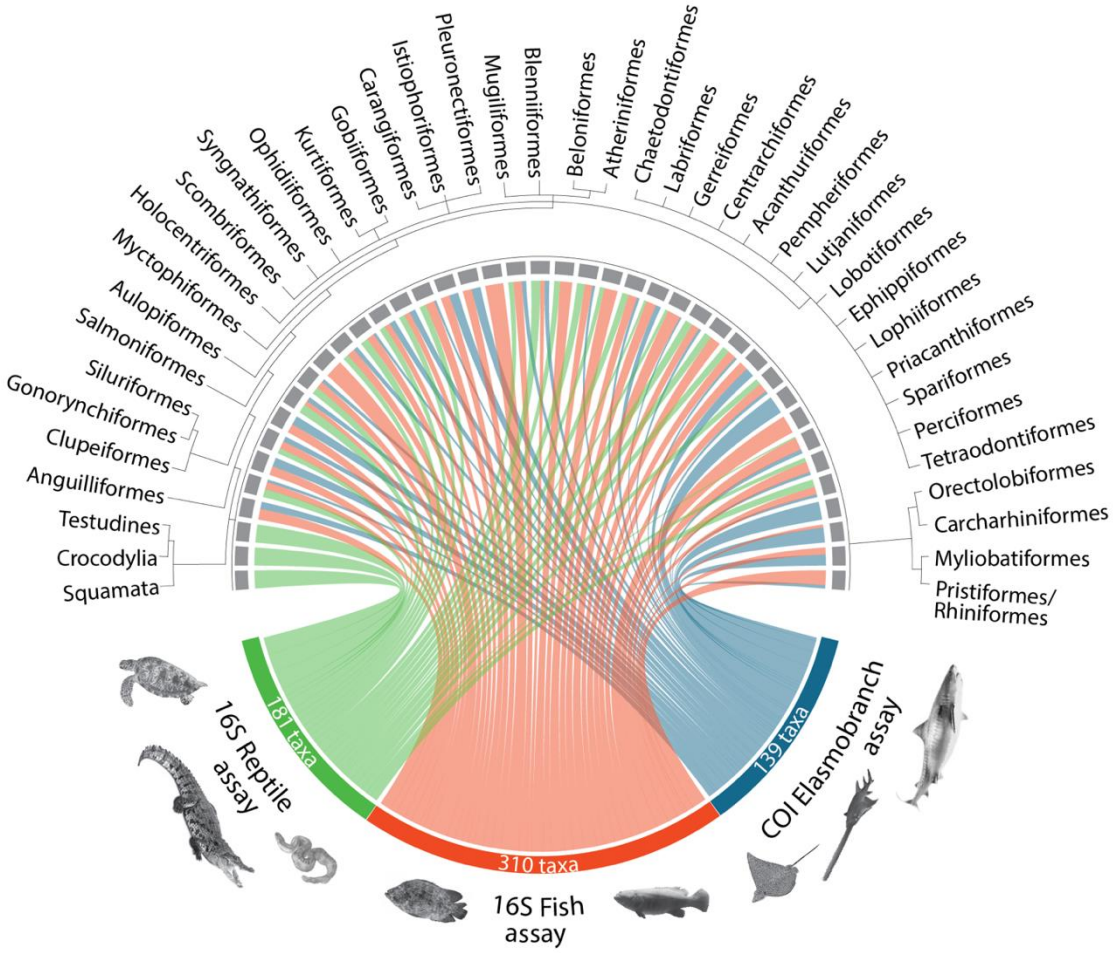


Figure 3.3 Order level dendrogram of bony fish, sharks and reptile diversity detected across the Canning and Kimberley bioregions using eDNA metabarcoding.

Three metabarcoding assays were applied (16S Reptile, 16S Fish and COI Elasmobranch) and produced 181, 310 and 139 unique taxonomic assignments, respectively. As depicted in the *circlize* plot, there were cross-amplification of taxonomic groups between the three assays.

Table 3.2 Selected taxa of conservation, biosecurity, cultural, recreation and/or commercial importance detected via eDNA.

Common Name	Scientific Name	Importance	Site/s detected	Assay	Associated habitat	Distribution
Bony fish						
Queensland groper	*Epinephelus lanceolatus*	Is a protected species in Western Australia, Queensland and New South Wales. Listed as 'data-deficient' by the IUCN.	11, 14, 17, 21-23, 25, 31-33, 38, 40, 41, 64, 67-69	16S Fish, 16S Reptile	Hard substrate	Indo-West Pacific
Barramundi	*Lates calcarifer*	Is of cultural, recreational and commercial fishing importance.	1, 7-9, 14, 19-23, 25-27, 30, 32-36, 38-44, 46, 47, 53, 59, 61-71	16S Fish, COI Elasmobranch	Pelagic, Estuarine	Indo-West Pacific
Shokihaze goby	*Tridentiger barbatus*	Possible invasive species – is a species of concern on the Western Australian Prevention List for Introduced Marine Pests (2016). This detection is a new occurrence record for the Kimberley region, however it cannot be verified as not all congeneric *Tridentiger* have been barcoded for the COI gene region.	38, 40, 41	COI Elasmobranch	Estuarine	Pacific
Freshwater sculpins	*Cottus*	Possible invasive species. This detection is a new occurrence record for the Kimberley region.	71	16S Fish	Freshwater	North America & Eurasia
Dusky spinefoot	*Siganus luridus*	Possible invasive species – is a species of concern on the Western Australian Prevention List for Introduced Marine Pests (2016). This detection is a new occurrence record for the Kimberley region, however it cannot be verified as not all congeneric *Siganus* have been barcoded for the 16S gene region.	17, 26, 34, 35, 37, 47	16S Reptile	Hard substrate	Indian Ocean
Sharks and rays						
Largetooth sawfish	*Pristis pristis*	Is a protected species across Australia. Listed as 'critically endangered' by the IUCN.	62, 68	16S Fish	Soft substrate, Estuarine, Freshwater	Circumglobal
Dwarf sawfish	*Pristis clavata*	Is a protected species across Australia. Listed as 'endangered' by the IUCN.	34, 38, 39	16S Fish	Soft substrate, Estuarine, Mangrove	Northern Australia
Knifetooth sawfish	*Anoxypristis cuspidata*	Is a protected species across Australia. Listed as 'endangered' by the IUCN.	51-54	16S Fish, COI Elasmobranch	Soft substrate, Estuarine, Freshwater	Indo-West Pacific

Wedgefishes	*Rhynchobatus*	Are heavily targeted in the international fin trade. Australia and Papua New Guinea provide a local refuge for *R. australiae*, however this is not enough to improve their global assessment (Kyne et al. 2019). All species in the *Rhynchobatus* genus are listed as 'critically endangered' by the IUCN.	17, 27, 31, 54, 59, 67	16S Fish	Soft substrate	Indo-West Pacific, Atlantic
Northern river shark	*Glyphis garricki*	Is a protected species in Western Australia and the Northern Territory. Listed as 'critically endangered' by the IUCN.	68, 71	COI Elasmobranch	Soft substrate, Estuarine, Freshwater	Indo-Australian
Scalloped hammerhead	*Sphyrna lewini*	Is a protected species in Western Australia and New South Wales. Listed as 'endangered' by the IUCN.	41, 43, 51, 53, 54	COI Elasmobranch	Pelagic	Circumglobal
Great hammerhead	*Sphyrna mokarran*	Is a protected species in Western Australia and New South Wales. Listed as 'endangered' by the IUCN.	16, 17, 22, 23, 25, 26, 30, 35, 38, 40, 43, 51, 65	COI Elasmobranch	Hard substrate, Pelagic	Circumglobal

required all congeneric taxa to have been barcoded for the targeted (in this case 16S and COI) gene regions. Therefore, we cannot rule out the possibility that our new occurrence records represent a closely-related (not yet barcoded) taxon.

Forty-four shark and ray taxa (class: Chondrichthyes) were detected from 11 families within four orders (Table S3.3). Predominant families were Carcharhinidae (requiem sharks; 15 taxa), Dasyatidae (stingrays; 14), Myliobatidae (eagle rays; 3) and Pristidae (sawfish; 3). We detected five shark and ray taxa that are listed as either 'endangered' or 'critically endangered' on the IUCN Red List and are under various national and state protection management (see Table 3.2); these taxa were the largetooth sawfish (*Pristis pristis*), the dwarf sawfish (*Pristis clavata*), the knifetooth sawfish (*Anoxypristis cuspidate*), wedgefishes (genus: *Rhynchobatus*), the northern river shark (*Glyphis garricki*), the scalloped hammerhead (*Sphyrna lewini*) and the great hammerhead (*Sphyrna mokarran*). Other rare taxa include the Australian weasel shark (*Hemigaleus australiensis*), the snaggletooth shark (*Hemipristis elongata*) and the pigeye shark (*Carcharhinus amboinensis*).

Only five reptile taxa (class: Reptilia) were detected from five families within three orders (Table S3.3): the saltwater crocodile (*Crocodylus porosus*), the black-headed python (*Aspidites melanocephalus*), Stokes's sea snake (*Hydrophis stokesii*), the white-bellied mangrove snake (*Fordonia leucobalia*) and the green turtle (*Chelonia mydas*). Given the low frequency of detection of these taxa, reptiles were excluded from all multivariate analyses.

3.5.3 Community composition

3.5.3.1 Bony fish composition

Bony fish composition was examined independently in each habitat type (inshore, coastal and nearshore estuarine), excluding the midshelf habitat which was only comprised of one site. In regards to inshore bony fish compositions, a distance based linear model (DistLM) analysis indicated that bioregion (i.e. the Canning and Kimberley bioregions) explained the highest proportion of fitted variance, followed

by site depth (Table 3.3; Table S3.4). Pairwise DistLM analyses between each of the subregions (i.e. Canning, South Kimberley and North Kimberley), again identified a highly significant bioregional influence on the inshore bony fish compositions between the Canning and adjacent South Kimberley region, and between the Canning and North Kimberley region. The inshore bony fish composition between the South Kimberley and North Kimberley regions however, were found to transition based on site depth and latitudinal gradients. Bony fish composition across all inshore sites is visualised in a PCO in Figure 3.4a. Cumulatively, the significant spatial predictor variables that formed models explained between 15.6-22.4% of total fitted variance between inshore fish assemblages (Table 3.3; Table S3.4).

Taxonomic richness of inshore bony fish was found not to significantly differ between the two bioregions (Table S3.5; Figure S3.13), indicating that the detected bioregional variation is not influenced by uneven taxonomic richness. Similarity percentage analysis (SIMPER) was used to identify prominent inshore bony fish taxa contributing most to pairwise dissimilarity between the Canning and Kimberley bioregions (Table S3.6). This indicated a higher detection rate of sardinella (genus: *Amblygaster*), purple tuskfish (*Choerodon cephalotes*) and chub mackerels (genus: *Rastrelliger*) in the Canning bioregion, whilst the Kimberley region had a higher detection rate of Spanish mackerel (genus: *Scomberomorus*), giant trevally (*Caranx ignobilis*) and blue tuskfish (*Choerodon cyanodus*).

In examining coastal fish assemblages (that were only surveyed in the South and North Kimberley subregions), a DistLM analysis indicated that subregion was a highly significant predictor variable, explaining 17.8% of fitted variance (Table 3.3; Table S3.7; see also Figure 3.4b PCO). For bony fish composition in nearshore estuarine sites (only surveyed in the South Kimberley), there were no significant tested predictor variables that could explain fitted variance (Table S3.8; Figure 3.4c PCO).

3.5.3.2 Shark and ray composition

Shark and ray composition across all inshore sites were found to be driven by subregion, explaining 18.1% of fitted variance (Table 3.4 DistLM; Figure 3.5a PCO;

Table 3.3 Summary table of the distance based linear model (DistLM) analyses for bony fish.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. The predictor variables highlighted in bold are significant (P<0.05). Full DistLM results, including marginal tests and best solutions, are provided in Tables S3.4, S3.7, S3.8.

Sites	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*	
Inshore						
All sites	**Bioregion**	3.281	0.086	0.086	0.000	***
	Depth	2.878	0.071	0.157	0.000	***
Canning & South Kimberley	**Bioregion**	2.913	0.122	0.122	0.000	***
	Depth	2.621	0.102	0.224	0.000	***
South & North Kimberley	**Depth**	2.960	0.096	0.096	0.000	***
	Latitude	1.934	0.060	0.156	0.000	***
Canning & North Kimberley	**Bioregion**	3.501	0.156	0.156	0.000	***
Coastal						
South & North Kimberley	**Subregion**	3.239	0.178	0.178	0.000	***

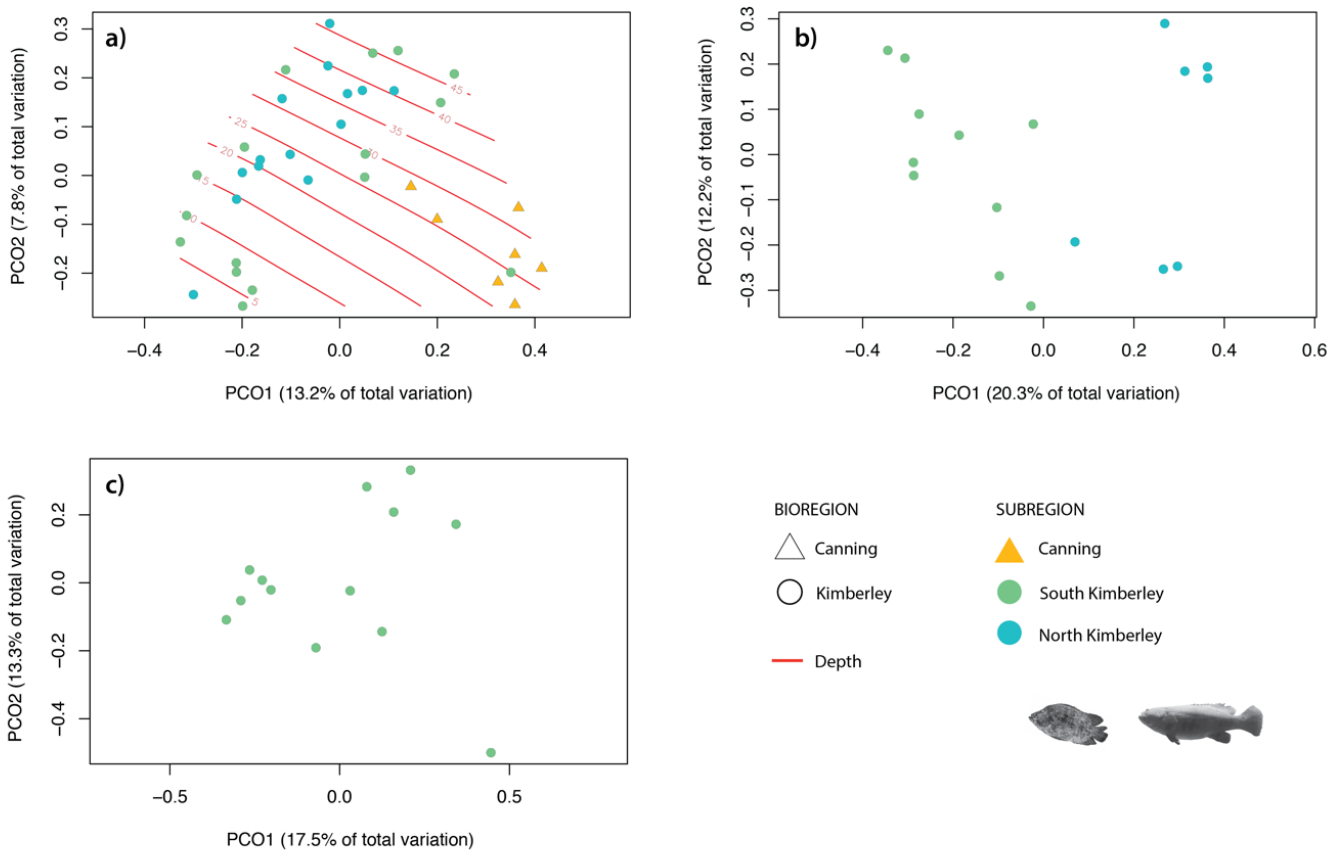


Figure 3.4 Principal coordinates analysis (PCO) of bony fish composition in a) inshore, b) coastal and c) nearshore estuarine sites. Depth gradients are plotted if they are a significant predictor variable in the corresponding DistLM analyses. The proportion of variation explained by each axis is shown on the axis labels.

Table 3.4 Summary table of the distance based linear model (DistLM) analyses for sharks and rays.
These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. The predictor variables highlighted in bold are significant (P<0.05). Full DistLM results, including marginal tests and best solutions, are provided in Tables S3.9, S3.12, S3.13.

Sites	Predictor	Adj R^2	Proportion	Cumulative Proportion	*P*	
Inshore						
All sites	**Subregion**	2.6572	0.181	0.181	0.000	***
Canning & South Kimberley	**Bioregion**	2.836	0.191	0.191	0.011	*
South & North Kimberley	**Subregion**	2.499	0.098	0.098	0.001	***
Canning & North Kimberley	**Bioregion**	2.784	0.176	0.176	0.010	*
Coastal						
South & North Kimberley	**Depth**	1.952	0.131	0.131	0.017	*

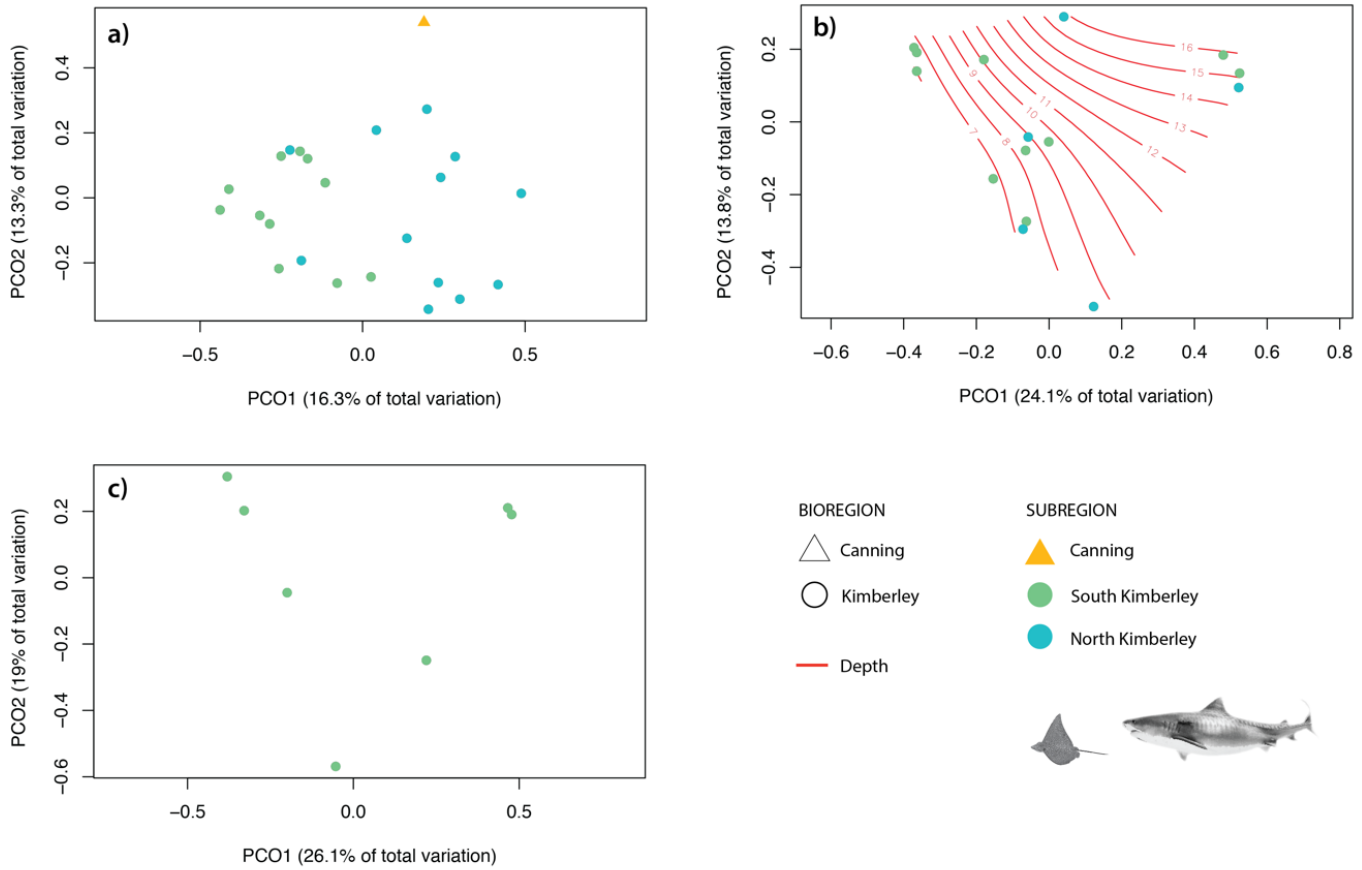


Figure 3.5 Principal coordinates analysis (PCO) of shark and ray composition in a) inshore, b) coastal and c) nearshore estuarine sites. Depth gradients are plotted if they are a significant predictor variable in the corresponding DistLM analyses. The proportion of variation explained by each axis is shown on the axis labels.

Table S3.9). Pairwise DistLM analyses between each of the three subregions revealed a bioregional effect between the Canning and South Kimberley, and between the Canning and North Kimberley, in addition to a subregional influence between the South and North Kimberley (Table 3.4; Table S3.9). Cumulatively, the significant spatial predictor variables that formed models explained between 9.8-19.1% of total fitted variance between inshore sites (Table 3.4). Taxonomic richness of inshore sharks and rays did not significantly differ between the three bioregions (Table S3.10; Figure S3.14), however within the Canning bioregion only two sites had detectable traces of shark and ray taxa. SIMPER analysis was used to identify prominent inshore shark and ray taxa contributing most to pairwise dissimilarity between the three subregions (Table S3.11). Bluespotted maskray (*Neotrygon kuhlii*) dominated detections in the Canning region, mangrove whipray (*Himantura granulate*), bluespotted ribbontail ray (*Taeniura lymma*) and brownbanded bamboo shark (*Chiloscyllium punctatum*) in the South Kimberley subregion, whilst the North Kimberley had a higher detection of requiem sharks such as grey reef shark (*Carcharhinus amblyrhynchos*), bull shark (*Carcharhinus leucas*), blacktip reef shark (*Carcharhinus melanopterus*) and spot-tail shark (*Carcharhinus sorrah*).

In examining shark and ray taxa composition within coastal sites (South and North Kimberley only), site depth was the only significant predictor variable, explaining 13.1% of total fitted variance (Table 3.4; Table S3.12; see also Figure 3.5b PCO). For the nearshore estuarine sites, there were no significant tested predictor variables that could explain fitted variance (Table 3.4; Table S3.13; Figure 3.5c).

3.5.4 Haplotype (ASV) diversity in stripey snapper (*L. carponotatus*)

Following data filtering with DADA2, a total of 22 unique ASVs were assigned to stripey snapper – these were detected in 40 sites across the Canning, South Kimberley and North Kimberley subregions. The most abundant ASV matched with 100% percent identity to a 16S rDNA reference sequence for stripey snapper in NCBI's GenBank repository (accession number: DQ784730.1) and had an overall abundance of 64,846 sequences across 39 sites. All other ASVs, corresponding to unique 16S rDNA haplotypes, radiated from this central haplotype by 1-3bp mutational steps as plotted in a median-joining network analysis (Figure 3.6). Sequencing abundance for

the radiating haplotypes ranged from 1089 to 6 sequences with the lowest abundant ASV found in more than one site, reducing the possibility that it was a false positive haplotype (as a result of sequencing error), but not entirely ruling it out. A DistLM analysis examining variation in the composition of stripey snapper 16S haplotypes across our surveyed sites did identify a bioregional effect, however this predictor variable was not significant (P=0.069; Table S3.14).

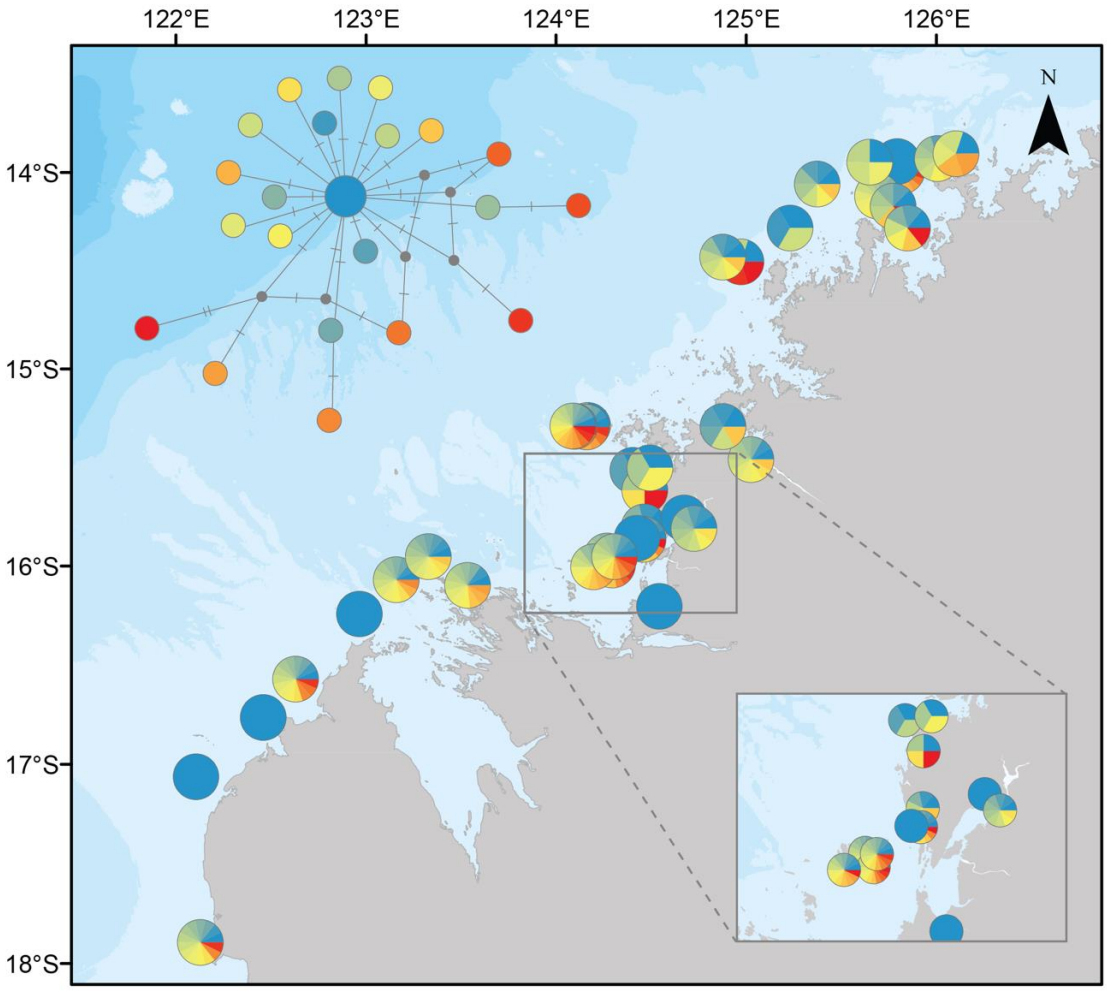


Figure 3.6 Median-joining (MJ) network of 16S rDNA haplotypes and corresponding haplotype distribution map for the stripey snapper (*Lutjanus carponotatus*) in NWA.
The haplotype network in the top left corner is comprised of individual ASVs that radiate 1-3bp mutational steps from a central haplotype (100% identity match to a 16S reference sequence for stripey snapper; NCBI accession number: DQ784730.1). The short slashes between haplotypes represent mutational steps. The distribution map illustrates the proportion of haplotypes identified at 40 sites where stripey snapper were detected by eDNA metabarcoding.

3.6 Discussion

3.6.1 Bony fish compositional transitions across NWA

The Canning and Kimberley bioregions have some of the least impacted marine and coastal ecosystems in the world (Halpern et al., 2008). There are over 1,500 reported species of bony fish across the Canning and Kimberley coastlines (Fox & Beckley, 2005; Moore, Morrison, & Johnson, 2020; Moore, Morrison, Hutchins, Allen, & Sampey, 2014). This synthesis is the result of numerous surveys and museum records since the 1880's (Hutchins, 2001; Moore et al., 2020; Moore et al., 2014; Paxton, Gates, & Bray, 2006). This singular eDNA study detected 404 bony fish taxa from 80 families across 71 sites, which comprises nearly a third of the total fish assemblage known from this region. This detection rate is comparable to a previous trawling survey of inshore fish fauna (up to 361 species from 85 families), revealing faunal transitions over NWA (Travers et al., 2010). With a similar high detection rate, we expected the eDNA site assemblages to be well-representative of the overall fish composition and reveal fine-scale changes across the Canning and Kimberley bioregions. Potential confounding effects of water movement on eDNA profiles between our sites (>4 km apart) is expected to be minimal, given a growing body of evidence indicating the localisation of eDNA over small spatial scales (<1km) (Jeunen et al., 2018; Koziol et al., 2018; Murakami et al., 2019; Stat et al., 2019; West et al., 2020).

The profound change in the underlying geomorphology and ensuing offshore, inshore and coastal habitats of the Canning and Kimberley bioregions corresponds to documented transitions in marine faunal composition across Cape Leveque. Trawling and trapping surveys of inshore fish fauna across NWA (Travers et al., 2006; 2010) revealed a latitudinal transition in addition to a depth effect, overlaid on a strong bioregional influence between the Canning and Kimberley bioregions. This built upon earlier meta-analyses and visual surveys of fish distributions across Western Australia which identified a distinction between the Kimberley and the Northwest shelf region (Fox & Beckley, 2005; Hutchins, 2001; Hutchins, 1997). In examining the inshore bony fish compositions that traverse the purported biogeographic break across Cape Leveque, we found that site dissimilarity is driven by a bioregional effect (in addition

to site depth) across the Canning and South Kimberley. This provides new evidence to support the existence of the biogeographic break on inshore fish composition. It also reveals the ability of eDNA to detect fine-scale changes across large-scale regions, even with a suboptimal level of replication to maximise observed species richness.

Surprisingly, we find no significant difference in inshore fish taxa richness (i.e. total number of species detected) between the Canning and Kimberley regions. This contradicts recent research (Travers, Clarke, Newman, Hall, & Potter, 2018; Travers, Potter, Clarke, & Newman, 2012) indicating regional variation in species richness and abundance between the Kimberley, Canning and Pilbara; the authors expecting to find a poleward latitudinal decline in species richness as demonstrated in numerous global studies (Hillebrand, 2004; Jablonski, Roy, & Valentine, 2006). This study found no evidence of either a bioregional or latitudinal decline in inshore richness across the Kimberley and Canning regions. However, it is possible that a circumtropical study would be required to discern a global latitudinal effect on species richness.

The coastal bony fish assemblages, surveyed in this study across the South and North Kimberley regions, were found to be influenced by a subregional effect. A recent compilation of shallow water fish across the Kimberley documents a total of 1,529 species (Moore et al., 2020). In agreement with Hutchins (2001) inshore survey, Hutchins (1997) identified discrete nearshore fish assemblages on either side of Cape Leveque and in the northeast Kimberley (east of Cape Londonderry), the latter typifying species found in the Northern Territory. Hutchins (1997) attributes the bioregional breaks in nearshore fish fauna to varying environmental conditions, such as increased turbidity in the King Sound area, which may prevent species from ranging further northwards. Unfortunately, the coastal sites surveyed in this study did not range beyond the South and North Kimberley, corresponding to the central Kimberley coast, therefore we could not examine whether this is a possible bioregional break in coastal fish between the Canning and South Kimberley, and a subregional break between the north and northeast Kimberley. However, this is the first report of a potential subregional break between coastal fish in the South Kimberley and North Kimberley regions.

3.6.2 Shark and ray compositional transitions across NWA

Shark and ray fauna are underrepresented in NWA marine surveys, with scant information on populations and species compositions across the region. Elasmobranchs are not regularly caught in trawl surveys (Travers et al., 2012) and can elude visual observations in NWA; particularly taxa such as sawfish, which are benthic in nature and prefer turbid environments (Simpfendorfer et al., 2016). In examining shark and ray eDNA-derived compositional data across the Canning and Kimberley regions, we detected a significant bioregional influence on inshore species on either side of the purported biogeographic break, and a subregional influence on species found across the South and North Kimberley regions.

This widespread bioregional and subregional influence on inshore shark and ray composition across NWA was surprising, given inshore shark species exhibit a wide variety of life-history strategies and are highly mobile (Harry et al., 2011). A global distribution study of elasmobranchs indicated that mean area of occupancy, temperature and bathymetry were the highest contributing predictors of species richness (Guisande et al., 2013)—the area of occupancy relating to Rapoport's rule where it is predicted that species occupy greater ranges with increasing latitude (Rapoport, 1982). Guisande et al., (2013) reported a highly significant inverse relationship between species richness and mean area of occupancy for both sharks and batoids. Therefore, we expected a significant latitudinal influence on the composition and species richness of our broad-scale inshore shark and ray assemblages. However, in addition to no latitudinal influence on the composition, we reported no significant difference in inshore shark and ray richness between the Canning, South Kimberley and North Kimberley regions; however, it should be noted that we only detected one ray species (bluespotted maskray; *N. kuhlii*) at two locations in the Canning region.

A similar elasmobranch survey in terms of latitudinal scope and position was conducted using BRUV technology across the entire Great Barrier Reef Marine Park (GBRMP). This ten-year survey reported that shark composition across the GBRMP did not follow the expected poleward decline of species richness described by Guisande et al., (2013); instead the outermost northern and southern localities exhibited the highest species richness, whilst the intermediate latitudes had the lowest

(Espinoza, Cappo, Heupel, Tobin, & Simpfendorfer, 2014). The distribution of a subset of indicator shark species were highly influenced by relative distance across the shelf, reef proximity and percent of hard coral cover (Espinoza et al., 2014). Acoustic array data from a series of surveys conducted at the Ningaloo Reef Marine Park (NMP), further south along the Western Australian coastline, indicated that several shark and ray species tended to remain in discrete regions of the NMP with little exchange, despite their apparent mobility (Stevens, Last, White, McAuley, & Meekan, 2009). It is apparent that inshore and offshore shark and ray composition and richness across both the north-western and north-eastern coastlines of Australia do not follow an incremental pattern of latitudinal change, but are influenced by external environmental, and in the case of NWA, bioregional and subregional conditions.

Interestingly, the coastal shark and ray assemblages in the South Kimberley and North Kimberley regions were solely influenced by site depth, unlike the inshore assemblage which exhibited subregional variation. The coastal sharks and rays detected in this study exhibit a wide range of associated depths, from those commonly found in shallow coral and sandy areas, e.g. the blacktip reef shark (*C. melanopterus*), bull shark (*C. leucas*) and tawny nurse shark (*Nebrius ferrugineus*); to deeper coral reef drop-offs, e.g. grey reef shark (*C. amblyrhynchos*) and whitetip reef shark (*Triaenodon obesus*); to pelagic sharks, e.g. the Australian sharpnose shark (*Rhizoprionodon taylori*), great hammerhead (*Sphyrna mokarran*) and scalloped hammerhead (*Sphyrna lewini*).

3.6.3 Haplotype community variation in stripey snapper across NWA

Intraspecific diversity analyses of target fish species have typically required the recovery of individual tissue samples using traditional capture methods, e.g. baited traps, casting nets, rotenone and electrofishing. This approach requires handling and sampling live animals, and typically focusses on a single target species at a time. The use of high-throughput sequencing in the generation of eDNA-derived data can, after the removal of erroneous sequences, comprehensively reveal intraspecific population structure (Baker, Steel, Nieukirk, & Klinck, 2018; Elbrecht, Vamos, Steinke, & Leese, 2018; Parsons et al., 2018; Sigsgaard et al., 2016; Stat et al., 2017). Amplicon sequence variants (ASVs; also referred to as zero-radius OTUs) that are confidently matched to

a known species are equivalent to unique haplotypes gained from individual tissue samples. While the use of mtDNA haplotypes will never have the resolution of GBS datasets it may facilitate a cost-effective way to identify genetic patterns in metabarcoding data that extend beyond taxon lists.

Stripey snapper (*L. carponotatus*) is a widely-distributed Indo-West Pacific fish species and is a highly abundant and targeted recreational fishing species on tropical inshore reefs across NWA (Anderson & Allen, 2001; Ryan et al., 2015). Hyper-variable region 1 (HVR 1 or control region) data indicate that there are two distinct lineages across NWA; the larger Clade A comprised of 83 haplotypes found across all three bioregions, whilst the smaller Clade B, comprised of 9 haplotypes, was largely from the Kimberley region. Based on a lack of control region phylogeographic structure however, it was suggested that stripey snapper form a single, panmictic population across NWA (Veilleux, van Herwerden, Evans, Travers, & Newman, 2011). However, a successive genome-wide survey of stripey snapper, which genotyped nuclear 4,402 SNP loci in 1,016 individual fish collected from 51 locations across NWA, identified distinct genetic clusters representing a Kimberley/Northern Territory and a Pilbara/Canning cluster, with an admixed transition zone spanning 180km across the border of the Kimberley and Canning bioregions (DiBattista et al., 2017). Low spatial autocorrelation across the transition region likely represents restricted dispersal across the tip of the Dampier Peninsula and Buccaneer Archipelago, coinciding with the purported bioregional break in marine communities (DiBattista et al., 2017).

Our study derived 22 16S rDNA ASVs that closely matched to stripey snapper reference sequences. These haplotypes, radiating from one central haplotype by 1 to 3bp, were deemed to be legitimate haplotypes as they had been conservatively filtered through the DADA2 pipeline (Callahan et al. 2016); which is effective in removing the majority of false positive haplotypes that are generated by sequencing artefacts (Tsuji et al., 2018). In examining the community composition of our stripey snapper haplotypes across 40 sites in the Canning, South Kimberley and North Kimberley regions, we report a bioregional influence on site haplotype composition, and although marginally significant (P=0.06), it did not meet the significance threshold. Whilst this corroborates the previous mitochondrial control region study in

informing no maternal population structure across NWA, it does not support the in-depth SNP research indicating distinct populations on either side of the biogeographic break. This difference in outcome likely reflects the lower variability of the 16S gene, in addition to the short length of the amplicon.

3.6.4 A new approach for surveying endangered, elusive and data-deficient taxa in NWA

The coastline of NWA exhibits high turbidity resulting from immense tidal action and seasonal discharge from major rivers (Semeniuk, 1993; Thackway & Cresswell, 1998). This limits visibility and subsequently the application of visual surveillance techniques, such as BRUVs and marine manta tows. Additionally, the widespread presence of saltwater crocodiles restricts diving even in inshore coral reef areas. eDNA metabarcoding has thus provided an alternative biomonitoring approach which bypasses many of the logistical and safety limitations of working in this marine region. Our multi-marker eDNA metabarcoding survey has successfully amplified a wide-range of bony fish, shark and ray taxa, including 21 putative new occurrence records (requiring further validation), a number of endangered, elusive and data-deficient taxa, and three potential biosecurity threats detected in the Kimberley for the first time. Aquatic reptiles were under-detected in this study based on low-detection rates, which was unexpected given the widespread distribution of saltwater crocodiles (*C. porosus*) and marine turtles (superfamily: Chelonioidea) across the Canning and Kimberley regions; this may reflect an emerging challenge in regards to the shedding of reptilian skin cells and subsequent detection with eDNA (Adams, Hoekstra, Muell, & Janzen, 2019; see S3.9.3 for further discussion). Despite a low detection rate, this is the first study to our knowledge, to have detected crocodiles and sea snakes using an eDNA approach under field conditions.

A species-specific eDNA assay has previously been developed to detect the largetooth sawfish (*P. pristis*) across northern Australia (Simpfendorfer et al., 2016). A significant finding in this study was the detection of three out of the four globally endangered sawfish taxa (family: Pristidae) found in Australia using a metabarcoding approach. The largetooth sawfish is a euryhaline elasmobranch species that was once globally distributed in tropical marine, estuarine and freshwater environments of the Eastern and Western Atlantic, Eastern Pacific and Indo-West Pacific, however,

population declines and extirpation have led to significant range contractions (Kyne, Carlson, & Smith, 2013). It is currently listed as Critically Endangered on the IUCN Red List of Threatened Species and is a protected species in Western Australia, the Northern Territory and Queensland; Northern Australia may be the last viable stronghold for the Indo-Pacific population and likely comprises a large proportion of the remaining global population (Kyne, Carlson, et al., 2013; Last & Stevens, 2009). In Western Australia, this species has been identified in the King Sound, Fitzroy, Durack, Robinson and Ord Rivers; with eDNA detections this distribution now extends to the Gairdner River in the Southern Kimberley, and Robroy Reef, an inshore site in the North Kimberley.

The dwarf sawfish (*P. clavata*) and the knifetooth sawfish (*A. cuspidate*) are both currently listed as Endangered on the IUCN Red List and like their confamiliars, have undergone significant, yet largely unquantified declines in and outside of Australia (D'Anastasi, Simpfendorfer, & Van Herwerden, 2013; Kyne, Rigby, & Simpfendorfer, 2013). Scattered records of the dwarf sawfish across the Indo-West Pacific indicate that it may have been widely distributed, however there has been a lack of confirmed records outside of Australia since the 1800s (Kyne, Rigby, et al., 2013). Within Australia, the Kimberley and northern Pilbara regions represent significant strongholds for the dwarf sawfish; these include sites in the King Sound, Fitzroy, May and Robinson Rivers, Hall Point and Cape Kerauden (Stevens, McAuley, Simpfendorfer, & Pillans, 2008; Thorburn, Morgan, Rowland, Gill, & Paling, 2008). eDNA detection extends the distribution of dwarf sawfish to Wildcat Reef, George Water and the Glenelg River in the South Kimberley. The knifetooth sawfish distribution extends across the Indo-West Pacific from the Persian Gulf to Japan and the central coasts of western and eastern Australia (Last & Stevens, 2009); eDNA of knifetooth sawfish was detected in Cape Bougainville, Troughton Island, Freshwater Bay and Vansittart Bay in the North Kimberley.

The northern river shark (*G. garricki*) is considered an extremely rare species with limited distribution information and no population estimates available (Field et al., 2013). All identified populations are considered to be of high conservation value and as such, recreational fishing of the species is banned under Australian federal law. The detection of the northern river shark in this study extends its current known distribution

in Western Australia from scattered sightings in the King Sound (Thorburn & Morgan, 2004, 2005), Doctors Creek (Compagno, White, & Last, 2008), Ord River, King River and Joseph Bonaparte Gulf (Pillans, Stevens, Kyne, & Salini, 2009) to the Gairdner River and the Walcott River in the South Kimberley region. These additional distribution records of endangered elasmobranchs will enable appropriate protections to be put in place and are already informing where additional surveys will occur.

The detection of potential invasive species in NWA demonstrates the ability of eDNA metabarcoding to derive both biodiversity and biosecurity information from the same samples and dataset. The Shokihaze goby (*Tridentiger barbatus*) is a brackish/demersal species native to the northwest Pacific region. Unfortunately, ballast water transport has enabled a successful invasion of the Shokihaze goby in San Francisco Bay, California (Wonham, Carlton, Ruiz, & Smith, 2000), where it is now a highly abundant species consuming a variety of benthic invertebrates (Slater, 2005). Whilst the detection of this species in three sites in the South Kimberley region cannot be entirely verified by eDNA (particularly as not all congeneric species of *Tridentiger* have been barcoded for COI), it does flag further investigation as a species of concern on the Western Australian Prevention List for Introduced Marine Pests (2016). This extends to the detection of the dusky spinefoot rabbifish (*Siganus luridus*) in this study; it is also flagged on the WA Prevention List but given incomplete barcoding of the *Siganus* genus, requires further verification. This potential invasive species was detected in six sites, widely extending across the Kimberley from Montgomery Reef to Long Reef. Lastly, the detection of freshwater sculpin (genus: *Cottus*) in the Walcott inlet, an estuary in the South Kimberley region, is a putative new occurrence record and may present a biosecurity concern. This detection is unusual given that sculpins are typically found in colder, temperate environments, however the genus *Antipodocottus* (of the encompassing Cottidae family) has previously been reported off the Pilbara coast. At this point in time, we cannot rule out that this new occurrence record for *Cottus* may actually represent *Antipodocottus*, as the latter genus does not yet have any accessible sequences on NCBI's GenBank nucleotide database.

3.7 Conclusion

We demonstrate in this study that eDNA metabarcoding is a highly sensitive detection tool, capable of producing large amounts of high-resolution (e.g. to a species level) presence-absence data that can discern fine-scale patterns across large geographic regions. Further broad-scale applications of this technique could be used to potentially reveal marine biogeographic breaks in other regions. For example, significant phylogeographic structure in mantis shrimp and seahorses in southeast Asia are purported to reflect historical oceanographic divisions; the former exhibits divergence along a sharp genetic break between the Indian and Pacific Ocean regions (previously separated during the Last Glacial Maximum), whilst the latter is separated into east and west lineages reminiscent of the terrestrial Wallace's Line (Barber, Palumbi, Erdmann, & Moosa, 2000; Lourie & Vincent, 2004). eDNA metabarcoding could be used to assess whether these phylogeographic breaks reflect wider biogeographic partitioning in marine community compositions.

The eDNA samples resulting from this study will be archived and available for further assay applications extending beyond the taxonomic groups targeted in this study. Additionally, our sequencing data can be retrospectively analysed with expanding databases to resolve ambiguous taxonomic assignments. Our georeferenced sites herein will be used as a baseline for future eDNA biomonitoring and notably will direct targeted surveying for the critically endangered elasmobranchs and potential invasive species across NWA. We anticipate that eDNA metabarcoding will be integrated into broad-scale monitoring tool kits, particularly in northern Australia, where it bypasses many of the logistical and safety limitations of visual surveillance. At present this technique is limited in its ability to provide quantitative data in relation to population sizes and biomass. However, its ability to produce multi-taxon data, potentially whole-ecosystem data, without the need for taxonomic expertise is both time and cost-efficient; innovative approaches to whole-ecosystem surveying is increasingly needed amidst global population declines and subsequent limited conservation/management resources. Accurate taxonomic information is required to implement best practice management programs tasked with preserving biodiversity, fisheries, tourism and cultural values in this remote region of Australia.

3.8 References

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Adams, C. I. M., Hoekstra, L. A., Muell, M. R., & Janzen, F. J. (2019). A Brief Review of Non-Avian Reptile Environmental DNA (eDNA), with a Case Study of Painted Turtle (*Chrysemys picta*) eDNA Under Field Conditions. *Diversity*, *11*(4), 50.

Akerman, K. (2015). A review of the Indigenous watercraft of the Kimberley region, Western Australia. *The Great Circle*, *37*(1), 82–111.

Anderson, M., Gorley, R. N., & Clarke, R. K. (2008). *Permanova+ for Primer: Guide to Software and Statisticl Methods*. Primer-E Limited.

Anderson, W. D., & Allen, G. R. (2001). Lutjanidae. In *FAO species identification guide for fishery purposes. The living marine resources of the Western Central Pacific. Volume 5, part 3.* (pp. 2840–2918).

Baker, C. S., Steel, D., Nieukirk, S., & Klinck, H. (2018). Environmental DNA (eDNA) from the wake of the whales: droplet digital PCR for detection and species identification. *Frontiers in Marine Science*, *5*, 133.

Barber, P. H., Palumbi, S. R., Erdmann, M. V, & Moosa, M. K. (2000). A marine Wallace's line? *Nature*, *406*(6797), 692–693.

Benson, D. A., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Wheeler, D. L. (2005). GenBank. *Nucleic Acids Research*, *33*, D34–D38. https://doi.org/10.1093/nar/gki063

Berry, T. E., Osterrieder, S. K., Murray, D. C., Coghlan, M. L., Richardson, A. J., Grealy, A. K., … Bunce, M. (2017). DNA metabarcoding for diet analysis and biodiversity: A case study using the endangered Australian sea lion (*Neophoca cinerea*). *Ecology and Evolution*, *7*(14), 5435–5453.

Berry, T. E., Saunders, B. J., Coghlan, M. L., Stat, M., Jarman, S., Richardson, A. J., … Bunce, M. (2019). Marine environmental DNA biomonitoring reveals seasonal patterns in biodiversity and identifies ecosystem responses to anomalous climatic events. *PLOS Genetics*, *15*(2), e1007943. https://doi.org/10.1371/journal.pgen.1007943

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a Unix-inspired software package for DNA metabarcoding. *Molecular Ecology Resources*, *16*(1), 176-182.

Callahan, B. J., McMurdie, P. J., Rosen, M. J., Han, A. W., Johnson, A. J. A., & Holmes, S. P. (2016). DADA2: high-resolution sample inference from Illumina amplicon data. *Nature Methods*, *13*(7), 581.

Caporaso, J. G., Kuczynski, J., Stombaugh, J., Bittinger, K., Bushman, F. D., Costello, E. K., … Gordon, J. I. (2010). QIIME allows analysis of high-throughput community sequencing data. *Nature Methods*, *7*(5), 335.

Clarke, K. R., & Gorley, R. N. (2015). *PRIMER v7: User Manual/Tutorial*. Plymouth, UK.

Compagno, L. J. V, White, W. T., & Last, P. R. (2008). *Glyphis garricki* sp. nov., a new species of river shark (Carcharhiniformes: Carcharhinidae) from northern Australia and Papua New Guinea, with a redescription of Glyphis glyphis (Müller & Henle, 1839). *Descriptions of New Australian Chondrichthyans*, CSIRO, Hobart, 203–225.

Cresswell, I. D., & Semeniuk, V. (2011). Mangroves of the Kimberley Coast: Ecological patterns in a tropical ria coast setting. *Journal of the Royal Society of Western Australia*, *94*(2), 213.

D'Anastasi, B., Simpfendorfer, C. A., & Van Herwerden, L. (2013). *Anoxypristis cuspidata* (errata varsion published in 2019), *The IUCN Red List of Threatened Species*.

Dafforn, K. A., Johnston, E. L., Ferguson, A., Humphrey, C. L., Monk, W., Nichols, S. J., … Baird, D. J. (2016). Big data opportunities and challenges for assessing multiple stressors across scales in aquatic ecosystems. *Marine and Freshwater Research*, *67*(4), 393–413.

Department of Sustainability, Environment, Water, Population and Communities. (2011). *The West Kimberley, Western Australia*.

DiBattista, J. D., Travers, M. J., Moore, G. I., Evans, R. D., Newman, S. J., Feng, M., … Berry, O. (2017). Seascape genomics reveals fine-scale patterns of dispersal for a reef fish along the ecologically divergent coast of Northwestern Australia. *Molecular Ecology*, *26*(22), 6206–6223.

Elbrecht, V., Vamos, E. E., Steinke, D., & Leese, F. (2018). Estimating intraspecific genetic diversity from community DNA metabarcoding data. *PeerJ*, *6*, e4644.

Ellis, S. L., Incze, L. S., Lawton, P., Ojaveer, H., MacKenzie, B. R., Pitcher, C. R., … Zeller, B. M. (2011). Four regional marine biodiversity studies: approaches and contributions to ecosystem-Based Management. *PLoS One*. https://doi.org/10.1371/journal.pone.0018997

Espinoza, M., Cappo, M., Heupel, M. R., Tobin, A. J., & Simpfendorfer, C. A. (2014). Quantifying shark distribution patterns and species-habitat associations: implications of marine park zoning. *PLoS One*, *9*(9), e106885.

Evans, K., Bax, N., & Smith, D. C. (2017). Australia State of the Environment 2016: Marine Environment, independent report to the Australian Government Minister for the Environment and Energy. *Australian Government Department of the Environment and Energy, Canberra*.

Feeken, E. H. J. (1973). First charting of the Northern Shores of Australia. *Cartography*, *8*(1), 34–39.

Field, I. C., Tillett, B. J., Charters, R., Johnson, G. J., Buckworth, R. C., Meekan, M. G., & Bradshaw, C. J. A. (2013). Distribution, relative abundance and risks from fisheries to threatened *Glyphis* sharks and sawfishes in northern Australia. *Endangered Species Research*, *21*(2), 171–180.

Fox, N. J., & Beckley, L. E. (2005). Priority areas for conservation of Western Australian coastal fishes: a comparison of hotspot, biogeographical and complementarity approaches. *Biological Conservation*, *125*(4), 399–410.

Fraija-Fernández, N., Bouquieaux, M.-C., Rey, A., Mendibil, I., Cotano, U., Irigoien, X., … Rodríguez-Ezpeleta, N. (2019). Marine water environmental DNA metabarcoding provides a comprehensive fish diversity assessment and reveals spatial patterns in a large oceanic area. *BioRxiv*, 864710. https://doi.org/10.1101/864710

Froese, R., & Pauly, D. (Eds.) (2018). FishBase. In Species 2000 & ITIS Catalogue of Life. Retrieved from http://www.fishbase.org.

Gaertner, J. C., Maiorano, P., Mérigot, B., Colloca, F., Politou, C. Y., Gil De Sola, L., … Mannini, A. (2013). Large-Scale Diversity of Slope Fishes: Pattern Inconsistency between Multiple Diversity Indices. *PLoS One*, *8*(7). https://doi.org/10.1371/journal.pone.0066753

George, A. S. (1999). *William Dampier in New Holland: Australia's first natural historian*. Bloomings Books Hawthorn, Australia.

Guisande, C., Patti, B., Vaamonde, A., Manjarrés-Hernández, A., Pelayo-Villamil, P.,

García-Roselló, E., … Granado-Lorencio, C. (2013). Factors affecting species richness of marine elasmobranchs. *Biodiversity and Conservation*, *22*(8), 1703–1714.

Halpern, B. S., Walbridge, S., Selkoe, K. A., Kappel, C. V, Micheli, F., D'Agrosa, C., … Fox, H. E. (2008). A global map of human impact on marine ecosystems. *Science*, *319*(5865), 948–952.

Harry, A. V, Tobin, A. J., Simpfendorfer, C. A., Welch, D. J., Mapleston, A., White, J., … Stapley, J. (2011). Evaluating catch and mitigating risk in a multispecies, tropical, inshore shark fishery within the Great Barrier Reef World Heritage Area. *Marine and Freshwater Research*, *62*(6), 710–721.

Hillebrand, H. (2004). On the generality of the latitudinal diversity gradient. *The American Naturalist*, *163*(2), 192–211.

Horton, T., Kroh, A., Ahyong, S., Bailly, N., Boyko, C. B., Brandão, S. N., … Zhao, Z. (2018). World Register of Marine Species (WoRMS). WoRMS Editorial Board. Retrieved from http://www.marinespecies.org

Hutchins, J. Barry. (2001). Biodiversity of shallow reef fish assemblages in Western Australia using a rapid censusing technique. *Records of the Western Australian Museum*, *20*, 247–270.

Hutchins, J B. (2001). Checklist of fishes of Western Australia. *Records of the Western Australian Museum Supplement*, *63*, 9–50.

Hutchins, J Barry. (1997). Biogeography of the nearshore marine fish fauna of the Kimberley, Western Australia. In *Proceedings of the Fifth Indo-Pacific Fish Conference, Noumea* (pp. 99–108).

Jablonski, D., Roy, K., & Valentine, J. W. (2006). Out of the tropics: evolutionary dynamics of the latitudinal diversity gradient. *Science*, *314*(5796), 102–106.

Jeunen, G., Knapp, M., Spencer, H. G., Lamare, M. D., Taylor, H. R., Stat, M., … Gemmell, N. J. (2018). Environmental DNA (eDNA) metabarcoding reveals strong discrimination among diverse marine habitats connected by water movement. *Molecular Ecology Resources*, *19*(2), 426-438.

Koziol, A., Stat, M., Simpson, T., Jarman, S., DiBattista, J. D., Harvey, E. S., … Bunce, M. (2018). Environmental DNA metabarcoding studies are critically affected by substrate selection. *Molecular Ecology Resources*, *19*(2), 366-376.

Kyne, P. M., Carlson, J., & Smith, K. (2013). *Pristis pristis* (errata version published in 2019). *The IUCN Red List of Threatened Species*.

Kyne, P. M., Rigby, C., & Simpfendorfer, C. A. (2013). *Pristis clavata* (errata version published in 2019). *The IUCN Red List of Threatened Species*.

Kyne, P.M, Rigby, C. ., Dharmadi, & Jabado, R. . (2019). *Rhynchobatus australiae*. The IUCN Red List of Threatened Species 2019: e.T41853A68643043.

Last, P. R., & Stevens, J. D. (2009). *Sharks and Rays of Australia. Second Edition.* Collingwood, Australia: CSIRO Publishing.

Leigh, J. W., & Bryant, D. (2015). PopART: Full-feature software for haplotype network construction. *Methods in Ecology and Evolution*, 6(9), 1110–1116.

Lourie, S. A., & Vincent, A. C. J. (2004). A marine fish follows Wallace's Line: the phylogeography of the three-spot seahorse (*Hippocampus trimaculatus*, Syngnathidae, Teleostei) in Southeast Asia. *Journal of Biogeography*, *31*(12), 1975–1985.

McMurdie, P. J., & Holmes, S. (2013). Phyloseq: an R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS One*, *8*(4), e61217.

Moore, G. I., Morrison, S. M., & Johnson, J. W. (2020). The distribution of shallow marine fishes of the Kimberley, Western Australia, based on a long-term dataset and multiple methods. *Records of the Western Australian Museum*, *105*, 115.

Moore, G., Morrison, S., Hutchins, J. B., Allen, G. R., & Sampey, A. (2014). Kimberley marine biota. Historical data: fishes. *Records of the Western Australian Museum Supplement*, *84*, 161–206.

Morgan, D. L., Whitty, J. M., Phillips, N. M., Thorburn, D. C., Chaplin, J. A., & McAuley, R. (2011). North-western Australia as a hotspot for endangered elasmobranchs with particular reference to sawfishes and the Northern River Shark. *Journal of the Royal Society of Western Australia*, *94*(2), 345–358.

Murakami, H., Yoon, S., Kasai, A., Minamoto, T., Yamamoto, S., Sakata, M. K., … Yamashita, Y. (2019). Dispersion and degradation of environmental DNA from caged fish in a marine environment. *Fisheries Science*, *85*(2), 327–337.

O'Connor, S. (2017). *30,000 Years of Aboriginal occupation: Kimberley, north west Australia*. Department of Archaeology and Natural History, The Australian National University, Canberra.

O'Donnell, J. L., Kelly, R. P., Shelton, A. O., Samhouri, J. F., Lowell, N. C., & Williams, G. D. (2017). Spatial distribution of environmental DNA in a nearshore marine habitat. *PeerJ*, *5*. https://doi.org/10.7717/peerj.3044

Oksanen, J., Blanchet, F. G., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., … Wagner, H. (2019). vegan: Community Ecology Package. R package version 2.5-5.

Parsons, K. M., Everett, M., Dahlheim, M., & Park, L. (2018). Water, water everywhere: Environmental DNA can unlock population structure in elusive marine species. *Royal Society Open Science*, *5*(8), 180537. https://doi.org/10.1098/rsos.180537

Paxton, J. R., Gates, J. E., & Bray, D. J. (2006). Zoological catalogue of Australia, fishes. *CSIRO Publishing*, Canberra, *35*, 1–3.

Piacenza, S. E., Barner, A. K., Benkwitt, C. E., Boersma, K. S., Cerny-Chipman, E. B., Ingeman, K. E., … Heppell, S. S. (2015). Patterns and variation in benthic biodiversity in a large marine ecosystem. *PLoS One*, *10*(8). https://doi.org/10.1371/journal.pone.0135135

Pillans, R. D., Stevens, J. D., Kyne, P. M., & Salini, J. (2009). Observations on the distribution, biology, short-term movements and habitat requirements of river sharks *Glyphis* spp. in northern Australia. *Endangered Species Research*, *10*, 321–332.

Port, J. A., O'Donnell, J. L., Romero-Maraccini, O. C., Leary, P. R., Litvin, S. Y., Nickols, K. J., … Kelly, R. P. (2016). Assessing vertebrate biodiversity in a kelp forest ecosystem using environmental DNA. *Molecular Ecology*, *25*(2), 527-541. https://doi.org/10.1111/mec.13481

Rapoport, E. H. (1982). *Areography: geographical strategies of species* (Vol. 1). Elsevier.

Richards, Z., Bryce, M., & Bryce, C. (2018). The composition and structure of shallow benthic reef communities in the Kimberley, north-west Australia. *Records of the Western Australian Museum Supplement*, *85*, 103.

Richards, Z., Sampey, A., & Marsh, L. (2014). Kimberley marine biota. Historical data: scleractinian corals. *Records of the Western Australian Museum Supplement*, *84*(11).

Ryan, K. L., Hall, N. G., Lai, E. K., Smallwood, C. B., Taylor, S. M., & Wise, B. S. (2015). *State-wide survey of boat-based recreational fishing in Western Australia 2013/14*. Fisheries Research Division, Western Australian Fisheries and Marine Research Laboratories.

R Studio Team (2015). RStudio: integrated development for R. *RStudio Inc*, Boston,

MA. Retrieved from *http://www.rstudio.com*

Semeniuk, V. (1993). The mangrove systems of Western Australia: 1993 presidential address. *Journal of the Royal Society of Western Australia, 76*(4), 99–122.

Sigsgaard, E. E., Nielsen, I. B., Bach, S. S., Lorenzen, E. D., Robinson, D. P., Knudsen, S. W., … Thomsen, P. F. (2016). Population characteristics of a large whale shark aggregation inferred from seawater environmental DNA. *Nature Ecology & Evolution, 1*(1), 0004. https://doi.org/10.1038/s41559-016-0004

Simpfendorfer, C. A., Kyne, P. M., Noble, T. H., Goldsbury, J., Basiita, R. K., Lindsay, R., … Jerry, D. R. (2016). Environmental DNA detects Critically Endangered largetooth sawfish in the wild. *Endangered Species Research, 30*, 109–116.

Slater, S. B. (2005). Life history and diet of the Shokihaze goby *Tridentiger barbatus* in the San Francisco Estuary. California State University, Sacramento.

Stat, M., Huggett, M. J., Bernasconi, R., DiBattista, J. D., Berry, T. E., Newman, S. J., … Bunce, M. (2017). Ecosystem biomonitoring with eDNA: metabarcoding across the tree of life in a tropical marine environment. *Scientific Reports, 7*(1), 1-11.

Stat, M., John, J., DiBattista, J. D., Newman, S. J., Bunce, M., & Harvey, E. S. (2019). Combined use of eDNA metabarcoding and video surveillance for the assessment of fish biodiversity. *Conservation Biology*, 33(1), 196-205.

Stevens, J. D., Last, P. R., White, W. T., McAuley, R. B., & Meekan, M. G. (2009). Diversity, abundance and habitat utilisation of sharks and rays. *Final Report, Western Australian Marine Science Institute, Perth, WA, Australia*.

Stevens, J. D., McAuley, R. B., Simpfendorfer, C. A., & Pillans, R. D. (2008). Spatial distribution and habitat utilisation of sawfish (*Pristis* spp) in relation to fishing in northern Australia. *A Report to Department of the Environment, Water, Heritage and the Arts, 31*.

Taillebois, L., Barton, D. P., Crook, D. A., Saunders, T., Taylor, J., Hearnden, M., … Welch, D. J. (2017). Strong population structure deduced from genetics, otolith chemistry and parasite abundances explains vulnerability to localized fishery collapse in a large Sciaenid fish, *Protonibea diacanthus*. *Evolutionary Applications*, *10*(10), 978–993.

Thackway, R., & Cresswell, I. . (Eds.). (1998). Interim marine and coastal regionalisation for Australia: an ecosystem-based classification for marine and

coastal environments. In *Interim Marine and Coastal Regionalisation for Australia Technical Group Report, version 3.3*. Canberra: Environment Australia, Commonwealth Department of the Environment.

Thomsen, P. F., Kielgast, J., Iversen, L. L., Møller, P. R., Rasmussen, M., & Willerslev, E. (2012). Detection of a diverse marine fish fauna using environmental DNA from seawater samples. *PLoS One*. https://doi.org/10.1371/journal.pone.0041732

Thomsen, P. F., Møller, P. R., Sigsgaard, E. E., Knudsen, S. W., Jørgensen, O. A., & Willerslev, E. (2016). Environmental DNA from Seawater Samples Correlate with Trawl Catches of Subarctic, Deepwater Fishes. *PLoS One*, 1–22. https://doi.org/10.1371/journal.pone.0165252

Thomsen, P. F., & Willerslev, E. (2015). Environmental DNA - An emerging tool in conservation for monitoring past and present biodiversity. *Biological Conservation, 183,* 4-18. https://doi.org/10.1016/j.biocon.2014.11.019

Thorburn, D. C., & Morgan, D. L. (2004). The northern river shark *Glyphis* sp. C (Carcharhinidae) discovered in Western Australia. *Zootaxa*, *685*(1), 1–8.

Thorburn, D. C., & Morgan, D. L. (2005). Threatened fishes of the world: *Glyphis* sp. C (Carcharhinidae). *Environmental Biology of Fishes*, *73*(2), 140.

Thorburn, D. C., Morgan, D. L., Rowland, A. J., Gill, H. S., & Paling, E. (2008). Life history notes of the critically endangered dwarf sawfish, *Pristis clavata*, Garman 1906 from the Kimberley region of Western Australia. *Environmental Biology of Fishes*, *83*(2), 139–145.

Tittensor, D. P., Mora, C., Jetz, W., Lotze, H. K., Ricard, D., Berghe, E. Vanden, & Worm, B. (2010). Global patterns and predictors of marine biodiversity across taxa. *Nature*, *466*(7310), 1098–1101.

Travers, M.J, Newman, S. ., & Potter, I. . (2006). Influence of latitude, water depth, day v. night and wet v. dry periods on the species composition of reef fish communities in tropical Western Australia. *Journal of Fish Biology*, *69*(4), 987–1017.

Travers, Michael J, Clarke, K. R., Newman, S. J., Hall, N. G., & Potter, I. C. (2018). To what extents are species richness and abundance of reef fishes along a tropical coast related to latitude and other factors? *Continental Shelf Research*, *167*, 99–110.

Travers, Michael J, Potter, I. C., Clarke, K. R., & Newman, S. J. (2012). Relationships

between latitude and environmental conditions and the species richness, abundance and composition of tropical fish assemblages over soft substrata. *Marine Ecology Progress Series*, *446*, 221–241.

Travers, Michael J, Potter, I. C., Clarke, K. R., Newman, S. J., & Hutchins, J. . (2010). The inshore fish faunas over soft substrates and reefs on the tropical west coast of Australia differ and change with latitude and bioregion. *Journal of Biogeography*, *37*(1), 148–169.

Tsuji, S., Miya, M., Ushio, M., Sato, H., Minamoto, T., & Yamanaka, H. (2018). Evaluating intraspecific diversity of a fish population using environmental DNA: An approach to distinguish true haplotypes from erroneous sequences. *BioRxiv*, 429993.

Underwood, J., Richards, Z., Berry, O., & Gilmour, J. (2017). Population connectivity and genetic diversity in brooding and broadcast spawning corals in the Kimberley. *Western Australian Marine Science Institution*, Perth.

Valentini, A., Taberlet, P., Miaud, C., Civade, R., Herder, J., Thomsen, P. F., … Dejean, T. (2016). Next-generation monitoring of aquatic biodiversity using environmental DNA metabarcoding. *Molecular Ecology, 25*(4), 929-942. https://doi.org/10.1111/mec.13428

Veilleux, H. D., van Herwerden, L., Evans, R. D., Travers, M. J., & Newman, S. J. (2011). Strong genetic subdivision generates high genetic variability among eastern and western Australian populations of *Lutjanus carponotatus* (Richardson). *Fisheries Research*, *108*(1), 74–80.

West, K. M., Stat, M., Harvey, E. S., Skepper, C. L., DiBattista, J. D., Richards, Z. T., … Bunce, M. (2020). eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem. *Molecular Ecology*. https://doi.org/10.1111/mec.15382

Whiteway, T. G. (2009). Australian bathymetry and topography grid. *Geoscience Australia, Canberra*. Retrieved from http://dx.doi.org/10.4225/25/53D99B6581B9A

Wickham, H. (2016). *ggplot2: elegant graphics for data analysis*. Springer.

Wilson, B. (2013). *The biogeography of the Australian North West Shelf: Environmental change and life's response*. Newnes.

Wilson, B. (2014). Kimberley marine biota. History and environment. *Records of the Western Australian Museum Supplement*, *84*, 1–18.

Wonham, M. J., Carlton, J. T., Ruiz, G. M., & Smith, L. D. (2000). Fish and ships: relating dispersal frequency to success in biological invasions. *Marine Biology*, *136*(6), 1111–1121.

3.9 Supplementary information

Table S3.1 Site location information across the Canning/Kimberley bioregions.
S Kimberley and N Kimberley refer to the Southern and Northern Kimberley bioregions, respectively.

Site No.	Coordinates Latitude	Longitude	Location	Bioregion	Subregion	Habitat	Site Depth (m)	Date Collected
1	-17.893883	122.128	Gantheaume Bay	Canning	Canning	Inshore	10	06/09/2017
2	-17.5261	122.06065	James Price Patches	Canning	Canning	Inshore	14	06/09/2017
3	-17.06235	122.105267	Baskerville Shoal	Canning	Canning	Inshore	12	06/09/2017
4	-16.763833	122.459083	Beagle Bay	Canning	Canning	Inshore	19	06/09/2017
5	-16.571317	122.629733	Lord Mayor Shoal	Canning	Canning	Inshore	15	06/09/2017
6	-16.426667	122.777033	Chile Head	Canning	Canning	Inshore	25	06/09/2017
7	-16.2419	122.9654	Pitt Shoals	Canning	Canning	Inshore	40	06/09/2017
8	-16.069217	123.158467	Pincombe Reef	Kimberley	S Kimberley	Inshore	58	06/09/2017
9	-15.952083	123.327983	Caffarelli North	Kimberley	S Kimberley	Inshore	55	06/09/2017
10	-15.85735	123.588633	King Island	Kimberley	S Kimberley	Inshore	44	06/09/2017
11	-15.595717	123.975133	Lorikeet Shoal North	Kimberley	S Kimberley	Inshore	36	07/09/2017
12	-15.51525	124.403483	Wailgwin Islands	Kimberley	S Kimberley	Inshore	18	08/09/2017
13	-15.721	124.409033	Prior Inlet	Kimberley	S Kimberley	Coastal	6.4	08/09/2017
14	-15.616583	124.466467	Deception Bay Inlet	Kimberley	S Kimberley	Coastal	5.5	08/09/2017
15	-15.591483	124.428067	Deception Bay	Kimberley	S Kimberley	Coastal	5.5	08/09/2017
16	-15.502567	124.493333	Sampson Inlet	Kimberley	S Kimberley	Coastal	13	08/09/2017
17	-15.505317	124.583117	Perseverance Inlet	Kimberley	S Kimberley	Coastal	2	08/09/2017
18	-15.621783	125.30905	Prince Regent Upper	Kimberley	S Kimberley	Nearshore Estuarine	5.1	10/09/2017
19	-15.470533	125.1048	Prince Regent Lower	Kimberley	S Kimberley	Nearshore Estuarine	27	10/09/2017
20	-15.46	125.025517	St George Basin	Kimberley	S Kimberley	Nearshore Estuarine	15	10/09/2017
21	-15.293433	124.878583	St George Entrance	Kimberley	S Kimberley	Coastal	20	10/09/2017
22	-15.789418	124.461558	Mud Bay Upper 1	Kimberley	S Kimberley	Coastal	4.1	13/09/2017
23	-15.802308	124.463535	Mud Bay Upper 2	Kimberley	S Kimberley	Coastal	5.4	13/09/2017

24	-15.863917	124.45719	Mud Bay Lower 1	Kimberley	S Kimberley	Coastal	10.2	13/09/2017
25	-15.859962	124.425363	Mud Bay Lower 2	Kimberley	S Kimberley	Coastal	3.6	13/09/2017
26	-15.948117	124.268833	Montgomery Reef Channel Upper	Kimberley	S Kimberley	Inshore	11.5	16/09/2017
27	-15.99485	124.29735	Montgomery Reef Channel Lower	Kimberley	S Kimberley	Inshore	9.5	16/09/2017
28	-16.003683	124.197033	Montgomery Reef South	Kimberley	S Kimberley	Inshore	7.3	16/09/2017
29	-15.952133	124.308383	Montgomery Reef East	Kimberley	S Kimberley	Inshore	4.2	16/09/2017
30	-16.19885	124.591783	3 Ways Upper	Kimberley	S Kimberley	Nearshore Estuarine	8.5	17/09/2017
31	-16.203333	124.543533	3 Ways Lower	Kimberley	S Kimberley	Nearshore Estuarine	20.9	17/09/2017
32	-16.010567	124.744533	Sale River Upper	Kimberley	S Kimberley	Nearshore Estuarine	3.4	17/09/2017
33	-15.971767	124.597933	Sale River Lower	Kimberley	S Kimberley	Nearshore Estuarine	33.5	17/09/2017
34	-15.286967	124.166383	Wildcat Reef East 1	Kimberley	S Kimberley	Inshore	5.4	18/09/2017
35	-15.295483	124.160483	Wildcat Reef East 2	Kimberley	S Kimberley	Inshore	5.3	18/09/2017
36	-15.291917	124.09755	Wildcat Reef West 1	Kimberley	S Kimberley	Inshore	8.8	18/09/2017
37	-15.2905	124.087867	Wildcat Reef West 2	Kimberley	S Kimberley	Inshore	7.2	18/09/2017
38	-15.756833	124.673917	George Water North	Kimberley	S Kimberley	Nearshore Estuarine	4.5	19/09/2017
39	-15.809647	124.725643	Glenelg Entrance	Kimberley	S Kimberley	Nearshore Estuarine	18.4	19/09/2017
40	-15.84928	124.670843	George Water South	Kimberley	S Kimberley	Nearshore Estuarine	3.2	19/09/2017
41	-15.924325	124.6034	George Water Entrance	Kimberley	S Kimberley	Nearshore Estuarine	45	19/09/2017
42	-16.11125	124.02462	Kingfisher Island	Kimberley	S Kimberley	Inshore	70	20/09/2017
43	-16.09698	123.53348	Irvine Island	Kimberley	S Kimberley	Inshore	60	20/09/2017
44	-16.05733	123.42498	Admiral Island	Kimberley	S Kimberley	Inshore	50	20/09/2017
45	-13.57631	125.75828	West Holothuria Reef	Kimberley	N Kimberley	Inshore	22	06/07/2018
46	-13.56566	126.05331	East Holothuria Reef	Kimberley	N Kimberley	Inshore	14	06/07/2018
47	-13.99230	125.81954	Long Reef South East	Kimberley	N Kimberley	Inshore	8	07/07/2018
48	-13.94425	125.79544	Long Reef North East	Kimberley	N Kimberley	Inshore	9	07/07/2018
49	-13.99408	125.97186	Gibson Point	Kimberley	N Kimberley	Coastal	21	08/07/2018
50	-12.92813	126.00680	Hat Point	Kimberley	N Kimberley	Coastal	9	08/07/2018
51	-13.90358	126.10430	Cape Bougainville	Kimberley	N Kimberley	Coastal	12	08/07/2018
52	-13.75881	126.12005	Troughton Island	Kimberley	N Kimberley	Inshore	33	09/07/2018
53	-13.97473	126.25411	Freshwater Bay	Kimberley	N Kimberley	Coastal	14	09/07/2018

54	-14.10344	126.26023	Vansittart Bay	Kimberley	N Kimberley	Coastal	16	09/07/2018
55	-14.12048	125.69103	Fenelon Island	Kimberley	N Kimberley	Inshore	20	10/07/2018
56	-14.28276	125.23090	West Montalivet Island	Kimberley	N Kimberley	Inshore	42	11/07/2018
57	-14.05870	125.37400	Jamieson Reef	Kimberley	N Kimberley	Inshore	54	11/07/2018
58	-13.94680	125.65028	Cassini Island	Kimberley	N Kimberley	Inshore	45	11/07/2018
59	-14.17184	125.77399	Lafontaine Island	Kimberley	N Kimberley	Inshore	18	11/07/2018
60	-14.37772	124.97697	North Maret Island	Kimberley	N Kimberley	Inshore	18	12/07/2018
61	-14.45515	124.97313	South Maret Island	Kimberley	N Kimberley	Inshore	21	12/07/2018
62	-14.43160	124.87730	Robroy Reef	Kimberley	N Kimberley	Inshore	34	13/07/2018
63	-14.28287	125.84876	Berthoud Island	Kimberley	N Kimberley	Inshore	26	12/07/2018
64	-14.43064	125.88385	Crystal Head	Kimberley	N Kimberley	Coastal	19	12/07/2018
65	-14.57449	125.87018	Port Warrender	Kimberley	N Kimberley	Coastal	11	12/07/2018
66	-14.11409	123.55982	Browse Island	Kimberley	N Kimberley	Midshelf	11	14/07/2018
67	-15.68540	124.74360	Gairdner River	Kimberley	S Kimberley	Nearshore Estuarine	3	07/09/2018
68	-15.65830	124.73840	Gairdner River	Kimberley	S Kimberley	Nearshore Estuarine	3	07/09/2018
69	-15.65650	124.81770	Glenelg	Kimberley	S Kimberley	Nearshore Estuarine	5	08/09/2018
70	-16.39240	124.69910	Walcott	Kimberley	S Kimberley	Nearshore Estuarine	2	12/09/2018
71	-16.40890	124.59060	Walcott	Kimberley	S Kimberley	Nearshore Estuarine	2	12/09/2018

S3.9.1 Laboratory processing

DNA was extracted from half of each filter membrane using a DNeasy Blood and Tissue Kit (Qiagen; Venlo, Netherlands) with the following modifications: 540µl of ATL lysis buffer and 60µl of Proteinase K. This was completed within four weeks of collection. Blank controls were processed in parallel with all samples to detect any cross-contamination. Three PCR assays were applied: 16S Fish, COI Elasmobranch and 16S Reptile (Table 3.1; see S3.9.3 Primer development note for Reptile 16S) that were designed to amplify bony fish, elasmobranchs and aquatic reptiles, respectively, from mixed environmental samples. The 16S Fish and COI Elasmobranch PCR assays have been optimised and successfully applied in previous eDNA metabarcoding studies examining fish diversity and trophic interactions (Berry et al., 2017; Stat et al., 2017, 2019; West et al., 2020).

Following quantitative PCR-based (qPCR) quantification to optimise levels of input DNA (Berry et al., 2017), final qPCR was performed in a single step using fusion tagged primer architecture consisting of an Illumina sequencing adaptor, a unique index (6-8bp in length) and a respective primer sequence for each assay. All qPCR reactions were prepared in a dedicated trace DNA facility at the TrEnD Laboratory, Curtin University. Each qPCR reaction was carried out in 25µl containing: 1X AmpliTaq Gold® PCR buffer (Life Technologies, Massachusetts, USA), 2mM $MgCl_2$, 0.1mM dNTPs, 0.2µM each of forward and reverse primers (Integrated DNA Technologies, Australia), 10ug BSA (Fisher Biotec, Australia), 0.6µl of 5X SYBR® Green (Life Technologies), 1U AmpliTaq Gold® DNA Polymerase (Life Technologies), 4µl of eDNA template (neat), and made to volume with Ultrapure™ Distilled Water (Life Technologies). For the multiplex COI Elasmobranch assay, containing two forward primers and one reverse primer (Fields et al., 2015), the volume of forward primers was adjusted to 0.1µM each, whilst the volume of reverse primer remained unchanged at 0.2µM. Each qPCR was performed on a StepOnePlus Real-Time PCR System (Applied Biosystems, Massachusetts, USA) under the following conditions: initial denaturation at 95 °C for 5 min, followed by 50 cycles of 30 s at 95 °C, 51-54 °C for 30 s (see respective annealing temperatures in Table 3.1) and 45 s at 72 °C, with a final extension for 10 min at 72 °C. Each eDNA sample

was amplified in triplicate PCRs, combined into a larger library of amplicons pooled at equimolar ratios based on qPCR ΔRn values and size-selected using a Pippin-Prep (Sage Science, Beverly, USA) to remove any amplicons outside of the target range. Size-selected libraries were then purified using the Qiaquick PCR Purification Kit (Qiagen, Venlo, Netherlands), quantified using a Qubit 4.0 Fluorometer (Invitrogen, Carlsbad, USA) and if necessary, diluted to 2nM, for loading onto a 300 cycle MiSeq® V2 Standard Flow Cell. Sequencing was conducted on an Illumina MiSeq platform (Illumina, San Diego, USA), housed in the TrEnD Laboratory at Curtin University, Western Australia.

S3.9.2 Bioinformatics processing

Sequencing reads were demultiplexed into their respective samples using the ngsfilter (allowing up to three mismatches in primer sequences) and obisplit commands in OBITools (v1.2.9; Boyer et al., 2014). Data was then quality filtered (minimum length=100, maximum expected errors=2, no ambiguous nucleotides), denoised, filtered for chimeras and dereplicated (pool=TRUE) in RStudio (v1.1.423; RStudio Team, 2015) using the DADA2 bioinformatics package (Callahan et al. 2016). This generated an amplicon sequence variant (ASV) fasta and count table; the former was queried against NCBI's GenBank nucleotide database (accessed in 2018/19; Benson et al., 2005) using BLASTn (minimum percentage identity of 90, maximum target sequences of 10, reward value of 1) and also queried against an in-house 16S rRNA Western Australian fish database (see Table S2.6 in chapter two) via Zeus, an SGI cluster, based at the Pawsey Supercomputing Centre in Kensington, Western Australia. Taxonomic assignments of ASVs were curated using a lowest common ancestor (LCA) approach (https://github.com/mahsa-mousavi/LCA_taxonomyAssignment). This script compared the top ten hits for each query, collapsing to the lowest common ancestor if the percentage identity between each consecutive hit differed by less than one (based on the same query coverage). Consolidated taxa assignments were then additionally categorised based on habitat association and biogeographic distribution obtained from Kimberley biodiversity checklists (Moore et al. 2014), FishBase (Froese & Pauly, 2018) and the World Register of Marine Species (WoRMS; Horton et al., 2018). The circlize package (Gu,

Gu, Eils, Schlesner, & Brors, 2014) in RStudio was used to plot the range of order level taxa detected by each assay.

ASV accumulation curves by the addition of sampling replicates and sites were plotted using the vegan 'specaccum' function (Oksanen et al., 2019) in RStudio. To determine an appropriate subsampling depth in order to normalise the dataset, we plotted rarefaction curves (after pooling sample reads based on site) using the 'rarecurve' function in vegan and conducted consecutive rounds of subsampling using the multiple_rarefactions function (for 16S Fish: min=25,000, max=200,000, step=25,000; for COI Elasmobranch and 16S Reptile: min=4,000, max=20,000, step=4,000) in Qiime (Caporaso et al., 2010). We aimed to maximise the number of sites retained, whilst minimising ASV loss; the final subsampling thresholds were implemented using the single_rarefactions function in Qiime. ASVs detected in filtration and/or extraction blanks were entirely removed, prior to ASVs being merged by taxonomy using the phyloseq 'tax_glom' function (McMurdie & Holmes, 2013) in RStudio. We then consolidated our assay data into three taxonomic-based datasets: Actinopterygii (bony fish), Chondrichthyes (sharks and rays) and Reptilia (reptiles), which allowed us to examine community composition by discrete taxonomic group, rather than by individual metabarcoding assays which contained some overlap in taxonomic detections. Additionally, we extracted the 16S Fish ASV data for stripey snapper to examine 16S rDNA haplotype variation for this species across the NWA.

Table S3.2 Mean reads per sample in bioinformatic processing.

Assay	Demultiplexed Input	DADA2 Quality filtered	DADA2 Denoised	Post-chimera removal	Dereplicated unique sequences	ASVs with BLAST Hit
Fish 16S	83,259	79,203	78,992	75,800	157	80
Elasmobranch COI	40,066	16,102	15,957	15,838	205	12
Reptile 16S	73,787	33,311	33,200	32,569	95	32

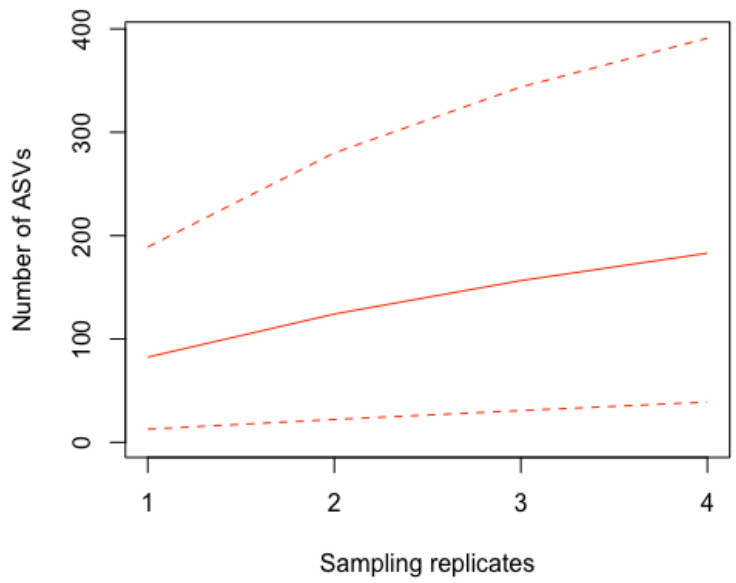


Figure S3.1 ASV accumulation curves by sampling replicate for the 16S Fish assay.
The ASV accumulation is based on the addition of each 1L sampling replicate (total of 4) at each site. Depicted in dotted lines are the maximum and minimum ASV accumulation curves; the solid line represents the median ASV accumulation curve. On fitting a polynomial curve to the median accumulation curve ($y = -3.8x^2 + 52.38x + 34.1$), it was extrapolated that an average of 6.9 1L water replicates would be required to maximise ASV richness for the 16S Fish assay.

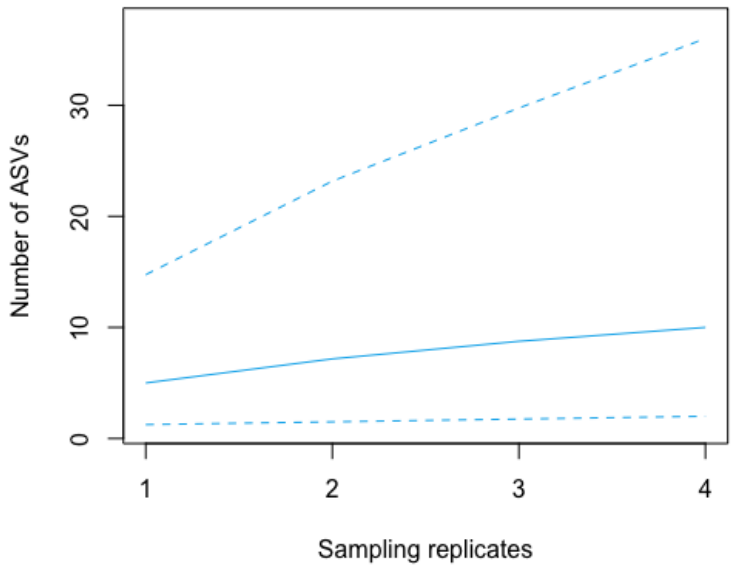


Figure S3.2 ASV accumulation curves by sampling replicate for the COI Elasmobranch assay.
The ASV accumulation is based on the addition of each 1L sampling replicate (total of 4) at each site. Depicted in dotted lines are the maximum and minimum ASV accumulation curves; the solid line represents the median ASV accumulation curve. On fitting a polynomial curve to the median accumulation curve ($y = -0.2275x^2 + 2.7965x + 2.4425$), it was extrapolated that an average of 6.1 1L water replicates would be required to maximise ASV richness for the COI Elasmobranch assay.

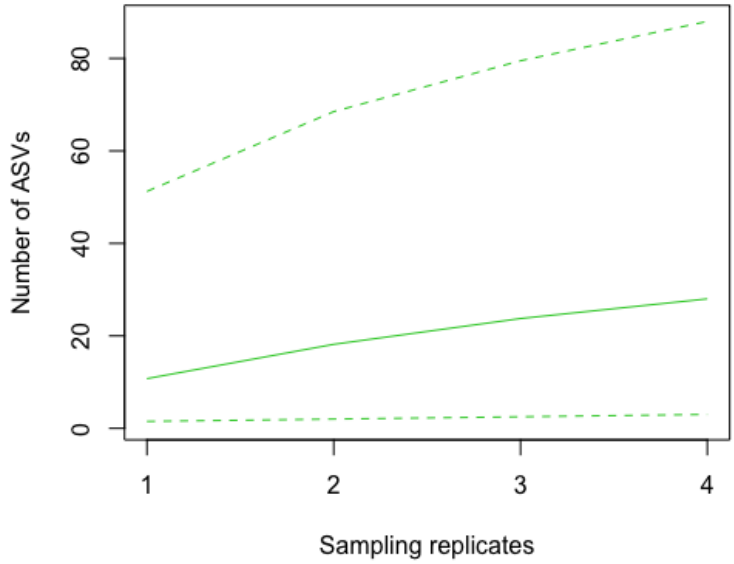


Figure S3.3 ASV accumulation curves by sampling replicate for the 16S Reptile assay.
The ASV accumulation is based on the addition of each 1L sampling replicate (total of 4) at each site. Depicted in dotted lines are the maximum and minimum ASV accumulation curves; the solid line represents the median ASV accumulation curve. On fitting a polynomial curve to the median accumulation curve ($y = -0.7918x^2 + 9.692x + 1.8748$), it was extrapolated that an average of 6.1 1L water replicates would be required to maximise ASV richness for the 16S Reptile assay.

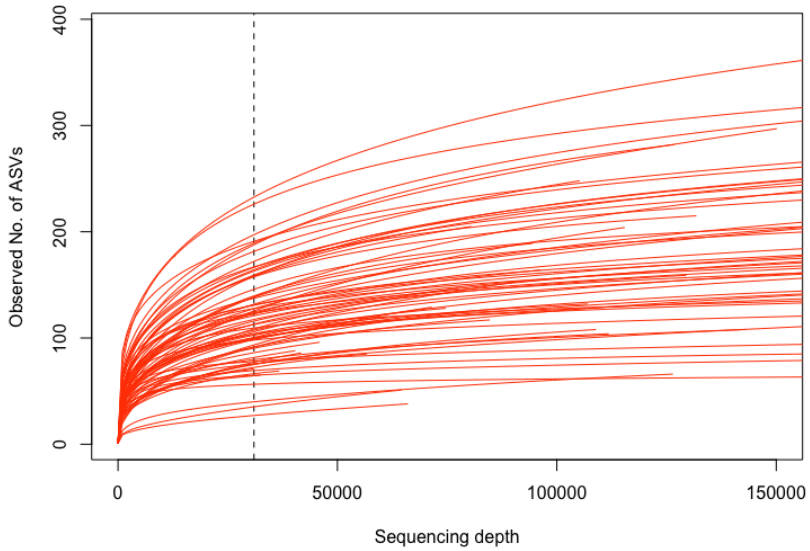


Figure S3.4 ASV rarefaction curve by sequencing depth for the Fish 16S assay. Dotted line refers to the subsampling cut-off of 30,977 reads (after pooling of the four replicates per site) for the 16S Fish assay. This subsampling cut-off was deemed appropriate as it represents the minimum depth across all the sites (thereby retaining all samples/sites in the dataset) and consecutive subsampling revealed minimal deviation in site dissimilarity patterns between the original, un-subsampled dataset and the 30,977 subsampled dataset. This outweighs the average loss of 58.7 ASVs per site - although subsequent loss of samples/sites - had I had set a higher subsampling cut-off.

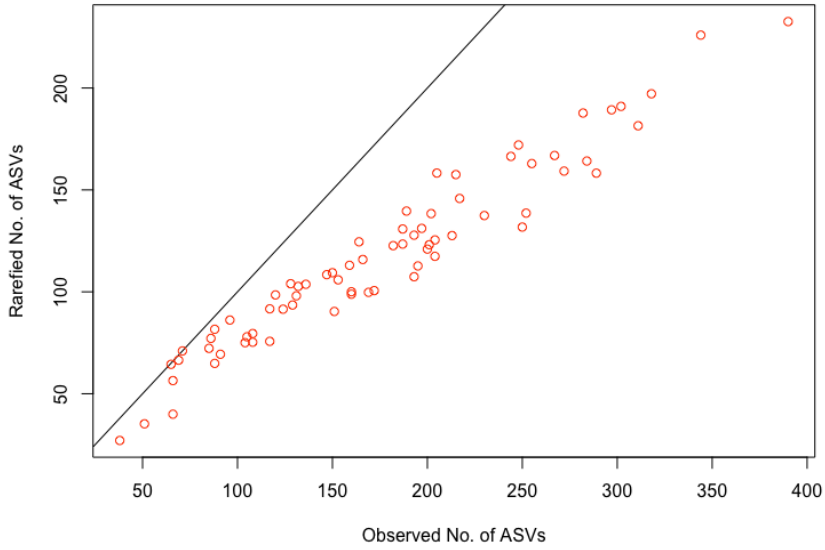


Figure S3.5 Ratio between rarefied and observed number of ASVs for the Fish 16S assay.
Open circles represent sites subsampled at the 30,977 threshold. As noted previously, there is an average loss of 57.8 ASVs per site when subsampling at this threshold. Therefore, there is a lower ratio between the rarefied number of ASVs (at 30,977 depth) and the observed number of ASVs (at each sites maximum depth).

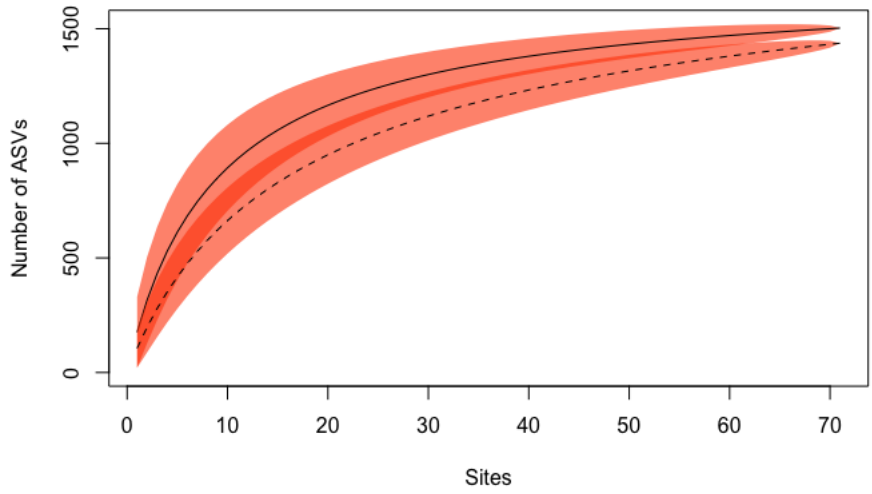


Figure S3.6 ASV accumulation curve by site for the 16S Fish assay.
The ASV accumulation is based on the addition of each site (after pooling of the four replicates per site). Depicted in the solid and dotted lines are the ASV curves pre- and post-subsampling of the 16S Fish data respectively, with 95% confidence intervals indicated in the shaded areas.

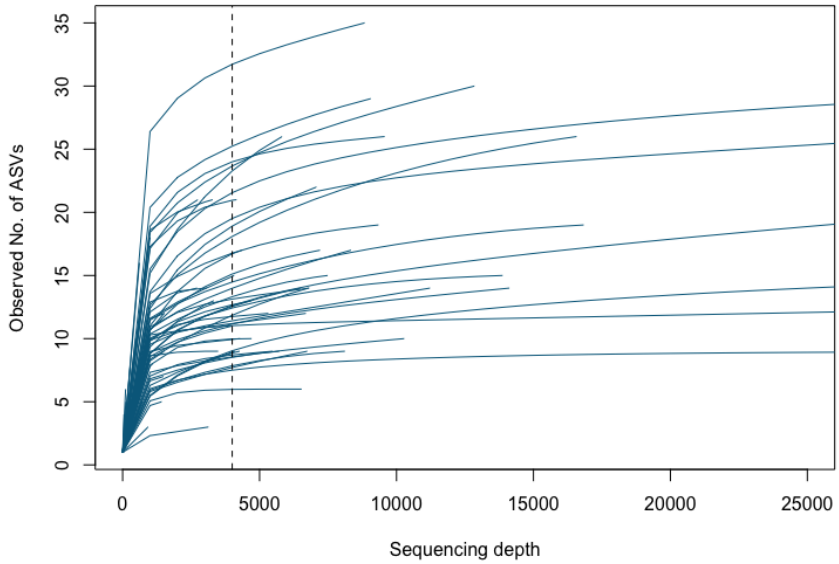


Figure S3.7 Accumulation curve of ASVs by sequencing depth for the Elasmobranch COI assay.
Dotted line refers to the subsampling cut-off of 4,000 reads (after pooling of the four replicates per site) for the COI Elasmobranch assay. This subsampling cut-off was deemed appropriate as it essentially required an average of 1000 reads per sample replicate, it retained the majority of sites (40 retained, 31 lost), and displayed minimal deviation in site dissimilarity patterns between the original, un-subsampled dataset and the 4,000 subsampled dataset. In addition, there was only an average loss of 2.5 ASVs per site - yet a rapid loss of sites/samples - had I had set a higher subsampling cut-off.

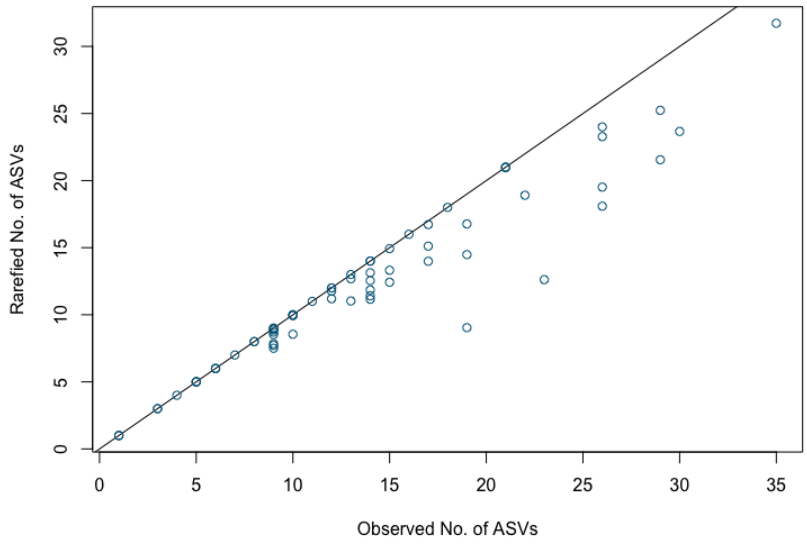


Figure S3.8 Ratio between rarefied and observed number of ASVs for the Elasmobranch COI assay.
Open circles represent sites subsampled at the 4,000 threshold. As noted previously, there is an average loss of 2.5 ASVs per site when subsampling at this threshold. Therefore, there is a lower ratio between the rarefied number of ASVs (at 4,000 depth) and the observed number of ASVs (at each sites maximum depth).

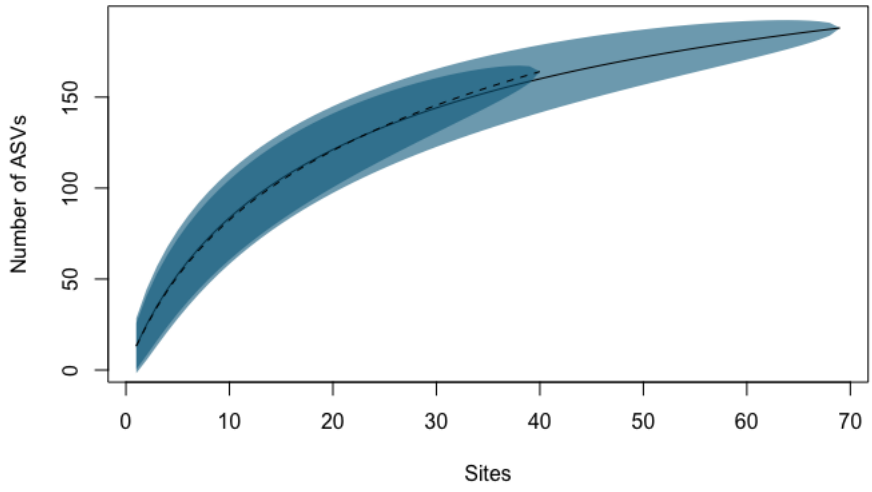


Figure S3.9 ASV accumulation curve by site for the COI Elasmobranch assay.
The ASV accumulation is based on the addition of each site (after pooling of the four replicates per site). Depicted in the solid and dotted lines are the ASV curves pre- and post-subsampling of the COI Elasmobranch data respectively, with 95% confidence intervals indicated in the shaded areas.

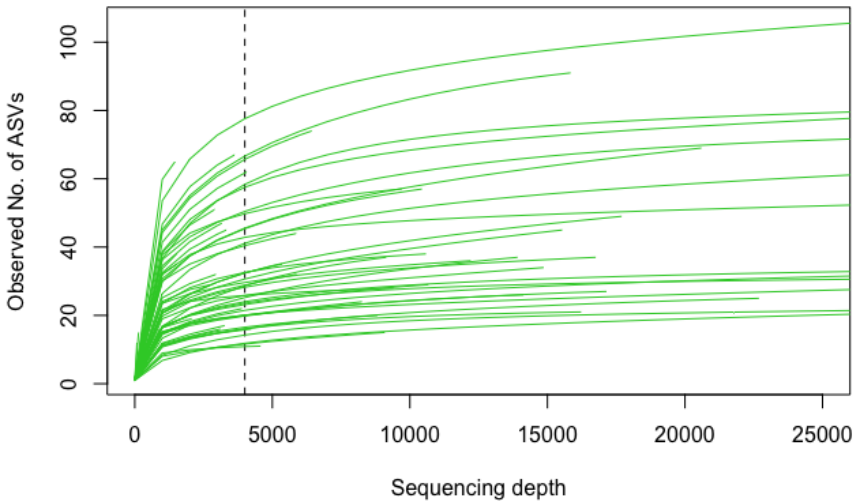


Figure S3.10 Accumulation curve of ASVs by sequencing depth for the Reptile 16S assay.
Dotted line refers to the subsampling cut-off of 4,000 reads (after pooling of the four replicates per site) for the 16S Reptile assay. This subsampling cut-off was deemed appropriate as it essentially required an average of 1000 reads per sample replicate, it retained the majority of sites (40 retained, 31 lost), and displayed minimal deviation in site dissimilarity patterns between the original, un-subsampled dataset and the 4,000 subsampled dataset. There was an average loss of 15.5 ASVs per site - yet a rapid loss of sites/samples - had I had set a higher subsampling cut-off.

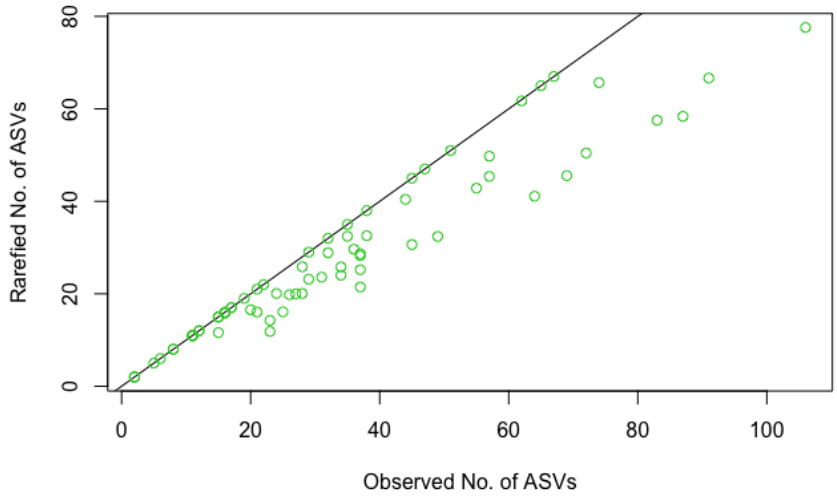


Figure S3.11 Ratio between rarefied and observed number of ASVs for the Reptile 16S assay.
Open circles represent sites subsampled at the 4,000 threshold. As noted previously, there is an average loss of 15.5 ASVs per site when subsampling at this threshold. Therefore, there is a lower ratio between the rarefied number of ASVs (at 4,000 depth) and the observed number of ASVs (at each sites maximum depth).

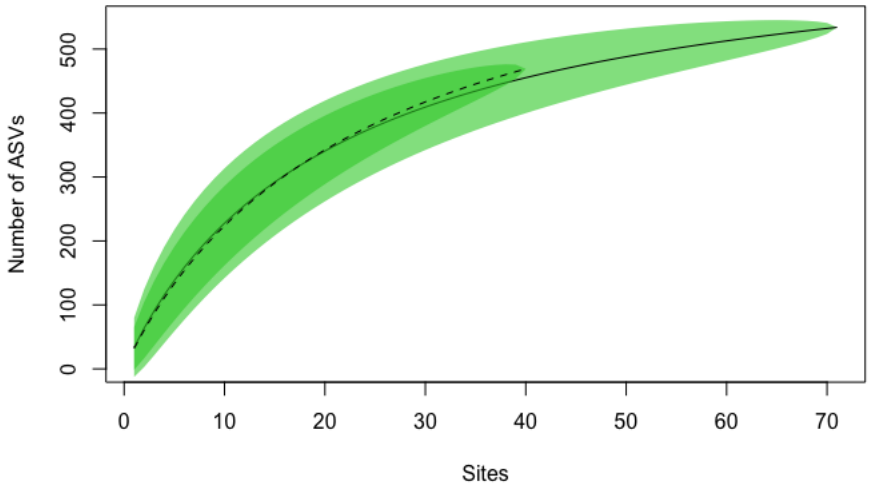


Figure S3.12 ASV accumulation curve by site for the 16S Reptile assay. The ASV accumulation is based on the addition of each site (after pooling of the four replicates per site). Depicted in the solid and dotted lines are the ASV curves pre- and post-subsampling of the 16S Reptile data respectively, with 95% confidence intervals indicated in the shaded areas.

Table S3.3 Consolidated taxa of eDNA detections across NWA.

Family	Genus	*Species*	Site/s detected	eDNA Assay
Zanclidae	*Zanclus*	*Zanclus cornutus*	37,47,66	16S Fish, 16S Reptile
Acanthuridae	*Acanthurus*	*Acanthurus olivaceus*	66	16S Fish
Acanthuridae	*Naso*	*Naso hexacanthus*	66	COI Elasmobranch
Acanthuridae	*Naso*	*Naso unicornis*	34,37,47	16S Reptile
Acanthuridae	*Naso*		65,66	16S Fish, 16S Reptile
Ophichthidae	*Neenchelys*		20	16S Fish
Ophichthidae	*Ophichthus*		9,12,16,17,24	16S Fish
Ophichthidae	*Pisodonophis*		22,46	16S Fish
Muraenidae	*Echidna*	*Echidna polyzona*	47	COI Elasmobranch
Muraenidae	*Gymnothorax*	*Gymnothorax griseus*	34,61,62	COI Elasmobranch
Muraenidae	*Gymnothorax*		61	COI Elasmobranch
Atherinidae	*Hypoatherina*	*Hypoatherina temminckii*	14,25,28,29,37,59	16S Reptile
Atherinidae	*Atherinomorus*		2,3,4,5,6,7,8,14,58	16S Fish, COI Elasmobranch, 16S Reptile
Atherinidae	*Hypoatherina*		24,27,28,29,30,31,34,37,49,59	16S Fish
Atherinidae	*Stenatherina*		28,29	16S Fish
Synodontidae	*Harpadon*	*Harpadon microchir*	11,42,43	16S Reptile
Synodontidae	*Saurida*	*Saurida wanieso*	48,49,63	16S Reptile
Synodontidae	*Trachinocephalus*	*Trachinocephalus myops*	52	COI Elasmobranch
Synodontidae	*Harpadon*		43	16S Fish
Zenarchopteridae	*Zenarchopterus*		14,16,17,32,70	16S Fish, 16S Reptile
Hemiramphidae	*Arrhamphus*	*Arrhamphus sclerolepis*	22	16S Reptile
Hemiramphidae	*Hemiramphus*	*Hemiramphus far*	14,16,26,27,29,32,37,40,43,45,46,47,48,49,50,58,60,61,62,64	16S Fish
Hemiramphidae	*Hyporhamphus*	*Hyporhamphus quoyi*	21,23,26,47,65	16S Fish,16S Reptile
Hemiramphidae	*Hyporhamphus*	*Hyporhamphus regularis*	25,27,36,65	16S Fish
Hemiramphidae	*Arrhamphus*		14,17,18,21,22,23,25,30,33,38	16S Fish
Hemiramphidae	*Hemiramphus*		11	16S Fish
Exocoetidae	*Parexocoetus*	*Parexocoetus brachypterus*	9	16S Reptile
Exocoetidae	*Parexocoetus*		1,2,3,4,5,7,8,10,11,12,44,45,46,48,54,58	16S Fish

Exocoetidae			3,6,8,55,58,62,70	16S Fish
Belonidae	*Strongylura*	*Strongylura incisa*	17,29,30	16S Fish
Belonidae	*Tylosurus*		14,15,17,20,23,24,26,29,30,31,34,35,38,40,46,49,51,53,66	16S Fish, 16S Reptile
Blenniidae	*Atrosalarias*	*Atrosalarias holomelas*	26,27,28,60,61	16S Fish, 16S Reptile
Blenniidae	*Blenniella*	*Blenniella bilitonensis*	60,61	16S Reptile
Blenniidae	*Cirripectes*	*Cirripectes stigmaticus*	34,60,66	COI Elasmobranch
Blenniidae	*Meiacanthus*	*Meiacanthus atrodorsalis*	9	16S Reptile
Blenniidae	*Salarias*	*Salarias fasciatus*	34,37,47,60	16S Fish, 16S Reptile
Blenniidae	*Omobranchus*		29	COI Elasmobranch
Rachycentridae	*Rachycentron*	*Rachycentron canadum*	2,5,7,9,13,37	16S Fish, COI Elasmobranch
Carangidae	*Alepes*	*Alepes apercna*	50,51,57,59,62,64,65,66	COI Elasmobranch
Carangidae	*Alepes*	*Alepes kleinii*	12,36,37,58,62,63,64	16S Fish
Carangidae	*Atule*	*Atule mate*	1,3,4,5,6,7,8,15,17,23,24,25,27,29,34,36,37,38,39,44,45,46,49,51,52,53,55,59,60,62,65,66	16S Fish, COI Elasmobranch
Carangidae	*Carangoides*	*Carangoides armatus*	16,37,41,43	16S Fish
Carangidae	*Carangoides*	*Carangoides malabaricus*	50,54,66	COI Elasmobranch
Carangidae	*Caranx*	*Caranx ignobilis*	10,11,12,13,14,15,16,17,18,21,22,23,24,25,26,27,29,30,31,32,33,34,35,36,37,38,40,41,43,47,51,55,56,57,58,59,60,61,62,63,64,65,66,67,68,70,71	16S Fish
Carangidae	*Caranx*	*Caranx melampygus*	16,34,66	16S Fish, COI Elasmobranch
Carangidae	*Gnathanodon*	*Gnathanodon speciosus*	1,3,12,13,15,16,17,24,25,26,29,34,37,47,62	COI Elasmobranch
Carangidae	*Megalaspis*	*Megalaspis cordyla*	11,12,13,15,16,48,49,56	16S Fish, COI Elasmobranch
Carangidae	*Parastromateus*	*Parastromateus niger*	31,51,54,64,67	16S Fish, COI Elasmobranch
Carangidae	*Scomberoides*	*Scomberoides commersonnianus*	22,29,39,50,59	COI Elasmobranch
Carangidae	*Scomberoides*	*Scomberoides lysan*	1,17,29,34,37,61,66	COI Elasmobranch
Carangidae	*Scomberoides*	*Scomberoides tol*	11,16,20,27,40,44,47,60,61,62,66	COI Elasmobranch
Carangidae	*Selar*	*Selar boops*	2,36,48,50,56,64	COI Elasmobranch
Carangidae	*Selar*	*Selar crumenophthalmus*	1,3,4,5,6,8,47,58	16S Fish, COI Elasmobranch
Carangidae	*Alepes*		12,37,58,63,64,70	16S Fish, COI Elasmobranch
Carangidae	*Atule*		29,34,35,37,45,46,48,49	16S Fish
Carangidae	*Carangoides*		14,35,49,50,51,58,59,60,65,66	16S Fish, COI Elasmobranch

Carangidae	*Caranx*		1,10,11,12,14,15,16,17,22,23,24,25,26,29,30,31,32,33,37,38,40,46,50,51,53,56,57,58,59,60,62,61,62,63,64,65,66,70	16S Fish, COI Elasmobranch
Carangidae	*Decapterus*		66	COI Elasmobranch
Carangidae	*Megalaspis*		12,13,16,49	16S Fish
Carangidae	*Parastromateus*		31,64,67	16S Fish
Carangidae	*Scomberoides*		1,2,5,7,13,14,15,16,17,21,22,23,25,27,29,30,31,32,33,35,38,44,46,50,59,67,68,70,71	16S Fish
Carangidae	*Selar*		3,6,8	16S Fish
Carangidae	*Trachinotus*		14	COI Elasmobranch
Carangidae			3,4,12,16,29,30,31,36,37,38,43,45,49,51,52,54,55,58,59,60,62,64,65,66,71	16S Fish, COI Elasmobranch
Terapontidae	*Leiopotherapon*	*Leiopotherapon aheneus*	14,15,16,17,23	16S Fish, 16S Reptile
Terapontidae	*Pelates*	*Pelates octolineatus*	51,53,54	16S Fish, 16S Reptile
Terapontidae	*Terapon*	*Terapon puta*	54	16S Fish
Terapontidae	*Terapon*	*Terapon theraps*	2,5,7,9,11,12,16,17,31,48,49,51,53,54,56,59,60,61,62,63,64,65,68	16S Fish, 16S Reptile
Terapontidae	*Leiopotherapon*		14,17	16S Fish
Terapontidae	*Pelates*		12,49,51,52,53,56	16S Fish
Terapontidae	*Terapon*		12,17,31,51,53,62,63,64	16S Fish, COI Elasmobranch
Kyphosidae	*Kyphosus*		59,60,61,62,63	16S Fish
Leiognathidae	*Aurigequula*	*Aurigequula fasciata*	14,15,17,60,62,63	16S Fish
Leiognathidae	*Deveximentum*	*Deveximentum indicium*	11,12,16,31,49,50,59,63,64	16S Fish
Leiognathidae	*Deveximentum*	*Deveximentum megalolepis*	14,21,30,31,32,35,43,48,51,52,53,65,67	16S Fish
Leiognathidae	*Equulites*	*Equulites laterofenestra*	1,4,6,13,49,55,57,58,59,60,63	16S Fish
Leiognathidae	*Equulites*	*Equulites leuciscus*	13,14,16,17,24,25,29,40,46,47,49,50,53,59,63	16S Reptile
Leiognathidae	*Leiognathus*	*Leiognathus equulus*	17,30,56	16S Fish
Leiognathidae	*Nuchequula*	*Nuchequula decora*	1,10,13,14,15,17,18,19,20,21,22,23,27,28,30,31,32,33,38,39,40,42,43,53,63,64,65,67,70	16S Fish
Leiognathidae	*Photopectoralis*	*Photopectoralis bindus*	11	16S Fish
Leiognathidae	*Deveximentum*		1,7,9,11,12,13,16,17,21,30,31,35,43,48,49,50,52,55,59,61,63,64	16S Fish
Leiognathidae	*Gazza*		15,49,51,60,63	16S Fish

Leiognathidae	*Photopectoralis*		11,12,16,35,36,37,44	16S Fish
Leiognathidae			1,5,6,7,8,9,10,11,12,13,14,15,16,17,18,21,22,25,30,32,35, 40,49,51,52,53,54,55,57,59	16S Fish
Chaetodontidae	*Chaetodon*	*Chaetodon adiergastos*	61	16S Reptile
Chaetodontidae	*Chaetodon*	*Chaetodon octofasciatus*	20,21,26,27,28,29,34,35,37,43,47,49,50	16S Reptile
Chaetodontidae	*Chelmon*		2,4,5,7,10,13,16,26,27,28,29,31,35,37,46,47,49,50,52,63	16S Fish, 16S Reptile
Engraulidae	*Encrasicholina*	*Encrasicholina punctifer*	1,34,35,36,37,38,45,47,48,51,52,53,55,56,57,58,59,60,61, 62,63,66	16S Fish, COI Elasmobranch, 16S Reptile
Engraulidae	*Stolephorus*	*Stolephorus indicus*	15,16	16S Reptile
Engraulidae	*Thryssa*	*Thryssa hamiltonii*	14,19,22,23,24,30,31,38,39,40,49,53,54,64,65,67,69	16S Reptile
Engraulidae	*Encrasicholina*		1,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,22,27,35,46,47, 48,49,50,51,52,53,54,55,56,58,59,60,61,62,63,64,65,66,6 7,68	16S Fish, COI Elasmobranch, 16S Reptile
Engraulidae	*Thryssa*		14,18,19,20,22,23,30,31,32,38,39,54,65	COI Elasmobranch
Dussumieriidae	*Dussumieria*	*Dussumieria elopsoides*	9,11,13,16,20,21,36,37,52,54,56,59,60,63,64	16S Fish, 16S Reptile
Dussumieriidae	*Etrumeus*	*Etrumeus teres*	58	16S Reptile
Clupeidae	*Anodontostoma*	*Anodontostoma chacunda*	11,14,17,19,21,23,26,28,30,32,33,35,36,38,39,65,67	16S Fish, 16S Reptile
Clupeidae	*Nematalosa*	*Nematalosa come*	12,14,15,50,51,65	16S Reptile
Clupeidae	*Spratelloides*	*Spratelloides delicatulus*	13,14,27,34,47,48,49,58,59	16S Reptile
Clupeidae	*Amblygaster*		1,2,3,4,5,6,7,8	16S Fish
Clupeidae	*Escualosa*		1,7,18,19,20,21,38,65,66,68	16S Fish
Clupeidae	*Nematalosa*		9,10,11,12,13,14,15,16,49,51,54,56	16S Fish
Clupeidae	*Sardinella*		1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,2 3,24,25,26,27,28,29,31,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,47,48,49,50,51,52,53,54,55,56,58,59,60,61,62,6 3,64,65,66,67,68,69,70	16S Fish, 16S Reptile
Clupeidae	*Sardinops*		1,8,14,21,23,24,25,26,27,28,29,30,31,33,34,35,36,37,38,3 9,40,41,43,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59, 60,61,62,63,64,65,66,68	16S Fish, 16S Reptile
Clupeidae	*Spratelloides*		12,13,14,15,16,20,25,34,36,37,47,48	16S Fish
Chirocentridae	*Chirocentrus*		11,12,13,14,15,16,17,19,22,24,25,27,31,32,33,35,36,39,4 3,48,50,51,54,55,59,60,61,65	COI Elasmobranch
Ephippidae	*Platax*	*Platax orbicularis*	29,50,61	16S Reptile

Ephippidae	*Platax*	*Platax teira*	26,27,28,29,30,31,36,45,47,50,59,60,62	16S Fish, 16S Reptile
Ephippidae	*Platax*		27,28,29,36,45,47,60	16S Fish
Drepaneidae	*Drepane*		1,9,10,12,13,14,15,17,18,19,21,22,23,24,25,26,27,28,30,31,32,33,38,39,40,43,48,51,67,70	16S Fish, COI Elasmobranch
Gerreidae	*Gerres*	*Gerres erythrourus*	13,14,15,16,17,27,33	16S Fish, COI Elasmobranch, 16S Reptile
Gerreidae	*Gerres*	*Gerres filamentosus*	1,3,4,5,6,8,14,15,17,27,29,30,32,35,38,40,42	16S Fish
Gerreidae	*Gerres*	*Gerres japonicus*	14,17	16S Fish
Gerreidae	*Gerres*	*Gerres oyena*	17,26,29	COI Elasmobranch
Gerreidae	*Gerres*		1,4,6,11,13,14,16,17,25,27,29,30,32,36,40,54	16S Fish
Gobiidae	*Acentrogobius*	*Acentrogobius caninus*	1,5,8,10,13,15,17,18,20,23,25,26,27,28,30,31,33,40,65	16S Fish, 16S Reptile
Gobiidae	*Acentrogobius*	*Acentrogobius janthinopterus*	12,17	16S Fish, 16S Reptile
Gobiidae	*Cryptocentrus*	*Cryptocentrus caeruleomaculatus*	28,33,48,50,51	16S Fish, 16S Reptile
Gobiidae	*Drombus*	*Drombus triangularis*	14,16,17,22,23,29,30,32,33,39,40,67,70	16S Reptile
Gobiidae	*Exyrias*	*Exyrias puntang*	14	COI Elasmobranch, 16S Reptile
Gobiidae	*Istigobius*	*Istigobius nigroocellatus*	2,5,6,7,8,47,60,62,63	16S Fish, 16S Reptile
Gobiidae	*Mahidolia*	*Mahidolia mystacina*	17	16S Reptile
Gobiidae	*Mugilogobius*	*Mugilogobius mertoni*	17,29,32,38,70	16S Fish, 16S Reptile
Gobiidae	*Oxyurichthys*	*Oxyurichthys ophthalmonema*	14	16S Reptile
Gobiidae	*Parachaeturichthys*	*Parachaeturichthys polynema*	10,12,40,51,53,54,59	16S Fish, 16S Reptile
Gobiidae	*Paragobiodon*	*Paragobiodon xanthosoma*	34	16S Fish, 16S Reptile
Gobiidae	*Paratrypauchen*	*Paratrypauchen microcephalus*	51,53,65	16S Reptile
Gobiidae	*Periophthalmus*	*Periophthalmus argentilineatus*	14,17,25,70	16S Fish, COI Elasmobranch, 16S Reptile
Gobiidae	*Periophthalmus*	*Periophthalmus minutus*	17	16S Fish
Gobiidae	*Psammogobius*	*Psammogobius biocellatus*	14,17	16S Fish
Gobiidae	*Pseudogobius*	*Pseudogobius javanicus*	17	16S Reptile
Gobiidae	*Ptereleotris*	*Ptereleotris microlepis*	34,35,36,37,60,61	16S Fish, 16S Reptile
Gobiidae	*Ptereleotris*	*Ptereleotris monoptera*	66	COI Elasmobranch
Gobiidae	*Ptereleotris*	*Ptereleotris rubristigma*	37	16S Fish
Gobiidae	*Tridentiger*	*Tridentiger barbatus*	38,40,41	COI Elasmobranch

Gobiidae	*Acentrogobius*		15,17,23,25,31	16S Fish
Gobiidae	*Amoya*		14,15,16,17,21,25,38,39,40,70	16S Fish, 16S Reptile
Gobiidae	*Drombus*		12,14,17,21,22,23,32	16S Fish
Gobiidae	*Eviota*		66	16S Fish
Gobiidae	*Favonigobius*		14,16,17,25,29	COI Elasmobranch
Gobiidae	*Gobiodon*		34,35,37,47	16S Fish, 16S Reptile
Gobiidae	*Istigobius*		21,29,37	16S Fish, 16S Reptile
Gobiidae	*Mahidolia*		15,40	16S Fish
Gobiidae	*Mugilogobius*		17	16S Fish
Gobiidae	*Oplopomus*		12	COI Elasmobranch
Gobiidae	*Parachaeturichthys*		40,51,59	16S Fish
Gobiidae	*Paratrypauchen*		12,21,22,23,38,40,49,51,54,64,65,68	16S Fish
Gobiidae	*Parioglossus*		34,37	16S Fish, 16S Reptile
Gobiidae	*Periophthalmus*		17,41	16S Reptile
Gobiidae	*Pseudogobius*		17,22,32,38,67	16S Fish, 16S Reptile
Gobiidae	*Ptereleotris*		34,37,60	COI Elasmobranch
Gobiidae	*Taenioides*		13,18,22,23,38,67,68,71	16S Fish
Gobiidae	*Yongeichthys*		29	16S Reptile
Gobiidae			17,19,20,24,25,27,38,40,41,67,68,71	16S Fish, 16S Reptile
Eleotridae	*Bostrychus*	*Bostrychus sinensis*	13,17	16S Reptile
Eleotridae	*Butis*	*Butis butis*	10,14,15,17,22,23,32	16S Fish
Eleotridae	*Butis*	*Butis koilomatodon*	11,25,39,41,67	16S Reptile
Eleotridae	*Calumia*	*Calumia godeffroyi*	66	COI Elasmobranch
Eleotridae	*Odonteleotris*	*Odonteleotris macrodon*	70	16S Fish
Eleotridae	*Ophiocara*	*Ophiocara porocephala*	17	16S Fish
Eleotridae	*Prionobutis*	*Prionobutis microps*	13,14,15,16,17,18,19,21,22,23,30,32,67,68,69	16S Fish
Eleotridae	*Bostrychus*		12,17,22,38,70	16S Fish
Eleotridae	*Butis*		25,30,36,68,70	16S Fish
Eleotridae	*Mogurnda*		71	16S Fish
Eleotridae	*Prionobutis*		14,17,23,67	16S Fish
Chanidae	*Chanos*	*Chanos chanos*	12,22,23,25,27,59,61	16S Fish, COI Elasmobranch
Holocentridae	*Sargocentron*	*Sargocentron rubrum*	1,4,7,26,28,34,35,45,46,47,59,60,61	16S Fish
Istiophoridae			1,3,6,8,57,58	16S Fish, COI Elasmobranch

Apogonidae	*Apogon*	*Apogon semilineatus*	66	16S Fish
Apogonidae	*Fowleria*		37,38	COI Elasmobranch
Apogonidae	*Jaydia*		53,60	16S Reptile
Labridae	*Cheilinus*	*Cheilinus chlorourus*	34,60,66	COI Elasmobranch
Labridae	*Cheilinus*	*Cheilinus fasciatus*	12	16S Fish
Labridae	*Cheilinus*	*Cheilinus trilobatus*	35,47,60,66	COI Elasmobranch
Labridae	*Choerodon*	*Choerodon cephalotes*	1,2,3,4,5,6,7,8,39,49,51	16S Fish, 16S Reptile
Labridae	*Choerodon*	*Choerodon cyanodus*	1,8,10,11,13,14,15,19,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,37,39,45,47,48,49,50,51,52,53,57,61	16S Fish, COI Elasmobranch
Labridae	*Choerodon*	*Choerodon monostigma*	1,11,12,13,24,33,34,35,36,39,43,46,50,52,55	16S Fish, 16S Reptile
Labridae	*Choerodon*	*Choerodon schoenleinii*	1,4,6,9,11,12,13,14,15,16,18,19,24,25,26,27,29,33,34,35,37,47,49,51,53,58,59,61,64	16S Fish
Labridae	*Choerodon*	*Choerodon vitta*	1,3,4,6,7,8,11,24,60,62	16S Fish, 16S Reptile
Labridae	*Cirrhilabrus*	*Cirrhilabrus exquisitus*	66	16S Fish, 16S Reptile
Labridae	*Diproctacanthus*	*Diproctacanthus xanthurus*	1,35,37	16S Fish, 16S Reptile
Labridae	*Halichoeres*	*Halichoeres argus*	24,34,37,55	16S Reptile
Labridae	*Halichoeres*	*Halichoeres leucurus*	59,60	16S Fish, 16S Reptile
Labridae	*Halichoeres*	*Halichoeres melanurus*	34,35,47,59,60	16S Fish, 16S Reptile
Labridae	*Halichoeres*	*Halichoeres nebulosus*	37,60	16S Fish, 16S Reptile
Labridae	*Halichoeres*	*Halichoeres nigrescens*	1,6,8,9,10,11,13,14,15,18,21,22,24,25,26,27,28,29,30,32,33,34,35,36,37,38,39,41,46,47,48,49,51,52,56,60	16S Fish, COI Elasmobranch, 16S Reptile
Labridae	*Hemigymnus*	*Hemigymnus fasciatus*	66	16S Reptile
Labridae	*Hemigymnus*	*Hemigymnus melapterus*	1,6,9,26,28,32,33,34,35,37,47,48,51,59,60,63	16S Fish, COI Elasmobranch, 16S Reptile
Labridae	*Leptoscarus*	*Leptoscarus vaigiensis*	34	16S Reptile
Labridae	*Novaculichthys*	*Novaculichthys taeniourus*	66	16S Fish
Labridae	*Pseudocheilinus*	*Pseudocheilinus evanidus*	66	16S Fish, 16S Reptile
Labridae	*Scarus*	*Scarus niger*	66	16S Fish
Labridae	*Scarus*	*Scarus psittacus*	66	16S Fish
Labridae	*Thalassoma*	*Thalassoma amblycephalum*	61,65,66,67,68,69	16S Fish, COI Elasmobranch, 16S Reptile
Labridae	*Thalassoma*	*Thalassoma jansenii*	66	16S Fish
Labridae	*Thalassoma*	*Thalassoma lunare*	47,60	16S Reptile

Labridae	*Thalassoma*	*Thalassoma purpureum*	66	16S Reptile
Labridae	*Cheilinus*		9,14,34,35,59,60,65,66	16S Fish, 16S Reptile
Labridae	*Choerodon*		3,6,8,13,14,21,24,26,27,28,29,34,37,39,47,49,51,61	16S Fish
Labridae	*Coris*		66	COI Elasmobranch
Labridae	*Gomphosus*		56,66	16S Fish, 16S Reptile
Labridae	*Halichoeres*		14,15,21,27,28,29,34,35,36,37,41,46,47,48,51,52	16S Fish
Labridae	*Scarus*		1,7,11,12,13,14,19,24,26,28,29,31,33,34,35,36,37,47,49,50,51,55,56,60,61,66	16S Fish, 16S Reptile
Labridae	*Thalassoma*		66	16S Fish
Lobotidae	*Lobotes*	*Lobotes surinamensis*	33,34,38	16S Fish, COI Elasmobranch
Antennariidae	*Lophiocharon*		26,36,43	COI Elasmobranch
Lutjanidae	*Aphareus*	*Aphareus furca*	61	16S Fish
Lutjanidae	*Caesio*	*Caesio caerulaurea*	60,61,65,66	16S Fish, 16S Reptile
Lutjanidae	*Caesio*	*Caesio cuning*	1,8,10,9,11,12,13,14,15,16,18,19,24,25,26,34,35,36,37,43,45,46,47,48,49,50,51,52,55,56,59,60,61,66	16S Fish, 16S Reptile
Lutjanidae	*Lutjanus*	*Lutjanus argentimaculatus*	12,13,14,15,17,18,22,23,25,29,30,31,32,33	16S Fish, 16S Reptile
Lutjanidae	*Lutjanus*	*Lutjanus carponotatus*	1,3,4,5,6,7,8,9,12,13,14,15,16,17,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,47,48,49,50,51,52,53,55,56,57,58,59,60,61,62,63,64,66	16S Fish, 16S Reptile
Lutjanidae	*Lutjanus*	*Lutjanus fulviflamma*	37,51,52,60	16S Fish
Lutjanidae	*Lutjanus*	*Lutjanus fulvus*	13,22,29,34,37,41,48,60	16S Reptile
Lutjanidae	*Lutjanus*	*Lutjanus johnii*	1,8,12,13,14,15,17,19,21,22,23,26,27,29,30,31,32,33,34,35,36,37,38,39,40,41,43,47,61,64,67,68,69,70,71	16S Fish, COI Elasmobranch, 16S Reptile
Lutjanidae	*Lutjanus*	*Lutjanus russellii*	14,15,16,30,38	16S Fish
Lutjanidae	*Lutjanus*	*Lutjanus vitta*	3,5,8,59,60,63	16S Fish, 16S Reptile
Lutjanidae	*Macolor*	*Macolor niger*	66	16S Fish, 16S Reptile
Lutjanidae	*Pristipomoides*	*Pristipomoides filamentosus*	57	COI Elasmobranch
Lutjanidae	*Pristipomoides*	*Pristipomoides multidens*	1,6,49,55,57	16S Fish, COI Elasmobranch
Lutjanidae	*Pristipomoides*	*Pristipomoides typus*	57,66	COI Elasmobranch
Lutjanidae	*Pterocaesio*	*Pterocaesio tile*	66	16S Reptile
Lutjanidae	*Symphorus*	*Symphorus nematophorus*	66	16S Reptile
Lutjanidae	*Caesio*		11,13,14,16,19,24,25,29,34,35,36,37,43,45,46,47,49,50,51,52,56,57,59,60,61,66	16S Fish

Lutjanidae	*Lutjanus*		1,3,4,5,6,7,8,9,11,12,13,14,15,16,17,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,43,45,46,47,50,51,55,57,58,59,60,61,63,66	16S Fish, COI Elasmobranch, 16S Reptile
Lutjanidae	*Pristipomoides*		57	16S Fish
Lutjanidae			12,13,14,17,19,21,22,25,26,29,31,32,34,35,36,37,38,41,43,45,46,47,49,50,53,60,61,64,66	16S Fish, 16S Reptile
Haemulidae	*Diagramma*	*Diagramma pictum*	12,13,21,25,26,43	COI Elasmobranch
Haemulidae	*Plectorhinchus*	*Plectorhinchus albovittatus*	1,9,12,13,14,15,17,18,24,25,26,27,28,30,33,34,37,40,47,48,50,51,53,59,60,61,63	16S Fish, 16S Reptile
Haemulidae	*Plectorhinchus*	*Plectorhinchus chaetodonoides*	12,13,43,47,49,50,51,61	16S Fish, COI Elasmobranch, 16S Reptile
Haemulidae	*Plectorhinchus*	*Plectorhinchus chrysotaenia*	11,12,46,47,50,52,53	16S Fish
Haemulidae	*Plectorhinchus*	*Plectorhinchus gibbosus*	1,7,8,9,11,12,13,14,15,17,19,20,21,22,23,25,26,27,29,30,31,32,33,36,37,38,40,41,42,50,67	16S Fish, 16S Reptile
Haemulidae	*Plectorhinchus*	*Plectorhinchus polytaenia*	3,5,8,45,46	16S Fish
Haemulidae	*Pomadasys*	*Pomadasys argenteus*	31,37,53,68,71	16S Reptile
Haemulidae	*Pomadasys*	*Pomadasys kaakan*	12,14,15,17,18,19,21,22,23,25,27,30,31,32,35,38,49,53,60,67,70,71	16S Fish, 16S Reptile
Haemulidae	*Pomadasys*	*Pomadasys maculatus*	6,12,18,23,39,60,62,63,64	16S Reptile
Haemulidae	*Diagramma*		1,12,21	16S Fish
Haemulidae	*Plectorhinchus*		12,13,14,17,18,19,21,24,25,26,27,28,29,30,31,32,37,38,40,43,47,48,49,50,53,59,60,61,63	16S Fish, 16S Reptile
Haemulidae	*Pomadasys*		12,14,21,22,30,31,32,33,38,40,53,56,67,71	16S Fish
Haemulidae			1,4,5,6,7,10,11,12,13,14,15,16,17,18,19,20,21,22,23,25,26,27,28,31,32,36,37,40,43,45,47,49,52,53,60,61,62,63,66	16S Fish, 16S Reptile
Mugilidae	*Chelon*	*Chelon planiceps*	14,23,30,31,38,70	16S Fish, 16S Reptile
Mugilidae	*Ellochelon*	*Ellochelon vaigiensis*	1,8,13,14,15,16,17,18,19,20,22,23,25,27,29,30,31,32,33,38,40,41,49,51,67,70,71	16S Fish, 16S Reptile
Mugilidae	*Moolgarda*	*Moolgarda seheli*	11,12,13,14,15,16,17,21,22,23,25,26,29,30,32,40	16S Fish
Mugilidae	*Oedalechilus*	*Oedalechilus labiosus*	1,29,55,60	16S Fish, 16S Reptile
Mugilidae	*Planiliza*	*Planiliza subviridis*	9,14,17,22,23,32,37,38,70	16S Fish, COI Elasmobranch, 16S Reptile
Mugilidae	*Squalomugil*	*Squalomugil nasutus*	71	16S Fish, COI Elasmobranch

Mugilidae	*Moolgarda*		13,14,17,22,23,25,26,32,38,70	16S Fish, COI Elasmobranch
Mugilidae	*Mugil*		19,21,22,23,25,30,31,32,34,38,39,40	COI Elasmobranch
Mugilidae	*Planiliza*		38,70,71	16S Reptile
Mugilidae			1,12,13,14,16,17,18,19,21,22,23,24,25,29,30,31,32,33,35, 38,40,42,53,67,68,70,71	16S Fish
Myctophidae	*Benthosema*	*Benthosema pterotum*	42,52,66	COI Elasmobranch, 16S Reptile
Myctophidae	*Benthosema*		11,12,41,42,43,52,66	16S Fish, 16S Reptile
Myctophidae	*Diaphus*		66	16S Fish
Bythitidae	*Diancistrus*	*Diancistrus jeffjohnsoni*	25,26,28	16S Fish, 16S Reptile
Bythitidae	*Dinematichthys*	*Dinematichthys iluocoeteoides*	15,32,34	COI Elasmobranch
Pempheridae	*Parapriacanthus*	*Parapriacanthus ransonneti*	66	16S Reptile
Pempheridae	*Pempheris*		49,51	16S Fish
Glaucosomatidae	*Glaucosoma*	*Glaucosoma magnificum*	2,3,5,7,8,13,19,24,25,26,30,40,45,47,58,60	16S Fish, 16S Reptile
Glaucosomatidae	*Glaucosoma*		3,5,8,25,60	16S Fish
Sphyraenidae	*Sphyraena*	*Sphyraena jello*	35	16S Fish
Sphyraenidae	*Sphyraena*	*Sphyraena obtusata*	1,4,5,6,7,10,12,47,49,63	16S Fish
Sillaginidae	*Sillago*	*Sillago aeolus*	2,5,12,13,14,17,22,23,25,40,51,53,59,63	16S Reptile
Sillaginidae	*Sillago*	*Sillago bassensis*	55,57,58	16S Fish
Sillaginidae	*Sillago*	*Sillago schomburgkii*	20,21,22,23	16S Reptile
Sillaginidae	*Sillago*	*Sillago sihama*	18,22,23,40,67	16S Reptile
Sillaginidae	*Sillago*		1,3,4,5,6,7,8,9,10,12,13,14,15,17,19,20,21,22,23,24,25,26, 27,28,29,30,32,35,38,40,45,49,50,51,53,54,55,59,60,67	16S Fish, 16S Reptile
Siganidae	*Siganus*	*Siganus luridus*	17,26,34,35,37,47	16S Reptile
Siganidae	*Siganus*	*Siganus stellatus*	26,28,34,37,38	COI Elasmobranch
Siganidae	*Siganus*	*Siganus virgatus*	28,34,43	16S Reptile
Siganidae	*Siganus*		1,5,7,8,9,10,11,12,14,15,16,17,18,25,26,27,28,29,31,33,34 ,35,36,37,38,43,45,46,47,48,49,51,52,59,60,63	16S Fish, COI Elasmobranch, 16S Reptile
Serranidae	*Aethaloperca*	*Aethaloperca rogaa*	66	16S Fish
Serranidae	*Cephalopholis*	*Cephalopholis boenak*	14,16,33,45,46,50,61,63	16S Fish, 16S Reptile
Serranidae	*Cephalopholis*	*Cephalopholis cyanostigma*	7,34,35,47,54,55,56,59,60,61,63	16S Fish, 16S Reptile
Serranidae	*Epinephelus*	*Epinephelus bleekeri*	9,10	16S Fish
Serranidae	*Epinephelus*	*Epinephelus corallicola*	34	16S Fish

Serranidae	*Epinephelus*	*Epinephelus fasciatus*	34,35,37,47,50,51,52,55,60,61,63	16S Fish, COI Elasmobranch, 16S Reptile
Serranidae	*Epinephelus*	*Epinephelus lanceolatus*	11,14,17,21,22,23,25,31,32,33,38,40,41,64,67,68,69	16S Fish, 16S Reptile
Serranidae	*Epinephelus*	*Epinephelus macrospilos*	66	16S Fish, 16S Reptile
Serranidae	*Epinephelus*	*Epinephelus quoyanus*	21,34,45,48	16S Fish, 16S Reptile
Serranidae	*Plectropomus*	*Plectropomus maculatus*	9,11,13,21,22,29,34,35,36,37,42,47,60,62,66,71	16S Fish, 16S Reptile
Serranidae	*Variola*	*Variola albimarginata*	66	COI Elasmobranch
Serranidae	*Variola*	*Variola louti*	66	16S Fish, COI Elasmobranch
Serranidae	*Cephalopholis*		14,15,21,22,26,28,33,35,36,46,47,49,50,51,61,63	16S Fish, 16S Reptile
Serranidae	*Epinephelus*		10,12,13,14,15,16,17,18,21,22,23,28,29,30,31,32,34,38,39,40,45,47,51,57,58,62,65,66,67,70	16S Fish, COI Elasmobranch, 16S Reptile
Serranidae	*Plectropomus*		66	COI Elasmobranch
Serranidae	*Variola*		66	16S Fish
Serranidae			16	16S Reptile
Scorpaenidae	*Scorpaenopsis*	*Scorpaenopsis macrochir*	55	16S Fish
Sciaenidae	*Protonibea*	*Protonibea diacanthus*	12,31,41,50,56,65,67,68,69,70,71	16S Fish
Sciaenidae	*Atrobucca*		71	16S Fish
Scatophagidae	*Scatophagus*	*Scatophagus argus*	14,15,17,20,21,22,23,30,31,32,38,70	16S Fish
Scatophagidae			14,17,22,23,25,30,38	16S Fish
Pseudochromidae	*Labracinus*	*Labracinus cyclophthalmus*	26,28,59	16S Reptile
Pseudochromidae	*Pseudochromis*		24,26,28,29,31,33,51	16S Fish, 16S Reptile
Pomacentridae	*Abudefduf*	*Abudefduf bengalensis*	13,34,35,37,47	16S Fish, 16S Reptile
Pomacentridae	*Abudefduf*	*Abudefduf sordidus*	60	16S Fish
Pomacentridae	*Amblyglyphidodon*	*Amblyglyphidodon ternatensis*	26,29	16S Fish
Pomacentridae	*Chromis*	*Chromis atripectoralis*	47,60	16S Fish
Pomacentridae	*Chrysiptera*	*Chrysiptera rollandi*	37,47	16S Fish
Pomacentridae	*Hemiglyphidodon*	*Hemiglyphidodon plagiometopon*	17	16S Fish
Pomacentridae	*Neoglyphidodon*	*Neoglyphidodon melas*	47	16S Fish
Pomacentridae	*Pomacentrus*	*Pomacentrus albicaudatus*	34,35,56,60,61,62,63	16S Fish
Pomacentridae	*Stegastes*	*Stegastes fasciolatus*	66	COI Elasmobranch
Pomacentridae	*Abudefduf*		60	16S Reptile
Pomacentridae	*Acanthochromis*		26,27,28,29,31,32,33,35,43,47,49,51,61,63	16S Fish

Pomacentridae	*Chromis*		66	16S Fish, 16S Reptile
Pomacentridae	*Dascyllus*		66	16S Fish
Pomacentridae	*Dischistodus*		49,50	16S Fish
Pomacentridae	*Neopomacentrus*		26,34,35,36,37,45,46,47,60	16S Fish, COI Elasmobranch
Pomacentridae	*Pomacentrus*		34,37,62,66	16S Fish
Pomacentridae	*Stegastes*		34,37,51,55,60,61	16S Fish, 16S Reptile
Pomacanthidae	*Chaetodontoplus*	*Chaetodontoplus duboulayi*	1,4,11,12,13,15,16,17,18,20,22,23,24,25,27,29,31,33,40,41,49,52,59,61,62,63,66	16S Fish, 16S Reptile
Pomacanthidae	*Chaetodontoplus*	*Chaetodontoplus mesoleucus*	29,51	16S Fish
Pomacanthidae	*Pomacanthus*	*Pomacanthus sexstriatus*	10,12,13,19,24,26,27,28,29,35,36,37,42,47,51,60,62	16S Fish, COI Elasmobranch, 16S Reptile
Polynemidae	*Eleutheronema*	*Eleutheronema tetradactylum*	1,8,9,11,14,17,18,19,20,21,22,23,24,25,27,30,32,33,38,65,66,67,68,69,70,71	16S Fish, COI Elasmobranch, 16S Reptile
Polynemidae	*Eleutheronema*		19,21,22,67,68,69,70,71	16S Fish
Platycephalidae	*Cymbacephalus*	*Cymbacephalus nematophthalmus*	14,17,37	COI Elasmobranch
Platycephalidae	*Cymbacephalus*	*Cymbacephalus staigeri*	21,23	16S Fish
Platycephalidae	*Elates*	*Elates ransonnettii*	12,59	16S Fish
Platycephalidae	*Inegocia*	*Inegocia harrisii*	36	16S Fish
Platycephalidae	*Inegocia*	*Inegocia japonica*	11,13,14,18,24,55,64,65	16S Fish, COI Elasmobranch
Platycephalidae	*Platycephalus*	*Platycephalus australis*	13,53,54,65	16S Fish
Platycephalidae	*Platycephalus*	*Platycephalus westraliae*	13	16S Fish
Platycephalidae	*Sorsogona*	*Sorsogona tuberculata*	1,4,7	16S Fish
Monodactylidae	*Monodactylus*	*Monodactylus argenteus*	13,14,21,31,41	16S Fish, 16S Reptile
Menidae	*Mene*	*Mene maculata*	12,46,53	16S Fish
Leptobramidae	*Leptobrama*	*Leptobrama muelleri*	67,68,69,70,71	16S Fish, COI Elasmobranch
Latidae	*Lates*	*Lates calcarifer*	1,7,8,9,14,19,20,21,22,23,25,26,27,30,32,33,34,35,36,38,39,40,41,42,43,44,46,47,53,59,61,62,63,64,65,66,67,68,69,70,71	16S Fish, COI Elasmobranch
Latidae	*Psammoperca*	*Psammoperca waigiensis*	1,5,6,8,9,14,16,17,24,26,28,29,31,33,34,47,48,51,59,61	16S Fish
Latidae	*Lates*		19,22,23,32,42,67,68,69,70,71	16S Fish
Emmelichthyidae	*Plagiogeneion*	*Plagiogeneion rubiginosum*	66	16S Reptile
Cottidae	*Cottus*		71	16S Fish

Centrogeniidae	*Centrogenys*	*Centrogenys vaigiensis*	13,18,24,25,26,27,28,29,31,32,33,51,59,60	16S Fish
Apistidae	*Apistus*	*Apistus carinatus*	49	16S Fish
Soleidae			2,15	16S Fish
Psettodidae	*Psettodes*		24,26	16S Fish
Paralichthyidae	*Pseudorhombus*	*Pseudorhombus jenynsii*	36	16S Fish
Priacanthidae	*Priacanthus*	*Priacanthus hamrur*	2,3,5,7,8	16S Fish
Salmonidae	*Salmo*		51,56,63	16S Fish, COI Elasmobranch
Trichiuridae	*Trichiurus*	*Trichiurus brevis*	35,36,37,59,60,61,62	16S Fish
Trichiuridae	*Lepturacanthus*		35	COI Elasmobranch
Trichiuridae	*Trichiurus*		36,37,59,61	16S Fish
Scombridae	*Acanthocybium*	*Acanthocybium solandri*	55	16S Reptile
Scombridae	*Gymnosarda*	*Gymnosarda unicolor*	66	16S Fish, 16S Reptile
Scombridae	*Rastrelliger*	*Rastrelliger kanagurta*	64	COI Elasmobranch
Scombridae	*Scomberomorus*	*Scomberomorus commerson*	1,2,5,7,8,9,11,12,13,15,17,18,24,28,36,37,39,41,42,47,55, 56,57,58,59,60,61,62,63,64,65,66,67,68,69	16S Fish, COI Elasmobranch, 16S Reptile
Scombridae	*Auxis*		38,49,66	16S Fish, 16S Reptile
Scombridae	*Euthynnus*		1,3,4,6,7,11,12,44,49,56,58,60,62,63,65,66	16S Fish, COI Elasmobranch, 16S Reptile
Scombridae	*Gymnosarda*		66	16S Fish
Scombridae	*Rastrelliger*		2,3,5,6,7,8,11,12,13,44,48,49,55,58,62,63,64	16S Fish
Scombridae	*Scomber*		22,46,56	16S Fish, COI Elasmobranch
Scombridae	*Scomberomorus*		5,6,8,9,10,11,12,13,14,15,16,17,19,20,21,22,23,25,27,29,3 0,32,33,35,36,38,39,41,42,43,44,45,47,51,55,58,59,60,61, 62,63,64,65,66,67	16S Fish, COI Elasmobranch, 16S Reptile
Scombridae	*Thunnus*		1,11,12,13,15,16,20,38,43,62,65,66	16S Fish, COI Elasmobranch, 16S Reptile
Scombridae			3	16S Reptile
Arripidae	*Arripis*	*Arripis georgianus*	46,48,49,50,53,54,55	16S Fish, 16S Reptile
Arripidae	*Arripis*		65	16S Fish
Ariidae	*Netuma*	*Netuma thalassina*	65	COI Elasmobranch
Ariidae	*Plicofollis*	*Plicofollis argyropleuron*	38,70	16S Fish
Ariidae	*Neoarius*		14,18,22	COI Elasmobranch
Ariidae	*Netuma*		2,5,7,13	16S Fish

Ariidae	*Plicofollis*		67,68,69,71	16S Reptile
Ariidae			18,22,23,32,38,67,68,69,70,71	16S Fish, 16S Reptile
Sparidae	*Rhabdosargus*	*Rhabdosargus sarba*	68	16S Fish
Sparidae	*Acanthopagrus*		9,12,13,14,15,17,19,20,21,22,23,25,27,28,29,30,31,32,33, 38,39,40,41,67,68,70,71	16S Fish
Sparidae	*Pagrus*		50,52	16S Fish
Nemipteridae	*Nemipterus*	*Nemipterus peronii*	12,25,27,29,58	COI Elasmobranch
Nemipteridae	*Scolopsis*	*Scolopsis monogramma*	1,2,4,5,12,14,16,17,28,29,34,35,42,45,47,59,61,63,64	16S Reptile
Nemipteridae	*Nemipterus*		12	COI Elasmobranch
Lethrinidae	*Lethrinus*	*Lethrinus atkinsoni*	66	16S Fish
Lethrinidae	*Lethrinus*	*Lethrinus laticaudis*	1,11,12,13,19,22,23,24,25,26,27,28,29,31,33,34,35,41,43, 46,51,58,59,60,62,65	16S Fish, COI Elasmobranch
Lethrinidae	*Lethrinus*	*Lethrinus lentjan*	8,14,15,17,18,29,30,47,51	16S Fish
Lethrinidae	*Lethrinus*	*Lethrinus nebulosus*	1,12,33,35,46,47,49,50,51,53,54,55,56,57,58,59,60,61,62, 63,64,65,66	16S Fish
Lethrinidae	*Monotaxis*	*Monotaxis grandoculis*	66	16S Fish
Lethrinidae	*Lethrinus*		9,12,13,14,17,22,24,26,28,29,33,34,35,41,43,46,47,51,54, 56,57,58,59,61,62,65	16S Fish
Syngnathidae	*Hippichthys*	*Hippichthys penicillus*	17,41	16S Reptile
Pegasidae	*Pegasus*	*Pegasus volitans*	17,19,22,24,25,27,28	16S Fish, 16S Reptile
Pegasidae	*Pegasus*		17,22,24,50,63	16S Fish
Mullidae	*Mulloidichthys*	*Mulloidichthys vanicolensis*	12,23,44	COI Elasmobranch
Mullidae	*Parupeneus*	*Parupeneus barberinus*	34,35,48	16S Fish, 16S Reptile
Mullidae	*Parupeneus*	*Parupeneus forsskali*	9,11,12,22,46,53,56,59,60,63	16S Reptile
Mullidae	*Parupeneus*	*Parupeneus multifasciatus*	66	16S Fish, 16S Reptile
Mullidae	*Upeneus*	*Upeneus tragula*	2,3,4,5,6,7,8,9,10,12,14,15,17,26,28,29,34,35,36,37,44,45, 47,48,49,50,51,52,53,55,58,59,60,61,62,63,64,65,66	16S Fish, COI Elasmobranch, 16S Reptile
Mullidae	*Upeneus*		1,3,4,5,6,7,8,10,12,15,36,47,49,53,58,60,61,62	16S Fish
Triacanthidae	*Trixiphichthys*	*Trixiphichthys weberi*	11,12,13,36	16S Fish, 16S Reptile
Tetraodontidae	*Arothron*	*Arothron manilensis*	1,7,9,12,13,14,15,16,17,18,19,22,23,27,29,30,31,32,38,40, 55,60,66,70	16S Fish, 16S Reptile
Tetraodontidae	*Arothron*	*Arothron stellatus*	13,23,67	16S Reptile

Tetraodontidae	*Chelonodontops*	*Chelonodontops patoca*	12,14,15,17,21,22,23,32,38,44,45,67,68,71	16S Fish, COI Elasmobranch, 16S Reptile
Tetraodontidae	*Lagocephalus*	*Lagocephalus sceleratus*	56,66	16S Fish, COI Elasmobranch
Tetraodontidae	*Marilyna*	*Marilyna darwinii*	9,14,15,16,18,22,23,32,33,38,39,40,70	16S Fish, COI Elasmobranch, 16S Reptile
Tetraodontidae	*Marilyna*	*Marilyna meraukensis*	68,69,71	16S Fish, 16S Reptile
Tetraodontidae	*Torquigener*	*Torquigener brevipinnis*	49	16S Reptile
Tetraodontidae	*Arothron*		14,16,17,18,19,22,27,29,30,31,39,41,47	16S Fish, COI Elasmobranch
Tetraodontidae	*Chelonodontops*		14,17,22,23	16S Fish
Tetraodontidae	*Lagocephalus*		43,38,44,48,53	16S Fish, COI Elasmobranch, 16S Reptile
Tetraodontidae	*Takifugu*		5,14,17	16S Reptile
Tetraodontidae	*Torquigener*		48	16S Fish
Tetraodontidae			21,54	16S Fish, 16S Reptile
Monacanthidae	*Aluterus*	*Aluterus monoceros*	66	16S Fish
Monacanthidae	*Monacanthus*	*Monacanthus chinensis*	1,4,6,9,14,17,24,34,39,47,50,52	16S Fish, 16S Reptile
Monacanthidae	*Paramonacanthus*	*Paramonacanthus choirocephalus*	5,17,27,49,58,60	16S Reptile
Monacanthidae	*Paramonacanthus*		5,14,17,52	COI Elasmobranch
Diodontidae	*Diodon*	*Diodon hystrix*	26	16S Reptile
Diodontidae	*Diodon*		2,3,4,5,58	16S Reptile
Balistidae	*Balistapus*	*Balistapus undulatus*	66	16S Fish
Balistidae	*Melichthys*	*Melichthys vidua*	66	16S Fish
Balistidae	*Odonus*	*Odonus niger*	66	16S Reptile
Balistidae			47,60,61,66	16S Fish, 16S Reptile
Hemigaleidae	*Hemigaleus*	*Hemigaleus australiensis*	27,35,52,59	COI Elasmobranch
Hemigaleidae	*Hemipristis*	*Hemipristis elongata*	9	COI Elasmobranch
Carcharhinidae	*Carcharhinus*	*Carcharhinus amblyrhynchos*	15,16,38,45,46,50,53,54,56,57,59,60,61,64,66	COI Elasmobranch
Carcharhinidae	*Carcharhinus*	*Carcharhinus amboinensis*	53,65,68,71	COI Elasmobranch
Carcharhinidae	*Carcharhinus*	*Carcharhinus leucas*	53,55,56,57,58,62,63	COI Elasmobranch
Carcharhinidae	*Carcharhinus*	*Carcharhinus melanopterus*	3,13,14,24,25,28,29,31,37,38,43,47,51,52,58,60	COI Elasmobranch
Carcharhinidae	*Carcharhinus*	*Carcharhinus sorrah*	17,43,48,51,53,59,60,61,63,64,66	COI Elasmobranch
Carcharhinidae	*Galeocerdo*	*Galeocerdo cuvier*	29,52	COI Elasmobranch

Carcharhinidae	*Glyphis*	*Glyphis garricki*	68,71	COI Elasmobranch
Carcharhinidae	*Negaprion*	*Negaprion acutidens*	14,15,17,22,38,43,63	COI Elasmobranch
Carcharhinidae	*Rhizoprionodon*	*Rhizoprionodon acutus*	13,14,15,42,44,51,56,59,60	COI Elasmobranch
Carcharhinidae	*Rhizoprionodon*	*Rhizoprionodon taylori*	11,33,51,54,71	COI Elasmobranch
Carcharhinidae	*Sphyrna*	*Sphyrna lewini*	41,43,51,53,54	COI Elasmobranch
Carcharhinidae	*Sphyrna*	*Sphyrna mokarran*	16,17,22,23,25,26,30,35,38,40,43,51,65	COI Elasmobranch
Carcharhinidae	*Triaenodon*	*Triaenodon obesus*	24,47,51,58,59	16S Fish, COI Elasmobranch
Carcharhinidae	*Carcharhinus*		4,5,12,13,14,15,16,17,21,22,23,24,25,26,27,28,29,32,33,34,36,37,38,48,49,50,51,53,54,55,56,57,58,60,61,62,63,64,65,67,68,69,70,71	16S Fish, COI Elasmobranch
Carcharhinidae			24,33,39,44,51,54,65,67,69	16S Fish
Potamotrygonidae	*Taeniura*	*Taeniura lymma*	14,26,27,29,34,35,40,48,50	16S Fish
Potamotrygonidae	*Taeniura*	*Taeniura meyeni*	34,41,47,66	16S Fish, COI Elasmobranch
Myliobatidae	*Aetobatus*	*Aetobatus narinari*	1,9,13,14,15,16,19,21,22,24,26,27,28,29,31,34,35,39,41,47,49,51,54,60,62,63,67,68	16S Fish
Myliobatidae	*Aetomylaeus*	*Aetomylaeus vespertilio*	15,34,39,59	COI Elasmobranch
Myliobatidae	*Mobula*		18,59	COI Elasmobranch
Dasyatidae	*Dasyatis*	*Dasyatis microps*	13,27	COI Elasmobranch
Dasyatidae	*Himantura*	*Himantura astra*	25,27,37,39,51,53,54,65	16S Fish
Dasyatidae	*Himantura*	*Himantura granulata*	4,14,15,16,17,24,25,26,27,28,29,34,35,37,43,46,47,55,59,62,64	COI Elasmobranch
Dasyatidae	*Himantura*	*Himantura leoparda*	14,15,16,17,18,22,23,25,27,29,32,39,40,49	COI Elasmobranch
Dasyatidae	*Himantura*	*Himantura uarnak*	50,55,66,68,71	16S Fish
Dasyatidae	*Neotrygon*	*Neotrygon kuhlii*	1,7,24,27,45,47,48,49,50,51,55,56,60,62	16S Fish, COI Elasmobranch
Dasyatidae	*Neotrygon*	*Neotrygon leylandi*	45,55	16S Fish
Dasyatidae	*Neotrygon*	*Neotrygon ningalooensis*	13,14,16,29,51	16S Fish
Dasyatidae	*Pastinachus*	*Pastinachus atrus*	3,13,14,15,22,51,65	16S Fish, COI Elasmobranch
Dasyatidae	*Pateobatis*	*Pateobatis fai*	3,17,32	COI Elasmobranch
Dasyatidae	*Pateobatis*	*Pateobatis jenkinsii*	16,25,40,50,51	COI Elasmobranch
Dasyatidae	*Urogymnus*	*Urogymnus asperrimus*	3,13,17,23,27,29,34,45,47,48,51,63	COI Elasmobranch
Dasyatidae	*Himantura*		11,22,23,25,27,30,37,38,44,50,51,52,65	COI Elasmobranch

Dasyatidae	*Neotrygon*		17	COI Elasmobranch
Stegostomatidae	*Stegostoma*	*Stegostoma fasciatum*	2,4,14	COI Elasmobranch
Orectolobidae	*Orectolobus*		47	COI Elasmobranch
Hemiscylliidae	*Chiloscyllium*	*Chiloscyllium punctatum*	3,4,13,29,34,35,36,42,43,46,47	COI Elasmobranch
Ginglymostomatidae	*Nebrius*	*Nebrius ferrugineus*	1,15,24,26,27,28,29,36,42,47,48,51	COI Elasmobranch
Rhynchobatidae	*Rhynchobatus*		17,27,31,54,59,67	16S Fish
Pristidae	*Anoxypristis*	*Anoxypristis cuspidata*	51,52,53,54	16S Fish, COI Elasmobranch
Pristidae	*Pristis*	*Pristis clavata*	34,38,39	16S Fish
Pristidae	*Pristis*	*Pristis pristis*	62,68	16S Fish
Crocodylidae	*Crocodylus*	*Crocodylus porosus*	69,71	16S Reptile
Pythonidae	*Aspidites*	*Aspidites melanocephalus*	1	16S Reptile
Hydrophiidae	*Hydrophis*	*Hydrophis stokesii*	51	16S Reptile
Homalopsidae	*Fordonia*	*Fordonia leucobalia*	67	16S Reptile
Cheloniidae	*Chelonia*	*Chelonia mydas*	25,28,29,33,34,37,47,49	16S Reptile

Table S3.4 Distance based linear model (DistLM) analysis of bony fish community variation across the inshore sites.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*	
All sites						
Marginal	**Latitude**	2.705	0.072		0.000	***
	Depth	2.364	0.063		0.000	***
	Subregion	2.535	0.130		0.000	***
	Bioregion	3.281	0.086		0.000	***
Sequential (step-wise)	**Bioregion**	3.281	0.086	0.086	0.000	***
	Depth	2.878	0.071	0.157	0.000	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		303	0.157	2	2, 4	
Canning & South Kimberley						
Marginal	**Latitude**	2.279	0.098		0.000	***
	Depth	2.055	0.089		0.005	**
	Bioregion	2.913	0.122		0.001	***
Sequential (step-wise)	**Bioregion**	2.913	0.122	0.122	0.000	***
	Depth	2.621	0.102	0.224	0.000	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		188.5	0.224	2	2, 3	
South & North Kimberley						
Marginal	**Latitude**	1.816	0.061		0.002	**
	Depth	2.960	0.096		0.000	***
	Subregion	1.654	0.056		0.007	**
Sequential (step-wise)	**Depth**	2.960	0.096	0.096	0.000	***
	Distance	1.934	0.060	0.156	0.000	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		246.08	0.156	2	1, 2	
Canning & North Kimberley						
Marginal	**Latitude**	3.053	0.138		0.000	***
	Depth	1.416	0.069		0.053	*
	Bioregion	3.501	0.156		0.000	***
Sequential (step-wise)	**Bioregion**	3.501	0.156	0.156	0.000	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		171.46	0.156	1	3	

Table S3.5 ANOVA of taxonomic richness of inshore bony fish between bioregions.
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Habitat	Groups	d.f	Sum of squares	Mean Sq	F value	*P*
Inshore	Bioregion	1	549	549.3	2.532	0.121
	Residuals	35	7594	217.0		

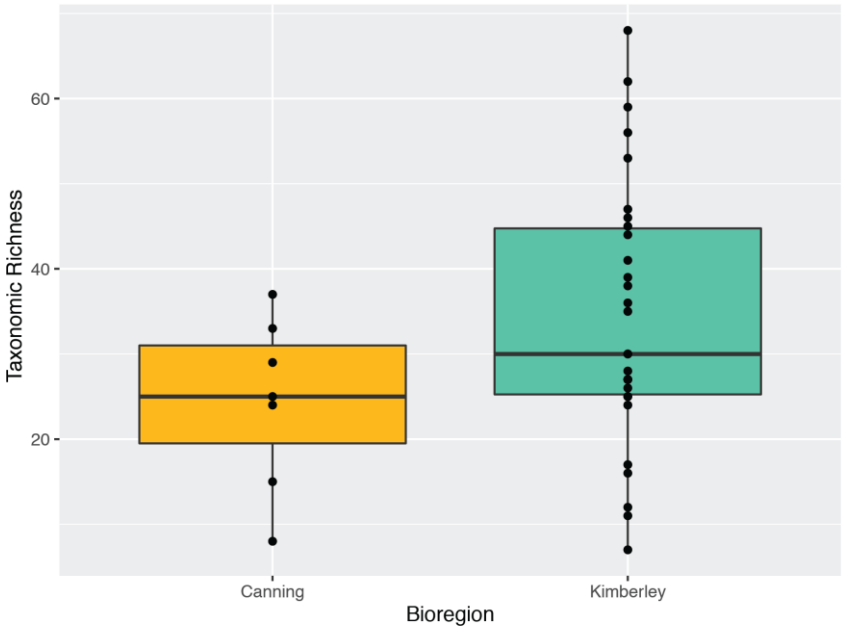


Figure S3.13 Taxonomic richness in inshore bony fish across the bioregions.

Table S3.6 Similarity percentage analysis (SIMPER) of bony fish in inshore sites. Pairwise dissimilarity between the bioregions (Canning and Kimberley).

Bioregion	Avg. Diss %	Taxa	Av. Abund in Group 1	Av. Abund in Group 2	Av. Diss	Diss/ SD	Contrib.%	Cum. %
Canning & Kimberley	83.05	*Amblygaster*	0.86	0.02	1.49	1.73	1.8	1.8
		Choerodon cephalotes	0.86	0.05	1.47	1.64	1.76	3.56
		Rastrelliger	0.71	0.16	1.18	1.13	1.42	4.99
		Parexocoetus	0.71	0.16	1.14	1.11	1.38	6.36
		Upeneus	0.71	0.13	1.13	1.21	1.36	7.72
		Chelmon	0.57	0.16	1.12	0.86	1.35	9.06
		Sillago	0.86	0.48	1.04	0.85	1.25	10.31
		Istigobius nigroocellatus	0.57	0.03	1.03	0.99	1.24	11.55
		Atherinomorus	0.57	0.03	1.02	0.99	1.23	12.78
		Upeneus tragula	0.57	0.43	1.01	0.82	1.21	13.99
		Glaucosoma magnificum	0.57	0.15	1.01	0.96	1.21	15.21
		Scomberomorus	0	0.54	1	0.9	1.21	16.41
		Choerodon vitta	0.57	0.05	0.98	1.03	1.18	17.59
		Scomberomorus commerson	0.57	0.31	0.96	0.92	1.16	18.75
		Caranx ignobilis	0	0.56	0.95	1.01	1.14	19.89
		Gerres filamentosus	0.57	0.13	0.94	1.02	1.14	21.03
		Atule mate	0.57	0.36	0.93	0.95	1.12	22.16
		Sphyraena obtusata	0.57	0.07	0.93	1.04	1.12	23.28
		Lutjanus	0.71	0.62	0.93	0.75	1.12	24.39
		Lutjanus carponotatus	0.57	0.64	0.93	0.88	1.12	25.51
		Haemulidae	0.57	0.3	0.92	0.98	1.11	26.62
		Encrasicholina	0.43	0.46	0.92	0.87	1.11	27.73
		Diodon	0.43	0	0.87	0.7	1.04	28.77
		Scomberoides	0.43	0.3	0.85	0.83	1.03	29.8
		Leiognathidae	0.43	0.33	0.85	0.86	1.03	30.83
		Caranx	0	0.49	0.84	0.88	1.01	31.84
		Terapon theraps	0.43	0.2	0.84	0.81	1.01	32.85
		Euthynnus	0.43	0.15	0.78	0.82	0.94	34.78
		Rachycentron canadum	0.43	0.03	0.78	0.77	0.94	35.72
		Chaetodontoplus duboulayi	0.29	0.26	0.78	0.65	0.94	36.66
		Netuma	0.43	0	0.77	0.77	0.93	37.59
		Siganus	0.29	0.38	0.77	0.81	0.92	38.51
		Choerodon cyanodus	0	0.46	0.76	0.85	0.91	39.42
		Selar crumenophthalmus	0.43	0.03	0.74	0.81	0.9	40.32
		Caesio cuning	0.14	0.41	0.74	0.81	0.89	41.2
		Priacanthus hamrur	0.43	0.02	0.71	0.81	0.86	42.06
		Drepane	0.14	0.34	0.67	0.72	0.81	42.87
		Plectorhinchus gibbosus	0.14	0.33	0.67	0.69	0.81	43.68
		Scolopsis monogramma	0.29	0.11	0.67	0.56	0.8	44.48
		Ellochelon vaigiensis	0.14	0.34	0.66	0.73	0.8	45.28
		Psammoperca waigiensis	0.29	0.23	0.66	0.73	0.79	46.07
		Choerodon	0.29	0.15	0.63	0.68	0.76	46.83
		Halichoeres nigrescens	0	0.39	0.61	0.75	0.74	47.57
		Lethrinus nebulosus	0.14	0.26	0.6	0.65	0.72	48.28
		Lutjanidae	0	0.39	0.6	0.75	0.72	49
		Choerodon schoenleinii	0.14	0.31	0.59	0.71	0.71	49.72
		Scarus	0.14	0.31	0.59	0.71	0.71	50.43

Table S3.7 Distance based linear model (DistLM) analysis of bony fish community variation across the coastal sites.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*	
All sites - South & North Kimberley only						
Marginal	**Latitude**	3.062	0.170		0.000	***
	Depth	2.483	0.142		0.001	**
	Subregion	3.239	0.178		0.000	***
Sequential (step-wise)	**Subregion**	3.239	0.178	0.178	0.000	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		138.22	0.178	1	3	

Table S3.8 Distance based linear model (DistLM) analysis of bony fish community variation across the nearshore estuarine sites.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*	
All sites - South Kimberley only						
Marginal	**Latitude**	1.651	0.130		0.006	**
	Depth	1.351	0.110		0.064	
Sequential (step-wise)	-	-	-	-	-	
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		107.71	0	0	-	

Table S3.9 Distance based linear model (DistLM) analysis of shark and ray community variation across the inshore sites.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*	
All sites						
Marginal	**Latitude**	1.636	0.061		0.039	*
	Depth	1.485	0.056		0.081	
	Subregion	2.657	0.181		0.000	***
	Bioregion	2.543	0.092		0.002	**
Sequential (step-wise)	**Subregion**	2.6572	0.181	0.181	0.000	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		224.16	0.181	1	3	
Canning & South Kimberley						
Marginal	**Latitude**	1.812	0.131		0.001	**
	Depth	1.225	0.093		0.211	
	Bioregion	2.836	0.191		0.013	*
Sequential (step-wise)	**Bioregion**	2.836	0.191	0.191	0.011	*
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		116.93	0.191	1	3	
South & North Kimberley						
Marginal	**Latitude**	1.869	0.075		0.017	*
	Depth	1.602	0.065		0.049	*
	Subregion	2.499	0.098		0.000	***
Sequential (step-wise)	**Subregion**	2.499	0.098	0.098	0.001	***
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		207.92	0.098	1	3	
Canning & North Kimberley						
Marginal	**Latitude**	2.160	0.142		0.008	**
	Depth	1.221	0.086		0.247	
	Bioregion	2.784	0.176		0.010	*
Sequential (step-wise)	**Bioregion**	2.784	0.176	0.176	0.010	*
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		123.27	0.176	1	3	

Table S3.10 ANOVA of taxonomic richness of inshore shark and ray between subregions.

Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Habitat	Groups	d.f	Sum of squares	Mean Sq	F value	*P*
Inshore	Subregion	2	22.18	11.092	1.519	0.239
	Residuals	24	175.22	7.301		

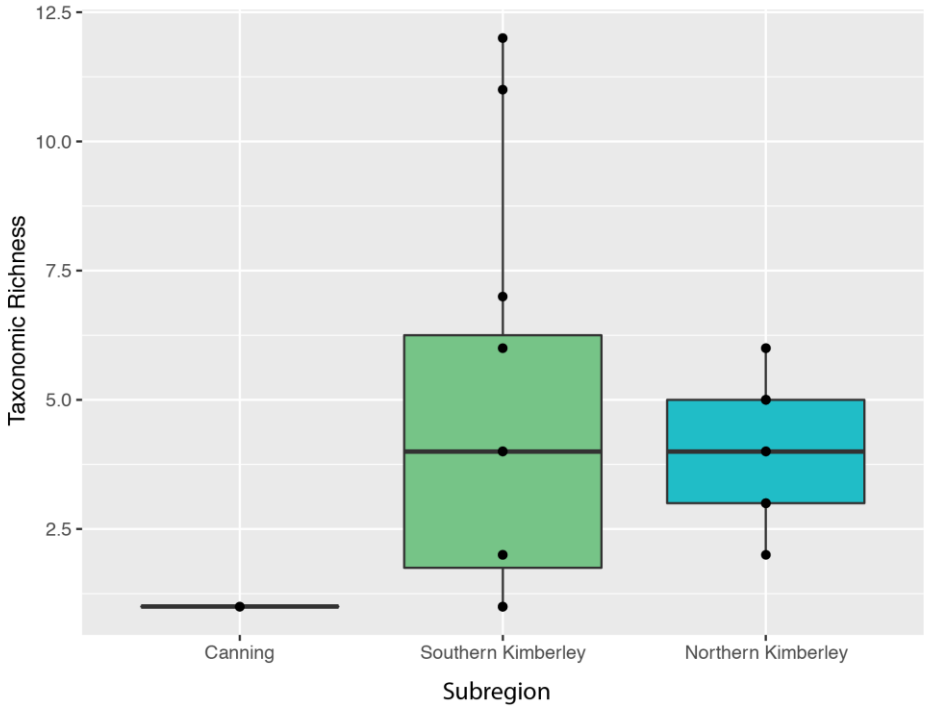


Figure S3.14 Taxonomic richness in inshore shark and ray taxa across the subregions.

Table S3.11 Similarity percentage analysis (SIMPER) of sharks and rays in inshore sites.

Pairwise dissimilarity between the subregions (Canning, South Kimberley and North Kimberley).

Subregion	Avg. Diss %	Taxa	Av. Abund in Group 1	Av. Abund in Group 2	Av. Diss	Diss/ SD	Contrib. %	Cum. %
Canning & South Kimberley	98.72	*Neotrygon kuhlii*	1	0.08	25.98	1.55	26.32	26.32
		Himantura granulata	0	0.58	8.57	1.02	8.68	35
		Taeniura lymma	0	0.42	8.36	0.58	8.47	43.47
		Chiloscyllium punctatum	0	0.42	7.37	0.7	7.47	50.93
		Aetobatus narinari	0	0.33	7.17	0.5	7.26	58.19
		Carcharhinus	0	0.33	5.78	0.56	5.85	64.05
		Himantura	0	0.25	5.09	0.49	5.15	69.2
		Carcharhinus melanopterus	0	0.33	5.07	0.64	5.14	74.34
South Kimberley & North Kimberley	86.15	*Carcharhinus*	0.33	0.69	7.75	0.99	9	9
		Carcharhinus amblyrhynchos	0	0.46	6.7	0.81	7.78	16.78
		Himantura granulata	0.58	0.15	6.05	0.97	7.03	23.81
		Carcharhinus leucas	0	0.46	5.84	0.82	6.78	30.58
		Aetobatus narinari	0.33	0.23	5.23	0.69	6.07	36.65
		Chiloscyllium punctatum	0.42	0.08	4.99	0.76	5.79	42.44
		Taeniura lymma	0.42	0.08	4.9	0.72	5.68	48.12
		Neotrygon kuhlii	0.08	0.38	4.84	0.74	5.62	53.74
		Carcharhinus melanopterus	0.33	0.23	4.76	0.76	5.53	59.27
		Urogymnus asperrimus	0.25	0.23	3.83	0.68	4.44	63.71
		Himantura	0.25	0.08	3.45	0.57	4	67.72
		Carcharhinus sorrah	0.08	0.23	3.36	0.55	3.9	71.62
Canning & North Kimberley	86.15	*Neotrygon kuhlii*	1	0.38	13.53	1.14	15.64	15.64
		Carcharhinus	0	0.69	12.97	1.38	14.99	30.63
		Carcharhinus amblyrhynchos	0	0.46	10.13	0.89	11.71	42.34
		Carcharhinus leucas	0	0.46	8.48	0.87	9.8	52.14
		Aetobatus narinari	0	0.23	4.95	0.49	5.72	57.85
		Carcharhinus sorrah	0	0.23	4.3	0.52	4.97	62.83
		Carcharhinus melanopterus	0	0.23	4.1	0.53	4.74	67.57
		Urogymnus asperrimus	0	0.23	3.92	0.53	4.53	72.1

Table S3.12 Distance based linear model (DistLM) analysis of shark and ray community variation across the coastal sites.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*	
All sites - South & North Kimberley only						
Marginal	Latitude	0.933	0.067		0.521	
	Depth	1.952	0.131		0.018	*
	Subregion	0.870	0.063		0.600	
Sequential (step-wise)	**Depth**	1.952	0.131	0.131	0.017	*
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**	
		124.33	0.131	1	2	

Table S3.13 Distance based linear model (DistLM) analysis of shark and ray community variation across the nearshore estuarine sites.
These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*
All sites - South Kimberley only					
Marginal	Latitude	0.519	0.080		0.964
	Depth	1.479	0.198		0.070
Sequential (step-wise)	-	-	-	-	-
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**
		68.296	0	0	

Table S3.14 Distance based linear model (DistLM) analysis of stripey snapper variation.

These were constructed using a sequential step-wise selection procedure and AIC criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Pseudo-F	Proportion	Cumulative Proportion	*P*
All sites					
Marginal	Latitude	1.311	0.034		0.221
	Depth	0.966	0.025		0.406
	Subregion	1.459	0.075		0.148
	Bioregion	2.020	0.052		0.069
Sequential (step-wise)	Bioregion	2.020	0.052	0.052	0.069
Best solution		**AIC**	**R^2**	**No. Groups**	**Selections**
		306.29	0.052	1	4

S3.9.3 Primer development note for 16S Reptile

S3.9.3.1 Introduction

The aquatic and semi-aquatic reptile fauna of northern Australia are highly distinctive, exhibiting life history traits and physical adaptations to extreme climates and seasonal, but increasingly inconsistent, water availability (Pusey, 2011). They form an integral component of the region's aquatic and riparian food webs across multiple trophic levels; the saltwater crocodile (*Crocodylus porosus*) often filling the role of apex predator in these ecosystems. There are currently 1092 recognised reptile (including terrestrial) species in Australia; 976 of which are endemic (Uetz, Freed, & Hošek, 2018). The majority of Australia's reptile species have not historically experienced population declines; this is purported to reflect their inhabitance of arid, inland areas where anthropogenic influence is minimal (Fox, 2008). However, a recent update to the IUCN Red List indicates that 7% of Australia's endemic reptiles are now threatened with extinction, whilst the majority are under threat from invasive species (e.g. feral cats and cane toad, *Rhinella marina*) and climate change; the latter affecting the intensity and frequency of bushfires (IUCN, 2018). In addition, latitudinal niche shifts to mitigate changing temperatures have prehistorically been much more challenging for ectotherms, such as reptiles, which have a lower climatic tolerance and reproductive ability (Rolland et al., 2018). The viability of reptile species that exhibit temperature-dependent sex determination (TSD) may be compromised with projected temperature levels in the next 50 years (Tomillo, Genovart, Paladino, Spotila, & Oro, 2015). Thus, the continual monitoring of reptile species, particularly in northern Australia, is an increasing necessity.

Typical survey techniques for the detection of aquatic reptiles include snorkelling, trapping and seining, although this can be limited in northern Australian waterbodies by the threat of saltwater crocodiles (Australian Government, 2011). In comparison to other faunal groups, there is a severe lack of data on aquatic reptile distributions across catchments in northern Australia (Fox, 2008). The majority of aquatic reptile surveys in this region have targeted the saltwater crocodile; two thirds of consolidated aquatic/semi-aquatic reptile distribution records are attributed to this species (Fox, 2008), leaving data deficiencies for other reptilian taxa.

The coastal and offshore waters of northern Australia are a biodiversity hotspot for sea snakes (Elapidae: Hydrophiinae) with at least 32 species, of which 4 are nationally endemic (Uetz et al., 2018). Two of these endemic species, *Aipysurus foliosquama* and *A. apraefrontalis*, were listed as Critically Endangered in 2011 under Australia's Environment Protection and Biodiversity Conservation (EPBC) Act 1999, given sustained declines on the Ashmore and Hibernia Reefs, with a total estimated area of occupancy of only 227 km^2 (Guinea, 2003; Lukoschek, Beger, Ceccarelli, Richards, & Pratchett, 2013). There is a lack of distribution data for many Australian sea snakes which reflects a global knowledge gap; 34% of all sea snakes are listed as Data Deficient (DD) on the IUCN Red List (Elfes et al., 2013). DD sea snakes in northern Australia include the Arafura sea snake (*Aipysurus tenuis*), Zweifel's sea snake (*Enhydrina zweifeli*), the faint-banded sea snake (*Hydrophis belcheri*), the fine-spined sea snake (*Hydrophis czeblukovi*), the slender-necked sea snake (*Hydrophis melanocephalus*) and the northern mangrove sea snake (*Parahydrophis mertoni*; Elfes et al., 2013). Other data deficient aquatic reptile taxa in northern Australia include freshwater turtles, such as Irwin's snapping turtle (*Elseya irwini*), the Gulf snapping turtle (*Elseya lavarackorum*), and the northern yellow-faced turtle (*Emydura tanybaraga*; Van Dyke, Ferronato, & Spencer, 2018), in addition to the marine flatback turtle (*Natator depressus*).

Whilst a high level of data deficiency does not necessarily correspond to a high extinction risk, insufficient information in regards to population trajectories, distribution and taxonomy creates uncertainty around extinction risk patterns, conservation progress, priorities and legislation (Bland & Böhm, 2016; Böhm et al., 2013). Robust distribution records provide the baseline upon which species ecology and conservation status can be developed. However, given the sheer number of DD reptiles, that may or may not be as threatened as assessed species, it may not be economically feasible to conduct in-depth distribution surveys of individual or small groups of DD taxa at a time. In addition, data-deficient taxa may require more specialised trapping techniques and/or taxonomic expertise to morphologically identify cryptic lineages.

Environmental DNA (eDNA) technologies offer a sensitive, cost-effective and non-invasive genetic alternative to individual species and multi-taxon surveying in marine,

freshwater and terrestrial environments (Bohmann et al., 2014; Evans et al., 2016; Olds et al., 2016; Thomsen et al., 2012). To date, reptile eDNA studies have largely applied species-specific markers to amplify individual species from mixed environmental samples (Baker et al., 2018; Davy, Kidd, & Wilson, 2015; De Souza, Godwin, Renshaw, & Larson, 2016; Feist et al., 2018; Halstead et al., 2017). The first reptile eDNA study, published in 2014, developed a diagnostic PCR to detect the Burmese python (*Python bivittatus*), a semi-aquatic, invasive species in Florida (Piaggio et al., 2014). The python's elusive nature, cryptic colouration and occupation of aquatic habitats that were logistically difficult to survey, prompted an eDNA approach. Piaggio et al., (2014) developed a *P. bivittatus*-specific mitochondrial cytochrome b assay that was applied to water samples from field sites in south Florida, successfully detecting the species where it had been previously observed.

Advances in high-throughput sequencing now allow the simultaneous amplification and sequencing of multiple taxa through universal or broad-taxonomic primer assays (referred to as eDNA metabarcoding), proving a more efficient approach to genetic surveying. However, the use of eDNA metabarcoding for the detection of multiple reptiles has not yet been thoroughly tested. Kelly, Port, Yamahara, & Crowder (2014) applied a metabarcoding assay (vertebrate-specific mitochondrial 12S rRNA primers) in order to detect green sea turtle (*Chelonia mydas*) in a large mesocosm, but were unsuccessful, despite successfully amplifying the intended target with species-specific primers. Lacoursière-Roussel, Dubois, Normandeau, & Bernatchez (2016) had more success in using COI metabarcoding assays to detect three species of snake and two species of turtle across lakes and rivers in Canada. Our main objectives in this study were to design a metabarcoding assay that simultaneously targets aquatic and semi-aquatic reptile groups in Australia and test the utility of this assay on 284 water samples collected across north-western Australia.

S3.9.3.2 Primer development and testing

Mitochondrial DNA is typically targeted for eDNA metabarcoding as there can be up to several thousand copies within a cell in comparison to one copy of nuclear DNA; subsequently mitochondrial DNA persists at a greater proportion than nuclear DNA in the environment (Rees, Maddison, Middleditch, Patmore, & Gough, 2014). Two Indo-

Pacific sea snake databases were curated for the mitochondrial 16S rRNA and cytochrome b gene regions by Sanders, Lee, Bertozzi, & Rasmussen (2013) and provided for this project. The cytochrome b database was substantially larger (361 sequences, 1104bp length) than the 16S database (50 sequences, 531bp length), however the cytochrome b region presented no conserved regions that were consistent across all sea snake sequences and such would be suitable for primer binding. Conversely, the 16S region exhibited a few conserved regions (approximately 30-50bp length) flanking larger hypervariable regions. Additional 16S rRNA sequences of a number of Australian aquatic and semi-aquatic reptiles (Table S3.15) were downloaded from NCBI GenBank (https://www.ncbi.nlm.nih.gov/genbank/) and aligned using the MUSCLE plugin in Geneious 10.0.6. Primer pairs were designed in conserved regions using the built-in primer design tool in Geneious; the final primer pair chosen by visually inspecting alignments for a target region with maximum variation and under a length of 280bp. The length constriction allows for the sequencing of barcode indexes (on the 5' end of a forward/reverse primer within a fusion-tagged primer setup) on a single-end Illumina MiSeq sequencing run (up to 325bp). The forward primer was modified to include a degenerate base allowing for greater binding across the reptile lineages. The resulting primer pair (herein referred to as the 16S Reptile assay) are AqReptileF-degenerate: 5'–AGACNAGAAGACCCTGTG-3' and AqReptileR: 5'–CCTGATCCAACATCGAGG-3'; with a G/T content between 50.0-55.6%, T_M between 52.0-55.5 and hairpin T_M of 32.4.

In-vitro testing of the primers to assess PCR amplification and optimise the annealing temperature was conducted using tissue extracts (1/10 dilution) from salt-water crocodile (*Crocodylus porosus)* and flatback turtle (*Natator depressus)*, and the 284 eDNA samples collected across northwestern Australia. A gradient PCR determined an optimum annealing temperature of 52°C. Each qPCR reaction was carried out in 25µl containing: 1X AmpliTaq Gold® PCR buffer (Life Technologies, Massachusetts, USA), 2mM $MgCl_2$, 0.1mM dNTPs, 0.2µM each of forward and reverse primers (Integrated DNA Technologies, Australia), 10ug BSA (Fisher Biotec, Australia), 0.6µl of 5X SYBR® Green (Life Technologies), 1U AmpliTaq Gold® DNA Polymerase (Life Technologies), 4µl of template DNA, and made to volume with Ultrapure™

Distilled Water (Life Technologies). Reptile extracts were amplified in duplicate on a StepOnePlus Real-Time PCR System (Applied Biosystems, Massachusetts, USA) under the following conditions: initial denaturation at 95 °C for 5 min, followed by 50 cycles of 30 s at 95 °C, 52°C for 30 s and 45 s at 72 °C, with a final extension for 10 min at 72 °C. Both tissue and eDNA samples were processed with the same library preparation, sequencing and bioinformatics workflow in S3.9.1 and S3.9.2.

S3.9.3.3 Results

S3.9.3.3.1 Primer design

The 16S Reptile assay was designed using sea snake, sea krait, crocodile, marine turtle, freshwater turtle and monitor lizard 16S rRNA reference sequences (Table S3.15). Target length of the amplified fragments ranged from 212bp (sea snake and sea krait) to 275bp (crocodile). There were no mismatches with the primers, except for the monitor lizard reference sequences (*Varanus indicus* and *V. panoptes*) which both exhibited one mismatch towards the 5' end of the forward primer. This should not hinder potential amplification of monitor lizards with the 16S Reptile assay.

The pairwise percent identity for sea snakes across the target region was 93.9%; the pairwise percent identity for congeneric sea snakes averaged 96.4% (min 93.4%, max 100%), providing enough variation to distinguish closely-related taxa. Across the two congeneric crocodiles and two congeneric monitor lizards, the pairwise percent identity was 95.3% and 88.8%, respectively. For marine turtles (superfamily: Chelonioidea) the average pairwise percent identity was 90.9%; and for freshwater turtles (family: Chelidae and Carettochelyidae) 74.0%.

S3.9.3.3.2 *In-vitro* primer testing

The 16S Reptile assay was firstly tested *in vitro* using tissue extractions of salt water crocodile and flatback turtle. These successfully amplified with an average cycle threshold (Ct) value of 27.1 and 21.9, respectively. The respective extracts matched to

NCBI reference sequences of saltwater crocodile and flatback turtle with percentage identities of 100% and 99.6%.

Primers were then tested *in vitro* with the 284 water samples collected across northwestern Australia. The majority of the eDNA extracts amplified with the 16S Reptile assay; albeit with high Ct values ranging from 28 to 39. The 16S Reptile assay yielded a total of 5,108,878 sequencing reads; the mean number of filtered sequences (post-quality, denoising and chimera filtering) was 32,569 ± 42,241 per sample (130,278 ± 114,036 per site). This resulted in a total of 181 taxa detected with the 16S Reptile assay (see Table S3.3; Figure 3.3), however the majority of taxa were not reptiles (see Figure S3.15). The highest average proportion of sequencing reads were attributed to mammals (class: Mammalia, 70.4%) largely *Homo sapien*, followed by bony fish (class: Actinopterygii, 19.3%), bivalve molluscs (class: Bivalvia, 8.6%), birds (class: Aves, 0.6%), brittle stars (class: Ophiuroidea, 0.4%) and reptiles (class: Reptilia, 0.4%). Only five reptile taxa were detected from 13 out of 71 sites (Table S3.16). Average read depth varies from 1 read (0.002% of average total reads) to 7652 reads (95.056% of average total reads). The percent identity match ranges from 95.8% to 100% (Table S3.16). The taxonomic assignment script (see S3.9.2 for more information) required a 98% match for a species level assignment, however the sequence that was matched to white-bellied mangrove snake (*Fordonia leucobalia)* at 95.8% identity, provided no other hits above 90%. Additionally, *F. leucobalia* belongs to a monotypic genus. Therefore, the sequence detected presents more than 4% intraspecific differentiation from the NCBI reference sequence of *F. leucobalia*, or represents a closely-related taxon that is yet to be taxonomically identified and barcoded.

S3.9.3.4 Discussion

The *in-vitro* eDNA testing of our 16S Reptile assay produced a relatively low detection rate of reptiles, with only five reptile taxa detected from 13 out of 71 sites. Saltwater crocodile (*C. porosus*) was only detected in two sites, despite its widespread distribution in coastal river systems across NWA. Additionally, we visually identified saltwater crocodiles in multiple nearshore estuarine sites, yet these sites did not return

a positive eDNA detection hit. In regards to sea snakes, we only detected two species (*H. stokesii* and *F. leucobalia*), both from one site each. The green sea turtle (*C. mydas*) was the only turtle species identified, however it exhibited a higher site abundance than the other reptiles. Given the widespread distribution of green sea turtles in the coastal and inshore areas of NWA, we still suspect that we under-detected turtles with our metabarcoding assay.

The poor performance of the 16S Reptile assay may be attributed to the non-specificity of the assay and/or a low shedding rate of reptiles in the environment. Reptiles only accounted for 0.4% of the average proportion of sequencing reads per sample, however this varied greatly between samples. The detection of Stokes's sea snake (*H. stokesii*) for example, accounted for 95.1% of the total sequencing reads from a site 51 replicate. Therefore, whilst the 16S Reptile assay does detect other taxonomic groups (mammals, bony fish, bivalve molluscs, birds and brittle stars), it is not exhibiting consistent preferential amplification of these groups above reptiles. Nonetheless, further optimisation of this assay to reduce non-target amplification is required; however, this may not be entirely possible for all non-reptile groups given that the primers are located in a highly-conserved region of the 16S rRNA gene. The inclusion of mismatches increases the risk that rare reptile variants will be excluded from amplification; additionally, the placement of mismatches on 3' ends may compromise the efficiency of qPCR assays (Wilcox, Schwartz, McKelvey, Young, & Lowe, 2014). Alternatively, the development of discrete assays for each major reptilian order (i.e. turtles, crocodiles, tuatara, and lizards and snakes) may be more effective in refining specificity, but retaining broad-taxonomic amplification. Another possibility is the use of a blocking primer, which preferentially binds and restricts amplification of a targeted taxonomic group. Blocking primers have been successfully used in conjunction with metabarcoding assays to increase the specificity of amphibian and bony fish amplicons by reducing the amplification of human DNA (Sasso et al., 2017; Valentini et al., 2016).

The shedding rate of reptiles may also undermine our ability to detect them via eDNA methods. It has recently been suggested that reptiles may have a lower shedding rate because of their keratinised scales and reduced urine production, and subsequently are less detectable than other mucus-covered organisms, such as fish and amphibians

(Adams, Hoekstra, Muell, & Janzen, 2019; Raemy & Ursenbacher, 2018). Whilst this is yet to be explicitly tested, it may explain the inconsistent amplification of reptiles in aquatic environments across multiple studies. For example, giant garter snake (*Thamnophis gigas*) mesocosm experiments reported positive eDNA detection in tanks with snake skin and snake feces, however no detection with live snakes in tanks, nor at field locations, despite capture of the species with traps (Halstead et al., 2017). The collection of water within one metre of eastern massasauga rattlesnakes (*Sistrurus catenatus*) only produced a positive eDNA detection for 2/100 water samples (Baker et al., 2018). Conversely, Lacoursière-Roussel, Dubois, Normandeau, & Bernatchez (2016) successfully used metabarcoding to detect redbelly snake (*Storeria occipitomaculata*), northern watersnake (*Nerodia sipedon*), milksnake (*Lampropeltis triangulum*), snapping turtle (*Chelydra serpentine*) and wood turtle (*Glyptemys insculpta*) in rivers and lakes in Canada; however the wood turtle was not detected in 4 rivers that produced positive detections via species-specific qPCR and visual surveying. Overall, there has been a lot more success in the detection of wild turtles with eDNA, primarily with species-specific assays (De Souza et al., 2016; Feist et al., 2018; Kelly et al., 2014), with a push to quantify turtle abundance and biomass using eDNA (Adams et al., 2019). In regards to crocodiles, there has only been one study that has attempted to amplify crocodile eDNA. However despite observing West African crocodile (*Crocodylus suchus*) and the Nile monitor (*Varanus niloticus*) in the water at their field sites in Mauritania, they were unable to amplify any crocodile or other reptilian eDNA using a metabarcoding approach (Egeter et al., 2018). A low shedding rate may therefore limit eDNA detection, despite a well-designed assay that is capable of amplifying the targeted taxa.

Another limitation to the implementation of eDNA metabarcoding for broad-reptile surveying is potential reference database gaps for data-deficient taxa. At present, only 27.6% of reptile species have been barcoded for the mitochondrial cytochrome *c* oxidase I (COI) gene; a short, standardised gene region that has historically formed the primary barcode sequence for animal species (Ratnasingham & Hebert, 2007). Only four sea snake species (Elapidae: Hydrophiinae) have been barcoded for this region; 2 of which are distributed in northern Australia. Our assay design targeted the mitochondrial 16S rRNA gene region, given the development of a 16S Indo-Pacific sea snake database with a high representation of northern Australian

sea snakes (Sanders et al., 2013). The development and implementation of eDNA metabarcoding for reptile species, particularly for data-deficient taxa, should ideally be tailored to available reference sequences for the targeted taxonomic group. As such, the development of discrete taxonomic assays, as discussed previously for refining specificity, may be a superior approach to ensure high-resolution assignments (i.e. to a species-level). Ultimately however, it will be easier to implement broad metabarcoding assays with either a standardised barcode region, a complete suite of barcoded gene regions, or ideally a complete mitochondrial genome for each representative species.

The inconsistent amplification and detection of aquatic reptilian eDNA, despite positive visual and trapping detections at survey sites, indicates that at present it provides an unreliable estimate of diversity and community composition between sites. We recommend that aquatic and semi-aquatic reptile shedding rates into various substrates (e.g. water, sediment, soil) are substantially tested before eDNA approaches, in particular metabarcoding, are further developed as reptile survey tools. This will provide greater insight into inconsistencies in amplification between taxonomic groups and whether assays need to be tailored to accommodate this i.e. the use of species-specific assays for taxa with low shedding rates. We further recommend that reptile eDNA metabarcoding assays are developed at an order level or lower in order to increase specificity and consider the coverage of reference databases for various gene regions.

Table S3.15 List of targeted Australian aquatic and semi-aquatic reptile species for in-silico primer analysis.

^ Indicates selection of reference sequences sourced from Sanders et al., (2013). * Indicates species whose tissue extracts were used to test primers *in vitro*. Under mismatches, F refers to forward primer.

Common name	Species name	Accession number	Target length (bp)	Mismatches
Sea snake				
^Short-nosed sea snake	*Aipysurus apraefrontalis*	JX423420	213	0
^Dusky sea snake	*Aipysurus fuscus*	JX423430	213	0
^Olive sea snake	*Aipysurus laevis*	EU547181	213	0
^Turtle-headed sea snake	*Emydocephalus annulatus*	EU547185	214	0
^Black-headed sea snake	*Hydrophis atriceps*	KC014320	212	0
^Blue-banded sea snake	*Hydrophis cyanocinctus*	KC014331	212	0
^Ornate sea snake	*Hydrophis ornatus*	KC014358	212	0
^Yellow-bellied sea snake	*Pelamis platurus*	KC014375	212	0
Stokes's sea snake	*Hydrophis stokesii*	JQ217146	212	0
Horned sea snake	*Hydrophis peronii*	KU323976	212	0
Shaw's sea snake	*Hydrophis curtus*	KX239662	212	0
Sea krait				
Yellow-lipped sea krait	*Laticauda colubrine*	NC_036054	212	0
Crocodile				
*Saltwater crocodile	*Crocodylus porosus*	NC_008143	275	0
Freshwater crocodile	*Crocodylus johnsoni*	NC_015238	275	0
Marine turtle				
*Flatback turtle	*Natator depressus*	NC_018550	259	0
Loggerhead turtle	*Caretta caretta*	MF579505	258	0
Olive ridley turtle	*Lepidochelys olivacea*	DQ486893	258	0
Leatherback turtle	*Dermochelys coriacea*	JX454992	259	0
Green turtle	*Chelonia mydas*	JX454990	260	0
Hawksbill turtle	*Eretmochelys imbricata*	MF571906	258	0
Freshwater turtle				
Pig-nosed turtle	*Carettochelys insculpta*	FJ862792	250	0
Cann's snake-necked turtle	*Chelodina canni*	NC_041286	254	0
Northern snake-necked turtle	*Chelodina rugosa*	KY776451	253	0
Northern snapping turtle	*Elseya dentata*	KY779844	251	0
Red-faced turtle	*Emydura victoriae*	NC_042473	253	0
Western swamp turtle	*Pseudemydura umbrina*	NC_035731	244	0
Monitor lizard				
Mangrove monitor	*Varanus indicus*	EF193674	219	1 (F)
Argus monitor	*Varanus panoptes*	EF193685	223	1 (F)

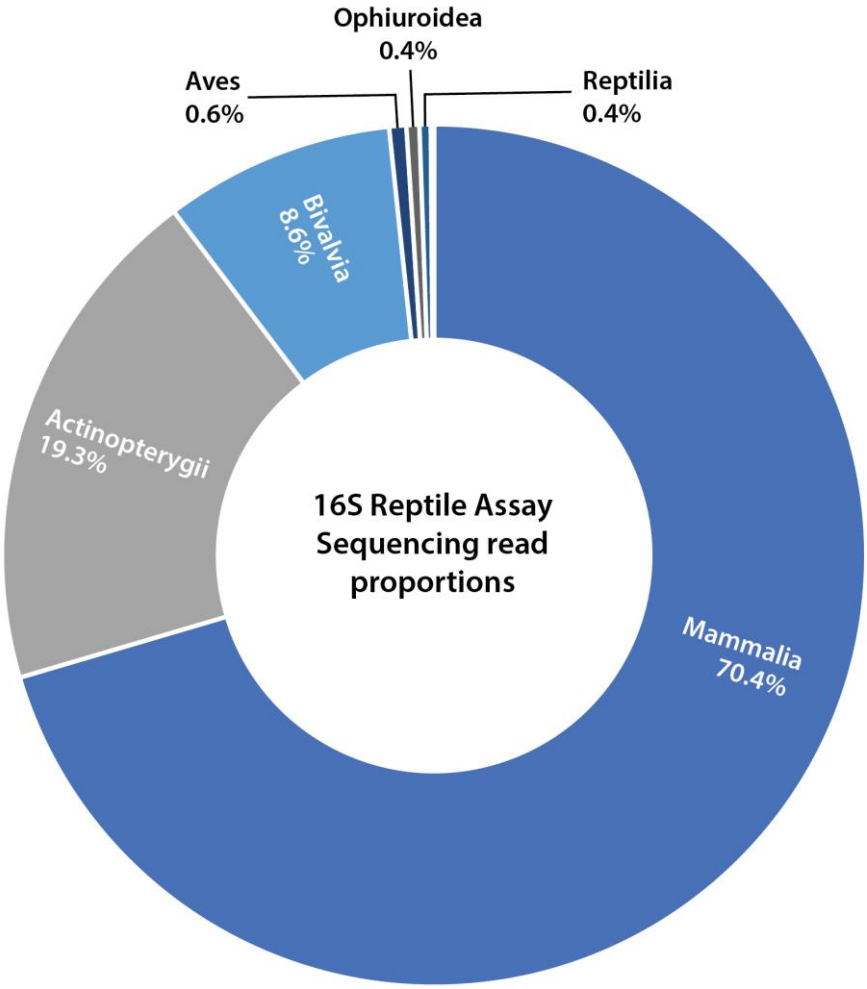


Figure S3.15 Average proportion of sequencing reads per sample for the 16S Reptile assay.
This is attributed at class level.

Table S3.16 Reptile taxa detected from environmental samples in northwestern Australia.

Common Name	Scientific Name	Distribution	Site/s detected	Average read depth	Average % of total reads	Average % identity
Saltwater crocodile	*Crocodylus porosus*	Indo-Australia. Located in soft substrate, estuarine and freshwater environments.	69, 71	9	0.680	100.0
Black-headed python	*Aspidites melanocephalus*	Northern Australia. Terrestrial species.	1	1	0.002	98.6
Stokes's sea snake	*Hydrophis stokesii*	Indo-Pacific. Located in soft and hard substrate environments.	51	7652	95.056	100.0
White-bellied mangrove snake	*Fordonia leucobalia*	Indo-Pacific. Located in soft substrate, estuarine and mangrove environments.	67	417	1.143	95.8
Green turtle	*Chelonia mydas*	Indo-Pacific and Atlantic. Located in hard and soft substrates, pelagic and seagrass environments.	25,28,29,33,3-4,37,47,49	1031	13.100	98.7

S3.9.4 Supplementary references

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Adams, C. I. M., Hoekstra, L. A., Muell, M. R., & Janzen, F. J. (2019). A Brief Review of Non-Avian Reptile Environmental DNA (eDNA), with a Case Study of Painted Turtle (*Chrysemys picta*) eDNA Under Field Conditions. *Diversity*, *11*(4), 50.

Australian Government. (2011). *Survey Guidelines for Australia's Threatened Reptiles, Guidelines for Detecting Reptiles Listed as Threatened Under the Environment Protection and Biodiversity Conservation Act 1999*.

Baker, S. J., Niemiller, M. L., Stites, A. J., Ash, K. T., Davis, M. A., Dreslik, M. J., & Phillips, C. A. (2018). Evaluation of environmental DNA to detect *Sistrurus catenatus* and *Ophidiomyces ophiodiicola* in crayfish burrows. *Conservation Genetics Resources*, 1–3.

Benson, D. A., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Wheeler, D. L. (2005). GenBank. *Nucleic Acids Research*, *33*, D34–D38. https://doi.org/10.1093/nar/gki063

Berry, T. E., Osterrieder, S. K., Murray, D. C., Coghlan, M. L., Richardson, A. J., Grealy, A. K., … Bunce, M. (2017). DNA metabarcoding for diet analysis and biodiversity: A case study using the endangered Australian sea lion (*Neophoca cinerea*). *Ecology and Evolution*, *7*(14), 5435–5453.

Bland, L. M., & Böhm, M. (2016). Overcoming data deficiency in reptiles. *Biological Conservation*, *204*, 16–22.

Böhm, M., Collen, B., Baillie, J. E. M., Bowles, P., Chanson, J., Cox, N., … Ram, M. (2013). The conservation status of the world's reptiles. *Biological Conservation*, *157*, 372–385.

Bohmann, K., Evans, A., Gilbert, M. T. P., Carvalho, G. R., Creer, S., Knapp, M., … De Bruyn, M. (2014). Environmental DNA for wildlife biology and biodiversity monitoring. *Trends in Ecology & Evolution*, *29*(6), 358–367.

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a Unix-inspired software package for DNA metabarcoding. *Molecular Ecology*

Resources, 16(1), 176-182.

Caporaso, J. G., Kuczynski, J., Stombaugh, J., Bittinger, K., Bushman, F. D., Costello, E. K., … Gordon, J. I. (2010). QIIME allows analysis of high-throughput community sequencing data. *Nature Methods*, *7*(5), 335.

Davy, C. M., Kidd, A. G., & Wilson, C. C. (2015). Development and validation of environmental DNA (eDNA) markers for detection of freshwater turtles. *PLoS One*, *10*(7), e0130965.

De Souza, L. S., Godwin, J. C., Renshaw, M. A., & Larson, E. (2016). Environmental DNA (eDNA) detection probability is influenced by seasonal activity of organisms. *PLoS One*, *11*(10), e0165273.

Egeter, B., Peixoto, S., Brito, J. C., Jarman, S., Puppo, P., & Velo-Antón, G. (2018). Challenges for assessing vertebrate diversity in turbid Saharan water-bodies using environmental DNA. *Genome, 61*(11), 807–814.

Elfes, C., Livingstone, S., Lane, A., Lukosche, V., Sanders, K., Courtney, A., … Milton, D. (2013). Fascinating and forgotten: the conservation status of marine elapid snakes. *Herpetological Conservation and Biology*, *8*(1), 37–52.

Evans, N. T., Olds, B. P., Renshaw, M. A., Turner, C. R., Li, Y., Jerde, C. L., … Lodge, D. M. (2016). Quantification of mesocosm fish and amphibian species diversity via environmental DNA metabarcoding. *Molecular Ecology Resources*, *16*(1), 29–41.

Feist, S. M., Jones, R. L., Copley, J. L., Pearson, L. S., Berry, G. A., & Qualls, C. P. (2018). Development and validation of an environmental DNA method for detection of the Alligator Snapping Turtle (*Macrochelys temminckii*). *Chelonian Conservation and Biology*, *17*(2), 271–279.

Fields, A. T., Abercrombie, D. L., Eng, R., Feldheim, K., & Chapman, D. D. (2015). A novel mini-DNA barcoding assay to identify processed fins from internationally protected shark species. *PLoS One*, *10*(2), e0114844.

Fox, G. (2008). Semi-aquatic and aquatic reptiles. In G. Lukacs & C. Finlayson (Eds.), *A Compendium of Ecological Information on Australia's Northern Tropical Rivers. Sub-project 1 of Australia's Tropical Rivers - an integrated data assessment and analysis (DET18)*. National Centre for Tropical Wetland Research, Townsville, Queensland.

Froese, R., & Pauly, D. (Eds.) (2018). FishBase. In Species 2000 & ITIS Catalogue of Life. Retrieved from http://www.fishbase.org.

Gu, Z., Gu, L., Eils, R., Schlesner, M., & Brors, B. (2014). circlize implements and enhances circular visualization in R. *Bioinformatics*, *30*(19), 2811–2812.

Guinea, M. L. (2003). *Ecology, systematics and biogeography of sea snakes*. Northern Territory University.

Halstead, B. J., Wood, D. A., Bowen, L., Waters, S. C., Vandergast, A. G., Ersan, J. S., … Casazza, M. L. (2017). *An evaluation of the efficacy of using environmental DNA (eDNA) to detect giant gartersnakes (Thamnophis gigas)*. US Geological Survey.

Horton, T., Kroh, A., Ahyong, S., Bailly, N., Boyko, C. B., Brandão, S. N., … Zhao, Z. (2018). World Register of Marine Species (WoRMS). WoRMS Editorial Board. Retrieved from http://www.marinespecies.org

IUCN. (2018). Australia's reptiles threatened by invasive species, climate change - IUCN Red List. Retrieved December 30, 2019, from https://www.iucn.org/news/species/201807/australias-reptiles-threatened-invasive-species-climate-change-iucn-red-list

Kelly, R. P., Port, J. A., Yamahara, K. M., & Crowder, L. B. (2014). Using environmental DNA to census marine fishes in a large mesocosm. *PLoS One*, *9*(1), e86175.

Lacoursière-Roussel, A., Dubois, Y., Normandeau, E., & Bernatchez, L. (2016). Improving herpetological surveys in eastern North America using the environmental DNA method. *Genome*, *59*(11), 991–1007.

Lukoschek, V., Beger, M., Ceccarelli, D., Richards, Z., & Pratchett, M. (2013). Enigmatic declines of Australia's sea snakes from a biodiversity hotspot. *Biological Conservation*, *166*, 191–202.

McMurdie, P. J., & Holmes, S. (2013). Phyloseq: an R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS One*, *8*(4), e61217.

Oksanen, J., Blanchet, F. G., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., … Wagner, H. (2019). vegan: Community Ecology Package. R package version 2.5-5.

Olds, B. P., Jerde, C. L., Renshaw, M. A., Li, Y., Evans, N. T., Turner, C. R., … Shirey, P. D. (2016). Estimating species richness using environmental DNA. *Ecology and Evolution*, *6*(12), 4214–4226.

Piaggio, A. J., Engeman, R. M., Hopken, M. W., Humphrey, J. S., Keacher, K. L.,

Bruce, W. E., & Avery, M. L. (2014). Detecting an elusive invasive species: a diagnostic PCR to detect Burmese python in Florida waters and an assessment of persistence of environmental DNA. *Molecular Ecology Resources*, *14*(2), 374–380.

Pusey, B. (2011). *Aquatic biodiversity in northern Australia: patterns, threats and future*. Charles Darwin University Press (CDU Press).

Raemy, M., & Ursenbacher, S. (2018). Detection of the European pond turtle (*Emys orbicularis*) by environmental DNA: is eDNA adequate for reptiles? *Amphibia-Reptilia*, *39*(2), 135–143.

Ratnasingham, S., & Hebert, P. D. N. (2007). BOLD: The Barcode of Life Data System (http://www. barcodinglife.org). *Molecular Ecology Notes*, *7*(3), 355–364.

Rees, H. C., Maddison, B. C., Middleditch, D. J., Patmore, J. R. M., & Gough, K. C. (2014). The detection of aquatic animal species using environmental DNA–a review of eDNA as a survey tool in ecology. *Journal of Applied Ecology*, *51*(5), 1450–1459.

Rolland, J., Silvestro, D., Schluter, D., Guisan, A., Broennimann, O., & Salamin, N. (2018). The impact of endothermy on the climatic niche evolution and the distribution of vertebrate diversity. *Nature Ecology & Evolution*, *2*(3), 459.

R Studio Team (2015). RStudio: integrated development for R. *RStudio Inc*, Boston, MA. Retrieved from *http://www.rstudio.co*

Sanders, K. L., Lee, M. S. Y., Bertozzi, T., & Rasmussen, A. R. (2013). Multilocus phylogeny and recent rapid radiation of the viviparous sea snakes (Elapidae: Hydrophiinae). *Molecular Phylogenetics and Evolution*, *66*(3), 575–591.

Sasso, T., Lopes, C. M., Valentini, A., Dejean, T., Zamudio, K. R., Haddad, C. F. B., & Martins, M. (2017). Environmental DNA characterization of amphibian communities in the Brazilian Atlantic forest: Potential application for conservation of a rich and threatened fauna. *Biological Conservation*, *215*, 225–232.

Stat, M., Huggett, M. J., Bernasconi, R., DiBattista, J. D., Berry, T. E., Newman, S. J., … Bunce, M. (2017). Ecosystem biomonitoring with eDNA: metabarcoding across the tree of life in a tropical marine environment. *Scientific Reports*, *7*(1), 1-11.

Stat, M., John, J., DiBattista, J. D., Newman, S. J., Bunce, M., & Harvey, E. S. (2019).

Combined use of eDNA metabarcoding and video surveillance for the assessment of fish biodiversity. *Conservation Biology*, *33*(1), 196-205.

Thomsen, P. F., Kielgast, J., Iversen, L. L., Møller, P. R., Rasmussen, M., & Willerslev, E. (2012). Detection of a diverse marine fish fauna using environmental DNA from seawater samples. *PloS One*, *7*(8). https://doi.org/10.1371/journal.pone.0041732

Tomillo, P. S., Genovart, M., Paladino, F. V, Spotila, J. R., & Oro, D. (2015). Climate change overruns resilience conferred by temperature-dependent sex determination in sea turtles and threatens their survival. *Global Change Biology*, *21*(8), 2980–2988.

Uetz, P., Freed, P., & Hošek, J. (2018). The Reptile Database.

Valentini, A., Taberlet, P., Miaud, C., Civade, R., Herder, J., Thomsen, P. F., … Dejean, T. (2016). Next-generation monitoring of aquatic biodiversity using environmental DNA metabarcoding. *Molecular Ecology*, *25*(4), 929-942. https://doi.org/10.1111/mec.13428

Van Dyke, J. U., Ferronato, B. de O., & Spencer, R.-J. (2018). Current conservation status of Australian freshwater turtles. *Australian Journal of Zoology*, *66*(1), 1–3.

West, K. M., Stat, M., Harvey, E. S., Skepper, C. L., DiBattista, J. D., Richards, Z. T., … Bunce, M. (2020). eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem. *Molecular Ecology*. https://doi.org/10.1111/mec.15382

Wilcox, T. M., Schwartz, M. K., McKelvey, K. S., Young, M. K., & Lowe, W. H. (2014). A blocking primer increases specificity in environmental DNA detection of bull trout (*Salvelinus confluentus*). *Conservation Genetics Resources*, *6*(2), 283–284.

Chapter 4

Under the karst: detecting hidden subterranean assemblages using eDNA metabarcoding in the caves of Christmas Island, Australia







4.1 Prologue

Chapters two and three have explored the applicability of marine eDNA metabarcoding across small and large spatial-scales. Both studies produced high-resolution data, i.e. to a species-level, which was afforded by well-developed marine reference databases. The use of eDNA metabarcoding in poorly-characterised ecosystems, e.g. subterranean aquatic, that subsequently present suboptimal reference databases, has not yet been thoroughly examined. Furthermore, it has not yet been tested whether eDNA-derived community composition data, in conjunction with environmental data, can be used to identify potential underground interconnectivity between subterranean systems. The following chapter addresses these topics by conducting a eukaryotic eDNA metabarcoding survey across the subterranean aquatic landscape of Christmas Island, a remote island situated in Australia's Indian Ocean region.

Christmas Island (10.4475° S, 105.6904° E) is an isolated oceanic island, located 350 km south of Java, Indonesia. It was first recorded by European explorers in 1615 and annexed by the British Crown in 1888, following the discovery of pure phosphate deposits (Burstyn, 1975). This soon led to the first settlement on the island by Andrew Clunies-Ross (the brother of George Clunies-Ross—who was ruling the Cocos (Keeling) Islands at that time) and a small group of Cocos Malay workers. Phosphate mining began in 1899 using indentured workers from the Cocos (Keeling) Islands and China (Hunt, 2011). This continued up until World War II when the Japanese invaded and forcefully occupied the island until 1945. Phosphate mining resumed in 1946 and has continued on the island ever since. Christmas Island was acquired by the Australian Government in 1958, following the payment of £2.9 million to the British Colony of Singapore, and is now managed as one of Australia's external Indian Ocean territories (Stanhope, Wettenhall, & Bhusal, 2016).

Christmas Island forms the pinnacle of a basaltic volcanic seamount, rising 4500 m above the ocean floor. It re-emerged above sea level approximately 5.66–4.49 Ma (Ali & Aitchison, 2019) and is characterised by a central plateau enclosed by terraces and sheer limestone cliffs, further surrounded by a narrow fringing coral reef shelf. The

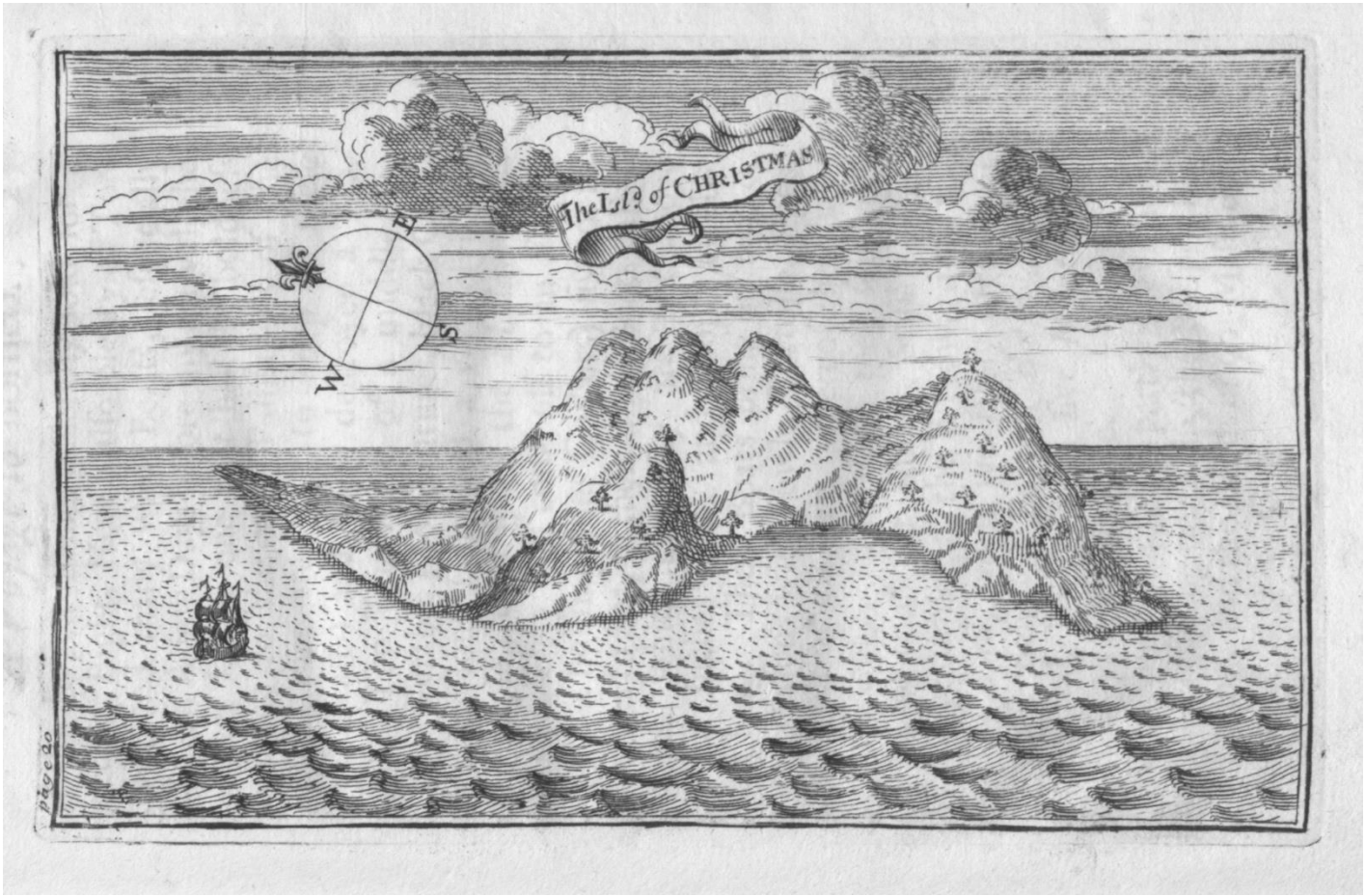


Figure 4.1 Depiction of Christmas Island by Captain Daniel Beeckman in 1714.
Illustrated in *A Voyage to and from the Island of Borneo in the East-Indies…* (1718), p 20. Image courtesy of Cornell University Library as part of The John M. Echols Collection on Southeast Asia. | http://seasiavisions.library.cornell.edu/

island has extensive tropical rainforests, wetlands and coral reefs, in addition to a highly developed karst (subterranean) system (Grimes, 2001). Approximately one-quarter of the island's rainforest has been cleared since 1888, largely for phosphate mining, but also settlement purposes (Director of National Parks, 2014b). Like other oceanic islands, the biodiversity of Christmas Island has been shaped by its isolation. It supports an important seabird and forest bird rookery including endemic bird species, such as the Christmas Island frigatebird (*Fregata andrewsi*) and Abbott's booby (*Papasula abbotti*). Other endemic terrestrial species include the well-known Christmas Island red crab (*Gecarcoidea natalis*), the Christmas Island flying-fox (*Pteropus melanotus natalis*) and the blue-tailed skink (*Cryptoblepharus egeriae*). The island's marine productivity and nutrients are influenced by the Indonesian Throughflow, bringing currents and upwelling from Java (Brewer et al., 2009). Over 600 marine fish species, including endemic and hybrid reef fish, have been recorded (Allen, Steene, & Orchard, 2007; Hobbs, Frisch, Allen, & Van Herwerden, 2008). Christmas Island's subterranean, in particular aquatic "stygofauna" are less well known, but are of high global conservation significance (Director of National Parks, 2014a). Notably, Christmas Island boasts one of only two anchialine systems that have been documented in the southern hemisphere (Humphreys & Eberhard, 2001).

4.1.1 Chapter acknowledgements

I would like to express my upmost gratitude to the Western Australian Speleological Group (WASG) for support throughout the entirety of the project, in particular to the following members who assisted on the field expedition which would never have been possible without their expertise and good humour: Robert Susac (trip leader), Weidi Koh, Danny Wilkinson, Brett Wiltshire, Sue Hamilton, Rod Hamilton, Ross Anderson, Jay Anderson, John Clarke and Anne Wood. I would also like thank the team at Parks Australia (Christmas Island) for logistical assistance, Robert Muller for his personal time as a skipper, Sandy and Hiro (Extra Divers Christmas Island) for guiding myself and the expedition team through Thundercliff Cave, Andrew and Mel from WaterCorp for allowing me to sample from the testing taps at Jedda and Jane-up Cave, Bill Humphreys (WAM), Nicole White (Curtin) and the entire TrEnD lab (Curtin University) for their support. I am grateful to Parks Australia headquarters for the following access and sampling permits: AU-COM2018-422 and AU-COM2018-428.

Lastly this project would not have been possible without financial support from the Holsworth Wildlife Research Endowment in association with the Ecological Society of Australia. Cover photos include those from Danny Wilkinson.

4.1.2 Data accessibility

Demultiplexed (unfiltered) metabarcoding sequencing data and taxonomic presence/absence matrices from this study are temporarily available from https://cloudstor.aarnet.edu.au/plus/s/mxE39tFaPW5sKYz and on request. The manuscript arising from this chapter is currently in preparation for submission to *Scientific Reports*. As such, this chapter has been formatted for *Scientific Reports*. Data will be uploaded onto Dryad Digital Repository coinciding with the manuscript submission.

4.2 Abstract

Environmental DNA (eDNA) metabarcoding is a highly sensitive biomonitoring tool that is increasingly being applied to survey the biodiversity of marine, freshwater and terrestrial fauna and flora. It has great potential to survey subterranean ecosystems which are understudied and challenging to assess because of the inaccessibility of underground voids and networks. We conducted a multi-marker eukaryotic eDNA metabarcoding survey across the karst landscape of Christmas Island, (Indian Ocean, Australia) in order to evaluate the potential of this technique for detecting subterranean aquatic 'stygofauna' assemblages. Three metabarcoding assays targeting the mitochondrial 16S rRNA and nuclear 18S genes were applied to 159 water and sediment samples collected from 23 caves and springs across the island. We anticipated that flowing stream and conduit water through the cave systems and springs would act as a conveyor belt of stygofauna DNA through voids that are inaccessible to humans. Taken together, our assays detected a wide diversity of chordates, cnidarians, porifera, arthropods, molluscs, annelids and bryozoans from 71 families across 60 orders. We report a high level of variation between cave and spring subterranean community compositions which are significantly influenced by varying levels of salinity. Additionally, we show that dissolved oxygen and longitudinal gradients significantly affect biotic assemblages within cave communities. We update distribution information and highlight subterranean taxa of conservation and biodiversity importance, such as spring-tails, Indonesian shortfin eel and cave gudgeon. Lastly, we combined eDNA-derived community composition and environmental (water quality) data to predict potential underground interconnectivity across Christmas Island. We identified three cave and spring groups that showed a high degree of biotic and abiotic similarity likely indicating local connectivity. This study demonstrates the high-sensitivity of eDNA metabarcoding in detecting the presence of subterranean eukaryotic organisms from multi-trophic levels. We anticipate that eDNA metabarcoding will increasingly become a key tool to reveal the unique biota within subterranean ecosystems.

4.3 Introduction

Subterranean environments are notoriously underexplored. It has been estimated that more than 80% of Australia's subterranean fauna have yet to be discovered (Guzik et al., 2011). Subterranean environments include those above (e.g. cave and cavities) and below the water table (e.g. aquifers and anchialine systems). Previous research demonstrates that subterranean environments exhibit a high diversity of (largely invertebrate) taxa that are adapted to a lack of light, and also to variable temperature, nutrient, salinity and dissolved oxygen levels and in some cases, water stratification (Gonzalez, Iliffe, Macalady, Schaperdoth, & Kakuk, 2011). Short-range, endemic species are common, as highly fragmented environments pose barriers to gene flow, fostering evolutionary drift over time (Gibert & Deharveng, 2002). Commonly surveyed aquatic 'stygofauna', found within aquifers and anchialine systems, include fresh and saltwater fish, eels, gastropods, salamanders, flatworms, and crustaceans such as amphipods, decapods, isopods, ostracods and syncarids (Moldovan, 2018).

The use of genetic techniques in conjunction with traditional biospeleological sampling and morphological assessment can provide in-depth information in relation to patterns of stygofauna diversity, discrete lineages, colonisation and speciation histories. Biospeleological studies that have incorporated genetic methodology have largely focused on the use of single-source "barcoding" or genome building, where individual specimens are collected and targeted for sequencing (Adams & Humphreys, 1993; Asmyhr, Linke, Hose, & Nipperess, 2014; Bradford, Adams, Humphreys, Austin, & Cooper, 2010; Juan, Guzik, Jaume, & Cooper, 2010; Matthews et al., 2020; Page, Humphreys, & Hughes, 2008). However, single-source sequencing remains reliant upon the use of capture-based sampling, which is intrinsically linked to the measurement of biodiversity. Whilst capture-based sampling is fundamental to biospeleological research, it can be hindered in subterranean environments that present difficult to access underground voids and networks. Biospeleological research could therefore benefit from a non-invasive biomonitoring tool that complements capture-based sampling in order to gauge stygofauna diversity and distribution.

Environmental DNA (eDNA) metabarcoding has been widely developed within the last few years for use in marine, freshwater and terrestrial environments and has

demonstrated its power as an efficient, non-invasive and highly sensitive taxa detection tool (Evans et al., 2016; Olds et al., 2016; Stat et al., 2017; Thomsen et al., 2012). However, its application in subterranean environments has not yet been thoroughly explored. To date, only a small number of biospeleological studies have utilised metabarcoding as a multi-species detection tool on subterranean environmental samples (Humphreys et al., 2012; Korbel, Chariton, Stephenson, Greenfield, & Hose, 2017; Ortiz et al., 2014; Smith et al., 2012; Wu et al., 2016). Preliminary research has demonstrated the viability of eDNA metabarcoding in detecting multi-species compositions from underground water samples, however these have largely focused on microbial communities (Humphreys et al., 2012; Korbel et al., 2017; Smith et al., 2012). Further research is needed to assess the applicability of eDNA metabarcoding for higher eukaryotic stygofauna detections in subterranean aquatic ecosystems, which through a lack of previous surveying and subsequent reference barcoding of biota, may be hindered by suboptimal reference databases. There is great potential to expand the use of eDNA metabarcoding, not only for the detection and monitoring of both described and undescribed eukaryotes in subterranean environments, but also to investigate community assemblages across trophic levels and haloclines, the evolution and population diversity of stygofauna, and the interconnectivity of underground ecosystems.

Christmas Island (CI; 10.4475° S, 105.6904° E), one of Australia's external Indian Ocean territories, forms the pinnacle of an isolated seamount that re-emerged above sea level approximately 5.66–4.49 Ma (Ali & Aitchison, 2019). A highly developed karst landscape has formed out of imbedded carbonates and approximately 30 accessible caves have been documented, ranging from plateau and freshwater stream caves; to fissure caves, collapsed caves, sea caves, and coastal caves with ocean access points (Grimes, 2001). Rainfall percolates through limestone fractures and solution holes and is largely discharged by coastal and offshore springs, however there are also major inland springs at Waterfall Spring, Ross Hill Gardens and The Dales (Figure 4.2; Grimes, 2001). Cave fauna are considered to form a significant component of this island's unique ecosystem (Beeton et al., 2010). However, the diversity and distribution of stygofauna, in particular anchialine fauna, across CI's extensive underground networks requires further research (Humphreys, 2014). This is of particular importance given the use of the karstic landscape for phosphate mining and

as a water supply for local households and businesses. The subterranean interconnectivity of the majority of the CI caves are unknown, but it is suspected that caves within close proximity (such as the Whip and Runaway Caves that incidentally also share many species of macrofauna) are connected underground (Davie & Ng, 2012). The need for a comprehensive biodiversity audit of CI's extensive subterranean habitats makes it an ideal location to conduct one of the first eDNA metabarcoding surveys of eukaryotic cave stygofauna. Our main objectives were to: (i) identify putative new occurrence and extend distribution records for CI's subterranean fauna using eDNA metabarcoding; (ii) assess variation in community composition of cave and spring sites, and; (iii) investigate potential underground interconnectivity across CI by combining biotic and abiotic (environmental parameter) data. Overall, we seek to evaluate the applicability of eDNA metabarcoding as a non-invasive, highly-sensitive biomonitoring tool for biospeleological assessment.

4.4 Methods

4.4.1 Field sampling

Six one-litre water replicates and one 50-ml sediment sample were collected from 23 cave and spring sites across CI (Figures 4.2 and 4.3; Table 4.1) in October 2018, totaling 159 samples across a 110 km^2 area. For public safety and to protect the integrity of the caves, the exact GPS coordinates for our sampled sites are not published here, however they can be requested from Parks Australia (Christmas Island). Sediment could not be taken from the Jedda and Jane-up Cave sites because conduit streams through these plateau caves provide the town water supply and hence access is restricted. However, water samples were taken with permission from the WaterCorp testing taps at these sites. Sediment was sampled at all remaining sites, by collecting a top layer of sediment (approximately 2 cm depth) from underwater sources. Water samples were collected at the surface of each site using bleach sterilised Nalgene bottles and then immediately stored on ice. Each sample was individually filtered across Pall 0.2μm Supor® polyethersulfone membranes using a Pall Sentino® Microbiology pump (Pall Corporation, Port Washington, USA), within two hours of collection. A bleach solution (approximately 10% household bleach) was used to clean filtration equipment between samples and a one litre sample of this was filtered at the end of each sampling day to serve as a filtration control throughout laboratory processing. Filter membranes and sediment samples were immediately frozen and stored at -20°C prior and post-transportation to a quarantine facility within the Trace & Environmental DNA (TrEnD) Laboratory in Perth, Western Australia.

A range of environmental parameters were also taken at the time of sampling from each respective site (Table S4.1). Measurements of water acidity (pH), temperature (℃), conductivity (mS) and salinity (ppt) were collected using a Hanna HI98129 tester (Hanna Instruments; Victoria, Australia). Air saturation and dissolved oxygen (mg/L) measurements were collected using an OxyGuard Handy Polaris 2 DO Meter (OxyGuard; Farum, Denmark).

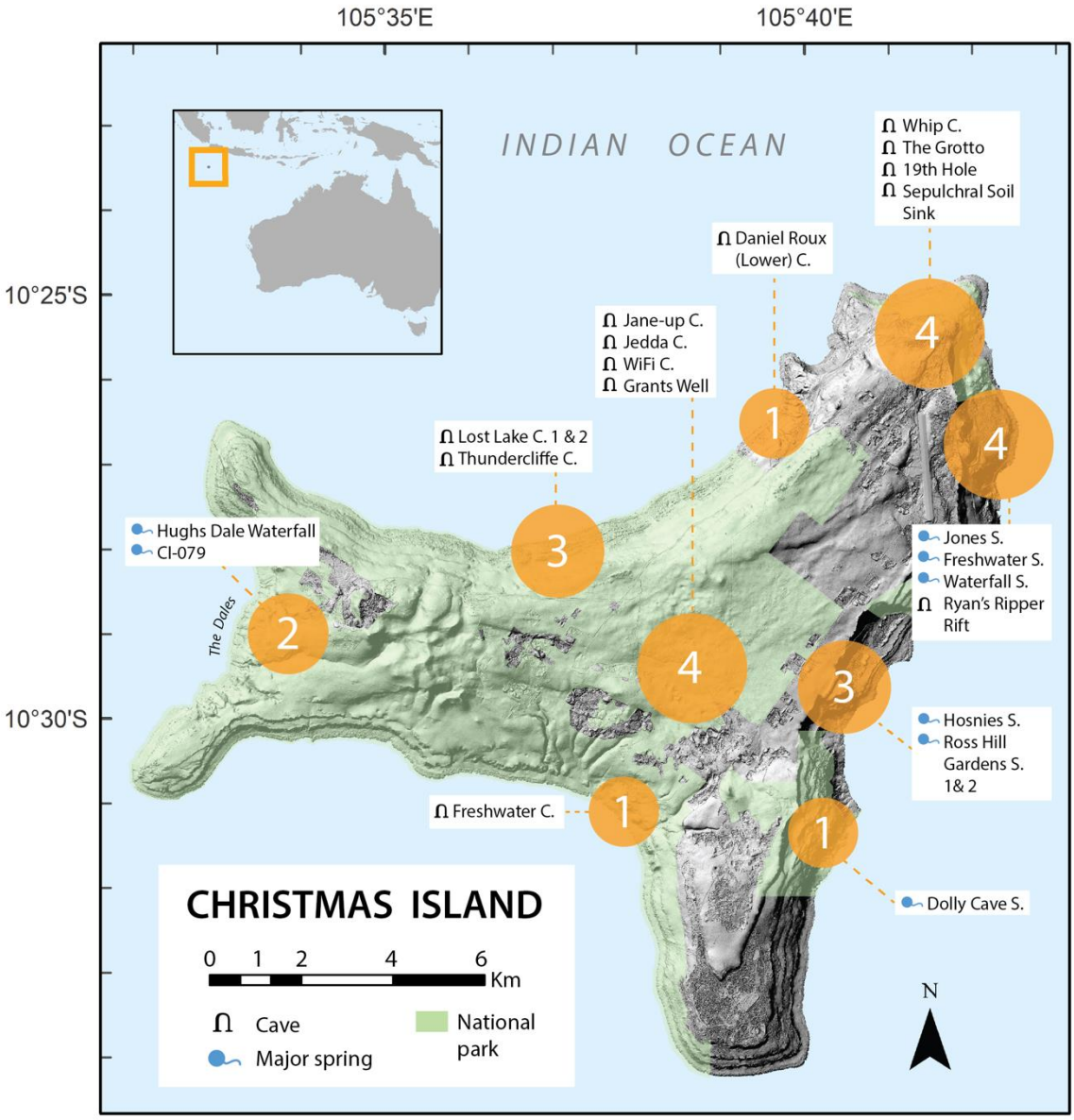


Figure 4.2 Location of eDNA sampling sites on Christmas Island.
Orange spheres give the approximate number and location of sampling sites; topographical symbols indicate whether samples from each respective site were taken from within a cave system or at a surface spring. Hillshade relief and national park boundary data was sourced from Geoscience Australia (Porritt & Walker, 2015).

Table 4.1 eDNA site information on Christmas Island.
Type refers to whether samples at each respective site were taken from within a cave system or at a surface spring. Salinity classifications for each site were assigned based on salinity readings at the time of sample collection (see Table S4.1 for full environmental parameter data and CI cave index numbering).

Site No.	Site name	Type	Brief description	Salinity groups
1	19th Hole	Cave	Anchialine cave with tidal influence. Water table accessed on land via a small cave mouth and down a slope.	Mesohaline
2	The Grotto	Cave	Marine coastal cave; strong sea currents and tidal influence. Accessed on land via Golf Course Road and open to the public. Possible connection to Whip Cave (Grimes, 2001).	Oligohaline
3	Whip Cave	Cave	Anchialine cave system with tidal influence. Possible connection to The Grotto (Grimes, 2001).	Oligohaline
4	Daniel Roux (Lower) Cave	Cave	Anchialine cave system with tidal influence. Water table accessed on land by the lower chamber entrance, down a series of steep rock-mounted steel ladders.	Freshwater
5	Freshwater Cave	Cave	Anchialine stream passage cave connecting several chambers.	Oligohaline
6	Hugh Dale Waterfall	Spring	Waterfall cascading from tufa spring.	Freshwater
7	CI-079 (unnamed)	Spring	Small low cave with spring outflow.	Freshwater
8	Sepulchral Soil Sink	Cave	Small entrance cave to deeper chambers with low pools.	Mesohaline
9	Waterfall Spring	Spring	Water supply spring south of CI Resort.	Freshwater
10	Freshwater Spring	Spring	Spring outflow north of CI Resort.	Freshwater
11	Jedda Cave	Cave	Cement steps into doline, pump station for town water supply.	Freshwater
12	Jane-up Cave	Cave	Muddy stream passage with high CO_2; used for town water supply.	Freshwater
13	Jones Spring	Spring	Spring with weir, water re-sinks nearby.	Freshwater
14	Grants Well	Cave	Well shaft intersecting 20 m below to cave stream passage; high CO_2.	Freshwater
15	Thundercliff Cave	Cave	Sea cave entrance leading to freshwater outflow.	Polyhaline
16	Ryan's Ripper Rift	Cave	Rift cave ending in very deep crystalline edged anchialine water.	Oligohaline
17	Hosnies Spring	Spring	Freshwater seepage on shore terrace that feeds a protected mangrove stand.	Freshwater
18	Ross Hill Gardens Spring 1	Spring	Captured spring once used for town water supply.	Freshwater
19	Ross Hill Gardens Spring 2	Spring	Captured spring once used for town water supply.	Freshwater
20	Dolly Cave Spring	Spring	Low muddy cave with spring outflow.	Freshwater
21	Lost Lake Cave 1	Cave	Extensive tidal system passages with various large breakdown chambers. Sampled from stream passage approximately 150m from sea entrance.	Oligohaline
22	Lost Lake Cave 2	Cave	Extensive tidal system passages with various large breakdown chambers. Sampled from stream passage far into the cave extent.	Oligohaline
23	WiFi Cave	Cave	Deep vertical shaft that is believed to intersect the muddy stream passage between Grants Well and Jane-up Cave.	Freshwater

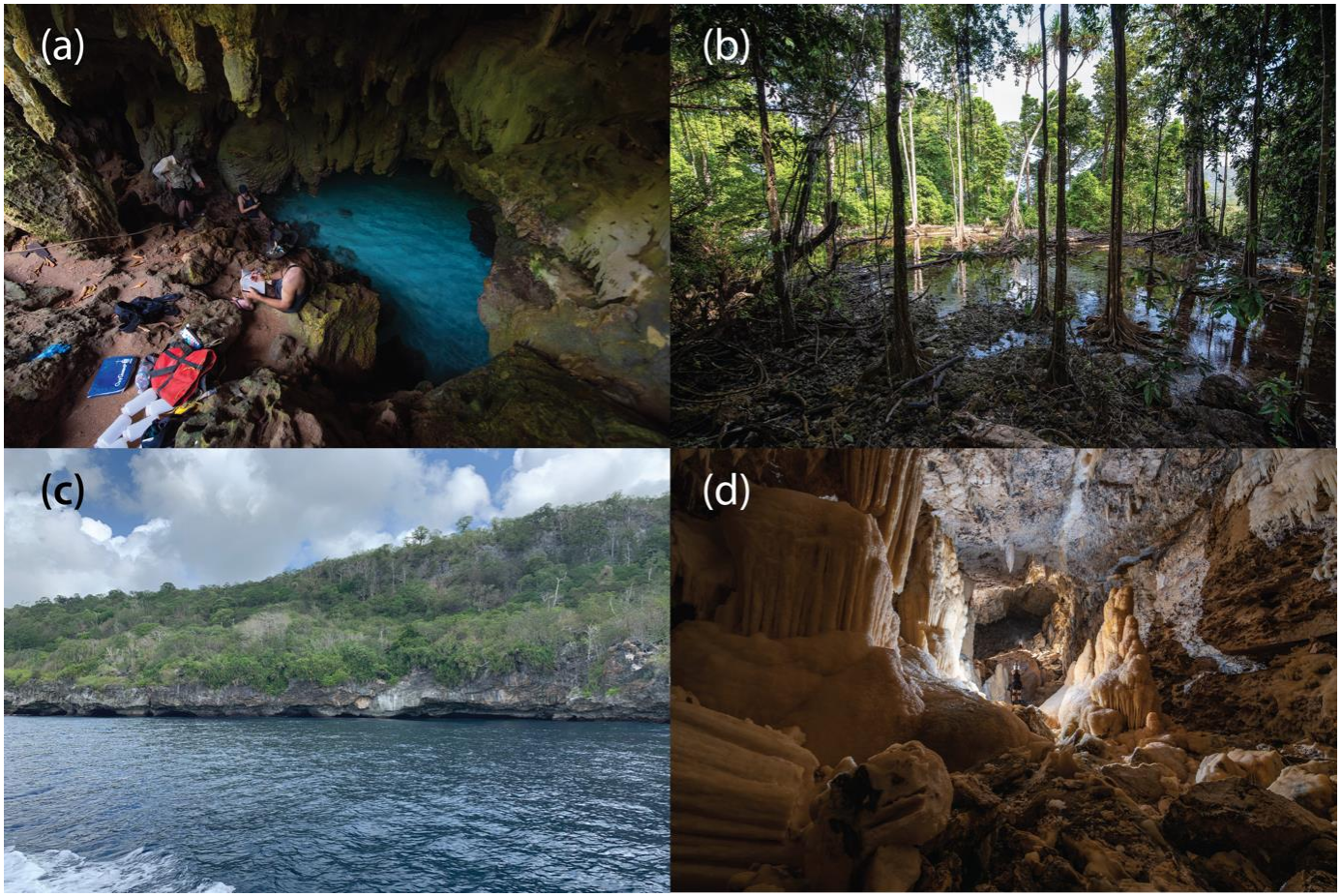


Figure 4.3 Imagery from eDNA sampling localities across Christmas Island.
(a) Water and sediment sampling in The Grotto cave site. b) The Dales wetland area where Hugh Dales Waterfall and CI-079 spring sites were sampled. The (c) ocean entrance and (d) a chamber passage through the extensive Lost Lake Cave system. Photos: © Danny Wilkinson and Weidi Koh http://wasg.org.au/

4.4.2 Laboratory processing

DNA was extracted from half of each filter membrane and 250 mg of each sediment sample using a DNeasy PowerLyzer PowerSoil Kit (Qiagen; Venlo, Netherlands) following manufacturer's instructions. This was completed within two weeks of collection. Filtration controls and extraction blanks, containing no sample, were extracted and processed alongside all samples in order to detect any cross-contamination introduced from the laboratory environment. DNA was amplified using three previously published PCR assays to largely target bony fish, molluscs and arthropods (such as crustaceans, ostracods and insects) from our mixed environmental samples (see Table 4.2 for details). Quantitative PCR (qPCR) amplification was performed using fusion tagged primers that consist of an Illumina sequencing adaptor, a unique multiplexing index (8bp in length) and a primer sequence from each respective assay. All qPCR reactions were prepared in an ultra-clean trace DNA facility and thermocycling was carried out in a physically separated laboratory (see S4.9.1 for qPCR reagents and conditions).

Each eDNA template was amplified in duplicate and pooled into larger amplicon libraries at equimolar ratios based on qPCR ΔRn values. Each library was size-selected (retaining amplicons between 160-450bp for the 16S assays, and 200-600bp for the 18S assay) using a Pippin Prep (Sage Science, Beverly, USA), and was then purified using the Qiaquick PCR Purification Kit (Qiagen, Venlo, Netherlands) following manufacturer's instructions. Final libraries were quantified using a Qubit 4.0 Fluorometer (Invitrogen, Carlsbad, USA) and if necessary were diluted to 2nM prior sequencing. Libraries were sequenced on either a 300 cycle (for unidirectional sequencing of the 16S amplicons) or 500 cycle (for paired-end sequencing of the 18S amplicons) MiSeq® V2 Standard Flow Cell on an Illumina MiSeq platform (Illumina, San Diego, USA), housed in the TrEnD Laboratory at Curtin University, Western Australia.

4.4.3 Bioinformatics

Unidirectional and unmerged paired-end sequencing reads were demultiplexed using

Table 4.2 Metabarcoding assay information for subterranean eDNA surveys on Christmas Island.
Three PCR primer sets: 16S Fish (short), 16S Crustacean and 18S Universal corresponding to the mitochondrial 16S rRNA and nuclear 18S regions, were applied to all collected water and sediment samples. In the primer name, "F" refers to the forward primer and "R" to the reverse primer.

PCR Assay	Target taxa	Primer Name	Oligonucleotide Sequence (5'-3')	Target Length (bp)	Annealing temp (°C)	Primer Reference
16S Fish (short)	Bony Fish	16S_FishSyn_Short forward	GACGAGAAGACCCTGTGGAGC	70-140	55	Nester et al. (unpublished)
		16S_FishSyn_Short reverse	CCGYGGTCGCCCCAAC			
16S Crustacean	Crustacean	Crust16S_F (short)	GGGACGATAAGACCCTATA	90-213	51	Berry et al., 2017
		Crust16S_R (short)	ATTACGCTGTTATCCCTAAAG			
18S Universal	Eukaryotes	18S_1F	GCCAGTAGTCATATGCTTGTCT	336-423	52	Pochon, Bott, Smith, & Wood, 2013
		18S_400R	GCCTGCTGCCTTCCTT			

OBITools (v1.2.9; Boyer et al., 2014) and the insect package (Wilkinson, Davy, Bunce, & Stat, 2018) in RStudio (v1.1.423; RStudio Team, 2015), respectively. Demultiplexed data was then quality filtered using the DADA2 pipeline in RStudio (Callahan et al., 2016; see S4.9.2 for bioinformatic parameter details). Resulting amplicon sequence variants (ASV) for each assay were then queried against NCBI's GenBank nucleotide database (accessed in 2019; Benson et al., 2005) using BLASTn and also against a curated 16S rDNA Western Australian fish database (see Table S2.6 in chapter two) via Zeus, an SGI cluster, based at the Pawsey Supercomputing Centre in Kensington, Western Australia. Taxonomic assignments of ASVs were curated using a lowest common ancestor approach (https://github.com/mahsa-mousavi/LCA_taxonomyAssignment; see S4.9.2); consolidated taxa assignments were then additionally categorised based on associated environment and biogeographic distribution obtained from CI subterranean biodiversity surveys (Hui, Naruse, Fujita, & Kiat, 2014; Humphreys, 2014) and the World Register of Marine Species (WoRMS; Horton et al., 2018). Putative new occurrence records were additionally assessed for whether all congeneric taxa have been barcoded for the targeted gene region. Any ASVs that were detected in filtration and/or extraction blanks were entirely removed; remaining ASVs that share the exact taxonomy assignment were then merged using the phyloseq 'tax_glom' function (McMurdie & Holmes, 2013) in RStudio. This produced a taxonomic-based matrix. Read abundance was converted to presence/absence data in PRIMER v7 (Clarke & Gorley, 2015) for subsequent statistical analyses.

4.4.4 Statistical analyses

Variation in the community composition was tested between the water and sediment samples to determine whether there is a difference in the type and number of taxa detected between the two substrates. The presence-absence data of taxa at each site was converted to a Jaccard similarity matrix and tested for the effect of substrate and site using a two-way crossed PERMANOVA in the PERMANOVA+ add-on (Anderson, Gorley, & Clarke, 2008) of PRIMER v7. Site variation by substrate was visualised by principal coordinates analysis (PCO) in PRIMER v7. Species accumulation per replicate (of the two substrates) and a comparison of the number of taxa per substrate replicate was graphed using the vegan 'specaccum' function

(Oksanen et al., 2019) and ggplot (Wickham, 2016), respectively, in RStudio. Additionally, a similarity percentage analysis (SIMPER) were conducted in PRIMER v7 to identify contributing taxa to pairwise dissimilarity between substrates.

Sampling replicates of both substrates were then merged per site and converted to a Jaccard similarity matrix to examine the overall community composition variation between sites. Taxa accumulation per site was graphed using vegan in RStudio. Distance-based linear model (DistLM) analyses were conducted in the PERMANOVA+ add-on using normalised measures of acidity, temperature, salinity and dissolved oxygen, in addition to latitude/longitude, as environmental and spatial predictor variables. These analyses were initially conducted across all site types (cave and spring), and then within cave and spring sites separately. The environmental parameters of conductivity and air saturation were omitted due to collinearity with salinity and dissolved oxygen, respectively. Site variation was visualised by PCO using the stats function 'cmdscale' in R Studio; significant predictor variables corresponding to the DistLM analyses were overlaid using the vegan 'ordisurf' function (Oksanen et al., 2019). A SIMPER analysis was conducted to identify contributing taxa to pairwise dissimilarity between site type (cave/spring) and salinity. Original salinity readings (ppt; Table S4.1) were categorised into the following salinity groups: freshwater ≤0.49 ppt, oligohaline 0.5-4.9 ppt, mesohaline 5.0-17.9 ppt, polyhaline 18.0-29.9 ppt, and euhaline (seawater) ≥30.0 ppt (Mitsch & Gosselink, 1986; see Table 4.1 for site salinity groupings). Hierarchical clustering, using group-averaging, and SIMPROF analyses were applied to both the community composition (biotic) and environmental parameter (abiotic; including latitude/longitude) datasets in PRIMER v7, to identify groupings of sites that may potentially reflect underground interconnectivity.

4.5 Results

4.5.1 Sampling and sequencing statistics

The three metabarcoding assays yielded a total of 35,698,221 sequencing reads across 159 samples. The mean number of filtered sequences (post-quality, denoising and chimera filtering) was 74,014 ± 73,211 per replicate sample for the 16S Fish (short) assay; 66,152 ± 115,417 per replicate sample for the 16S Crustacean assay; and 22,968 ± 16,704 per replicate sample for the 18S Universal assay (Table S4.2). ASVs that were detected in filtration and extraction blanks and/or are common laboratory contaminants were omitted from all samples and subsequent analyses; this included ASVs for minnows (genus: *Phoxinus*), a branching bryozoan (*Fredericella sultana*), human (*Homo sapiens*), junglefowl (*Gallus gallus*), cat (*Felis catus*), pig (*Sus scrofa*), cattle (*Bos taurus*) and turkey (*Meleagris gallopavo*). We also omitted all ASVs for taxa outside of our study scope; this included any ASV in the domain Bacteria, hairybacks (phylum: Gastrotricha), the kingdom Fungi, ciliates (phylum: Ciliophora), nematodes (phylum: Nematoda), flatworms (phylum: Platyhelminthes), plants (clade: Streptophyta) and algae (phylum: Cryptophyta).

Taxa accumulation curves based on the addition of each sampling replicate per site (Figure S4.1) indicated that six 1 L water replicates (chosen a priori to sampling) was not completely sufficient to maximise the observed taxonomic richness at each site. The addition of a sediment replicate also provided an increment in taxa diversity beyond those detected with water. Therefore, more taxa were likely to be detected if further water and sediment replicates were examined per site. Variation in the composition of taxa detected between the water and sediment substrates was highly significant ($P = 0.000$, $df = 1$; Table S4.3, Figure S4.2). A single water sample detected on average a higher number of taxa than a single sediment sample (Figure S4.3), however, this difference was not statistically significant. A SIMPER analysis of pairwise dissimilarity between the substrates indicated that the water samples were able to detect a large proportion of the overall detected taxa (Table S4.4), however sediment provided a greater detection rate for yellow nipper crab (*Geograpsus crinipes*) and whiteleg shrimp (*Penaeus vannamei*).

4.5.2 Overall diversity

A total of 25 taxa (1.5 ± 1.0 ASVs per taxa) were detected with the 16S Fish (short) assay, 15 taxa (5.1 ± 7.8 ASVs per taxa) with the 16S Crustacean assay, and 77 taxa (2.0 ± 2.5 ASVs per taxa) with the 18S Universal assay (Table S4.5). Overall, the three metabarcoding assays yielded 115 identifiable taxa, representing 71 families within 60 orders of the phylums Chordata, Cnidaria, Porifera, Arthropoda, Mollusca, Annelida and Bryozoa (Figure 4.4, Table S4.5). The majority of these taxonomic assignments were resolved to a species level (37.4%), followed by order level (22.6%), genus level (20.8%), family level (17.4%) and class level only (1.7%). The detected taxa were found to be largely associated with marine environments (53.9%), followed by terrestrial (37.4%), freshwater (20.9%) and brackish (14.8%) environments (Table S4.5). Of these taxa, 64.3% are circumglobal, 6.1% are distributed across the Indo-West Pacific and 5.2% more broadly across the Indo-Pacific, with smaller distributions in Africa, Asia and the eastern Indian Ocean. Taxa accumulation based on the addition of cave/spring sites did not plateau, indicating more taxa are likely to be detected if further sites are examined (Figure S4.4).

Thirteen bony fish taxa (class: Actinopterygii) were detected (three at family level only, three at genus level only, and seven at a species level) from 11 families within eight orders (Table S4.5). The assemblage was predominantly comprised of marine fish detected in the anchialine caves; this included snooks (family: Centropomidae), giant trevally (*Caranx ignobilis*), flying fish (family: Exocoetidae), oriental trumpeter whiting (*Sillago aeolus*), black triggerfish (*Melichthys niger*) and halfbeak (*Oxyporhamphus micropterus*). Three notable detections in the freshwater caves were Indonesian shortfin eel (*Anguilla bicolor bicolor*), gudgeon (genus: *Eleotris*) and carp/minnows (family: Cyprinidae); see Table 4.3 for more information on taxa of conservation, biodiversity and biosecurity importance. We also report three putative new fish occurrence records: Cyprinidae, *Cottus* and *Gobio gobio*. However, the latter species cannot be verified to a species level as not all congeneric taxa for *Gobio* are represented by a reference barcode in the database. Therefore, we cannot rule out that our assignment of *Gobio gobio* may represent a closely-related taxon.

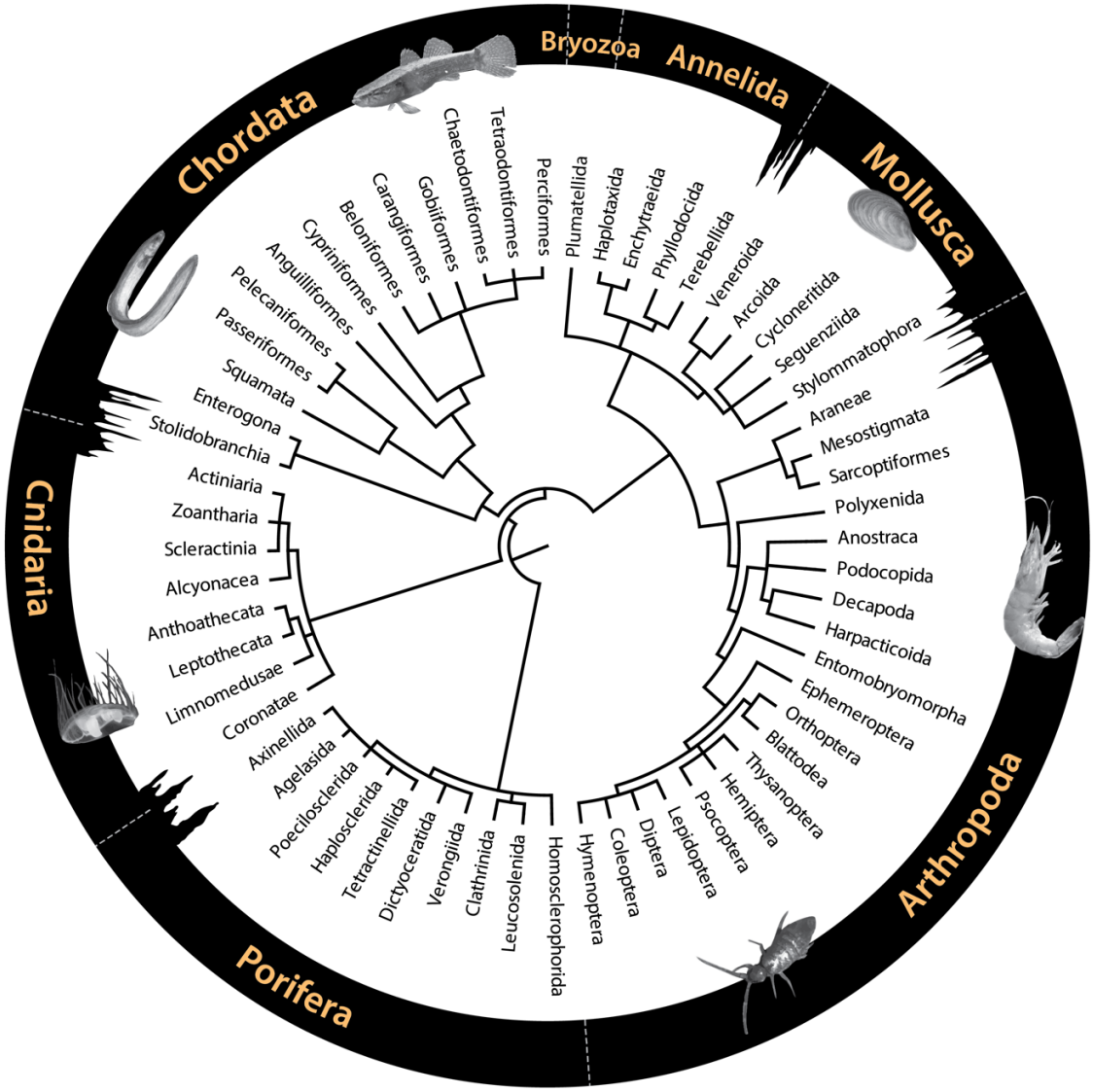


Figure 4.4 A total ordinal-level dendrogram of chordate, cnidarian, porifera, arthropod, mollusc, annelid and bryozoan taxa detected by multi-assay metabarcoding on 159 eDNA samples collected across Christmas Island.

Table 4.3 Taxa of conservation, biodiversity and biosecurity importance.
Site number is given in parentheses next to site name.

Common Name	Scientific Name	Importance	Site/s detected	Assay	Associated environment	Biogeographic region
Indonesian shortfin eel	*Anguilla bicolor bicolor*	A single juvenile was previously obtained in Daniel Roux Cave, with larger adults reported by the public in the Dales (Hui et al., 2014). Our detection at Dolly Cave Spring in the south-east of CI indicates that this elusive species is more widely distributed across the island.	Dolly Cave Spring (20)	16S Fish (short)	Marine, brackish, freshwater	Indo-West Pacific
Gudgeon	*Eleotris*	*Eleotris fusca* has previously been reported in Daniel Roux Cave, Whip Cave, Runaway Cave and Freshwater Cave; these populations exhibit cave adaptations such as a pale body colour, in addition to enlarged pectoral and caudal fins (Hui et al., 2014; Humphreys & Eberhard, 2001). Also occur in caves on many Indo-Pacific Islands (Humphreys, 2014).	Freshwater Cave (5), Waterfall Spring (9), Freshwater Spring (10), Ryan's Ripper Rift (16), Hosnies Spring (17)	16S Fish (short)	Freshwater	Circumglobal
Carps, minnows & relatives	Cyprinidae	There were a number of hits for Cyprinidae, however they could not be resolved beyond a family level, exempting the genus *Gobio*. Nonetheless the detection of Cyprinidae is a new occurrence record for CI; this family is not native to Australia and may represent an introduction such as *Cyprinus carpio* in the Murray-Darling River system of eastern Australia. Alternatively, it could represent an unreferenced species in southeast Asia, given the proximity of CI to Indonesia.	Hosnies Spring (17), Ross Hill Gardens (18), Dolly Cave Spring (20), Lost Lake Cave (21,22), WiFi Cave (23)	16S Fish (short)	Freshwater	Circumglobal
Freshwater jellyfish	*Craspedacusta sowerbii*	Putative new occurrence record for CI; this freshwater jellyfish is typically found in calm freshwater lakes and reservoirs around the world.	Waterfall Spring (9), Dolly Cave Spring (20)	16S Fish (short), 16S Crustacean	Freshwater	Circumglobal
Prawn	*Litopenaeus*	*Litopenaeus vannamei* was detected in this study, however we cannot verify this new occurrence record to a species level, as not all *Litopenaeus* have been barcoded for 18S. Nonetheless, the detection of this genus at CI expands upon previous surveys which give reference to unidentified species of the encompassing family Penaeidae (Morgan, 2000).	Whip Cave (3), Lost Lake Cave (21), WiFi Cave (23)	18S Universal	Marine, brackish	Circumglobal
Shrimp	Atyidae	A member of this family, *Antecaridina lauensis*, has previously been identified in Whip Cave, 19th Hole and Runaway Cave (Bruce & Davie, 2006; Davie & Ng, 2012; Hui et al., 2014; Namiotko, Wouters,	Ryan's Ripper Rift (16)	18S Universal	Freshwater, brackish	Circumglobal

		Danielopol, & Humphreys, 2004). The distribution of this coastal anchialine shrimp may extend to Ryan's Ripper Rift.				
Millipedes	Lophoproctidae	Two species of this family, *Lophoturus speophilus* and *L. humphreysi,* have previously been identified in Jedda Cave and Jane-up Cave, and 19th Hole, respectively (Duy-Jacquemin, 2014). The *Lophoproctidae* millipedes are commonly found in low light environments such as cave habitats (Huynh & Veenstra, 2018).	Jedda Cave (11)	18S Universal	Terrestrial	Circumglobal
Harpacticoid copepod	*Nitokra*	*Nitokra* sp. has previously been identified in Whip Cave and Hendersons Spring (Humphreys & Eberhard, 2001; Namiotko et al., 2004). The distribution of this copepod now extends to Waterfall Spring. Two species of Nitokra have been previously reported in Western Australia; one in the anchialine groundwaters of the Cape Range karst area (Karanovic & Pesce, 2002) and the other in calcrete aquifers broadly across arid WA (Karanovic, 2004).	Waterfall Spring (9)	18S Universal	Freshwater, brackish	Circumglobal
Slender springtails (collembola)	*Willowsia*	*Willowsia nigromaculata* was detected in this study, however we cannot verify this new occurrence record to a species level, as not all *Willowsia* have been barcoded for 16S. Nonetheless, this genus is a putative new occurrence record for CI; *Willowsia* sp. have previously been recorded in caves on mainland Western Australia (Greenslade, 2002). Unidentified species from the encompassing subclass Collembola have previously been identified in Runaway Cave, Jane-up Cave, The Grotto, Grants Well, 19th Hole and Whip Cave (Humphreys & Eberhard, 2001).	Jane-up Cave (12), Grants Well (14), Thundercliff Cave (15), Ryan's Ripper Rift (16), Hosnies Spring (17)	16S Crustacean	Terrestrial	Circumglobal

Forty-seven arthropods (phylum: Arthropoda) were detected (10 at order level only, 11 at family level only, 10 at genus level only and 16 at a species level; Table S4.5). The assemblage was largely comprised of insects, in addition to arachnids, crustaceans, collembola and millipedes. The majority of the detected arthropods in this study are known to be terrestrial ground-dwelling taxa; the detection of their DNA in water/sediment substrates is not unsurprising given that rainwater percolates through the karst landscape, potentially carrying organisms or traces of terrestrial DNA into the system. We detected a range of land crabs that are largely know to inhabit the forest floor on CI, but were also spotted in this study within entrances of cave systems; these included the orange-legged crab (*Tuerkayana magnum*), the yellow nipper (*Geograpsus crinipes*), the little nipper (*Geograpsus grayi*) and the purple crab (*Gecarcoidea lalandii*). Of the aquatic taxa, we detected freshwater brine shrimp (*Artemia franciscana*), copepod (*Nitokra*), shrimp (family: Atyidae), ostracods (*Darwinula stevensoni* and *Schlerochilus*) and whiteleg shrimp (*Penaeus vannamei*; see Table 4.3).

4.5.3 Community composition and clustering

A distance-based linear model (DistLM) analysis of community composition across all sites, indicated that site type (cave/spring) explained the highest proportion of fitted variance (9%), followed by salinity (6.1%; Table 4.4, Table S4.6). This is visualised in the PCO in Figure 4.5a. Taxa richness per site was found not to significantly differ between cave and spring sites ($P = 0.840$, $df = 1$, Table S4.7, Figure S4.5), indicating that compositional variation between the site types is not driven by unequal taxa richness. Similarity percentage analysis (SIMPER) was used to identify which subterranean taxa contributed most to pairwise dissimilarity between the site types and between salinity groupings (Tables S4.8 and S4.9). In examining cave sites only, we found that compositional dissimilarity is driven by a longitudinal transition, in addition to dissolved oxygen. Cumulatively, these two significant predictor variables explain 21.7 % of the total fitted variance between cave assemblages (Figure 4.5b, Table 4.4, Table S4.6). Within spring sites only, the DistLM identified a latitudinal and dissolved oxygen effect on compositional dissimilarity, however, these were not significant ($P = 0.103$ and $P = 0.298$ respectively, Table 4.4, Figure 4.5c, Table S4.6). Hierarchical clustering based on the community composition (Jaccard similarity) of all sites

revealed a number of discrete groupings; eight were significantly separated by SIMPROF ($P< 0.05$, Figure 4.6). For the abiotic environmental data (Euclidean distance), SIMPROF identified three groupings that were significantly different from each other ($P< 0.05$, Figure 4.7). A high possibility of local interconnection was attributed to three cave and spring groups which exhibited both biotic and abiotic clustering. These sites were Whip Cave and The Grotto, Jones Spring and Waterfall Spring, and Lost Lake Cave site 1 and site 2.

Table 4.4 Summary table of the distance based linear model (DistLM) analyses for subterranean fauna.

These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'. The predictor variables highlighted in bold are significant (P<0.05). Full DistLM results, including marginal tests and best solutions, are provided in Table S4.6.

Sites	Predictor	Adj R^2	Proportion	Cumulative Proportion	*P*	
All sites	**Site type (cave/spring)**	0.047	0.090	0.090	0.002	**
	Salinity	0.066	0.061	0.151	0.041	*
	Latitude	0.073	0.048	0.199	0.257	
Cave sites only	**Longitude**	0.045	0.119	0.119	0.029	*
	Dissolved oxygen	0.074	0.098	0.217	0.052	*
	Salinity	0.108	0.097	0.314	0.063	
	Acidity	0.138	0.089	0.403	0.129	
	Temperature	0.176	0.090	0.493	0.127	
Spring sites only	Latitude	0.046	0.165	0.165	0.103	
	Dissolved oxygen	0.064	0.133	0.298	0.298	

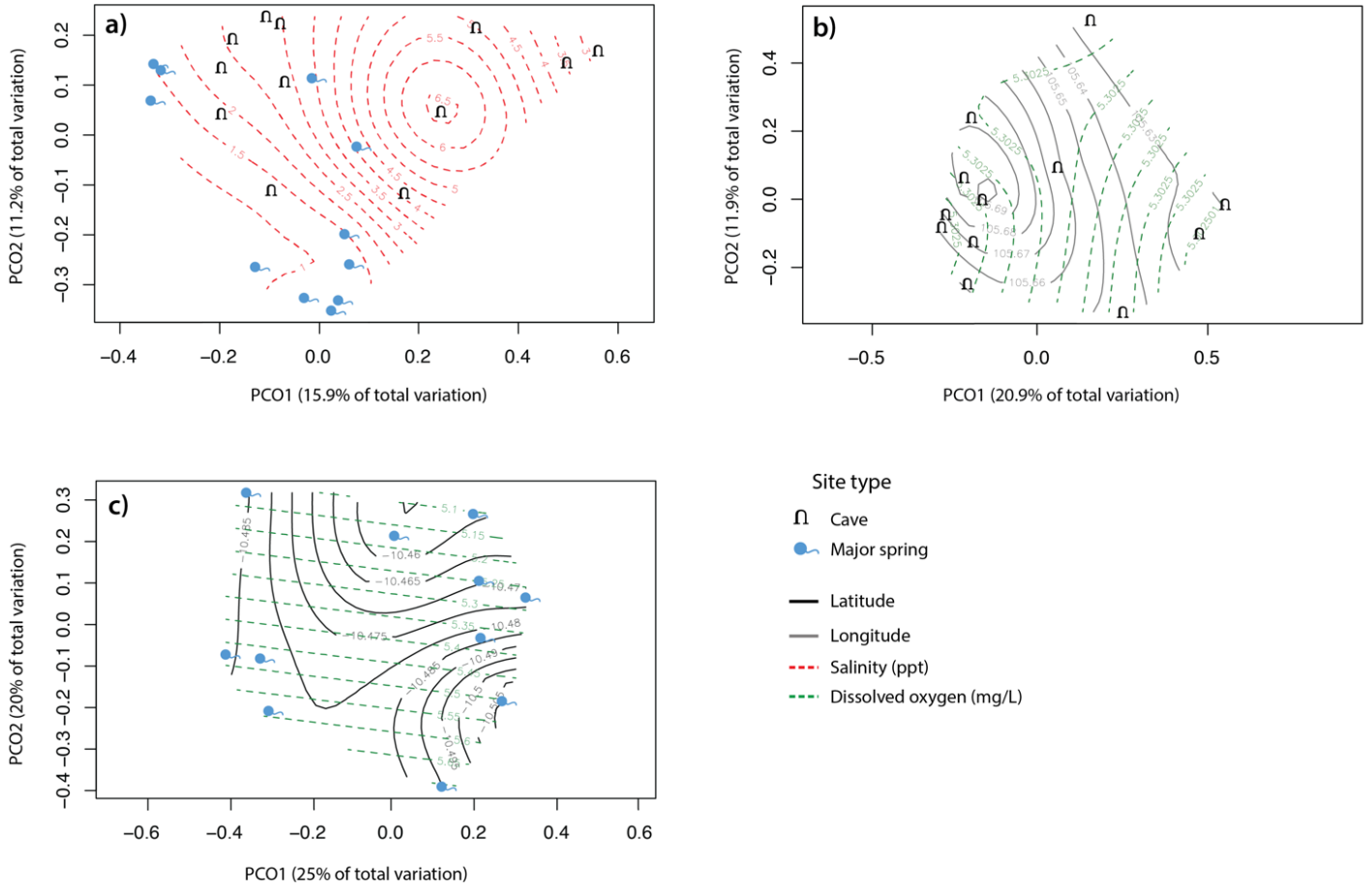


Figure 4.5 Principal coordinates analysis (PCO) of bony fish composition in a) all sites, b) cave sites only and c) spring sites only. Latitudinal, longitudinal, salinity and dissolved oxygen gradients are overlaid if identified as a predictor variable in corresponding DistLM analyses. The proportion of variation explained by each axis is shown on the axis labels.

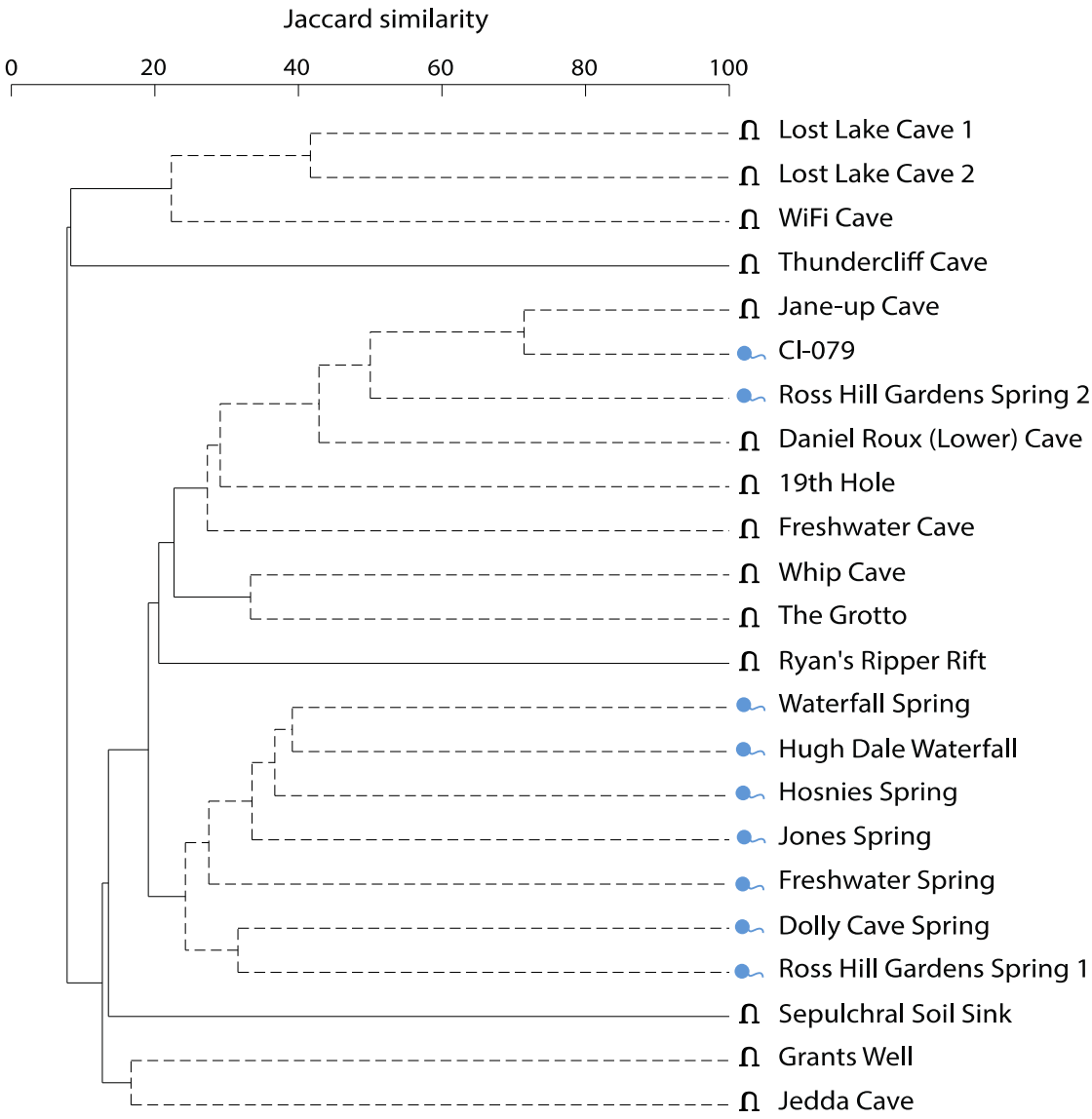


Figure 4.6 Cluster analysis of CI community composition similarity.
Site composition is comprised of all assigned taxa resulting from the three metabarcoding assays. Solid lines indicate groups that the SIMPROF analysis identified were significantly different from each other ($P < 0.05$).

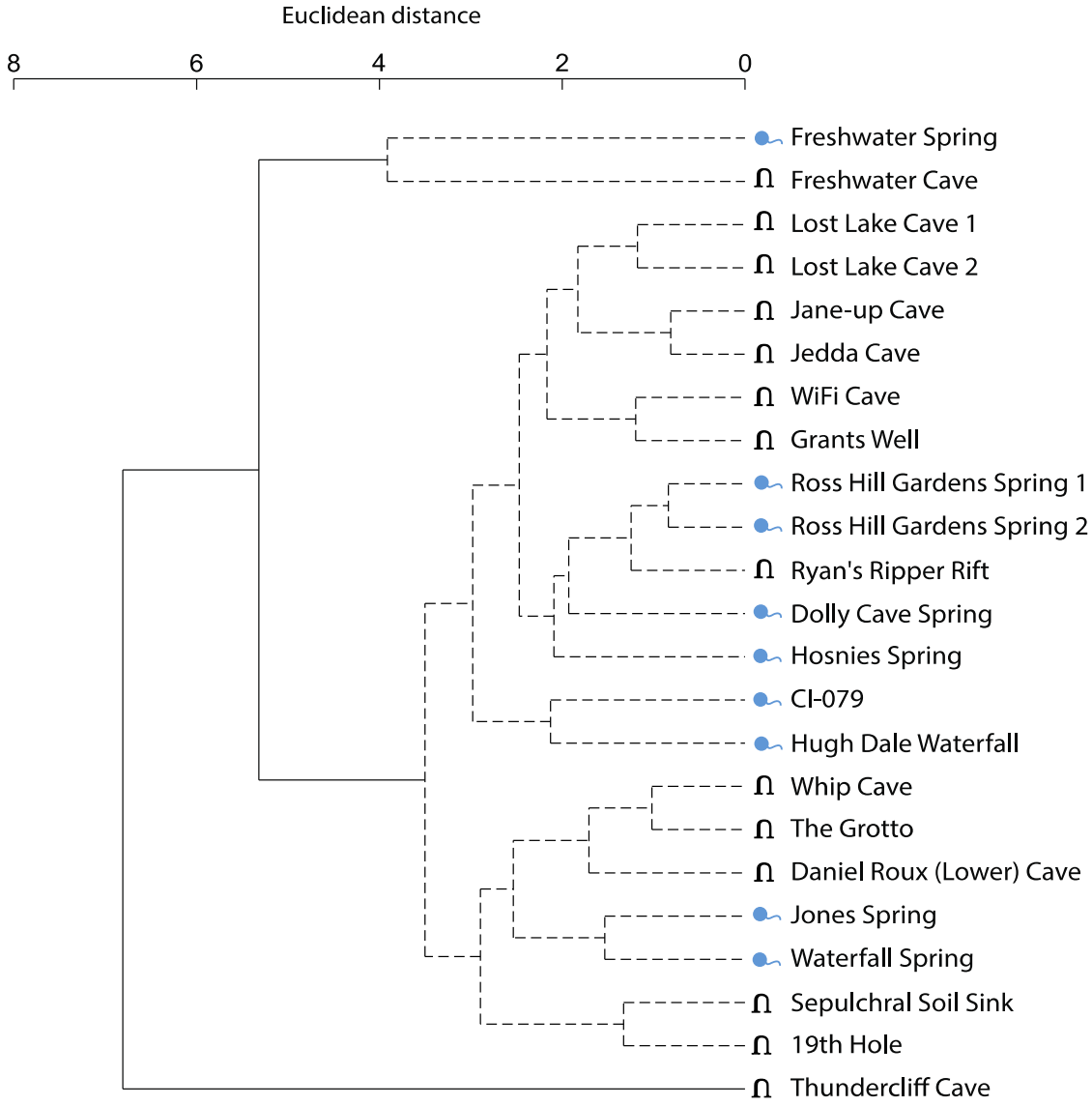


Figure 4.7 Cluster analysis of CI abiotic environmental similarity.
Environmental data incorporated into this analysis included acidity, temperature, salinity, dissolved oxygen, latitude and longitude readings taken from each site at the time of sample collection. Solid lines indicate groups that the SIMPROF analysis identified were significantly different from each other ($P < 0.05$).

4.6 Discussion

4.6.1 A new approach to biospeleological surveying

Our multi-assay eDNA metabarcoding approach successfully detected a wide range of chordates, cnidarians, porifera, arthropods, molluscs, annelids and bryozoans from CI's subterranean habitats. Despite targeting CI's aquatic stygofauna, our water and sediment samples also produced detection hits for troglofauna (terrestrial cave fauna) that have likely shed DNA into the water below. This demonstrates that water and sediment can be used to detect troglofauna. However, it is likely that other substrates (e.g. soil samples) may provide a greater detection rate for subterranean terrestrial species. Notable detections included a putative new occurrence record of Collembola (genus: *Willowsia*) across five sites, which potentially resolves the classification of unidentified Collembola specimens previously collected in Runaway Cave, Jane-up Cave, The Grotto, Grants Well, 19th Hole and Whip Cave (Humphreys & Eberhard, 2001), although a specimen of the genus *Cyphoderopsis* has also been identified from an unreferenced cave on the island (Greenslade, 2002). Likewise, the detection of prawn (genus: *Litopenaeus*) at three cave sites may also resolve unidentified specimens collected on CI of the encompassing family Penaeidae (Morgan, 2000). An extension in the distribution records of Indonesian shortfin eel (*Anguilla bicolor bicolor*) and gudgeon (genus: *Eleotris*) across CI's subterranean habitats aids management purposes, particularly as the former is an elusive species that is rarely seen and only reported in two locations (Hui et al., 2014), whilst the latter may represent a highly unique lineage exhibiting cave adaptations such as a pale body colour and enlarged pectoral and caudal fins (Hui et al., 2014; Humphreys & Eberhard, 2001).

The overall assemblage also included a total of 21 putative new occurrence records, however their verification requires additional reference barcodes of subterranean fauna in order to rule out incorrect assignments to a closely-related (not yet barcoded) taxon. Subterranean environments, and by extension their fauna, are notoriously under surveyed. As such, only a small number of vouchered specimens have been barcoded for commonly targeted gene regions (i.e. COI, 16S, 12S and 18S rRNA). Differences between sexes and life-stages that are classified as different species can also

complicate the picture when assembling reference barcode material. One further complication is the high likelihood that vouchered subterranean specimens have been preserved using formalin, such as the two specimens of the rare cave-dwelling flashlight fish (*Photoblepharon palpebratum*) collected from Thundercliff Cave on CI (J. DiBattista, personal communication, January 2019). The preservative unfortunately fragments DNA, modifies bases and creates crosslinks within the DNA (Campos & Gilbert, 2012), making it challenging to extract and piece together reference sequences. In addition, subterranean environments are highly fragmented habitats, facilitating the evolution of short-range endemic species (Gibert & Deharveng, 2002). These unique lineages require representation in reference databases in order to make robust (species-level) assignments. However, despite a suboptimal reference database, this study has demonstrated that eDNA metabarcoding can still reveal a wide diversity of subterranean taxon detections, including some at a species level. Additionally, the detection of unknown species using eDNA (such as the 22.6% of assignments that could not be resolved beyond order level) can be used to direct traditional sampling efforts for specimen acquisition, taxonomic classification and barcoding.

4.6.2 Subterranean composition

The cave and spring systems on CI are distinct and host a different complement of subterranean community assemblages. Given spring water naturally flows through underground and near-surface conduits, we expected spring sites to contain subterranean and potentially surface taxa. Therefore, we anticipated additional taxa in spring sites compared to those detected in the cave sites. However, variation in taxa richness between cave and spring sites was not significant. The spring sites typically produced more sub-surface and terrestrial taxa, whilst the caves sites produced subterranean, largely aquatic, taxa. For example, spring sites were typified by freshwater ostracods (family: Darwinulidae, including *Darwinula stevensoni*), land crabs (*Discoplax magna*, *Gecarcoidea lalandii* and *Geograpsus crinipes*), ticks (order: Sarcoptiformes) and freshwater jellyfish (*Craspedacusta sowerbii*), whilst the cave sites was characterised by slender springtails (*Willowsia*), demosponges (order: Haplosclerida), whiteleg shrimp (*Penaeus vannamei*), and carp/minnows (family: Cyprinidae).

Variation in salinity between the freshwater springs (0.2-0.4ppt) and a combination of freshwater, brackish and saltwater cave sites (0.3-26.0ppt) also had a strong influence on the overall composition. Distinct fauna were found to be associated with each salinity level, for example freshwater and oligohaline sites (0-4.9ppt) were characterised by freshwater ostracods, land crabs, gudgeon (*Eleotris*) and clitellate oligochaete worms (family: Naididae); mesohaline sites (5.0-17.9ppt) by brine shrimp (*Artemia franciscana*), sea sponge (*Iophon*) and giant trevally (*Caranx ignobilis*); and lastly the polyhaline site (18.0-29.9ppt; Thundercliff Cave only) by demosponge (*Callyspongia*), black triggerfish (*Melichthys niger*) and brown tubular sponge (*Agelas schmidti*).

In examining the variation between cave and spring sites separately, the effect of salinity is no longer significant as there is a reduction in salinity variation within each site type. Cave sites were then found to vary on a longitudinal transition across the island—indicating that site dissimilarity increases with distance—in addition to a dissolved oxygen influence. Surface water at most of the cave sites was well oxygenated (ranging between 4.0-5.7 mg/L), although Freshwater Cave (4.0 mg/L) exhibited a lower concentration than previous reports on the island (Humphreys & Eberhard, 2001). Spring sites also varied on a distance-based (albeit latitudinal) and dissolved oxygen transition across the island, although this was not statistically significant. Notably, however, Freshwater Spring exhibited a low dissolved oxygen level of 2.5 mg/L, well below that of any other sampled site. Despite these low oxygen levels, which can induce hypoxia in some freshwater fish species (Franklin, 2014), two bony fish taxa (genus: *Eleotris* and family: Leiognathidae) were detected.

4.6.3 Subterranean connectivity

Potential underground connectivity was assessed across biotic (community composition) and abiotic (environmental parameter) hierarchical cluster analyses. Sites which exhibit both biotic and abiotic clustering and are presumed to have a high possibility of connection include Whip Cave and The Grotto, Jones Spring and Waterfall Spring, and Lost Lake Cave Site 1 and Site 2. Whip Cave, an anchialine cave, and The Grotto, a coastal marine cave with strong sea currents, are accessed at a distance of 80m apart. They exhibit highly similar environmental parameters and

community composition, and were both noted onsite to be tidally influenced. A possible hydrological connection between Whip Cave and The Grotto has been previously reported in Grimes (2001) and Humphreys & Eberhard (2001), the latter also indicating a connection with Runaway Cave. However, we were unable to sample Runaway Cave as part of this project. Our data supports the premise that Whip Cave and The Grotto are connected.

There are no previous reports of connectivity between Jones Spring and Waterfall Spring, however, given they occur in close proximity (1.07 km apart) it is a possibility that they are fed by shared flow systems. Freshwater Spring is situated approximately halfway between Jones and Waterfall Spring and was therefore expected to cluster with these spring sites. Whilst Freshwater Spring is comprised of a similar community composition, it exhibited very different abiotic readings in relation to dissolved oxygen and air saturation. Therefore, we can only suspect based on this evidence that Jones Spring and Waterfall Spring have a high possibility of interconnection. Lost Lake Cave Site 1 and Site 2 were sampled from a continuous stream passage; the former taken approximately 150 m from the sea entrance and the latter sampled far into the cave extent (more than 500 m from sea entrance). It was therefore expected that these two sites would be classified as having a high possibility of interconnection.

Sites with a medium possibility of interconnection (i.e. sites which exhibit either biotic or abiotic clustering) include the plateau sites of Jane-up Cave, Jedda Cave, WiFi Cave and Grants Well, CI-079 and the Hugh Dale Waterfall, and Sepulchral Soil Sink and 19th Hole. Jedda Cave is the mainstay of CI's water supply and is purported to access a subterranean flow between Grants Well and Jane-up Cave (Barrett, 2001; Humphreys, 2014). In order to trace the potential flow of water between Grants Well and Jane-up Cave (1.3 km apart), stream water was previously spiked with salt and measured downstream; the through-flow time between the two sites was three hours (400 m/h), confirming their interconnectivity (Anon, 1971). Abiotic clustering in this study indicates that Jane-up Cave, Jedda Cave, WiFi Cave and Grants Well (located within 1.29 km) are potentially all connected underground, with almost exact water quality readings across the four sites. The spring sites of CI-079 and the Hugh Dale Waterfall, located within The Dales wetland area, also exhibited abiotic clustering and are within a 0.6km distance of each other. Likewise, the cave sites of Sepulchral Soil

Sink and 19th Hole, located east of Flying Fish Cove, exhibited abiotic clustering and are located approximately 1.9 km apart.

The detection of marine taxa can also elucidate sites with an ocean connections and therefore potential tidal influence; this included Thundercliff Cave, Lost Lake Cave, The Grotto, 19th Hole, Whip Cave, Sepulchral Soil Sink, Ryan's Ripper Rift, Hosnies Spring and Ross Hill Gardens Site 1 and 2. All of the cave sites, except for Sepulchral Soil Sink, were noted onsite to have had hourly changes in water level which was attributed to tidal influence. Sepulchral Soil Sink, however, produced salinity readings that were in the range of other tidal influenced sites on CI indicating that it also had an ocean connection. The detection of marine taxa from the three spring sites of Hosnies Spring and Ross Hill Gardens Site 1 and 2 was surprising given they were classified as freshwater based on the surface readings. However, these spring sites are all within 1 km of the coastline; it is therefore possible that they have an ocean connection but exhibit a halocline (salinity) gradient between the groundwater and surface spring.

4.7 Conclusion

The use of eDNA sampling as a biomonitoring tool in caves where population dynamics may be extremely fragile to external pressures circumvents the impacts from traditional biospeleological approaches, where specimens must be continually captured for verification beyond initial vouchering and barcoding. Such processes require multiple visits to caves compounding the impacts that are exerted to conservation sensitive areas and fauna. This not only increases levels of impacts exposed to the cave, but increases risk to researchers, and requires additional permits, time and resources to obtain samples. Whilst sampling cave water or sediment for eDNA analyses is logistically easier, karst groundwater is subject to extreme fluctuations in water level during wet and dry seasons/years. This must be taken into consideration when eDNA sampling over successive periods (Gorički, 2019). The power of eDNA metabarcoding lies in its ability to widely amplify any target taxonomic group without specific taxonomic expertise to morphologically classify taxa. This is particularly beneficial in subterranean monitoring given the prominence of endemics and taxa that exhibit cave adaptations (e.g. pale body morphs, increased sensory organs, elongated appendages and reduced eyesight).

Subterranean ecosystems are notoriously under surveyed, largely because of logistical difficulties in accessing underground voids and networks. We demonstrated that the application of eDNA metabarcoding assays to water and sediment collected from cave and spring conduits can be used to characterise multi-trophic eukaryotic subterranean diversity from voids that may be inaccessible to conventional biospeleological surveying. We detected a wide range of chordates, cnidarians, porifera, arthropods, molluscs, annelids and bryozoans from freshwater and anchialine spring and cave sites across CI. Community composition was found to vary based on site type (i.e. cave or spring) and salinity; cave sites were additionally influenced by dissolved oxygen and longitudinal gradients. We update distribution information for taxa of biodiversity importance, such as the Indonesian shortfin eel and cave-adapted gudgeon, and potentially resolve unidentified specimen classifications for Collembola and prawn that were previously reported on CI. Lastly, we combined eDNA-derived eukaryotic community composition and environmental (water quality) data to investigate potential underground interconnectivity across CI; based on hierarchical clustering we

identified three groups with a high possibility of interconnection. We strongly advocate for ongoing development of subterranean reference databases to facilitate the implementation of eDNA metabarcoding as a biospeleological survey tool, particularly for stygofauna. With this development, we anticipate that eDNA metabarcoding will be increasingly employed for subterranean multi-trophic surveying, which may reveal food webs and enlighten subterranean ecosystem functioning. In conjunction with marine, freshwater and terrestrial developments, eDNA may soon be able to provide quantitative estimates e.g. current development of ddPCR for the cave-dwelling olm *Proteus anguinus* (Gorički, 2019), and population genetic information. The latter may provide greater insight into interconnectivity and isolation of entire subterranean communities. We anticipate that this study demonstrates the potential for using multi-marker eDNA metabarcoding approaches for subterranean stygofauna surveying and exploration.

4.8 References

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Adams, M., & Humphreys, W. F. (1993). Patterns of genetic diversity within selected subterranean fauna of the Cape Range peninsula, Western Australia: systematic and biogeographic implications. *Records of the Western Australian Museum Supplement*, *45*, 145–164.

Ali, J. R., & Aitchison, J. C. (2019). Time of re-emergence of Christmas Island and its biogeographical significance. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *537*, 109396.

Allen, G. R., Steene, R. C., & Orchard, M. (2007). *Fishes of Christmas Island*. Christmas Island Natural History Association.

Anderson, M., Gorley, R. N., & Clarke, R. K. (2008). *Permanova+ for Primer: Guide to Software and Statistical Methods*. Primer-E Limited.

Anon. (1971). Water Search - Christmas Island. In unpublished report held in records of Phosphate Mining Company of Christmas Island (p. 9).

Asmyhr, M. G., Linke, S., Hose, G., & Nipperess, D. A. (2014). Systematic conservation planning for groundwater ecosystems using phylogenetic diversity. *PLoS One*, *9*(12), e115132.

Barrett, P. J. (2001). Searching for water on Christmas Island. *Helictite*, *37*(2), 37–39.

Beeton, B., Burbidge, A., Grigg, G., Harrison, P., How, R., Humphries, B., … Woinarski, J. (2010). *Final Report Christmas Island Expert Working Group to Minister for the Environment, Heritage and the Arts*.

Benson, D. A., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Wheeler, D. L. (2005). GenBank. *Nucleic Acids Research*, *33*, D34–D38. https://doi.org/10.1093/nar/gki063

Berry, T. E., Osterrieder, S. K., Murray, D. C., Coghlan, M. L., Richardson, A. J., Grealy, A. K., … Bunce, M. (2017). DNA metabarcoding for diet analysis and biodiversity: A case study using the endangered Australian sea lion (*Neophoca cinerea*). *Ecology and Evolution*, *7*(14), 5435–5453.

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a

Unix-inspired software package for DNA metabarcoding. *Molecular Ecology Resources, 16*(1), 176–182.

Bradford, T., Adams, M., Humphreys, W. F., Austin, A. D., & Cooper, S. J. B. (2010). DNA barcoding of stygofauna uncovers cryptic amphipod diversity in a calcrete aquifer in Western Australia's arid zone. *Molecular Ecology Resources*, *10*(1), 41–50.

Brewer, D. T., Skewes, T. D., Lyne, V. D., Davies, C., Taranto, T. J., Fuller, M., … Andersen, J. (2009). Conservation values in Commonwealth waters of the Christmas and Cocos (Keeling) Island remote Australian territories. Report to Department of Environment and Water Resources, CSIRO, Cleveland.

Bruce, A. J., & Davie, P. J. F. (2006). A new anchialine shrimp of the genus *Procaris* from Christmas Island: the first occurrence of the Procarididae in the Indian Ocean (Crustacea: Decapoda: Caridea). *Zootaxa, 1238*, 23–33.

Burstyn, H. L. (1975). Science pays off: Sir John Murray and the Christmas Island phosphate industry, 1886-1914. *Social Studies of Science*, *5*(1), 5–34.

Callahan, B. J., McMurdie, P. J., Rosen, M. J., Han, A. W., Johnson, A. J. A., & Holmes, S. P. (2016). DADA2: high-resolution sample inference from Illumina amplicon data. *Nature Methods*, *13*(7), 581.

Campos, P. F., & Gilbert, T. M. P. (2012). DNA extraction from formalin-fixed material. In *Ancient DNA* (pp. 81–85). Springer.

Clarke, K. R., & Gorley, R. N. (2015). *PRIMER v7: User Manual/Tutorial*. Plymouth, UK.

Davie, P. J. F., & Ng, P. K. L. (2012). Two new species of Orcovita (Crustacea: Decapoda: Brachyura: Varunidae) from anchialine caves on Christmas Island, eastern Indian Ocean. *Raffles Bulletin of Zoology*, *60*(1), 57-70.

Director of National Parks. (2014a). *Christmas Island Biodiversity Conservation*, Canberra.

Director of National Parks. (2014b). *Christmas Island National Park Management Plan 2014-2024*. Canberra.

Duy-Jacquemin, M. N. (2014). Two new species of Lophoturus (Diplopoda, Penicillata, Lophoproctidae) from caves in Christmas Island, Australia, including the second troglomorph in Penicillata. *Zoosystema, 36*(1), 29–40.

Evans, N. T., Olds, B. P., Renshaw, M. A., Turner, C. R., Li, Y., Jerde, C. L., … Lodge, D. M. (2016). Quantification of mesocosm fish and amphibian species diversity

via environmental DNA metabarcoding. *Molecular Ecology Resources*, *16*(1), 29–41.

Franklin, P. A. (2014). Dissolved oxygen criteria for freshwater fish in New Zealand: a revised approach. *New Zealand Journal of Marine and Freshwater Research*, *48*(1), 112–126.

Gibert, J., & Deharveng, L. (2002). Subterranean Ecosystems: A Truncated Functional Biodiversity: This article emphasizes the truncated nature of subterranean biodiversity at both the bottom (no primary producers) and the top (very few strict predators) of food webs and discusses the implications of this truncation both from functional and evolutionary perspectives. *BioScience*, *52*(6), 473–481.

Gonzalez, B. C., Iliffe, T. M., Macalady, J. L., Schaperdoth, I., & Kakuk, B. (2011). Microbial hotspots in anchialine blue holes: initial discoveries from the Bahamas. *Hydrobiologia*, *677*(1), 149–156.

Gorički, Š. (2019). Environmental DNA as a conservation tool. In *Encyclopedia of Caves* (pp. 387–393). Elsevier.

Greenslade, P. (2002). Systematic composition and distribution of Australian cave collembolan faunas with notes on exotic taxa. *Helictite*, *38*(1), 11–15.

Grimes, K. G. (2001). Karst features of Christmas Island (Indian Ocean). *Helictite*, *37*(2), 41–58.

Guzik, M. T., Austin, A. D., Cooper, S. J. B., Harvey, M. S., Humphreys, W. F., Bradford, T., … Muirhead, K. A. (2011). Is the Australian subterranean fauna uniquely diverse? *Invertebrate Systematics*, *24*(5), 407–418.

Hobbs, J.-P. A., Frisch, A. J., Allen, G. R., & Van Herwerden, L. (2008). Marine hybrid hotspot at Indo-Pacific biogeographic border. *Biology Letters*, *5*(2), 258–261.

Horton, T., Kroh, A., Ahyong, S., Bailly, N., Boyko, C. B., Brandão, S. N., … Zhao, Z. (2018). World Register of Marine Species (WoRMS). WoRMS Editorial Board. Retrieved from http://www.marinespecies.org

Hui, T. H., Naruse, T., Fujita, Y., & Kiat, T. S. (2014). Observations on the fauna from submarine and associated anchialine caves in Christmas Island, Indian Ocean Territory, Australia. *Raffles Bulletin of Zoology Supplement,* 30, 406-418.

Humphreys, W. F. (2014). Subterranean fauna of Christmas Island: habitats and salient features. *Raffles Bulletin of Zoology Supplement,* 30, 29-44.

Humphreys, W. F., & Eberhard, S. (2001). Subterranean fauna of Christmas Island,

Indian Ocean. *Helictite, 37*(2), 59–74.

Humphreys, W., Tetu, S., Elbourne, L., Gillings, M., Seymour, J., Mitchell, J., & Paulsen, I. (2012). Geochemical and microbial diversity of Bundera sinkhole, an anchialine system in the eastern Indian ocean. *Natura Croatica: Periodicum Musei Historiae Naturalis Croatici*, *21*(1), 59–63.

Hunt, J. G. (2011). *Suffering through strength: The men who made Christmas Island*. Published by the author J. Hunt, Canberra.

Huynh, C., & Veenstra, A. A. (2018). Two new Lophoturus species (Diplopoda, Polyxenida, Lophoproctidae) from Queensland, Australia. *ZooKeys*, *741*, 133.

Juan, C., Guzik, M. T., Jaume, D., & Cooper, S. J. B. (2010). Evolution in caves: Darwin's 'wrecks of ancient life' in the molecular era. *Molecular Ecology*, *19*(18), 3865–3880.

Karanovic, T. (2004). Subterranean Copepoda from arid Western Australia. Crustaceana Monographs. Brill, Leiden.

Karanovic, Tomislav, & Pesce, G. (2002). Copepods from ground waters of Western Australia, VII. *Nitokra humphreysi* sp. nov. (Crustacea: Copepoda: Harpacticoida). *Hydrobiologia, 470*(1–3), 5–12.

Korbel, K., Chariton, A., Stephenson, S., Greenfield, P., & Hose, G. C. (2017). Wells provide a distorted view of life in the aquifer: implications for sampling, monitoring and assessment of groundwater ecosystems. *Scientific Reports*, *7*, 40702.

Matthews, E. F., Abrams, K. M., Cooper, S. J. B., Huey, J. A., Hillyer, M. J., Humphreys, W. F., … Guzik, M. T. (2020). Scratching the surface of subterranean biodiversity: Molecular analysis reveals a diverse and previously unknown fauna of Parabathynellidae (Crustacea: Bathynellacea) from the Pilbara, Western Australia. *Molecular Phylogenetics and Evolution*, *142*, 106643.

McMurdie, P. J., & Holmes, S. (2013). Phyloseq: an R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS One*, *8*(4), e61217.

Mitsch, W. J., & Gosselink, J. G. (1986). *Wetlands, 539 pp*. Van Nostrand Reinhold, New York.

Moldovan, O. T. (2018). An Overview on the Aquatic Cave Fauna. In *Cave Ecology* (pp. 173–194). Springer.

Morgan. (2000). Decapod Crustacea of Christmas Island. *History*, *10*(16), 629–637.

Namiotko, T., Wouters, K., Danielopol, D. L., & Humphreys, W. F. (2004). On the origin and evolution of a new anchialine stygobitic Microceratina species (Crustacea, Ostracoda) from Christmas Island (Indian Ocean). *Journal of Micropalaeontology*, *23*(1), 49–59.

Oksanen, J., Blanchet, F. G., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., … Wagner, H. (2019). vegan: Community Ecology Package. R package version 2.5-5.

Olds, B. P., Jerde, C. L., Renshaw, M. A., Li, Y., Evans, N. T., Turner, C. R., … Shirey, P. D. (2016). Estimating species richness using environmental DNA. *Ecology and Evolution*, *6*(12), 4214–4226.

Ortiz, M., Legatzki, A., Neilson, J. W., Fryslie, B., Nelson, W. M., Wing, R. A., … Maier, R. M. (2014). Making a living while starving in the dark: metagenomic insights into the energy dynamics of a carbonate cave. *The ISME Journal*, *8*(2), 478.

Page, T. J., Humphreys, W. F., & Hughes, J. M. (2008). Shrimps down under: evolutionary relationships of subterranean crustaceans from Western Australia (Decapoda: Atyidae: Stygiocaris). *PLoS One*, *3*(2), e1618.

Pochon, X., Bott, N. J., Smith, K. F., & Wood, S. A. (2013). Evaluating detection limits of next-generation sequencing for the surveillance and monitoring of international marine pests. *PLoS One*, *8*(9), e73935.

Porritt, K., & Walker, K. (2015). Christmas Island GIS 2013 release. Retrieved from http://pid.geoscience.gov.au/dataset/ga/89783, https://researchdata.ands.org.au/christmas-island-gis-2013-release

R Studio Team (2015). RStudio: integrated development for R. *RStudio Inc*, Boston, MA. Retrieved from *http://www.rstudio.com*

Smith, R. J., Jeffries, T. C., Roudnew, B., Fitch, A. J., Seymour, J. R., Delpin, M. W., … Mitchell, J. G. (2012). Metagenomic comparison of microbial communities inhabiting confined and unconfined aquifer ecosystems. *Environmental Microbiology*, *14*(1), 240–253.

Stanhope, J., Wettenhall, R., & Bhusal, T. (2016). Governance challenge: Australia's Indian Ocean island territories. *Asia Pacific Journal of Public Administration*, *38*(2), 87–102.

Stat, M., Huggett, M. J., Bernasconi, R., DiBattista, J. D., Berry, T. E., Newman, S. J.,

… Bunce, M. (2017). Ecosystem biomonitoring with eDNA: metabarcoding across the tree of life in a tropical marine environment. *Scientific Reports*, *7*(1), 1-11.

Thomsen, P. F., Kielgast, J., Iversen, L. L., Møller, P. R., Rasmussen, M., & Willerslev, E. (2012). Detection of a diverse marine fish fauna using environmental DNA from seawater samples. *PLoS One*, *7*(8). https://doi.org/10.1371/journal.pone.0041732

Wickham, H. (2016). *ggplot2: elegant graphics for data analysis*. Springer.

Wilkinson, S. P., Davy, S. K., Bunce, M., & Stat, M. (2018). Taxonomic identification of environmental DNA with informatic sequence classification trees. *PeerJ Preprints*. DOI: 10.7287/peerj.preprints.26812v1.

Wu, X., Holmfeldt, K., Hubalek, V., Lundin, D., Åström, M., Bertilsson, S., & Dopson, M. (2016). Microbial metagenomes from three aquifers in the Fennoscandian shield terrestrial deep biosphere reveal metabolic partitioning among populations. *The ISME Journal*, *10*(5), 1192.

4.9 Supplementary information

Table S4.1 Environmental parameter data collected at the time of eDNA sampling.

Site No.	Cave index No.	Site name	Acidity (pH)	Temperature (°C)	Conductivity (mS)	Salinity (ppt)	Air saturation (% sat)	Dissolved Oxygen (mg/L)
1	CI-19	19th Hole	7.9	26.6	14.1	7.1	86.4	5.4
2	CI-1	The Grotto	7.1	26.1	6.4	3.2	87.7	5.7
3	CI-54	Whip Cave	7.0	25.9	5.2	2.6	83.0	5.2
4	CI-3	Daniel Roux (Lower) Cave	7.3	25.7	0.8	0.4	85.5	5.6
5	CI-10	Freshwater Cave	7.2	26.1	1.0	0.5	61.8	4.0
6	-	Hugh Dale Waterfall	7.4	25.2	0.6	0.3	87.3	5.7
7	CI-79	CI-079 (unnamed)	7.5	26.0	0.6	0.3	92.4	5.9
8	CI-103	Sepulchral Soil Sink	7.6	26.3	12.6	6.4	86.0	5.5
9	-	Waterfall Spring	7.5	26.5	0.5	0.2	95.6	6.1
10	-	Freshwater Spring	7.1	26.2	0.8	0.4	39.5	2.5
11	CI-5	Jedda Cave	7.7	26.1	0.5	0.3	80.0	5.1
12	CI-6	Jane-up Cave	7.6	25.8	0.5	0.3	80.4	5.1
13	CI-86	Jones Spring	7.3	26.6	0.7	0.3	86.1	5.5
14	CI-11	Grants Well	7.3	25.1	0.6	0.3	79.5	5.1
15	CI-90	Thundercliff Cave	7.7	26.1	41.6	26.0	79.5	5.0
16	CI-142	Ryan's Ripper Rift	7.7	25.7	3.3	1.6	87.2	5.7
17	CI-75	Hosnies Spring	8.2	25.6	0.6	0.3	87.9	5.7
18	-	Ross Hill Gardens Spring 1	7.5	25.4	0.6	0.3	90.4	5.8
19	-	Ross Hill Gardens Spring 2	7.7	25.6	0.6	0.3	92.5	6.0
20	CI-143	Dolly Cave Spring	7.9	25.8	0.5	0.3	87.5	5.6
21	CI-7	Lost Lake Cave 1	7.4	26.0	9.7	4.8	85.0	5.5
22	CI-7	Lost Lake Cave 2	7.5	26.0	1.2	1.3	86.2	5.6
23	CI-141	WiFi Cave	7.4	25.5	0.6	0.3	82.9	5.3

S4.9.1 qPCR reagents and conditions

Quantitative PCR (qPCR) amplification was performed using fusion tagged primers that consist of an Illumina sequencing adaptor, a unique index (8bp in length) and a primer sequence from each respective assay (see Table 4.2 for assay details). All qPCR reactions were prepared in an ultra-clean trace DNA facility, with each 25 µl reaction containing final concentrations of: 1X AmpliTaq Gold® PCR buffer (Life Technologies, Massachusetts, USA), 2mM $MgCl_2$, 0.1mM dNTPs, 0.2µM each of forward and reverse primers (Integrated DNA Technologies, Australia), 10ug BSA (Fisher Biotec, Australia), 0.6µl of 5X SYBR® Green (Life Technologies), 1U AmpliTaq Gold® DNA Polymerase (Life Technologies), 4µl of eDNA template, and made to volume with Ultrapure™ Distilled Water (Life Technologies). Each qPCR was performed on a StepOnePlus Real-Time PCR System (Applied Biosystems, Massachusetts, USA) under the following conditions: initial denaturation at 95 °C for 5 min, followed by 50 cycles of 30 s at 95 °C, 51-54 °C for 30 s (see respective annealing temperatures in Table 4.2) and 45 s at 72 °C, with a final extension for 10 min at 72 °C.

S4.9.2 Bioinformatic parameters

Unidirectional sequencing reads were demultiplexed using the ngsfilter (allowing up to three mismatches in primer sequences, zero mismatches in unique indexes) and obisplit commands in OBITools (v1.2.9; Boyer et al., 2014). Unmerged paired-end sequencing reads were demultiplexed using default parameters in the insect package (Wilkinson, Davy, Bunce, & Stat, 2018) in RStudio (v1.1.423; RStudio Team, 2015). Demultiplexed data was then quality filtered (minimum length=100, maximum expected errors=2, no ambiguous nucleotides), denoised, filtered for chimeras and dereplicated (pool=TRUE) using the DADA2 pipeline in RStudio (Callahan et al. 2016). The resulting amplicon sequence variant (ASV) fasta file for each assay was then queried against NCBI's GenBank nucleotide database (accessed in 2019; Benson et al., 2005) using BLASTn (minimum percentage identity of 90, maximum target sequences of 10, reward value of 1) and also against a curated 16S rDNA Western Australian fish database (see Table S2.6 in chapter two) via Zeus, an SGI cluster, based at the Pawsey Supercomputing Centre in Kensington, Western

Australia. Taxonomic assignments of ASVs were curated using a lowest common ancestor (LCA) approach (https://github.com/mahsa-mousavi/LCA_taxonomyAssignment). This collapses assignments to their LCA if the percentage identity between each consecutive hit (set to the top 10 hits) differed by less than one (based on 100% query coverage). All taxonomic assignments were classified based on associated environment and biogeographic distribution obtained from CI subterranean biodiversity surveys (Hui, Naruse, Fujita, & Kiat, 2014; Humphreys, 2014) and the World Register of Marine Species (WoRMS; Horton *et al.*, 2018). Putative new occurrence records were additionally assessed for whether all congeneric taxa have been barcoded for the targeted gene region. Any ASVs that were detected in filtration and/or extraction blanks were entirely removed; remaining ASVs that share the exact taxonomy assignment were then merged using the phyloseq 'tax_glom' function (McMurdie & Holmes, 2013) in RStudio. This produced a taxonomic-based matrix; read abundance was converted to presence/absence data in PRIMER v7 (Clarke & Gorley, 2015) for subsequent statistical analyses.

Table S4.2 Mean reads per replicate sample in bioinformatic processing.

Assay	Demultiplexed Input	DADA2 Quality filtered	DADA2 Denoised	Post-chimera removal	Dereplicated unique sequences	ASVs with BLAST Hit
16S Fish (short)	118,973	74,309	74,014	74,014	36	9
16S Crustacean	70,860	66,743	66,159	66,152	15	6
18S Universal	31,833	28,895	27,079	22,968	215	143

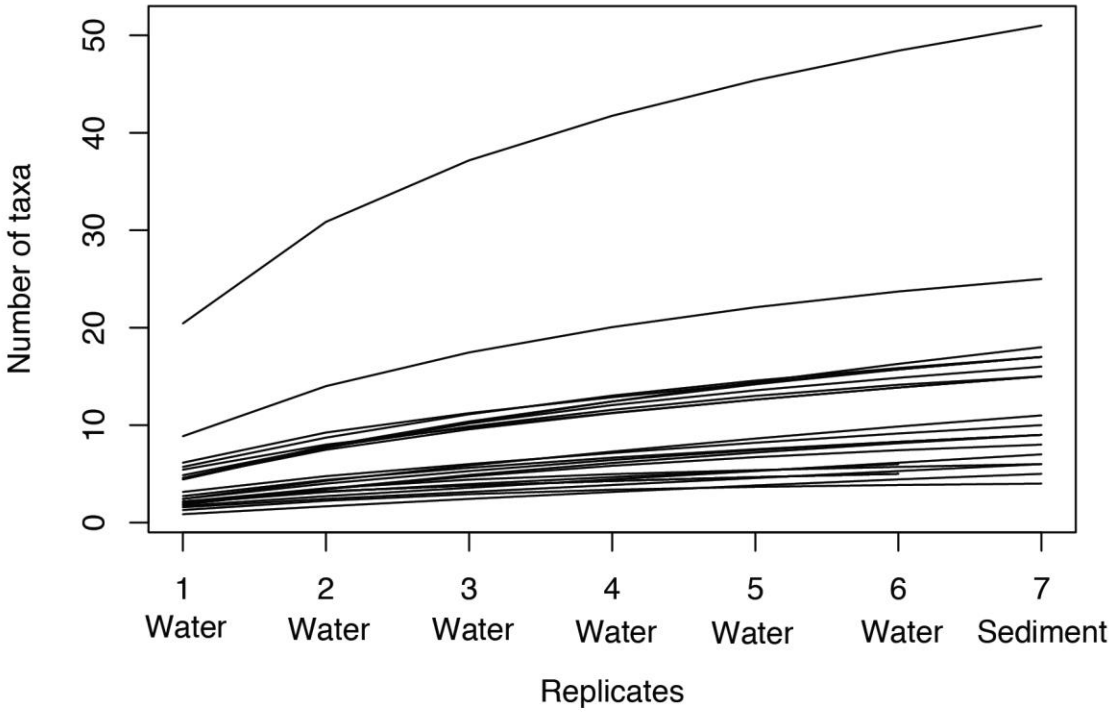


Figure S4.1 Taxa accumulation curve by the addition of replicates (six water and one sediment).
Each line represents the addition of replicates at an individual site.

Table S4.3 Two-way crossed PERMANOVA testing for the effect of site and substrate (water and sediment).
Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Source	d.f	Sum of squares	Mean Sq	Pseudo-F	Unique perms	*P*	
Site	22	81952	3725.1	1.3235	9716	**0.0034**	**
Substrate	1	18805	18805	6.6813	9943	**0.0001**	***
Residual	16	45033	2814.6				

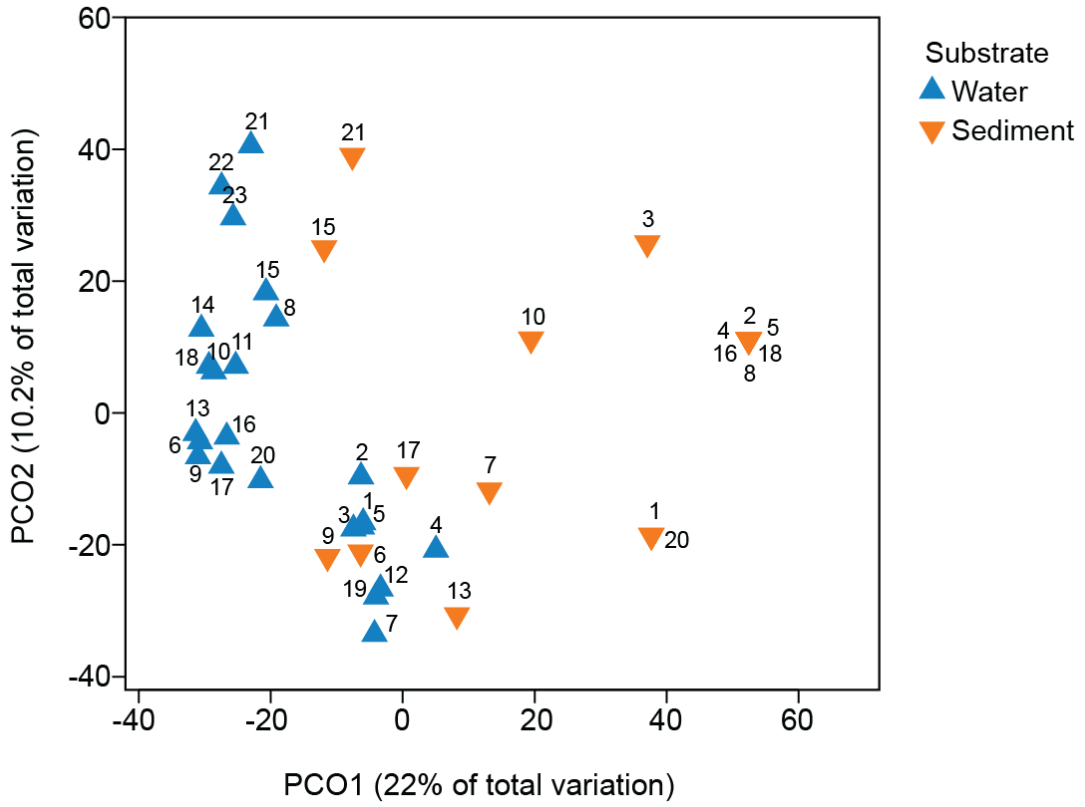


Figure S4.2 Principal coordinates analysis (PCO) of cave faunal composition by substrate (water and sediment).
Sites are numbered next to their respective water (pooled replicates) and sediment samples.

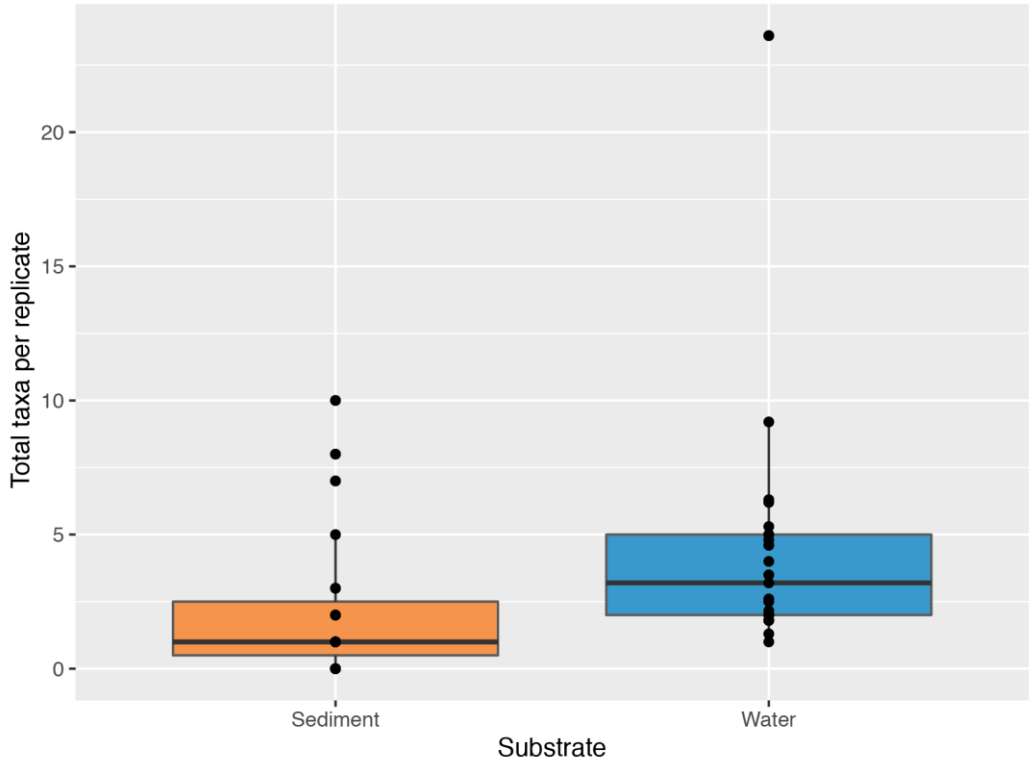


Figure S4.3 Boxplots of total taxa per sediment and water replicate.

Table S4.4 Similarity percentage analysis (SIMPER) of CI subterranean taxa. Pairwise dissimilarity between substrates.

Groups	Avg. Diss %	Taxon	Av. Abund in Water	Av. Abund in Sediment	Av. Diss	Diss/ SD	Contrib.%	Cum. %
Water & Sediment	85.85	*Discoplax*	0.96	0.06	7.95	1.68	9.26	9.26
		Gecarcoidea lalandii	0.61	0.29	4.77	0.89	5.55	14.81
		Geograpsus crinipes	0.43	0.88	4.39	0.89	5.11	19.92
		Naididae	0.52	0.18	3.96	0.81	4.62	24.54
		Discoplax magna	0.35	0	3.18	0.64	3.7	28.24
		Darwinulidae	0.3	0.29	2.99	0.76	3.48	31.72
		Sarcoptiformes	0.3	0.24	2.44	0.74	2.85	34.57
		Decapoda	0.26	0.06	2.14	0.58	2.49	37.06
		Anoplolepis gracilipes	0.3	0	1.97	0.54	2.3	39.36
		Gobio gobio	0.26	0	1.85	0.51	2.15	41.52
		Formicidae	0.26	0	1.69	0.56	1.96	43.48
		Psocoptera	0.22	0	1.66	0.5	1.93	45.41
		Shelfordella	0.22	0	1.65	0.44	1.92	47.33
		Penaeus vannamei	0.04	0.12	1.56	0.38	1.81	49.14
		Coptotermes gestroi	0.22	0.06	1.5	0.5	1.75	50.89
		Willowsia nigromaculata	0.22	0	1.41	0.44	1.65	52.54
		Eleotris	0.22	0	1.36	0.49	1.59	54.13
		Exocoetus	0.13	0.06	1.33	0.41	1.55	55.68
		Araneae	0.22	0	1.32	0.51	1.53	57.22
		Demospongiae	0.13	0.06	1.21	0.38	1.41	58.63
		Haplosclerida	0.17	0	1.05	0.4	1.23	59.86
		Cyprinidae	0.13	0	1.05	0.33	1.23	61.08
		Amynthas	0.13	0.06	1.04	0.43	1.21	62.3
		Entomobryomorpha	0.17	0	1.01	0.45	1.18	63.47
		Haplotaxida	0.09	0.06	0.92	0.37	1.07	64.54
		Darwinula stevensoni	0.04	0.12	0.91	0.4	1.06	65.6
		Hemidactylus frenatus	0.13	0	0.86	0.37	1	66.6
		Lepidopsocidae	0.09	0	0.86	0.28	1	67.6
		Lasioseius	0.09	0	0.78	0.28	0.91	68.52
		Craspedacusta sowerbii	0.09	0.06	0.78	0.39	0.91	69.42
		Haliclona cinerea	0.09	0	0.74	0.3	0.86	70.29

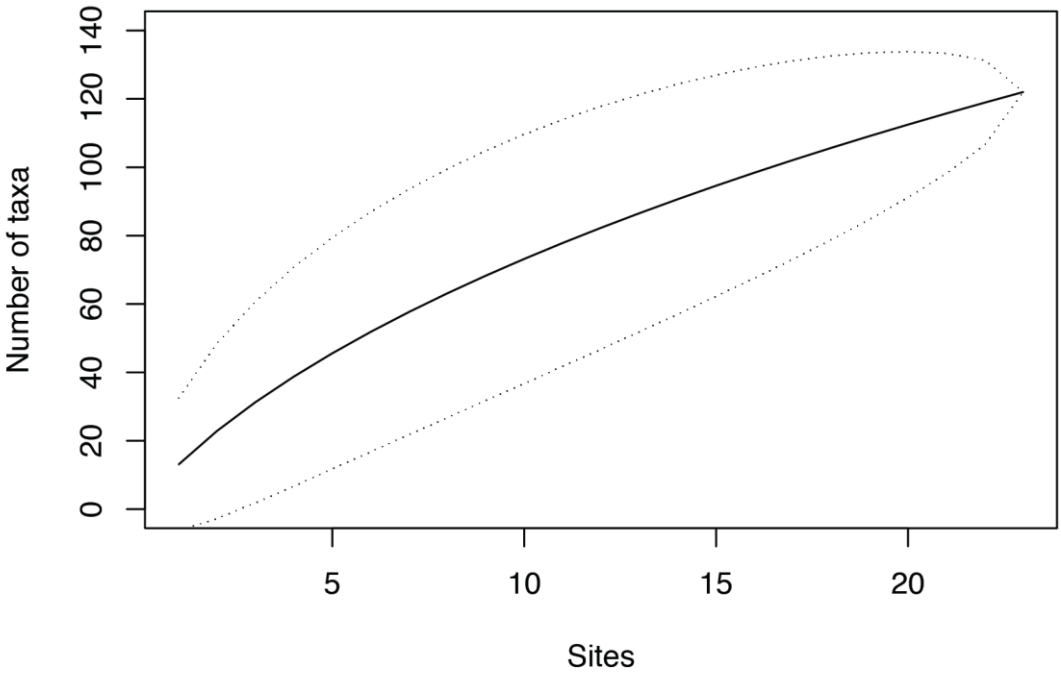


Figure S4.4 Taxa accumulation curve by the addition of sites (cave/spring). Dotted line represents 95% confidence interval.

Table S4.5 Consolidated taxa of eDNA detections across Christmas Island.

Class	Order	Family	Genus	Species	Site/s detected	eDNA Assay
Clitellata	Haplotaxida				10,13	Universal18S
Clitellata	Haplotaxida	Megascolecidae	Amynthas		6,13,17	Universal18S
Clitellata	Haplotaxida	Naididae			4,6,9,10,14,15,17,18,20,21,22,23	Universal18S
Clitellata	Haplotaxida	Naididae	Limnodrilus		6,10	Universal18S
Clitellata	Enchytraeida	Enchytraeidae	Marionina		17	Universal18S
Polychaeta		Nerillidae	Micronerilla	Micronerilla minuta	15,21	Universal18S
Polychaeta		Nerillidae			15	Universal18S
Polychaeta	Terebellida	Terebellidae			2	Universal18S
Polychaeta	Phyllodocida	Polynoidae			21	Universal18S
Arachnida	Sarcoptiformes				3,6,9,10,13,15,16,17,20	Universal18S
Arachnida	Araneae				6,9,10,21,22	Universal18S
Arachnida	Mesostigmata	Blattisociidae	Lasioseius		8,16	Universal18S
Arachnida	Sarcoptiformes	Glycyphagidae			8	Universal18S
Arachnida	Sarcoptiformes	Acaridae	Tyrophagus		22	Universal18S
Arachnida	Mesostigmata	Laelapidae	Stratiolaelaps		16	Universal18S
Arachnida	Sarcoptiformes	Oribatulidae	Oribatula	Oribatula sakamorii	10	Universal18S
Arachnida	Araneae	Araneidae			6,15,17	Universal18S
Arachnida	Sarcoptiformes	Brachychthoniidae	Liochthonius		10	Universal18S
Branchiopoda	Anostraca	Artemiidae	Artemia	Artemia franciscana	1	Crust16S
Collembola	Entomobryomorpha	Entomobryidae	Willowsia	Willowsia nigromaculata	12,14,15,16,17	Crust16S
Collembola	Entomobryomorpha				9,10,13,22	Universal18S
Diplopoda	Polyxenida	Lophoproctidae			11	Universal18S
Diplopoda	Polyxenida				10,16	Universal18S
Hexanauplia	Harpacticoida				1,20	Universal18S
Hexanauplia	Harpacticoida	Ameiridae	Nitokra	Nitokra hibernica	9	Universal18S
Insecta	Hymenoptera	Formicidae	Anoplolepis	Anoplolepis gracilipes	11,13,14,15,16,17,21	Crust16S
Insecta	Blattodea	Rhinotermitidae	Coptotermes	Coptotermes gestroi	3,9,13,14,15,17	Crust16S
Insecta	Blattodea	Blattidae	Shelfordella		7,12,13,15,16	Crust16S
Insecta	Orthoptera	Gryllidae	Gryllodes	Gryllodes sigillatus	5	Crust16S
Insecta	Hymenoptera	Formicidae	Cardiocondyla		21	Crust16S
Insecta	Hemiptera	Aphididae	Aphis		5,13	Crust16S
Insecta	Blattodea	Blattidae	Periplaneta	Periplaneta americana	17	Crust16S
Insecta	Psocoptera				5,6,13,14,20	Universal18S

Insecta	Lepidoptera				5	Universal18S
Insecta	Orthoptera	Pyrgomorphidae	Atractomorpha		8	Universal18S
Insecta	Hymenoptera	Formicidae			6,9,14,16,17,20	Universal18S
Insecta	Psocoptera	Lepidopsocidae			11,13	Universal18S
Insecta	Blattodea				14	Universal18S
Insecta	Ephemeroptera	Caenidae			17	Universal18S
Insecta	Thysanoptera	Thripidae			10	Universal18S
Insecta	Coleoptera	Chrysomelidae			6	Universal18S
Insecta	Diptera				10	Universal18S
Insecta	Orthoptera	Gryllidae			20	Universal18S
Malacostraca	Decapoda	Gecarcinidae	Discoplax		1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,22,23	Crust16S
Malacostraca	Decapoda	Grapsidae	Geograpsus	Geograpsus crinipes	1,2,3,4,5,6,7,8,9,10,12,13,16,17,18,19,20	Crust16S
Malacostraca	Decapoda	Gecarcinidae	Gecarcoidea	Gecarcoidea lalandii	1,2,3,4,5,6,7,9,12,13,15,16,17,19,20	Crust16S
Malacostraca	Decapoda	Grapsidae	Geograpsus	Geograpsus grayi	9	Crust16S
Malacostraca	Decapoda	Gecarcinidae	Discoplax	Discoplax magna	6,7,10,11,12,13,17,19	Crust16S
Malacostraca	Decapoda				1,9,13,17,18,19	Universal18S
Malacostraca	Decapoda	Penaeidae	Penaeus	Penaeus vannamei	3,21,23	Universal18S
Malacostraca	Decapoda	Atyidae			16	Universal18S
Ostracoda	Podocopida	Darwinulidae			6,7,9,10,13,17,18,20	Universal18S
Ostracoda	Podocopida	Darwinulidae	Darwinula	Darwinula stevensoni	6,9,17	Universal18S
Ostracoda	Podocopida	Cytheruridae	Semicytherura	Semicytherura striata	15	Universal18S
Ostracoda	Podocopida	Bythocytheridae	Schlerochilus	Schlerochilus oshoroensis	15	Universal18S
Ostracoda	Podocopida	Candonidae	Mungava		20	Universal18S
Phylactolaemata	Plumatellida	Fredericellidae	Fredericella	Fredericella sultana	15	Fish16S (short)
Actinopterygii	Cypriniformes	Cyprinidae	Gobio	Gobio gobio	17,18,20,21,22,23	Fish16S (short)
Actinopterygii	Gobiiformes	Eleotridae	Eleotris		5,9,10,16,17	Fish16S (short)
Actinopterygii	Cypriniformes	Cyprinidae			21,22,23	Fish16S (short)
Actinopterygii	Perciformes	Centropomidae			1	Fish16S (short)
Actinopterygii	Carangiformes	Carangidae	Caranx	Caranx ignobilis	8	Fish16S (short)
Actinopterygii	Beloniformes	Exocoetidae	Exocoetus		17,18,19	Fish16S (short)
Actinopterygii	Perciformes	Sillaginidae	Sillago	Sillago aeolus	16	Fish16S (short)
Actinopterygii	Tetraodontiformes	Balistidae	Melichthys	Melichthys niger	15	Fish16S (short)
Actinopterygii	Beloniformes	Hemiramphidae	Oxyporhamphus	Oxyporhamphus micropterus	17,18	Fish16S (short)
Actinopterygii	Anguilliformes	Anguillidae	Anguilla	Anguilla bicolor bicolor	20	Fish16S (short)

Actinopterygii	Beloniformes	Exocoetidae	Exocoetus	Exocoetus monocirrhus	17	Fish16S (short)
Actinopterygii	Perciformes	Cottidae	Cottus		18	Fish16S (short)
Actinopterygii	Chaetodontiformes	Leiognathidae			10	Fish16S (short)
Ascidiacea	Stolidobranchia	Styelidae			2,15	Universal18S
Ascidiacea	Enterogona	Perophoridae	Perophora	Perophora japonica	15	Universal18S
Aves	Passeriformes	Turdidae	Turdus		9,13	Fish16S (short)
Aves	Pelecaniformes	Fregatidae	Fregata		17	Fish16S (short)
Reptilia	Squamata	Gekkonidae	Hemidactylus	Hemidactylus frenatus	18,20,22	Fish16S (short)
Anthozoa	Scleractinia				15	Universal18S
Anthozoa	Alcyonacea				15,21,22	Universal18S
Anthozoa	Actiniaria				21,22	Universal18S
Anthozoa	Zoantharia	Parazoanthidae			15	Universal18S
Hydrozoa	Limnomedusae	Olindiidae	Craspedacusta	Craspedacusta sowerbii	9,20	Fish16S (short), Crust 16S
Hydrozoa	Anthoathecata	Moerisiidae	Moerisia	Moerisia inkermanica	2	Fish16S (short)
Hydrozoa	Leptothecata	Hebellidae	Hebella	Hebella contorta	15	Fish16S (short)
Hydrozoa	Anthoathecata				21,22	Universal18S
Hydrozoa	Leptothecata	Sertulariidae	Dynamena	Dynamena crisioides	15	Universal18S, Fish16S (short)
Hydrozoa	Leptothecata	Campanulariidae	Eucheilota	Eucheilota maculata	21,22	Universal18S
Hydrozoa	Leptothecata	Campanulariidae	Billardia	Billardia subrufa	15	Universal18S
Hydrozoa	Leptothecata	Sertulariidae	Thyroscyphus	Thyroscyphus ramosus	15	Universal18S
Hydrozoa	Leptothecata	Lafoeidae	Lafoea	Lafoea dumosa	15	Universal18S
Hydrozoa	Leptothecata				15,22	Universal18S
Scyphozoa	Coronatae				21	Universal18S
Bivalvia	Arcoida				15	Universal18S
Bivalvia	Veneroida				15	Universal18S
Gastropoda	Stylommatophora				8	Universal18S
Gastropoda	Cycloneritida	Hydrocenidae	Georissa	Georissa bangueyensis	14	Universal18S
Gastropoda	Seguenziida				15	Universal18S
Calcarea	Clathrinida				15	Universal18S
Calcarea	Leucosolenida				2,15	Universal18S
Demospongiae	Haplosclerida	Callyspongiidae	Callyspongia		2,15	Fish16S (short)
Demospongiae	Poecilosclerida	Acarnidae	Iophon		1,15,16	Fish16S (short)
Demospongiae	Agelasida	Agelasidae	Agelas	Agelas schmidti	15	Fish16S (short)
Demospongiae	Haplosclerida				2,3,15,22	Universal18S
Demospongiae	Haplosclerida	Chalinidae	Haliclona		2	Universal18S
Demospongiae	Dictyoceratida				15	Universal18S

Demospongiae	Poecilosclerida	Microcionidae	Clathria	Clathria venosa	15	Universal18S
Demospongiae	Axinellida	Heteroxyidae	Myrmekioderma	Myrmekioderma granulatum	15	Universal18S
Demospongiae	Poecilosclerida				15,21	Universal18S
Demospongiae	Haplosclerida	Chalinidae	Haliclona	Haliclona cinerea	2,3	Universal18S
Demospongiae	Dictyoceratida	Thorectidae	Cacospongia		15	Universal18S
Demospongiae	Tetractinellida				15	Universal18S
Demospongiae	Poecilosclerida	Guitarridae	Guitarra	Guitarra antarctica	15	Universal18S
Demospongiae	Axinellida	Raspailiidae	Janulum	Janulum spinispiculum	15	Universal18S
Demospongiae	Verongiida				15	Universal18S
Demospongiae	Tetractinellida	Tetillidae	Cinachyrella		15	Universal18S
Homoscleromorpha	Homosclerophorida	Plakinidae	Oscarella		15	Fish16S (short)
Homoscleromorpha	Homosclerophorida	Plakinidae	Oscarella	Oscarella carmela	15	Universal18S

Table S4.6 Distance based linear model (DistLM) analysis of subterranean faunal community variation.

These were constructed using a sequential step-wise selection procedure and adjusted R^2 criterion. Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Type of test	Predictor	Adj R^2	Proportion	Cumulative Proportion	*P*	
All sites						
Marginal	Latitude		0.051		0.285	
	Longitude		0.042		0.584	
	Acidity		0.048		0.374	
	Temperature		0.049		0.350	
	Salinity		0.073		0.015	*
	Dissolved oxygen		0.039		0.724	
	Site type (cave/spring)		0.090		0.003	**
Sequential (step-wise)	**Site type (cave/spring)**	0.047	0.090	0.090	0.002	**
	Salinity	0.066	0.061	0.151	0.041	*
	Latitude	0.073	0.048	0.199	0.257	
Best solution		**Adj R^2**	**R^2**	**No. Groups**	**Selections**	
		0.073	0.199	3	1, 6, 9	
Cave sites only						
Marginal	Latitude		0.084		0.280	
	Longitude		0.119		0.030	*
	Acidity		0.083		0.295	
	Temperature		0.090		0.197	
	Salinity		0.100		0.057	
	Dissolved oxygen		0.065		0.776	
Sequential (step-wise)	**Longitude**	0.045	0.119	0.119	0.029	*
	Dissolved oxygen	0.074	0.098	0.217	0.053	*
	Salinity	0.108	0.097	0.314	0.063	
	Acidity	0.138	0.089	0.403	0.129	
	Temperature	0.176	0.090	0.493	0.127	
Best solution		**Adj R^2**	**R^2**	**No. Groups**	**Selections**	
		0.176	0.493	5	2-4, 6, 8	
Spring sites only						
Marginal	Latitude		0.165		0.113	
	Longitude		0.108		0.717	
	Acidity		0.133		0.377	
	Temperature		0.131		0.397	
	Salinity		0.138		0.317	
	Dissolved oxygen		0.155		0.076	
Sequential (step-wise)	Latitude	0.046	0.165	0.165	0.103	
	Dissolved oxygen	0.064	0.133	0.298	0.311	
Best solution		**Adj R^2**	**R^2**	**No. Groups**	**Selections**	
		0.064	0.298	2	1, 8	

Table S4.7 One-way ANOVA testing for the effect of site type (cave or spring) on site richness.

Significant *P* value codes are as follows: 0 < 0.001 '***', 0.001 < 0.01 '**', 0.01 < 0.05 '*'.

Source	d.f	Sum of squares	Mean Sq	F value	*P*
Site type	1	4.3	4.28	0.042	0.84
Residual	21	2155.5	102.65		

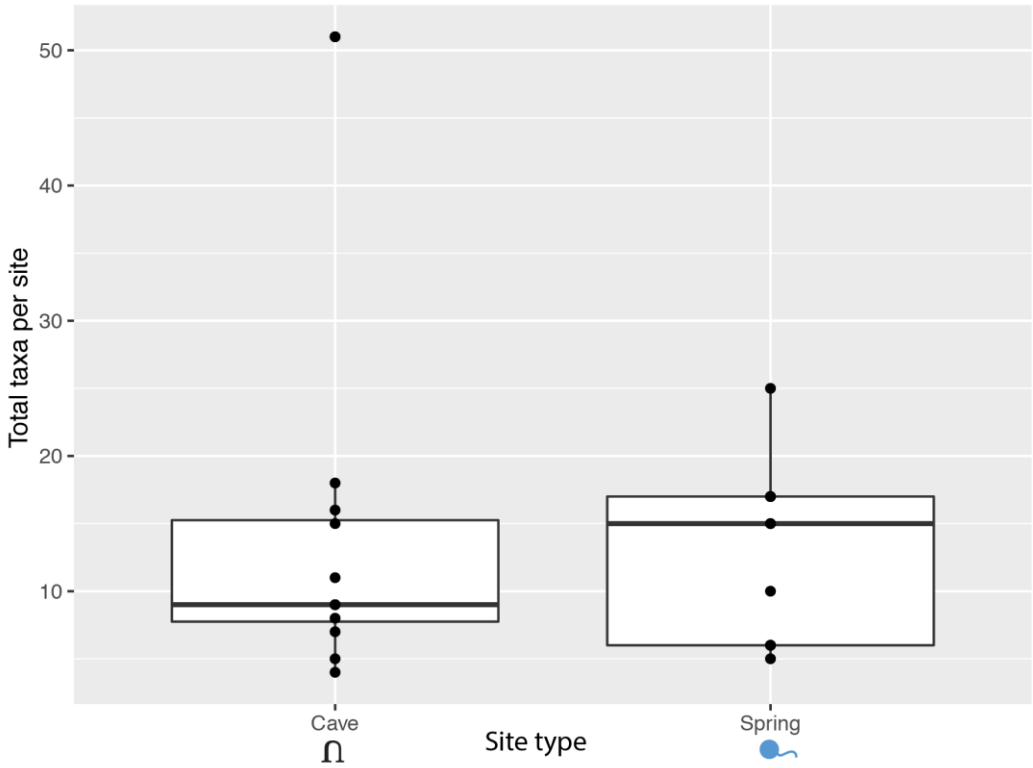


Figure S4.5 Taxa richness in subterranean fauna across the two site types.

Table S4.8 Similarity percentage analysis (SIMPER) of CI subterranean taxa. Pairwise dissimilarity between the site types (cave and spring sites).

Groups	Avg. Diss %	Taxon	Av. Abund in Caves	Av. Abund in Springs	Av. Diss	Diss/ SD	Contrib.%	Cum. %
Cave & Spring sites	73.89	Darwinulidae	0	0.89	3.75	1.87	5.08	5.08
		Discoplax magna	0.14	0.67	2.76	1.04	3.73	8.81
		Decapoda	0.07	0.56	2.41	0.93	3.27	12.07
		Sarcoptiformes	0.21	0.67	2.34	1.13	3.17	15.25
		Naididae	0.43	0.67	2.32	0.92	3.13	18.38
		Gecarcoidea lalandii	0.57	0.78	2.1	0.81	2.84	21.22
		Formicidae	0.14	0.44	1.77	0.86	2.4	23.62
		Geograpsus crinipes	0.57	1	1.74	0.72	2.35	25.97
		Gobio gobio	0.21	0.33	1.69	0.74	2.28	28.25
		Exocoetus	0	0.33	1.62	0.62	2.19	30.44
		Anoplolepis gracilipes	0.36	0.22	1.61	0.74	2.18	32.62
		Psocoptera	0.14	0.33	1.59	0.73	2.15	34.77
		Shelfordella	0.21	0.22	1.52	0.61	2.06	36.83
		Coptotermes gestroi	0.21	0.33	1.5	0.75	2.03	38.86
		Araneae	0.14	0.33	1.46	0.75	1.98	40.84
		Eleotris	0.14	0.33	1.43	0.73	1.93	42.77
		Entomobryomorpha	0.07	0.33	1.34	0.72	1.81	44.58
		Willowsia nigromaculata	0.29	0.11	1.24	0.62	1.67	46.25
		Amynthas	0	0.33	1.17	0.67	1.59	47.84
		Darwinula stevensoni	0	0.33	1.17	0.67	1.59	49.43
		Hemidactylus frenatus	0.07	0.22	1.15	0.56	1.56	50.99
		Haplosclerida	0.29	0	0.98	0.56	1.33	52.32
		Penaeus vannamei	0.21	0	0.97	0.48	1.31	53.63
		Cyprinidae	0.21	0	0.89	0.47	1.2	54.83
		Oxyporhamphus micropterus	0	0.22	0.88	0.49	1.2	56.02
		Craspedacusta sowerbii	0	0.22	0.86	0.51	1.16	57.19
		Limnodrilus	0	0.22	0.86	0.51	1.16	58.35
		Araneidae	0.07	0.22	0.83	0.55	1.12	59.47
		Turdus	0	0.22	0.82	0.52	1.12	60.59
		Haplotaxida	0	0.22	0.82	0.52	1.12	61.7
		Lepidopsocidae	0.07	0.11	0.75	0.41	1.02	62.72
		Harpacticoida	0.07	0.11	0.71	0.43	0.96	63.68
		Aphis	0.07	0.11	0.69	0.42	0.94	64.62
		Iophon	0.21	0	0.69	0.46	0.94	65.56
		Demospongiae	0.21	0	0.67	0.46	0.9	66.46
		Haliclona cinerea	0.14	0	0.63	0.39	0.85	67.31
		Lasioseius	0.14	0	0.62	0.38	0.84	68.15
		Polyxenida	0.07	0.11	0.62	0.43	0.84	68.99
		Anthozoa	0.21	0	0.58	0.48	0.79	69.78
		Alcyonacea	0.21	0	0.58	0.48	0.79	70.57

Table S4.9 Similarity percentage analysis (SIMPER) of CI subterranean taxa. Pairwise dissimilarity between the salinity groups (freshwater, oligohaline, mesohaline & polyhaline).

Groups	Avg. Diss %	Taxon	Av. Abund in Group 1	Av. Abund in Group 2	Av. Diss	Diss/ SD	Contrib.%	Cum. %
Mesohaline & Oligohaline	79.18	*Caranx ignobilis*	0.5	0	2.61	0.92	3.3	3.3
		Atractomorpha	0.5	0	2.61	0.92	3.3	6.6
		Glycyphagidae	0.5	0	2.61	0.92	3.3	9.9
		Stylommatophora	0.5	0	2.61	0.92	3.3	13.2
		Lasioseius	0.5	0.17	2.58	0.92	3.26	16.47
		Gecarcoidea lalandii	0.5	0.67	2.56	0.92	3.23	19.7
		Haplosclerida	0	0.5	2.56	0.94	3.23	22.93
		Iophon	0.5	0.17	2.39	0.93	3.02	25.95
		Artemia franciscana	0.5	0	2.36	0.93	2.98	28.93
Mesohaline & Freshwater	74.77	Naididae	0	0.64	3.38	1.15	4.52	4.52
		Discoplax magna	0	0.57	3.19	1.02	4.27	8.79
		Caranx ignobilis	0.5	0	3.06	0.9	4.09	12.89
		Lasioseius	0.5	0	3.06	0.9	4.09	16.98
		Atractomorpha	0.5	0	3.06	0.9	4.09	21.07
		Glycyphagidae	0.5	0	3.06	0.9	4.09	25.16
		Stylommatophora	0.5	0	3.06	0.9	4.09	29.26
		Gecarcoidea lalandii	0.5	0.64	2.92	0.9	3.9	33.16
		Decapoda	0.5	0.36	2.79	0.91	3.73	36.89
Oligohaline & Freshwater	75.46	*Discoplax magna*	0	0.57	2.54	1.02	3.37	3.37
		Naididae	0.33	0.64	2.47	1	3.27	6.64
		Haplosclerida	0.5	0	2.35	0.92	3.11	9.75
		Darwinulidae	0	0.57	2.16	1.06	2.86	12.61
		Gecarcoidea lalandii	0.67	0.64	2.04	0.83	2.71	15.32
		Sarcoptiformes	0.33	0.43	2.03	0.86	2.69	18.01
		Eleotris	0.33	0.21	1.83	0.74	2.43	20.43
		Geograpsus crinipes	0.67	0.79	1.8	0.75	2.38	22.81
		Anoplolepis gracilipes	0.33	0.29	1.79	0.8	2.37	25.18
Mesohaline & Polyhaline	93.28	*Geograpsus crinipes*	1	0	1.7	41.72	1.82	1.82
		Anoplolepis gracilipes	0	1	1.7	41.72	1.82	3.64
		Willowsia nigromaculata	0	1	1.7	41.72	1.82	5.45
		Coptotermes gestroi	0	1	1.7	41.72	1.82	7.27
		Shelfordella	0	1	1.7	41.72	1.82	9.09
		Melichthys niger	0	1	1.7	41.72	1.82	10.91
		Callyspongia	0	1	1.7	41.72	1.82	12.72
		Fredericella sultana	0	1	1.7	41.72	1.82	14.54
		Agelas schmidti	0	1	1.7	41.72	1.82	16.36
		Oscarella	0	1	1.7	41.72	1.82	18.18
Oligohaline & Polyhaline	82.41	*Melichthys niger*	0	1	1.57	15.66	1.91	1.91
		Fredericella sultana	0	1	1.57	15.66	1.91	3.81
		Agelas schmidti	0	1	1.57	15.66	1.91	5.72
		Oscarella	0	1	1.57	15.66	1.91	7.63
		Hebella contorta	0	1	1.57	15.66	1.91	9.54
		Dynamena crisioides	0	1	1.57	15.66	1.91	11.44
		Scleractinia	0	1	1.57	15.66	1.91	13.35
		Nerillidae	0	1	1.57	15.66	1.91	15.26
		Billardia subrufa	0	1	1.57	15.66	1.91	17.17

Freshwater & Polyhaline	87.89	*Melichthys niger*	0	1	1.62	10.15	1.85	1.85
		Callyspongia	0	1	1.62	10.15	1.85	3.69
		Iophon	0	1	1.62	10.15	1.85	5.54
		Fredericella sultana	0	1	1.62	10.15	1.85	7.38
		Agelas schmidti	0	1	1.62	10.15	1.85	9.23
		Oscarella	0	1	1.62	10.15	1.85	11.08
		Hebella contorta	0	1	1.62	10.15	1.85	12.92
		Haplosclerida	0	1	1.62	10.15	1.85	14.77
		Micronerilla minuta	0	1	1.62	10.15	1.85	16.62

S4.9.3 Supplementary references

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Benson, D. A., Karsch-Mizrachi, I., Lipman, D. J., Ostell, J., & Wheeler, D. L. (2005). GenBank. *Nucleic Acids Research*, (*33*), D34–D38. https://doi.org/10.1093/nar/gki063

Boyer, F., Mercier, C., Bonin, A., Taberlet, P., & Coissac, E. (2014). OBITools: a Unix-inspired software package for DNA metabarcoding. *Molecular Ecology Resources, 16*(1), 176–182.

Clarke, K. R., & Gorley, R. N. (2015). *PRIMER v7: User Manual/Tutorial*. Plymouth, UK.

Horton, T., Kroh, A., Ahyong, S., Bailly, N., Boyko, C. B., Brandão, S. N., … Zhao, Z. (2018). World Register of Marine Species (WoRMS). WoRMS Editorial Board. Retrieved from http://www.marinespecies.org

Hui, T. H., Naruse, T., Fujita, Y., & Kiat, T. S. (2014). Observations on the fauna from submarine and associated anchialine caves in Christmas Island, Indian Ocean Territory, Australia. *Raffles Bulletin of Zoology Supplement, 30,* 406-418.

Humphreys, W. F. (2014). Subterranean fauna of Christmas Island: habitats and salient features. *Raffles Bulletin of Zoology Supplement, 30*, 29-44.

McMurdie, P. J., & Holmes, S. (2013). Phyloseq: an R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS One, 8*(4), e61217.

R Studio Team (2015). RStudio: integrated development for R. *RStudio Inc*, Boston, MA. Retrieved from *http://www.rstudio.com*

Wilkinson, S. P., Davy, S. K., Bunce, M., & Stat, M. (2018). Taxonomic identification of environmental DNA with informatic sequence classification trees. *PeerJ Preprints*. DOI: 10.7287/peerj.preprints.26812v1.

Chapter 5

Discussion







5.1 Summary of findings

This thesis research has applied eDNA metabarcoding technologies to a number of aquatic environments in Australia's Indian Ocean region. The outputs provide greater insight into the applicability of eDNA metabarcoding as an aquatic biomonitoring tool for species discovery, monitoring and sustainable management. This final discussion chapter draws together the main concepts and findings from each of the data chapters, and addresses the limitations and the significance of this thesis for applied eDNA metabarcoding. Finally, I discuss future research directions that have arisen as the direct result of thesis outputs, as well as speculate on future progressions of eDNA metabarcoding.

This thesis demonstrated the utility of multi-marker eDNA metabarcoding across a variety of aquatic ecosystems (i.e. marine, coastal, estuarine and subterranean) and spatial scales within Australia's Indian Ocean region. This resulted in the assignment of approximately 1000 taxa covering a wide-range of taxonomic groups, from bony fish, elasmobranchs, arthropods, molluscs, echinoderms, aquatic reptiles, cnidarians, porifera, annelids and bryozoans. It is my hope that the data that I have collected can be used to establish baselines for future eDNA biomonitoring in this region. Most importantly, this thesis adds to a growing body of literature that demonstrates the utility of eDNA as a tool to study aquatic biodiversity and potentially collect data across a whole-ecosystem. Several main findings from this thesis are summarised below:

#1. eDNA metabarcoding is a highly-sensitive detection tool that is complementary to conventional biomonitoring. A multi-marker metabarcoding approach detected a high number of aquatic taxa, including putative new occurrence records, in each of the three chapters. In chapter two, a new eDNA baseline of 372 bony fish, elasmobranch, crustacean, mollusc and echinoderm taxa was established at the Cocos (Keeling) Islands, with 46 putative new occurrence records. This single eDNA survey captured 37.3% of the total fish fauna previously documented at this locality. The yield and efficiency of this technique was comparable to recent BRUV surveys at the Cocos (Keeling) Islands, however was not as effective as invasive techniques (e.g. rotenone, explosives). Interestingly, there was no overlap in new

occurrence records with the BRUV surveys, indicating that the combined use of eDNA metabarcoding and BRUVs may be able to provide complementary baseline data. This has previously been reported by Stat et al., (2019), where the combination of eDNA and BRUVs yielded a 30% increase in generic richness than either method used alone.

The eDNA yield of 404 bony fish in chapter three was on par with a previous trawling survey of inshore fish fauna across the same region (Travers et al., 2010). Both of these approaches captured nearly a third of the total fish assemblage of north-western Australia. The eDNA yield in chapter four was not directly compared to conventional biospeleological approaches, given a lack of previous surveying across Christmas Island's aquatic subterranean ecosystems. Nevertheless, the Christmas Island subterranean survey set a new eDNA baseline of 115 taxa from a wide variety of eukaryotic groups, including 21 putative new occurrence records. All three chapters produced significant yields from a single biological survey within their respective ecosystems, demonstrating the sensitivity of applied eDNA metabarcoding. Additionally, this thesis adds to a growing body of literature advocating for the integrated use, rather than alternative application, of eDNA metabarcoding with conventional survey approaches (Kelly et al., 2017; Port et al., 2016; Stat et al., 2019; von Ammon et al., 2018).

#2. eDNA metabarcoding can reveal fine-scale habitat partitioning, indicative of eDNA localisation on a small spatial scale. In chapter two, the marine community composition varied significantly over a small spatial scale (140 km^2), reflecting habitat partitioning across the Cocos (Keeling) Islands. This was amidst extensive tidal and oceanic movements in and around the main atoll. Some of the sites sampled in adjacent habitats were within a 1km distance, indicating that eDNA signals from seawater are localised within a small area. Preliminary studies have reported varying eDNA decay rates in seawater (Dell'Anno & Corinaldesi, 2004; Thomsen, Kielgast, Iversen, Møller, et al., 2012), which is expected given variability in environmental abiotic and biotic factors which degrade DNA (Barnes et al., 2014). The lowest current estimate of eDNA decay in seawater in the field, is 10 hours in the Adriatic Sea (Dell'Anno & Corinaldesi, 2004). Given extreme tidal and oceanic movements however, it is possible that eDNA would have moved through the Cocos atoll, confounding site profiles, before it degraded beyond the detection threshold. The detection of discrete

community assemblages reflecting habitat partitioning over this small spatial scale, indicates however, that the eDNA shed at a given site did not confound site profiles in another habitat. This supports the hypothesis that eDNA localisation is largely influenced by diffusion (and dilution) in open water environments and likely dissipates beyond the detection threshold before eDNA decay naturally occurs (Stat et al., 2019; Yamamoto et al., 2016).

#3. eDNA metabarcoding can reveal fine-scale patterns over large-scale regions. It follows that if eDNA can statistically separate habitats over small spatial scales then this pattern should hold true over much broader spatial scales. Despite this, relatively little research has explored eDNA over a wide geographical area (see Fraija-Fernández et al., 2019). In chapter three, a broad-scale eDNA metabarcoding survey across approximately 700 km of north-western Australian coastline, was able to discern a previously reported marine biogeographic break, demarcating the border between the Kimberley and Canning bioregions. (Travers, Potter, Clarke, Newman, & Hutchins, 2010; Wilson, 2014). In addition, it also detected previously unrecognised subregional differentiation for bony fish and shark/ray community compositions across the Kimberley region. This chapter demonstrated the applicability of eDNA metabarcoding not only for detecting diversity and distributions of marine taxa across large-scale regions, but also for testing biogeographic hypotheses. A preliminary cost-efficiency analysis indicates that eDNA metabarcoding becomes more cost-efficient over conventional methods when scaled up in regions of high biodiversity (Bálint et al., 2018). Taken together, this preliminary work indicates that eDNA metabarcoding may be particularly well-suited for broad-scale aquatic surveying.

#4. eDNA metabarcoding can detect endangered and/or elusive aquatic taxa. eDNA detected a number of endangered and/or elusive taxa across all three chapters. In chapter two, eDNA metabarcoding detected a range of deep-sea lanternfish and dragonfish from surface seawater samples, illustrating the potential of this technique to identify cryptic biodiversity, particularly of taxonomic groups that inhabit pelagic zones and are largely inaccessible with other monitoring techniques. In chapter three, three globally endangered sawfish taxa (family: Pristidae), wedgefish (genus: *Rhynchobatus*) and northern river shark (*Glyphis garricki*) were detected with eDNA metabarcoding. The detection of numerous sawfish species was particularly exciting,

given that visual observations of this benthic taxa is limited, due to extreme turbidity in the waterways of northern Australia. Sawfish have been detected using eDNA previously, albeit with the development and application of species-specific primers (Simpfendorfer et al., 2016). The simultaneous detection of these endangered elasmobranchs demonstrates the efficiency of applying a metabarcoding assay, rather than developing species-specific assays. However, a positive qPCR detection with a metabarcoding assay would require further sequencing in order to gain taxonomic identifications. The data collected across north-western Australia in chapter three, detected three potential invasive taxa (Cyprinidae, *Cottus* and *Gobio gobio*), that if physically validated, pose significant biosecurity threats to this region. Two of these taxa are flagged on the Western Australian Prevention List for Introduced Marine Pests (2016). Their detection demonstrates the ability of eDNA metabarcoding to detect both elusive native and invasive species from the same samples and dataset. Lastly, the subterranean survey in chapter four, produced detection hits for Indonesian shortfin eel (*Anguilla bicolor bicolor*), a highly elusive species that is rarely seen across Christmas Island and only previously reported in two locations (Hui, Naruse, Fujita, & Kiat, 2014). This thesis adds to a growing body of literature demonstrating the utility of eDNA metabarcoding, in addition to species-specific approaches, in identifying endangered and elusive taxa (Day et al., 2019; Piaggio et al., 2014; Thomsen, Kielgast, Iversen, Wiuf, et al., 2012; Weltz et al., 2017).

#5. eDNA metabarcoding may not reliably inform aquatic reptile distributions. Chapter three described the design and optimisation of a metabarcoding assay that simultaneously targets aquatic and semi-aquatic reptile groups in northern Australia. Only five reptile taxa were detected from 13 out of 71 sites, despite visually observing a higher occurrence of reptiles, particularly of saltwater crocodiles (*Crocodylus porosus*), across this vast region. The under-detection of reptiles in this study may be attributed to the non-specificity of the designed assay and/or a low shedding rate of reptiles in the environment. Aquatic reptiles have been inconsistently amplified in species-specific and metabarcoding studies (Baker et al., 2018; Davy, Kidd, & Wilson, 2015; De Souza, Godwin, Renshaw, & Larson, 2016; Feist et al., 2018; Halstead et al., 2017; Lacoursière-Roussel, Dubois, Normandeau, & Bernatchez, 2016). This may reflect an emerging challenge resulting from lower shedding rates and/or reduced waste production in reptiles, which would undermine our ability to detect them via

eDNA methods (Adams, Hoekstra, Muell, & Janzen, 2019). At present, eDNA approaches cannot reliably inform aquatic reptile detections and community compositions (see section 5.4.1).

#6. eDNA metabarcoding can produce high-resolution taxonomic assignments in poorly-surveyed ecosystems, such as subterranean aquatic ecosystems. Subterranean aquatic ecosystems are understudied and subsequently present suboptimal reference databases. This may hinder the ability of eDNA metabarcoding to provide high-resolution assignments and compositional data. In chapter four, I applied eDNA metabarcoding to water samples collected from 23 cave and spring sites across Christmas Island. This produced 115 taxonomic assignments, the majority of which were resolved to species, followed by order level. This study has demonstrated that despite suboptimal databases, eDNA metabarcoding can still reveal a wide diversity of taxon detections in poorly-surveyed ecosystems. There remains a need for further subterranean surveying and barcoding initiatives to facilitate eDNA metabarcoding. However, a benefit of applying it early is that it can be used to direct traditional sampling efforts to localities with a high proportion of unresolved (e.g. order-level) assignments, for specimen acquisition and taxonomic classification. Whilst eDNA metabarcoding remains reliant upon the acquisition of specimens for barcoding purposes, it does provide an alternative to capture-based biospeleological surveying in logistically difficult to access locations.

#7. eDNA-derived compositional data can be used in conjunction with environmental data to identify potential underground interconnectivity. In chapter four, hierarchical clustering of eDNA-derived biotic and environmental abiotic data identified three cave and spring groups across Christmas Island with a high possibility of interconnection. This validated a previously hypothesised hydrological connection between cave systems. Alleged subterranean interconnectivity is often based on proximity and the sharing of select macrofauna and/or populations between cave systems (Davie & Ng, 2012; Mascareñas, de Leon, & Delos Reyes, 2019). The combination of multi-trophic community composition data, as provided by eDNA metabarcoding, with environmental (in this case, water quality) data, allows for a more objective analysis of interconnectivity. However, potential interconnectivity will require further testing to validate.

5.2 Limitations

This thesis demonstrates the potential for eDNA metabarcoding to complement conventional aquatic surveying approaches, yet several limitations to the eDNA metabarcoding method remain. Firstly, PCR bias remains a problem in the DNA amplification process, whereby DNA from a species that presents perfect homology (or no mismatches) to the primers, is preferentially amplified over other species that present mismatches (Schloss, Gevers, & Westcott, 2011). Hence, DNA sequences with a lower homology to the primers amplify at a lower efficiency which subsequently skews PCR efficiency and ultimately impacts the downstream read abundance data. It has been recommended that there should be no more than two mismatches between a primer and targeted taxa (Taberlet, Bonin, Zinger, & Coissac, 2018). Additionally, there should be minimal mismatches towards the 3' end of a primer to ensure annealing and elongation can proceed (Dieffenbach, Lowe, & Dveksler, 1993). The effect of PCR bias is hard to identify and reduce bioinformatically, as opposed to other artifacts such as sequencing error. Therefore, it is important to design primers in highly conserved areas for the target taxonomic group. This becomes more difficult when designing primers for broad or multiple taxonomic groups, given a degree of variability between highly developed lineages, even in evolutionary conserved regions (Schmalenberger, Schwieger, & Tebbe, 2001). In this thesis, I used degenerate bases in places with a high proportion of mismatches to widen PCR amplification across broad taxonomic groups. However, as seen in chapter two's supplementary info (SI Figure 2), this can facilitate an increase in non-specific amplification. An alternative approach to reducing PCR bias would have been to reduce annealing temperatures (see: Ishii & Fukui, 2001; Sipos et al., 2007), however, this can also increase the risk of non-specific amplification. PCR bias will likely remain a limitation of the eDNA metabarcoding process until the development and/or adoption of non PCR-based approaches, such as target (or capture) enrichment, which can use an array of synthesised baits to hybridise target taxa from eDNA (Dowle, Pochon, Banks, Shearer, & Wood, 2016; Gauthier et al., 2020; Wilcox et al., 2018) (see section 5.4.2 for new technologies).

As expected, the incompleteness of genetic reference databases, and also the length and variability within the chosen molecular markers, meant that a range of taxonomic

assignments were made across all three chapters that could not be resolved to a species-level. It is anticipated that as the extent and quality of reference databases expand, this limitation will lessen over time. In this regard, an additional benefit of eDNA metabarcoding is the ability to retrospectively analyse sequencing data, meaning that as reference databases develop in the future, data can be reanalyzed to resolve previous taxonomic assignments. In order to ensure that incomplete reference databases did not greatly affect my taxonomic assignments at this point in time, I only reported putative new occurrence records if all congeneric species (of the putative taxa in question) had been barcoded for the molecular marker in use. However, this did not extend to all taxonomic assignments, if they had been detected in that location before. Ideally, taxonomic assignments would be collapsed to a lowest common ancestor if all congeneric species had not been barcoded, which rules out the possibility of incorrectly assigning to a closely-related species. However, at present, that strict approach would be severely limiting if applied, as there are a high number of faunal groups and ecosystem types (e.g. crustaceans and subterranean ecosystems) that are data-limited and require significant future investment in systematic and barcoding research.

A third limitation of this thesis is related to the need for short barcoding regions and subsequent detection accuracy. The degradation and subsequent fragmentation of eDNA in the environment (e.g. from physical degradation, exogenous enzymes and/or chemical reactions) dictates the use of short barcoding regions (usually under 500bp) (Taberlet et al., 2018). At present, this constrains the use of larger barcoding regions, or complete mitochondrial genomes, to distinguish closely-related species. It also limits the use of eDNA-derived haplotypes (such as in chapter three), which with increasingly length and subsequent resolution, could be used to examine population genetics. However, further developments, such as single-molecule sequencing which isolate intact cells for sequencing, could circumvent the use of fragmented eDNA (Liu et al., 2012) (see section 5.4.2).

Lastly, a significant limitation for eDNA metabarcoding is in its inability to reliably provide quantitative estimates of population size and/or biomass. At present, eDNA (both species-specific and metabarcoding) can only reliably provide presence/absence data (Lamb et al., 2019), which was solely used in this thesis. PCR-bias, as discussed

above, skews downstream read abundance data and therefore can only be circumvented if all target taxa present the same homology to the primers. Relative sequencing depth could then be used as a proxy for population size and/or biomass. However, another issue arises from variation in shedding rates between different species (Sassoubre, Yamahara, Gardner, Block, & Boehm, 2016). Quantification research has therefore largely focused on individual species and utilised specific-specific primers (De Ventura, Kopp, Seppälä, & Jokela, 2017; Klymus, Richter, Chapman, & Paukert, 2015; Lacoursière-Roussel, Côté, Leclerc, & Bernatchez, 2016; Sassoubre et al., 2016). Droplet digital PCR (ddPCR) has been increasingly utilised for absolute quantification of eDNA (Baker, Steel, Nieukirk, & Klinck, 2018; Doi et al., 2015; Wood et al., 2019). It does this by partitioning template DNA molecules into approximately 20,000 water-oil droplets, each ideally containing either zero or one copy of DNA (Hindson et al., 2011). Each droplet essentially functions as a PCR well or tube, and is a medium for where the PCR takes place, albeit in a smaller format. Thousands of individual amplification events can be then measured from a single sample and it subsequently provides an absolute count of target DNA copies per input sample/per microlitre (Hindson et al., 2011). However, given that it relies upon PCR amplification to measure abundance, it can only effectively be applied for quantifying an individual species with a species-specific assay. The simultaneous amplification of multiple taxa with a metabarcoding assay, would lead to the quantification of all amplified taxa within a single sample. Therefore, alternative, PCR-free approaches are ideal for multi-species quantification estimates (see section 5.4.2).

5.3 Significance of thesis

The research conducted in this thesis provides an insight into the use of aquatic eDNA metabarcoding in Australia's Indian Ocean region and addresses a number of applied questions for further implementation in these ecosystems. My findings in regards to the localisation of eDNA in seawater (chapter two) validates previous coastal research (Jeunen et al., 2018; Koziol et al., 2018; Murakami et al., 2019; Stat et al., 2019) and provides further evidence to substantiate that marine eDNA profiles reliably represent their immediate surroundings. The ability of eDNA metabarcoding to detect fine-scale biogeographic patterns across a large-scale region further validates its use as a broad-scale biomonitoring tool. This thesis also concurs with another study (Stat et al., 2019) that eDNA metabarcoding and BRUV technologies provide complementary datasets, warranting a combined approach for future biomonitoring initiatives.

The main findings of this thesis also have direct implications for management and conservation of endangered and elusive taxa, including potential biosecurity threats to Australia's Indian Ocean region. All three chapters set new eDNA baselines in their respective study regions. Notably, chapter two is the first study to conduct a multi-trophic eDNA metabarcoding survey across an entire coral reef atoll. Chapter three, at present, provides one of the largest (on a geographical scale) continuous eDNA metabarcoding surveys that has been globally conducted. The success of eDNA metabarcoding to detect fine-scale patterns across a large-scale region validates its use as a broad-scale biomonitoring tool. The detection of multiple endangered sawfish species and the northern river shark in this chapter is particularly significant, and is available to guide management efforts, e.g. catch and tag surveys, which provide estimates of population numbers, age and growth profiles (Kyne & Pillans, 2014; Lear et al., 2019). It is anticipated that eDNA metabarcoding will be more widely applied for aquatic surveying in northern Australia, given that it can circumvent logistical issues (including turbidity and diver safety) associated with conventional surveying across this vast, remote region. Its use for aquatic reptiles however, is at present, not reliable for monitoring applications.

Lastly, chapter four is the first eukaryotic eDNA metabarcoding survey conducted in cave systems. It extended the distribution for elusive cave fauna across Christmas

Island, detected a number of putative new occurrence records and is the first to use multi-trophic eDNA data to investigate potential underground interconnectivity. Christmas Island's national parks and by extension, their cave systems, are managed by Parks Australia. The only biospeleological surveys conducted in these cave systems, however, are by research academics periodically. New eDNA baselines providing multi-tropic community composition data for 12 cave systems will aid subterranean management across Christmas Island. This survey will provide a benchmark and demonstrate the applicability of eDNA metabarcoding for further biospeleological surveys in caves, and subterranean ecosystems more broadly.

5.4 Future directions

5.4.1 New aims and questions arising from this thesis

With a developing technology like eDNA metabarcoding, there remains a vast number of potential additional experiments, validations and optimisations that could be implemented. Whilst not a complete list of possible directions, the list below represents what I consider to be some of the logical extensions to this thesis work and/or some of the additional technological improvements.

Future direction #1. Further optimisation of metabarcoding assays is required for target species detections. The COI Elasmobranch assay (applied in chapters two and three), the 16S Crustacean assay (applied in chapters two and four), and the 16S Reptile assay (applied in chapter three) all had strengths but need further refinement. All three assays amplified a high proportion of reads that were classified as "no hit" (i.e. reads for which no sequence match was found during the BLAST analysis) or amplified taxa outside of the target taxonomic groups. The assays could therefore be fine-tuned to increase specificity to the target taxonomic groups; a byproduct of which, would be an increase in the amplification of low-abundant target taxa, as there is reduced non-target template competition (Wilcox et al., 2013).

Future direction #2. Most metabarcoding studies lament the lack of complete reference databases—this thesis is no exception. As previously discussed in the limitations section, there is a need for continual expansion of genetic reference databases. Whilst chapter four demonstrated that high-resolution data (i.e. to a species level) can be achieved despite a suboptimal database, there were still many taxonomic assignments that could not be resolved beyond an order-level. This was apparent not only for the subterranean aquatic taxa, but for many marine crustaceans in chapter two which could not be resolved to a species-level. This may be attributed to low-resolution in the 16S region for crustaceans, however is likely to be influenced by the need for further surveying and subsequent systematics/barcoding of decapod and stomatopod crustaceans in north-western Australia (Titelius, Sampey, & Hass, 2009). Additionally, putative new occurrence records in all three chapters require further

validation through the acquisition of specimens. This is particularly crucial for the potential invasive species detections in the Kimberley region.

Future direction #3. The broad-scale application of eDNA metabarcoding in chapter three identified a subregional influence on coastal fish and inshore shark/ray compositions across the Kimberley. This has not yet been detected with conventional techniques and may warrant further surveying in this remote region. The ability of eDNA metabarcoding to detect fine-scale patterns across large-scale regions opens up new doors to potentially revealing marine biogeographic breaks in other regions e.g. to investigate previous reports of phylogeographic structure between the Indian and Pacific Ocean regions in southeast Asia (Barber, Palumbi, Erdmann, & Moosa, 2000; Lourie & Vincent, 2004). Given broad-scale metabarcoding successfully detected a suspected biogeographic break, it is anticipated that this research will instigate further broad-scale applications across catchments and regions. However, a cost-benefit analysis is needed to further explore the efficacy of scaling up eDNA metabarcoding in comparison to the use of conventional aquatic surveying methods. One study already suggests eDNA metabarcoding is more cost-efficient when scaled up in regions of high biodiversity (Bálint et al., 2018), however this needs to be further examined before eDNA metabarcoding is rolled out and scaled-up by management authorities, across large-scale regions.

Future direction #4. Poor detection frequencies of aquatic reptile taxa in chapter three may be attributed to both the non-specificity of the assay and/or a low shedding rate of reptiles in the environment. It is recommended that discrete assays for each major reptilian order are developed, given the difficulty in locating conserved regions across these highly developed lineages that are not also conserved in other taxonomic groups. Inconsistent amplification of aquatic reptiles across multiple studies however, suggests that under detection is widespread and may not necessarily be attributed to non-specificity in this thesis. This gives credence to the hypothesis of lower shedding rates in reptiles compared to other mucus-covered organisms, such as fish and amphibians (Adams et al., 2019; Raemy & Ursenbacher, 2018). Lower shedding rates may also extend to crustaceans, with preliminary research reporting inconsistent amplification (Forsström & Vasemägi, 2016; Tréguier et al., 2014). Lower shedding rates in both

reptiles and crustaceans however, has not yet been explicitly tested and therefore requires further experimental research.

Future direction #5. Lastly, the novel application of eDNA metabarcoding in cave systems presents an array of questions that have already been addressed in marine, freshwater and terrestrial applications. For example, is substrate selection critical in detecting various groups of subterranean fauna, e.g. the sampling of cave water for aquatic "stygofauna", and soil and guano for air-breathing "troglofauna"? What level of replication and volume of samples is required in subterranean systems, particularly as caves present variable energy dynamics and subsequent species diversity, in the absence of phototrophic production (Ortiz et al., 2014)? Does eDNA persist longer in cave substrates, given that cave conditions are already conducive to ancient DNA preservation (Gamba et al., 2008; Grealy et al., 2016)? It is anticipated that eDNA metabarcoding will be increasingly used in subterranean aquatic ecosystems, particularly given logistical difficulties in conventional surveying, and so will require further methodological and applied research.

5.4.2 New technologies for eDNA metabarcoding

Over the duration of this thesis, the fields of eDNA and third generation sequencing have progressed rapidly. Here, I propose some future directions in the technologies that underpin eDNA approaches.

Future direction #6. Target (capture) enrichment has been extensively used in human clinical, ecological and ancient DNA research to capture, sequence and reconstruct contiguous DNA fragments (Ávila-Arcos et al., 2011; Horn, 2012; McCormack et al., 2013). The process requires the development of baits (synthesised biotinylated oligonucleotides that are complementary to the targeted gene region/s) which can then be used to hybridise and capture fragmented genomic DNA of the target specimen (Maricic, Whitten, & Pääbo, 2010). Captured DNA is then sequenced and assembled *de novo* or mapped to a reference genome, bypassing the need for primers or intact DNA. Target enrichment has largely been applied to individual specimens (particularly those that exhibit DNA fragmentation), to reconstruct genomes and/or phylogenetic histories of modern and ancient species. Recently, target capture from

individual faeces has been used as a non-invasive approach to study the population genetics of endangered great ape species in central Africa (van der Valk, Lona Durazo, Dalén, & Guschanski, 2017). Its use in eDNA metabarcoding is an emerging alternative to PCR-based approaches (Dowle et al., 2016; Gauthier et al., 2020; Shokralla et al., 2016; Wilcox et al., 2018). What makes the application of target enrichment to complex environmental samples more challenging than individual specimens, is that an eDNA sample contains a myriad of species. This subsequently requires the development of thousands of baits and *a priori* knowledge of the types of taxa present in an environmental sample. Target enrichment may therefore be less cost-efficient, relative to a PCR-based approach. However, the preliminary application of target enrichment in eDNA metabarcoding has provided a higher and more reliable detection rate compared to a PCR-based approach (Dowle et al., 2016; Gauthier et al., 2020; Shokralla et al., 2016; Wilcox et al., 2018). Importantly, it provides an avenue to eDNA quantification, given that target enrichment circumvents PCR-bias. Post-capture relative sequence abundance reflects initial template concentrations and biomass (Gauthier et al., 2020; Wilcox et al., 2018) and therefore could be used as a proxy for population size. Target enrichment could additionally be applied to capture larger or overlapping fragments of eDNA from a targeted taxonomic group, to potentially be used in a population genetic approach.

Future direction #7. The development of third-generation sequencing technologies (TGS) allows for the direct sequencing of DNA molecules, bypassing the need for PCR and associated amplification errors and bias. Two TGS sequencing approaches are currently available: single molecule real-time (SMRT) sequencing developed by Pacific Biosciences (Eid et al., 2009) and nanopore sequencing (Branton et al., 2010), now commercialised through Oxford Nanopore Technologies.

SMRT sequencing occurs within a SMRT cell, which contains up to 1 million observation structures, referred to as Zero Mode Waveguides (ZMWs). Polymerase bound DNA molecules are then loaded across the ZMSs. A sequencing reaction occurs in each ZMW, whereby the polymerase incorporates fluorescently labelled nucleotides to the complementary DNA strand. In incorporating each successive nucleotide, it also cleaves the attached fluorescent tag, emitting a fluorescent signal which is then recorded by a camera in real-time observation. These signals are then sequentially

converted to sequencing reads (Eid et al., 2009). Library preparation for SMRT sequencing is fast (approximately 4-6 hours), given that it doesn't require any PCR amplification. The sequencing run itself is also fast (within one day) and produces an average read length of 1300 bp, which is longer than that produced by next-generation sequencing (Liu et al., 2012).

Nanopore sequencing occurs by embedding a α-haemolysin (αHL) nanopore into a lipid membrane that facilitates ion movement (Branton et al., 2010; Deamer, Akeson, & Branton, 2016). An electrical current is then continuously applied across the membrane and subsequently through the nanopore. As an individual single-stranded DNA is drawn through the tiny (~1.4 nm) nanopore channel, it creates short-term base-specific current fluctuations which are sequentially converted into a DNA sequencing read in real-time (Branton et al., 2010; Deamer et al., 2016). Nanopore sequencing is not constricted by read length, as they read the length of DNA presented. Oxford Nanopore Technologies developed the handheld MinION device in 2012 (Deamer et al., 2016). The portability of this nanopore sequencer allows for high-throughput long-read sequencing out in the field. With recent developments in rapid, short-read DNA library preparation and sequencing for MinIONs (Wei, Weiss, & Williams, 2018), this sequencing platform holds great promise for rapid eDNA applications in remote locations (Sigsgaard et al., 2019). Recently, a MinION device was used to rapidly identify great white shark (*Carcharodon carcharias*) from seawater samples, whilst onboard an oceanographic vessel (Truelove, Andruszkiewicz, & Block, 2019).

Whilst SMRT and nanopore sequencing of eDNA is still in a preliminary stage (Brown, Watson, Minot, Rivera, & Franklin, 2017; Truelove et al., 2019), they could potentially be used to target longer eDNA fragments. This would likely provide greater resolution within eDNA-derived haplotype data, which could subsequently be used for population genetic analyses. An effective population size derived from eDNA haplotype data could provide a more reliable quantitative estimate than relative sequencing depth and/or number of copies per sample (Sigsgaard et al., 2019). Nevertheless, these direct sequencing approaches may provide a more robust avenue to quantification through non-biased relative sequencing abundance. However, these approaches, including target enrichment, do not account for variation in shedding rates, and so quantification estimates derived from eDNA must be cautiously

presented. Additionally, SMRT and nanopore sequencing are (at present) significantly more costly than next-generation sequencing.

Future direction #8. The use of fluorescence-activated cell sorting (FACS) to differentiate cells within complex environmental samples. FACS is a specialised form of flow cytometry that is used to isolate variable cells within a heterogeneous mixture, even if they occur in low abundance (Cormack, Valdivia, & Falkow, 1996). Cells can be sorted based on cell size and surface protein expression; the latter through the use of fluorescence. Selected fluorescently-tagged antibodies are applied to the heterogenous cell mixture, staining cells of interest by binding to targeted cell surface molecules that exist only in the desired cell population. FACs applies a laser beam and fluorescence detection system to a cell stream to identify and sort between cells that meet the desired criteria (Basu, Campbell, Dittel, & Ray, 2010). This approach could be used to purify complex environmental samples and increase targeted PCR, target enrichment and/or TGS sequencing efficiency. For example, FACs can be used to size-select or fluorescently tag bacterial cells, through the use of labelled DNA probes (Batani, Bayer, Böge, Hentschel, & Thomas, 2019). Chlorophyll fluorescence in photosynthetic cells can also be used to identify and sort algae, plant and cyanobacterial cells (Best et al., 2016). The possibility of distinguishing and retaining cells of target taxonomic groups will fundamentally change eDNA sample composition. This may remove inhibitors, reduce PCR biases and facilitate the detection of low-abundant taxa.

Future direction #9. Lastly, the analogous development of environmental RNA (eRNA) metabarcoding is a rapidly expanding area of research that can differentiate 'living' vs 'legacy' signals (Pochon, Zaiko, Fletcher, Laroche & Wood 2017). This field has largely grown in response to biosecurity concerns where it is necessary to know if an eDNA detection reflects a living or deceased organism. Single-stranded ribonucleic acid (RNA) degrades relatively quickly after cell death, with detections up to 13 hours after cellular deposition (Wood et al. 2020). As such, an eRNA detection may indicate a more recent cellular deposition than a persisting eDNA signal. Environmental RNA signals are detected via a slightly modified eDNA protocol. Trace DNA in an extract is eliminated using DNase treatments, with the remaining RNA then reverse transcribed into complementary DNA (cDNA; Laroche et al. 2016, 2017,

2018; Pochon et al. 2017; Wood et al. 2020; von Ammon et al. 2019). PCR primers are then applied to the cDNA extracts to amplify targeted gene regions and taxonomic groups of interest. Environmental RNA approaches have been applied to identify living organisms (of biosecurity concern) in bilge water samples (Pochon et al. 2017), assess foraminiferal communities at impacted sites adjacent to offshore oil and gas platforms (Laroche et al. 2016, 2017, 2018), and identify invasive Mediterranean fanworm (*Sabella spallanzanii*) from settlement plates and water samples in New Zealand (von Ammon et al. 2019).

5.5 Thesis conclusion

The emergence of eDNA metabarcoding, facilitated by recent advances in sequencing technologies, offers a sensitive, high-resolution and importantly, a non-invasive approach, to aquatic biomonitoring. This technology provides a genetic avenue to broad taxonomic detections, and ultimately whole-ecosystem surveying. The integration of this technique across Australia's Indian Ocean region would be particularly beneficial, as the region presents high-biodiversity and economic importance across a large geographical scale. The overarching question of this thesis was: "Is eDNA metabarcoding a viable aquatic biomonitoring tool for use in Australia's Indian Ocean ecosystems?". This thesis research demonstrates the viability of eDNA metabarcoding as a biomonitoring tool in this region. It shows the degree of eDNA localisation across small-spatial scales, its ability to provide complementary baseline data with a routinely applied visual surveillance approach, and its applicability to be scaled up, revealing marine biogeographic breaks across large-scale regions. eDNA metabarcoding detected a range of endangered and elusive taxa, including potential invasive species, in each of the chapters of this thesis. It also highlights the need for further research into reptilian DNA shedding, before eDNA approaches can be utilised for this taxonomic group. Lastly, this thesis presents a novel application of eDNA metabarcoding into cave ecosystems and the ability to use eDNA-derived compositional data to explore underground interconnectivity. The results of this thesis support the integration of applied eDNA metabarcoding into institutional, environmental consultancy and government monitoring and management approaches in Australia's Indian Ocean region. The ability of eDNA metabarcoding to provide multi-trophic distribution data may be key to facilitating spatial cost-benefit analyses to ultimately better manage key conservation areas and populations in Australia's Indian Ocean region, and more broadly, across the globe.

5.6 References

Every reasonable effort has been made to acknowledge the owners of copyright material. I would be pleased to hear from any copyright owner who has been omitted or incorrectly acknowledged.

Adams, C. I. M., Hoekstra, L. A., Muell, M. R., & Janzen, F. J. (2019). A brief review of non-avian reptile environmental DNA (eDNA), with a case study of Painted Turtle (*Chrysemys picta*) eDNA under field conditions. *Diversity*, *11*(4), 50.

Ávila-Arcos, M. C., Cappellini, E., Romero-Navarro, J. A., Wales, N., Moreno-Mayar, J. V., Rasmussen, M., … Willerslev, E. (2011). Application and comparison of large-scale solution-based DNA capture-enrichment methods on ancient DNA. *Scientific Reports*, *1*(1), 1–5.

Baker, C. S., Steel, D., Nieukirk, S., & Klinck, H. (2018). Environmental DNA (eDNA) from the wake of the whales: droplet digital PCR for detection and species identification. *Frontiers in Marine Science*, *5*, 133.

Baker, S. J., Niemiller, M. L., Stites, A. J., Ash, K. T., Davis, M. A., Dreslik, M. J., & Phillips, C. A. (2018). Evaluation of environmental DNA to detect *Sistrurus catenatus* and *Ophidiomyces ophiodiicola* in crayfish burrows. *Conservation Genetics Resources*, 1–3.

Bálint, M., Nowak, C., Márton, O., Pauls, S. U., Wittwer, C., Aramayo, J. L., … Jansen, M. (2018). Accuracy, limitations and cost efficiency of eDNA-based community survey in tropical frogs. *Molecular Ecology Resources*, *18*(6), 1415–1426.

Barber, P. H., Palumbi, S. R., Erdmann, M. V, & Moosa, M. K. (2000). A marine Wallace's line? *Nature*, *406*(6797), 692–693.

Barnes, M. A., Turner, C. R., Jerde, C. L., Renshaw, M. A., Chadderton, W. L., & Lodge, D. M. (2014). Environmental conditions influence eDNA persistence in aquatic systems. *Environmental Science & Technology*, *48*(3), 1819–1827.

Basu, S., Campbell, H. M., Dittel, B. N., & Ray, A. (2010). Purification of specific cell population by fluorescence activated cell sorting (FACS). *JoVE (Journal of Visualized Experiments)*, *41*, e1546.

Batani, G., Bayer, K., Böge, J., Hentschel, U., & Thomas, T. (2019). Fluorescence in situ hybridization (FISH) and cell sorting of living bacteria. *Scientific Reports*,

9(1), 1–13.

Best, R. J., Lyczakowski, J. J., Abalde-Cela, S., Yu, Z., Abell, C., & Smith, A. G. (2016). Label-free analysis and sorting of Microalgae and Cyanobacteria in microdroplets by intrinsic chlorophyll fluorescence for the identification of fast growing strains. *Analytical Chemistry*, *88*(21), 10445–10451.

Branton, D., Deamer, D. W., Marziali, A., Bayley, H., Benner, S. A., Butler, T., … Huang, X. (2010). The potential and challenges of nanopore sequencing. In *Nanoscience and technology: A collection of reviews from Nature Journals* (pp. 261–268). World Scientific.

Brown, B. L., Watson, M., Minot, S. S., Rivera, M. C., & Franklin, R. B. (2017). MinION^{TM} nanopore sequencing of environmental metagenomes: a synthetic approach. *Gigascience*, *6*(3), gix007.

Cormack, B. P., Valdivia, R. H., & Falkow, S. (1996). FACS-optimized mutants of the green fluorescent protein (GFP). *Gene*, *173*(1), 33–38.

Davie, P. J. F., & Ng, P. K. L. (2012). Two new species of Orcovita (Crustacea: Decapoda: Brachyura: Varunidae) from anchialine caves on Christmas Island, eastern Indian Ocean. *Raffles Bulletin of Zoology*, *60*(1).

Davy, C. M., Kidd, A. G., & Wilson, C. C. (2015). Development and validation of environmental DNA (eDNA) markers for detection of freshwater turtles. *PLoS One*, *10*(7), e0130965.

Day, K., Campbell, H., Fisher, A., Gibb, K., Hill, B., Rose, A., & Jarman, S. N. (2019). Development and validation of an environmental DNA test for the endangered Gouldian finch. *Endangered Species Research*, *40*, 171–182.

De Souza, L. S., Godwin, J. C., Renshaw, M. A., & Larson, E. (2016). Environmental DNA (eDNA) detection probability is influenced by seasonal activity of organisms. *PLoS One*, *11*(10), e0165273.

De Ventura, L., Kopp, K., Seppälä, K., & Jokela, J. (2017). Tracing the quagga mussel invasion along the Rhine river system using eDNA markers: early detection and surveillance of invasive zebra and quagga mussels. *Management of Biological Invasions*, *8*(1), 101.

Deamer, D., Akeson, M., & Branton, D. (2016). Three decades of nanopore sequencing. *Nature Biotechnology*, *34*(5), 518.

Dell'Anno, A., & Corinaldesi, C. (2004). Degradation and turnover of extracellular DNA in marine sediments: ecological and methodological considerations.

Applied and Environmental Microbiology., *70*(7), 4384–4386.

Dieffenbach, C. W., Lowe, T. M., & Dveksler, G. S. (1993). General concepts for PCR primer design. *PCR Methods and Applications*, *3*(3), S30–S37.

Doi, H., Uchii, K., Takahara, T., Matsuhashi, S., Yamanaka, H., & Minamoto, T. (2015). Use of droplet digital PCR for estimation of fish abundance and biomass in environmental DNA surveys. *PLoS One*, *10*(3).

Dowle, E. J., Pochon, X., C. Banks, J., Shearer, K., & Wood, S. A. (2016). Targeted gene enrichment and high-throughput sequencing for environmental biomonitoring: A case study using freshwater macroinvertebrates. *Molecular Ecology Resources*, *16*(5), 1240–1254.

Eid, J., Fehr, A., Gray, J., Luong, K., Lyle, J., Otto, G., … Bettman, B. (2009). Real-time DNA sequencing from single polymerase molecules. *Science*, *323*(5910), 133–138.

Feist, S. M., Jones, R. L., Copley, J. L., Pearson, L. S., Berry, G. A., & Qualls, C. P. (2018). Development and validation of an environmental DNA method for detection of the Alligator Snapping Turtle (*Macrochelys temminckii*). *Chelonian Conservation and Biology*, *17*(2), 271–279.

Forsström, T., & Vasemägi, A. (2016). Can environmental DNA (eDNA) be used for detection and monitoring of introduced crab species in the Baltic Sea? *Marine Pollution Bulletin*, *109*(1), 350–355.

Fraija-Fernández, N., Bouquieaux, M.-C., Rey, A., Mendibil, I., Cotano, U., Irigoien, X., … Rodríguez-Ezpeleta, N. (2019). Marine water environmental DNA metabarcoding provides a comprehensive fish diversity assessment and reveals spatial patterns in a large oceanic area. *BioRxiv*, 864710. https://doi.org/10.1101/864710

Gamba, C., Fernández, E., Oliver, A., Tirado, M., Baeza, C., López-Parra, A. M., & Arroyo-Pardo, E. (2008). Population genetics and DNA preservation in ancient human remains from Eastern Spain. *Forensic Science International: Genetics Supplement Series*, *1*(1), 462–464.

Gauthier, M., Konecny-Dupré, L., Nguyen, A., Elbrecht, V., Datry, T., Douady, C., & Lefebure, T. (2020). Enhancing DNA metabarcoding performance and applicability with bait capture enrichment and DNA from conservative ethanol. *Molecular Ecology Resources*, *20*(1), 79–96.

Grealy, A., Macken, A., Allentoft, M. E., Rawlence, N. J., Reed, E., & Bunce, M.

(2016). An assessment of ancient DNA preservation in Holocene–Pleistocene fossil bone excavated from the world heritage Naracoorte Caves, South Australia. *Journal of Quaternary Science, 31*(1), 33–45.

Halstead, B. J., Wood, D. A., Bowen, L., Waters, S. C., Vandergast, A. G., Ersan, J. S., … Casazza, M. L. (2017). *An evaluation of the efficacy of using environmental DNA (eDNA) to detect giant gartersnakes (Thamnophis gigas)*. US Geological Survey.

Hindson, B. J., Ness, K. D., Masquelier, D. A., Belgrader, P., Heredia, N. J., Makarewicz, A. J., … Legler, T. C. (2011). High-throughput droplet digital PCR system for absolute quantitation of DNA copy number. *Analytical Chemistry, 83*(22), 8604–8610.

Horn, S. (2012). Target enrichment via DNA hybridization capture. In *Ancient DNA* (pp. 177–188). Springer.

Hui, T. H., Naruse, T., Fujita, Y., & Kiat, T. S. (2014). Observations on the fauna from submarine and associated anchialine caves in Christmas Island, Indian Ocean Territory, Australia. *Raffles Bulletin of Zoology, 30,* 406-418.

Ishii, K., & Fukui, M. (2001). Optimization of annealing temperature to reduce bias caused by a primer mismatch in multitemplate PCR. *Applied and Environmental Microbiology, 67*(8), 3753–3755.

Jeunen, G., Knapp, M., Spencer, H. G., Lamare, M. D., Taylor, H. R., Stat, M., … Gemmell, N. J. (2018). Environmental DNA (eDNA) metabarcoding reveals strong discrimination among diverse marine habitats connected by water movement. *Molecular Ecology Resources, 19*(2), 426-438.

Kelly, R. P., Closek, C. J., O'Donnell, J. L., Kralj, J. E., Shelton, A. O., & Samhouri, J. F. (2017). Genetic and manual survey methods yield different and complementary views of an ecosystem. *Frontiers in Marine Science, 3*, 283.

Klymus, K. E., Richter, C. A., Chapman, D. C., & Paukert, C. (2015). Quantification of eDNA shedding rates from invasive bighead carp *Hypophthalmichthys nobilis* and silver carp *Hypophthalmichthys molitrix*. *Biological Conservation, 183,* 77–84.

Koziol, A., Stat, M., Simpson, T., Jarman, S., DiBattista, J. D., Harvey, E. S., … Bunce, M. (2018). Environmental DNA metabarcoding studies are critically affected by substrate selection. *Molecular Ecology Resources, 19*(2), 366-376.

Kyne, P. M., & Pillans, R. D. (2014). Protocols for Surveying and Tagging Sawfishes

and River Sharks. *National Environmental Research Program Marine Biodiversity Hub*, Hobart.

Lacoursière-Roussel, A., Dubois, Y., Normandeau, E., & Bernatchez, L. (2016). Improving herpetological surveys in eastern North America using the environmental DNA method. *Genome*, *59*(11), 991–1007.

Lacoursière-Roussel, A., Côté, G., Leclerc, V., & Bernatchez, L. (2016). Quantifying relative fish abundance with eDNA: a promising tool for fisheries management. *Journal of Applied Ecology*, *53*(4), 1148–1157.

Lamb, P. D., Hunter, E., Pinnegar, J. K., Creer, S., Davies, R. G., & Taylor, M. I. (2019). How quantitative is metabarcoding: A meta-analytical approach. *Molecular Ecology*, *28*(2), 420–430.

Laroche, O., Wood, S. A., Tremblay, L. A., Ellis, J. I., Lejzerowicz, F., Pawlowski, J., Lear, G., Atalah, J., & Pochon, X. (2016). First evaluation of foraminiferal metabarcoding for monitoring environmental impact from an offshore oil drilling site. *Marine Environmental Research*, *120*, 225-235.

Laroche, O., Wood, S. A., Tremblay, L. A., Lear, G., Ellis, J. I., & Pochon, X. (2017). Metabarcoding monitoring analysis: the pros and cons of using co-extracted environmental DNA and RNA data to assess offshore oil production impacts on benthic communities. *PeerJ*, *5*, e3347.

Laroche, O., Wood, S. A., Tremblay, L. A., Ellis, J. I., Lear, G., & Pochon, X. (2018). A cross-taxa study using environmental DNA/RNA metabarcoding to measure biological impacts of offshore oil and gas drilling and production operations. *Marine Pollution Bulletin*, *127*, 97-107.

Lear, K. O., Gleiss, A. C., Whitty, J. M., Fazeldean, T., Albert, J. R., Green, N., … Morgan, D. L. (2019). Recruitment of a critically endangered sawfish into a riverine nursery depends on natural flow regimes. *Scientific Reports*, *9*(1), 17071. https://doi.org/10.1038/s41598-019-53511-9

Liu, L., Li, Y., Li, S., Hu, N., He, Y., Pong, R., … Law, M. (2012). Comparison of next-generation sequencing systems. *BioMed Research International*, *2012*.

Lourie, S. A., & Vincent, A. C. J. (2004). A marine fish follows Wallace's Line: the phylogeography of the three-spot seahorse (*Hippocampus trimaculatus*, Syngnathidae, Teleostei) in Southeast Asia. *Journal of Biogeography*, *31*(12), 1975–1985.

Maricic, T., Whitten, M., & Pääbo, S. (2010). Multiplexed DNA sequence capture of

mitochondrial genomes using PCR products. *PLoS One*, *5*(11).

Mascareñas, D. R., de Leon, R. O. D., & Delos Reyes, E. L. (2019). Watershed and Pahu-an cave stream interconnectivity in Bonliw, Torrijos, Marinduque, Philippines. *Environments*, *6*(2), 11.

McCormack, J. E., Harvey, M. G., Faircloth, B. C., Crawford, N. G., Glenn, T. C., & Brumfield, R. T. (2013). A phylogeny of birds based on over 1,500 loci collected by target enrichment and high-throughput sequencing. *PLoS One*, *8*(1).

Murakami, H., Yoon, S., Kasai, A., Minamoto, T., Yamamoto, S., Sakata, M. K., … Yamashita, Y. (2019). Dispersion and degradation of environmental DNA from caged fish in a marine environment. *Fisheries Science*, *85*(2), 327-337.

Ortiz, M., Legatzki, A., Neilson, J. W., Fryslie, B., Nelson, W. M., Wing, R. A., … Maier, R. M. (2014). Making a living while starving in the dark: metagenomic insights into the energy dynamics of a carbonate cave. *The ISME Journal*, *8*(2), 478.

Piaggio, A. J., Engeman, R. M., Hopken, M. W., Humphrey, J. S., Keacher, K. L., Bruce, W. E., & Avery, M. L. (2014). Detecting an elusive invasive species: a diagnostic PCR to detect Burmese python in Florida waters and an assessment of persistence of environmental DNA. *Molecular Ecology Resources*, *14*(2), 374–380.

Pochon, X., Zaiko, A., Fletcher, L. M., Laroche, O., & Wood, S. A. (2017). Wanted dead or alive? Using metabarcoding of environmental DNA and RNA to distinguish living assemblages for biosecurity applications. *PLoS One*, *12*(11), e0187636.

Port, J. A., O'Donnell, J. L., Romero-Maraccini, O. C., Leary, P. R., Litvin, S. Y., Nickols, K. J., … Kelly, R. P. (2016). Assessing vertebrate biodiversity in a kelp forest ecosystem using environmental DNA. *Molecular Ecology*, *25*(2), 527–541.

Raemy, M., & Ursenbacher, S. (2018). Detection of the European pond turtle (*Emys orbicularis*) by environmental DNA: is eDNA adequate for reptiles? *Amphibia-Reptilia*, *39*(2), 135–143.

Sassoubre, L. M., Yamahara, K. M., Gardner, L. D., Block, B. A., & Boehm, A. B. (2016). Quantification of environmental DNA (eDNA) shedding and decay rates for three marine fish. *Environmental Science & Technology*, *50*(19), 10456–10464.

Schloss, P. D., Gevers, D., & Westcott, S. L. (2011). Reducing the effects of PCR

amplification and sequencing artifacts on 16S rRNA-based studies. *PLoS One*, *6*(12).

Schmalenberger, A., Schwieger, F., & Tebbe, C. C. (2001). Effect of primers hybridizing to different evolutionarily conserved regions of the small-subunit rRNA gene in PCR-based microbial community analyses and genetic profiling. *Applied and Environmental Microbiology*, *67*(8), 3557–3563.

Shokralla, S., Gibson, J., King, I., Baird, D., Janzen, D., Hallwachs, W., & Hajibabaei, M. (2016). Environmental DNA barcode sequence capture: targeted, PCR-free sequence capture for biodiversity analysis from bulk environmental samples. *BioRxiv*, 87437.

Sigsgaard, E. E., Jensen, M. R., Winkelmann, I. E., Møller, P. R., Hansen, M. M., & Thomsen, P. F. (2019). Population-level inferences from environmental DNA—Current status and future perspectives. *Evolutionary Applications*, *13*(2), 245-262.

Simpfendorfer, C. A., Kyne, P. M., Noble, T. H., Goldsbury, J., Basiita, R. K., Lindsay, R., … Jerry, D. R. (2016). Environmental DNA detects Critically Endangered largetooth sawfish in the wild. *Endangered Species Research*, *30*, 109–116.

Sipos, R., Székely, A. J., Palatinszky, M., Révész, S., Márialigeti, K., & Nikolausz, M. (2007). Effect of primer mismatch, annealing temperature and PCR cycle number on 16S rRNA gene-targetting bacterial community analysis. *FEMS Microbiology Ecology*, *60*(2), 341–350.

Stat, M., John, J., DiBattista, J. D., Newman, S. J., Bunce, M., & Harvey, E. S. (2019). Combined use of eDNA metabarcoding and video surveillance for the assessment of fish biodiversity. *Conservation Biology*, *33*(1), 196-205.

Taberlet, P., Bonin, A., Zinger, L., & Coissac, E. (2018). *Environmental DNA: For Biodiversity Research and Monitoring*. Oxford University Press.

Thomsen, P. F., Kielgast, J., Iversen, L. L., Møller, P. R., Rasmussen, M., & Willerslev, E. (2012). Detection of a diverse marine fish fauna using environmental DNA from seawater samples. *PLoS One*, *7*(8). https://doi.org/10.1371/journal.pone.0041732

Thomsen, P. F., Kielgast, J. O. S., Iversen, L. L., Wiuf, C., Rasmussen, M., Gilbert, M. T. P., … Willerslev, E. (2012). Monitoring endangered freshwater biodiversity using environmental DNA. *Molecular Ecology*, *21*(11), 2565–2573.

Titelius, M. A., Sampey, A., & Hass, C. G. (2009). Crustaceans of Mermaid (Rowley Shoals), Scott and Seringapatam Reefs, north Western Australia. *Records of the Western Australian Museum Supplement*, *77*, 145–176.

Travers, M. J., Potter, I. C., Clarke, K. R., Newman, S. J., & Hutchins, J. . (2010). The inshore fish faunas over soft substrates and reefs on the tropical west coast of Australia differ and change with latitude and bioregion. *Journal of Biogeography*, *37*(1), 148–169.

Tréguier, A., Paillisson, J., Dejean, T., Valentini, A., Schlaepfer, M. A., & Roussel, J. (2014). Environmental DNA surveillance for invertebrate species: advantages and technical limitations to detect invasive crayfish *Procambarus clarkii* in freshwater ponds. *Journal of Applied Ecology*, *51*(4), 871–879.

Truelove, N. K., Andruszkiewicz, E. A., & Block, B. A. (2019). A rapid environmental DNA method for detecting white sharks in the open ocean. *Methods in Ecology and Evolution*, *10*(8), 1128–1135.

van der Valk, T., Lona Durazo, F., Dalén, L., & Guschanski, K. (2017). Whole mitochondrial genome capture from faecal samples and museum-preserved specimens. *Molecular Ecology Resources*, *17*(6), e111–e121.

von Ammon, U., Wood, S. A., Laroche, O., Zaiko, A., Tait, L., Lavery, S., ... & Pochon, X. (2018). Combining morpho-taxonomy and metabarcoding enhances the detection of non-indigenous marine pests in biofouling communities. *Scientific Reports*, *8*(1), 1-11.

von Ammon, U., Wood, S. A., Laroche, O., Zaiko, A., Lavery, S. D., Inglis, G. J., & Pochon, X. (2019). Linking environmental DNA and RNA for improved detection of the marine invasive fanworm Sabella spallanzanii. *Frontiers in Marine Science*, *6*, 621.

Wei, S., Weiss, Z. R., & Williams, Z. (2018). Rapid multiplex small DNA sequencing on the MinION nanopore sequencing platform. *G3: Genes, Genomes, Genetics*, *8*(5), 1649–1657.

Weltz, K., Lyle, J. M., Ovenden, J., Morgan, J. A. T., Moreno, D. A., & Semmens, J. M. (2017). Application of environmental DNA to detect an endangered marine skate species in the wild. *PLoS One*, *12*(6).

Wilcox, T. M., McKelvey, K. S., Young, M. K., Jane, S. F., Lowe, W. H., Whiteley, A. R., & Schwartz, M. K. (2013). Robust detection of rare species using environmental DNA: the importance of primer specificity. *PLoS One*, *8*(3).

Wilcox, T. M., Zarn, K. E., Piggott, M. P., Young, M. K., McKelvey, K. S., & Schwartz, M. K. (2018). Capture enrichment of aquatic environmental DNA: A first proof of concept. *Molecular Ecology Resources*, *18*(6), 1392–1401.

Wilson, B. (2014). Kimberley marine biota. History and environment. *Records of the Western Australian Museum Supplement*, *84*, 1–18.

Wood, S. A., Pochon, X., Laroche, O., von Ammon, U., Adamson, J., & Zaiko, A. (2019). A comparison of droplet digital polymerase chain reaction (PCR), quantitative PCR and metabarcoding for species-specific detection in environmental DNA. *Molecular Ecology Resources*, *19*(6), 1407–1419.

Wood, S. A., Biessy, L., Latchford, J. L., Zaiko, A., von Ammon, U., Audrezet, F., Cristescu, M. E., & Pochon, X. (2020). Release and degradation of environmental DNA and RNA in a marine system. *Science of The Total Environment*, *704*, 135314.

Yamamoto, S., Minami, K., Fukaya, K., Takahashi, K., Sawada, H., Murakami, H., … Horiuchi, T. (2016). Environmental DNA as a 'snapshot'of fish distribution: A case study of Japanese jack mackerel in Maizuru Bay, Sea of Japan. *PLoS One*, *11*(3).

— Appendices —

Appendix I. Copyright statements

Chapter 2: Published in the journal Molecular Ecology. As first author, permission is automatically granted to reproduce this copyright material if it constitutes less than half of the total material in such publication.

To Whom It May Concern, I, Katrina West, contributed to the conceptualisation of the study, collected and processed the samples, analysed the data, and wrote and edited the resulting manuscript:

West K.M, Stat M, Harvey E.S, Skepper C.L, DiBattista J.D, Richards Z.T, Travers M.J, Newman S.J, Bunce M. (2020). eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem. *Molecular Ecology*. https://doi.org/10.1111/mec.15382.

_____

I, as a co-author, endorse that this level of contribution by the candidate indicated above is appropriate.

Michael Stat _____

Euan Harvey _____

Craig Skepper _____

Joseph DiBattista _____

Zoe Richards _____

Michael Travers _____

Stephen Newman _____

Michael Bunce _____

Chapter 3: In preparation for publication

To Whom It May Concern, I, Katrina West, contributed to the conceptualisation of the study, collected and processed the samples, analysed the data, and wrote and edited the resulting manuscript:

West K.M, Travers M.J, Stat M, Harvey E.S, Richards Z.T, DiBattista J.D, Skepper C.L, Heydenrych M, Newman S.J, Bunce M. Large-scale eDNA metabarcoding survey reveals marine biogeographic break and transitions over tropical north-western Australia. *[in preparation]*

_____

I, as a co-author, endorse that this level of contribution by the candidate indicated above is appropriate.

Michael Travers _____

Michael Stat _____

Euan Harvey _____

Zoe Richards _____

Joseph DiBattista _____

Craig Skepper _____

Matthew Heydenrych _____

Stephen Newman _____

Michael Bunce _____

Chapter 4: In preparation for publication

To Whom It May Concern, I, Katrina West, contributed to funding acquisition, the conceptualisation of the study, collected and processed the samples, analysed the data, and wrote and edited the resulting manuscript:

West K.M, Richards Z.T, Harvey E.S, Susac R, Grealy A, Bunce M. Under the karst: detecting hidden subterranean assemblages using eDNA metabarcoding in the caves of Christmas Island, Australia. *[in preparation]*

_____

I, as a co-author, endorse that this level of contribution by the candidate indicated above is appropriate.

Zoe Richards _____

Euan Harvey _____

Robert Susac _____

Alicia Grealy _____

Michael Bunce _____

Appendix II. Research output

II.I Publications

West K.M, Stat M, Harvey E.S, Skepper C.L, DiBattista J.D, Richards Z.T, Travers M.J, Newman S.J, Bunce M. (2020). eDNA metabarcoding survey reveals fine-scale coral reef community variation across a remote, tropical island ecosystem. *Molecular Ecology*. https://doi.org/10.1111/mec.15382.

II.II Awards and conference presentations

2019 *eDNA metabarcoding in the offshore and coastal waterways of the Kimberley, WA* [oral presentation]
Australian Society of Fish Biology (ASFB) conference, Canberra.

2018 Curtin HDR Mobility Scholarship
Facilitated travel to the Hawaii Institute of Marine Biology (HIMB) for eDNA collaboration.

2018 *Comprehensive eDNA metabarcoding survey reveals localised signals & spatially-driven marine community variation across the tropical Cocos (Keeling) Islands, Australia* [oral presentation]
Combined Biological Sciences Meeting (CBSM) conference, Perth.

2018 Two-week placement in the European Molecular Laboratory (EMBL) Australia PhD Course, Sydney.

2017/18 Holsworth Wildlife Research Endowment [grant]
Funded eDNA Christmas Island project (Chapter 4).

2017 *The application of eDNA metabarcoding for marine biodiversity monitoring at the Cocos-Keeling Islands* [oral presentation]
Australian Marine Science Association (AMSA) conference, Darwin.

2017 *The application of eDNA metabarcoding for marine biodiversity monitoring at the Cocos-Keeling Islands* [oral presentation]
International Barcode of Life (IBOL) conference, South Africa.

2017 Australian Government Research Training Program Stipend Scholarship

II.III Public outreach

2019 *eDNA metabarcoding for subterranean applications* [oral presentation]
Australasian Land and Groundwater Association (ALGA)

2019 *Into Darkness: Detecting cave and deep-sea animals with DNA* [oral presentation]
The Aquarium of Western Australia (AQWA)